The Wempe Einheits-Chronometer: a build book

Generated from the working model of Gerhard D. Wempe's Einheits-Chronometer, as its works drawings were last revised: every part card's lines of its parts list (by drawing number), each line's fit as measured on the model, its material and treatment as its own drawing gives them (the title block's Werkstoff, the notes' hardening and tempering, the surface marks; nothing from practice: where a sheet gives none, or the part has no sheet in the set, it says so), how it is made (CNC milled or turned, by hand, or bought in), and the part's drawing from the model's solids, in millimetres. The works drawings themselves are Johannes Altmeppen's photographs (dg-chrono.de), not reproduced here; the transcriptions in the repository give what is read off each. What the model estimates is listed in the Wempe README's "Estimated, not from the drawings".

Two builds are worked out to navigate by (the README's "The build"): the 1940s build keeps the drawings' carbon-steel mainspring, the 1950s build takes a Nivaflex strip; both thin the gold spring at its bend. This book shows the 1950s build. Wind it as both builds are judged: turn the key to the end of each stroke, hold it and let it back; let go outright, the fusee's blow dents its click and bends the maintaining ratchet's teeth (tools/wempe_letgo_fix.js).

The gold spring thinned at its bend: the 1950s build's, a deviation from the drawings (the drawn part is beside it on its sheet); Mainspring in Nivaflex 45/5: the 1950s build's, a deviation from the drawings (the drawn part is beside it on its sheet). Each part card switches between the two.

Each part has its drawing, laid out as the Wempe's own works drawings are (first-angle projection, sizes in mm), drawn from the working model's solids by tools/drawbook.py, with a second sheet of its parts-list lines, holes and contour; each assembly a sheet with its pieces ballooned and listed. The works drawings themselves are not reproduced.

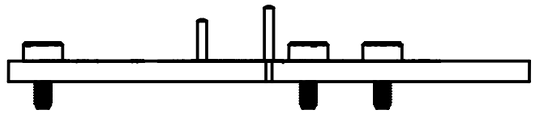
Contents

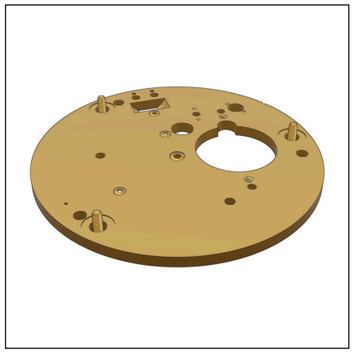
1. Pillar plate (Unterplatte)
   2. Pillar plate
   3. Pillar screws, dial side
   4. Centre arbor's bush (Messinglager WN-302)
   5. Stud bushes (Futter)
   6. Hand-setting pinion's bush
   7. Studs (Anrichtstifte)
8. Top plate (Oberplatte)
   9. Top plate
   10. Pillar screws, train side
   11. Stop pin (Anschlagstift)
   12. Train bushes (Messinglager WN-300, WN-301)
13. Pillars (Pfeiler)
14. Movement ring (Werkring)
   15. Movement ring
   16. Movement ring screws
   17. Ring's pin to the case (Stellstift 3)
   18. Plate's pin
19. Barrel bridge (Federhausbrücke)
   20. Barrel bridge
   21. Click (Sperrkegel)
   22. Click screw
   23. Barrel bridge screws
   24. Barrel bridge steady pins
25. Fusee bush (Schneckenfutter)
   26. Fusee bush
   27. Fusee bush screws
28. Dust bush (Staubfutter)
29. Cork ring (Korkring)
30. Upper balance cock (Unruhkloben oben)
   31. Upper balance cock
   32. Upper cock screw
   33. Upper cock steady pins
34. Lower balance cock (Unruhkloben unten)
   35. Lower balance cock
   36. Lower cock screw
   37. Lower cock steady pins
38. Intermediate bridge (Zwischenbrücke)
   39. Intermediate bridge
   40. Intermediate bridge screws
   41. Intermediate bridge steady pins
   42. Intermediate wheel's lower bush (Messinglager WN-300)
   43. Seconds jewel's setting (Fassung)
   44. Seconds jewel (Lagerstein f. Sek.)
45. Centre wheel (Minutenrad)
   46. Centre pinion and arbor (Minutentrieb)
   47. Centre wheel (Minutenrad)
48. Intermediate wheel (Zwischenrad)
   49. Intermediate pinion and arbor (Zwischentrieb)
   50. Intermediate wheel (Zwischenrad)
51. Seconds wheel (Sekundenrad)
   52. Seconds pinion and arbor (Sekundentrieb)
   53. Seconds wheel (Sekundenrad)
54. Balance (Unruh)
   55. Balance rim and arm
   56. Compensation masses
   57. Timing screws and weights
   58. Balance staff
   59. Balance hub (Unruhbutzen)
   60. Hub screws
61. Hairspring (Spiralfeder)
   62. Hairspring
   63. Hairspring collet (Spiralrolle)
64. Hairspring stud (Klötzchen)
   65. Stud
   66. Stud screw
67. Impulse and discharge rollers (Heberolle, Auslöserolle)
   68. Impulse roller and stone
   69. Discharge roller and stone
70. Escape wheel (Gangrad)
   71. Escape wheel
   72. Escape pinion and arbor

73. Escape wheel collet (Gangradbutzen)
74. Spring detent (Gangfeder) and gold spring
75. Detent with its locking stone
76. Gold spring (Goldfeder)
77. Detent foot screw and pins
78. Detent block (Gangfederkloben)
79. Detent block
80. Block screws and pins
81. Balance jewels, upper
82. Upper hole jewel
83. Upper endstone
84. Setting screws
85. Balance jewels, lower
86. Lower hole jewel
87. Lower endstone
88. Setting screws
89. Escape wheel endstone, upper
90. Endstone and setting
91. Setting screws
92. Escape wheel jewel, upper
93. Escape wheel jewel, lower
94. Escape wheel endstone, lower
95. Endstone and setting
96. Setting screws
97. Fusee (Schnecke)
98. Fusee body (Schneckenkörper)
99. Fusee arbor (Schneckenwelle)
100. Click wheel (Schneckensperrrad)
101. Click wheel screws (WN-19)
102. Chain pin (Kettenstift)
103. Cover plate (Deckplatte)
104. Cover plate
105. Cover plate screw (WN-20)
106. Vorreiber
107. Vorreiber
108. Vorreiber pin (Vorreiberstift)
109. Dust plate (Staubplatte)
110. Fusee wheel (Schneckenrad)
111. Maintaining work (Gegensperrrad, Gegenfeder)
112. Maintaining ratchet (Gegensperrrad)
113. Clicks (Sperrkegel)
114. Click screws (WN-21)
115. Click springs (Sperrfeder)
116. Click spring pins
117. Maintaining spring (Gegenfeder)
118. Maintaining spring's pins (Stellstifte)
119. Maintaining detent (Gegensperrkegel)
120. Detent click
121. Detent arbor
122. Detent spring (Feder) and its stud
123. Stop work (Stellungssteg)
124. Stop bar (Stellungssteg)
125. Stop bar spring (Stegfeder)
126. Stop bar's pin (Stift)
127. Stop bar spring's pin (Stift)
128. Barrel (Federhaus)
129. Barrel drum (Federhaustrommel)
130. Barrel lid (Federhausdeckel)
131. Spring hook block (Federhaushaken)
132. Steg (spring brace)
133. Barrel arbor (Federkern)
134. Barrel arbor (Federkern)
135. Arbor hook (Federhaken)
136. Set-up ratchet (Sperrad)
137. Mainspring (Zugfeder)
138. Fusee chain (Kette)
139. Chain links
140. Fusee band (Schneckenband)
141. Band (Schneckenband)
142. Band's round hook (Bandhaken rund)
143. Band's drum hook (Bandhaken f. Trommel)
144. Cannon tube (Minutenrohr)
145. Change wheel (Wechselrad)
146. Change wheel
147. Change wheel pipe (Wechselbutzen)
148. Change pinion (Wechseltrieb)
149. Hour wheel (Stundenrad)
150. Hour wheel
151. Hour wheel pipe (Stundenbutzen)
152. Up-and-down wheel (Auf- und Abrad)
153. Up-and-down wheel
154. Up-and-down wheel pipe
155. Up-and-down pinion (Auf- und Abtrieb)
156. Hands
157. Minute hand (Minutenzeiger)
158. Hour hand (Stundenzeiger)
159. Up-and-down hand (Auf- und Abzeiger)
160. Seconds hand (Sekundenzeiger) and collet
161. Dial (Zifferblatt)
162. Dial
163. Dial screws
164. Dial pin
165. Bowl (Gehäusetopf)
166. Bowl
167. Case brackets (Gehäusekloben)
168. Locking bracket (Feststellkloben)
169. Bracket screws (WN-78)
170. Closing plate (Verschlußplatte)
171. Closing plate
172. Closing plate screw
173. Closing plate knob
174. Closing spring (Verschlußfeder)
175. Bezel and glass (Glasrand)
176. Bezel (Glasrand)
177. Glass
178. Gimbal ring (Kardanring)
179. Gimbal ring
180. Ring bracket (Kardanringkloben)
181. Ring bracket screws
182. Inner pivot screws (WN-79)
183. Lock nuts (WN-400)
184. Wooden box (Holzkasten)
185. Box
186. Outer pivot screws (WN-80)
187. Bearing discs
188. Lid stay
189. Handles (Griff f. Chronometerkasten)
190. Handle screws
191. Hinge
192. Pins over the pivots
193. Keeper
194. Key holder
195. Key holder's fitting
196. Key holder's screws
197. Lid (Deckel)
198. Lid
199. Lid glass
200. Hinge
201. Lid stay
202. Maker's plate (Firmenschild)
203. Latch tongue (Zunge)
204. Tongue screw
205. Tongue knob
206. Outer box (Überkasten)
207. Outer box
208. Rim strip
209. Front and back cushions
210. End cushions
211. Floor cushion
212. Strap
213. Strap screws
214. Eyes
215. Eye screws
216. Hinges
217. Hook eye
218. Outer box lid

219. Outer box lid
220. Lid cushion
221. Hinge leaves
222. Hook
223. Transit lock (Feststeller)
224. Winding key (Chronometerschlüssel)
225. Key handle (Schlüsselgriff)
226. Key body (Schlüsselmittelteil)
227. Key tube (Schlüsselrohr)
228. Coil spring (Wendelfeder)
229. Grooved pin (Kerbstift)

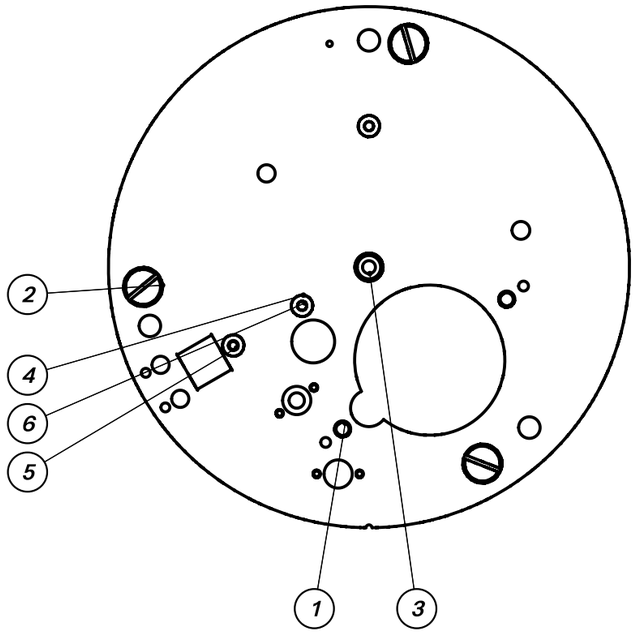
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
6	1	Studs (Anrichtstifte)	05 02 00-6	not on the drawing
5	1	Hand-setting pinion's bush	WN-hs	not on the drawing
4	2	Stud bushes (Futter)	05 08 02-2	not on the drawing
3	1	Centre arbor's bush (Messinglager WN-302)	WN-302	not on the drawing
2	3	Pillar screws, dial side	WN-56	not on the drawing
1	1	Pillar plate	05 08 02-1	„Messingblech 4 / SonderMsα-β

Werkstoff / Material: not on the drawing

Maßstab / Scale: **1:1**

Behandlung / Treatment:

Blatt / Sheet: **1 / 1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

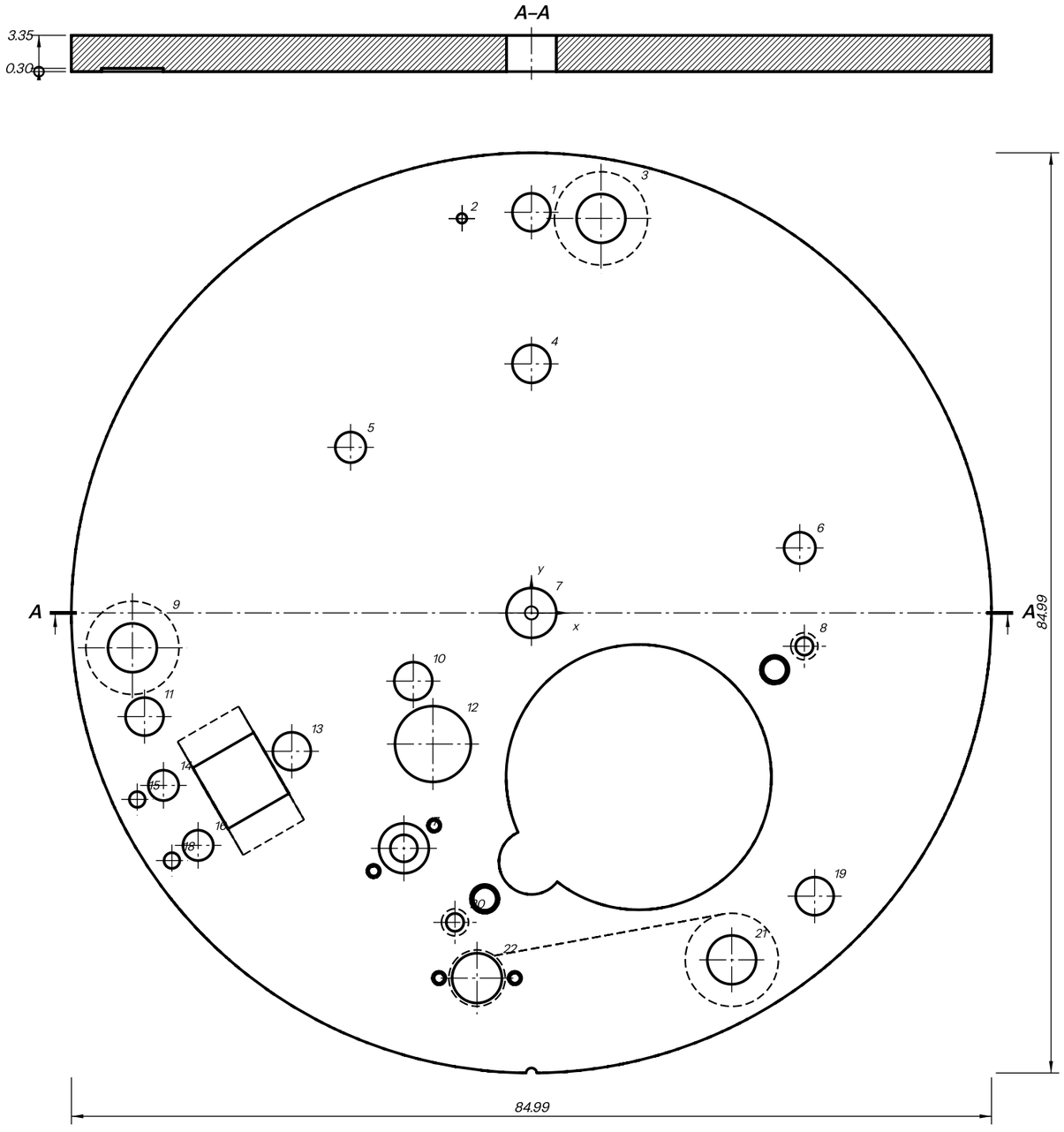
Zeichnung Nr. / Drawing no.: **05 02 00-6**

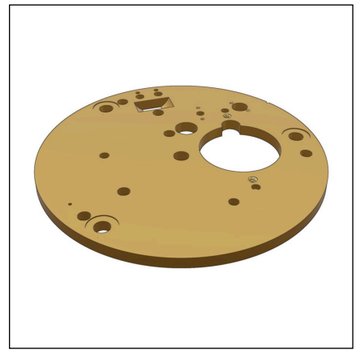
Teil / Part: pillarPlate

Benennung / Name: **Pillar plate (Unterplatte)**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Messingblech 4 / SonderMs α-βGef."*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 02-1***
Behandlung / Treatment *none on the drawing*			*Teil / Part* *pillarPlate.plate*
		Benennung / Name ***Pillar plate***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Pillar plate — sheet 2

The lower plate, under the dial: the foundation of the movement. The three pillars stand in seats turned in its inner face, each held by a screw from the dial side; it sits in the movement ring. It carries the train's lower pivots, the intermediate bridge on its dial side across the opening for the intermediate wheel, and the motion work under the dial. Every hole of its table is cut but the seconds contact's ("only when needed"). Near its rim at 4 o'clock seen from the dial, the window and four holes for the hand-setting work, which set the hands from outside the case: its sheet strikes the holes through ("only when the hand-setting is fitted"), but Wempe 3955's plate is cut and drilled for it with nothing fitted but the bush in its pinion's hole, and so it is drawn. The drawings are the sheets as last revised (about 1946-48).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 02-1	Unterplatte (pillar plate)	1	„Messingblech 4 / SonderMs α-βGef."	none on the drawing	takes Pfeiler (pillar): in it (run): 2.348 mm, least gap 0.0109; Pfeiler (pillar): on it: 0 mm; Werkring (movement ring): on it: 0 mm; Zwischenbrücke (intermediate bridge; its sheet not in the set, the assembly 05 08 04): on it: 0 mm; Futter (stud bush): in it (press): 3.352 mm, least gap 0.0019; Pillar screw, dial side (WN-56, by the coloured section's label): on it: 0 mm; and 12 more

Test on the bench (05 08 02-1). The train's depths, the plates bored and reamed, before the bushes and jewels go in: Centres 22.650, 18.120, 12.630, 11.850 mm (fusee-centre, centre-intermediate, intermediate-seconds, seconds-escape: the hole tables' coordinates), each within ±25 µm (gearing.js --model wempe: the seconds mesh's window less its pivots' shake; the tables give the places to 0.001 and no tolerance). A hole that errs should err wide: each mesh takes far more wide than close (-240/+436, -136/+355, -112/+281, -100/+263 µm). Closer than its window less the pivots' shake (220, 101, 62, 50 µm close): re-bush the hole, bored on the drawn centre.

What it does (05 08 02-1). The lower plate, under the dial: the foundation of the movement (its assembly, Unterplatte kompl., is 05 08 02).

Coordinates from the datum, the movement's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.35.

Holes

Hole	x	y	Size
1	0.00	37.00	Ø 3.51 through
2	-6.42	36.44	Ø 0.90 through
3	6.43	36.44	Ø 4.51 through (3.05 thick at 0.30-3.35); sink Ø 8.60 × 0.30 from the bottom
4	0.00	23.00	Ø 3.51 through
5	-16.71	15.29	Ø 2.82 through
6	24.78	6.00	Ø 2.90 through
7	0.00	0.00	Ø 4.60 through
8	25.21	-3.08	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
9	-36.86	-3.22	Ø 4.51 through (3.05 thick at 0.30-3.35); sink Ø 8.60 × 0.30 from the bottom
10	-10.91	-6.30	Ø 3.51 through
11	-35.74	-9.58	Ø 3.51 through
12	-9.10	-12.11	Ø 7.00 through
13	-22.14	-12.78	Ø 3.51 through
14	-33.99	-15.93	Ø 2.80 through
15	-36.41	-17.23	Ø 1.45 through
16	-30.79	-21.47	Ø 2.80 through
17	-11.78	-21.74	Ø 2.50 through (3.10 thick at 0.00-3.10); sink Ø 4.60 × 0.25 from the top
18	-33.21	-22.87	Ø 1.45 through

19	26.16	-26.16	Ø 3.51 through
20	-7.06	-28.58	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
21	18.50	-32.04	Ø 4.51 through (3.05 thick at 0.30-3.35)
22	-5.03	-33.73	Ø 4.60 through (0.50 thick at 2.85-3.35); sink Ø 5.21 × 2.85 from the bottom

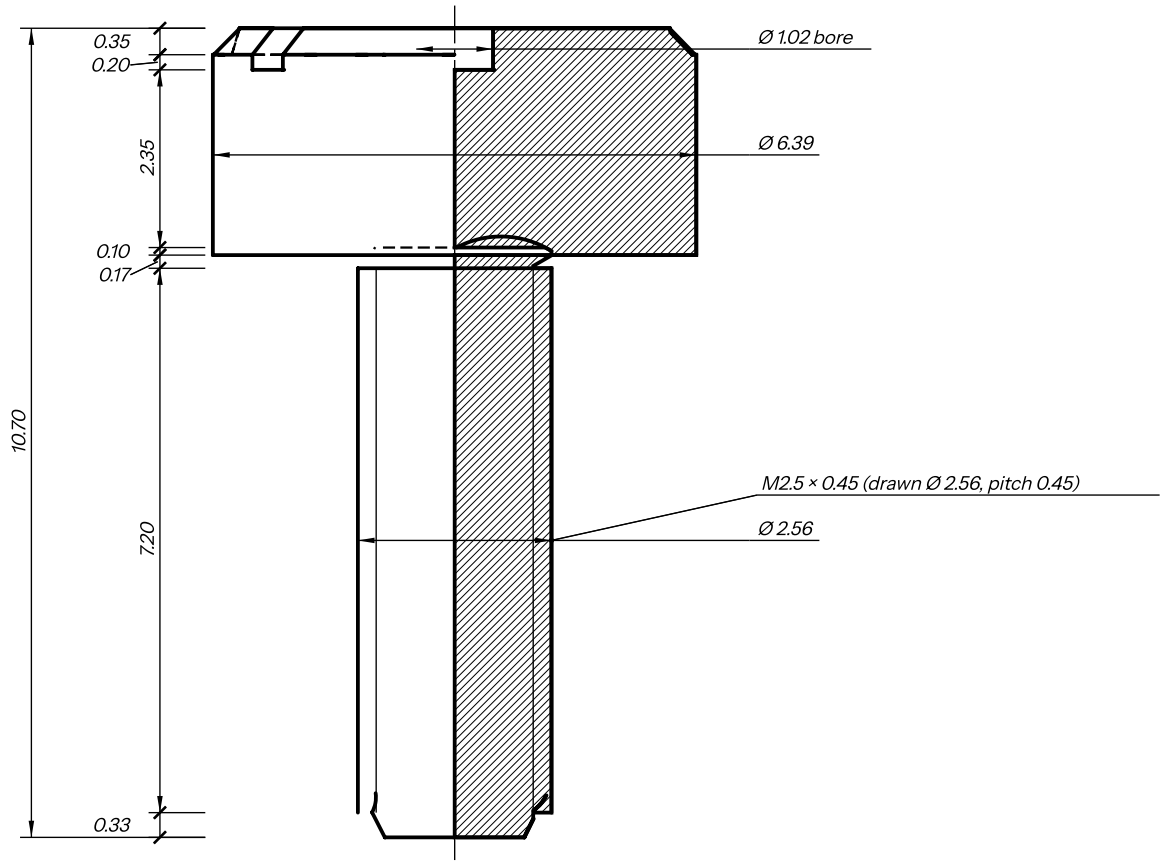
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

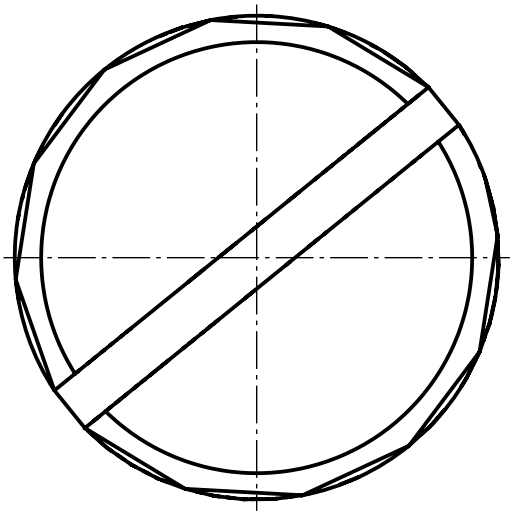
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-41.05, -11.00	0.50, -42.49	0.00, 0.00	42.49
Outline	2	arc anticlockwise	0.50, -42.49	-0.50, -42.49	0.00, -42.49	0.50
Outline	3	arc clockwise	-0.50, -42.49	-41.05, -11.00	0.00, 0.00	42.49
Cut-out 5	4	arc anticlockwise	-1.51, -10.75	-1.23, -20.26	9.91, -15.17	12.25
Cut-out 5	5	arc anticlockwise	-1.23, -20.26	2.38, -24.83	0.00, -23.00	3.00
Cut-out 5	6	arc anticlockwise	2.38, -24.83	-1.51, -10.75	9.91, -15.17	12.25
Cut-out 14	7	line	-5.08, -27.48	-5.08, -27.48		
Cut-out 14	8	line	-5.08, -27.48	-5.08, -27.48		
Cut-out 14	9	line	-5.08, -27.48	-5.08, -27.48		
Cut-out 14	10	line	-5.08, -27.48	-5.08, -27.48		
Cut-out 19	11	line	-8.50, -20.01	-8.50, -20.01		
Cut-out 19	12	arc anticlockwise	-8.50, -20.01	-8.50, -20.01	-1.77, -11.05	11.20
Cut-out 19	13	line	-8.50, -20.01	-8.50, -20.01		
Cut-out 21	14	line	-8.55, -34.35	-8.55, -34.35		
Cut-out 21	15	arc anticlockwise	-8.55, -34.35	-8.55, -34.35	80.25, -38.30	88.89
Cut-out 21	16	line	-8.55, -34.35	-8.55, -34.35		
Cut-out 23	17	line	-10.26, -23.46	-10.26, -23.46		
Cut-out 23	18	line	-10.26, -23.46	-10.26, -23.46		
Cut-out 23	19	line	-10.26, -23.46	-10.26, -23.46		
Cut-out 28	20	line	-28.03, -19.94	-22.40, -16.69		
Cut-out 28	21	line	-22.40, -16.69	-25.65, -11.06		
Cut-out 28	22	line	-25.65, -11.06	-31.28, -14.31		
Cut-out 28	23	line	-31.28, -14.31	-28.03, -19.94		

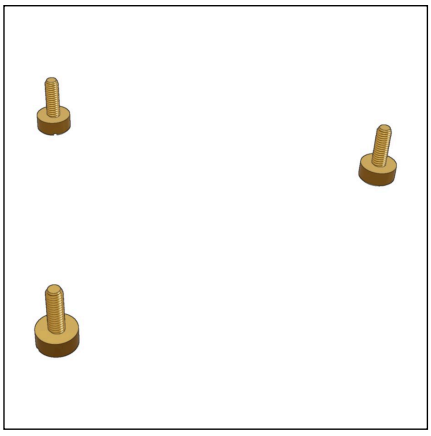
Levels

Faces in section A-A, from the bottom face: 0.30, 3.35.



View from above





Isometric view, not to scale

3 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-56*** *Teil / Part* *pillarPlate.screws*
Behandlung / Treatment		*Benennung / Name* ***Pillar screws, dial side***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Pillar screws, dial side — sheet 2

Three pillar screws (WN-56) from the dial side into the pillars' tapped ends, brass as Wempe 3955's.

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

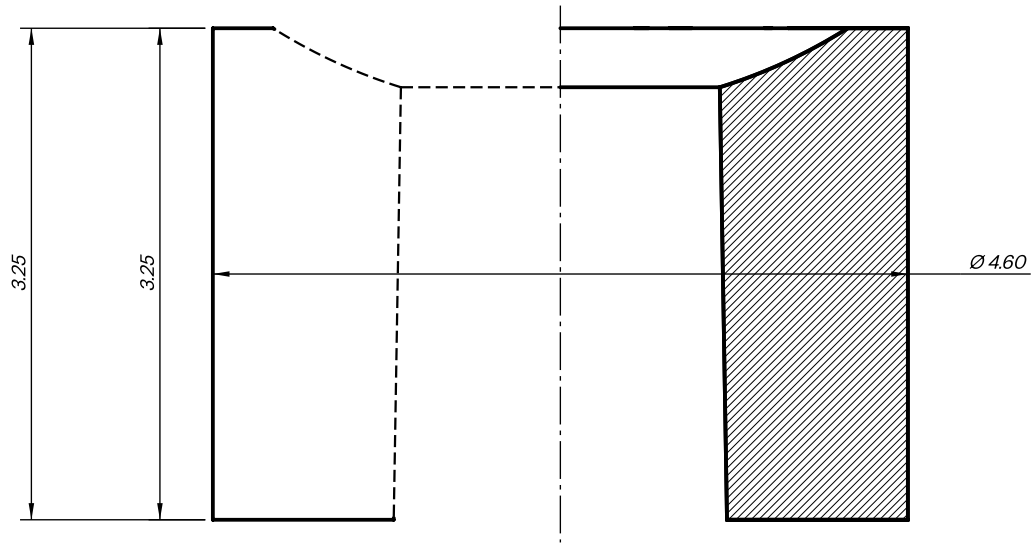
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-56	Pillar screw, dial side (WN-56, by the coloured section's label)	3	not on the drawing		in 050800-1 (tap): 7.448 mm, least gap 0.0253; on 050802-1: 0 mm

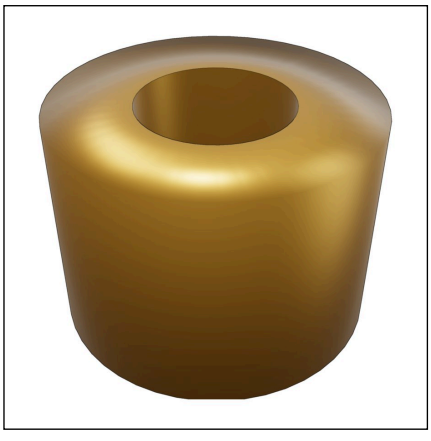
What it does (WN-56). Hold each pillar to the pillar plate, from the dial side.

Turned about its axis; length 10.70, largest Ø 6.40. Diameters: Ø 2.56, Ø 6.39; bores Ø 1.02. Shoulders from the lower end: 0.00, 0.33, 7.53, 7.70, 7.80, 10.15, 10.35, 10.70.

Thread

Outside thread 0.33–7.53 from the lower end, 7.20 long: drawn Ø 2.56 over its crests, 2.08 at its roots, pitch 0.45. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-302*** *Teil / Part* *pillarPlate.bush*
Behandlung / Treatment		*Benennung / Name* ***Centre arbor's bush (Messinglager WN-302)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Centre arbor’s bush (Messinglager WN-302) — sheet 2

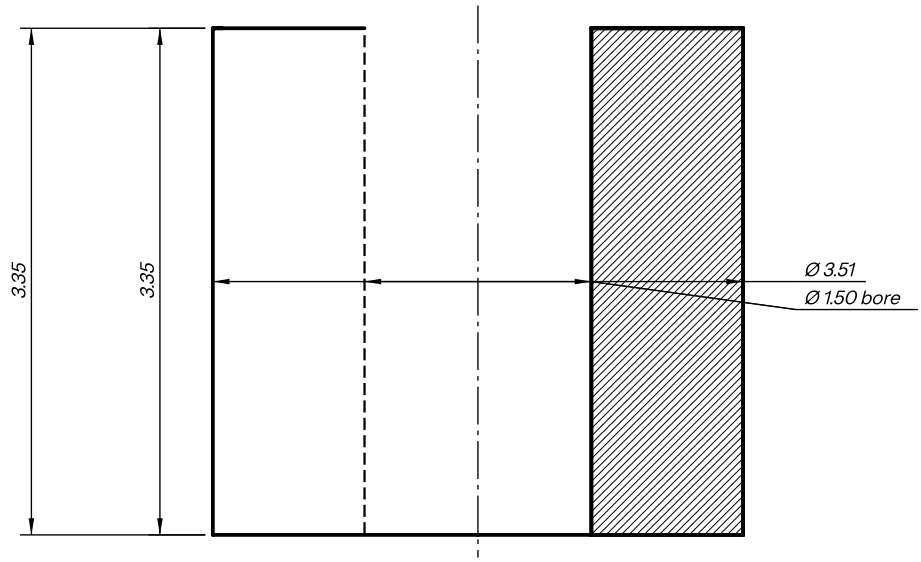
A brass bush pressed into the pillar plate, in which the centre arbor’s long journal turns; its inner face is set 0.1 under the plate’s, and the collar at the end of the centre wheel’s seat bears on it.

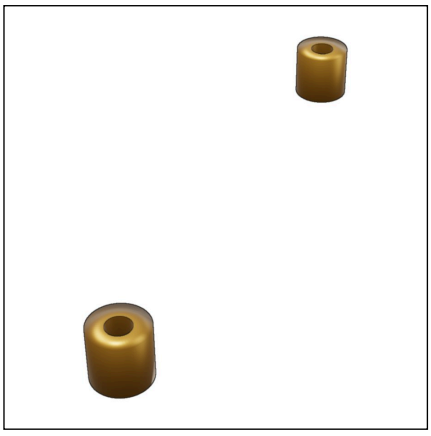
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-302	Messinglager WN-302 (brass bush, centre arbor's journal)	1	not on the drawing		in 050802-1 (press): 3.247 mm, least gap 0.0028

What it does (WN-302). The centre arbor's lower bearing, pressed into the pillar plate's hole 1, 0.1 under its inner face.

Turned about its axis; length 3.25, largest Ø 4.60. Diameters: Ø 4.60. Shoulders from the lower end: 0.00, 3.25.





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 08 02-2** *Teil / Part* *pillarPlate.studBush*
Behandlung / Treatment		*Benennung / Name* ***Stud bushes (Futter)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Stud bushes (Futter) — sheet 2

Two brass bushes pressed into the pillar plate from its inner face, holding the studs on which the up-and-down wheel and the motion work's change wheel turn on the dial side.

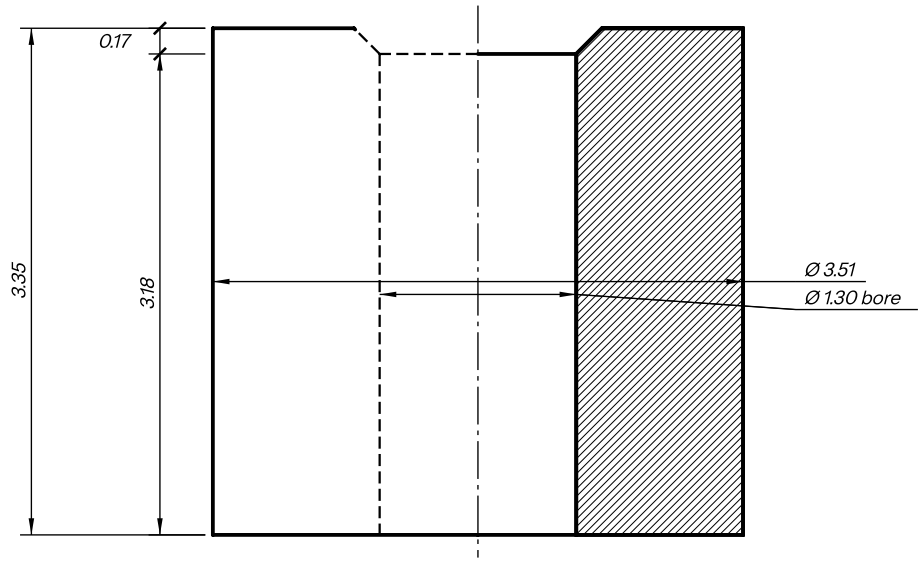
2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

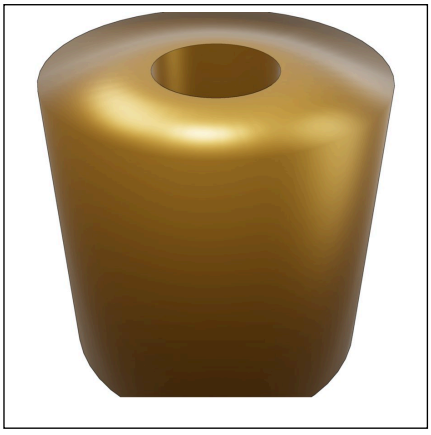
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 02-2	Futter (stud bush)	2	not on the drawing		in 050802-1 (press): 3.352 mm, least gap 0.0019

What it does (05 08 02-2). Pressed into the pillar plate from its inner face at holes 2 and 19, holding the motion work's studs (the one at 19 drawn dashed on 05 08 02: estimated).

Turned about its axis; length 3.35, largest Ø 3.51. Diameters: Ø 3.51; bores Ø 1.50. Shoulders from the lower end: 0.00, 3.35.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-hs** *Teil / Part* *pillarPlate.hsBush*
Behandlung / Treatment		*Benennung / Name* ***Hand-setting pinion's bush***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Hand-setting pinion’s bush — sheet 2

A brass bush pressed into the pillar plate’s hole 18, beside the hand-setting window: the bearing of the hand-setting work’s pinion. On Wempe 3955 it is the only part of that work fitted, and it is empty.

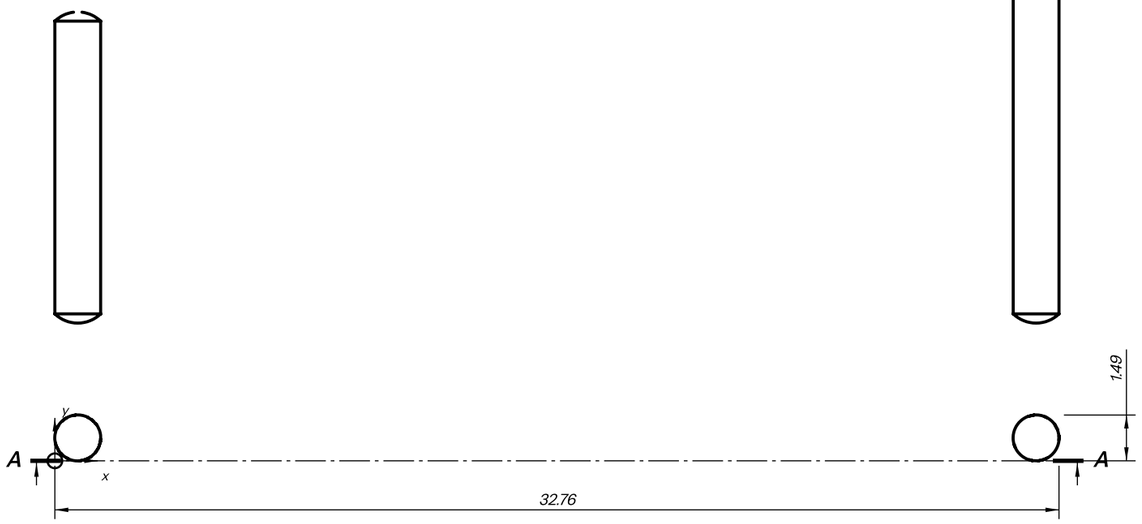
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-hs	Bush in hole 18, the hand-setting pinion's (not named on 05 08 02)	1	not on the drawing		in 050802-1 (press): 3.352 mm, least gap 0.0019

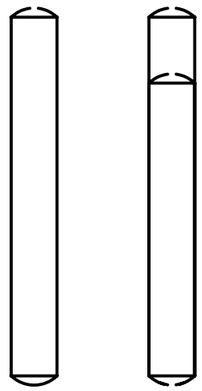
What it does (WN-hs). The bearing of the hand-setting work's pinion (Zeigerstelltrieb), pressed into the pillar plate's hole 18; on Wempe 3955 the only part of that work fitted, empty.

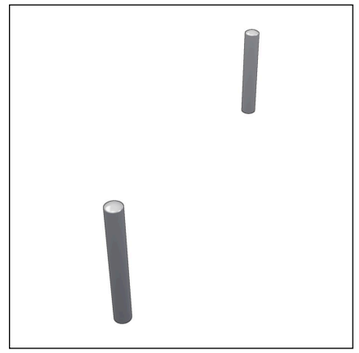
Turned about its axis; length 3.35, largest Ø 3.51. Diameters: Ø 3.51; bores Ø 1.30. Shoulders from the lower end: 0.00, 3.18, 3.35.

A–A



View from the left





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 02 00-6**
Behandlung / Treatment			Teil / Part pillarPlate.studs
Blatt / Sheet **1 / 2**		Benennung / Name **Studs (Anrichtstifte)**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Studs (Anrichtstifte) — sheet 2

Two steel studs pressed into the bushes: the long one, at 12 o'clock, carries the up-and-down wheel; the short one the change wheel of the motion work.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 00-6	Anrichtstift lg (long stud)	1	not on the drawing		in 050802-2 (press): 3.132 mm, least gap 0.0014
05 02 00-7	Anrichtstift kurz (short stud)	1	not on the drawing		in 050802-2 (press): 3.087 mm, least gap 0.0014

What it does (05 02 00-6). The up-and-down wheel's stud, at hole 2, 12.3 long from the plate's inner face.

What it does (05 02 00-7). The motion work's change wheel's stud, at hole 19, 6.8 out of the plate's dial face.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 12.30.

Contour in the plan

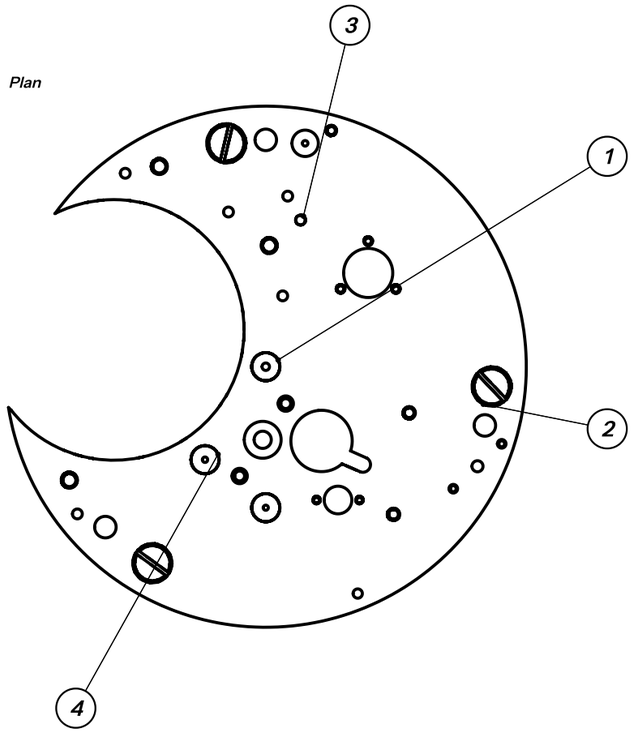
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			32.01, 0.75	0.75
Outline	2	circle			0.75, 0.75	0.75

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Train bushes (Messinglager WN-300, WN-301)	WN-300.t35	not on the drawing
3	1	Stop pin (Anschlagstift)	05 08 01-5	not on the drawing
2	3	Pillar screws, train side	WN-57	not on the drawing
1	1	Top plate	05 09 01	„Messingblech 4 / Sonder Ms α-

Werkstoff / Material: not on the drawing

Maßstab / Scale: 1:1

Behandlung / Treatment:

Blatt / Sheet: 1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

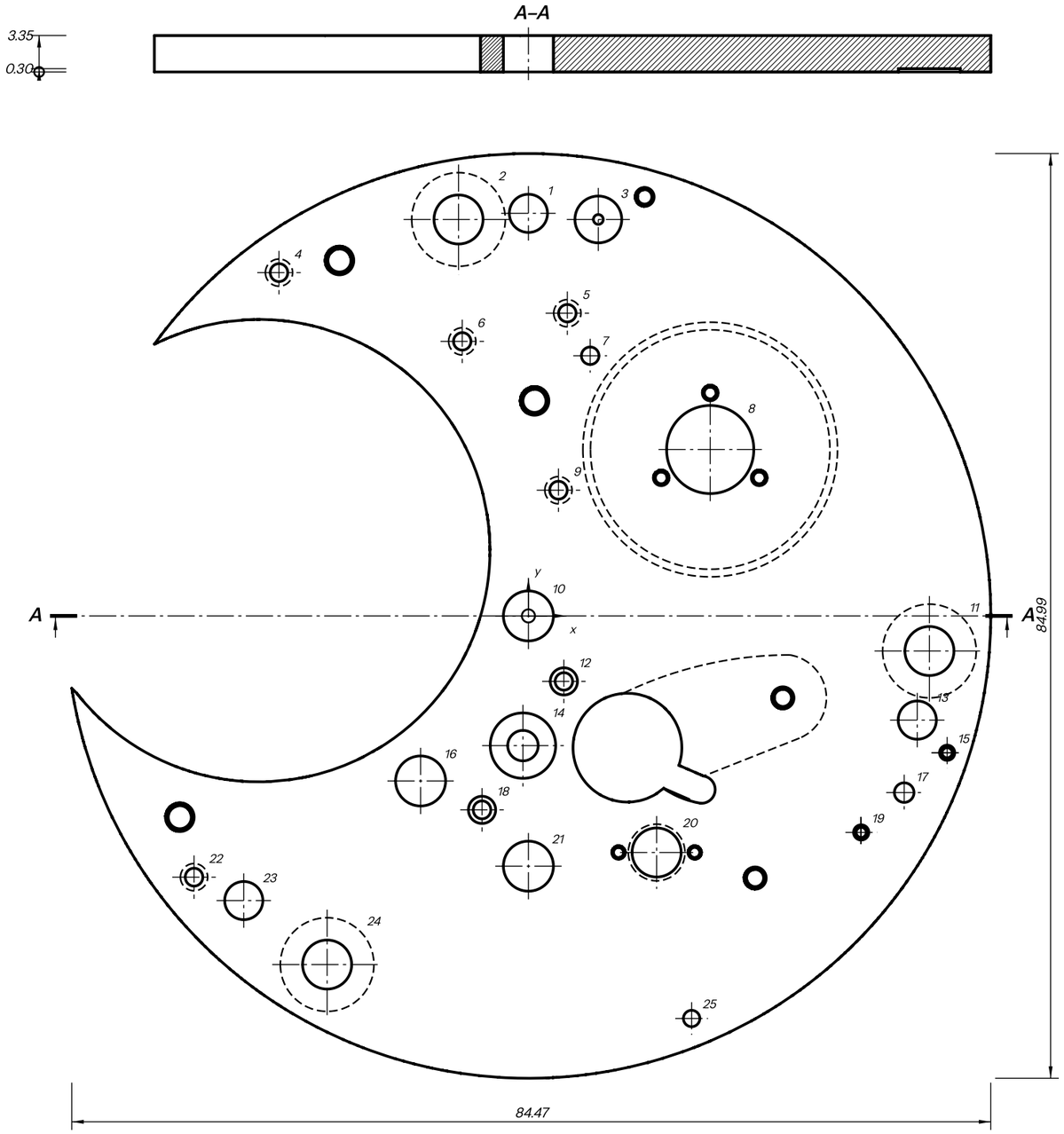
Zeichnung Nr. / Drawing no.: **05 08 01-5**

Teil / Part: topPlate

Benennung / Name: **Top plate (Oberplatte)**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Messingblech 4 / Sonder Ms α-β Gef." (special brass, α-β structure)*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 09 01*** *Teil / Part* *topPlate.plate*
Behandlung / Treatment *none on the drawing*		*Benennung / Name* ***Top plate***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Top plate — sheet 2

The upper plate, on the train side, a crescent: it is cut away round the barrel, which the barrel bridge holds. It carries the train's upper pivots, the fusee's bush, the two balance cocks (the lower one hangs under it) and the balance holding work in a pocket in its inner face. Its auxiliary holes 3, 12 and 40 stand beside the pillars. The drawings are the sheets as last revised (about 1946-48).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 09 01	Oberplatte (top plate)	1	„Messingblech 4 / Sonder Ms α-βGef.“ (special brass, α-β structure)	none on the drawing	takes Pfeiler (pillar): in it (run): 2.348 mm, least gap 0.0109; Pfeiler (pillar): on it: 0 mm; Federhausbrücke (barrel bridge): on it: 0 mm; Schneckenfutter (fusee bush): in it (free): 2.579 mm, least gap 0.009; Schneckenfutter (fusee bush): on it: 0 mm; Unruhkloben oben (upper balance cock): on it: 0 mm; and 22 more

Test on the bench (05 09 01). The train's depths, the plates bored and reamed, before the bushes and jewels go in: Centres 22.650, 18.120, 12.630, 11.850 mm (fusee-centre, centre-intermediate, intermediate-seconds, seconds-escape: the hole tables' coordinates), each within ±25 µm (gearing.js --model wempe: the seconds mesh's window less its pivots' shake; the tables give the places to 0.001 and no tolerance). A hole that errs should err wide: each mesh takes far more wide than close (-240/+436, -136/+355, -112/+281, -100/+263 µm). Closer than its window less the pivots' shake (220, 101, 62, 50 µm close): re-bush the hole, bored on the drawn centre.

What it does (05 09 01). The upper plate, on the train side (its sheet's title block numbers it 05 09 01; its assembly, Oberplatte kompl., is 05 08 01).

Coordinates from the datum, the movement's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.35.

Holes

Hole	x	y	Size
1	0.00	37.00	Ø 3.51 through
2	-6.43	36.44	Ø 4.51 through (3.05 thick at 0.30-3.35); sink Ø 8.60 × 0.30 from the bottom
3	6.43	36.44	Ø 0.90 through (2.41 thick at 0.00-2.41); countersunk to Ø 4.30
4	-22.92	31.55	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
5	3.58	27.82	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
6	-6.08	25.23	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
7	5.68	23.91	Ø 1.61 through
8	16.71	15.29	Ø 8.01 through (2.65 thick at 0.70-3.35)
9	2.76	11.56	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
10	0.00	0.00	Ø 4.60 through
11	36.86	-3.22	Ø 4.51 through (3.05 thick at 0.30-3.35); sink Ø 8.60 × 0.30 from the bottom
12	3.26	-6.02	Ø 1.60 through (1.75 thick at 0.00-1.75); sink Ø 2.50 × 1.60 from the top
13	35.74	-9.58	Ø 3.51 through
14	-0.50	-11.92	Ø 2.80 through (1.55 thick at 0.00-1.55); sink Ø 6.01 × 1.80 from the top
15	38.49	-12.58	Ø 0.93 through (3.13 thick at 0.00-3.13); countersunk to Ø 1.40
16	-9.91	-15.17	Ø 4.60 through
17	34.54	-16.24	Ø 1.80 through
18	-4.26	-17.83	Ø 1.60 through (1.75 thick at 0.00-1.75); sink Ø 2.50 × 1.60 from the top
19	30.59	-19.90	Ø 0.93 through (3.13 thick at 0.00-3.13); countersunk to Ø 1.40
20	11.78	-21.74	Ø 4.51 through (0.50 thick at 2.85-3.35); sink Ø 5.21 × 2.85 from the bottom
21	0.00	-23.00	Ø 4.60 through

22	-30.73	-24.01	Ø 1.60 through (1.75 thick at 1.60-3.35); sink Ø 2.50 × 1.60 from the bottom
23	-26.16	-26.16	Ø 3.51 through
24	-18.50	-32.04	Ø 4.51 through (3.05 thick at 0.30-3.35); sink Ø 8.60 × 0.30 from the bottom
25	15.00	-37.00	Ø 1.50 through

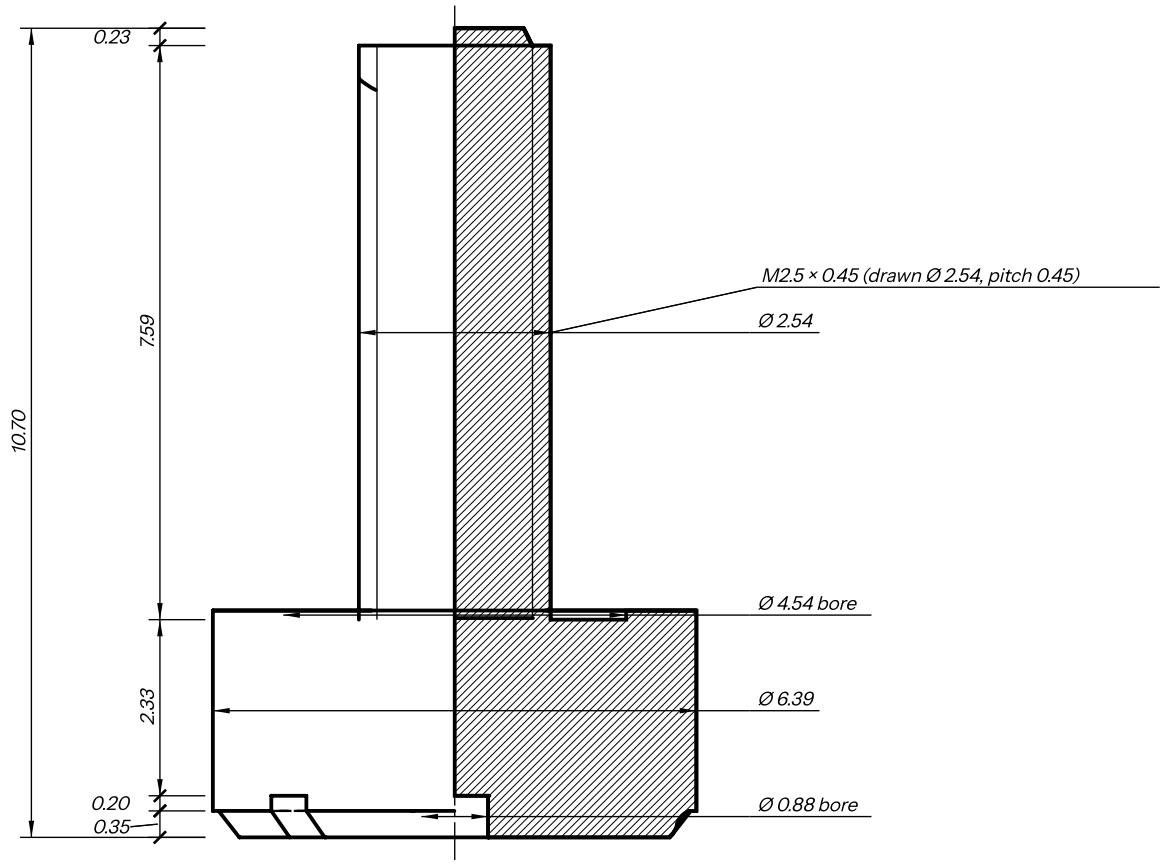
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

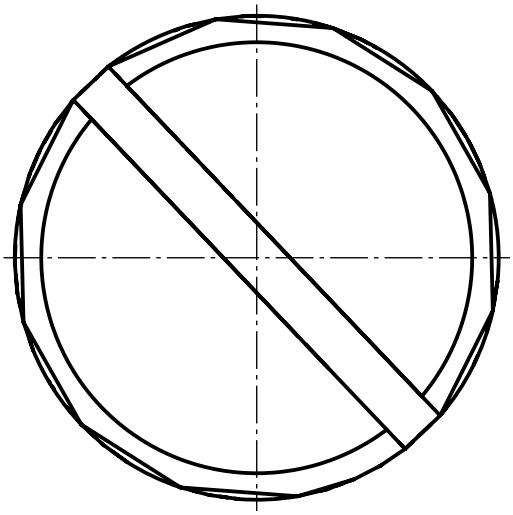
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-34.12, -25.34	-41.97, -6.65	0.00, 0.00	42.49
Outline	2	arc clockwise	-41.97, -6.65	-41.12, -7.60	-45.86, -11.02	5.85
Outline	3	arc anticlockwise	-41.12, -7.60	-33.80, 25.24	-24.78, 6.00	21.25
Outline	4	line	-33.80, 25.24	-34.38, 24.98		
Outline	5	arc clockwise	-34.38, 24.98	-34.12, -25.34	0.00, 0.00	42.49
Cut-out 11	6	arc anticlockwise	9.41, -7.12	8.57, -7.14	9.10, -12.04	4.93
Cut-out 11	7	arc anticlockwise	8.57, -7.14	12.42, -15.85	9.10, -12.11	5.00
Cut-out 11	8	arc anticlockwise	12.42, -15.85	15.57, -17.19	20.80, -0.56	17.43
Cut-out 11	9	arc anticlockwise	15.57, -17.19	16.33, -14.82	15.83, -15.96	1.25
Cut-out 11	10	line	16.33, -14.82	13.83, -13.73		
Cut-out 11	11	arc anticlockwise	13.83, -13.73	9.41, -7.12	9.10, -12.11	5.00
Cut-out 17	12	line	9.91, 38.89	9.91, 38.89		
Cut-out 17	13	line	9.91, 38.89	9.91, 38.89		
Cut-out 17	14	line	9.91, 38.89	9.91, 38.89		
Cut-out 17	15	line	9.91, 38.89	9.91, 38.89		
Cut-out 18	16	line	9.02, -41.38	9.02, -41.38		
Cut-out 18	17	arc anticlockwise	9.02, -41.38	9.02, -41.38	4.89, -20.63	21.15
Cut-out 18	18	line	9.02, -41.38	9.02, -41.38		

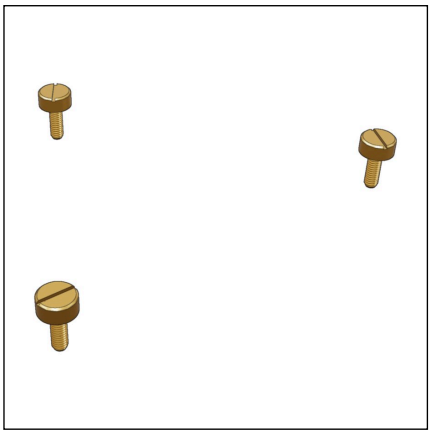
Levels

Faces in section A-A, from the bottom face: 0.30, 3.35.



View from below (seen from beneath, x reversed)





Isometric view, not to scale

3 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-57*** *Teil / Part* *topPlate.screws*
Behandlung / Treatment		*Benennung / Name* ***Pillar screws, train side***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Pillar screws, train side — sheet 2

Three pillar screws (WN-57) from the top plate's outer face into the pillars' tapped ends, brass as Wempe 3955's.

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

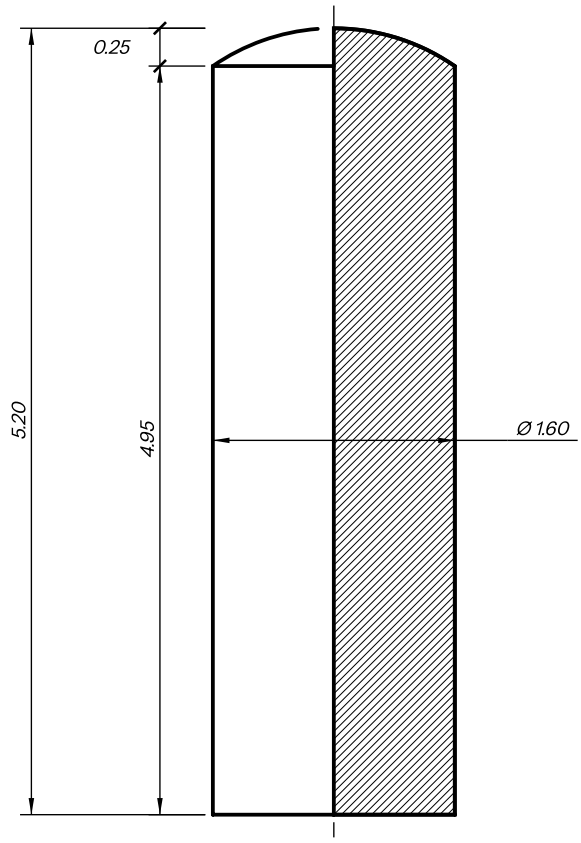
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-57	Pillar screw, train side (WN-57, by the coloured section's label)	3	not on the drawing		in 050800-1 (tap): 7.448 mm, least gap 0.0253; on 050901: 0 mm

What it does (WN-57). Hold the top plate to each pillar, from its outer face.

Turned about its axis; length 10.70, largest Ø 6.40. Diameters: Ø 2.54, Ø 6.39; bores Ø 0.88, Ø 4.54. Shoulders from the lower end: 0.00, 0.35, 0.55, 2.88, 10.47, 10.70.

Thread

Outside thread 2.88–10.47 from the lower end, 7.59 long: drawn Ø 2.54 over its crests, 2.06 at its roots, pitch 0.45. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 01-5***
Behandlung / Treatment			*Teil / Part* *topPlate.stopPin*
		Benennung / Name ***Stop pin (Anschlagstift)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Stop pin (Anschlagstift) — sheet 2

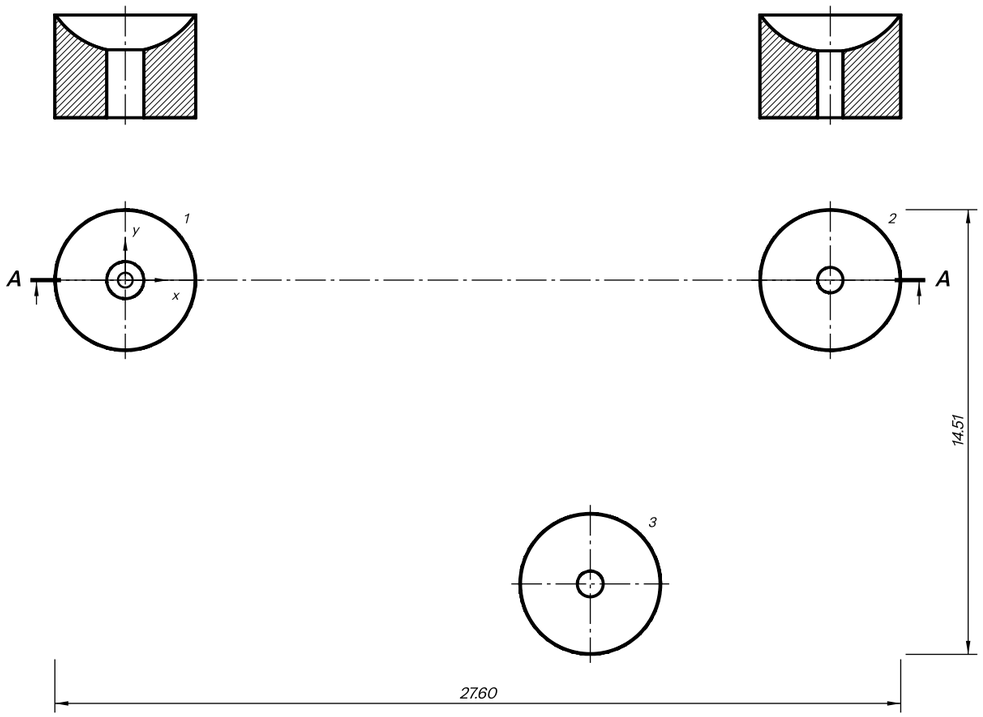
A steel pin in the top plate's hole 7, standing out under the plate ("unten vorstehend"): the fusee's stop bar meets it at full wind.

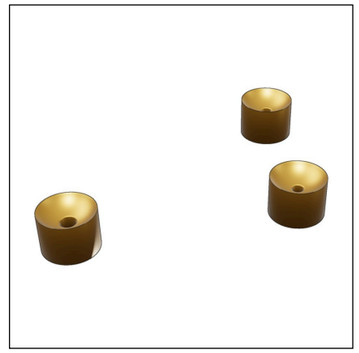
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 01-5	Anschlagstift (stop pin)	1	not on the drawing		in 050901 (press): 3.339 mm, least gap 0.0058

What it does (05 08 01-5). Pressed in the top plate's hole 7, standing out under it: the stop bar meets it at full wind.

Turned about its axis; length 5.20, largest Ø 1.60. Diameters: Ø 1.60. Shoulders from the lower end: 0.00, 4.95, 5.20.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-300.t35*** *Teil / Part* *topPlate.bushes*
Behandlung / Treatment		*Benennung / Name* ***Train bushes (Messinglager WN-300, WN-301)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Train bushes (Messinglager WN-300, WN-301) — sheet 2

Three brass bushes pressed into the top plate, flush with both faces, in which the centre, intermediate and seconds arbors' upper pivots run, each with an oil sink in its outer face (the 6MX ran the seconds and intermediate wheels in jewels in chatons instead).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-300.t35	Messinglager WN-300 (brass bush, seconds arbor's upper pivot)	1	not on the drawing		in 050901 (press): 3.353 mm, least gap 0.0028
WN-300.t38	Messinglager WN-300 (brass bush, intermediate arbor's upper pivot)	1	not on the drawing		in 050901 (press): 3.353 mm, least gap 0.0028
WN-301	Messinglager WN-301 (brass bush, centre arbor's upper pivot)	1	not on the drawing		in 050901 (press): 3.353 mm, least gap 0.0028

What it does (WN-300.t35). The seconds arbor's upper bearing, pressed into the top plate's hole 35.

What it does (WN-300.t38). The intermediate arbor's upper bearing, pressed into the top plate's hole 38.

What it does (WN-301). The centre arbor's upper bearing, pressed into the top plate's hole 1.

Coordinates from the datum, the movement's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.35.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.21 through (2.22 thick at 0.00-2.22); countersunk to Ø 4.55
2	23.00	0.00	Ø 0.83 through (2.18 thick at 0.00-2.18); countersunk to Ø 4.55
3	15.17	-9.91	Ø 0.85 through (2.19 thick at 0.00-2.19); countersunk to Ø 4.55

Contour in the plan

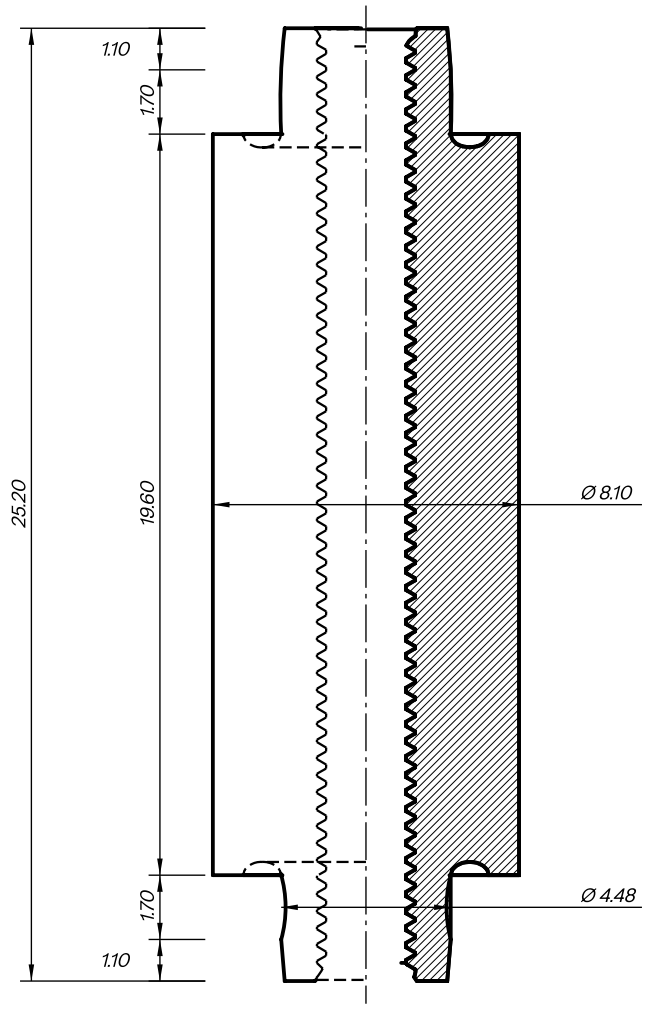
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

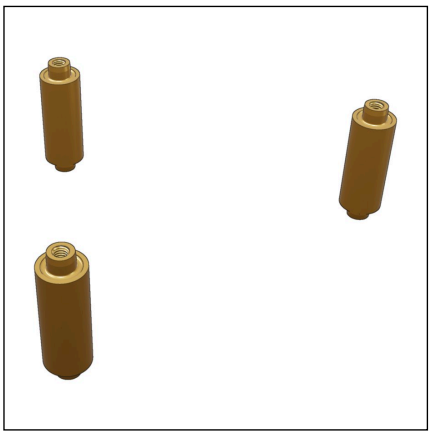
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			23.00, 0.00	2.30
Cut-out 1	2	line	24.45, -0.19	24.45, -0.19		
Cut-out 1	3	line	24.45, -0.19	24.45, -0.19		
Cut-out 1	4	line	24.45, -0.19	24.45, -0.19		
Cut-out 2	5	line	23.83, -2.00	23.83, -2.00		
Cut-out 2	6	line	23.83, -2.00	23.83, -2.00		
Cut-out 2	7	line	23.83, -2.00	23.83, -2.00		
Cut-out 2	8	line	23.83, -2.00	23.83, -2.00		
Cut-out 3	9	line	23.62, -1.50	23.62, -1.50		
Cut-out 3	10	line	23.62, -1.50	23.62, -1.50		
Cut-out 3	11	line	23.62, -1.50	23.62, -1.50		
Cut-out 5	12	line	23.27, 2.07	23.27, 2.07		
Cut-out 5	13	line	23.27, 2.07	23.27, 2.07		
Cut-out 5	14	line	23.27, 2.07	23.27, 2.07		
Cut-out 5	15	line	23.27, 2.07	23.27, 2.07		
Cut-out 6	16	line	22.85, -1.17	22.85, -1.17		
Cut-out 6	17	line	22.85, -1.17	22.85, -1.17		

Cut-out 6	18	line	22.85, -1.17	22.85, -1.17		
Cut-out 6	19	line	22.85, -1.17	22.85, -1.17		
Cut-out 7	20	line	22.81, -1.47	22.81, -1.47		
Cut-out 7	21	line	22.81, -1.47	22.81, -1.47		
Cut-out 7	22	line	22.81, -1.47	22.81, -1.47		
Cut-out 7	23	line	22.81, -1.47	22.81, -1.47		
Cut-out 8	24	line	22.75, -1.88	22.75, -1.88		
Cut-out 8	25	line	22.75, -1.88	22.75, -1.88		
Cut-out 8	26	line	22.75, -1.88	22.75, -1.88		
Cut-out 8	27	line	22.75, -1.88	22.75, -1.88		
Cut-out 9	28	line	22.22, 1.35	22.22, 1.35		
Cut-out 9	29	line	22.22, 1.35	22.22, 1.35		
Cut-out 9	30	line	22.22, 1.35	22.22, 1.35		
Cut-out 9	31	line	22.22, 1.35	22.22, 1.35		
Cut-out 10	32	line	21.67, 1.33	21.67, 1.33		
Cut-out 10	33	line	21.67, 1.33	21.67, 1.33		
Cut-out 10	34	line	21.67, 1.33	21.67, 1.33		
Cut-out 10	35	line	21.67, 1.33	21.67, 1.33		
Cut-out 11	36	line	21.64, 1.36	21.64, 1.36		
Cut-out 11	37	line	21.64, 1.36	21.64, 1.36		
Cut-out 11	38	line	21.64, 1.36	21.64, 1.36		
Cut-out 11	39	line	21.64, 1.36	21.64, 1.36		
Outline	40	circle			15.17, -9.91	2.30
Cut-out 2	41	line	14.91, -11.87	14.91, -11.87		
Cut-out 2	42	line	14.91, -11.87	14.91, -11.87		
Cut-out 2	43	line	14.91, -11.87	14.91, -11.87		
Cut-out 2	44	line	14.91, -11.87	14.91, -11.87		
Cut-out 3	45	line	14.46, -11.19	14.46, -11.19		
Cut-out 3	46	line	14.46, -11.19	14.46, -11.19		
Cut-out 3	47	line	14.46, -11.19	14.46, -11.19		
Cut-out 3	48	line	14.46, -11.19	14.46, -11.19		
Outline	49	circle			0.00, 0.00	2.30
Cut-out 1	50	line	2.22, -0.29	2.22, -0.29		
Cut-out 1	51	line	2.22, -0.29	2.22, -0.29		
Cut-out 2	52	line	0.90, 0.37	0.90, 0.37		
Cut-out 2	53	line	0.90, 0.37	0.90, 0.37		
Cut-out 3	54	line	0.81, 1.40	0.81, 1.40		
Cut-out 3	55	line	0.81, 1.40	0.81, 1.40		

Levels

Faces in section A-A, from the bottom face: .





Isometric view, not to scale

3 made; one drawn.

Werkstoff / Material *„Rundmessing 8,5 Din 1758 Ms 58" ("hart" struck through)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 00-1*** *Teil / Part* *pillars*
Behandlung / Treatment *none on the drawing*		*Benennung / Name* ***Pillars (Pfeiler)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Pillars (Pfeiler) — sheet 2

Three turned brass pillars hold the plates 19 mm apart. Each sits by its shoulders in a seat in each plate, its spigots in the plates' reamed holes, a screw from each plate's outer face in its tapped ends.

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

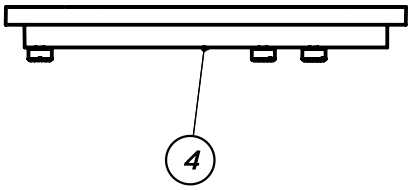
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 00-1	Pfeiler (pillar)	3	„Rundmessing 8,5 Din 1758 Ms 58“ ("hart" struck through)	none on the drawing	in 050802-1 (run): 2.348 mm, least gap 0.0109; in 050901 (run): 2.348 mm, least gap 0.0109; on 050802-1: 0 mm; on 050901: 0 mm

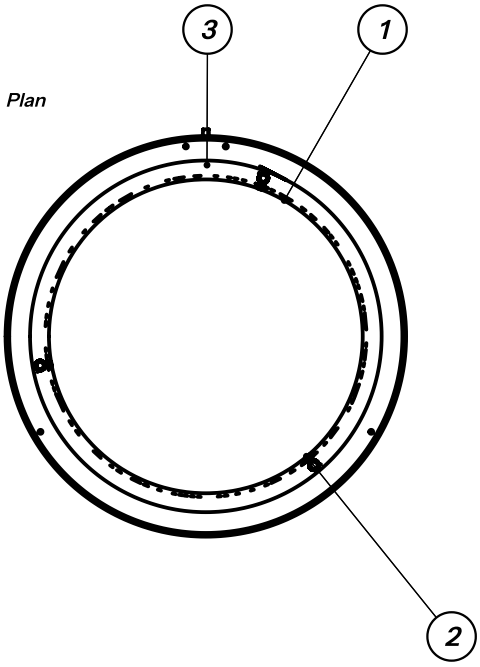
What it does (05 08 00-1). Hold the plates 19 mm apart, their shoulders in seats in the plates, their spigots in the plates' reamed holes.

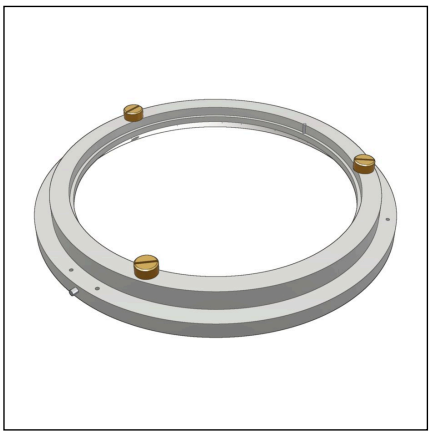
Turned about its axis; length 25.20, largest Ø 8.10. Diameters: Ø 4.48, Ø 8.10. Shoulders from the lower end: 0.00, 1.10, 2.80, 22.40, 24.10, 25.20.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Plate's pin	WN-201	not on the drawing
3	1	Ring's pin to the case (Stellstift 3)	WN-202	not on the drawing
2	3	Movement ring screws	WN-rs	not on the drawing
1	1	Movement ring	05 08 00-3	not on the drawing

Werkstoff / Material
not on the drawing

Maßstab / Scale
1:2

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 08 00-3

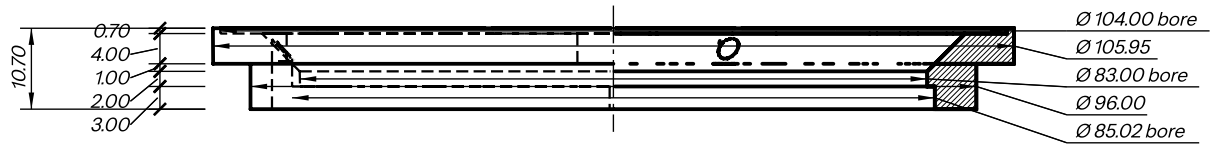
Teil / Part
ring

Behandlung / Treatment

Benennung / Name
Movement ring (Werkring)

Blatt / Sheet
1 / 1

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 08 00-3** *Teil / Part* *ring.ring*
Behandlung / Treatment		*Benennung / Name* ***Movement ring***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Movement ring — sheet 2

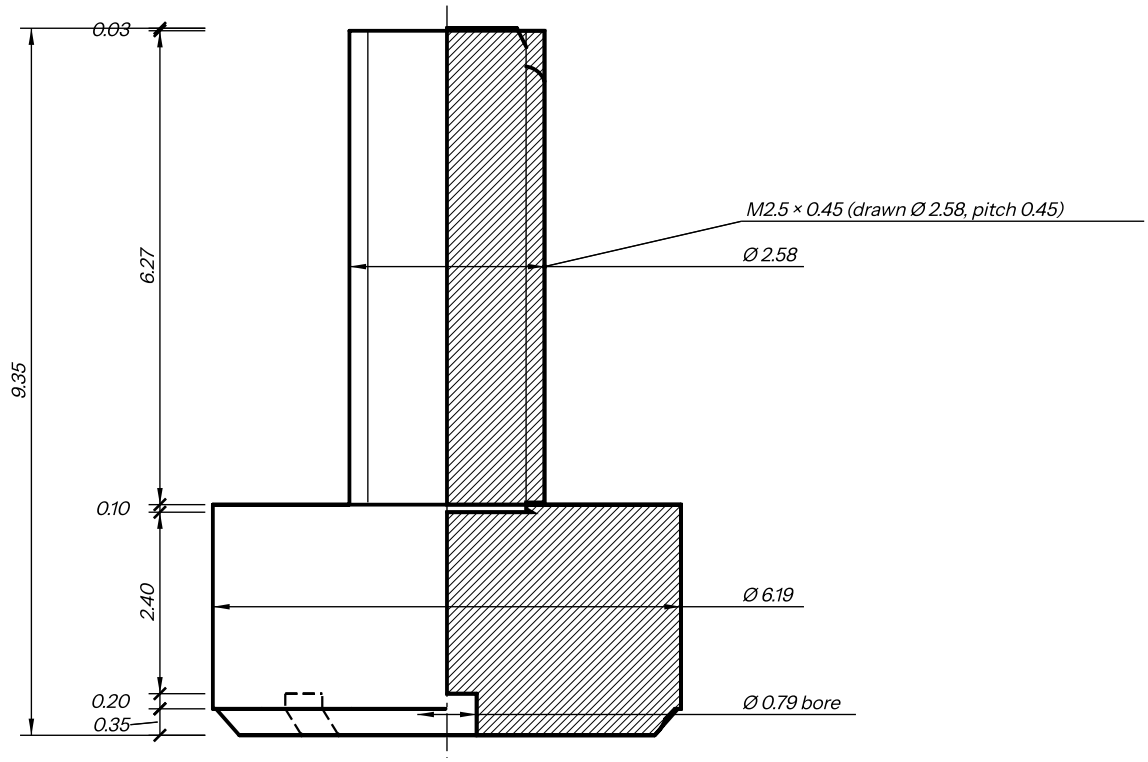
The ring the movement hangs in, in the case: the pillar plate sits in its seat on a lip, held by three screws whose heads bear on the plate's rim, and the dial lies in a recess in its flange. Below the lip the ring opens out at 45° toward the dial.

Parts list

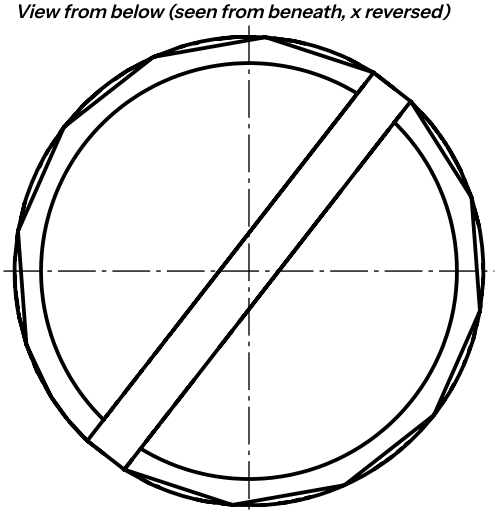
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 00-3	Werkring (movement ring)	1	not on the drawing		on 050802-1: 0 mm

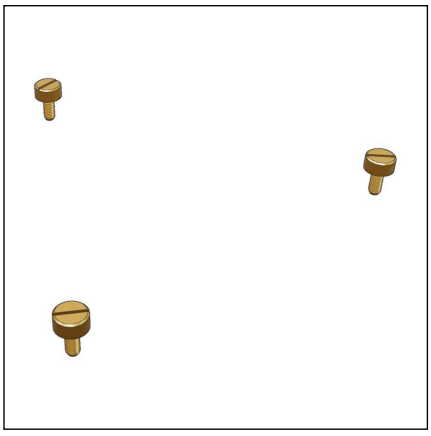
What it does (05 08 00-3). The ring the pillar plate sits in, which carries the dial and hangs in the case.

Turned about its axis; length 10.70, largest Ø 105.95. Diameters: Ø 96.00, Ø 105.95; bores Ø 83.00, Ø 85.02, Ø 104.00. Shoulders from the lower end: 0.00, 3.00, 5.00, 6.00, 10.00, 10.70.



View from below (seen from beneath, x reversed)





Isometric view, not to scale

3 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-rs*** *Teil / Part* *ring.screws*
Behandlung / Treatment		*Benennung / Name* ***Movement ring screws***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Movement ring screws — sheet 2

Three screws in the ring whose heads bear on the pillar plate’s rim, holding it in its seat (their WN number not read).

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

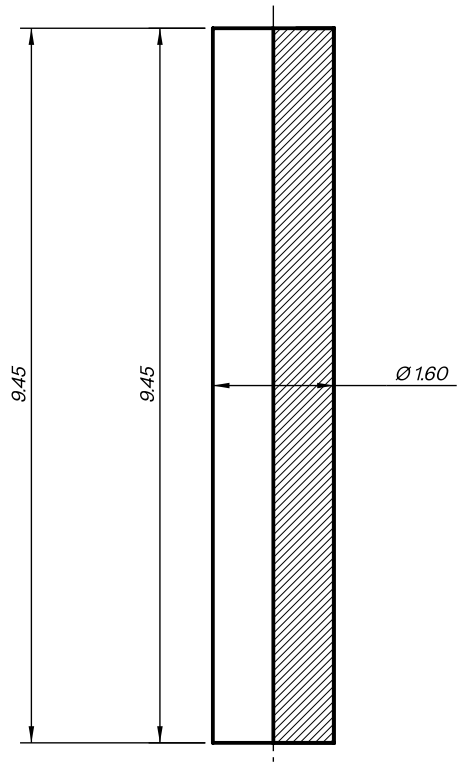
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-rs	Movement ring screw (WN number not read)	3	not on the drawing		in 050800-3 (tap): 5.926 mm, least gap 0.0334; on 050802-1: 0 mm

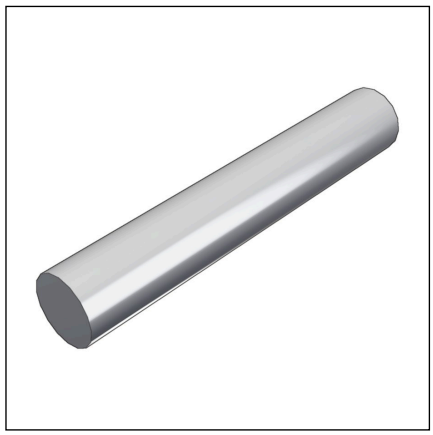
What it does (WN-rs). Its head bears on the pillar plate's rim and holds it in the ring's seat.

Turned about its axis; length 9.35, largest Ø 6.19. Diameters: Ø 2.58, Ø 6.19; bores Ø 0.79. Shoulders from the lower end: 0.00, 0.35, 0.55, 2.95, 3.05, 9.32, 9.35.

Thread

Outside thread 3.08–9.32 from the lower end, 6.24 long: drawn Ø 2.58 over its crests, 2.09 at its roots, pitch 0.45. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-202*** *Teil / Part* *ring.casePin*
Behandlung / Treatment		*Benennung / Name* ***Ring's pin to the case (Stellstift 3)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Ring’s pin to the case (Stellstift 3) — sheet 2

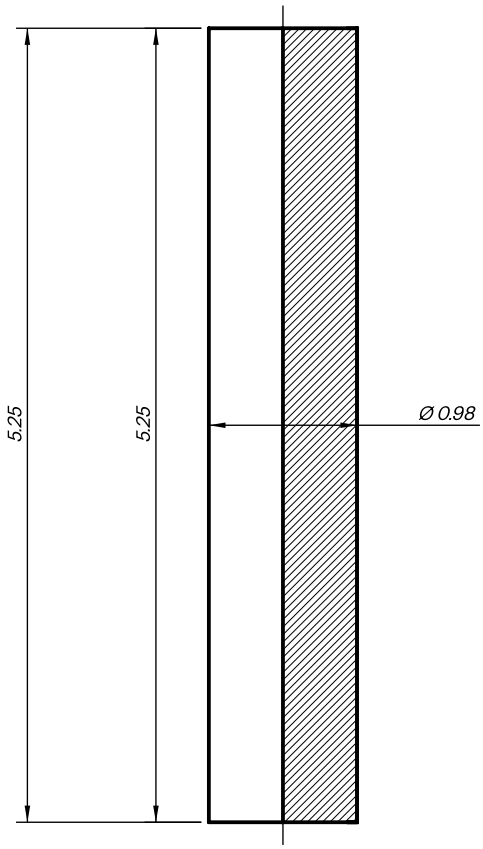
A pin Ø 1.6 standing 1.8 mm out of the ring’s flange at 12 o’clock into a notch cut through the bowl’s threaded rim: it sets the movement’s turn in the case, so the winding hole is over the fusee’s square, and holds it while the bezel is screwed down (the notch isn’t on the casting’s sheet: the 6MX’s bowls have it).

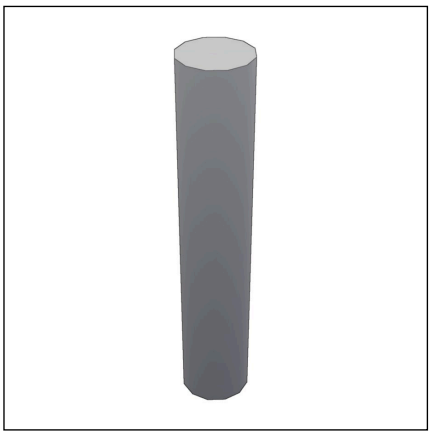
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-202	Stellstift 3 (WN-202): the movement ring’s radial pin to the case	1	not on the drawing		in 050800-3 (embed): 7.69 mm, least gap -0.7881

What it does (WN-202). Stands 1.78 out of the ring's flange at 12 o'clock into the notch in the bowl's threaded rim: it sets the movement's turn in the case and holds it while the bezel is screwed down.

Turned about its axis; length 9.45, largest Ø 1.60. Diameters: Ø 1.60. Shoulders from the lower end: 0.00, 9.45.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-201***
Behandlung / Treatment			*Teil / Part* *ring.pin*
		Benennung / Name ***Plate's pin***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Plate’s pin — sheet 2

A pin (WN-201) Ø 1 drilled in across the joint of the plate and the ring at assembly, which sets the movement’s turn in the ring.

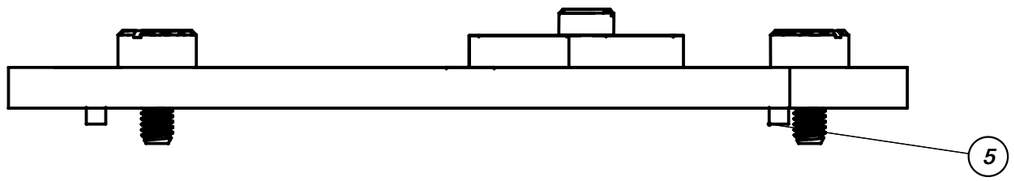
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-201	Stellstift (WN-201): the plate's pin in the movement ring	1	not on the drawing		in 050800-3 (press): 1.98 mm, least gap 0.0121

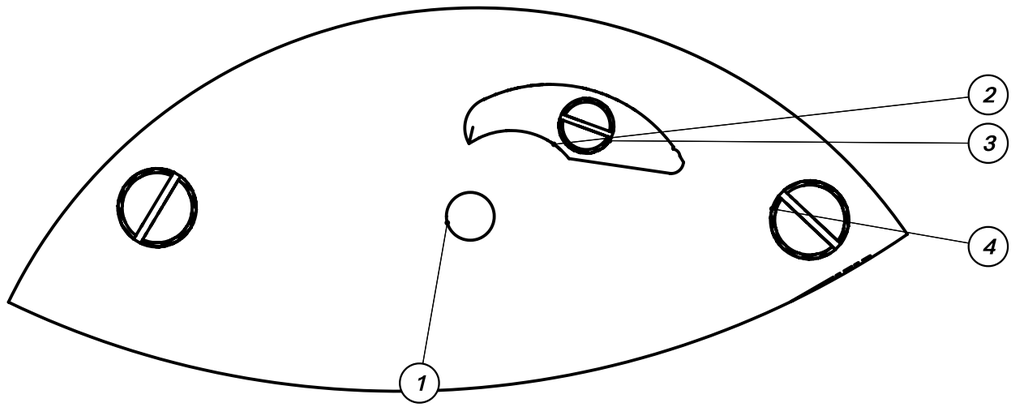
What it does (WN-201). Drilled across the joint of the pillar plate and the ring at assembly, it sets the movement's turn in the ring.

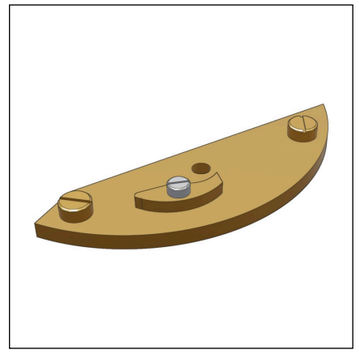
Turned about its axis; length 5.25, largest Ø 0.98. Diameters: Ø 0.98. Shoulders from the lower end: 0.00, 5.25.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
5	2	Barrel bridge steady pins	WN-203.bb	not on the drawing
4	2	Barrel bridge screws	WN-55	not on the drawing
3	1	Click screw	WN-22	not on the drawing
2	1	Click (Sperrkegel)	05 08 05-2	„Messingblech 3 Din 1751 Ms 58
1	1	Barrel bridge	05 08 05-1	„Messing blech 3,5 Din 1751 Ms

Werkstoff / Material
„Messing blech 3,5 Din 1751 Ms 58"

Maßstab / Scale
2:1

Behandlung / Treatment
untoleranced sizes ±0,1

Blatt / Sheet
1 / 1

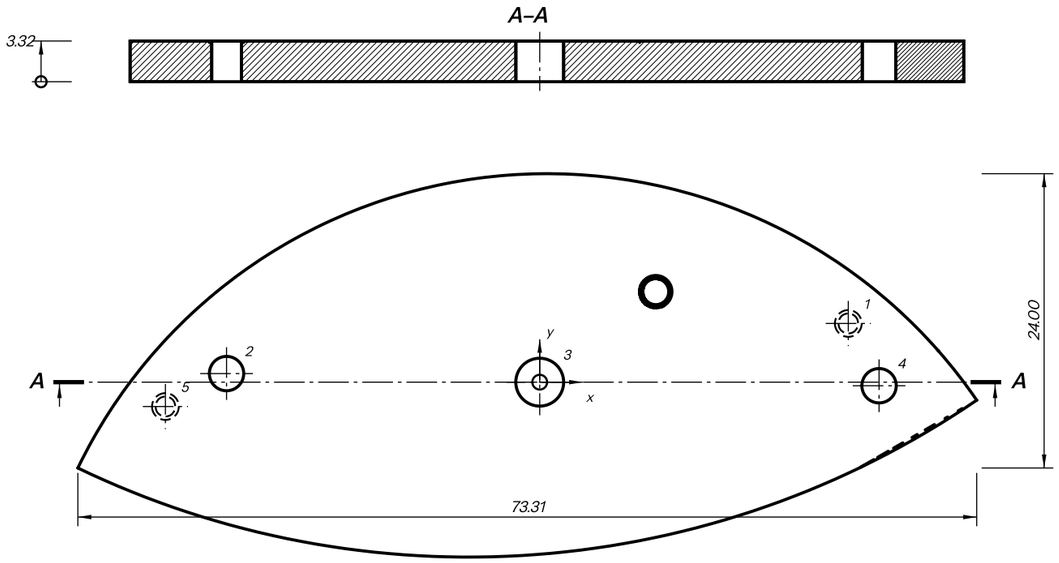
Wempe Einheits-Chronometer
The Marine Chronometer working model
drawn from its solids by tools/drawbook.py
Model v3.35.00 · 2026-10-26

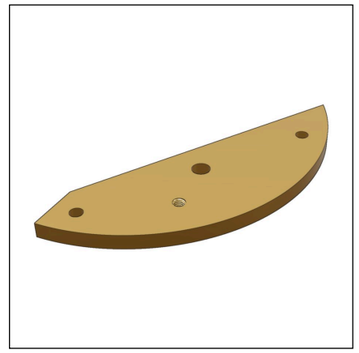
Zeichnung Nr. / Drawing no.
05 08 05-1

Teil / Part
barrelBridge

Benennung / Name
Barrel bridge (Federhausbrücke)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Messing blech 3,5 Din 1751 Ms 58"*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 05-1***
Behandlung / Treatment *untoleranced sizes ±0,1*			*Teil / Part* *barrelBridge.bridge*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Barrel bridge***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Barrel bridge — sheet 2

A crescent bridge on the top plate over the barrel, its outer edge turned with the plate's rim, carrying the barrel arbor's upper pivot and the click that holds the mainspring's set-up.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 05-1	Federhausbrücke (barrel bridge)	1	„Messing blech 3,5 Din 1751 Ms 58“	untoleranced sizes ±0,1	on 050901: 0 mm

What it does (05 08 05-1). Holds the barrel arbor's upper pivot over the top plate's cut (its assembly is 05 08 05).

Coordinates from the datum, hole 3's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.32.

Holes

Hole	x	y	Size
1	25.17	4.79	Ø 1.58 × 2.22 deep from the face at 0.28; countersunk to Ø 2.18
2	-25.54	0.70	Ø 2.80 through
3	0.00	0.00	Ø 3.90 through
4	27.68	-0.29	Ø 2.80 through
5	-30.53	-1.99	Ø 1.58 × 2.22 deep from the face at 0.28; countersunk to Ø 2.18

Contour in the plan

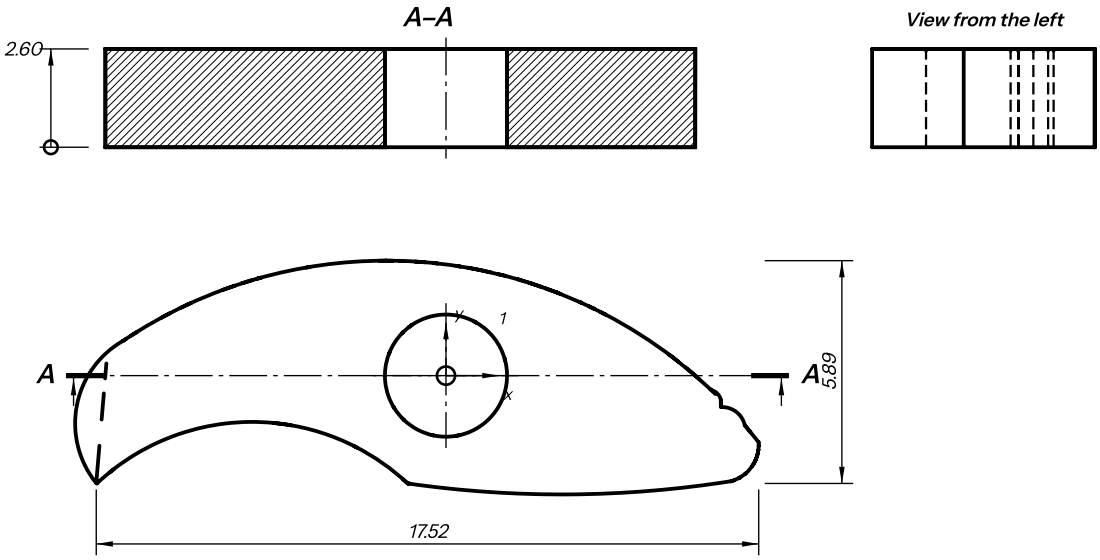
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

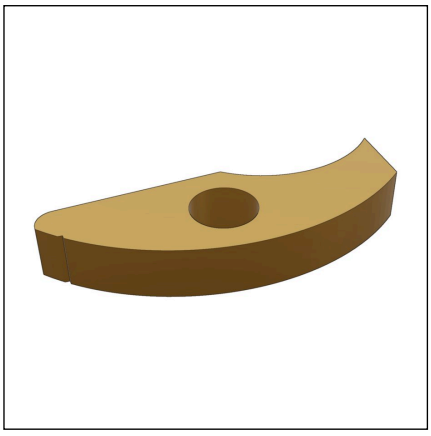
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-37.32, -6.28	35.64, -1.47	0.59, -25.49	42.49
Outline	2	arc clockwise	35.64, -1.47	-37.67, -7.00	-5.81, 59.25	73.52
Outline	3	line	-37.67, -7.00	-37.32, -6.28		
Cut-out 3	4	line	10.29, 6.36	10.29, 6.36		
Cut-out 3	5	line	10.29, 6.36	10.29, 6.36		
Cut-out 3	6	line	10.29, 6.36	10.29, 6.36		
Cut-out 4	7	line	10.21, 6.30	10.21, 6.30		
Cut-out 4	8	line	10.21, 6.30	10.21, 6.30		
Cut-out 4	9	line	10.21, 6.30	10.21, 6.30		
Cut-out 4	10	line	10.21, 6.30	10.21, 6.30		
Cut-out 5	11	line	9.56, 6.06	9.56, 6.06		
Cut-out 5	12	line	9.56, 6.06	9.56, 6.06		
Cut-out 5	13	line	9.56, 6.06	9.56, 6.06		
Cut-out 6	14	line	9.16, 6.10	9.16, 6.10		
Cut-out 6	15	line	9.16, 6.10	9.16, 6.10		
Cut-out 6	16	line	9.16, 6.10	9.16, 6.10		
Cut-out 7	17	arc anticlockwise	8.97, 6.16	8.97, 6.16	0.99, 9.35	8.60
Cut-out 7	18	arc anticlockwise	8.97, 6.16	8.97, 6.16	0.99, 9.35	8.60
Cut-out 8	19	line	8.85, 6.21	8.85, 6.21		
Cut-out 8	20	arc anticlockwise	8.85, 6.21	8.85, 6.21	3.14, 3.86	6.17
Cut-out 8	21	line	8.85, 6.21	8.85, 6.21		

Cut-out 9	22	line	8.64, 6.36	8.64, 6.36		
Cut-out 9	23	arc anticlockwise	8.64, 6.36	8.64, 6.36	4.45, 0.68	7.06
Cut-out 9	24	line	8.64, 6.36	8.64, 6.36		
Cut-out 10	25	arc anticlockwise	8.36, 6.66	8.36, 6.66	3.26, 3.32	6.10
Cut-out 10	26	arc anticlockwise	8.36, 6.66	8.36, 6.66	3.23, 3.38	6.09
Cut-out 11	27	arc anticlockwise	8.33, 6.71	8.33, 6.71	2.89, 4.10	6.04
Cut-out 11	28	arc anticlockwise	8.33, 6.71	8.33, 6.71	3.48, 2.86	6.19
Cut-out 12	29	line	8.25, 7.90	8.25, 7.90		
Cut-out 12	30	arc anticlockwise	8.25, 7.90	8.25, 7.90	2.84, 4.70	6.28
Cut-out 12	31	line	8.25, 7.90	8.25, 7.90		
Cut-out 13	32	line	8.25, 7.90	8.25, 7.90		
Cut-out 13	33	arc anticlockwise	8.25, 7.90	8.25, 7.90	2.84, 4.70	6.28
Cut-out 13	34	line	8.25, 7.90	8.25, 7.90		
Cut-out 16	35	arc anticlockwise	-25.13, 2.06	-25.13, 2.06	-13.83, 1.78	11.30
Cut-out 16	36	arc anticlockwise	-25.13, 2.06	-25.13, 2.06	-13.83, 1.78	11.30
Cut-out 17	37	line	-29.43, 0.39	-29.43, 0.39		
Cut-out 17	38	line	-29.43, 0.39	-29.43, 0.39		
Cut-out 17	39	line	-29.43, 0.39	-29.43, 0.39		
Cut-out 17	40	line	-29.43, 0.39	-29.43, 0.39		
Cut-out 18	41	line	-29.55, 0.27	-29.55, 0.27		
Cut-out 18	42	line	-29.55, 0.27	-29.55, 0.27		
Cut-out 18	43	line	-29.55, 0.27	-29.55, 0.27		
Cut-out 18	44	line	-29.55, 0.27	-29.55, 0.27		

Levels

Faces in section A-A, from the bottom face: 3.32.





Isometric view, not to scale

Werkstoff / Material „Messingblech 3 Din 1751 Ms 58 federhart"	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 05-2*** *Teil / Part* *barrelBridge.click*
Behandlung / Treatment untoleranced sizes ±0,1		*Benennung / Name* ***Click (Sperrkegel)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Click (Sperrkegel) — sheet 2

The click on the barrel's ratchet, which holds the mainspring's set-up. It is cut from spring-hard brass, so it is its own spring: no separate spring is drawn.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 05-2	Sperrkegel (click)	1	„Messingblech 3 Din 1751 Ms 58 federhart“	untoleranced sizes ±0,1	on 050805-1: 0 mm; round WN-22 (free): 2.634 mm, least gap 0.0706

What it does (05 08 05-2). The click on the barrel's ratchet, spring-hard brass, its own spring.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.60.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 3.23 through

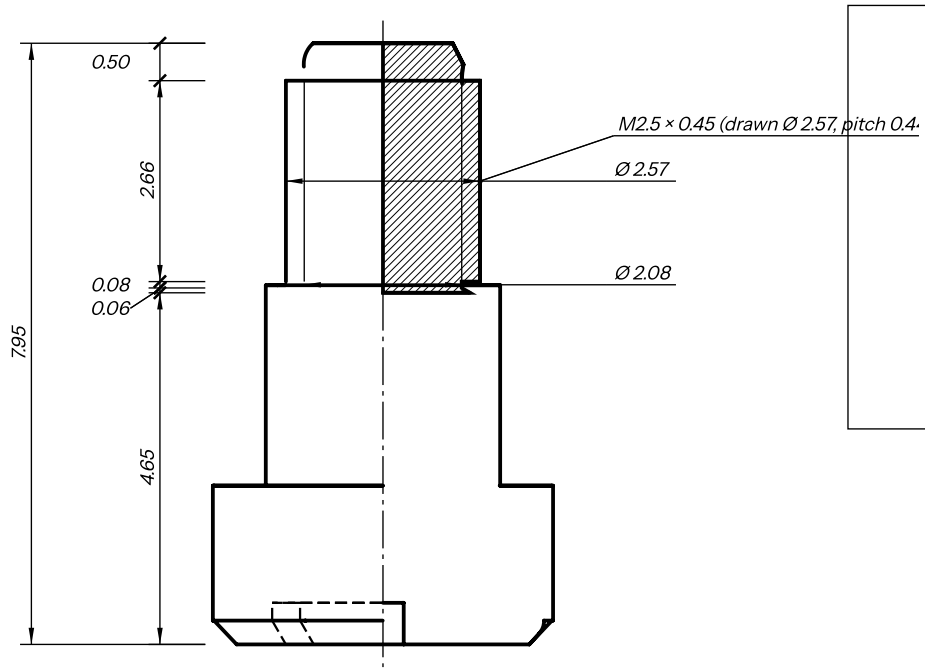
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-7.70, 1.44	7.27, -0.64	-1.58, -9.45	12.49
Outline	2	line	7.27, -0.64	7.27, -0.82		
Outline	3	line	7.27, -0.82	7.28, -0.83		
Outline	4	arc clockwise	7.28, -0.83	7.90, -1.31	7.30, -1.45	0.61
Outline	5	line	7.90, -1.31	8.27, -1.76		
Outline	6	arc clockwise	8.27, -1.76	7.25, -2.83	7.27, -1.83	1.00
Outline	7	line	7.25, -2.83	-1.01, -2.85		
Outline	8	arc anticlockwise	-1.01, -2.85	-9.23, -2.85	-5.12, -7.23	6.00
Outline	9	line	-9.23, -2.85	-9.25, -2.84		
Outline	10	arc clockwise	-9.25, -2.84	-8.54, 0.92	-7.30, -1.26	2.51
Outline	11	arc clockwise	-8.54, 0.92	-7.70, 1.44	-1.58, -9.46	12.50

Levels

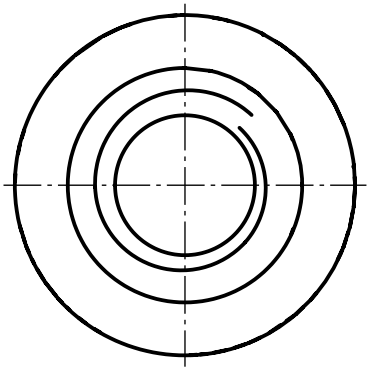
Faces in section A-A, from the bottom face: 2.60.





Isometric view, not to scale

View from above



Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-22** *Teil / Part* *barrelBridge.clickScrew*
Behandlung / Treatment		*Benennung / Name* ***Click screw***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Click screw — sheet 2

The shouldered screw the click turns on.

Parts list

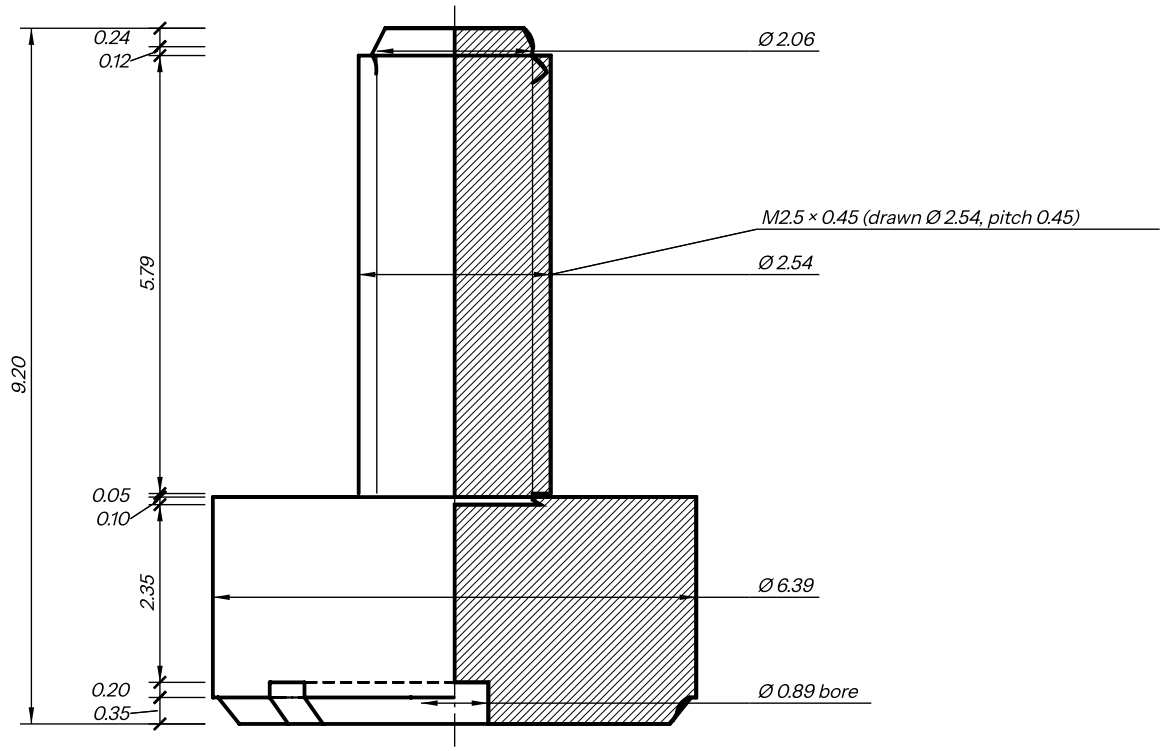
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-22	Click screw, shouldered (WN-22)	1	not on the drawing		in 050805-1 (tap): 3.181 mm, least gap 0.0244

What it does (WN-22). The click turns on its shoulder.

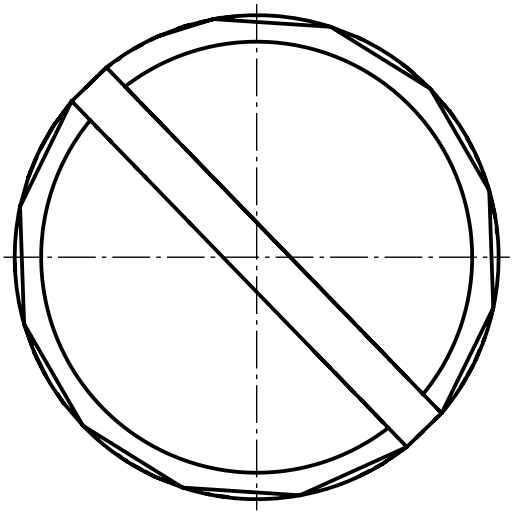
Turned about its axis; length 7.95, largest Ø 4.50. Diameters: Ø 2.08, Ø 2.57. Shoulders from the lower end: 0.00, 4.65, 4.71, 4.80, 7.46, 7.95.

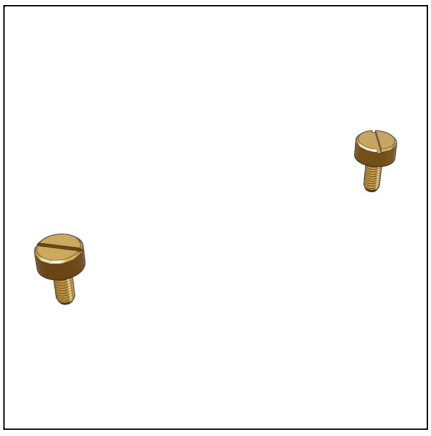
Thread

Outside thread 4.80–7.45 from the lower end, 2.66 long: drawn Ø 2.57 over its crests, 2.08 at its roots, pitch 0.44. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).



View from below (seen from beneath, x reversed)





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-55*** *Teil / Part* *barrelBridge.screws*
Behandlung / Treatment		*Benennung / Name* ***Barrel bridge screws***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Barrel bridge screws — sheet 2

Two screws holding the bridge to the top plate.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

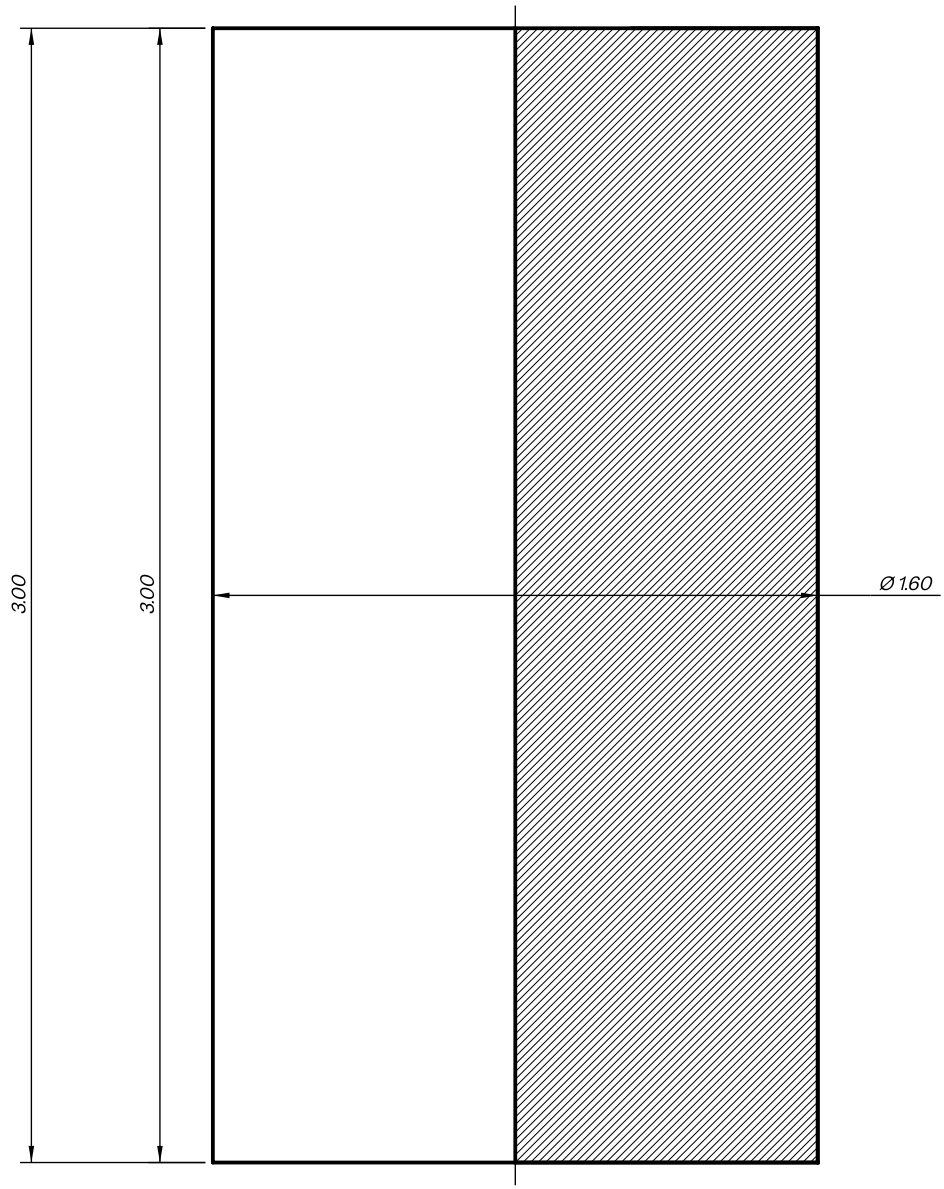
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-55	Barrel bridge screw (WN-55, by the coloured section's label)	2	not on the drawing		in 050805-1 (clear): 3.369 mm, least gap 0.1051; in 050901 (tap): 2.855 mm, least gap 0.027

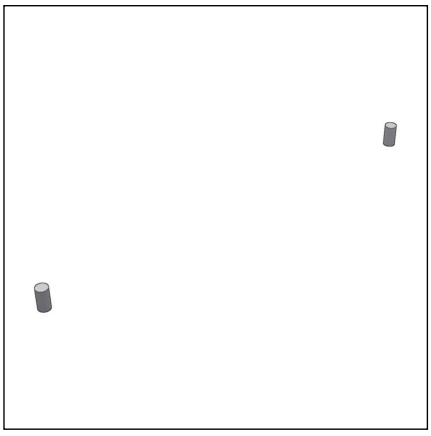
What it does (WN-55). Hold the barrel bridge to the top plate.

Turned about its axis; length 9.20, largest Ø 6.40. Diameters: Ø 2.06, Ø 2.54, Ø 6.39; bores Ø 0.89. Shoulders from the lower end: 0.00, 0.35, 0.55, 2.90, 3.00, 3.05, 8.84, 8.96, 9.20.

Thread

Outside thread 3.05–8.84 from the lower end, 5.79 long: drawn Ø 2.54 over its crests, 2.06 at its roots, pitch 0.45. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **50:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-203.bb** Teil / Part barrelBridge.pins
Behandlung / Treatment		Benennung / Name **Barrel bridge steady pins**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Barrel bridge steady pins — sheet 2

A steady pin (Stellstift), Ø 1.6, pressed into the part and standing 1.3 out of it into a reamed hole in the plate, which places the part. WN-203.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

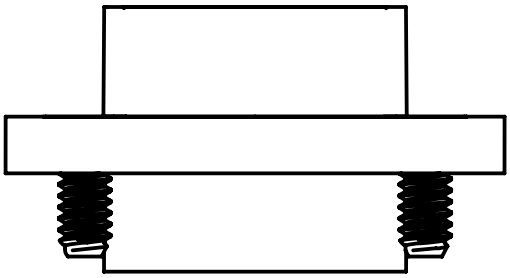
Parts list

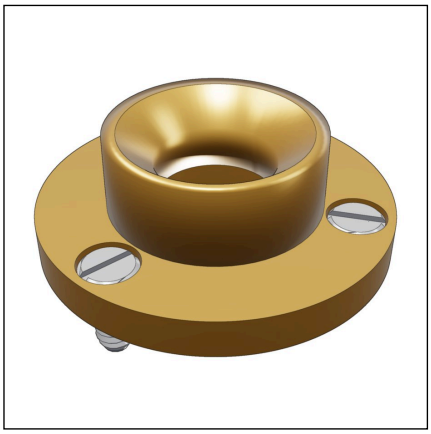
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-203.bb	Stellstift (WN-203), barrel bridge	2	not on the drawing		in 050805-1 (press): 1.427 mm, least gap 0.0019; in 050901 (clear): 1.279 mm, least gap 0.0116

What it does (WN-203.bb). Place the barrel bridge on the top plate.

Turned about its axis; length 3.00, largest Ø 1.60. Diameters: Ø 1.60. Shoulders from the lower end: 0.00, 3.00.

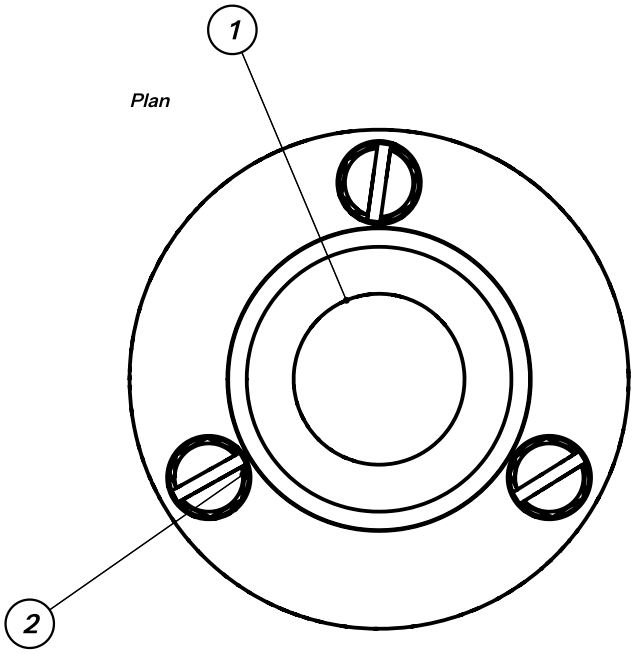
Front view





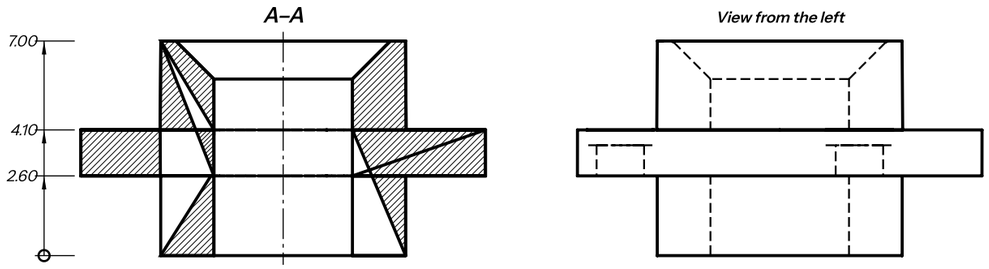
Isometric view, not to scale

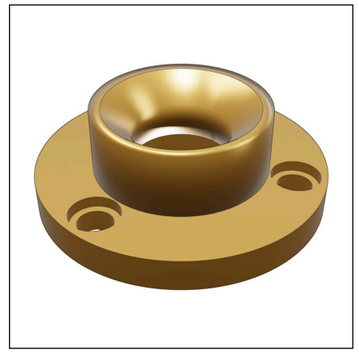
Plan



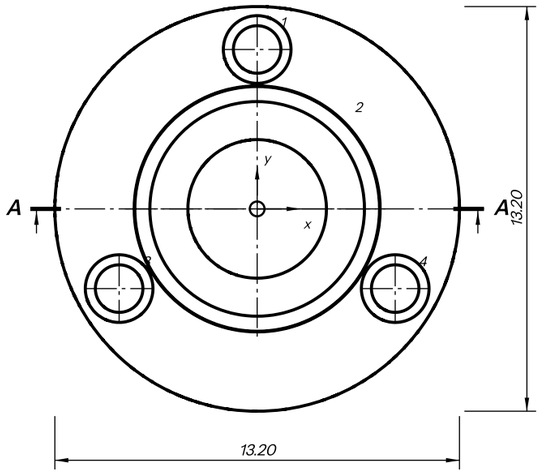
Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	3	Fusee bush screws	WN-34	not on the drawing
1	1	Fusee bush	05 08 01-2	„Rundmessing 13,5 Din 1758 Ms

Werkstoff / Material „Rundmessing 13,5 Din 1758 Ms 58"	Maßstab / Scale **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	Zeichnung Nr. / Drawing no. **05 08 01-2** Teil / Part *fuseeBush*
Behandlung / Treatment none on the drawing		Benennung / Name ***Fusee bush (Schneckenfutter)***	
Blatt / Sheet **1 / 1**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	





Isometric view, not to scale



Werkstoff / Material *„Rundmessing 13,5 Din 1758 Ms 58"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 01-2***
Behandlung / Treatment *none on the drawing*			*Teil / Part* *fuseeBush.bush*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Fusee bush***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Fusee bush — sheet 2

A flanged brass bush pressed into the top plate and screwed to it, in which the fusee arbor’s winding end turns.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 01-2	Schneckenfutter (fusee bush)	1	„Rundmessing 13,5 Din 1758 Ms 58“	none on the drawing	in 050901 (free): 2.579 mm, least gap 0.009; on 050901: 0 mm

What it does (05 08 01-2). Its spigot slides into the top plate's hole (8 g6 in 8.0 +0.015) and its flange is screwed to it: the fusee arbor's winding end turns in it.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 7.00.

Holes

Hole	x	y	Size
1	0.00	5.20	Ø 1.55 through (1.00 thick at 2.60–3.60)
2	0.00	0.00	Ø 4.52 through (1.68 thick at 4.10–5.78); sink Ø 4.52 × 2.60 from the face at 2.60; countersunk to Ø 7.00; sink Ø 8.02 × 1.50 from the face at 4.10
3	-4.50	-2.60	Ø 1.55 through (1.00 thick at 2.60–3.60)
4	4.50	-2.60	Ø 1.55 through (1.00 thick at 2.60–3.60)

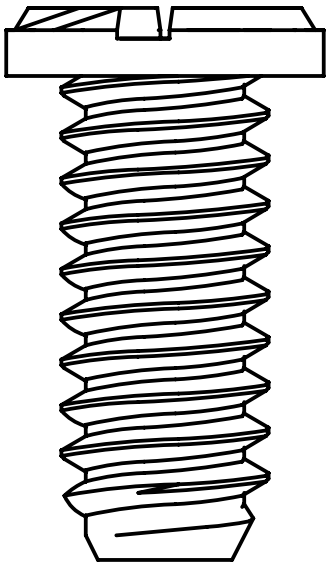
Contour in the plan

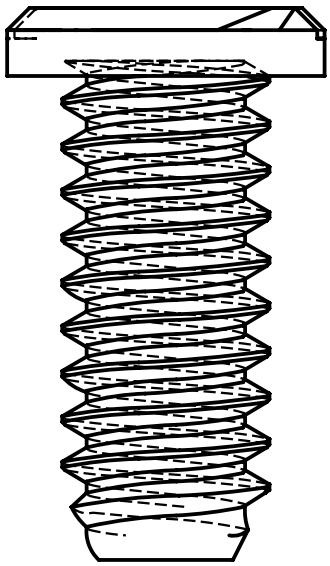
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

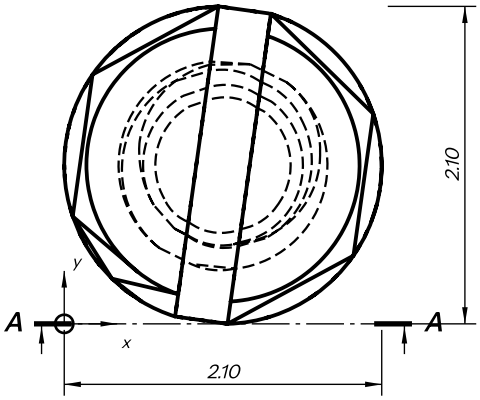
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			0.00, 0.00	6.60

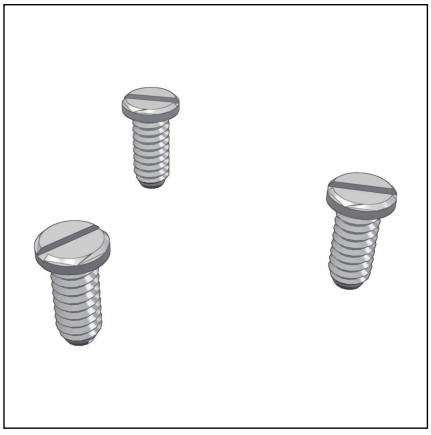
Levels

Faces in section A–A, from the bottom face: 2.60, 4.10, 7.00.









Isometric view, not to scale

3 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-34** *Teil / Part* *fuseeBush.screws*
Behandlung / Treatment		*Benennung / Name* ***Fusee bush screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Fusee bush screws — sheet 2

Three small screws holding the bush's flange to the top plate.

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-34	Fusee bush screw (WN-34)	3	not on the drawing		in 050801-2 (clear): 1.461 mm, least gap 0.0535; in 050901 (tap): 2.192 mm, least gap 0.0316

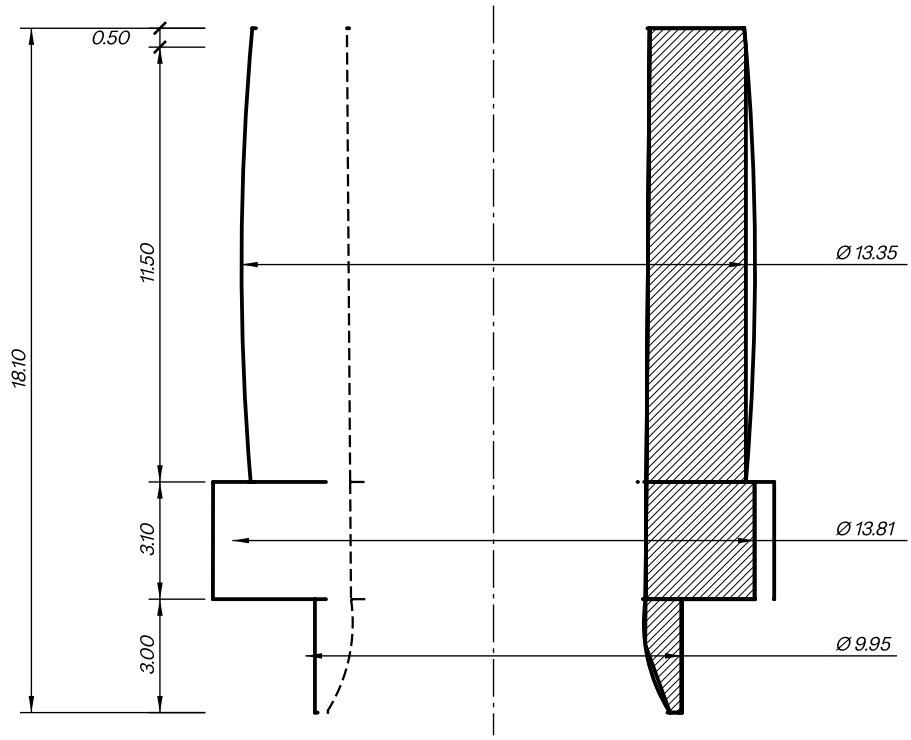
What it does (WN-34). Hold the fusee bush's flange to the top plate.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.65.

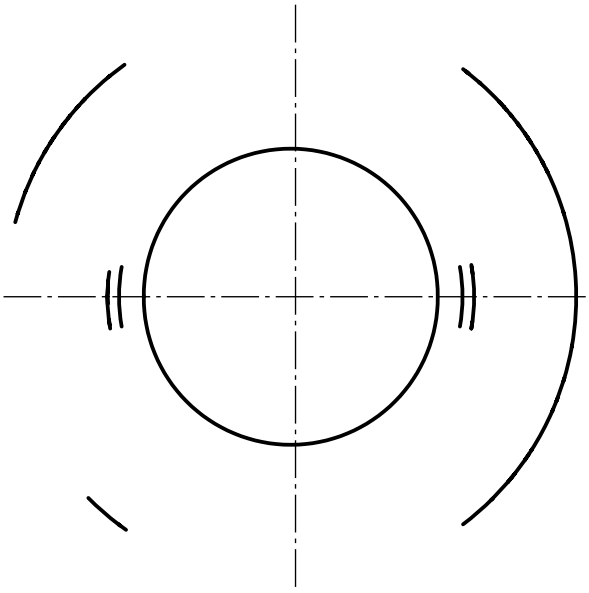
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			1.05, 1.05	1.05
Cut-out 1	2	line	1.08, 0.00	1.06, 0.37		
Cut-out 1	3	line	1.06, 0.37	1.08, 0.00		
Cut-out 2	4	line	1.73, 1.07	1.91, 1.33		
Cut-out 2	5	line	1.91, 1.33	1.73, 1.07		
Cut-out 3	6	line	1.60, 1.45	1.69, 1.68		
Cut-out 3	7	line	1.69, 1.68	1.60, 1.45		
Cut-out 4	8	line	1.28, 0.41	1.10, 0.15		
Cut-out 4	9	line	1.10, 0.15	1.28, 0.41		
Cut-out 5	10	line	0.82, 1.69	1.00, 1.95		
Cut-out 5	11	line	1.00, 1.95	0.82, 1.69		
Cut-out 6	12	line	1.00, 1.95	0.96, 1.73		
Cut-out 6	13	line	0.96, 1.73	1.00, 1.95		



View from below (seen from beneath, x reversed)





Isometric view, not to scale

Werkstoff / Material *„Rundm s. 15 Din 1758 Ms58"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 08 01-3*** *Teil / Part* *dustBush*
Behandlung / Treatment *none on the drawing*		*Benennung / Name* ***Dust bush (Staubfutter)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Dust bush (Staubfutter) — sheet 2

A tall brass bush on the fusee bush round the winding square, up to the case's winding hole, keeping dust out of the movement. A flat on its collar clears the balance cock.

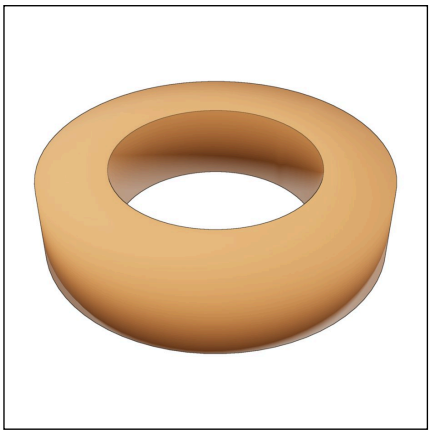
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 01-3	Staubfutter (dust bush)	1	„Rundm s. 15 Din 1758 Ms58“	none on the drawing	round 050801-2 (press): 2.877 mm, least gap -0.0106; on 050801-2: 0 mm

What it does (05 08 01-3). Pressed on the fusee bush's tapered top (8 H7 on 8 T6, both 1:70), up to the case's winding hole.

Turned about its axis; length 18.10, largest Ø 14.85. Diameters: Ø 9.95, Ø 13.35, Ø 13.81. Shoulders from the lower end: 0.00, 3.00, 6.10, 17.60, 18.10.





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 08 01-4** Teil / Part corkRing
Behandlung / Treatment		Benennung / Name **Cork ring (Korkring)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Cork ring (Korkring) — sheet 2

A cork ring on the dust bush's top, standing a little over it round the winding square. In the case it meets a raised seat round the winding hole on the bowl's bottom, so dust that comes in through the hole can't reach the movement: it goes only down the bush to the winding square.

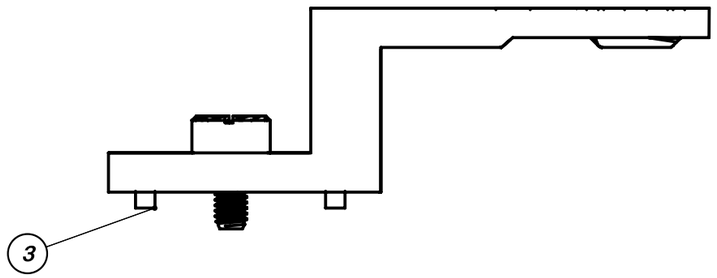
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 01-4	Korkring (cork ring)	1	not on the drawing		round 050801-3 (press): 3.035 mm, least gap 0.01; on 050801-3: 0 mm

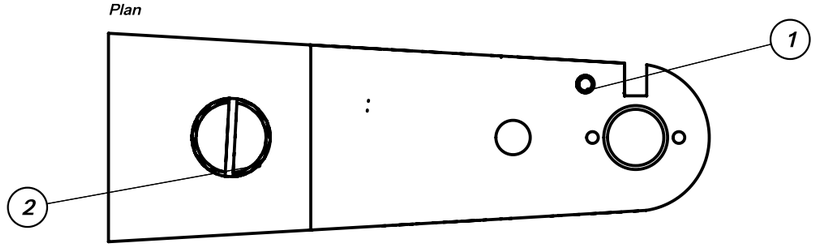
What it does (05 08 01-4). Pressed on the dust bush's neck, on its collar, standing 0.95 over it: the case's closing plate meets it round the winding hole, keeping dust out (no sheet; scaled on the coloured section).

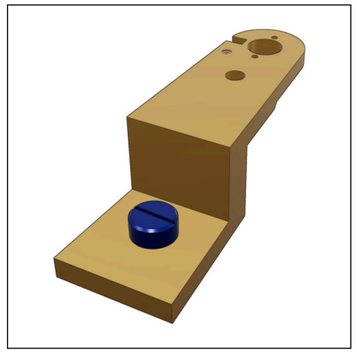
Turned about its axis; length 3.70, largest Ø 16.10. Diameters: Ø 16.10; bores Ø 9.72. Shoulders from the lower end: 0.00, 3.70.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	3	Upper cock steady pins	WN-203.uc	not on the drawing
2	1	Upper cock screw	WN-55	not on the drawing
1	1	Upper balance cock	05 08 03-1	„Profilmessing 50×15×3,3×5,5 M

Werkstoff / Material
„Profilmessing 50×15×3,3×5,5 Ms58" (the "5,5" likely)

Maßstab / Scale
2:1

Wempe Einheits-Chronometer
The Marine Chronometer working model
drawn from its solids by tools/drawbook.py
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 08 03-1

Teil / Part
upperCock

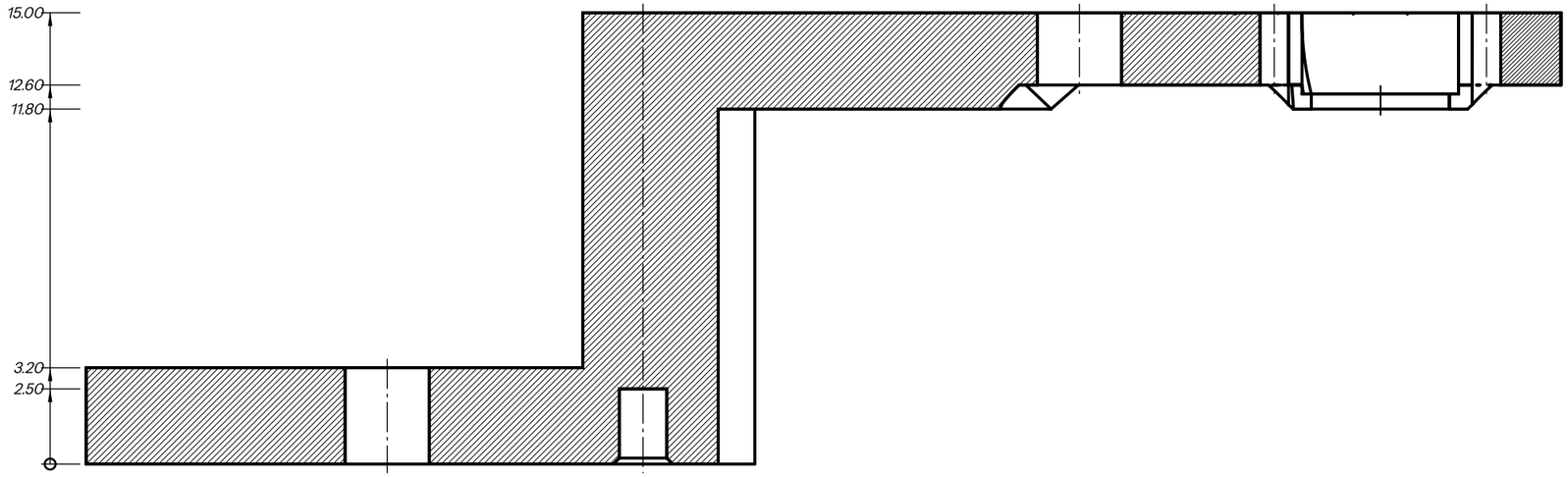
Behandlung / Treatment
blank (Rohmaß) Profil ×18; untoleranced sizes ±0.1

Benennung / Name
Upper balance cock (Unruhkloben oben)

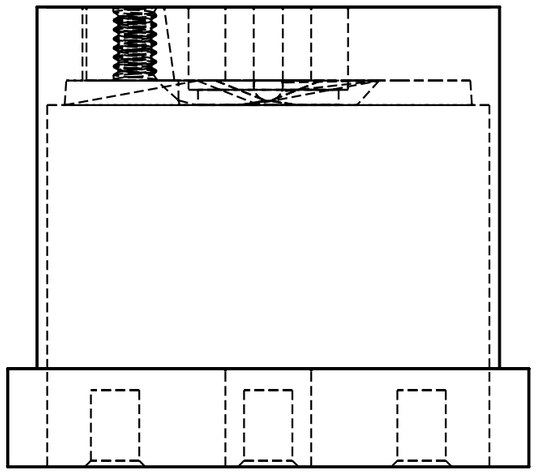
Blatt / Sheet
1 / 1

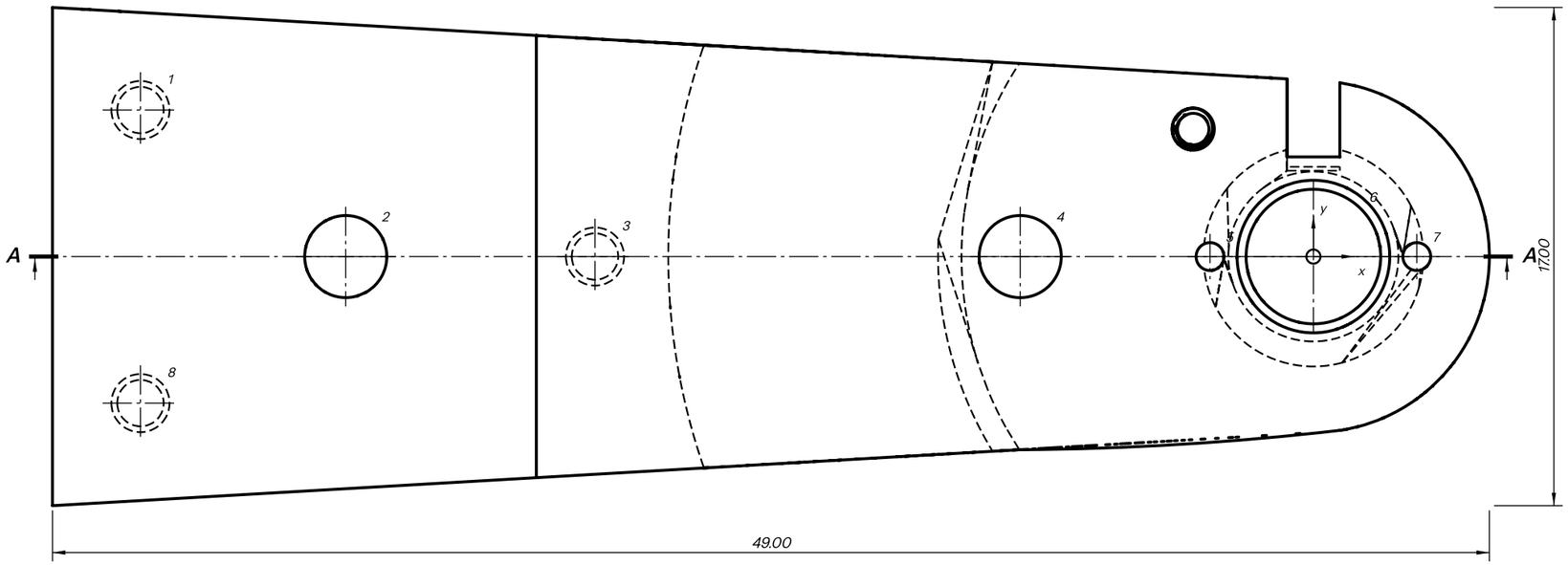
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).

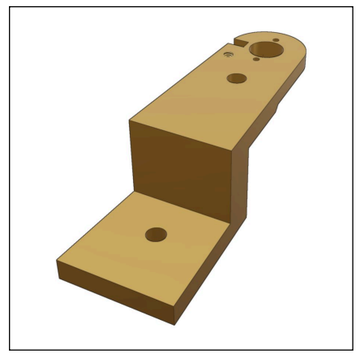
A–A



View from the left







Isometric view, not to scale

Werkstoff / Material „Profilmessing 50×15×3,3×5,5 Ms58" (the "5,5" likely)	Maßstab / Scale **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	Zeichnung Nr. / Drawing no. ***05 08 03-1***
Behandlung / Treatment *blank (Rohmaß) Profil ×18; untoleranced sizes ±0,1*			Teil / Part *upperCock.cock*
Blatt / Sheet **1 / 2**		Benennung / Name ***Upper balance cock***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Upper balance cock — sheet 2

The cranked cock over the balance, screwed and pinned to the top plate, holding the balance staff's upper jewel setting over the opening in the plate.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 03-1	Unruhkloben oben (upper balance cock)	1	„Profilmessing 50×15×3,3×5,5 Ms58“ (the "5,5" likely)	blank (Rohmaß) Profil ×18; untoleranced sizes ±0,1	on 050901: 0 mm

What it does (05 08 03-1). Holds the balance staff's upper jewel setting over the top plate.

Coordinates from the datum, hole 6's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 15.00.

Holes

Hole	x	y	Size
1	-40.00	5.00	Ø 1.58 × 2.31 deep from the face at 0.19; countersunk to Ø 1.98
2	-33.00	0.00	Ø 2.80 through (3.20 thick at 0.00–3.20)
3	-24.50	0.00	Ø 1.58 × 2.31 deep from the face at 0.19; countersunk to Ø 1.98
4	-10.00	0.00	Ø 2.80 through (2.40 thick at 12.60–15.00)
5	-3.53	0.00	Ø 0.95 through (2.40 thick at 12.60–15.00)
6	0.00	0.00	Ø 4.59 through (0.50 thick at 11.80–12.30)
7	3.53	0.00	Ø 0.95 through (2.40 thick at 12.60–15.00)
8	-40.00	-5.00	Ø 1.58 × 2.31 deep from the face at 0.19; countersunk to Ø 1.98

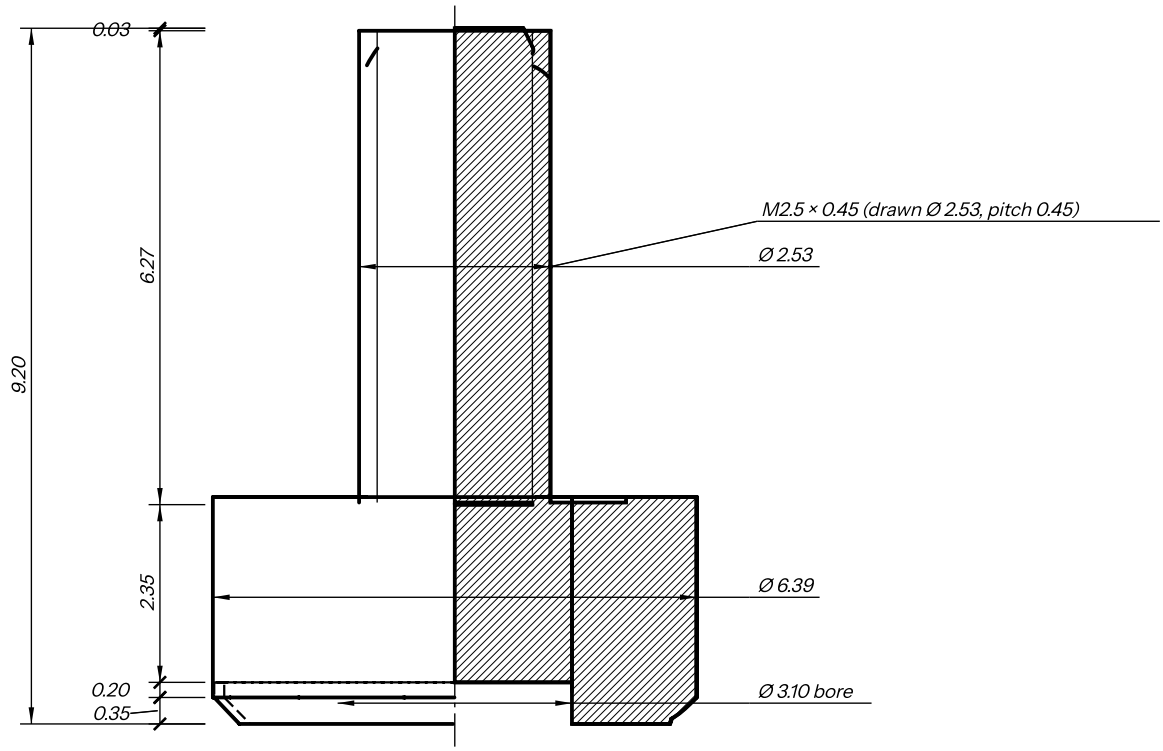
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

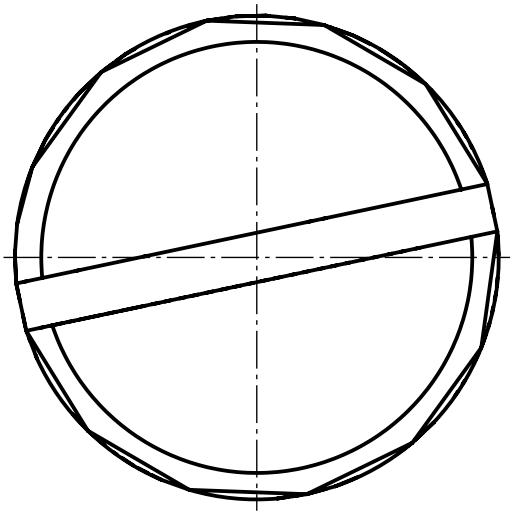
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-20.78, 7.21	-0.90, 6.06		
Outline	2	line	-0.90, 6.06	-0.90, 3.40		
Outline	3	line	-0.90, 3.40	0.90, 3.40		
Outline	4	line	0.90, 3.40	0.90, 5.93		
Outline	5	arc clockwise	0.90, 5.93	0.35, -5.99	0.00, 0.00	6.00
Outline	6	line	0.35, -5.99	-43.00, -8.50		
Outline	7	line	-43.00, -8.50	-43.00, 8.50		
Outline	8	line	-43.00, 8.50	-20.78, 7.21		

Levels

Faces in section A–A, from the bottom face: 2.50, 3.20, 11.80, 12.60, 15.00.



View from below (seen from beneath, x reversed)





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-55** *Teil / Part* *upperCock.screw*
Behandlung / Treatment		*Benennung / Name* ***Upper cock screw***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Upper cock screw — sheet 2

The blued screw holding the cock to the top plate.

Parts list

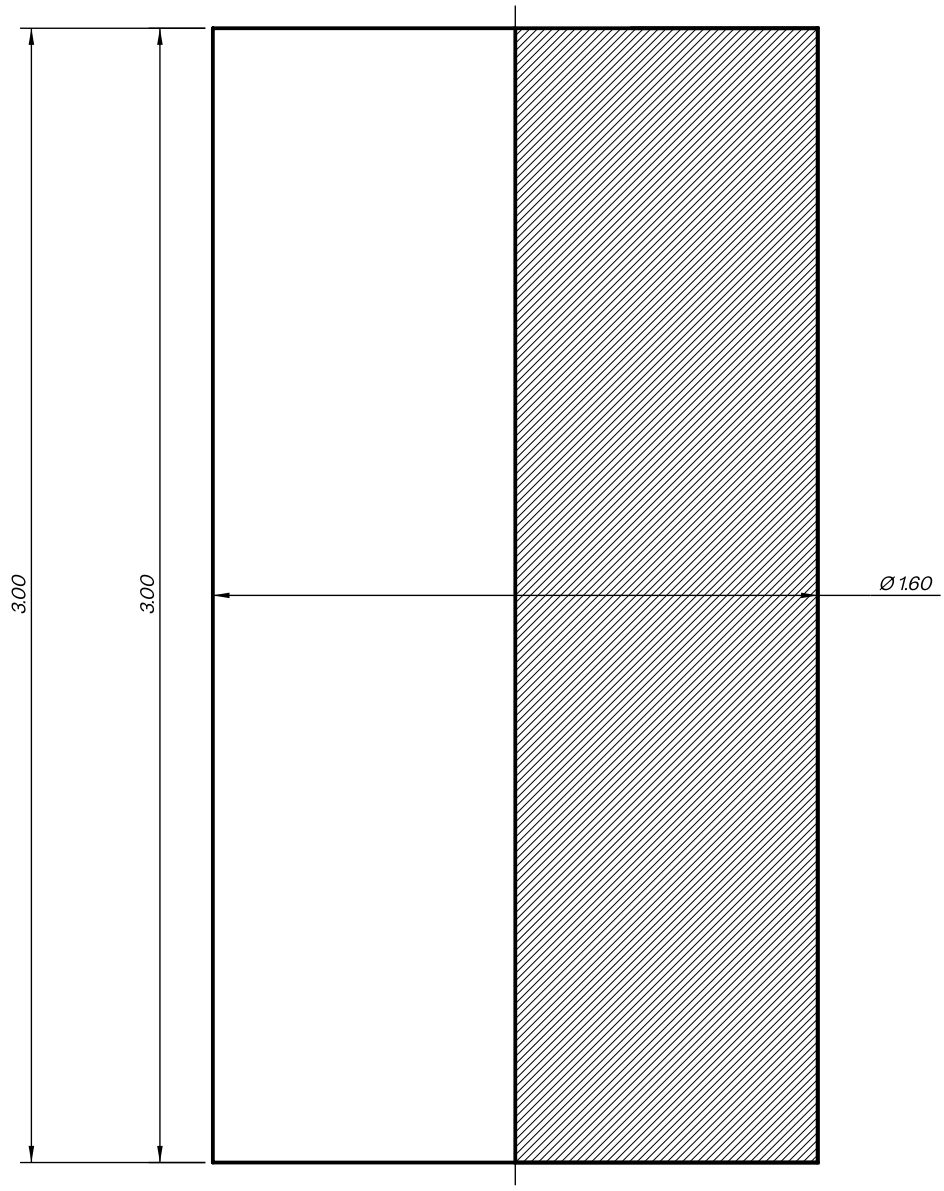
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-55	Upper balance cock screw (hole 8; WN-55 by the coloured section's label)	1	not on the drawing		in 050803-1 (clear): 3.14 mm, least gap 0.1051; in 050901 (tap): 3.026 mm, least gap 0.027

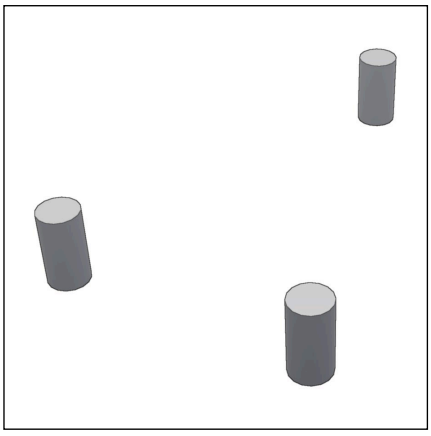
What it does (WN-55). Holds the upper cock to the top plate.

Turned about its axis; length 9.20, largest Ø 6.40. Diameters: Ø 2.53, Ø 6.39; bores Ø 3.10. Shoulders from the lower end: 0.00, 0.35, 0.55, 2.90, 9.17, 9.20.

Thread

Outside thread 2.93–9.17 from the lower end, 6.24 long: drawn Ø 2.53 over its crests, 2.05 at its roots, pitch 0.45. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

3 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **50:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-203.uc** Teil / Part upperCock.pins
Behandlung / Treatment		Benennung / Name **Upper cock steady pins**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Upper cock steady pins — sheet 2

A steady pin (Stellstift), Ø 1.6, pressed into the part and standing 1.3 out of it into a reamed hole in the plate, which places the part.

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

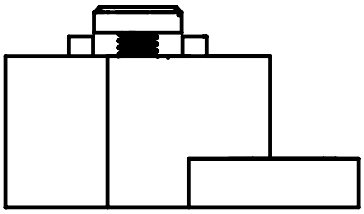
Parts list

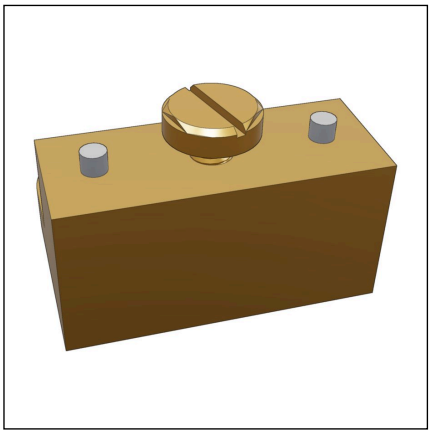
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-203.uc	Stellstift, upper balance cock (holes 6, 9, 50; WN not given, as WN-203)	3	not on the drawing		in 050803-1 (press): 1.525 mm, least gap 0.0019; in 050901 (clear): 1.279 mm, least gap 0.0116

What it does (WN-203.uc). Place the upper cock on the top plate.

Turned about its axis; length 3.00, largest Ø 1.60. Diameters: Ø 1.60. Shoulders from the lower end: 0.00, 3.00.

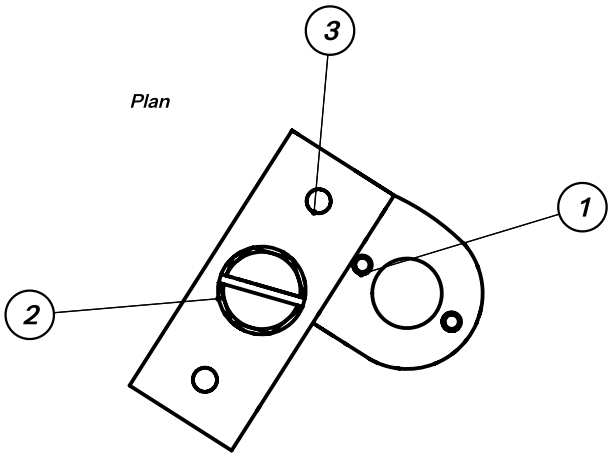
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	2	Lower cock steady pins	WN-203.lc	not on the drawing
2	1	Lower cock screw	WN-53	not on the drawing
1	1	Lower balance cock	05 08 00-2	„Profilmessing 20×10×8×3,3 Ms

Werkstoff / Material
„Profilmessing 20×10×8×3,3 Ms 58"

Maßstab / Scale
2:1

Behandlung / Treatment
blank (Rohmaß) Profil ×18

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

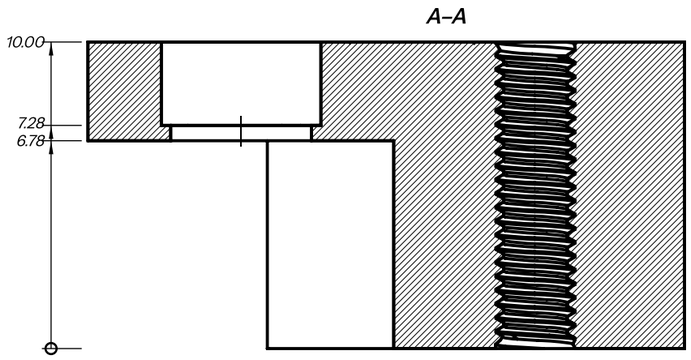
Model v3.35.00 · 2026-10-26

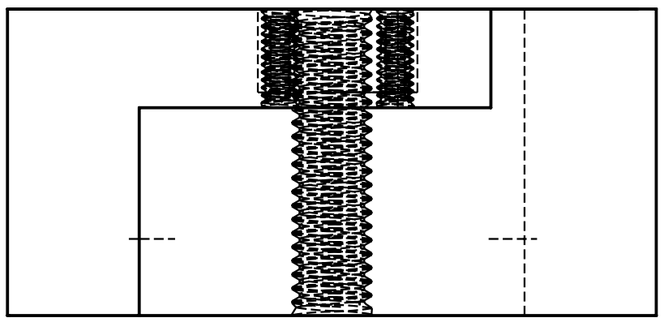
Zeichnung Nr. / Drawing no.
05 08 00-2

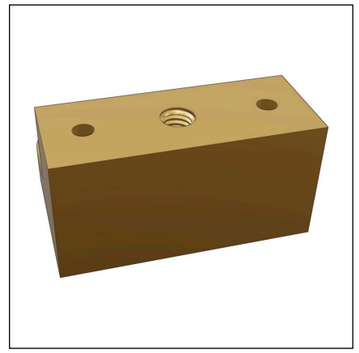
Teil / Part
lowerCock

Benennung / Name
Lower balance cock (Unruhkloben unten)

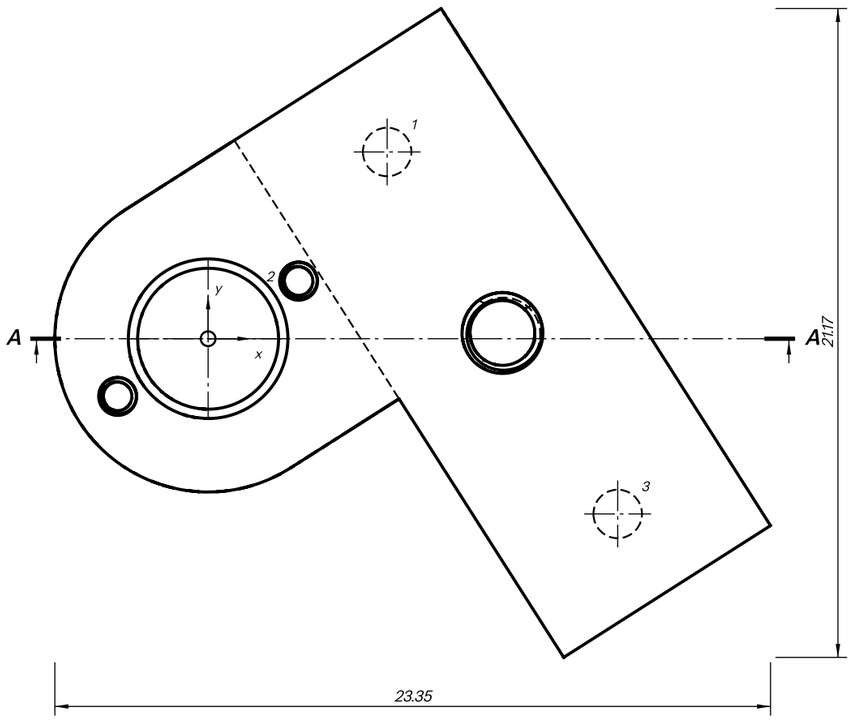
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).







Isometric view, not to scale



Werkstoff / Material „Profilmessing 20×10×8×3,3 Ms 58"	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 08 00-2**
Behandlung / Treatment blank (Rohmaß) Profil ×18			Teil / Part lowerCock.cock
Blatt / Sheet **1 / 2**		Benennung / Name **Lower balance cock**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Lower balance cock — sheet 2

A block screwed and pinned under the top plate, inside the frame, whose lug holds the balance staff's lower jewel setting about 10 mm under the plate: the balance swings above the plate, its escapement under it.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 00-2	Unruhkloben unten (lower balance cock)	1	„Profilmessing 20×10×8×3,3 Ms 58“	blank (Rohmaß) Profil ×18	on 050901: 0 mm

What it does (05 08 00-2). Hangs under the top plate and holds the balance staff's lower jewel setting.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 10.00.

Holes

Hole	x	y	Size
1	5.84	6.09	Ø 1.58 × 2.50 deep from the bottom
2	0.00	0.00	Ø 4.60 through (0.50 thick at 6.78–7.28)
3	13.36	-5.72	Ø 1.58 × 2.50 deep from the bottom

Contour in the plan

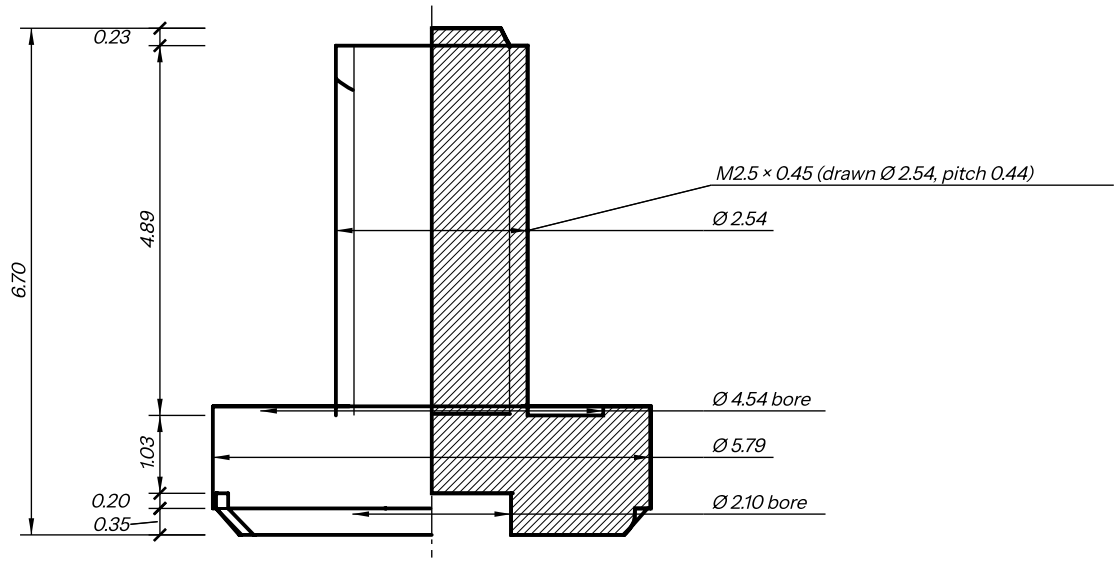
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-4.84, -1.25	-2.69, 4.22	0.00, 0.00	5.00
Outline	2	line	-2.69, 4.22	7.60, 10.77		
Outline	3	line	7.60, 10.77	18.35, -6.10		
Outline	4	line	18.35, -6.10	11.60, -10.40		
Outline	5	line	11.60, -10.40	6.23, -1.96		
Outline	6	arc clockwise	6.23, -1.96	1.99, -4.59	-11.12, 21.27	28.99
Outline	7	arc clockwise	1.99, -4.59	-4.84, -1.25	0.00, 0.00	5.00
Cut-out 2	8	line	10.56, 1.09	10.56, 1.09		
Cut-out 2	9	arc anticlockwise	10.56, 1.09	10.56, 1.09	7.92, 3.88	3.83
Cut-out 2	10	line	10.56, 1.09	10.56, 1.09		
Cut-out 3	11	line	10.40, 1.23	10.40, 1.23		
Cut-out 3	12	arc anticlockwise	10.40, 1.23	10.40, 1.23	8.51, 0.70	1.97
Cut-out 3	13	line	10.40, 1.23	10.40, 1.23		
Cut-out 4	14	line	10.13, -1.56	10.13, -1.56		
Cut-out 4	15	arc anticlockwise	10.13, -1.56	10.13, -1.56	8.38, -0.67	1.97
Cut-out 4	16	line	10.13, -1.56	10.13, -1.56		
Cut-out 5	17	line	9.28, 1.46	9.28, 1.46		
Cut-out 5	18	line	9.28, 1.46	9.28, 1.46		
Cut-out 5	19	line	9.28, 1.46	9.28, 1.46		
Cut-out 6	20	line	9.19, 1.43	9.19, 1.43		
Cut-out 6	21	line	9.19, 1.43	9.19, 1.43		
Cut-out 6	22	line	9.19, 1.43	9.19, 1.43		
Cut-out 6	23	line	9.19, 1.43	9.19, 1.43		

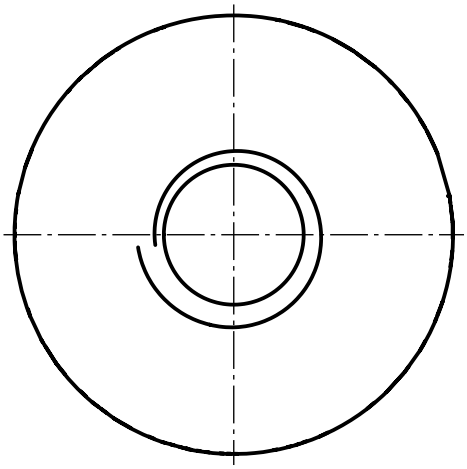
Cut-out 7	24	arc anticlockwise	8.83, 1.26	8.83, 1.26	7.80, 0.52	1.27
Cut-out 7	25	arc anticlockwise	8.83, 1.26	8.83, 1.26	7.80, 0.52	1.27
Cut-out 8	26	line	8.61, -0.67	8.61, -0.67		
Cut-out 8	27	arc anticlockwise	8.61, -0.67	8.61, -0.67	7.59, -0.22	1.11
Cut-out 8	28	line	8.61, -0.67	8.61, -0.67		

Levels

Faces in section A-A, from the bottom face: 6.78, 7.28, 10.00.



View from above





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-53** *Teil / Part* *lowerCock.screw*
Behandlung / Treatment		*Benennung / Name* ***Lower cock screw***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Lower cock screw — sheet 2

The screw from the top plate's outer face, its head sunk in the plate, into the cock.

Parts list

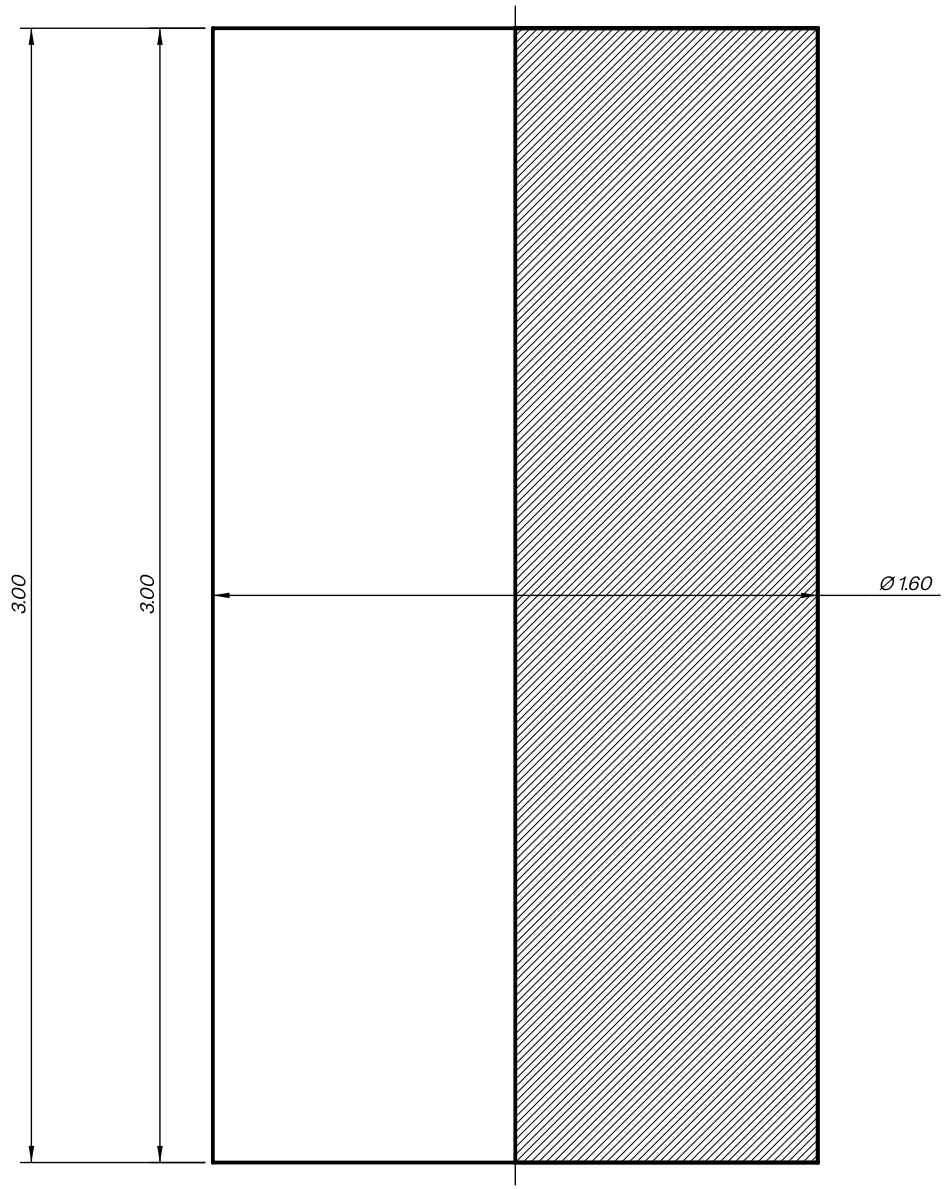
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-53	Lower balance cock screw (WN-53, by the coloured section's label)	1	not on the drawing		in 050901 (clear): 3.254 mm, least gap 0.1051; in 050800-2 (tap): 3.451 mm, least gap 0.0257

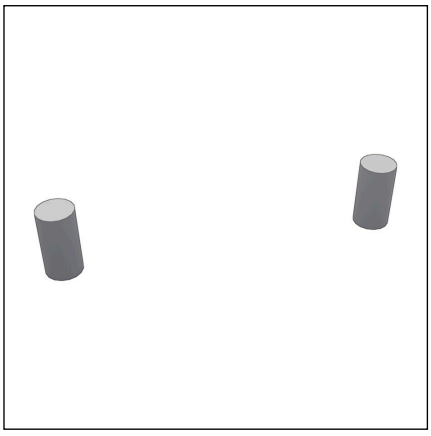
What it does (WN-53). From the top plate's outer face, its head sunk in the plate, into the cock.

Turned about its axis; length 6.70, largest Ø 5.79. Diameters: Ø 2.54, Ø 5.79; bores Ø 2.10, Ø 4.54. Shoulders from the lower end: 0.00, 0.35, 0.55, 1.58, 6.47, 6.70.

Thread

Outside thread 1.58–6.47 from the lower end, 4.89 long: drawn Ø 2.54 over its crests, 2.06 at its roots, pitch 0.44. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **50:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-203.lc** Teil / Part lowerCock.pins
Behandlung / Treatment		Benennung / Name **Lower cock steady pins**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Lower cock steady pins — sheet 2

A steady pin (Stellstift), Ø 1.6, pressed into the part and standing 1.3 out of it into a reamed hole in the plate, which places the part.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

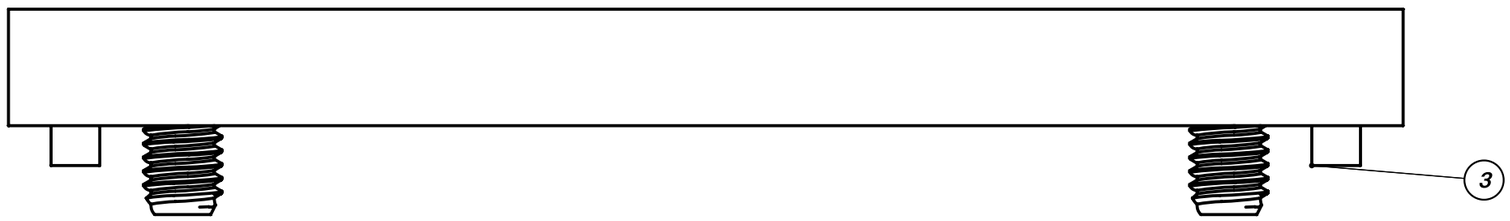
Parts list

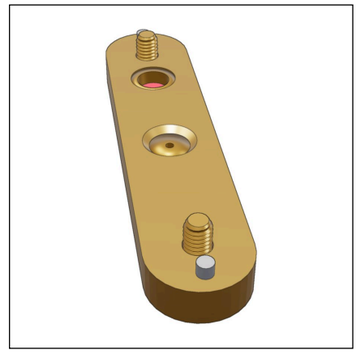
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-203.lc	Stellstift, lower balance cock (holes 22, 37; WN not given, as WN-203)	2	not on the drawing		in 050800-2 (press): 1.722 mm, least gap 0.0016; in 050901 (clear): 1.279 mm, least gap 0.0116

What it does (WN-203.lc). Place the lower cock under the top plate.

Turned about its axis; length 3.00, largest Ø 1.60. Diameters: Ø 1.60. Shoulders from the lower end: 0.00, 3.00.

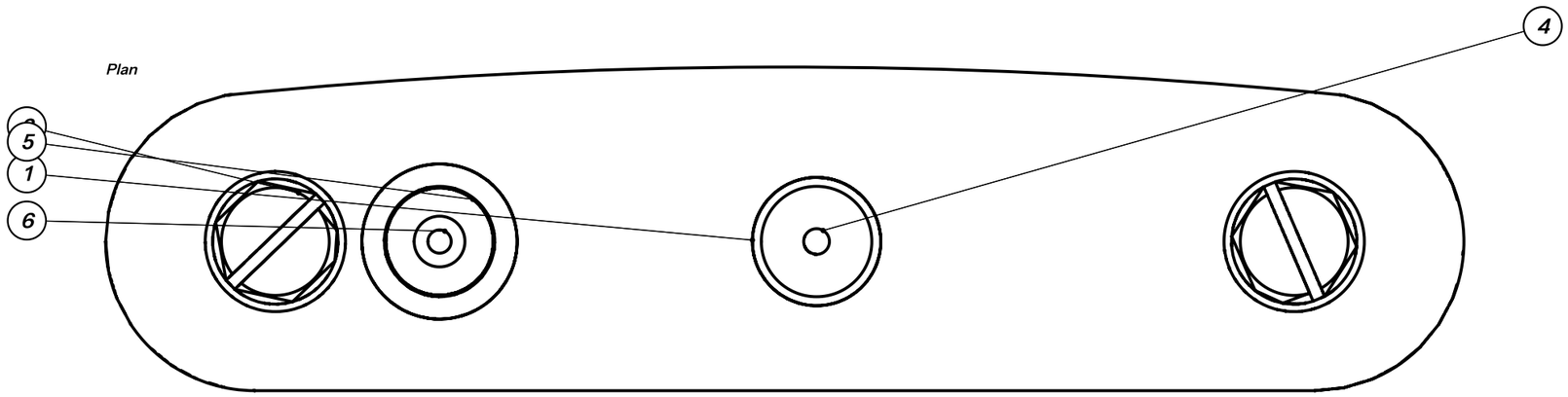
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
6	1	Seconds jewel (Lagerstein f. Sek.)	05 05 04-1	not on the drawing
5	1	Seconds jewel's setting (Fassung)	05 05 04-2	not on the drawing
4	1	Intermediate wheel's lower bush (Messinglager WN-300)	WN-300.zb	not on the drawing
3	2	Intermediate bridge steady pins	WN-203.zb	not on the drawing
2	2	Intermediate bridge screws	WN-52	not on the drawing
1	1	Intermediate bridge	05 08 04-1	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: **5:1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **05 05 04-1**

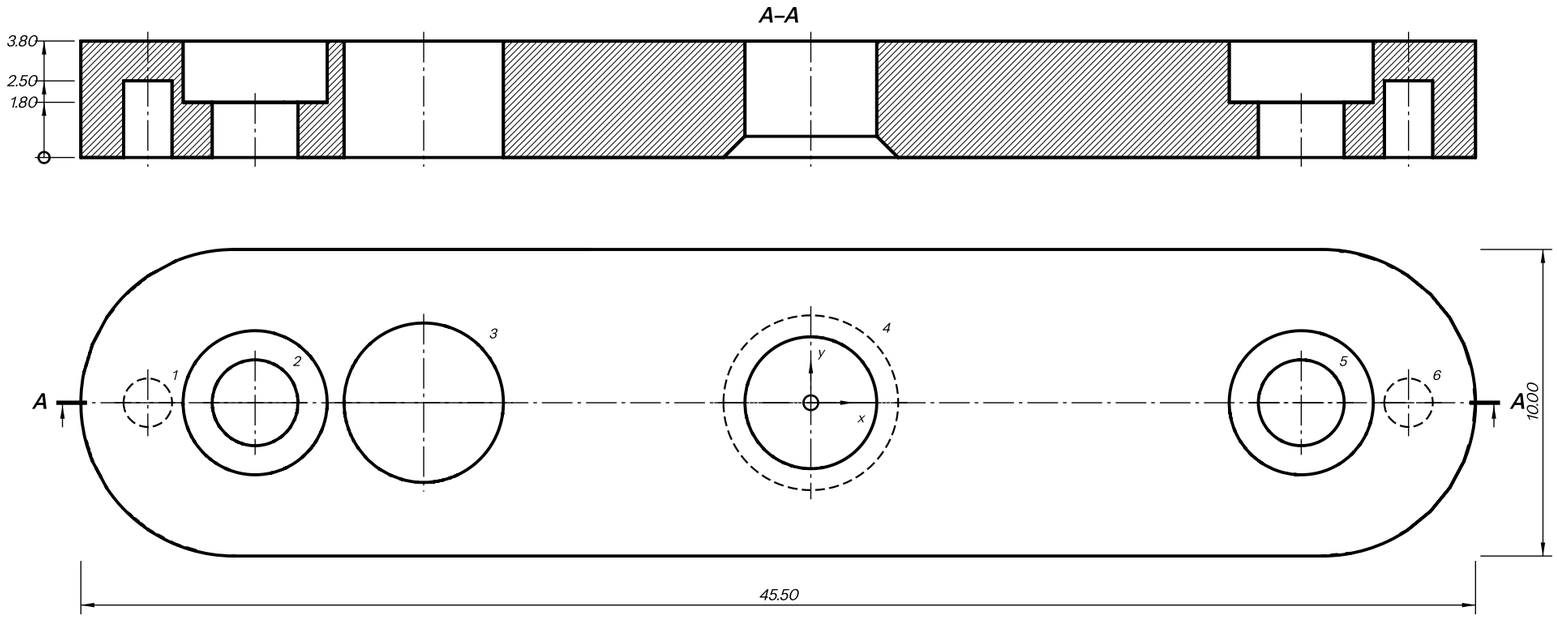
Teil / Part: intermediateBridge

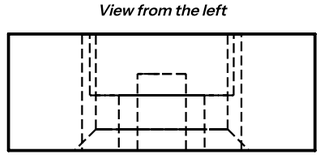
Behandlung / Treatment:

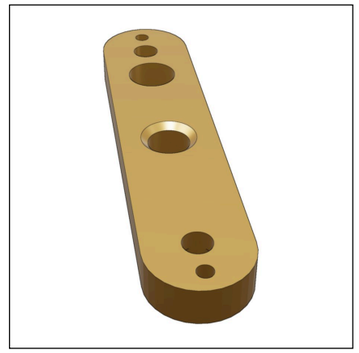
Benennung / Name: **Intermediate bridge (Zwischenbrücke)**

Blatt / Sheet: **1 / 1**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).







Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 08 04-1**
Behandlung / Treatment			Teil / Part intermediateBridge.bridge
Blatt / Sheet **1 / 2**		Benennung / Name **Intermediate bridge**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Intermediate bridge — sheet 2

A straight bridge on the pillar plate's dial side, across the opening for the intermediate wheel: it carries that wheel's lower bearing, a brass bush, and the seconds wheel's lower jewel in its setting.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 08 04-1	Zwischenbrücke (intermediate bridge; its sheet not in the set, the assembly 05 08 04)	1	not on the drawing		on 050802-1: 0 mm

What it does (05 08 04-1). On the pillar plate's dial side: the intermediate wheel's lower jewel and the seconds wheel's lower setting.

Coordinates from the datum, hole 4's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.80.

Holes

Hole	x	y	Size
1	-21.63	0.00	Ø 1.58 × 2.50 deep from the bottom
2	-18.13	0.00	Ø 2.80 through (1.80 thick at 0.00–1.80)
3	-12.63	0.00	Ø 5.20 through
4	0.00	0.00	Ø 4.30 through (3.12 thick at 0.68–3.80); countersunk to Ø 5.70
5	16.00	0.00	Ø 2.80 through (1.80 thick at 0.00–1.80)
6	19.50	0.00	Ø 1.58 × 2.50 deep from the bottom

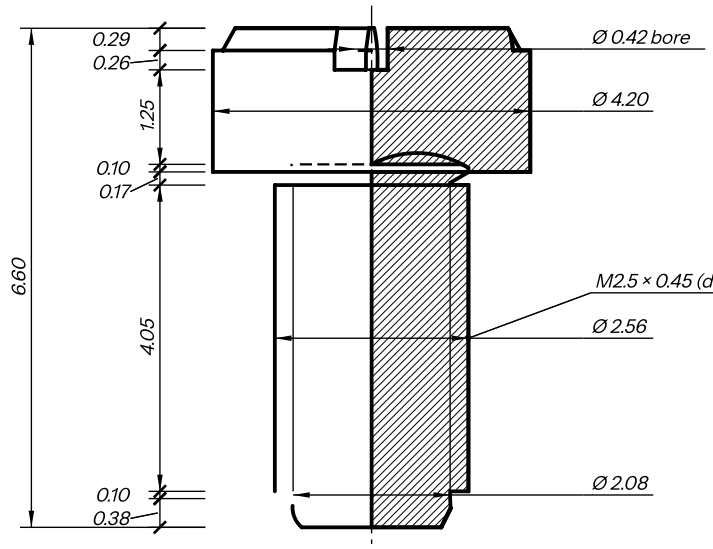
Contour in the plan

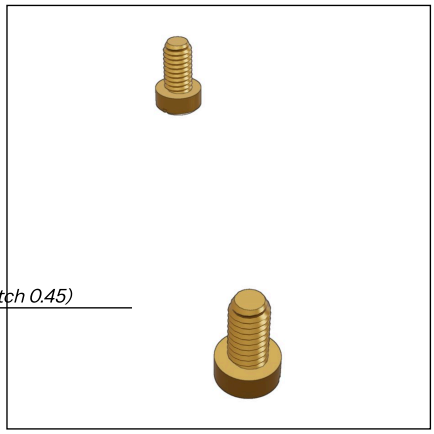
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-23.72, -0.98	-18.82, 5.00	-18.82, 0.00	5.00
Outline	2	arc clockwise	-18.82, 5.00	17.66, 4.90	-1.07, -180.22	186.07
Outline	3	arc clockwise	17.66, 4.90	16.68, -5.00	16.68, 0.00	5.00
Outline	4	arc clockwise	16.68, -5.00	-19.79, -4.90	-1.07, 180.22	186.07
Outline	5	arc clockwise	-19.79, -4.90	-23.72, -0.98	-18.82, 0.00	5.00

Levels

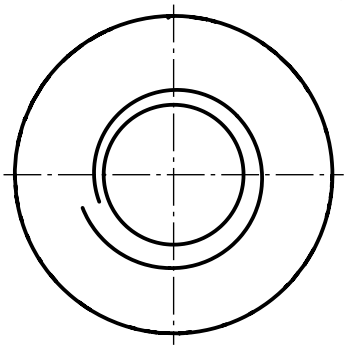
Faces in section A–A, from the bottom face: 1.80, 2.50, 3.80.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-52** *Teil / Part* *intermediateBridge.screws*
Behandlung / Treatment		*Benennung / Name* ***Intermediate bridge screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Intermediate bridge screws — sheet 2

Two screws from the dial side, their heads sunk in the bridge, into the pillar plate.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

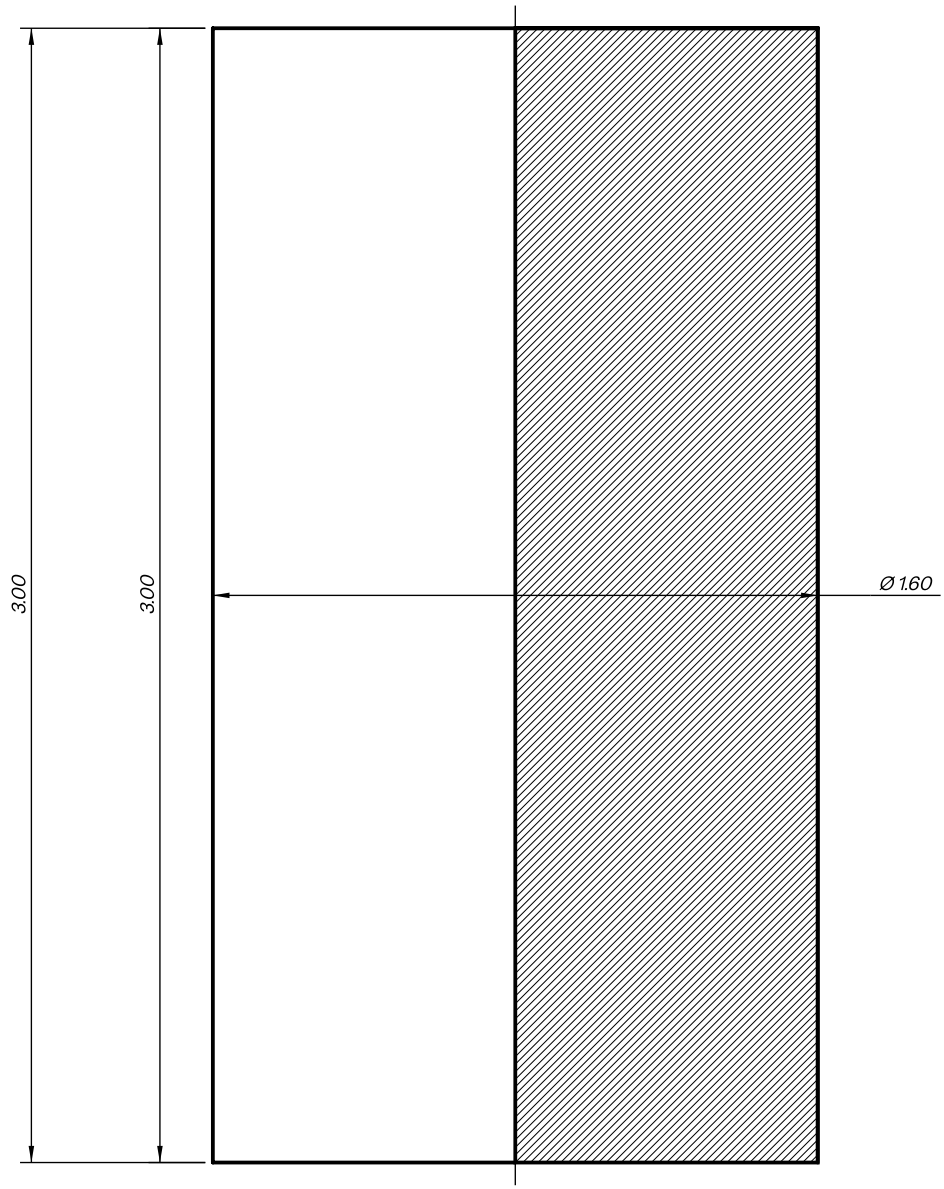
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-52	Intermediate bridge screw (WN-52)	2	not on the drawing		in 050804-1 (clear): 3.72 mm, least gap 0.1051; in 050802-1 (tap): 2.877 mm, least gap 0.0326

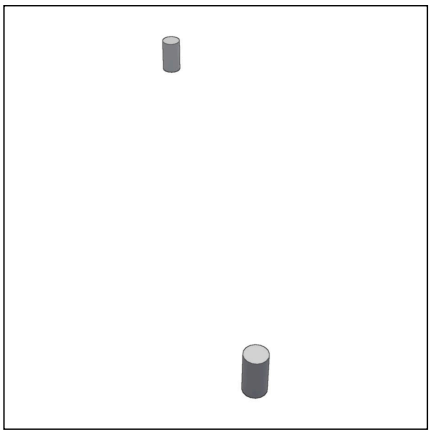
What it does (WN-52). Hold the intermediate bridge to the pillar plate, from the dial side.

Turned about its axis; length 6.60, largest Ø 4.20. Diameters: Ø 2.08, Ø 2.56, Ø 4.20; bores Ø 0.42. Shoulders from the lower end: 0.00, 0.38, 0.48, 4.53, 4.70, 4.80, 6.05, 6.31, 6.60.

Thread

Outside thread 0.48–4.53 from the lower end, 4.05 long: drawn Ø 2.56 over its crests, 2.08 at its roots, pitch 0.45. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **50:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-203.zb** Teil / Part intermediateBridge.pins
Behandlung / Treatment		Benennung / Name **Intermediate bridge steady pins**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Intermediate bridge steady pins — sheet 2

A steady pin (Stellstift), Ø 1.6, pressed into the part and standing 1.3 out of it into a reamed hole in the plate, which places the part. WN-203.

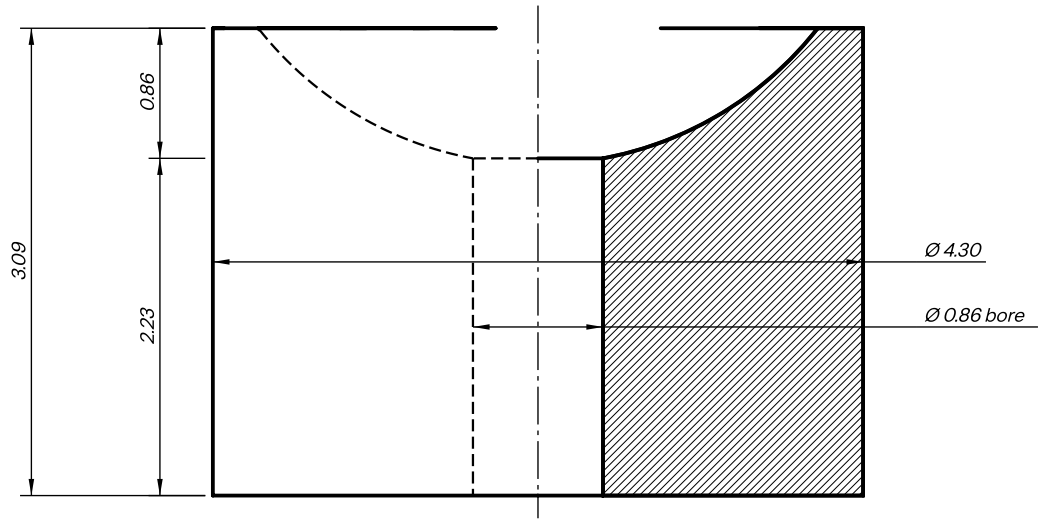
2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

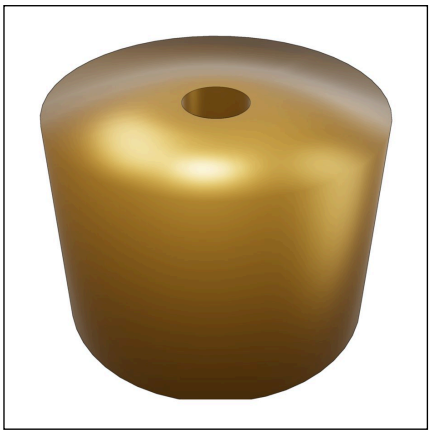
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-203.zb	Stellstift (WN-203), intermediate bridge	2	not on the drawing		in 050804-1 (press): 1.722 mm, least gap 0.0016; in 050802-1 (clear): 1.279 mm, least gap 0.0116

What it does (WN-203.zb). Place the intermediate bridge on the pillar plate.

Turned about its axis; length 3.00, largest Ø 1.60. Diameters: Ø 1.60. Shoulders from the lower end: 0.00, 3.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-300.zb*** *Teil / Part* *intermediateBridge.bush*
Behandlung / Treatment		*Benennung / Name* ***Intermediate wheel's lower bush (Messinglager WN-300)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Intermediate wheel’s lower bush (Messinglager WN-300) — sheet 2

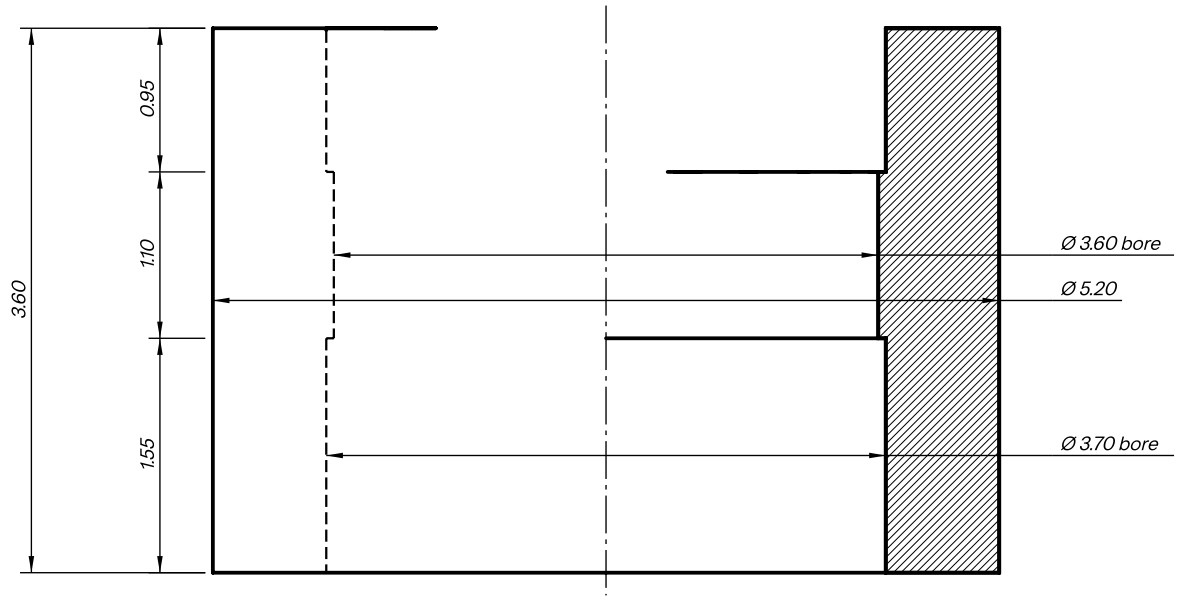
A brass bush pressed into the bridge, its face 0.7 in from the plate’s side, in which the intermediate arbor’s lower pivot runs; an oil sink in its dial-side face.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-300.zb	Messinglager WN-300 (brass bush, intermediate arbor's lower pivot)	1	not on the drawing		in 050804-1 (press): 3.087 mm, least gap 0.0027

What it does (WN-300.zb). The intermediate arbor's lower bearing, pressed into the intermediate bridge, 0.7 in from its plate side.

Turned about its axis; length 3.09, largest Ø 4.30. Diameters: Ø 4.30; bores Ø 0.86. Shoulders from the lower end: 0.00, 2.23, 3.09.





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **20:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 05 04-2**
Behandlung / Treatment			Teil / Part intermediateBridge.setting
		Benennung / Name **Seconds jewel's setting (Fassung)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Seconds jewel’s setting (Fassung) — sheet 2

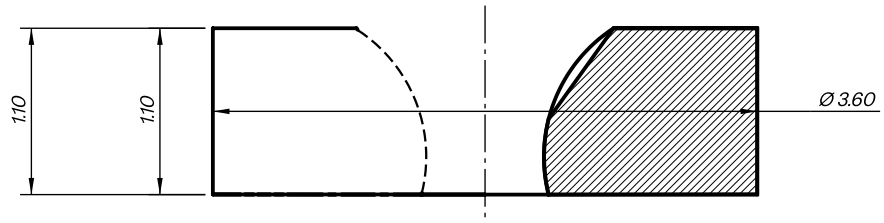
The brass setting of the seconds arbor’s lower jewel, pressed into the bridge; under the jewel a floor with a hole for the seconds post.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 04-2	Fassung (the seconds jewel’s setting)	1	not on the drawing		in 050804-1 (press): 3.596 mm, least gap 0.0034

What it does (05 05 04-2). Holds the seconds arbor's lower jewel in the intermediate bridge (its sheet isn't in the set; the coloured section names it).

Turned about its axis; length 3.60, largest Ø 5.20. Diameters: Ø 5.20; bores Ø 3.60, Ø 3.70. Shoulders from the lower end: 0.00, 1.55, 2.65, 3.60.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 05 04-1**
Behandlung / Treatment			*Teil / Part* *intermediateBridge.jewel*
		Benennung / Name ***Seconds jewel (Lagerstein f. Sek.)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Seconds jewel (Lagerstein f. Sek.) — sheet 2

The seconds arbor’s lower jewel, the train’s only one in the German original: its face 1.7 in from the bridge’s plate side, so the arbor’s shoulder takes its endshake on its flat face; an oil sink opens in its dial side.

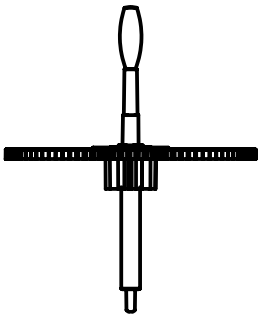
Parts list

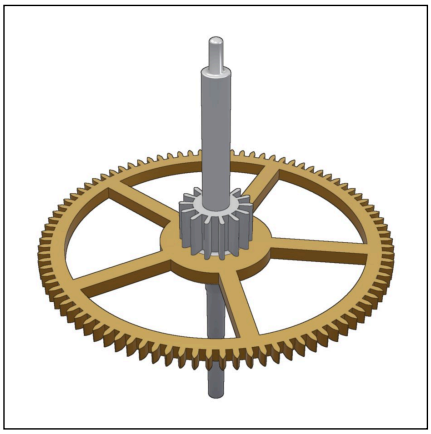
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 04-1	Lagerstein f. Sek. (the seconds arbor's lower jewel)	1	not on the drawing		takes Sekundentrieb (seconds pinion and arbor): in it (run): 0.676 mm, least gap 0.0178

What it does (05 05 04-1). The seconds arbor's lower pivot runs in it; its face takes the arbor's endshake (its sheet isn't in the set; the coloured section names it).

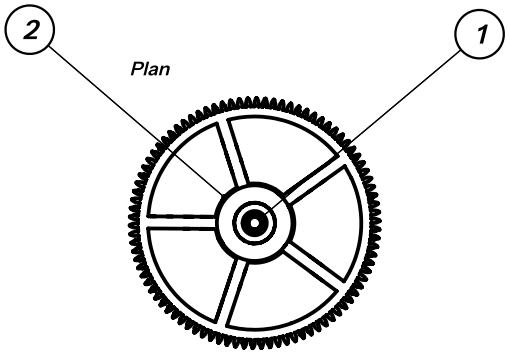
Turned about its axis; length 1.10, largest Ø 3.60. Diameters: Ø 3.60. Shoulders from the lower end: 0.00, 1.10.

Front view





Isometric view, not to scale



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Centre wheel (Minutenrad)	05 05 03-1	„Messingblech 1,5 Din 1751 Ms
1	1	Centre pinion and arbor (Minutentrieb)	05 05 03-2	„Präz. Rundstahl 7,2 Din 175“

Werkstoff / Material
„Messingblech 1,5 Din 1751 Ms 58 federhart“

Maßstab / Scale
1:1

Wempe Einheits-Chronometer
The Marine Chronometer working model
drawn from its solids by tools/drawbook.py
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 05 03-1

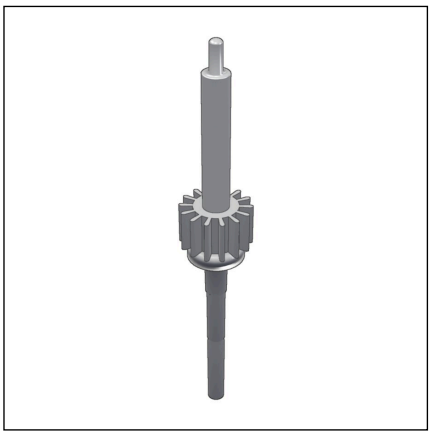
Teil / Part
centreWheel

Behandlung / Treatment
surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01

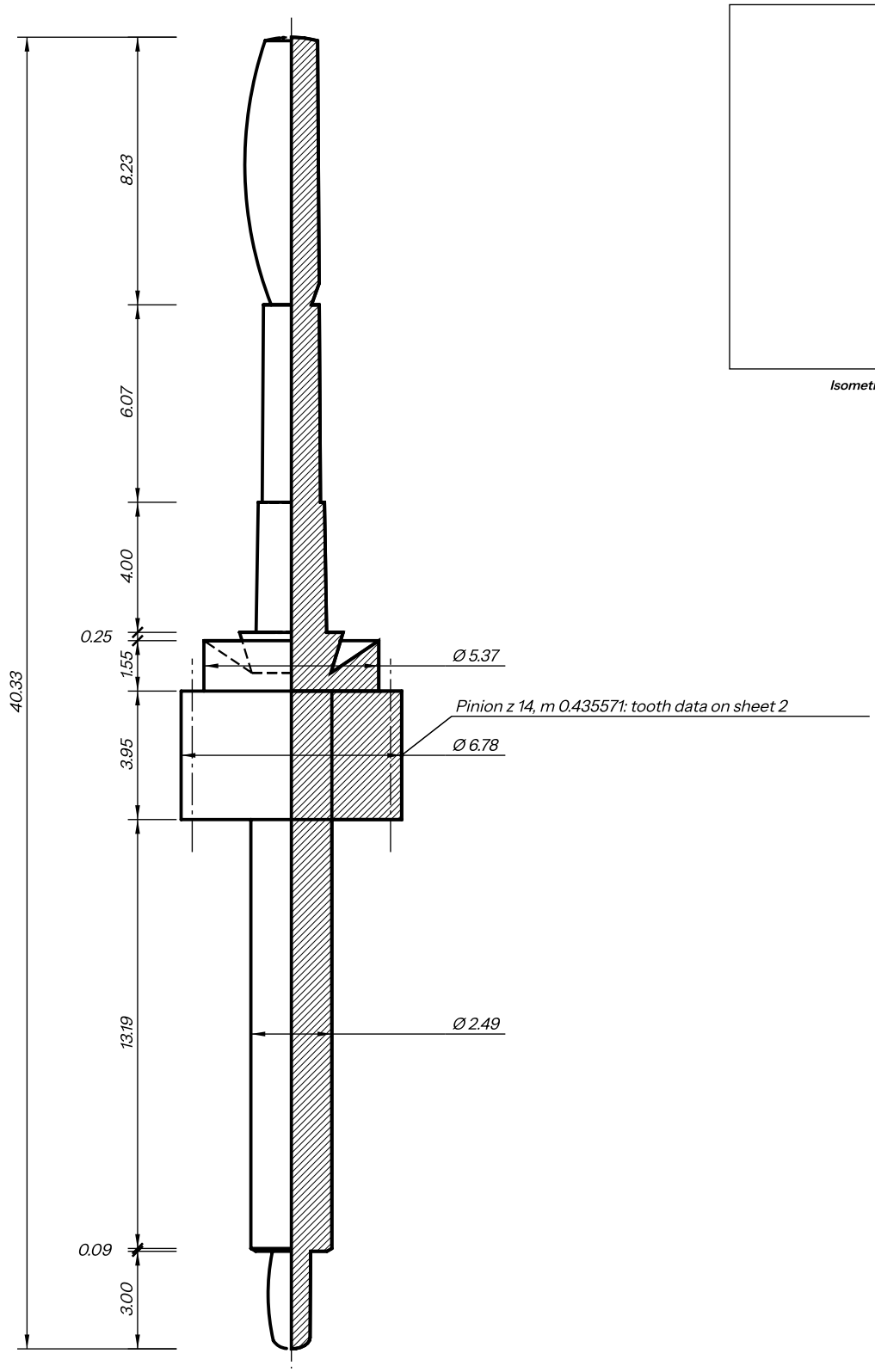
Benennung / Name
Centre wheel (Minutenrad)

Blatt / Sheet
1 / 1

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).



Isometric view, not to scale



Werkstoff / Material „Präz. Rundstahl 7,2 Din 175"	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 03-2*** *Teil / Part* *centreWheel.arbor*
Behandlung / Treatment „härten, 295° anlassen" (harden, temper at 295°); surface ▽▽▽ ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1		*Benennung / Name* ***Centre pinion and arbor (Minutentrieb)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Centre pinion and arbor (Minutentrieb) — sheet 2

The centre arbor's steel pinion and arbor, one piece: the upper pivot, the 14-leaf pinion the fusee wheel drives, the seat the wheel is riveted on, the collar that bears on the pillar plate's bush, the journal that runs in it, and the tapered shank for the cannon tube.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 03-2	Minutentrieb (centre pinion and arbor)	1	„Präz. Rundstahl 7,2 Din 175“	„härten, 295° anlassen“ (harden, temper at 295°); surface ▽▽▽ ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	in WN-301 (run): 2.255 mm, least gap 0.0106; in WN-302 (run): 3.006 mm, least gap 0.0128; endshake 0.070 mm (WN-302 / WN-301)

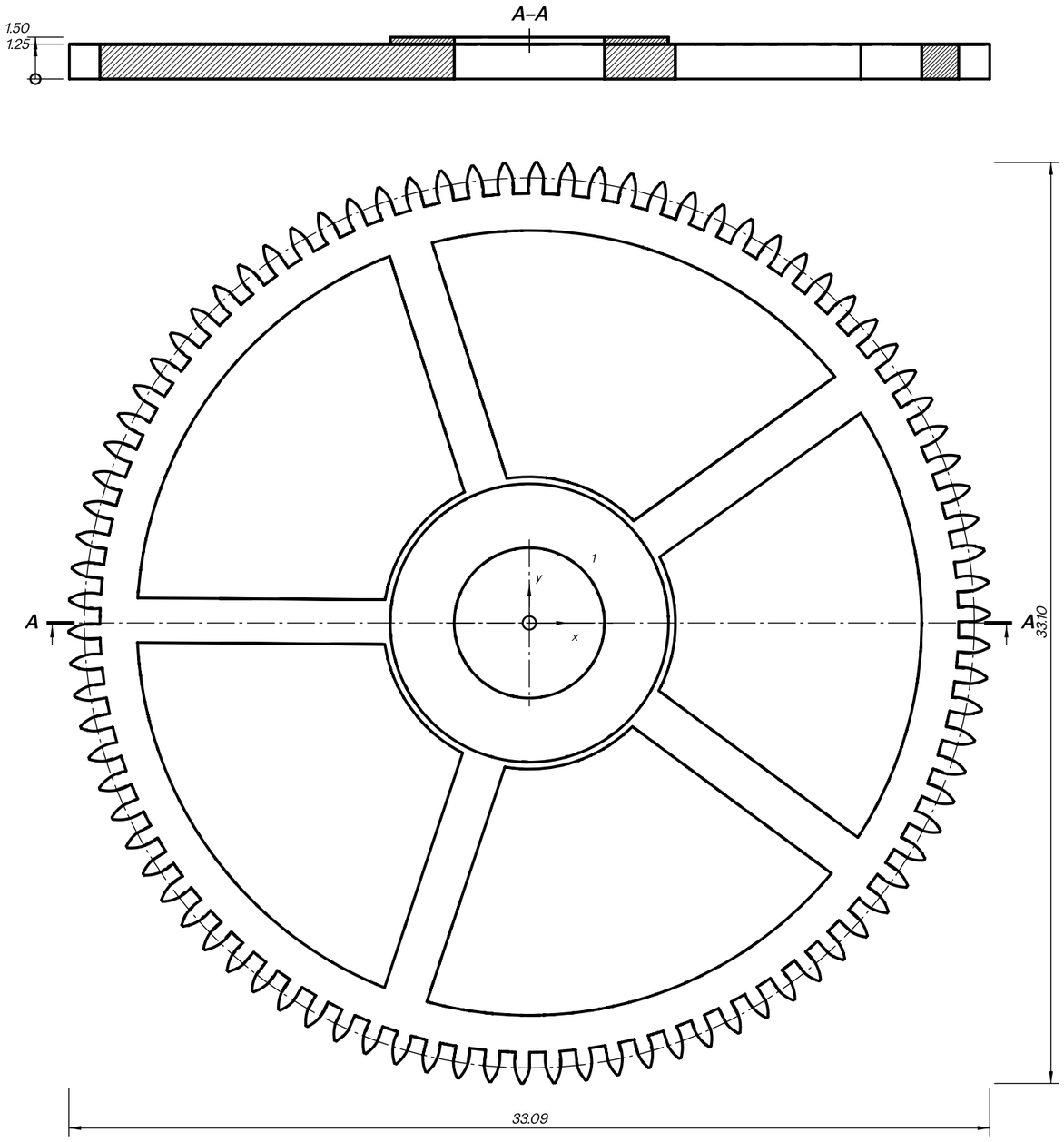
What it does (05 05 03-2). Driven by the fusee wheel; turns once an hour; runs in the top plate's bush WN-301 and, by its journal, in the pillar plate's WN-302, and carries the cannon tube (05 05 03-3) on its shank.

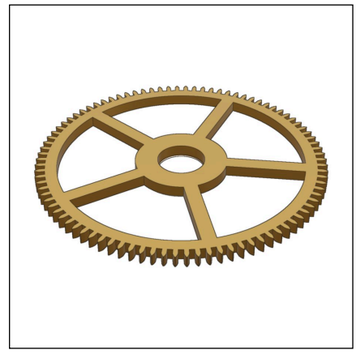
Turned about its axis; length 40.33, largest Ø 6.66. Diameters: Ø 2.49, Ø 5.37, Ø 6.78. Shoulders from the lower end: 0.00, 3.00, 3.09, 16.28, 20.23, 21.78, 22.03, 26.03, 32.10, 40.33.

Leaves 16.28–20.23 from the lower end, pitch Ø 6.10; their form is the view from the end, drawn at the sheet's scale.

Teeth

Number	Module	Tip Ø	Root Ø
14	0.43557142857142855	6.78	





Isometric view, not to scale

Werkstoff / Material *„Messingblech 1,5 Din 1751 Ms 58 federhart"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 03-1***
Behandlung / Treatment *surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01*			*Teil / Part* *centreWheel.wheel*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Centre wheel (Minutenrad)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Centre wheel (Minutenrad) — sheet 2

The centre wheel, five-spoked, riveted on the pinion's seat; it drives the intermediate pinion. Its hub keeps the sheet's thickness and stands 0.2 proud of the rim on the dial side.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 03-1	Minutenrad (centre wheel)	1	„Messingblech 1,5 Din 1751 Ms 58 federhart“	surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01	-7.5

What it does (05 05 03-1). Riveted on the pinion's seat; drives the intermediate pinion.

Coordinates from the datum, the movement's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.50.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 5.40 through (0.25 thick at 1.25–1.50)

The contour has 566 elements (its teeth): the drawing and the tooth data define it.

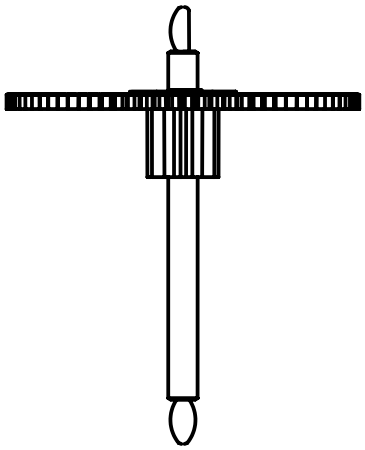
Levels

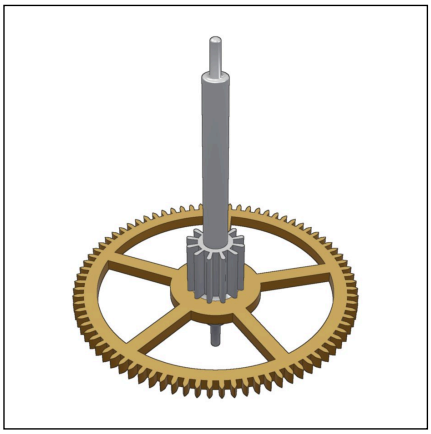
Faces in section A–A, from the bottom face: 1.25, 1.50.

Teeth

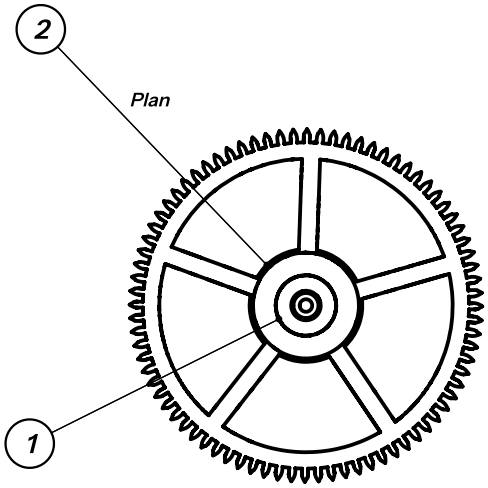
Number	Module	Tip Ø	Root Ø
90	0.3552888888888889	33.10	30.86

Front view





Isometric view, not to scale



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Intermediate wheel (Zwischenrad)	05 05 02-1	„Messingblech 1,2 Din 1751 Ms
1	1	Intermediate pinion and arbor (Zwischentrieb)	05 05 02-2	„Präz. Rundstahl 5,2 Din 175“

Werkstoff / Material
„Messingblech 1,2 Din 1751 Ms 58 federhart“

Maßstab / Scale
2:1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 05 02-1

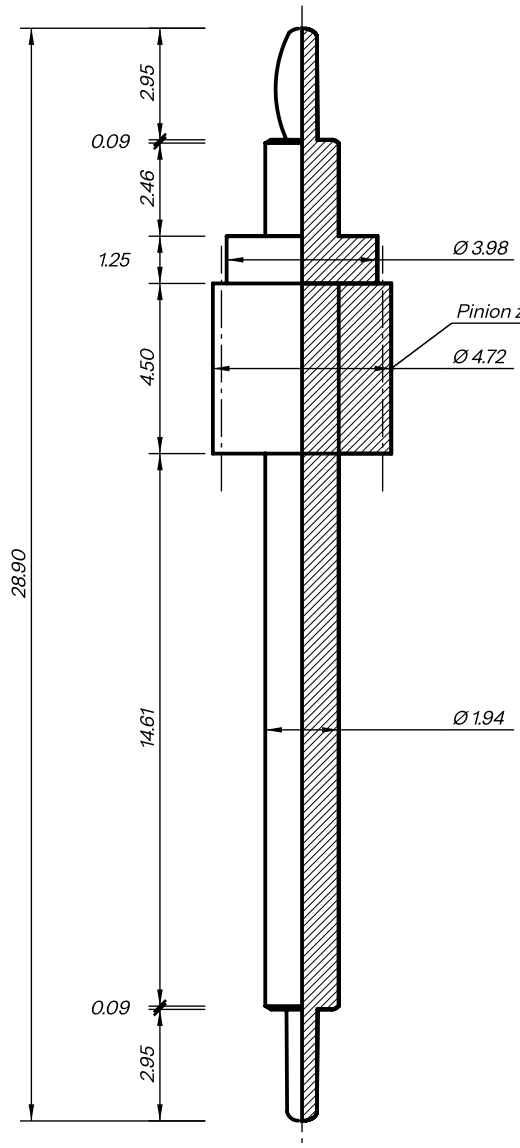
Teil / Part
intermediateWheel

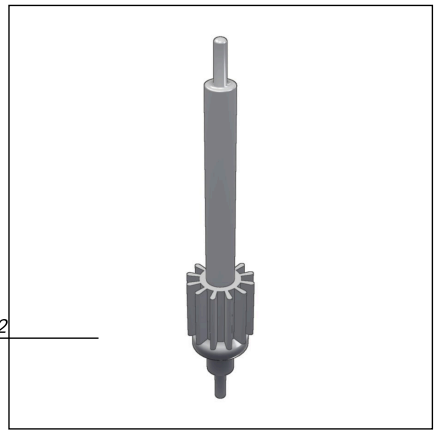
Behandlung / Treatment
surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01

Benennung / Name
Intermediate wheel (Zwischenrad)

Blatt / Sheet
1 / 1

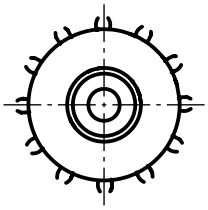
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

View from above



Werkstoff / Material „Präz. Rundstahl 5,2 Din 175“	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 02-2***
			Teil / Part *intermediateWheel.arbor*
Behandlung / Treatment *surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*		*Benennung / Name* ***Intermediate pinion and arbor (Zwischentrieb)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Intermediate pinion and arbor (Zwischentrieb) — sheet 2

The intermediate arbor's steel pinion and arbor, one piece: 12 leaves, driven by the centre wheel, and the seat its wheel is riveted on, between its two pivots.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 02-2	Zwischentrieb (intermediate pinion and arbor)	1	„Präz. Rundstahl 5,2 Din 175“	surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	in WN-300.t38/6MX.jw38u (run): 2.154 mm, least gap 0.0112; in WN-300.zb/6MX.jw38l (run): 2.154 mm, least gap 0.0187; endshake 0.060 mm (WN-300.zb / WN-300.t38)

What it does (05 05 02-2). Driven by the centre wheel; runs in the top plate's bush WN-300 and the intermediate bridge's. In the Soviet 6MX (the page's Variants) it runs in jewels in chatons at both ends.

Turned about its axis; length 28.90, largest Ø 4.72. Diameters: Ø 1.94, Ø 3.98, Ø 4.72. Shoulders from the lower end: 0.00, 2.95, 3.04, 17.65, 22.15, 23.40, 25.86, 25.95, 28.90.

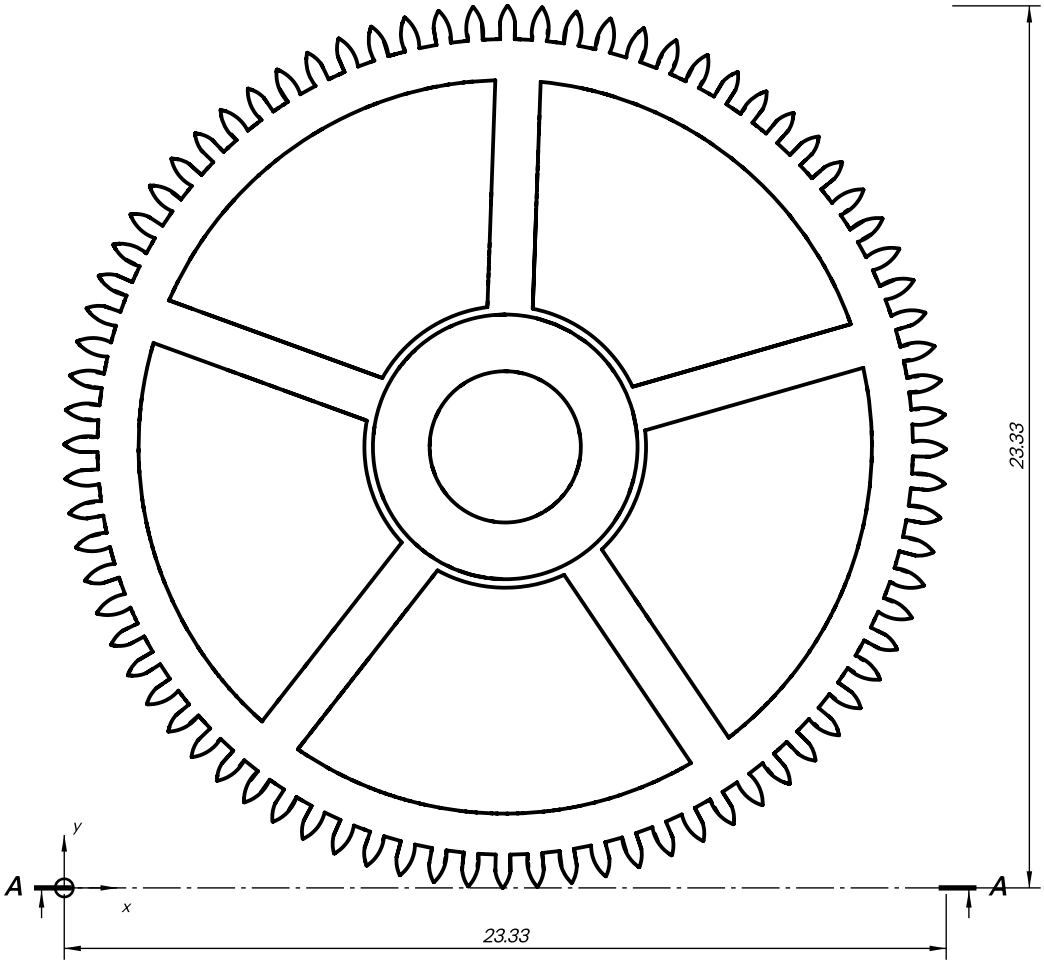
Leaves 17.65-22.15 from the lower end, pitch Ø 4.26; their form is the view from the end, drawn at the sheet's scale.

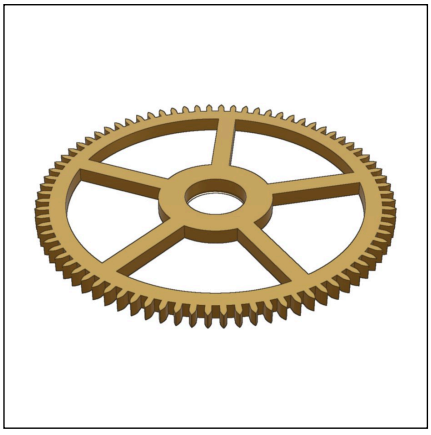
Teeth

Number	Module	Tip Ø	Root Ø
12	0.35533333333333333	4.72	

A–A







Isometric view, not to scale

Werkstoff / Material „Messingblech 1,2 Din 1751 Ms 58 federhart"	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 05 02-1**
Behandlung / Treatment surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01			Teil / Part intermediateWheel.wheel
		Benennung / Name **Intermediate wheel (Zwischenrad)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Intermediate wheel (Zwischenrad) — sheet 2

The intermediate wheel, five-spoked, riveted on the pinion's seat; it drives the seconds pinion. Its hub stands 0.2 proud on the dial side.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 02-1	Zwischenrad (intermediate wheel)	1	„Messingblech 1,2 Din 1751 Ms 58 federhart“	surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01	-8

What it does (05 05 02-1). Riveted on the pinion's seat; drives the seconds pinion.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.15.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.34, 9.05	1.12, 9.39	1.11, 8.36	1.03
Outline	2	line	1.12, 9.39	1.03, 9.85		
Outline	3	arc clockwise	1.03, 9.85	0.13, 9.90	0.64, 10.80	1.03
Outline	4	line	0.13, 9.90	0.13, 9.91		
Outline	5	arc clockwise	0.13, 9.91	0.98, 10.23	0.88, 9.19	1.03
Outline	6	line	0.98, 10.23	0.92, 10.69		
Outline	7	arc clockwise	0.92, 10.69	0.03, 10.81	0.60, 11.67	1.03
Outline	8	line	0.03, 10.81	0.03, 10.82		
Outline	9	arc clockwise	0.03, 10.82	0.90, 11.07	0.72, 10.05	1.03
Outline	10	line	0.90, 11.07	0.88, 11.53		
Outline	11	arc clockwise	0.88, 11.53	0.00, 11.72	0.64, 12.54	1.03
Outline	12	line	0.00, 11.72	0.00, 11.73		
Outline	13	arc clockwise	0.00, 11.73	0.88, 11.92	0.63, 10.91	1.03
Outline	14	line	0.88, 11.92	0.90, 12.38		
Outline	15	arc clockwise	0.90, 12.38	0.04, 12.64	0.74, 13.40	1.03
Outline	16	line	0.04, 12.64	0.04, 12.65		
Outline	17	arc clockwise	0.04, 12.65	0.94, 12.76	0.60, 11.78	1.03
Outline	18	line	0.94, 12.76	0.99, 13.22		
Outline	19	arc clockwise	0.99, 13.22	0.15, 13.55	0.91, 14.25	1.03
Outline	20	line	0.15, 13.55	0.15, 13.56		
Outline	21	arc clockwise	0.15, 13.56	1.05, 13.60	0.65, 12.65	1.03
Outline	22	line	1.05, 13.60	1.15, 14.05		
Outline	23	arc clockwise	1.15, 14.05	0.34, 14.44	1.14, 15.09	1.03
Outline	24	line	0.34, 14.44	0.34, 14.46		
Outline	25	arc clockwise	0.34, 14.46	1.24, 14.43	0.76, 13.51	1.03
Outline	26	line	1.24, 14.43	1.37, 14.87		
Outline	27	arc clockwise	1.37, 14.87	0.59, 15.32	1.45, 15.90	1.03
Outline	28	line	0.59, 15.32	0.59, 15.34		

Outline	29	arc clockwise	0.59, 15.34	1.49, 15.23	0.93, 14.36	1.03
Outline	30	line	1.49, 15.23	1.65, 15.67		
Outline	31	arc clockwise	1.65, 15.67	0.91, 16.18	1.81, 16.69	1.03
Outline	32	line	0.91, 16.18	0.91, 16.19		
Outline	33	arc clockwise	0.91, 16.19	1.80, 16.02	1.18, 15.19	1.03
Outline	34	line	1.80, 16.02	2.00, 16.44		
Outline	35	arc clockwise	2.00, 16.44	1.30, 17.01	2.23, 17.45	1.03
Outline	36	line	1.30, 17.01	1.30, 17.02		
Outline	37	arc clockwise	1.30, 17.02	2.17, 16.78	1.49, 16.00	1.03
Outline	38	line	2.17, 16.78	2.40, 17.19		
Outline	39	arc clockwise	2.40, 17.19	1.75, 17.81	2.72, 18.17	1.03
Outline	40	line	1.75, 17.81	1.76, 17.82		
Outline	41	arc clockwise	1.76, 17.82	2.60, 17.51	1.86, 16.79	1.03
Outline	42	line	2.60, 17.51	2.86, 17.90		
Outline	43	arc clockwise	2.86, 17.90	2.26, 18.57	3.26, 18.85	1.03
Outline	44	line	2.26, 18.57	2.27, 18.58		
Outline	45	arc clockwise	2.27, 18.58	3.09, 18.21	2.29, 17.54	1.03
Outline	46	line	3.09, 18.21	3.38, 18.57		
Outline	47	arc clockwise	3.38, 18.57	2.83, 19.28	3.85, 19.49	1.03
Outline	48	line	2.83, 19.28	2.84, 19.29		
Outline	49	arc clockwise	2.84, 19.29	3.63, 18.86	2.78, 18.26	1.03
Outline	50	line	3.63, 18.86	3.94, 19.20		
Outline	51	arc clockwise	3.94, 19.20	3.46, 19.95	4.48, 20.08	1.03
Outline	52	line	3.46, 19.95	3.47, 19.96		
Outline	53	arc clockwise	3.47, 19.96	4.22, 19.47	3.33, 18.94	1.03
Outline	54	line	4.22, 19.47	4.56, 19.78		
Outline	55	arc clockwise	4.56, 19.78	4.13, 20.57	5.17, 20.62	1.03
Outline	56	line	4.13, 20.57	4.14, 20.58		
Outline	57	arc clockwise	4.14, 20.58	4.85, 20.03	3.92, 19.57	1.03
Outline	58	line	4.85, 20.03	5.22, 20.31		
Outline	59	arc clockwise	5.22, 20.31	4.85, 21.13	5.89, 21.10	1.03
Outline	60	line	4.85, 21.13	4.86, 21.14		
Outline	61	arc clockwise	4.86, 21.14	5.53, 20.53	4.57, 20.15	1.03
Outline	62	line	5.53, 20.53	5.92, 20.79		
Outline	63	arc clockwise	5.92, 20.79	5.62, 21.64	6.65, 21.52	1.03
Outline	64	line	5.62, 21.64	5.63, 21.65		
Outline	65	arc clockwise	5.63, 21.65	6.24, 20.99	5.26, 20.68	1.03
Outline	66	line	6.24, 20.99	6.65, 21.21		
Outline	67	arc clockwise	6.65, 21.21	6.42, 22.08	7.44, 21.89	1.03
Outline	68	line	6.42, 22.08	6.43, 22.09		
Outline	69	arc clockwise	6.43, 22.09	6.99, 21.39	5.98, 21.16	1.03
Outline	70	line	6.99, 21.39	7.41, 21.58		
Outline	71	arc clockwise	7.41, 21.58	7.25, 22.46	8.25, 22.19	1.03
Outline	72	line	7.25, 22.46	7.27, 22.47		
Outline	73	arc clockwise	7.27, 22.47	7.77, 21.72	6.75, 21.57	1.03
Outline	74	line	7.77, 21.72	8.21, 21.88		
Outline	75	arc clockwise	8.21, 21.88	8.11, 22.78	9.09, 22.42	1.03
Outline	76	line	8.11, 22.78	8.13, 22.78		

Outline	77	arc clockwise	8.13, 22.78	8.57, 22.00	7.54, 21.93	1.03
Outline	78	line	8.57, 22.00	9.02, 22.12		
Outline	79	arc clockwise	9.02, 22.12	9.00, 23.02	9.94, 22.59	1.03
Outline	80	line	9.00, 23.02	9.01, 23.02		
Outline	81	arc clockwise	9.01, 23.02	9.39, 22.21	8.36, 22.22	1.03
Outline	82	line	9.39, 22.21	9.85, 22.30		
Outline	83	arc clockwise	9.85, 22.30	9.90, 23.19	10.80, 22.69	1.03
Outline	84	line	9.90, 23.19	9.91, 23.20		
Outline	85	arc clockwise	9.91, 23.20	10.23, 22.35	9.19, 22.45	1.03
Outline	86	line	10.23, 22.35	10.69, 22.40		
Outline	87	arc clockwise	10.69, 22.40	10.81, 23.30	11.67, 22.73	1.03
Outline	88	line	10.81, 23.30	10.82, 23.30		
Outline	89	arc clockwise	10.82, 23.30	11.07, 22.43	10.05, 22.61	1.03
Outline	90	line	11.07, 22.43	11.53, 22.45		
Outline	91	arc clockwise	11.53, 22.45	11.72, 23.33	12.54, 22.69	1.03
Outline	92	line	11.72, 23.33	11.73, 23.33		
Outline	93	arc clockwise	11.73, 23.33	11.92, 22.45	10.91, 22.70	1.03
Outline	94	line	11.92, 22.45	12.38, 22.43		
Outline	95	arc clockwise	12.38, 22.43	12.64, 23.29	13.40, 22.59	1.03
Outline	96	line	12.64, 23.29	12.65, 23.29		
Outline	97	arc clockwise	12.65, 23.29	12.76, 22.39	11.78, 22.73	1.03
Outline	98	line	12.76, 22.39	13.22, 22.34		
Outline	99	arc clockwise	13.22, 22.34	13.55, 23.18	14.25, 22.42	1.03
Outline	100	line	13.55, 23.18	13.56, 23.17		
Outline	101	arc clockwise	13.56, 23.17	13.60, 22.27	12.65, 22.68	1.03
Outline	102	line	13.60, 22.27	14.05, 22.18		
Outline	103	arc clockwise	14.05, 22.18	14.44, 22.99	15.09, 22.18	1.03
Outline	104	line	14.44, 22.99	14.46, 22.99		
Outline	105	arc clockwise	14.46, 22.99	14.43, 22.09	13.51, 22.57	1.03
Outline	106	line	14.43, 22.09	14.87, 21.96		
Outline	107	arc clockwise	14.87, 21.96	15.32, 22.74	15.90, 21.88	1.03
Outline	108	line	15.32, 22.74	15.34, 22.74		
Outline	109	arc clockwise	15.34, 22.74	15.23, 21.84	14.36, 22.39	1.03
Outline	110	line	15.23, 21.84	15.67, 21.68		
Outline	111	arc clockwise	15.67, 21.68	16.18, 22.42	16.69, 21.52	1.03
Outline	112	line	16.18, 22.42	16.19, 22.41		
Outline	113	arc clockwise	16.19, 22.41	16.02, 21.53	15.19, 22.15	1.03
Outline	114	line	16.02, 21.53	16.44, 21.33		
Outline	115	arc clockwise	16.44, 21.33	17.01, 22.03	17.45, 21.09	1.03
Outline	116	line	17.01, 22.03	17.02, 22.03		
Outline	117	arc clockwise	17.02, 22.03	16.78, 21.16	16.00, 21.84	1.03
Outline	118	line	16.78, 21.16	17.19, 20.93		
Outline	119	arc clockwise	17.19, 20.93	17.81, 21.58	18.17, 20.61	1.03
Outline	120	line	17.81, 21.58	17.82, 21.57		
Outline	121	arc clockwise	17.82, 21.57	17.51, 20.73	16.79, 21.47	1.03
Outline	122	line	17.51, 20.73	17.90, 20.47		
Outline	123	arc clockwise	17.90, 20.47	18.57, 21.07	18.85, 20.07	1.03
Outline	124	line	18.57, 21.07	18.58, 21.06		

Outline	125	arc clockwise	18.58, 21.06	18.21, 20.24	17.54, 21.04	1.03
Outline	126	line	18.21, 20.24	18.57, 19.95		
Outline	127	arc clockwise	18.57, 19.95	19.28, 20.50	19.49, 19.48	1.03
Outline	128	line	19.28, 20.50	19.29, 20.49		
Outline	129	arc clockwise	19.29, 20.49	18.86, 19.70	18.26, 20.55	1.03
Outline	130	line	18.86, 19.70	19.20, 19.38		
Outline	131	arc clockwise	19.20, 19.38	19.95, 19.87	20.08, 18.85	1.03
Outline	132	line	19.95, 19.87	19.96, 19.86		
Outline	133	arc clockwise	19.96, 19.86	19.47, 19.11	18.94, 20.00	1.03
Outline	134	line	19.47, 19.11	19.78, 18.77		
Outline	135	arc clockwise	19.78, 18.77	20.57, 19.20	20.62, 18.16	1.03
Outline	136	line	20.57, 19.20	20.58, 19.19		
Outline	137	arc clockwise	20.58, 19.19	20.03, 18.48	19.57, 19.41	1.03
Outline	138	line	20.03, 18.48	20.31, 18.11		
Outline	139	arc clockwise	20.31, 18.11	21.13, 18.47	21.10, 17.44	1.03
Outline	140	line	21.13, 18.47	21.14, 18.46		
Outline	141	arc clockwise	21.14, 18.46	20.53, 17.80	20.15, 18.76	1.03
Outline	142	line	20.53, 17.80	20.79, 17.41		
Outline	143	arc clockwise	20.79, 17.41	21.64, 17.71	21.52, 16.68	1.03
Outline	144	line	21.64, 17.71	21.65, 17.70		
Outline	145	arc clockwise	21.65, 17.70	20.99, 17.08	20.68, 18.07	1.03
Outline	146	line	20.99, 17.08	21.21, 16.68		
Outline	147	arc clockwise	21.21, 16.68	22.08, 16.91	21.89, 15.89	1.03
Outline	148	line	22.08, 16.91	22.09, 16.90		
Outline	149	arc clockwise	22.09, 16.90	21.39, 16.34	21.16, 17.35	1.03
Outline	150	line	21.39, 16.34	21.58, 15.91		
Outline	151	arc clockwise	21.58, 15.91	22.46, 16.08	22.19, 15.08	1.03
Outline	152	line	22.46, 16.08	22.47, 16.06		
Outline	153	arc clockwise	22.47, 16.06	21.72, 15.56	21.57, 16.58	1.03
Outline	154	line	21.72, 15.56	21.88, 15.12		
Outline	155	arc clockwise	21.88, 15.12	22.78, 15.21	22.42, 14.24	1.03
Outline	156	line	22.78, 15.21	22.78, 15.20		
Outline	157	arc clockwise	22.78, 15.20	22.00, 14.76	21.93, 15.79	1.03
Outline	158	line	22.00, 14.76	22.12, 14.31		
Outline	159	arc clockwise	22.12, 14.31	23.02, 14.33	22.59, 13.39	1.03
Outline	160	line	23.02, 14.33	23.02, 14.32		
Outline	161	arc clockwise	23.02, 14.32	22.21, 13.94	22.22, 14.97	1.03
Outline	162	line	22.21, 13.94	22.30, 13.48		
Outline	163	arc clockwise	22.30, 13.48	23.19, 13.43	22.69, 12.53	1.03
Outline	164	line	23.19, 13.43	23.20, 13.42		
Outline	165	arc clockwise	23.20, 13.42	22.35, 13.10	22.45, 14.13	1.03
Outline	166	line	22.35, 13.10	22.40, 12.64		
Outline	167	arc clockwise	22.40, 12.64	23.30, 12.52	22.73, 11.66	1.03
Outline	168	line	23.30, 12.52	23.30, 12.51		
Outline	169	arc clockwise	23.30, 12.51	22.43, 12.26	22.61, 13.28	1.03
Outline	170	line	22.43, 12.26	22.45, 11.80		
Outline	171	arc clockwise	22.45, 11.80	23.33, 11.61	22.69, 10.79	1.03
Outline	172	line	23.33, 11.61	23.33, 11.59		

Outline	173	arc clockwise	23.33, 11.59	22.45, 11.41	22.70, 12.42	1.03
Outline	174	line	22.45, 11.41	22.43, 10.95		
Outline	175	arc clockwise	22.43, 10.95	23.29, 10.69	22.59, 9.93	1.03
Outline	176	line	23.29, 10.69	23.29, 10.68		
Outline	177	arc clockwise	23.29, 10.68	22.39, 10.57	22.73, 11.55	1.03
Outline	178	line	22.39, 10.57	22.34, 10.11		
Outline	179	arc clockwise	22.34, 10.11	23.18, 9.78	22.42, 9.08	1.03
Outline	180	line	23.18, 9.78	23.17, 9.77		
Outline	181	arc clockwise	23.17, 9.77	22.27, 9.73	22.68, 10.68	1.03
Outline	182	line	22.27, 9.73	22.18, 9.28		
Outline	183	arc clockwise	22.18, 9.28	22.99, 8.89	22.18, 8.24	1.03
Outline	184	line	22.99, 8.89	22.99, 8.87		
Outline	185	arc clockwise	22.99, 8.87	22.09, 8.90	22.57, 9.82	1.03
Outline	186	line	22.09, 8.90	21.96, 8.46		
Outline	187	arc clockwise	21.96, 8.46	22.74, 8.01	21.88, 7.43	1.03
Outline	188	line	22.74, 8.01	22.74, 7.99		
Outline	189	arc clockwise	22.74, 7.99	21.84, 8.09	22.39, 8.97	1.03
Outline	190	line	21.84, 8.09	21.68, 7.66		
Outline	191	arc clockwise	21.68, 7.66	22.42, 7.15	21.52, 6.64	1.03
Outline	192	line	22.42, 7.15	22.41, 7.14		
Outline	193	arc clockwise	22.41, 7.14	21.53, 7.31	22.15, 8.14	1.03
Outline	194	line	21.53, 7.31	21.33, 6.89		
Outline	195	arc clockwise	21.33, 6.89	22.03, 6.32	21.09, 5.88	1.03
Outline	196	line	22.03, 6.32	22.03, 6.31		
Outline	197	arc clockwise	22.03, 6.31	21.16, 6.55	21.84, 7.32	1.03
Outline	198	line	21.16, 6.55	20.93, 6.14		
Outline	199	arc clockwise	20.93, 6.14	21.58, 5.52	20.61, 5.16	1.03
Outline	200	line	21.58, 5.52	21.57, 5.51		
Outline	201	arc clockwise	21.57, 5.51	20.73, 5.82	21.47, 6.54	1.03
Outline	202	line	20.73, 5.82	20.47, 5.43		
Outline	203	arc clockwise	20.47, 5.43	21.07, 4.76	20.07, 4.48	1.03
Outline	204	line	21.07, 4.76	21.06, 4.75		
Outline	205	arc clockwise	21.06, 4.75	20.24, 5.12	21.04, 5.79	1.03
Outline	206	line	20.24, 5.12	19.95, 4.76		
Outline	207	arc clockwise	19.95, 4.76	20.50, 4.05	19.48, 3.84	1.03
Outline	208	line	20.50, 4.05	20.49, 4.04		
Outline	209	arc clockwise	20.49, 4.04	19.70, 4.47	20.55, 5.07	1.03
Outline	210	line	19.70, 4.47	19.38, 4.13		
Outline	211	arc clockwise	19.38, 4.13	19.87, 3.38	18.85, 3.25	1.03
Outline	212	line	19.87, 3.38	19.86, 3.37		
Outline	213	arc clockwise	19.86, 3.37	19.11, 3.86	20.00, 4.39	1.03
Outline	214	line	19.11, 3.86	18.77, 3.55		
Outline	215	arc clockwise	18.77, 3.55	19.20, 2.76	18.16, 2.71	1.03
Outline	216	line	19.20, 2.76	19.19, 2.75		
Outline	217	arc clockwise	19.19, 2.75	18.48, 3.30	19.41, 3.76	1.03
Outline	218	line	18.48, 3.30	18.11, 3.02		
Outline	219	arc clockwise	18.11, 3.02	18.47, 2.19	17.44, 2.23	1.03
Outline	220	line	18.47, 2.19	18.46, 2.19		

Outline	221	arc clockwise	18.46, 2.19	17.80, 2.79	18.76, 3.18	1.03
Outline	222	line	17.80, 2.79	17.41, 2.54		
Outline	223	arc clockwise	17.41, 2.54	17.71, 1.69	16.68, 1.80	1.03
Outline	224	line	17.71, 1.69	17.70, 1.68		
Outline	225	arc clockwise	17.70, 1.68	17.08, 2.34	18.07, 2.65	1.03
Outline	226	line	17.08, 2.34	16.68, 2.12		
Outline	227	arc clockwise	16.68, 2.12	16.91, 1.25	15.89, 1.44	1.03
Outline	228	line	16.91, 1.25	16.90, 1.24		
Outline	229	arc clockwise	16.90, 1.24	16.34, 1.94	17.35, 2.17	1.03
Outline	230	line	16.34, 1.94	15.91, 1.75		
Outline	231	arc clockwise	15.91, 1.75	16.08, 0.87	15.08, 1.14	1.03
Outline	232	line	16.08, 0.87	16.06, 0.86		
Outline	233	arc clockwise	16.06, 0.86	15.56, 1.61	16.58, 1.76	1.03
Outline	234	line	15.56, 1.61	15.12, 1.45		
Outline	235	arc clockwise	15.12, 1.45	15.21, 0.55	14.24, 0.91	1.03
Outline	236	line	15.21, 0.55	15.20, 0.55		
Outline	237	arc clockwise	15.20, 0.55	14.76, 1.33	15.79, 1.40	1.03
Outline	238	line	14.76, 1.33	14.31, 1.21		
Outline	239	arc clockwise	14.31, 1.21	14.33, 0.31	13.39, 0.74	1.03
Outline	240	line	14.33, 0.31	14.32, 0.31		
Outline	241	arc clockwise	14.32, 0.31	13.94, 1.12	14.97, 1.11	1.03
Outline	242	line	13.94, 1.12	13.48, 1.03		
Outline	243	arc clockwise	13.48, 1.03	13.43, 0.13	12.53, 0.64	1.03
Outline	244	line	13.43, 0.13	13.42, 0.13		
Outline	245	arc clockwise	13.42, 0.13	13.10, 0.98	14.13, 0.88	1.03
Outline	246	line	13.10, 0.98	12.64, 0.92		
Outline	247	arc clockwise	12.64, 0.92	12.52, 0.03	11.66, 0.60	1.03
Outline	248	line	12.52, 0.03	12.51, 0.03		
Outline	249	arc clockwise	12.51, 0.03	12.26, 0.90	13.28, 0.72	1.03
Outline	250	line	12.26, 0.90	11.80, 0.88		
Outline	251	arc clockwise	11.80, 0.88	11.61, 0.00	10.79, 0.64	1.03
Outline	252	line	11.61, 0.00	11.59, 0.00		
Outline	253	arc clockwise	11.59, 0.00	11.41, 0.88	12.42, 0.63	1.03
Outline	254	line	11.41, 0.88	10.95, 0.90		
Outline	255	arc clockwise	10.95, 0.90	10.69, 0.04	9.93, 0.74	1.03
Outline	256	line	10.69, 0.04	10.68, 0.04		
Outline	257	arc clockwise	10.68, 0.04	10.57, 0.94	11.55, 0.60	1.03
Outline	258	line	10.57, 0.94	10.11, 0.99		
Outline	259	arc clockwise	10.11, 0.99	9.78, 0.15	9.08, 0.91	1.03
Outline	260	line	9.78, 0.15	9.77, 0.15		
Outline	261	arc clockwise	9.77, 0.15	9.73, 1.05	10.68, 0.65	1.03
Outline	262	line	9.73, 1.05	9.28, 1.15		
Outline	263	arc clockwise	9.28, 1.15	8.89, 0.34	8.24, 1.14	1.03
Outline	264	line	8.89, 0.34	8.87, 0.34		
Outline	265	arc clockwise	8.87, 0.34	8.90, 1.24	9.82, 0.76	1.03
Outline	266	line	8.90, 1.24	8.46, 1.37		
Outline	267	arc clockwise	8.46, 1.37	8.01, 0.59	7.43, 1.45	1.03
Outline	268	line	8.01, 0.59	7.99, 0.59		

Outline	269	arc clockwise	7.99, 0.59	8.09, 1.49	8.97, 0.93	1.03
Outline	270	line	8.09, 1.49	7.66, 1.65		
Outline	271	arc clockwise	7.66, 1.65	7.15, 0.91	6.64, 1.81	1.03
Outline	272	line	7.15, 0.91	7.14, 0.91		
Outline	273	arc clockwise	7.14, 0.91	7.31, 1.80	8.14, 1.18	1.03
Outline	274	line	7.31, 1.80	6.89, 2.00		
Outline	275	arc clockwise	6.89, 2.00	6.32, 1.30	5.88, 2.23	1.03
Outline	276	line	6.32, 1.30	6.31, 1.30		
Outline	277	arc clockwise	6.31, 1.30	6.55, 2.17	7.32, 1.49	1.03
Outline	278	line	6.55, 2.17	6.14, 2.40		
Outline	279	arc clockwise	6.14, 2.40	5.52, 1.75	5.16, 2.72	1.03
Outline	280	line	5.52, 1.75	5.51, 1.76		
Outline	281	arc clockwise	5.51, 1.76	5.82, 2.60	6.54, 1.86	1.03
Outline	282	line	5.82, 2.60	5.43, 2.86		
Outline	283	arc clockwise	5.43, 2.86	4.76, 2.26	4.48, 3.26	1.03
Outline	284	line	4.76, 2.26	4.75, 2.27		
Outline	285	arc clockwise	4.75, 2.27	5.12, 3.09	5.79, 2.29	1.03
Outline	286	line	5.12, 3.09	4.76, 3.38		
Outline	287	arc clockwise	4.76, 3.38	4.05, 2.83	3.84, 3.85	1.03
Outline	288	line	4.05, 2.83	4.04, 2.84		
Outline	289	arc clockwise	4.04, 2.84	4.47, 3.63	5.07, 2.78	1.03
Outline	290	line	4.47, 3.63	4.13, 3.94		
Outline	291	arc clockwise	4.13, 3.94	3.38, 3.46	3.25, 4.48	1.03
Outline	292	line	3.38, 3.46	3.37, 3.47		
Outline	293	arc clockwise	3.37, 3.47	3.86, 4.22	4.39, 3.33	1.03
Outline	294	line	3.86, 4.22	3.55, 4.56		
Outline	295	arc clockwise	3.55, 4.56	2.76, 4.13	2.71, 5.17	1.03
Outline	296	line	2.76, 4.13	2.75, 4.14		
Outline	297	arc clockwise	2.75, 4.14	3.30, 4.85	3.76, 3.92	1.03
Outline	298	line	3.30, 4.85	3.02, 5.22		
Outline	299	arc clockwise	3.02, 5.22	2.19, 4.85	2.23, 5.89	1.03
Outline	300	line	2.19, 4.85	2.19, 4.86		
Outline	301	arc clockwise	2.19, 4.86	2.79, 5.53	3.18, 4.57	1.03
Outline	302	line	2.79, 5.53	2.54, 5.92		
Outline	303	arc clockwise	2.54, 5.92	1.69, 5.62	1.80, 6.65	1.03
Outline	304	line	1.69, 5.62	1.68, 5.63		
Outline	305	arc clockwise	1.68, 5.63	2.34, 6.24	2.65, 5.26	1.03
Outline	306	line	2.34, 6.24	2.12, 6.65		
Outline	307	arc clockwise	2.12, 6.65	1.25, 6.42	1.44, 7.44	1.03
Outline	308	line	1.25, 6.42	1.24, 6.43		
Outline	309	arc clockwise	1.24, 6.43	1.94, 6.99	2.17, 5.98	1.03
Outline	310	line	1.94, 6.99	1.75, 7.41		
Outline	311	arc clockwise	1.75, 7.41	0.87, 7.25	1.14, 8.25	1.03
Outline	312	line	0.87, 7.25	0.86, 7.27		
Outline	313	arc clockwise	0.86, 7.27	1.61, 7.77	1.76, 6.75	1.03
Outline	314	line	1.61, 7.77	1.45, 8.21		
Outline	315	arc clockwise	1.45, 8.21	0.55, 8.11	0.91, 9.09	1.03
Outline	316	line	0.55, 8.11	0.55, 8.13		

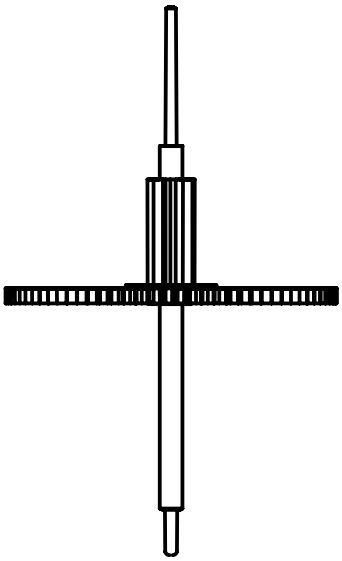
Outline	317	arc clockwise	0.55, 8.13	1.33, 8.57	1.40, 7.54	1.03
Outline	318	line	1.33, 8.57	1.21, 9.02		
Outline	319	arc clockwise	1.21, 9.02	0.31, 9.00	0.74, 9.94	1.03
Outline	320	line	0.31, 9.00	0.31, 9.01		
Outline	321	line	0.31, 9.01	0.34, 9.05		
Cut-out 1	322	line	15.14, 11.24	15.39, 11.62		
Cut-out 1	323	line	15.39, 11.62	15.14, 11.24		
Cut-out 2	324	arc clockwise	15.39, 11.62	14.22, 8.95	11.66, 11.66	3.73
Cut-out 2	325	line	14.22, 8.95	17.58, 3.97		
Cut-out 2	326	arc anticlockwise	17.58, 3.97	21.14, 13.76	11.66, 11.66	9.70
Cut-out 2	327	line	21.14, 13.76	15.36, 12.10		
Cut-out 2	328	arc clockwise	15.36, 12.10	15.39, 11.62	11.66, 11.66	3.72
Cut-out 3	329	line	15.39, 11.62	15.14, 11.31		
Cut-out 3	330	line	15.14, 11.31	15.39, 11.62		
Cut-out 4	331	line	12.22, 15.11	12.39, 15.32		
Cut-out 4	332	line	12.39, 15.32	12.22, 15.11		
Cut-out 5	333	arc clockwise	12.39, 15.32	15.03, 13.25	11.66, 11.66	3.72
Cut-out 5	334	line	15.03, 13.25	20.81, 14.91		
Cut-out 5	335	arc anticlockwise	20.81, 14.91	12.60, 21.32	11.66, 11.66	9.70
Cut-out 5	336	line	12.60, 21.32	12.39, 15.32		
Cut-out 6	337	arc clockwise	11.15, 7.98	9.87, 8.40	11.66, 11.66	3.73
Cut-out 6	338	line	9.87, 8.40	6.18, 3.67		
Cut-out 6	339	arc anticlockwise	6.18, 3.67	16.58, 3.30	11.66, 11.66	9.70
Cut-out 6	340	line	16.58, 3.30	13.22, 8.28		
Cut-out 6	341	arc clockwise	13.22, 8.28	11.15, 7.98	11.66, 11.66	3.72
Cut-out 7	342	arc anticlockwise	11.32, 13.63	10.68, 9.92	11.66, 11.66	2.00
Cut-out 7	343	line	10.68, 9.92	10.93, 9.80		
Cut-out 7	344	arc anticlockwise	10.93, 9.80	13.15, 13.00	11.66, 11.66	2.00
Cut-out 7	345	arc anticlockwise	13.15, 13.00	12.78, 13.32	11.94, 11.99	1.57
Cut-out 7	346	arc anticlockwise	12.78, 13.32	12.55, 13.45	12.12, 12.45	1.09
Cut-out 7	347	arc anticlockwise	12.55, 13.45	11.32, 13.63	11.67, 11.70	1.96
Cut-out 8	348	arc anticlockwise	10.04, 21.23	2.77, 15.54	11.66, 11.66	9.70
Cut-out 8	349	line	2.77, 15.54	8.42, 13.49		
Cut-out 8	350	arc clockwise	8.42, 13.49	11.19, 15.36	11.66, 11.66	3.72
Cut-out 8	351	line	11.19, 15.36	11.40, 21.36		
Cut-out 8	352	arc anticlockwise	11.40, 21.36	10.04, 21.23	11.66, 11.66	9.70
Cut-out 9	353	arc anticlockwise	2.29, 14.15	5.23, 4.40	11.66, 11.66	9.70
Cut-out 9	354	line	5.23, 4.40	8.93, 9.14		
Cut-out 9	355	arc clockwise	8.93, 9.14	8.00, 12.36	11.66, 11.66	3.72
Cut-out 9	356	line	8.00, 12.36	2.36, 14.41		
Cut-out 9	357	line	2.36, 14.41	2.29, 14.15		
Cut-out 10	358	line	8.05, 10.76	8.25, 10.95		
Cut-out 10	359	line	8.25, 10.95	8.05, 10.76		
Cut-out 11	360	line	7.97, 12.14	8.25, 12.43		
Cut-out 11	361	line	8.25, 12.43	7.97, 12.14		

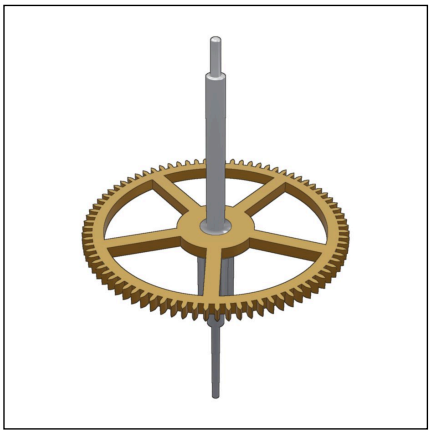
Teeth

Number	Module	Tip Ø	Root Ø

80	0.2806625		

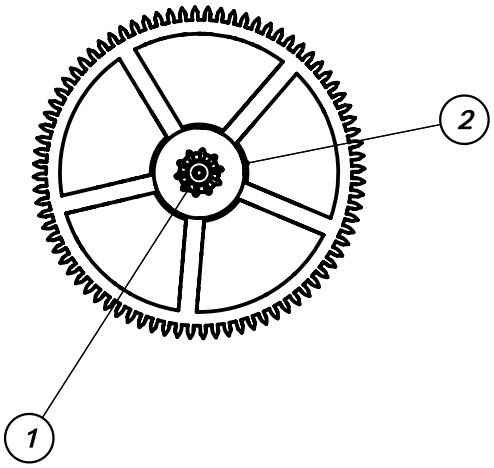
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Seconds wheel (Sekundenrad)	05 05 01-1	„Messingblech 1,2 Din 1751 Ms
1	1	Seconds pinion and arbor (Sekundentrieb)	05 05 01-2	„Präz. Rundstahl 3,5 Din 175“

Werkstoff / Material

„Messingblech 1,2 Din 1751 Ms 58 federhart“ (brass sheet 1.2 mm DIN 1751, Ms 58 spring-hard)

Maßstab / Scale

2:1

Behandlung / Treatment

surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01

Blatt / Sheet

1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.

05 05 01-1

Teil / Part

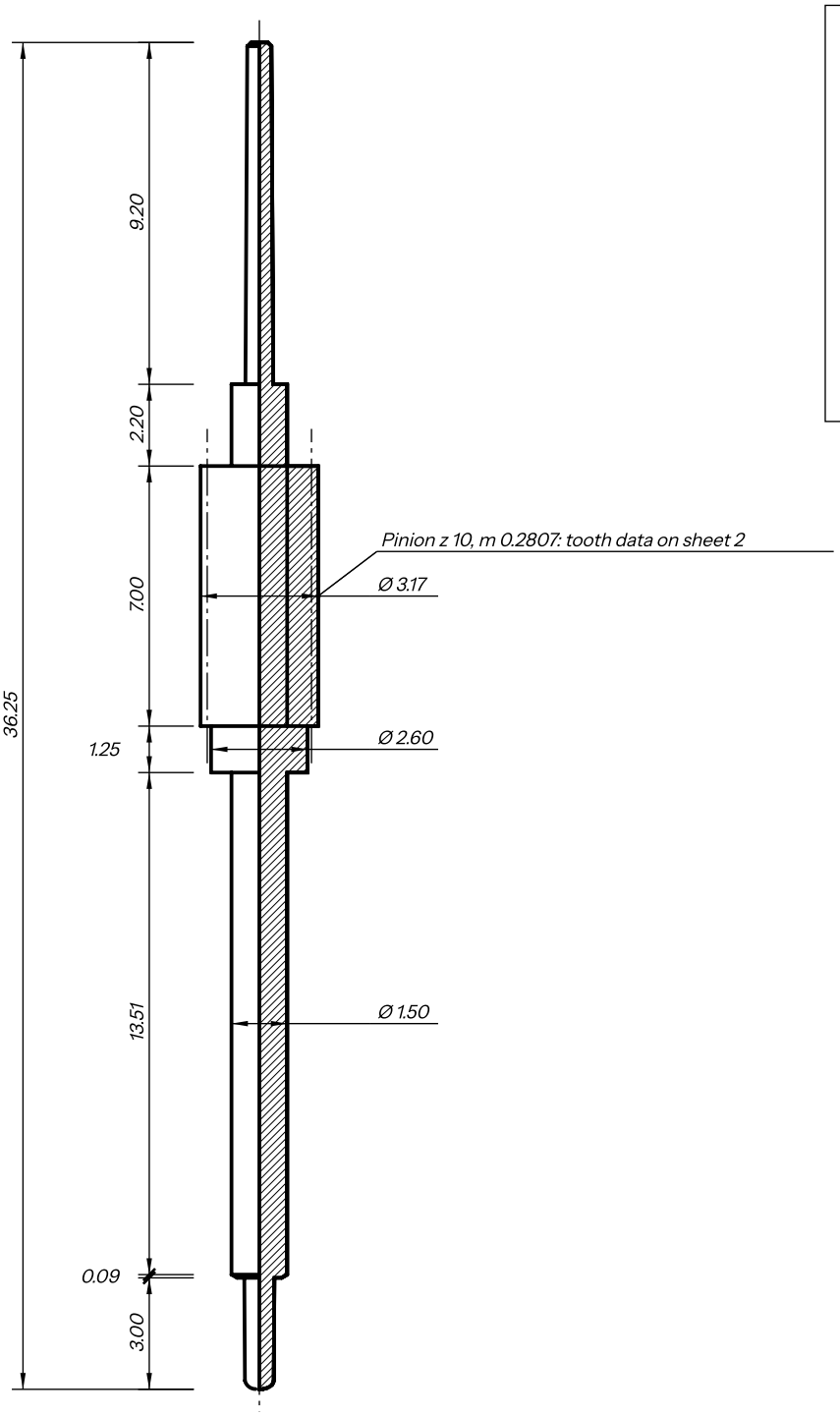
secondsWheel

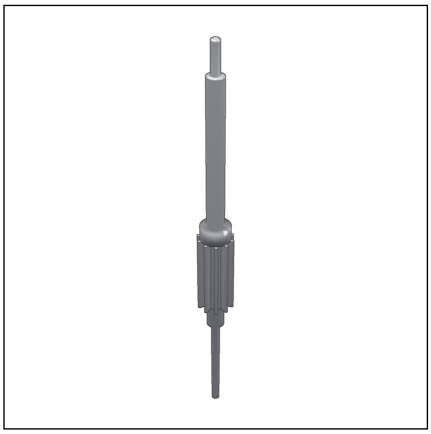
Benennung / Name

Seconds wheel (Sekundenrad)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

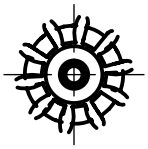
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

View from above



Werkstoff / Material „Präz. Rundstahl 3,5 Din 175" (precision round steel bar 3.5 mm, DIN 175)	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 01-2***
			Teil / Part *secondsWheel.arbor*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*		*Benennung / Name* ***Seconds pinion and arbor (Sekundentrieb)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Seconds pinion and arbor (Sekundentrieb) — sheet 2

The seconds arbor's steel pinion and arbor, one piece: its 10 leaves driven by the intermediate wheel, the seat its wheel is riveted on, and its lower pivot, which runs on as the post for the seconds hand.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 01-2	Sekundentrieb (seconds pinion and arbor)	1	„Präz. Rundstahl 3,5 Din 175“ (precision round steel bar 3.5 mm, DIN 175)	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	in WN-300.t35/6MX.jw35u (run): 2.252 mm, least gap 0.0124; in 050504-1 (run): 0.676 mm, least gap 0.0178; endshake 0.050 mm (050504-1 / WN-300.t35)

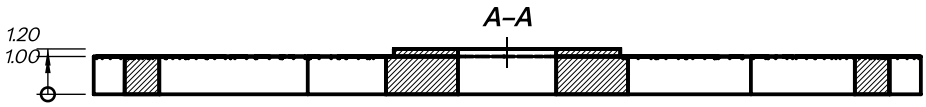
What it does (05 05 01-2). Driven by the intermediate wheel; turns once a minute; runs in the top plate's bush WN-300 and the jewel 05 05 04-1 in the intermediate bridge; its lower pivot goes on as the seconds hand's post. In the Soviet 6MX its upper pivot runs in a jewel in a chaton.

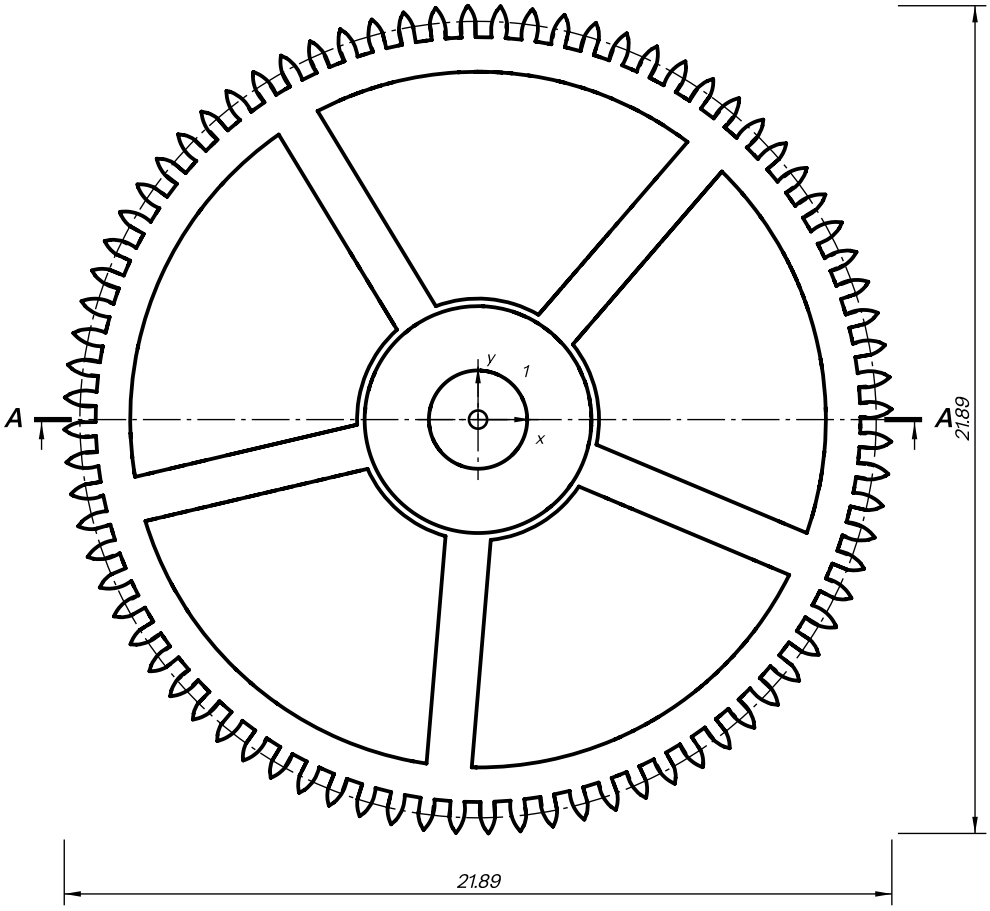
Turned about its axis; length 36.25, largest Ø 3.13. Diameters: Ø 1.50, Ø 2.60, Ø 3.17. Shoulders from the lower end: 0.00, 3.00, 3.09, 16.60, 17.85, 24.85, 27.05, 36.25.

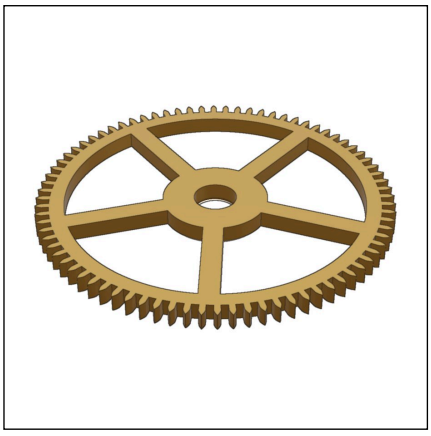
Leaves 17.85–24.85 from the lower end, pitch Ø 2.81; their form is the view from the end, drawn at the sheet's scale.

Teeth

Number	Module	Tip Ø	Root Ø
10	0.2807	3.17	







Isometric view, not to scale

Werkstoff / Material „Messingblech 1,2 Din 1751 Ms 58 federhart" (brass sheet 1.2 mm DIN 1751, Ms 58 spring-hard)	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 01-1*** *Teil / Part* *secondsWheel.wheel*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01*		*Benennung / Name* ***Seconds wheel (Sekundenrad)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Seconds wheel (Sekundenrad) — sheet 2

The seconds wheel, five-spoked, riveted on the pinion's seat; it drives the escape pinion. Its hub stands 0.2 proud toward the pinion.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 01-1	Sekundenrad (seconds wheel)	1	„Messingblech 1,2 Din 1751 Ms 58 federhart“ (brass sheet 1.2 mm DIN 1751, Ms 58 spring-hard)	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,01	-8

What it does (05 05 01-1). Riveted on the pinion's seat; drives the escape pinion (05 04 05-2).

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.20.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 2.60 through (0.20 thick at 1.00-1.20)

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-10.14, -2.90	-10.58, -2.82	-10.21, -2.01	0.89
Outline	2	line	-10.58, -2.82	-10.58, -2.81		
Outline	3	arc clockwise	-10.58, -2.81	-9.82, -2.43	-9.80, -3.40	0.97
Outline	4	line	-9.82, -2.43	-9.91, -2.01		
Outline	5	arc clockwise	-9.91, -2.01	-10.77, -1.98	-10.32, -1.12	0.97
Outline	6	line	-10.77, -1.98	-10.77, -1.97		
Outline	7	arc clockwise	-10.77, -1.97	-9.98, -1.65	-10.04, -2.62	0.97
Outline	8	line	-9.98, -1.65	-10.04, -1.22		
Outline	9	arc clockwise	-10.04, -1.22	-10.89, -1.13	-10.37, -0.31	0.97
Outline	10	line	-10.89, -1.13	-10.89, -1.12		
Outline	11	arc clockwise	-10.89, -1.12	-10.08, -0.86	-10.22, -1.83	0.97
Outline	12	line	-10.08, -0.86	-10.11, -0.43		
Outline	13	arc clockwise	-10.11, -0.43	-10.95, -0.27	-10.37, 0.51	0.97
Outline	14	line	-10.95, -0.27	-10.95, -0.26		
Outline	15	arc clockwise	-10.95, -0.26	-10.11, -0.07	-10.33, -1.02	0.97
Outline	16	line	-10.11, -0.07	-10.11, 0.36		
Outline	17	arc clockwise	-10.11, 0.36	-10.93, 0.59	-10.29, 1.32	0.97
Outline	18	line	-10.93, 0.59	-10.93, 0.59		
Outline	19	arc clockwise	-10.93, 0.59	-10.09, 0.73	-10.38, -0.21	0.97
Outline	20	line	-10.09, 0.73	-10.05, 1.15		
Outline	21	arc clockwise	-10.05, 1.15	-10.85, 1.44	-10.16, 2.12	0.97
Outline	22	line	-10.85, 1.44	-10.85, 1.45		
Outline	23	arc clockwise	-10.85, 1.45	-10.00, 1.52	-10.36, 0.61	0.97

Outline	24	line	-10.00, 1.52	-9.93, 1.94		
Outline	25	arc clockwise	-9.93, 1.94	-10.71, 2.29	-9.96, 2.92	0.97
Outline	26	line	-10.71, 2.29	-10.71, 2.30		
Outline	27	arc clockwise	-10.71, 2.30	-9.85, 2.30	-10.28, 1.42	0.97
Outline	28	line	-9.85, 2.30	-9.74, 2.71		
Outline	29	arc clockwise	-9.74, 2.71	-10.49, 3.12	-9.70, 3.69	0.97
Outline	30	line	-10.49, 3.12	-10.49, 3.13		
Outline	31	arc clockwise	-10.49, 3.13	-9.64, 3.06	-10.14, 2.22	0.97
Outline	32	line	-9.64, 3.06	-9.50, 3.47		
Outline	33	arc clockwise	-9.50, 3.47	-10.22, 3.94	-9.38, 4.44	0.97
Outline	34	line	-10.22, 3.94	-10.21, 3.94		
Outline	35	arc clockwise	-10.21, 3.94	-9.37, 3.81	-9.93, 3.01	0.97
Outline	36	line	-9.37, 3.81	-9.20, 4.20		
Outline	37	arc clockwise	-9.20, 4.20	-9.88, 4.72	-9.00, 5.16	0.97
Outline	38	line	-9.88, 4.72	-9.87, 4.73		
Outline	39	arc clockwise	-9.87, 4.73	-9.04, 4.53	-9.67, 3.78	0.97
Outline	40	line	-9.04, 4.53	-8.84, 4.91		
Outline	41	arc clockwise	-8.84, 4.91	-9.48, 5.49	-8.57, 5.85	0.97
Outline	42	line	-9.48, 5.49	-9.47, 5.49		
Outline	43	arc clockwise	-9.47, 5.49	-8.66, 5.23	-9.34, 4.53	0.97
Outline	44	line	-8.66, 5.23	-8.43, 5.59		
Outline	45	arc clockwise	-8.43, 5.59	-9.02, 6.21	-8.09, 6.51	0.97
Outline	46	line	-9.02, 6.21	-9.01, 6.22		
Outline	47	arc clockwise	-9.01, 6.22	-8.22, 5.89	-8.95, 5.25	0.97
Outline	48	line	-8.22, 5.89	-7.96, 6.24		
Outline	49	arc clockwise	-7.96, 6.24	-8.50, 6.90	-7.55, 7.12	0.97
Outline	50	line	-8.50, 6.90	-8.50, 6.91		
Outline	51	arc clockwise	-8.50, 6.91	-7.73, 6.52	-8.52, 5.93	0.97
Outline	52	line	-7.73, 6.52	-7.45, 6.84		
Outline	53	arc clockwise	-7.45, 6.84	-7.93, 7.55	-6.97, 7.69	0.97
Outline	54	line	-7.93, 7.55	-7.93, 7.55		
Outline	55	arc clockwise	-7.93, 7.55	-7.20, 7.11	-8.02, 6.58	0.97
Outline	56	line	-7.20, 7.11	-6.89, 7.40		
Outline	57	arc clockwise	-6.89, 7.40	-7.32, 8.14	-6.34, 8.21	0.97
Outline	58	line	-7.32, 8.14	-7.31, 8.15		
Outline	59	arc clockwise	-7.31, 8.15	-6.62, 7.65	-7.48, 7.19	0.97
Outline	60	line	-6.62, 7.65	-6.29, 7.92		
Outline	61	arc clockwise	-6.29, 7.92	-6.66, 8.69	-5.68, 8.69	0.97
Outline	62	line	-6.66, 8.69	-6.65, 8.70		
Outline	63	arc clockwise	-6.65, 8.70	-6.00, 8.14	-6.90, 7.76	0.97
Outline	64	line	-6.00, 8.14	-5.65, 8.39		
Outline	65	arc clockwise	-5.65, 8.39	-5.95, 9.19	-4.98, 9.10	0.97
Outline	66	line	-5.95, 9.19	-5.94, 9.19		
Outline	67	arc clockwise	-5.94, 9.19	-5.34, 8.59	-6.27, 8.27	0.97
Outline	68	line	-5.34, 8.59	-4.97, 8.81		
Outline	69	arc clockwise	-4.97, 8.81	-5.21, 9.63	-4.25, 9.47	0.97
Outline	70	line	-5.21, 9.63	-5.21, 9.63		
Outline	71	arc clockwise	-5.21, 9.63	-4.65, 8.98	-5.60, 8.74	0.97

Outline	72	line	-4.65, 8.98	-4.27, 9.17		
Outline	73	arc clockwise	-4.27, 9.17	-4.44, 10.01	-3.50, 9.77	0.97
Outline	74	line	-4.44, 10.01	-4.43, 10.01		
Outline	75	arc clockwise	-4.43, 10.01	-3.93, 9.32	-4.89, 9.15	0.97
Outline	76	line	-3.93, 9.32	-3.53, 9.48		
Outline	77	arc clockwise	-3.53, 9.48	-3.64, 10.33	-2.72, 10.02	0.97
Outline	78	line	-3.64, 10.33	-3.63, 10.33		
Outline	79	arc clockwise	-3.63, 10.33	-3.19, 9.60	-4.16, 9.51	0.97
Outline	80	line	-3.19, 9.60	-2.78, 9.73		
Outline	81	arc clockwise	-2.78, 9.73	-2.82, 10.58	-1.92, 10.20	0.97
Outline	82	line	-2.82, 10.58	-2.81, 10.58		
Outline	83	arc clockwise	-2.81, 10.58	-2.43, 9.82	-3.40, 9.81	0.97
Outline	84	line	-2.43, 9.82	-2.01, 9.91		
Outline	85	arc clockwise	-2.01, 9.91	-1.98, 10.77	-1.12, 10.32	0.97
Outline	86	line	-1.98, 10.77	-1.97, 10.77		
Outline	87	arc clockwise	-1.97, 10.77	-1.65, 9.98	-2.62, 10.04	0.97
Outline	88	line	-1.65, 9.98	-1.22, 10.04		
Outline	89	arc clockwise	-1.22, 10.04	-1.13, 10.89	-0.31, 10.37	0.97
Outline	90	line	-1.13, 10.89	-1.12, 10.89		
Outline	91	arc clockwise	-1.12, 10.89	-0.86, 10.08	-1.83, 10.22	0.97
Outline	92	line	-0.86, 10.08	-0.43, 10.11		
Outline	93	arc clockwise	-0.43, 10.11	-0.27, 10.95	0.51, 10.37	0.97
Outline	94	line	-0.27, 10.95	-0.26, 10.95		
Outline	95	arc clockwise	-0.26, 10.95	-0.07, 10.11	-1.02, 10.33	0.97
Outline	96	line	-0.07, 10.11	0.36, 10.11		
Outline	97	arc clockwise	0.36, 10.11	0.59, 10.93	1.32, 10.29	0.97
Outline	98	line	0.59, 10.93	0.59, 10.93		
Outline	99	arc clockwise	0.59, 10.93	0.73, 10.09	-0.21, 10.38	0.97
Outline	100	line	0.73, 10.09	1.15, 10.05		
Outline	101	arc clockwise	1.15, 10.05	1.44, 10.85	2.12, 10.16	0.97
Outline	102	line	1.44, 10.85	1.45, 10.85		
Outline	103	arc clockwise	1.45, 10.85	1.52, 10.00	0.61, 10.36	0.97
Outline	104	line	1.52, 10.00	1.94, 9.93		
Outline	105	arc clockwise	1.94, 9.93	2.29, 10.71	2.92, 9.96	0.97
Outline	106	line	2.29, 10.71	2.30, 10.71		
Outline	107	arc clockwise	2.30, 10.71	2.30, 9.85	1.42, 10.28	0.97
Outline	108	line	2.30, 9.85	2.71, 9.74		
Outline	109	arc clockwise	2.71, 9.74	3.12, 10.49	3.69, 9.70	0.97
Outline	110	line	3.12, 10.49	3.13, 10.49		
Outline	111	arc clockwise	3.13, 10.49	3.06, 9.64	2.22, 10.14	0.97
Outline	112	line	3.06, 9.64	3.47, 9.50		
Outline	113	arc clockwise	3.47, 9.50	3.94, 10.22	4.44, 9.38	0.97
Outline	114	line	3.94, 10.22	3.94, 10.21		
Outline	115	arc clockwise	3.94, 10.21	3.81, 9.37	3.01, 9.93	0.97
Outline	116	line	3.81, 9.37	4.20, 9.20		
Outline	117	arc clockwise	4.20, 9.20	4.72, 9.88	5.16, 9.00	0.97
Outline	118	line	4.72, 9.88	4.73, 9.87		
Outline	119	arc clockwise	4.73, 9.87	4.53, 9.04	3.78, 9.67	0.97

Outline	120	line	4.53, 9.04	4.91, 8.84		
Outline	121	arc clockwise	4.91, 8.84	5.49, 9.48	5.85, 8.57	0.97
Outline	122	line	5.49, 9.48	5.49, 9.47		
Outline	123	arc clockwise	5.49, 9.47	5.23, 8.66	4.53, 9.34	0.97
Outline	124	line	5.23, 8.66	5.59, 8.43		
Outline	125	arc clockwise	5.59, 8.43	6.21, 9.02	6.51, 8.09	0.97
Outline	126	line	6.21, 9.02	6.22, 9.01		
Outline	127	arc clockwise	6.22, 9.01	5.89, 8.22	5.25, 8.95	0.97
Outline	128	line	5.89, 8.22	6.24, 7.96		
Outline	129	arc clockwise	6.24, 7.96	6.90, 8.50	7.12, 7.55	0.97
Outline	130	line	6.90, 8.50	6.91, 8.50		
Outline	131	arc clockwise	6.91, 8.50	6.52, 7.73	5.93, 8.52	0.97
Outline	132	line	6.52, 7.73	6.84, 7.45		
Outline	133	arc clockwise	6.84, 7.45	7.55, 7.93	7.69, 6.97	0.97
Outline	134	line	7.55, 7.93	7.55, 7.93		
Outline	135	arc clockwise	7.55, 7.93	7.11, 7.20	6.58, 8.02	0.97
Outline	136	line	7.11, 7.20	7.40, 6.89		
Outline	137	arc clockwise	7.40, 6.89	8.14, 7.32	8.21, 6.34	0.97
Outline	138	line	8.14, 7.32	8.15, 7.31		
Outline	139	arc clockwise	8.15, 7.31	7.65, 6.62	7.19, 7.48	0.97
Outline	140	line	7.65, 6.62	7.92, 6.29		
Outline	141	arc clockwise	7.92, 6.29	8.69, 6.66	8.69, 5.68	0.97
Outline	142	line	8.69, 6.66	8.70, 6.65		
Outline	143	arc clockwise	8.70, 6.65	8.14, 6.00	7.76, 6.90	0.97
Outline	144	line	8.14, 6.00	8.39, 5.65		
Outline	145	arc clockwise	8.39, 5.65	9.19, 5.95	9.10, 4.98	0.97
Outline	146	line	9.19, 5.95	9.19, 5.94		
Outline	147	arc clockwise	9.19, 5.94	8.59, 5.34	8.27, 6.27	0.97
Outline	148	line	8.59, 5.34	8.81, 4.97		
Outline	149	arc clockwise	8.81, 4.97	9.63, 5.21	9.47, 4.25	0.97
Outline	150	line	9.63, 5.21	9.63, 5.21		
Outline	151	arc clockwise	9.63, 5.21	8.98, 4.65	8.74, 5.60	0.97
Outline	152	line	8.98, 4.65	9.17, 4.27		
Outline	153	arc clockwise	9.17, 4.27	10.01, 4.44	9.77, 3.50	0.97
Outline	154	line	10.01, 4.44	10.01, 4.43		
Outline	155	arc clockwise	10.01, 4.43	9.32, 3.93	9.15, 4.89	0.97
Outline	156	line	9.32, 3.93	9.48, 3.53		
Outline	157	arc clockwise	9.48, 3.53	10.33, 3.64	10.02, 2.72	0.97
Outline	158	line	10.33, 3.64	10.33, 3.63		
Outline	159	arc clockwise	10.33, 3.63	9.60, 3.19	9.51, 4.16	0.97
Outline	160	line	9.60, 3.19	9.73, 2.78		
Outline	161	arc clockwise	9.73, 2.78	10.58, 2.82	10.20, 1.92	0.97
Outline	162	line	10.58, 2.82	10.58, 2.81		
Outline	163	arc clockwise	10.58, 2.81	9.82, 2.43	9.80, 3.40	0.97
Outline	164	line	9.82, 2.43	9.91, 2.01		
Outline	165	arc clockwise	9.91, 2.01	10.77, 1.98	10.32, 1.12	0.97
Outline	166	line	10.77, 1.98	10.77, 1.97		
Outline	167	arc clockwise	10.77, 1.97	9.98, 1.65	10.04, 2.62	0.97

Outline	168	line	9.98, 1.65	10.04, 1.22		
Outline	169	arc clockwise	10.04, 1.22	10.89, 1.13	10.37, 0.31	0.97
Outline	170	line	10.89, 1.13	10.89, 1.12		
Outline	171	arc clockwise	10.89, 1.12	10.08, 0.86	10.22, 1.83	0.97
Outline	172	line	10.08, 0.86	10.11, 0.43		
Outline	173	arc clockwise	10.11, 0.43	10.95, 0.27	10.37, -0.51	0.97
Outline	174	line	10.95, 0.27	10.95, 0.26		
Outline	175	arc clockwise	10.95, 0.26	10.11, 0.07	10.33, 1.02	0.97
Outline	176	line	10.11, 0.07	10.11, -0.36		
Outline	177	arc clockwise	10.11, -0.36	10.93, -0.59	10.29, -1.32	0.97
Outline	178	line	10.93, -0.59	10.93, -0.59		
Outline	179	arc clockwise	10.93, -0.59	10.09, -0.73	10.38, 0.21	0.97
Outline	180	line	10.09, -0.73	10.05, -1.15		
Outline	181	arc clockwise	10.05, -1.15	10.85, -1.44	10.16, -2.12	0.97
Outline	182	line	10.85, -1.44	10.85, -1.45		
Outline	183	arc clockwise	10.85, -1.45	10.00, -1.52	10.36, -0.61	0.97
Outline	184	line	10.00, -1.52	9.93, -1.94		
Outline	185	arc clockwise	9.93, -1.94	10.71, -2.29	9.96, -2.92	0.97
Outline	186	line	10.71, -2.29	10.71, -2.30		
Outline	187	arc clockwise	10.71, -2.30	9.85, -2.30	10.28, -1.42	0.97
Outline	188	line	9.85, -2.30	9.74, -2.71		
Outline	189	arc clockwise	9.74, -2.71	10.49, -3.12	9.70, -3.69	0.97
Outline	190	line	10.49, -3.12	10.49, -3.13		
Outline	191	arc clockwise	10.49, -3.13	9.64, -3.06	10.14, -2.22	0.97
Outline	192	line	9.64, -3.06	9.50, -3.47		
Outline	193	arc clockwise	9.50, -3.47	10.22, -3.94	9.38, -4.44	0.97
Outline	194	line	10.22, -3.94	10.21, -3.94		
Outline	195	arc clockwise	10.21, -3.94	9.37, -3.81	9.93, -3.01	0.97
Outline	196	line	9.37, -3.81	9.20, -4.20		
Outline	197	arc clockwise	9.20, -4.20	9.88, -4.72	9.00, -5.16	0.97
Outline	198	line	9.88, -4.72	9.87, -4.73		
Outline	199	arc clockwise	9.87, -4.73	9.04, -4.53	9.67, -3.78	0.97
Outline	200	line	9.04, -4.53	8.84, -4.91		
Outline	201	arc clockwise	8.84, -4.91	9.48, -5.49	8.57, -5.85	0.97
Outline	202	line	9.48, -5.49	9.47, -5.49		
Outline	203	arc clockwise	9.47, -5.49	8.66, -5.23	9.34, -4.53	0.97
Outline	204	line	8.66, -5.23	8.43, -5.59		
Outline	205	arc clockwise	8.43, -5.59	9.02, -6.21	8.09, -6.51	0.97
Outline	206	line	9.02, -6.21	9.01, -6.22		
Outline	207	arc clockwise	9.01, -6.22	8.22, -5.89	8.95, -5.25	0.97
Outline	208	line	8.22, -5.89	7.96, -6.24		
Outline	209	arc clockwise	7.96, -6.24	8.50, -6.90	7.55, -7.12	0.97
Outline	210	line	8.50, -6.90	8.50, -6.91		
Outline	211	arc clockwise	8.50, -6.91	7.73, -6.52	8.52, -5.93	0.97
Outline	212	line	7.73, -6.52	7.45, -6.84		
Outline	213	arc clockwise	7.45, -6.84	7.93, -7.55	6.97, -7.69	0.97
Outline	214	line	7.93, -7.55	7.93, -7.55		
Outline	215	arc clockwise	7.93, -7.55	7.20, -7.11	8.02, -6.58	0.97

Outline	216	line	7.20, -7.11	6.89, -7.40		
Outline	217	arc clockwise	6.89, -7.40	7.32, -8.14	6.34, -8.21	0.97
Outline	218	line	7.32, -8.14	7.31, -8.15		
Outline	219	arc clockwise	7.31, -8.15	6.62, -7.65	7.48, -7.19	0.97
Outline	220	line	6.62, -7.65	6.29, -7.92		
Outline	221	arc clockwise	6.29, -7.92	6.66, -8.69	5.68, -8.69	0.97
Outline	222	line	6.66, -8.69	6.65, -8.70		
Outline	223	arc clockwise	6.65, -8.70	6.00, -8.14	6.90, -7.76	0.97
Outline	224	line	6.00, -8.14	5.65, -8.39		
Outline	225	arc clockwise	5.65, -8.39	5.95, -9.19	4.98, -9.10	0.97
Outline	226	line	5.95, -9.19	5.94, -9.19		
Outline	227	arc clockwise	5.94, -9.19	5.34, -8.59	6.27, -8.27	0.97
Outline	228	line	5.34, -8.59	4.97, -8.81		
Outline	229	arc clockwise	4.97, -8.81	5.21, -9.63	4.25, -9.47	0.97
Outline	230	line	5.21, -9.63	5.21, -9.63		
Outline	231	arc clockwise	5.21, -9.63	4.65, -8.98	5.60, -8.74	0.97
Outline	232	line	4.65, -8.98	4.27, -9.17		
Outline	233	arc clockwise	4.27, -9.17	4.44, -10.01	3.50, -9.77	0.97
Outline	234	line	4.44, -10.01	4.43, -10.01		
Outline	235	arc clockwise	4.43, -10.01	3.93, -9.32	4.89, -9.15	0.97
Outline	236	line	3.93, -9.32	3.53, -9.48		
Outline	237	arc clockwise	3.53, -9.48	3.64, -10.33	2.72, -10.02	0.97
Outline	238	line	3.64, -10.33	3.63, -10.33		
Outline	239	arc clockwise	3.63, -10.33	3.19, -9.60	4.16, -9.51	0.97
Outline	240	line	3.19, -9.60	2.78, -9.73		
Outline	241	arc clockwise	2.78, -9.73	2.82, -10.58	1.92, -10.20	0.97
Outline	242	line	2.82, -10.58	2.81, -10.58		
Outline	243	arc clockwise	2.81, -10.58	2.43, -9.82	3.40, -9.80	0.97
Outline	244	line	2.43, -9.82	2.01, -9.91		
Outline	245	arc clockwise	2.01, -9.91	1.98, -10.77	1.12, -10.32	0.97
Outline	246	line	1.98, -10.77	1.97, -10.77		
Outline	247	arc clockwise	1.97, -10.77	1.65, -9.98	2.62, -10.04	0.97
Outline	248	line	1.65, -9.98	1.22, -10.04		
Outline	249	arc clockwise	1.22, -10.04	1.13, -10.89	0.31, -10.37	0.97
Outline	250	line	1.13, -10.89	1.12, -10.89		
Outline	251	arc clockwise	1.12, -10.89	0.86, -10.08	1.83, -10.22	0.97
Outline	252	line	0.86, -10.08	0.43, -10.11		
Outline	253	arc clockwise	0.43, -10.11	0.27, -10.95	-0.51, -10.37	0.97
Outline	254	line	0.27, -10.95	0.26, -10.95		
Outline	255	arc clockwise	0.26, -10.95	0.07, -10.11	1.02, -10.33	0.97
Outline	256	line	0.07, -10.11	-0.36, -10.11		
Outline	257	arc clockwise	-0.36, -10.11	-0.59, -10.93	-1.32, -10.29	0.97
Outline	258	line	-0.59, -10.93	-0.59, -10.93		
Outline	259	arc clockwise	-0.59, -10.93	-0.73, -10.09	0.21, -10.38	0.97
Outline	260	line	-0.73, -10.09	-1.15, -10.05		
Outline	261	arc clockwise	-1.15, -10.05	-1.44, -10.85	-2.12, -10.16	0.97
Outline	262	line	-1.44, -10.85	-1.45, -10.85		
Outline	263	arc clockwise	-1.45, -10.85	-1.52, -10.00	-0.61, -10.36	0.97

Outline	264	line	-1.52, -10.00	-1.94, -9.93		
Outline	265	arc clockwise	-1.94, -9.93	-2.29, -10.71	-2.92, -9.96	0.97
Outline	266	line	-2.29, -10.71	-2.30, -10.71		
Outline	267	arc clockwise	-2.30, -10.71	-2.30, -9.85	-1.42, -10.28	0.97
Outline	268	line	-2.30, -9.85	-2.71, -9.74		
Outline	269	arc clockwise	-2.71, -9.74	-3.12, -10.49	-3.69, -9.70	0.97
Outline	270	line	-3.12, -10.49	-3.13, -10.49		
Outline	271	arc clockwise	-3.13, -10.49	-3.06, -9.64	-2.22, -10.14	0.97
Outline	272	line	-3.06, -9.64	-3.47, -9.50		
Outline	273	arc clockwise	-3.47, -9.50	-3.94, -10.22	-4.44, -9.38	0.97
Outline	274	line	-3.94, -10.22	-3.94, -10.21		
Outline	275	arc clockwise	-3.94, -10.21	-3.81, -9.37	-3.01, -9.93	0.97
Outline	276	line	-3.81, -9.37	-4.20, -9.20		
Outline	277	arc clockwise	-4.20, -9.20	-4.72, -9.88	-5.16, -9.00	0.97
Outline	278	line	-4.72, -9.88	-4.73, -9.87		
Outline	279	arc clockwise	-4.73, -9.87	-4.53, -9.04	-3.78, -9.67	0.97
Outline	280	line	-4.53, -9.04	-4.91, -8.84		
Outline	281	arc clockwise	-4.91, -8.84	-5.49, -9.48	-5.85, -8.57	0.97
Outline	282	line	-5.49, -9.48	-5.49, -9.47		
Outline	283	arc clockwise	-5.49, -9.47	-5.23, -8.66	-4.53, -9.34	0.97
Outline	284	line	-5.23, -8.66	-5.59, -8.43		
Outline	285	arc clockwise	-5.59, -8.43	-6.21, -9.02	-6.51, -8.09	0.97
Outline	286	line	-6.21, -9.02	-6.22, -9.01		
Outline	287	arc clockwise	-6.22, -9.01	-5.89, -8.22	-5.25, -8.95	0.97
Outline	288	line	-5.89, -8.22	-6.24, -7.96		
Outline	289	arc clockwise	-6.24, -7.96	-6.90, -8.50	-7.12, -7.55	0.97
Outline	290	line	-6.90, -8.50	-6.91, -8.50		
Outline	291	arc clockwise	-6.91, -8.50	-6.52, -7.73	-5.93, -8.52	0.97
Outline	292	line	-6.52, -7.73	-6.84, -7.45		
Outline	293	arc clockwise	-6.84, -7.45	-7.55, -7.93	-7.69, -6.97	0.97
Outline	294	line	-7.55, -7.93	-7.55, -7.93		
Outline	295	arc clockwise	-7.55, -7.93	-7.11, -7.20	-6.58, -8.02	0.97
Outline	296	line	-7.11, -7.20	-7.40, -6.89		
Outline	297	arc clockwise	-7.40, -6.89	-8.14, -7.32	-8.21, -6.34	0.97
Outline	298	line	-8.14, -7.32	-8.15, -7.31		
Outline	299	arc clockwise	-8.15, -7.31	-7.65, -6.62	-7.19, -7.48	0.97
Outline	300	line	-7.65, -6.62	-7.92, -6.29		
Outline	301	arc clockwise	-7.92, -6.29	-8.69, -6.66	-8.69, -5.68	0.97
Outline	302	line	-8.69, -6.66	-8.70, -6.65		
Outline	303	arc clockwise	-8.70, -6.65	-8.14, -6.00	-7.76, -6.90	0.97
Outline	304	line	-8.14, -6.00	-8.39, -5.65		
Outline	305	arc clockwise	-8.39, -5.65	-9.19, -5.95	-9.10, -4.98	0.97
Outline	306	line	-9.19, -5.95	-9.19, -5.94		
Outline	307	arc clockwise	-9.19, -5.94	-8.59, -5.34	-8.27, -6.27	0.97
Outline	308	line	-8.59, -5.34	-8.81, -4.97		
Outline	309	arc clockwise	-8.81, -4.97	-9.63, -5.21	-9.47, -4.25	0.97
Outline	310	line	-9.63, -5.21	-9.63, -5.21		
Outline	311	arc clockwise	-9.63, -5.21	-8.98, -4.65	-8.74, -5.60	0.97

Outline	312	line	-8.98, -4.65	-9.17, -4.27		
Outline	313	arc clockwise	-9.17, -4.27	-10.01, -4.44	-9.77, -3.50	0.97
Outline	314	line	-10.01, -4.44	-10.01, -4.43		
Outline	315	arc clockwise	-10.01, -4.43	-9.32, -3.93	-9.15, -4.89	0.97
Outline	316	line	-9.32, -3.93	-9.48, -3.53		
Outline	317	arc clockwise	-9.48, -3.53	-10.33, -3.64	-10.02, -2.72	0.97
Outline	318	line	-10.33, -3.64	-10.33, -3.63		
Outline	319	arc clockwise	-10.33, -3.63	-9.60, -3.19	-9.51, -4.16	0.97
Outline	320	line	-9.60, -3.19	-9.73, -2.78		
Outline	321	line	-9.73, -2.78	-10.14, -2.90		
Cut-out 1	322	arc clockwise	2.53, 1.96	3.13, -0.66	0.00, 0.00	3.20
Cut-out 1	323	line	3.13, -0.66	8.70, -3.00		
Cut-out 1	324	arc anticlockwise	8.70, -3.00	6.45, 6.56	0.00, 0.00	9.20
Cut-out 1	325	line	6.45, 6.56	2.51, 1.99		
Cut-out 1	326	line	2.51, 1.99	2.53, 1.96		
Cut-out 2	327	arc anticlockwise	-0.17, -9.20	8.23, -4.11	0.00, 0.00	9.20
Cut-out 2	328	line	8.23, -4.11	2.67, -1.77		
Cut-out 2	329	arc clockwise	2.67, -1.77	0.34, -3.18	0.00, 0.00	3.20
Cut-out 2	330	line	0.34, -3.18	-0.17, -9.20		
Cut-out 3	331	arc anticlockwise	-1.94, 8.99	-4.25, 8.16	0.00, 0.00	9.20
Cut-out 3	332	line	-4.25, 8.16	-1.12, 3.00		
Cut-out 3	333	arc clockwise	-1.12, 3.00	1.60, 2.77	0.00, 0.00	3.20
Cut-out 3	334	line	1.60, 2.77	5.54, 7.34		
Cut-out 3	335	arc anticlockwise	5.54, 7.34	-1.94, 8.99	0.00, 0.00	9.20
Cut-out 5	336	arc anticlockwise	-8.72, -2.93	-1.36, -9.10	0.00, 0.00	9.20
Cut-out 5	337	line	-1.36, -9.10	-0.86, -3.08		
Cut-out 5	338	arc clockwise	-0.86, -3.08	-2.92, -1.30	0.00, 0.00	3.20
Cut-out 5	339	line	-2.92, -1.30	-8.80, -2.68		
Cut-out 5	340	line	-8.80, -2.68	-8.72, -2.93		
Cut-out 6	341	arc anticlockwise	-7.62, 5.15	-9.07, -1.52	0.00, 0.00	9.20
Cut-out 6	342	line	-9.07, -1.52	-3.20, -0.13		
Cut-out 6	343	arc clockwise	-3.20, -0.13	-2.14, 2.38	0.00, 0.00	3.20
Cut-out 6	344	line	-2.14, 2.38	-5.27, 7.54		
Cut-out 6	345	arc anticlockwise	-5.27, 7.54	-7.62, 5.15	0.00, 0.00	9.20

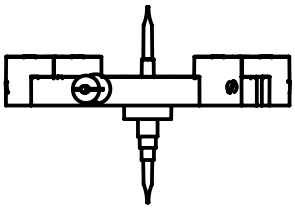
Levels

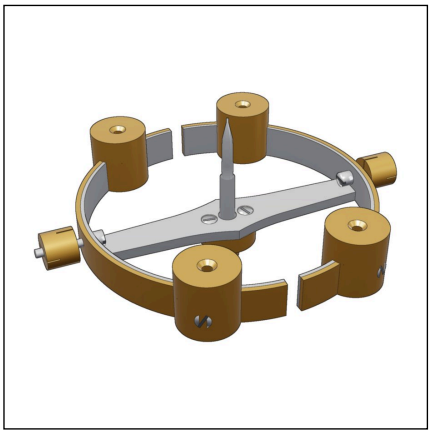
Faces in section A-A, from the bottom face: 1.00, 1.20.

Teeth

Number	Module	Tip Ø	Root Ø
80	0.263325	21.90	20.23

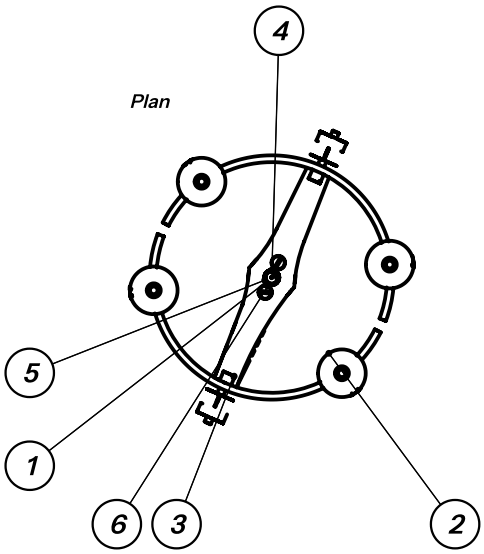
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
6	[0, 2]	Hub screws	WN-10	not on the drawing
5	1	Balance hub (Unruhbutzen)	05 04 02-2	„Ms.“ (brass)
4	1	Balance staff	05 04 02-1	not on the drawing
3	2	Timing screws and weights	05 03 01-2	not on the drawing
2	4	Compensation masses	05 03 01-4	not on the drawing
1	1	Balance rim and arm	05 03 01-1	„Nickelstahl Sonderlegierung u

Werkstoff / Material

„Nickelstahl Sonderlegierung u. Messing“ (nickel steel, special alloy, and brass)

Maßstab / Scale

1:1

Behandlung / Treatment

„▽▽▽ poliert“; surface ▽▽▽ ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0.03

Blatt / Sheet

1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.

05 03 01-1

Teil / Part

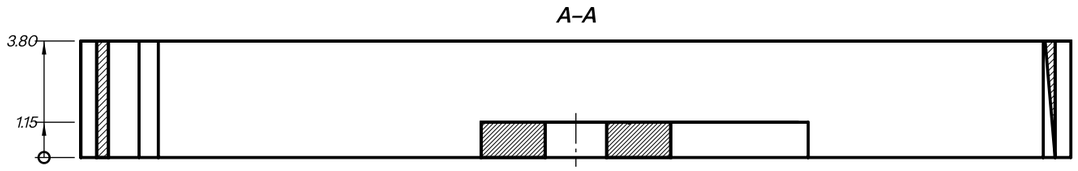
balance

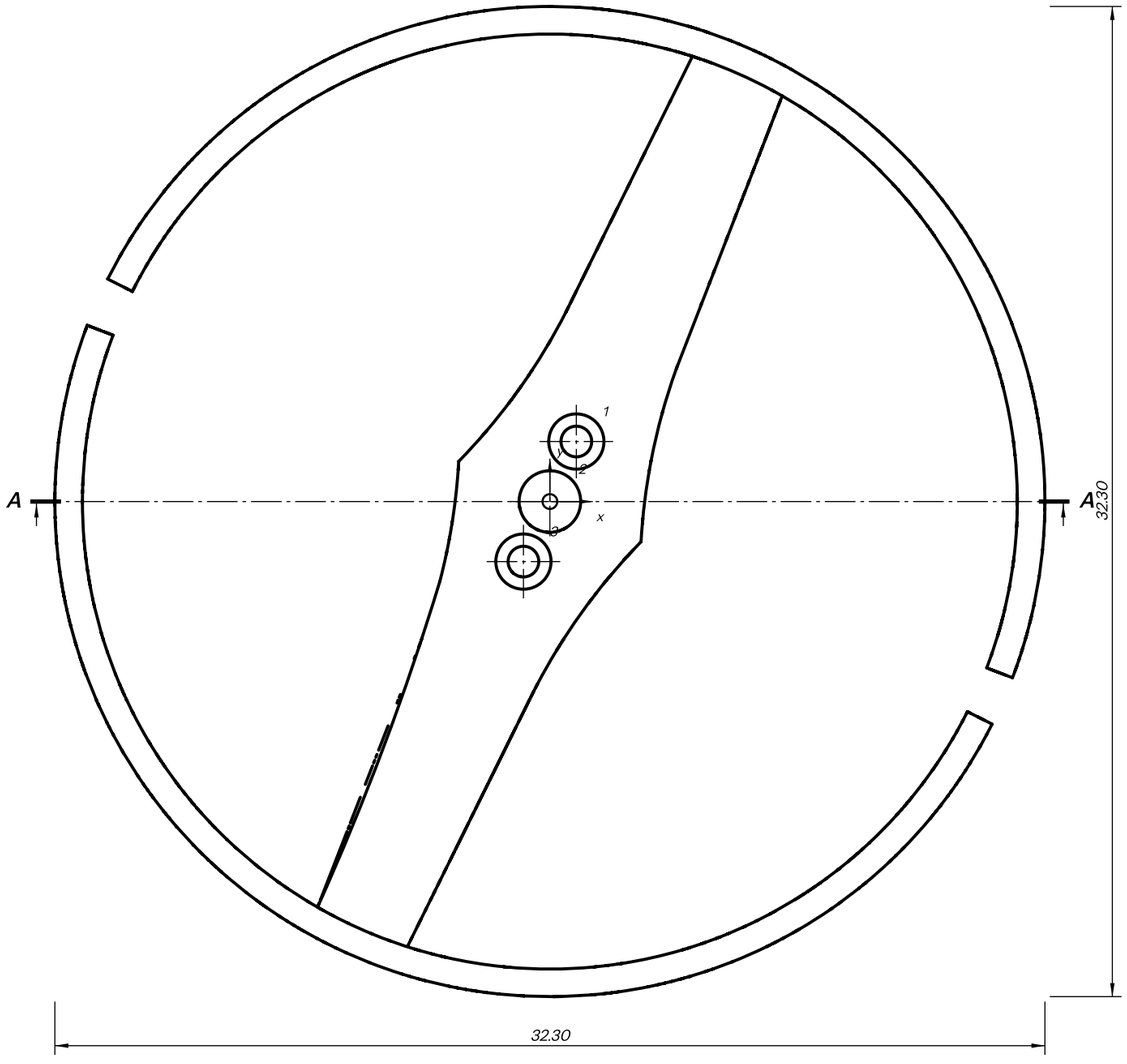
Benennung / Name

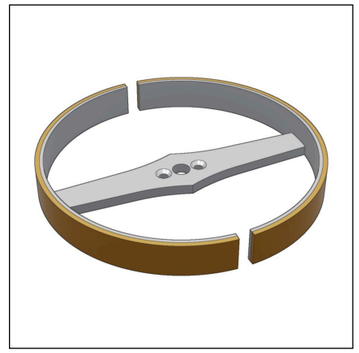
Balance (Unruh)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).







Isometric view, not to scale

Werkstoff / Material *„Nickelstahl Sonderlegierung u. Messing" (nickel steel, special alloy, and brass)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 03 01-1***
Behandlung / Treatment *„▽▽▽ poliert"; surface ▽▽▽ ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0.03*			*Teil / Part* *balance.rim*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Balance rim and arm***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Balance rim and arm — sheet 2

The balance: a cut bimetallic rim, nickel steel inside and brass outside, made in one with its straight steel arm (stamped RGG in a triangle, as on Wempe No. 3955), so each quarter of the rim curls in as it warms and carries its mass nearer the staff. Four round brass masses sit across the rim, two timing weights on screws at the arm's ends. It sits on the brass hub of its staff just above the top plate, its rollers under the plate. It swings about 1¼ turns, twice a second.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 03 01-1	Unruhreifen (balance rim with arm, nickel steel and brass)	1	„Nickelstahl Sonderlegierung u. Messing“ (nickel steel, special alloy, and brass)	„▽▽▽ poliert“; surface ▽▽▽ ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,03	round 050402-1 (press): 1.124 mm, least gap 0.008; on 050402-2: 0 mm

Test on the bench (05 03 01-1). The balance runs down free, the balance, hairspring and detent in, the train held still at a lock; before the mainspring is chosen: The model's balance halves its swing in 60 s (its TF 111 s; a running 6MX on video, moMvyx8eXfo, has TF 102–125 s at it: wempe_startup.js, which the model's TF is set from). Halving in 46, 55, 60, 66, 79, 93 s (TF 80, 100, 111, 125, 160, 200): 05 06_20's drive swings it 202, 221, 230, 241, 265, 286° at 12 h; the guard's 180° at the wind's lowest wants ×0.860, 0.734, 0.684, 0.633, 0.544, 0.480 of it, the sources' 225° ×1.288, 1.090, 1.012, 0.932, 0.793, 0.693 (the strip as the drive's cube root). Shorter than 46 s, look for a pivot, the endstone or the detent's springs dragging.

What it does (05 03 01-1). The cut bimetallic rim and its arm, on the hub of the staff.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.80.

Holes

Hole	x	y	Size
1	0.86	1.96	Ø 1.00 through (0.77 thick at 0.00–0.77); countersunk to Ø 1.80
2	0.00	0.00	Ø 2.01 through (1.15 thick at 0.00–1.15)
3	-0.86	-1.96	Ø 1.00 through (0.77 thick at 0.00–0.77); countersunk to Ø 1.80

Contour in the plan

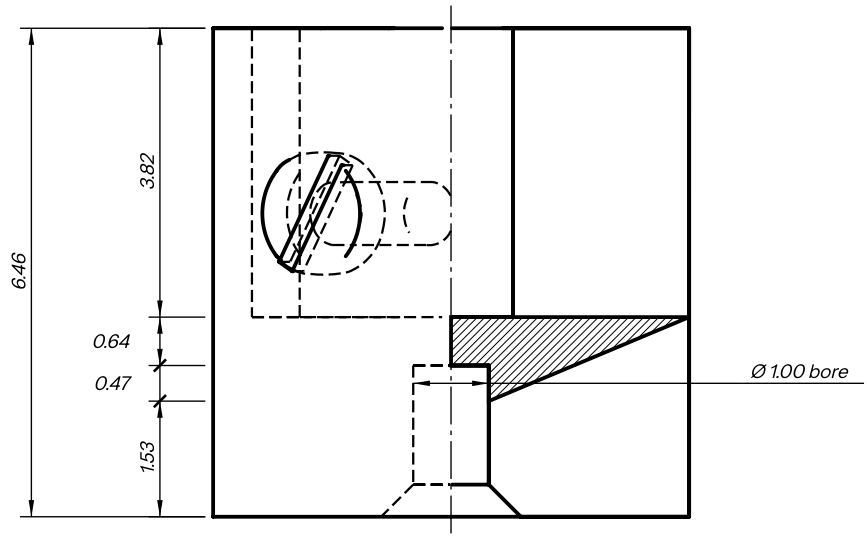
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-15.70, -3.77	-15.09, 5.75	0.00, 0.00	16.15
Outline	2	line	-15.09, 5.75	-14.25, 5.43		
Outline	3	arc anticlockwise	-14.25, 5.43	-7.58, -13.23	0.00, 0.00	15.25
Outline	4	arc anticlockwise	-7.58, -13.23	-3.80, -3.35	-122.28, 36.35	124.96
Outline	5	arc anticlockwise	-3.80, -3.35	-2.97, 1.31	-22.94, 2.45	20.00
Outline	6	arc anticlockwise	-2.97, 1.31	0.64, 6.41	-17.29, 15.28	20.00
Outline	7	line	0.64, 6.41	4.65, 14.52		
Outline	8	arc anticlockwise	4.65, 14.52	-13.62, 6.85	0.00, 0.00	15.25
Outline	9	line	-13.62, 6.85	-14.43, 7.26		
Outline	10	arc clockwise	-14.43, 7.26	15.09, -5.75	0.00, 0.00	16.15
Outline	11	line	15.09, -5.75	14.25, -5.43		
Outline	12	arc anticlockwise	14.25, -5.43	7.58, 13.23	0.00, 0.00	15.25
Outline	13	arc anticlockwise	7.58, 13.23	3.80, 3.35	122.28, -36.36	124.96
Outline	14	arc anticlockwise	3.80, 3.35	2.97, -1.31	22.94, -2.45	20.00
Outline	15	arc anticlockwise	2.97, -1.31	-0.64, -6.41	17.29, -15.28	20.00

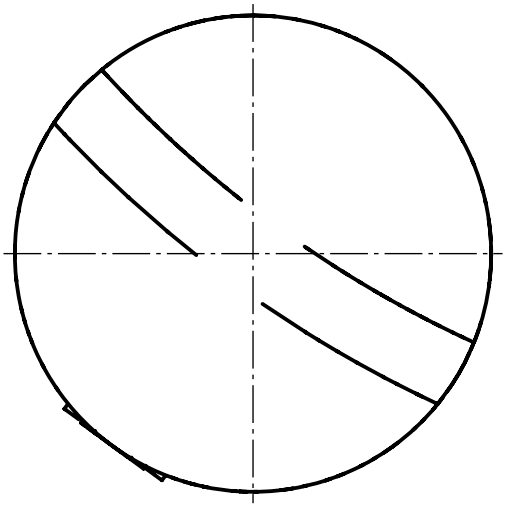
Outline	16	line	-0.64, -6.41	-4.65, -14.52		
Outline	17	arc anticlockwise	-4.65, -14.52	13.62, -6.85	0.00, 0.00	15.25
Outline	18	line	13.62, -6.85	14.43, -7.26		
Outline	19	arc clockwise	14.43, -7.26	-15.70, -3.77	0.00, 0.00	16.15
Cut-out 4	20	line	-0.69, -4.23	-0.69, -4.23		
Cut-out 4	21	line	-0.69, -4.23	-0.69, -4.23		
Cut-out 4	22	line	-0.69, -4.23	-0.69, -4.23		

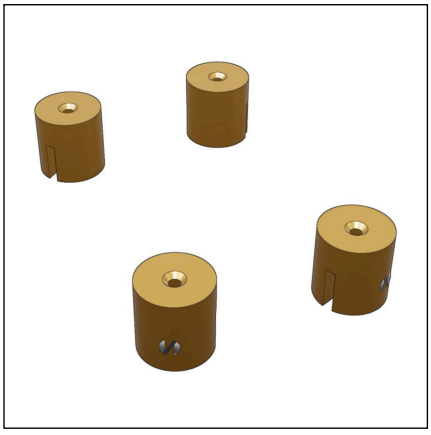
Levels

Faces in section A-A, from the bottom face: 1.15, 3.80.



View from above





Isometric view, not to scale

4 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 03 01-4** *Teil / Part* *balance.masses*
Behandlung / Treatment		*Benennung / Name* ***Compensation masses***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Compensation masses — sheet 2

Four round brass masses across the rim, each held by a screw (WN-12) through its outer jaw onto the rim; in each top a countersunk M1.2 hole for WN-12a, as the drawings have it, left empty (photographed Einheits chronometers show plain tops).

4 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

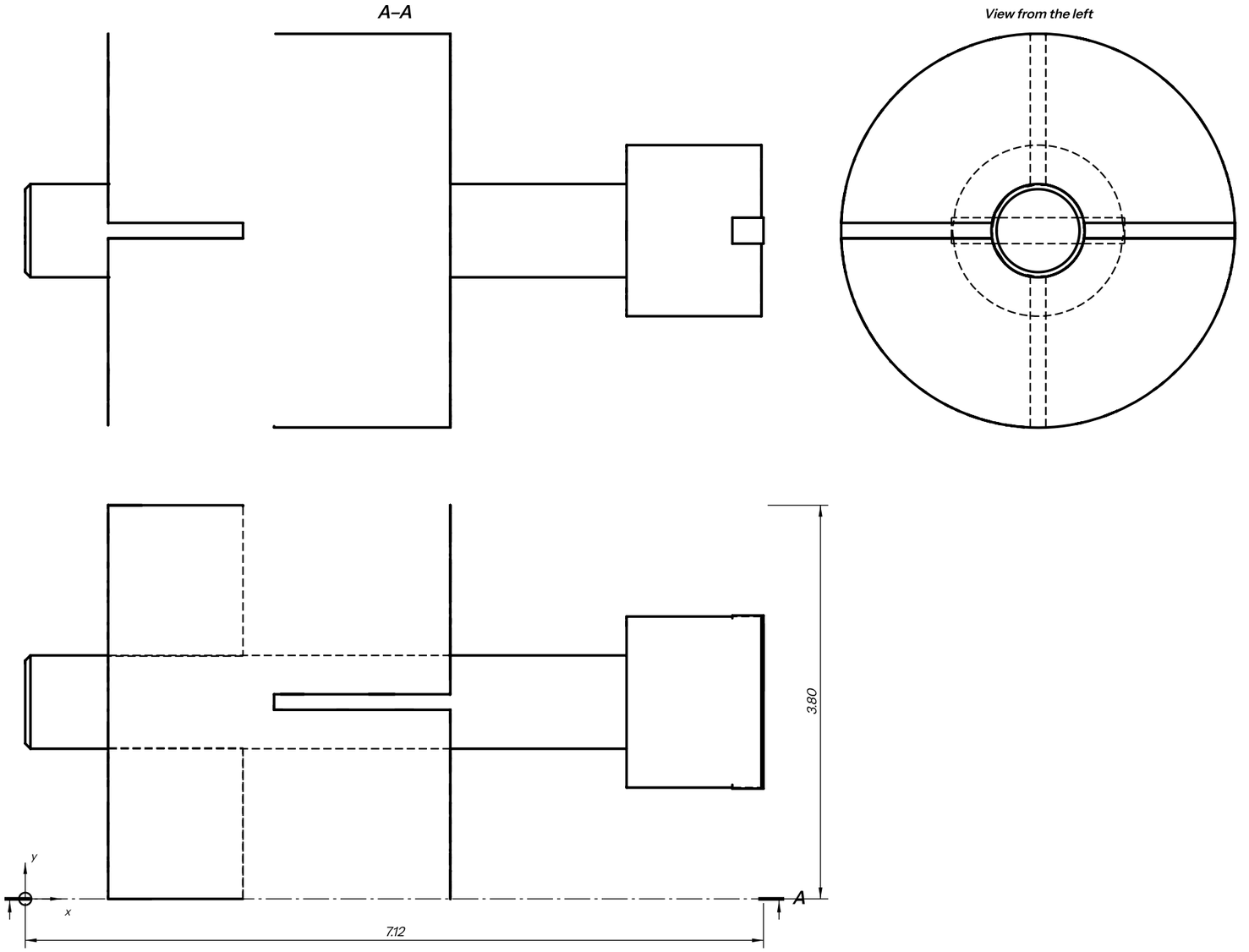
Parts list

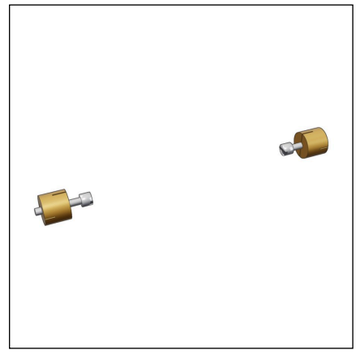
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 03 01-4	Masse (compensation mass)	4	not on the drawing		on 050301-1: 0.0199 mm
WN-12	Schraube (WN-12), the mass screw	4	not on the drawing		in 050301-4 (embed): 2.662 mm, least gap -0.7973

What it does (05 03 01-4). Round brass masses across the rim, which the rim carries in as it warms.

What it does (WN-12). Holds a mass on the rim: radially through its outer jaw, its tip on the rim's outer face (the 6MX's book, Fig. 5.62).

Turned about its axis; length 6.46, largest Ø 6.30. Diameters: ; bores Ø 1.00. Shoulders from the lower end: 0.00, 1.53, 2.00, 2.64, 6.46.





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***20:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 03 01-2***
Behandlung / Treatment			*Teil / Part* *balance.timing*
		Benennung / Name ***Timing screws and weights***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Timing screws and weights — sheet 2

A timing screw across the rim at each end of the arm, in a tapped hole Ø 0.9 × 0.16, carrying a brass weight: turned in or out, they set the rate.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 03 01-2	Zeitschraube (timing screw)	2	not on the drawing		in 050301-1 (embed): 0.902 mm, least gap -0.4433
05 03 01-3	Zeitmasse (timing weight)	2	not on the drawing		round 050301-2 (tap): 3.256 mm, least gap 0.0063

What it does (05 03 01-2). Across the rim at the arm's end in its tapped hole Ø 0.9 × 0.16, carrying the timing weight.

What it does (05 03 01-3). On the timing screw: turned in or out, sets the rate.

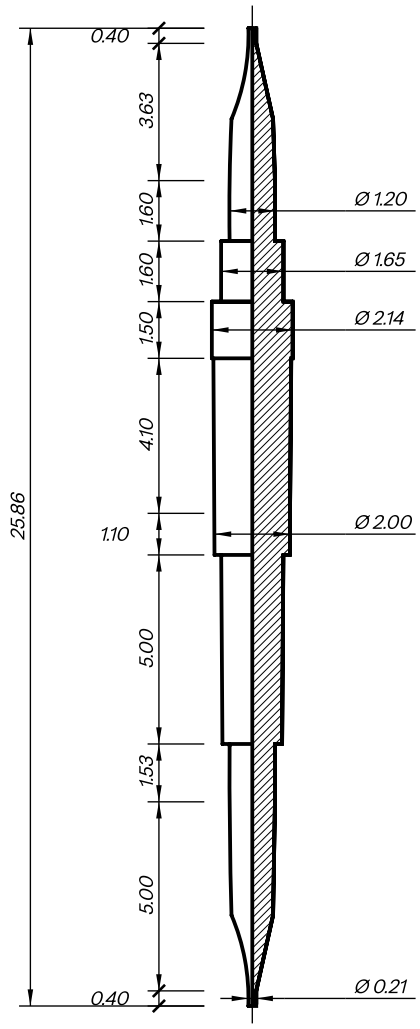
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.80.

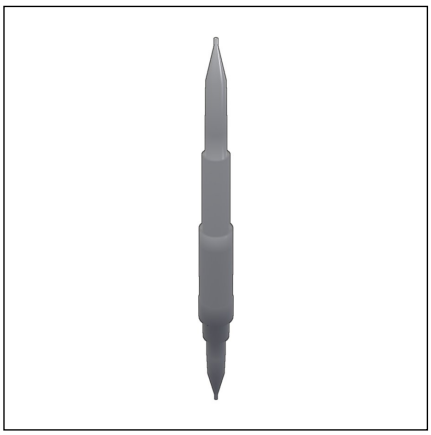
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	4.10, 0.47	4.10, 0.00		
Outline	2	line	4.10, 0.00	0.80, 0.00		
Outline	3	line	0.80, 0.00	0.80, 1.45		
Outline	4	line	0.80, 1.45	0.05, 1.45		
Outline	5	line	0.05, 1.45	0.00, 1.50		
Outline	6	line	0.00, 1.50	0.00, 2.30		
Outline	7	line	0.00, 2.30	0.05, 2.35		
Outline	8	line	0.05, 2.35	0.80, 2.35		
Outline	9	line	0.80, 2.35	0.80, 3.80		
Outline	10	line	0.80, 3.80	4.10, 3.80		
Outline	11	line	4.10, 3.80	4.10, 2.35		
Outline	12	line	4.10, 2.35	5.80, 2.35		
Outline	13	line	5.80, 2.35	5.80, 2.72		
Outline	14	line	5.80, 2.72	6.82, 2.72		
Outline	15	line	6.82, 2.72	6.82, 2.73		
Outline	16	line	6.82, 2.73	7.12, 2.73		
Outline	17	line	7.12, 2.73	7.12, 1.06		
Outline	18	line	7.12, 1.06	6.82, 1.06		
Outline	19	line	6.82, 1.06	6.82, 1.07		
Outline	20	line	6.82, 1.07	5.80, 1.07		
Outline	21	line	5.80, 1.07	5.80, 1.45		
Outline	22	line	5.80, 1.45	4.10, 1.45		

Outline	23	line	4.10, 1.45	4.10, 0.47		





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 02-1** *Teil / Part* *balance.staff*
Behandlung / Treatment „gehärtet, anlassen 240°“ (hardened, temper at 240°)		*Benennung / Name* ***Balance staff***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Balance staff — sheet 2

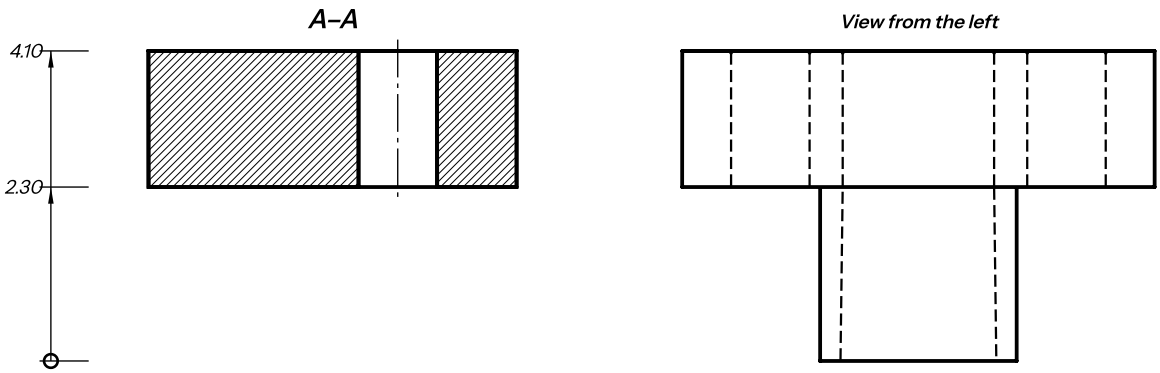
The hardened steel staff: its pivots run in the jewels in the two cocks; it carries the collet, the arm on the hub, and the rollers.

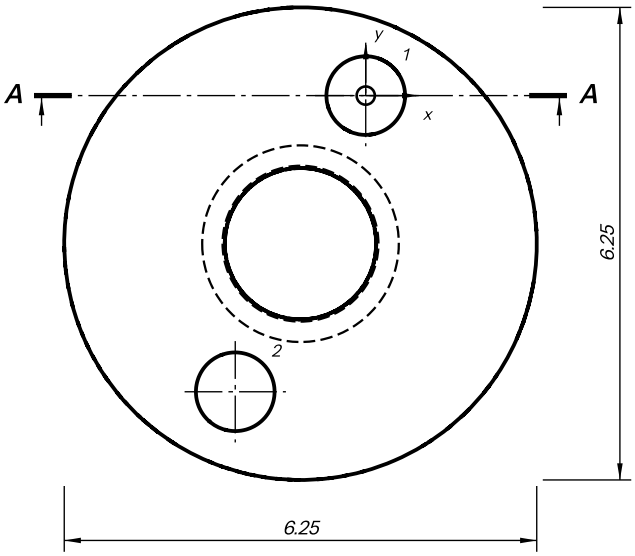
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 02-1	Unruhwelle (balance staff)	1	not on the drawing	„gehärtet, anlassen 240°“ (hardened, temper at 240°)	in 050407-1.bu (run): 0.321 mm, least gap 0.0163; in 050407-1.bl (run): 0.321 mm, least gap 0.0163; endshake 0.040 mm (050408-1.bl / 050408-1.bu)

What it does (05 04 02-1). The balance staff, its pivots in the jewels of the two cocks. As last revised 05 04 02 (29.5.46); as first drawn (the page's Variants) 05 04_5's staff of 4.4.41, which carries this number in pencil.

Turned about its axis; length 25.86, largest Ø 2.14. Diameters: Ø 0.21, Ø 1.20, Ø 1.65, Ø 2.00, Ø 2.14. Shoulders from the lower end: 0.00, 0.40, 5.40, 6.93, 11.93, 13.03, 17.13, 18.63, 20.23, 21.83, 25.46, 25.86.







Isometric view, not to scale

Werkstoff / Material „Ms." (brass)	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 02-2** *Teil / Part* *balance.hub*
Behandlung / Treatment		*Benennung / Name* ***Balance hub (Unruhbutzen)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Balance hub (Unruhbutzen) — sheet 2

The brass hub pressed on the staff, its flange carrying the arm.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 02-2	Unruhbutzen (balance hub, brass)	1	„Ms.“ (brass)		round 050402-1 (press): 4.176 mm, least gap -0.007

What it does (05 04 02-2). Pressed on the staff, its flange carries the arm. As first drawn (the page's Variants) 05 04_5's brass collar (040401-2), the arm riveted on its neck, with no screws.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 4.10.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.04 through (1.80 thick at 2.30–4.10)
2	-1.73	-3.92	Ø 1.04 through (1.80 thick at 2.30–4.10)

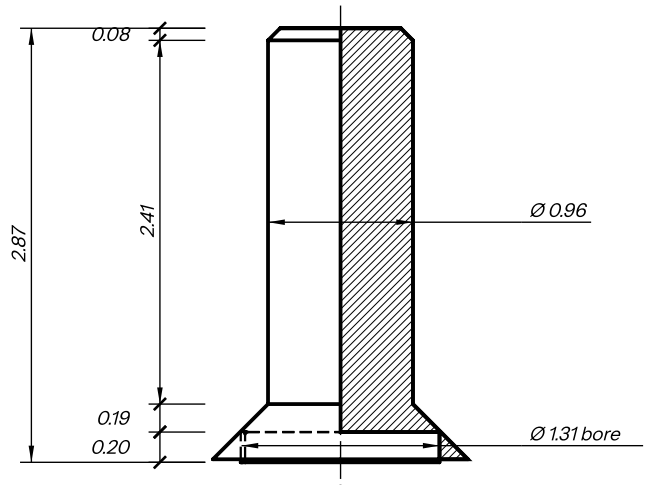
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			-0.86, -1.96	3.13
Cut-out 1	2	line	1.02, -4.46	-0.07, -2.99		
Cut-out 1	3	line	-0.07, -2.99	1.02, -4.46		
Cut-out 2	4	line	-0.67, -0.67	-0.28, -0.44		
Cut-out 2	5	line	-0.28, -0.44	-0.67, -0.67		
Cut-out 4	6	arc anticlockwise	-0.96, -0.96	-1.86, -1.99	-0.86, -1.96	1.00
Cut-out 4	7	arc anticlockwise	-1.86, -1.99	-1.72, -2.48	-0.90, -1.97	0.96
Cut-out 4	8	arc anticlockwise	-1.72, -2.48	-1.30, -2.86	-0.88, -1.98	0.97
Cut-out 4	9	arc anticlockwise	-1.30, -2.86	-0.88, -2.96	-0.86, -1.94	1.02
Cut-out 4	10	line	-0.88, -2.96	-0.73, -2.95		
Cut-out 4	11	arc anticlockwise	-0.73, -2.95	0.14, -1.97	-0.86, -1.96	1.00
Cut-out 4	12	line	0.14, -1.97	0.13, -1.83		
Cut-out 4	13	arc anticlockwise	0.13, -1.83	-0.83, -0.96	-0.86, -1.96	1.00
Cut-out 4	14	line	-0.83, -0.96	-0.96, -0.96		

Levels

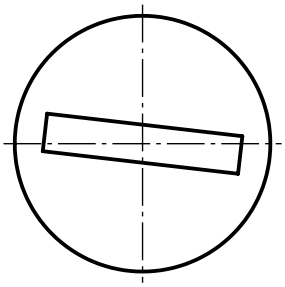
Faces in section A–A, from the bottom face: 2.30, 4.10.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **20:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-10** Teil / Part balance.screws
Behandlung / Treatment		Benennung / Name **Hub screws**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Hub screws — sheet 2

Two countersunk screws from the arm's top, through the arm into the hub's flange.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

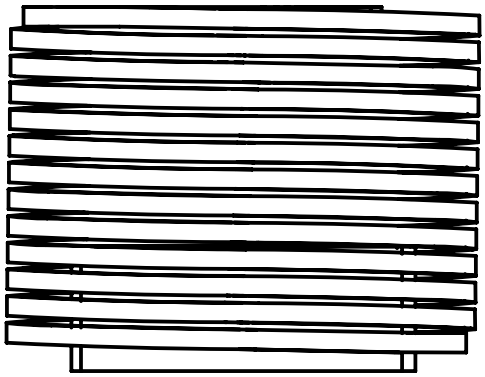
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-10	Schraube (WN-10), the hub's screws	[0, 2]	not on the drawing		in 050402-2 (tap): 1.75 mm, least gap 0.0396; in 050301-1 (clear): 1.118 mm, least gap 0.0045

What it does (WN-10). From under the hub's flange into the arm. Only on the 1946 hub: the 1941 collar has no holes for them.

Turned about its axis; length 2.87, largest Ø 1.69. Diameters: Ø 0.96; bores Ø 1.31. Shoulders from the lower end: 0.00, 0.20, 0.39, 2.79, 2.87.

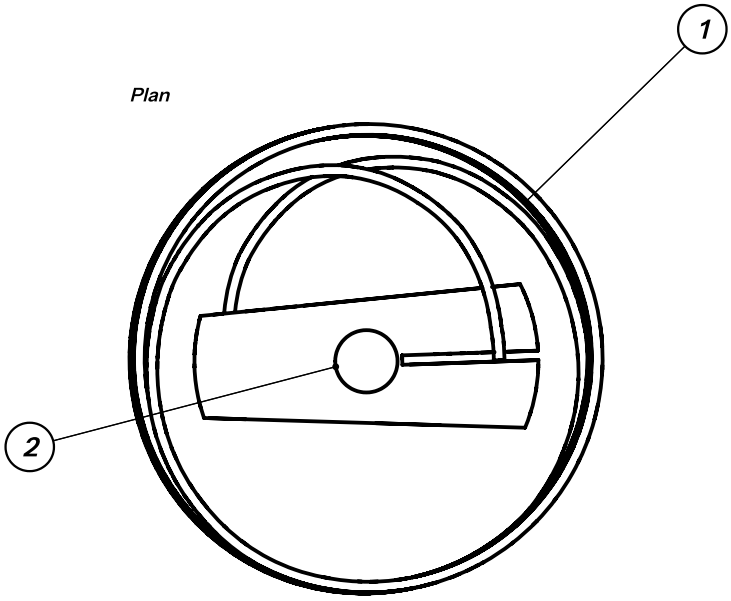
Front view





Isometric view, not to scale

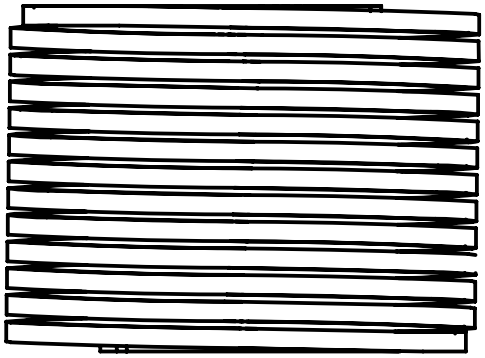
Plan



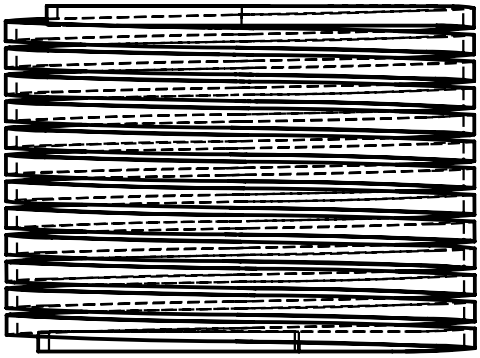
Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Hairspring collet (Spiralrolle)	05 03 02-2	„Messingblech 3,5 (overwritten
1	1	Hairspring	05 03 02-1	„Bandstahl 0,52×0,285 poliert“

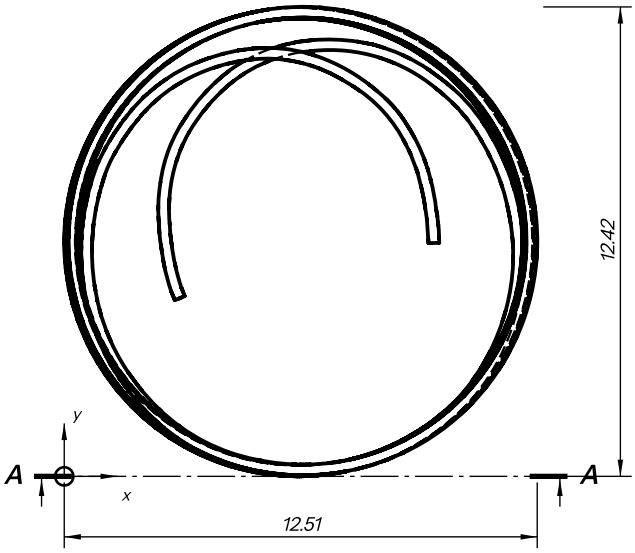
Werkstoff / Material „Bandstahl 0,52×0,285 poliert“ (strip steel, polished)	Maßstab / Scale **5:1**	***Wempe Einheits-Chronometer*** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 03 02-1** Teil / Part hairspring
Behandlung / Treatment „härten, dunkelblau anlassen“ (harden, temper to dark blue); blank (Rohmaß) 0,52 × 0,285 × 603		Benennung / Name ***Hairspring (Spiralfeder)***	
Blatt / Sheet **1 / 1**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

A–A



View from the left







Isometric view, not to scale

Werkstoff / Material „Bandstahl 0,52×0,285 poliert“ (strip steel, polished)	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 03 02-1** *Teil / Part* *hairspring.spring*
Behandlung / Treatment „härten, dunkelblau anlassen“ (harden, temper to dark blue); blank (Rohmaß) 0,52 × 0,285 × 603		*Benennung / Name* ***Hairspring***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Hairspring — sheet 2

The helical balance spring: a polished steel strip 0.52 × 0.285 wound in 13 turns, its two ends curved in to the centre on the drawn terminal curve, so the coils open and close concentrically about the staff. Its inner end is pinned in the brass collet on the staff, its outer end in the stud on the cock.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 03 02-1	Spiralfeder (helical hairspring)	1	„Bandstahl 0,52×0,285 poliert“ (strip steel, polished)	„härten, dunkelblau anlassen“ (harden, temper to dark blue); blank (Rohmaß) 0,52 × 0,285 × 603	not measured on the model: see its function, The balance spring: 13 turns between its two terminal curves.

Tolerance (05 03 02-1). The hairspring: as drawn its period is 0.4937 s (1098 s a day fast), so it is matched to the balance, not made to size: 0.5 s wants the strip 0.2826 mm thick at 206 GPa (0.2808–0.2854 over 200–210 GPa); a µm of thickness is 455 s a day, the timing weights' reach 293 s a day. Vibrate it against the balance and choose or grind it (the bench's test).

Test on the bench (05 03 02-1). The balance and its hairspring vibrated, the spring pinned to its collet and stud, its curves bent to 05 03_7, before it goes in: 2 oscillations a second (7,200 an hour), or slow by up to 293 s a day: the timing weights stand against their screws' heads as drawn and can only be screwed in, 5.0 turns of 58.7 s a day each (wempe_tolerances.js; their size estimated). The drawn spring on the weighed balance (2,501 g·mm²) is 18 min a day fast (0.4937 s): the strip wants 0.2826 mm, not 0.285 (0.2808–0.2854 over the steel's 200–210 GPa). No section tolerance holds the period (a µm of the strip is 455 s a day, the coil's ±0.05 ±173): fast, grind the strip thinner or choose a weaker spring; slow past the weights' reach, a stronger one.

What it does (05 03 02-1). The balance spring: 13 turns between its two terminal curves.

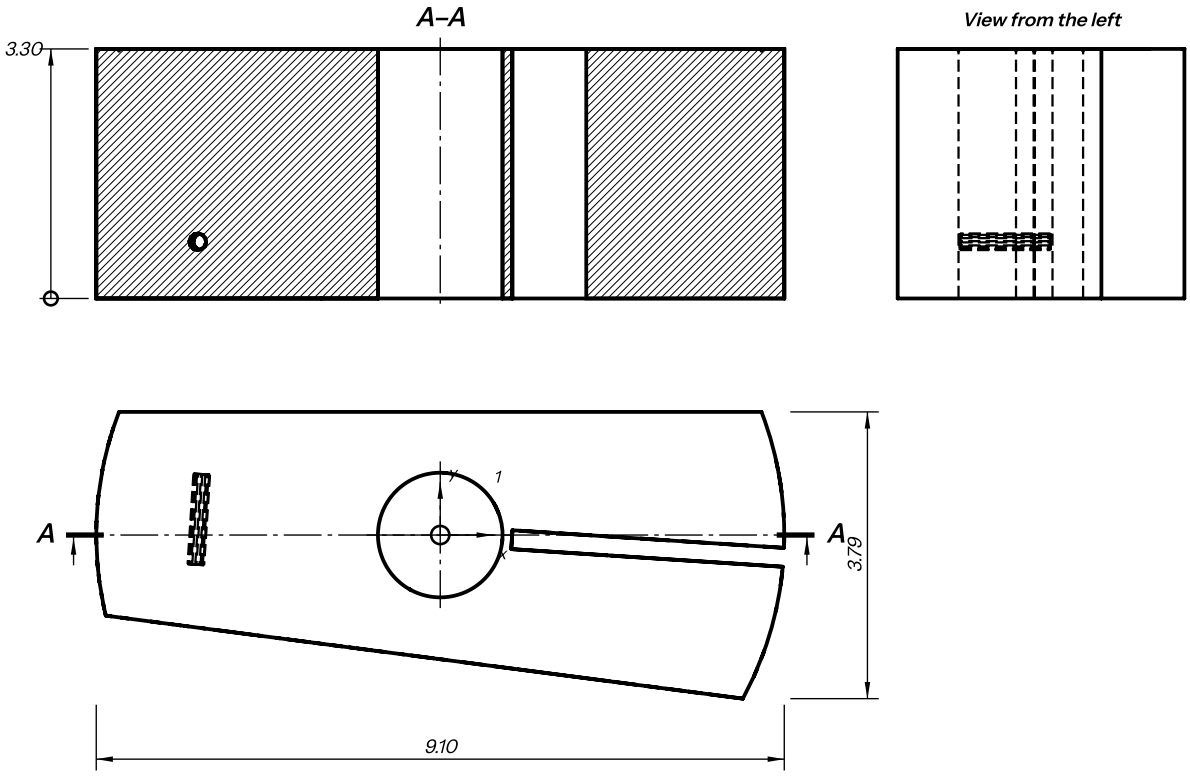
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 9.14.

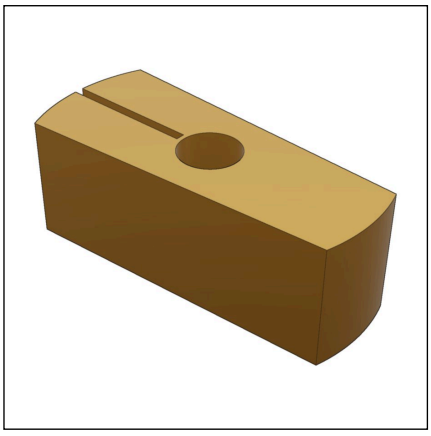
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.00, 6.32	4.51, 12.16	6.19, 6.20	6.19
Outline	2	line	4.51, 12.16	4.65, 12.20		
Outline	3	line	4.65, 12.20	5.17, 12.31		
Outline	4	line	5.17, 12.31	5.65, 12.38		
Outline	5	arc clockwise	5.65, 12.38	8.51, 0.43	6.32, 6.22	6.20
Outline	6	line	8.51, 0.43	8.08, 0.28		
Outline	7	line	8.08, 0.28	7.60, 0.16		
Outline	8	line	7.60, 0.16	7.06, 0.06		
Outline	9	line	7.06, 0.06	6.62, 0.01		
Outline	10	arc clockwise	6.62, 0.01	0.00, 6.32	6.19, 6.19	6.19
Cut-out 1	11	line	0.12, 4.99	0.13, 4.99		
Cut-out 1	12	line	0.13, 4.99	0.12, 4.99		
Cut-out 2	13	line	0.00, 6.32	0.01, 6.32		
Cut-out 2	14	line	0.01, 6.32	0.00, 6.32		
Cut-out 3	15	arc anticlockwise	5.99, 12.11	5.23, 12.03	6.15, 6.89	5.21
Cut-out 3	16	arc anticlockwise	5.23, 12.03	4.80, 11.94	6.04, 7.19	4.91
Cut-out 3	17	line	4.80, 11.94	4.40, 11.81		
Cut-out 3	18	arc anticlockwise	4.40, 11.81	0.51, 5.08	6.31, 6.22	5.91
Cut-out 3	19	arc clockwise	0.51, 5.08	1.10, 8.70	6.53, 5.98	6.08

Cut-out 3	20	arc clockwise	1.10, 8.70	5.52, 11.32	5.39, 6.53	4.80
Cut-out 3	21	arc clockwise	5.52, 11.32	10.31, 10.23	7.00, 6.78	4.78
Cut-out 3	22	arc clockwise	10.31, 10.23	12.09, 6.73	6.16, 5.95	5.97
Cut-out 3	23	arc anticlockwise	12.09, 6.73	5.99, 12.11	6.20, 6.20	5.91
Cut-out 4	24	line	11.32, 9.16	11.32, 9.16		
Cut-out 4	25	line	11.32, 9.16	11.32, 9.16		
Cut-out 5	26	arc anticlockwise	6.07, 10.98	5.53, 11.03	5.64, 9.39	1.65
Cut-out 5	27	arc anticlockwise	5.53, 11.03	3.30, 9.10	7.08, 7.01	4.31
Cut-out 5	28	arc anticlockwise	3.30, 9.10	2.79, 6.66	7.01, 7.06	4.24
Cut-out 5	29	arc anticlockwise	2.79, 6.66	3.19, 4.77	8.68, 6.88	5.89
Cut-out 5	30	line	3.19, 4.77	2.92, 4.66		
Cut-out 5	31	arc clockwise	2.92, 4.66	2.51, 6.64	8.67, 6.88	6.17
Cut-out 5	32	arc clockwise	2.51, 6.64	3.88, 10.31	6.91, 7.08	4.42
Cut-out 5	33	arc clockwise	3.88, 10.31	4.80, 11.00	6.65, 7.62	3.86
Cut-out 5	34	arc anticlockwise	4.80, 11.00	1.35, 8.57	5.38, 6.53	4.51
Cut-out 5	35	arc anticlockwise	1.35, 8.57	2.21, 2.09	6.55, 5.96	5.82
Cut-out 5	36	arc anticlockwise	2.21, 2.09	5.95, 0.32	6.16, 5.64	5.32
Cut-out 5	37	arc anticlockwise	5.95, 0.32	8.26, 0.66	6.33, 5.83	5.52
Cut-out 5	38	arc anticlockwise	8.26, 0.66	11.78, 4.83	6.47, 5.73	5.38
Cut-out 5	39	arc anticlockwise	11.78, 4.83	10.41, 9.74	5.91, 5.84	5.96
Cut-out 5	40	arc anticlockwise	10.41, 9.74	7.72, 11.21	6.89, 6.57	4.73
Cut-out 5	41	arc anticlockwise	7.72, 11.21	6.29, 11.22	6.99, 6.28	4.99
Cut-out 5	42	arc clockwise	6.29, 11.22	8.97, 9.44	5.12, 6.53	4.84
Cut-out 5	43	arc clockwise	8.97, 9.44	9.83, 7.08	4.72, 6.56	5.14
Cut-out 5	44	arc clockwise	9.83, 7.08	9.92, 6.18	4.88, 6.17	5.04
Cut-out 5	45	line	9.92, 6.18	9.63, 6.17		
Cut-out 5	46	arc anticlockwise	9.63, 6.17	9.55, 7.04	4.85, 6.17	4.79
Cut-out 5	47	arc anticlockwise	9.55, 7.04	8.36, 9.74	4.94, 6.63	4.63
Cut-out 5	48	arc anticlockwise	8.36, 9.74	6.07, 10.98	5.11, 6.52	4.57





Isometric view, not to scale

Werkstoff / Material „Messingblech 3,5 (overwritten, likely 3,5) Din 1751 Ms 58"	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 03 02-2** *Teil / Part* *hairspring.collet*
Behandlung / Treatment surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)		*Benennung / Name* ***Hairspring collet (Spiralrolle)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Hairspring collet (Spiralrolle) — sheet 2

The split brass collet on the staff, with the hole across its end where the spring's inner end is pinned (Spiralstift 05 03 02-4).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 03 02-2	Spiralrolle (hairspring collet)	1	„Messingblech 3,5 (overwritten, likely 3,5) Din 1751 Ms 58“	surface ∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished)	round 050402-1 (press): 3.373 mm, least gap 0.0052
05 03 02-4	Spiralstift (hairspring pin, at the collet)	1	not on the drawing		in 050302-2 (embed): 1.22 mm, least gap -0.1076

What it does (05 03 02-2). On the staff's 1:70 seat; the spring's inner end pinned in the hole across its end.

What it does (05 03 02-4). Wedges the spring's inner end in the collet's hole.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.30.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.65 through

Contour in the plan

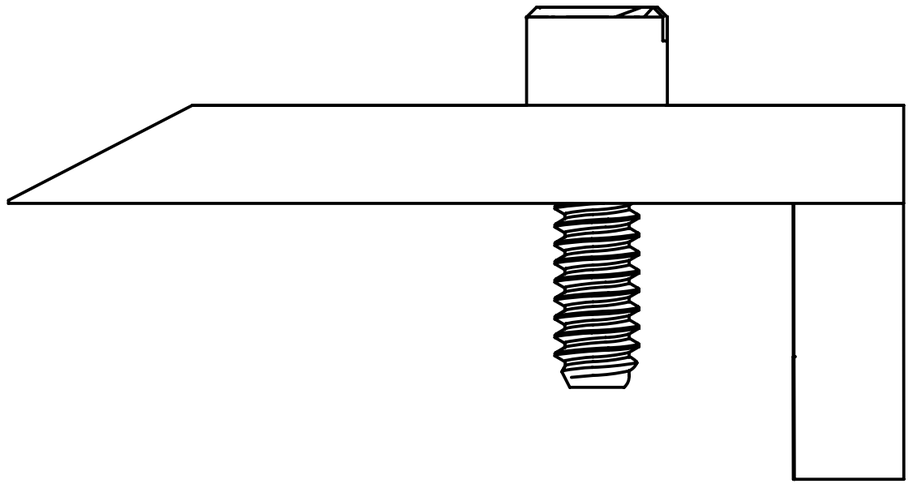
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-4.54, -0.28	-4.25, 1.63	0.00, 0.00	4.55
Outline	2	line	-4.25, 1.63	4.25, 1.63		
Outline	3	arc clockwise	4.25, 1.63	4.55, -0.17	0.01, 0.00	4.54
Outline	4	line	4.55, -0.17	0.96, 0.06		
Outline	5	line	0.96, 0.06	0.94, -0.19		
Outline	6	line	0.94, -0.19	4.53, -0.42		
Outline	7	arc clockwise	4.53, -0.42	4.00, -2.16	0.01, 0.00	4.54
Outline	8	line	4.00, -2.16	-4.42, -1.07		
Outline	9	arc clockwise	-4.42, -1.07	-4.54, -0.28	0.00, 0.00	4.55

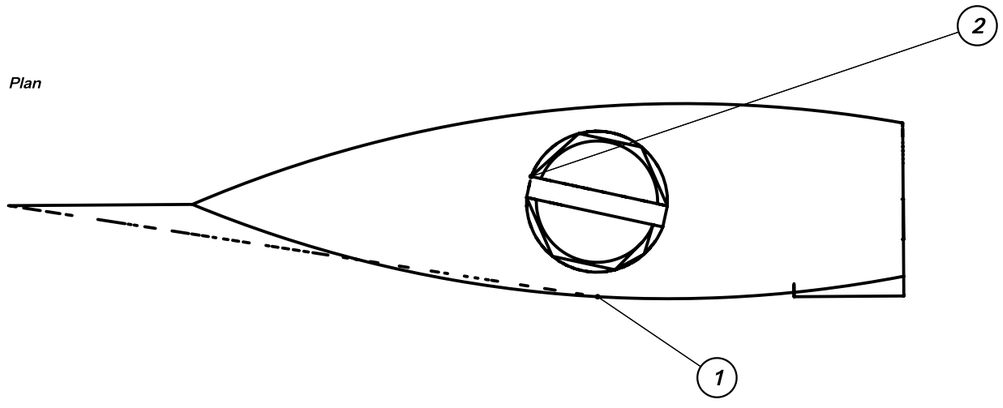
Levels

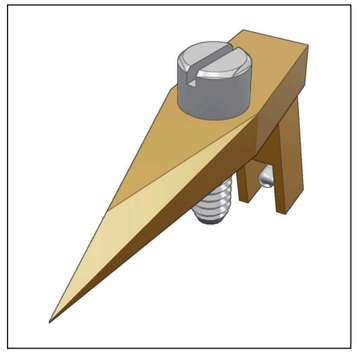
Faces in section A-A, from the bottom face: 3.30.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Stud screw	WN-11	not on the drawing
1	1	Stud	05 03 02-3	„Flachmessing 15×7 Din 1759 Ms

Werkstoff / Material: „Flachmessing 15×7 Din 1759 Ms 58"

Maßstab / Scale: **10:1**

Behandlung / Treatment: surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)

Blatt / Sheet: 1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

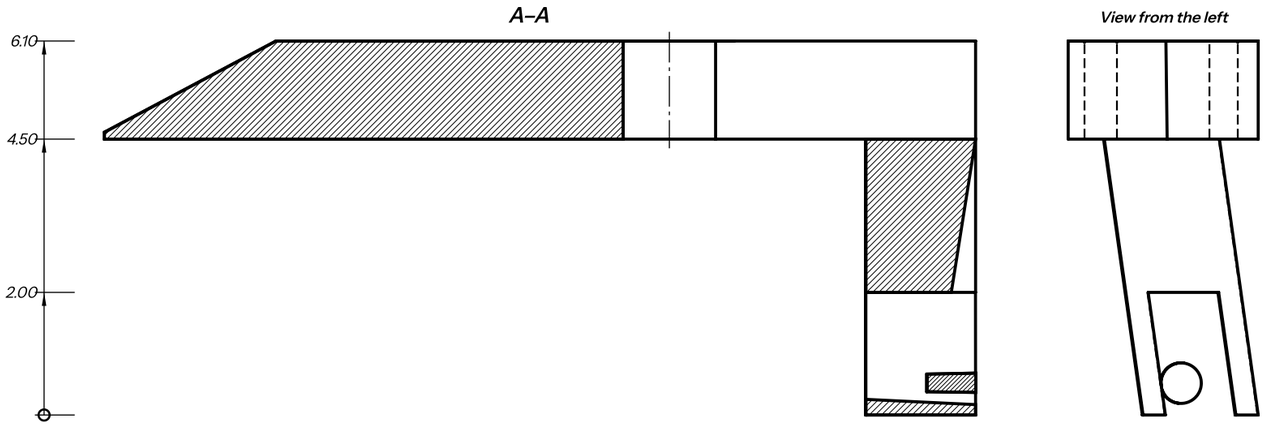
Zeichnung Nr. / Drawing no.: **05 03 02-3**

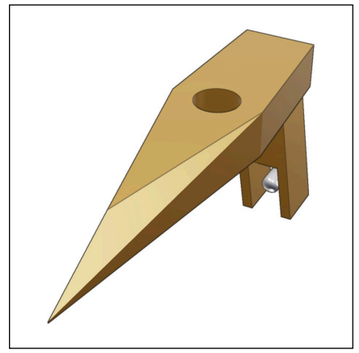
Teil / Part: stud

Benennung / Name: **Hairspring stud (Klötzchen)**

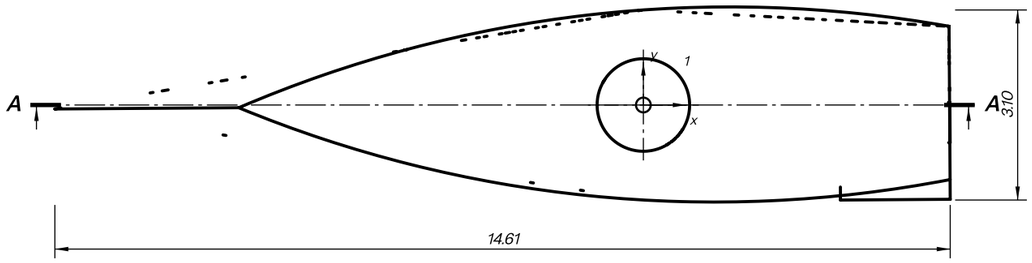
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale



Hidden lines are left out of the plan (too many to read); the section and the hole table give the depths.

Werkstoff / Material „Flachmessing 15×7 Din 1759 Ms 58"	Maßstab / Scale **10:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 03 02-3**
Behandlung / Treatment surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)			Teil / Part stud.stud
Blatt / Sheet **1 / 2**		Benennung / Name **Stud**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Stud — sheet 2

A small brass L screwed on the balance cock: its upright drops through the slot in the cock's arm and holds the hairspring's outer end, pinned in a slot at its foot. Slid along the cock's slot it sets the spring's pinning point.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 03 02-3	Klötzchen (hairspring stud)	1	„Flachmessing 15×7 Din 1759 Ms 58“	surface ∇∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	on 050803-1: 0 mm
05 03 02-4	Stift (hairspring pin, at the stud)	1	not on the drawing		on 050302-3: 0.0048 mm

What it does (05 03 02-3). On the cock, its upright through the cock's slot, holding the spring's outer end.

What it does (05 03 02-4). Wedges the spring's outer end in the stud's slot.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 6.10.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.51 through (1.60 thick at 4.50–6.10)

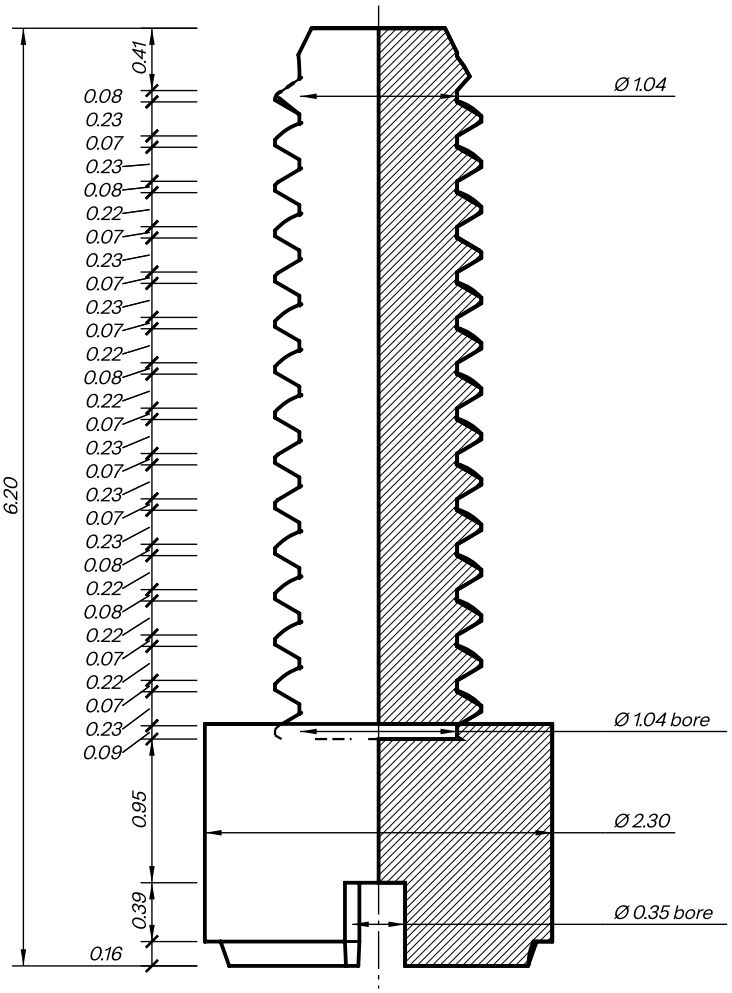
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	5.00, -1.22	5.01, -1.54		
Outline	2	line	5.01, -1.54	3.22, -1.55		
Outline	3	line	3.22, -1.55	3.21, -1.34		
Outline	4	arc clockwise	3.21, -1.34	-9.60, -0.06	-0.35, 27.94	29.49
Outline	5	arc clockwise	-9.60, -0.06	4.99, 1.28	0.71, -31.99	33.55
Outline	6	line	4.99, 1.28	4.99, 0.04		
Outline	7	line	4.99, 0.04	5.00, 0.04		
Outline	8	line	5.00, 0.04	5.00, -0.61		
Outline	9	line	5.00, -0.61	5.00, -0.61		
Outline	10	line	5.00, -0.61	5.00, -1.22		

Levels

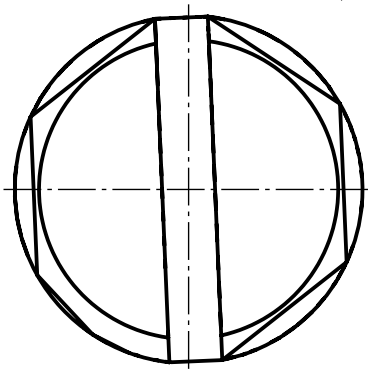
Faces in section A-A, from the bottom face: 2.00, 4.50, 6.10.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-11** *Teil / Part* *stud.screw*
Behandlung / Treatment		*Benennung / Name* ***Stud screw***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Stud screw — sheet 2

The screw holding the stud to the cock.

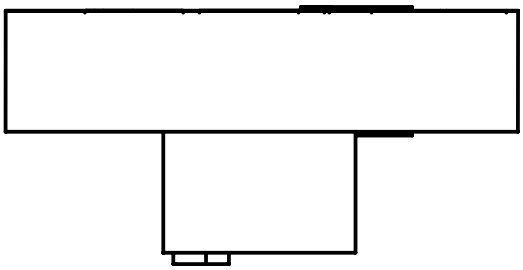
Parts list

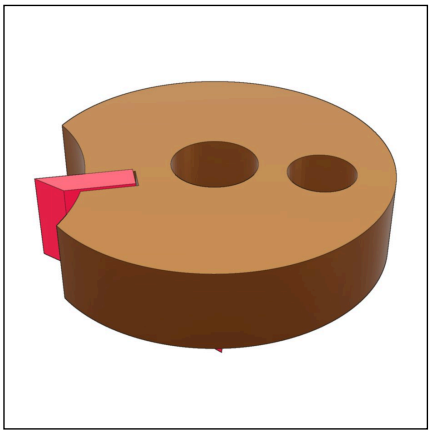
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-11	Schraube (WN-11), the stud's screw	1	not on the drawing		in 050302-3 (clear): 1.637 mm, least gap 0.0563; in 050803-1 (tap): 2.381 mm, least gap 0.0217

What it does (WN-11). Holds the stud to the cock.

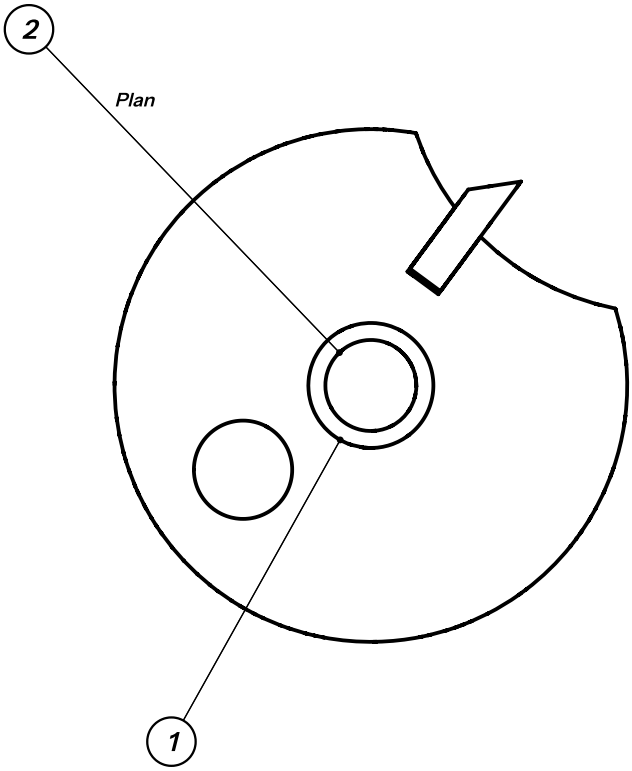
Turned about its axis; length 6.20, largest Ø 2.30. Diameters: Ø 1.04, Ø 2.30; bores Ø 0.35, Ø 1.04. Shoulders from the lower end: 0.00, 0.16, 0.55, 1.50, 1.59, 1.81, 1.89, 2.11, 2.19, 2.41, 2.49, 2.71, 2.79, 3.01, 3.09, 3.31, 3.39, 3.61, 3.69, 3.91, 3.99, 4.21, 4.29, 4.51, 4.59, 4.81, 4.89, 5.11, 5.19, 5.41, 5.49, 5.71, 5.79, 6.20.

Front view





Isometric view, not to scale



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Discharge roller and stone	05 04 04-1	not on the drawing
1	1	Impulse roller and stone	02 04 02	„Kupfer Beryllium, Saphir“ (be

Werkstoff / Material
„Kupfer Beryllium, Saphir“ (beryllium copper; sapphire)

Maßstab / Scale
10:1

Behandlung / Treatment
none on the drawing

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

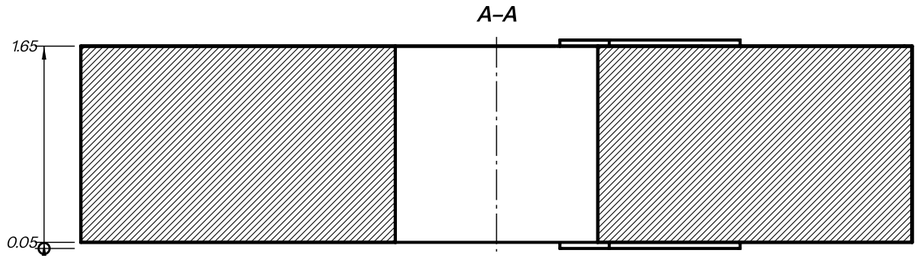
Model v3.35.00 · 2026-10-26

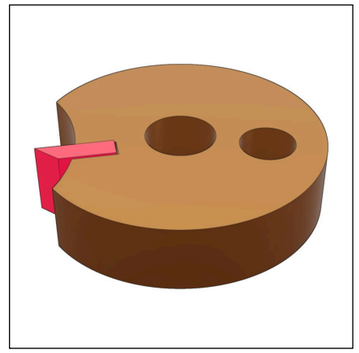
Zeichnung Nr. / Drawing no.
02 04 02

Teil / Part
rollers

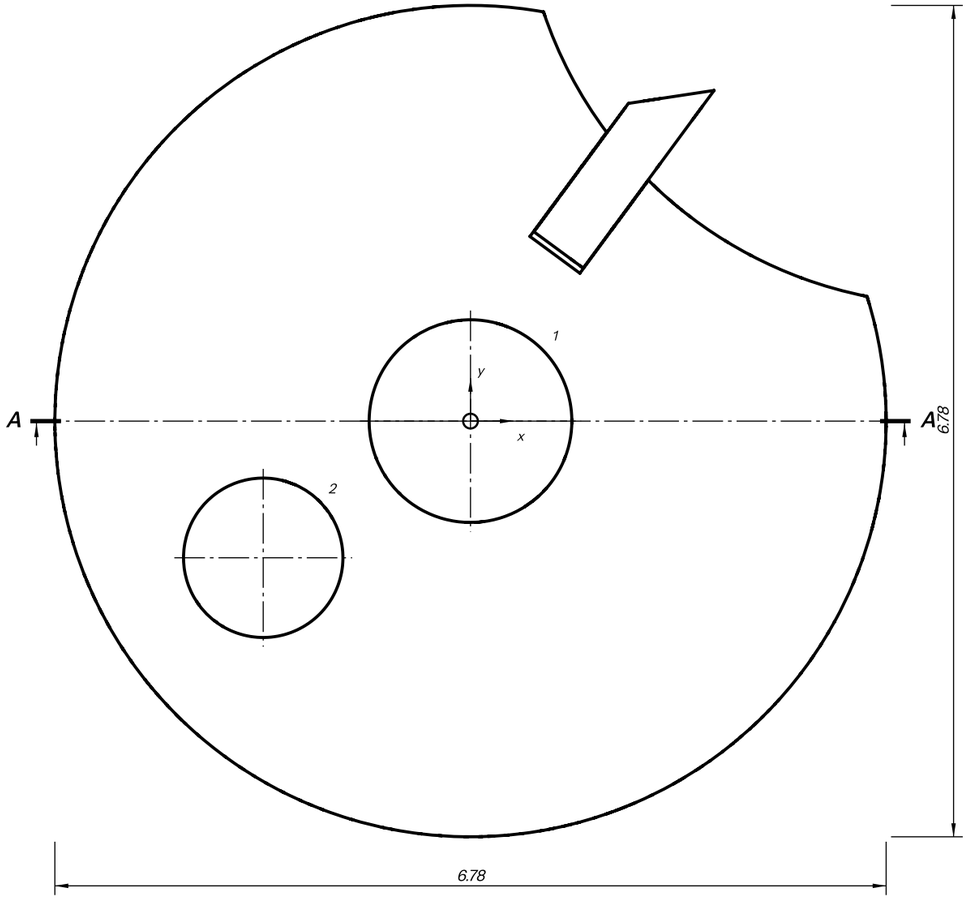
Benennung / Name
Impulse and discharge rollers (Heberolle, Auslöserolle)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale



Werkstoff / Material „Kupfer Beryllium, Saphir" (beryllium copper; sapphire)	Maßstab / Scale **20:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **02 04 02**
Behandlung / Treatment none on the drawing			Teil / Part rollers.impulse
Blatt / Sheet **1 / 2**		Benennung / Name **Impulse roller and stone**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Impulse roller and stone — sheet 2

The beryllium-copper impulse roller and its sapphire impulse stone, set in shellac.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
02 04 02	Heberolle (impulse roller, beryllium copper; 05 04 03-1 in the coloured section)	1	„Kupfer Beryllium, Saphir“ (beryllium copper; sapphire)	none on the drawing	round 050402-1 (press): 1.606 mm, least gap 0.0044
05 04 03-2	Hebestein (impulse stone, sapphire)	1	not on the drawing		not measured on the model: see its function, Receives the escape wheel's impulse.

What it does (02 04 02). Carries the impulse stone, which the escape wheel pushes.

What it does (05 04 03-2). Receives the escape wheel's impulse.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.70.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.65 through (1.60 thick at 0.05-1.65)
2	-1.69	-1.11	Ø 1.30 through (1.60 thick at 0.05-1.65)

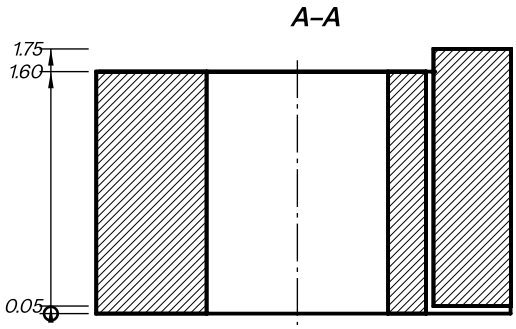
Contour in the plan

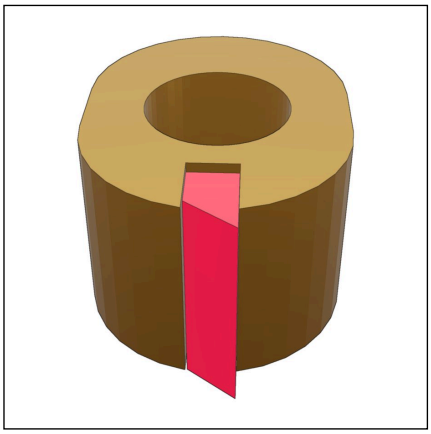
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-3.12, -1.34	0.60, 3.34	0.00, 0.00	3.39
Outline	2	arc anticlockwise	0.60, 3.34	1.11, 2.36	3.91, 4.45	3.50
Outline	3	line	1.11, 2.36	0.48, 1.51		
Outline	4	line	0.48, 1.51	0.89, 1.21		
Outline	5	line	0.89, 1.21	1.45, 1.96		
Outline	6	arc anticlockwise	1.45, 1.96	3.23, 1.02	3.91, 4.45	3.50
Outline	7	arc clockwise	3.23, 1.02	-3.12, -1.34	0.00, 0.00	3.39
Outline	8	line	1.99, 2.70	0.92, 1.25		
Outline	9	line	0.92, 1.25	0.52, 1.54		
Outline	10	line	0.52, 1.54	1.29, 2.59		
Outline	11	line	1.29, 2.59	1.99, 2.70		

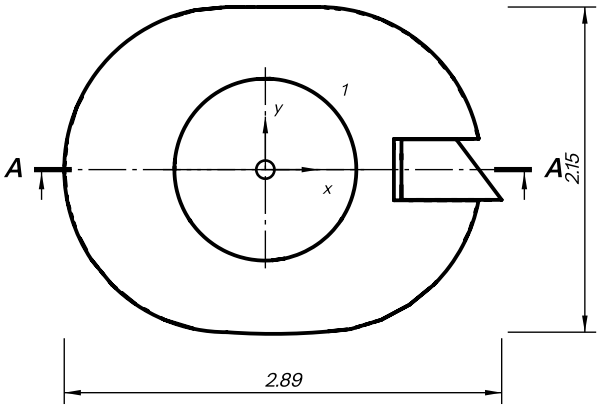
Levels

Faces in section A-A, from the bottom face: 0.05, 1.65.





Isometric view, not to scale



Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 04-1** *Teil / Part* *rollers.discharge*
Behandlung / Treatment		*Benennung / Name* ***Discharge roller and stone***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Discharge roller and stone — sheet 2

The small collar and its discharge stone that work the gold spring.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 04-1	Auslöserolle (discharge roller)	1	not on the drawing		round 050402-1 (press): 1.606 mm, least gap 0.0039
05 04 04-2	Auslösestein (discharge stone)	1	not on the drawing		not measured on the model: see its function, Lifts the detent through the gold spring on one swing; passes it on the other.

What it does (05 04 04-1). Carries the discharge stone.

What it does (05 04 04-2). Lifts the detent through the gold spring on one swing; passes it on the other.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.75.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.20 through (1.60 thick at 0.00–1.60)

Contour in the plan

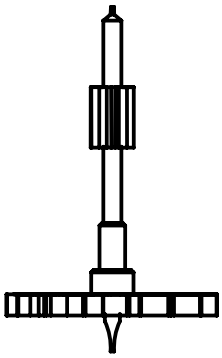
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

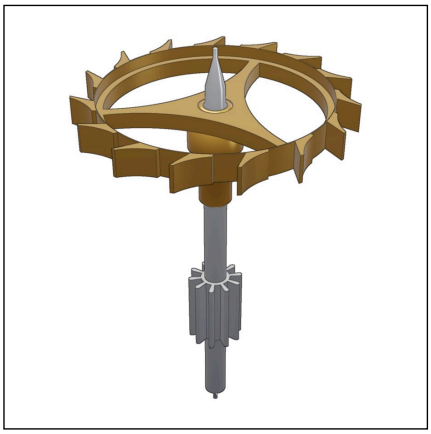
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.90, 0.20	1.26, 0.20		
Outline	2	line	1.26, 0.20	1.56, -0.20		
Outline	3	line	1.56, -0.20	0.90, -0.20		
Outline	4	line	0.90, -0.20	0.90, 0.20		
Outline	5	arc clockwise	-1.31, -0.21	-0.25, 1.08	-0.25, 0.00	1.08
Outline	6	arc clockwise	-0.25, 1.08	0.56, 1.05	0.05, -3.10	4.18
Outline	7	arc clockwise	0.56, 1.05	1.41, 0.20	0.36, 0.00	1.07
Outline	8	line	1.41, 0.20	0.85, 0.20		
Outline	9	line	0.85, 0.20	0.85, -0.20		
Outline	10	line	0.85, -0.20	1.41, -0.20		
Outline	11	arc clockwise	1.41, -0.20	0.35, -1.07	0.36, 0.00	1.07
Outline	12	arc clockwise	0.35, -1.07	-0.46, -1.05	0.05, 3.10	4.18
Outline	13	arc clockwise	-0.46, -1.05	-1.31, -0.21	-0.26, 0.00	1.07

Levels

Faces in section A–A, from the bottom face: 0.05, 1.60, 1.75.

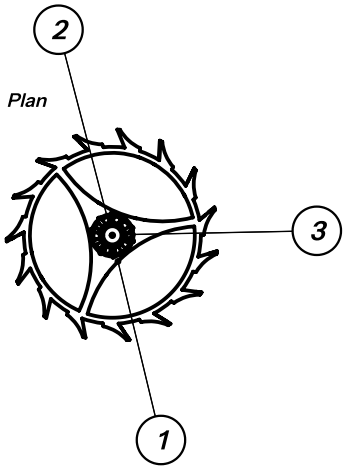
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	1	Escape wheel collet (Gangradbutzen)	05 04 05-3	not on the drawing
2	1	Escape pinion and arbor	05 04 05-2	„Präz. Rundstahl 3,4 Din 175“
1	1	Escape wheel	05 04 05-1	„Messingblech 1,6 Din 1751 Ms

Werkstoff / Material
„Messingblech 1,6 Din 1751 Ms 63 federhart“ (brass sheet 1.6, spring-hard)

Maßstab / Scale
2:1

Behandlung / Treatment
none on the drawing

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

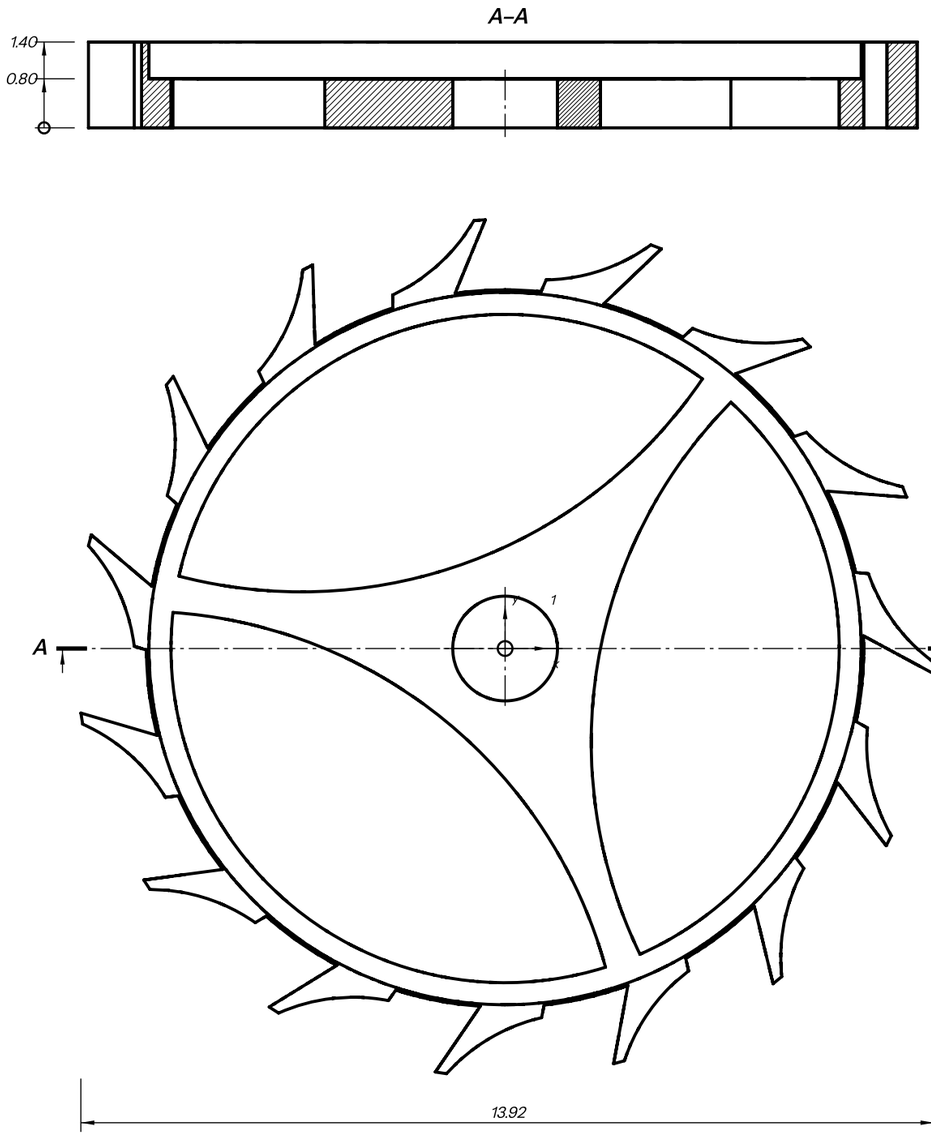
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 04 05-1

Teil / Part
escapeWheel

Benennung / Name
Escape wheel (Gangrad)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Messingblech 1.6 Din 1751 Ms 63 federhart" (brass sheet 1.6, spring-hard)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 04 05-1*** *Teil / Part* *escapeWheel.wheel*
Behandlung / Treatment *none on the drawing*		*Benennung / Name* ***Escape wheel***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Escape wheel — sheet 2

The 15-tooth escape wheel, just under the top plate, held by the locking stone between impulses. Released, it gives the balance's impulse stone a push through about 20 degrees and comes to rest on the next tooth: a tooth every half second, so its pinion turns the seconds wheel once a minute.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 05-1	Gangrad (escape wheel, 15 teeth)	1	„Messingblech 1,6 Din 1751 Ms 63 federhart“ (brass sheet 1,6, spring-hard)	none on the drawing	round 050405-3 (press): 0.792 mm, least gap 0.0065

What it does (05 04 05-1). Locked on the detent's stone between impulses, it gives the balance its impulse.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.40.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.71 through (0.80 thick at 0.00–0.80)

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

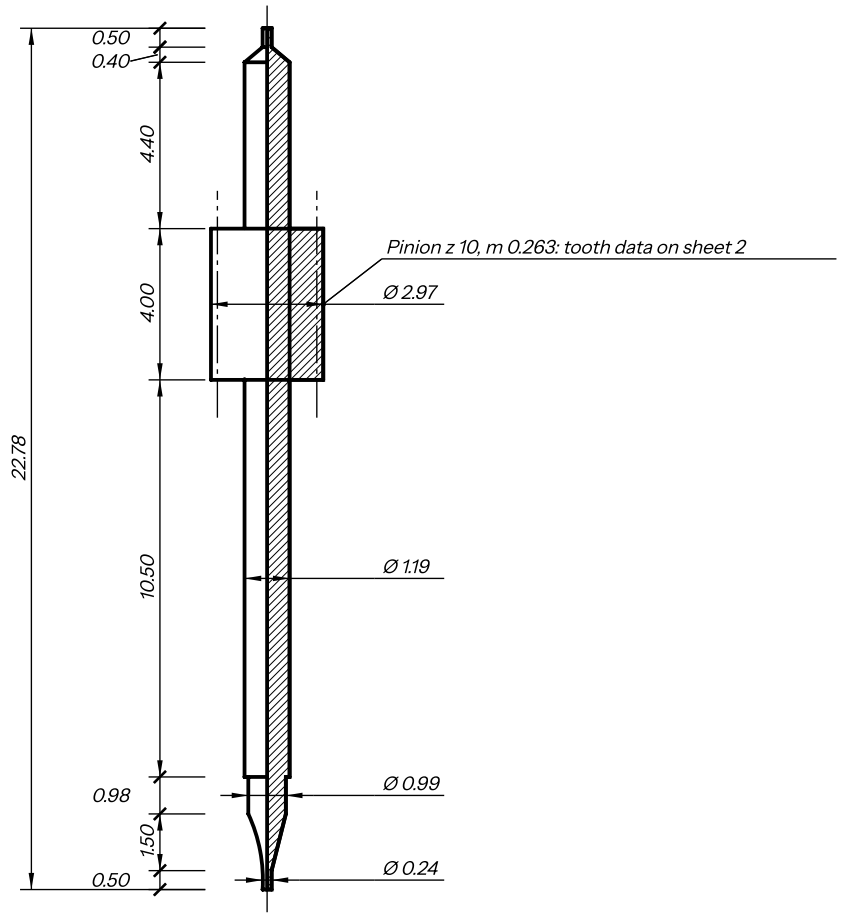
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc anticlockwise	-6.55, -1.41	-6.89, -1.23	-8.11, -3.97	3.00
Outline	2	line	-6.89, -1.23	-6.92, -1.05		
Outline	3	line	-6.92, -1.05	-5.68, -1.42		
Outline	4	arc clockwise	-5.68, -1.42	-5.85, -0.03	0.00, 0.00	5.85
Outline	5	line	-5.85, -0.03	-6.05, 0.04		
Outline	6	arc anticlockwise	-6.05, 0.04	-6.80, 1.68	-9.03, -0.33	3.00
Outline	7	line	-6.80, 1.68	-6.75, 1.85		
Outline	8	line	-6.75, 1.85	-5.76, 1.01		
Outline	9	arc clockwise	-5.76, 1.01	-5.35, 2.36	0.00, 0.00	5.85
Outline	10	line	-5.35, 2.36	-5.51, 2.49		
Outline	11	arc anticlockwise	-5.51, 2.49	-5.53, 4.30	-8.38, 3.37	3.00
Outline	12	line	-5.53, 4.30	-5.41, 4.44		
Outline	13	line	-5.41, 4.44	-4.85, 3.27		
Outline	14	arc clockwise	-4.85, 3.27	-3.93, 4.33	0.00, 0.00	5.85
Outline	15	line	-3.93, 4.33	-4.02, 4.52		
Outline	16	arc anticlockwise	-4.02, 4.52	-3.30, 6.17	-6.29, 6.49	3.00
Outline	17	line	-3.30, 6.17	-3.14, 6.26		
Outline	18	line	-3.14, 6.26	-3.10, 4.96		
Outline	19	arc clockwise	-3.10, 4.96	-1.83, 5.56	0.00, 0.00	5.85
Outline	20	line	-1.83, 5.56	-1.83, 5.77		
Outline	21	arc anticlockwise	-1.83, 5.77	-0.51, 6.98	-3.10, 8.48	3.00
Outline	22	line	-0.51, 6.98	-0.32, 6.99		
Outline	23	line	-0.32, 6.99	-0.82, 5.79		
Outline	24	arc clockwise	-0.82, 5.79	0.59, 5.82	0.00, 0.00	5.85

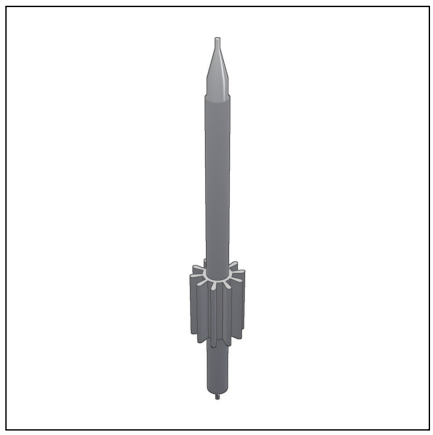
Outline	25	line	0.59, 5.82	0.67, 6.01		
Outline	26	arc anticlockwise	0.67, 6.01	2.38, 6.58	0.62, 9.01	3.00
Outline	27	line	2.38, 6.58	2.55, 6.52		
Outline	28	line	2.55, 6.52	1.61, 5.62		
Outline	29	arc clockwise	1.61, 5.62	2.90, 5.08	0.00, 0.00	5.85
Outline	30	line	2.90, 5.08	3.06, 5.22		
Outline	31	arc anticlockwise	3.06, 5.22	4.85, 5.05	4.23, 7.98	3.00
Outline	32	line	4.85, 5.05	4.98, 4.92		
Outline	33	line	4.98, 4.92	3.76, 4.48		
Outline	34	arc clockwise	3.76, 4.48	4.72, 3.46	0.00, 0.00	5.85
Outline	35	line	4.72, 3.46	4.92, 3.53		
Outline	36	arc anticlockwise	4.92, 3.53	6.48, 2.64	7.11, 5.57	3.00
Outline	37	line	6.48, 2.64	6.55, 2.47		
Outline	38	line	6.55, 2.47	5.26, 2.57		
Outline	39	arc clockwise	5.26, 2.57	5.72, 1.24	0.00, 0.00	5.85
Outline	40	line	5.72, 1.24	5.93, 1.22		
Outline	41	arc anticlockwise	5.93, 1.22	7.00, -0.23	8.76, 2.20	3.00
Outline	42	line	7.00, -0.23	6.99, -0.41		
Outline	43	line	6.99, -0.41	5.85, 0.21		
Outline	44	arc clockwise	5.85, 0.21	5.73, -1.19	0.00, 0.00	5.85
Outline	45	line	5.73, -1.19	5.91, -1.29		
Outline	46	arc anticlockwise	5.91, -1.29	6.30, -3.05	8.90, -1.55	3.00
Outline	47	line	6.30, -3.05	6.22, -3.22		
Outline	48	line	6.22, -3.22	5.42, -2.19		
Outline	49	arc clockwise	5.42, -2.19	4.75, -3.42	0.00, 0.00	5.85
Outline	50	line	4.75, -3.42	4.87, -3.59		
Outline	51	arc anticlockwise	4.87, -3.59	4.51, -5.35	7.50, -5.04	3.00
Outline	52	line	4.51, -5.35	4.37, -5.47		
Outline	53	line	4.37, -5.47	4.07, -4.21		
Outline	54	arc clockwise	4.07, -4.21	2.95, -5.05	0.00, 0.00	5.85
Outline	55	line	2.95, -5.05	2.99, -5.26		
Outline	56	arc anticlockwise	2.99, -5.26	1.95, -6.72	4.80, -7.65	3.00
Outline	57	line	1.95, -6.72	1.77, -6.77		
Outline	58	line	1.77, -6.77	2.00, -5.50		
Outline	59	arc clockwise	2.00, -5.50	0.64, -5.82	0.00, 0.00	5.85
Outline	60	line	0.64, -5.82	0.60, -6.02		
Outline	61	arc anticlockwise	0.60, -6.02	-0.96, -6.93	1.27, -8.94	3.00
Outline	62	line	-0.96, -6.93	-1.14, -6.91		
Outline	63	line	-1.14, -6.91	-0.41, -5.84		
Outline	64	arc clockwise	-0.41, -5.84	-1.78, -5.57	0.00, 0.00	5.85
Outline	65	line	-1.78, -5.57	-1.91, -5.74		
Outline	66	arc anticlockwise	-1.91, -5.74	-3.69, -5.95	-2.48, -8.69	3.00
Outline	67	line	-3.69, -5.95	-3.85, -5.85		
Outline	68	line	-3.85, -5.85	-2.74, -5.17		
Outline	69	arc clockwise	-2.74, -5.17	-3.90, -4.36	0.00, 0.00	5.85
Outline	70	line	-3.90, -4.36	-4.08, -4.47		
Outline	71	arc anticlockwise	-4.08, -4.47	-5.79, -3.93	-5.80, -6.93	3.00
Outline	72	line	-5.79, -3.93	-5.89, -3.78		

Outline	73	line	-5.89, -3.78	-4.61, -3.60		
Outline	74	arc clockwise	-4.61, -3.60	-5.33, -2.40	0.00, 0.00	5.85
Outline	75	line	-5.33, -2.40	-5.54, -2.43		
Outline	76	arc anticlockwise	-5.54, -2.43	-6.55, -1.41	-8.11, -3.97	3.00
Cut-out 1	77	arc anticlockwise	5.72, -0.05	5.72, -0.05	2.86, -0.01	2.86
Cut-out 1	78	arc anticlockwise	5.72, -0.05	5.72, -0.05	2.86, -0.01	2.86
Cut-out 2	79	arc anticlockwise	3.27, 3.57	2.20, -4.99	9.30, -1.53	7.90
Cut-out 2	80	arc anticlockwise	2.20, -4.99	3.68, 4.02	0.00, 0.00	5.45
Cut-out 2	81	arc anticlockwise	3.68, 4.02	3.27, 3.57	9.30, -1.53	7.90
Cut-out 3	82	arc anticlockwise	-0.16, 5.45	-5.32, 1.18	0.00, 0.00	5.45
Cut-out 3	83	arc anticlockwise	-5.32, 1.18	3.22, 4.40	-3.32, 8.82	7.90
Cut-out 3	84	arc anticlockwise	3.22, 4.40	-0.16, 5.45	0.00, 0.00	5.45
Cut-out 4	85	arc anticlockwise	0.04, -2.18	-5.42, 0.59	-5.98, -7.29	7.90
Cut-out 4	86	arc anticlockwise	-5.42, 0.59	1.64, -5.20	0.00, 0.00	5.45
Cut-out 4	87	arc anticlockwise	1.64, -5.20	0.04, -2.18	-5.98, -7.29	7.90

Levels

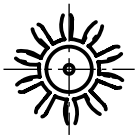
Faces in section A-A, from the bottom face: 0.80, 1.40.





Isometric view, not to scale

View from above



Werkstoff / Material „Präz. Rundstahl 3,4 Din 175" (precision round steel ⌀3,4)	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 04 05-2*** *Teil / Part* *escapeWheel.pinion*
Behandlung / Treatment none on the drawing		*Benennung / Name* ***Escape pinion and arbor***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Escape pinion and arbor — sheet 2

The steel arbor with its pinion of 10 leaves, which the seconds wheel drives.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 05-2	Gangtrieb (escape pinion, z 10, m 0.263, and arbor)	1	„Präz. Rundstahl 3,4 Din 175“ (precision round steel ø3,4)	none on the drawing	in 050406-1.eu (run): 0.141 mm, least gap 0.0189; in 050406-1.el (run): 0.141 mm, least gap 0.0121; endshake 0.060 mm (050408-1.el / 050408-1.eu)

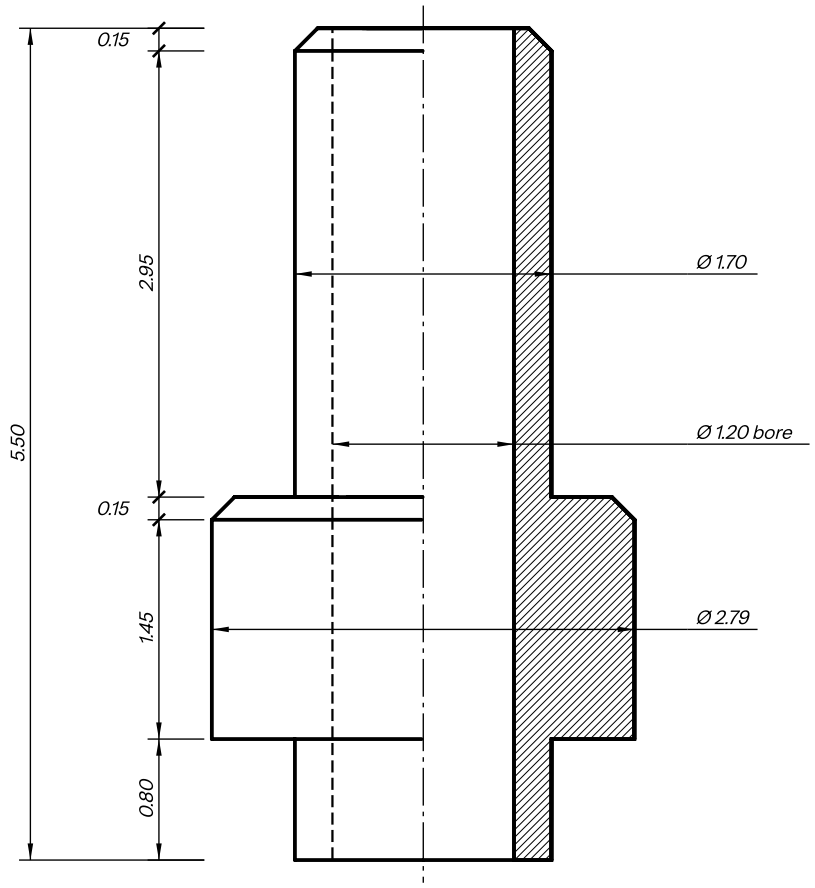
What it does (05 04 05-2). The arbor and pinion; driven by the seconds wheel. As first drawn (the page's Variants) 05 04_7.1's arbor of 4.4.41 (050408 bis 10, renumbered 05 04 05 on 05 04_7.2), the wheel 1.4 nearer the top plate.

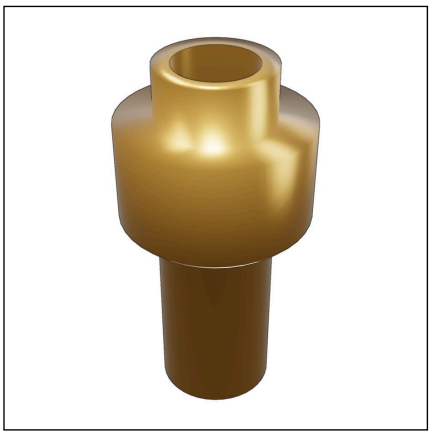
Turned about its axis; length 22.78, largest Ø 2.87. Diameters: Ø 0.24, Ø 0.99, Ø 1.19, Ø 2.97. Shoulders from the lower end: 0.00, 0.50, 2.00, 2.98, 13.48, 17.48, 21.88, 22.28, 22.78.

Leaves 13.48–17.48 from the lower end, pitch Ø 2.63; their form is the view from the end, drawn at the sheet's scale.

Teeth

Number	Module	Tip Ø	Root Ø
10	0.263	2.97	





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 05-3** *Teil / Part* *escapeWheel.collet*
Behandlung / Treatment		*Benennung / Name* ***Escape wheel collet (Gangradbutzen)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Escape wheel collet (Gangradbutzen) — sheet 2

The brass collet the wheel is pressed on.

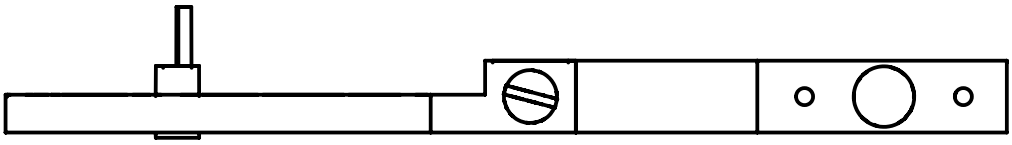
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 05-3	Gangradbutzen (escape wheel collet; 05 04 10 on 05 04_7.2)	1	not on the drawing		round 050405-2 (press): 5.518 mm, least gap 0.0048

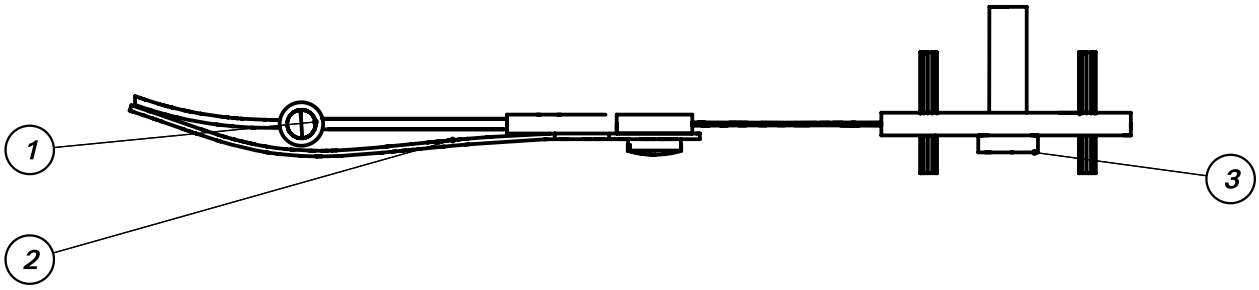
What it does (05 04 05-3). The brass collet the wheel is pressed on. As first drawn 05 04_7.1's: a cone Ø 3.0, then a pipe Ø 1.8 to 5 over the wheel.

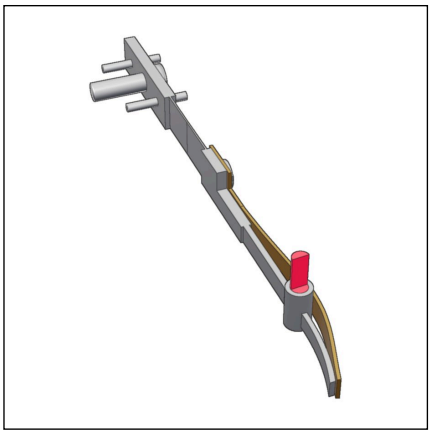
Turned about its axis; length 5.50, largest Ø 2.80. Diameters: Ø 1.70, Ø 2.79; bores Ø 1.20. Shoulders from the lower end: 0.00, 0.80, 2.25, 2.40, 5.35, 5.50.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	2	Detent foot screw and pins	WN-205	not on the drawing
2	1	Gold spring (Goldfeder)	05 04 01-2	„Goldblech 333 federhart“ (gol
1	1	Detent with its locking stone	05 04 01-1	„Stahl“

Werkstoff / Material
„Stahl“

Maßstab / Scale
5:1

Behandlung / Treatment
none on the drawing

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

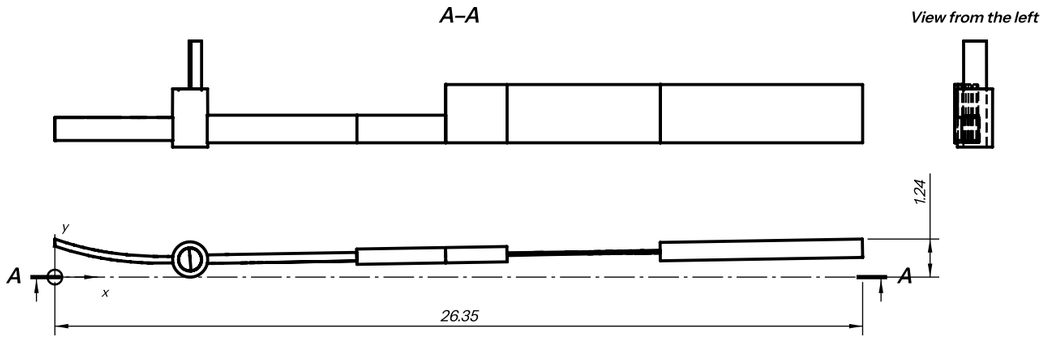
Model v3.35.00 · 2026-10-26

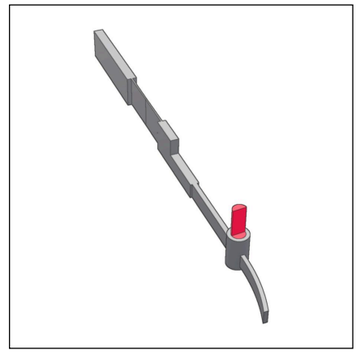
Zeichnung Nr. / Drawing no.
05 04 01-1

Teil / Part
detent

Benennung / Name
Spring detent (Gangfeder) and gold spring

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Stahl"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 04 01-1***
Behandlung / Treatment *none on the drawing*			*Teil / Part* *detent.detent*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Detent with its locking stone***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Detent with its locking stone — sheet 2

The detent: a steel blade on a thin spring, clamped by its foot to the block. The locking stone in its pipe holds a tooth of the escape wheel. On one swing the discharge stone pushes the gold spring against the detent's horn and lifts the detent through a quarter of a millimetre, freeing the wheel; on the other it only bends the gold spring aside.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 01-1	Gangfeder (spring detent, drawing 1436)	1	„Stahl“	none on the drawing	on 050401-4: 0 mm
05 04 01-3	Ruhestein (locking stone)	1	not on the drawing		not measured on the model: see its function, Locks each tooth of the escape wheel.

Tolerance (05 04 01-1). The detent spring (1436's 0.07, 0.09 at its ends): any thickness to 0.193 mm keeps the swing in the band; in time (its fall back before the next tooth) 0.063–0.100 mm as the uniform equivalent of the drawn 0.079. Its preload against the banking 0.8 g, at least 0.64 g (a knock) and under 0.85 g (past it the stone catches the escaping tooth's back).

Test on the bench (05 04 01-1). The detent rings, the detent made with its gold spring on, before it goes in: 93 Hz with the spring as drawn (0.070 mm, 1436's "0,07"); 80–134 Hz for a spring 0.063–0.100 mm thick, half to twice the drawn stiffness, over which the detent is home before the next tooth (dynamics.js --model wempe). The pitch goes nearly as the thickness to the 1.5 (0.050 mm 58 Hz, 0.063 mm 80 Hz, 0.060 mm 75 Hz, 0.070 mm 93 Hz, 0.080 mm 109 Hz, 0.100 mm 134 Hz, 0.090 mm 123 Hz: wempe_tripping.js's frame clamped at the foot), so it measures the spring's section, which no figure on 1436 settles (its "0,09" would ring 123 Hz). The gold spring rings near 213 Hz: the detent's own is the lower.

What it does (05 04 01-1). Holds the escape wheel on its locking stone; lifted on one swing by the discharge stone through the gold spring.

Tolerance (05 04 01-3). The depth of lock: 0.235 mm as drawn (05 04_1); it unlocks to 0.285 and holds against the stone's ringing down to 0.089 (wempe_tolerances.js, the build's ring 0.044).

What it does (05 04 01-3). Locks each tooth of the escape wheel.

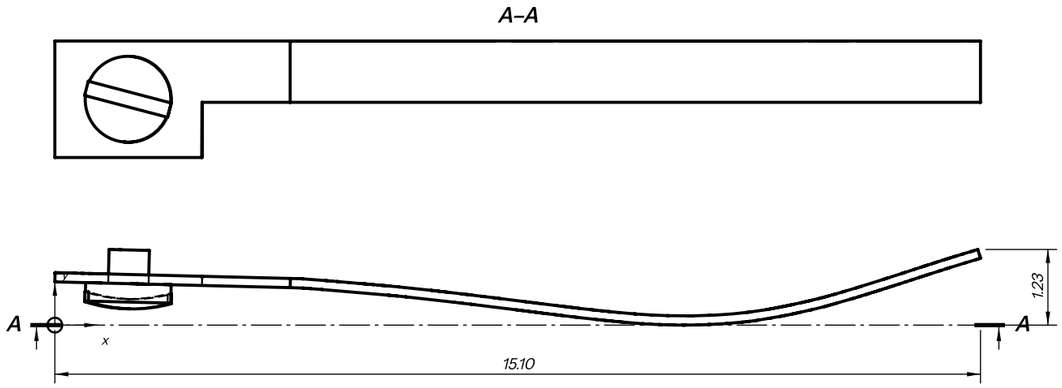
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.46.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	19.74, 1.13	26.34, 1.24		
Outline	2	line	26.34, 1.24	26.35, 0.64		
Outline	3	line	26.35, 0.64	19.75, 0.53		
Outline	4	line	19.75, 0.53	19.74, 1.13		
Outline	5	line	17.25, 0.75	14.75, 0.70		
Outline	6	line	14.75, 0.70	14.75, 0.79		
Outline	7	arc anticlockwise	14.75, 0.79	19.75, 0.88	12.03, 313.23	312.45
Outline	8	line	19.75, 0.88	19.75, 0.79		
Outline	9	line	19.75, 0.79	17.25, 0.75		
Outline	10	arc clockwise	1.99, 0.55	0.00, 1.01	3.29, 10.70	10.24
Outline	11	line	0.00, 1.01	0.00, 1.24		
Outline	12	arc anticlockwise	0.00, 1.24	3.85, 0.68	3.44, 11.40	10.73
Outline	13	line	3.85, 0.68	3.88, 0.79		
Outline	14	arc clockwise	3.88, 0.79	4.97, 0.72	4.41, 0.58	0.57
Outline	15	line	4.97, 0.72	9.85, 0.81		

Outline	16	line	9.85, 0.81	9.85, 0.92		
Outline	17	line	9.85, 0.92	14.75, 1.00		
Outline	18	line	14.75, 1.00	14.75, 0.50		
Outline	19	line	14.75, 0.50	9.85, 0.42		
Outline	20	line	9.85, 0.42	9.85, 0.53		
Outline	21	line	9.85, 0.53	4.97, 0.44		
Outline	22	line	4.97, 0.44	4.93, 0.32		
Outline	23	arc clockwise	4.93, 0.32	3.85, 0.46	4.41, 0.57	0.57
Outline	24	arc clockwise	3.85, 0.46	1.99, 0.55	3.52, 12.66	12.21
Outline	25	arc clockwise	4.06, 0.70	4.38, 0.95	4.41, 0.58	0.37
Outline	26	line	4.38, 0.95	4.48, 0.94		
Outline	27	line	4.48, 0.94	4.58, 0.91		
Outline	28	line	4.58, 0.91	4.66, 0.86		
Outline	29	line	4.66, 0.86	4.73, 0.77		
Outline	30	line	4.73, 0.77	4.77, 0.67		
Outline	31	line	4.77, 0.67	4.79, 0.56		
Outline	32	line	4.79, 0.56	4.77, 0.46		
Outline	33	line	4.77, 0.46	4.72, 0.37		
Outline	34	line	4.72, 0.37	4.66, 0.29		
Outline	35	line	4.66, 0.29	4.58, 0.24		
Outline	36	line	4.58, 0.24	4.47, 0.21		
Outline	37	line	4.47, 0.21	4.36, 0.20		
Outline	38	arc clockwise	4.36, 0.20	4.06, 0.70	4.41, 0.58	0.38





Isometric view, not to scale

Werkstoff / Material *„Goldblech 333 federhart" (gold sheet, 333 fine (8 ct), spring-hard)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 04 01-2*** *Teil / Part* *detent.gold*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*		*Benennung / Name* ***Gold spring (Goldfeder)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Gold spring (Goldfeder) — sheet 2

The gold passing spring, screwed to the detent, resting on its horn.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 01-2	Goldfeder (gold passing spring)	1	„Goldblech 333 federhart“ (gold sheet, 333 fine (8 ct), spring-hard)	surface ∇∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished)	on 050401-1: 0 mm
WN-14	Schraube (WN-14), the gold spring's screw	1	not on the drawing		in 050401-1 (embed): 0.489 mm, least gap -0.3282

Tolerance (05 04 01-2). The gold spring: its stiffness hardly moves the swing (to 22.2 times the drawn, 0.422 mm); it decides the lock's window instead, so make it to the build's thinned bend (its card).

What it does (05 04 01-2). Screwed to the detent, resting on its horn: the discharge stone lifts the detent through it on one swing and bends it aside on the other.

What it does (WN-14). Holds the gold spring to the detent's square.

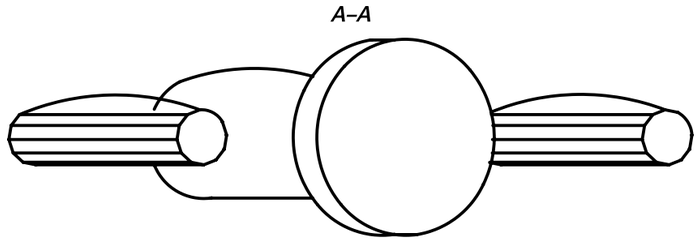
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.90.

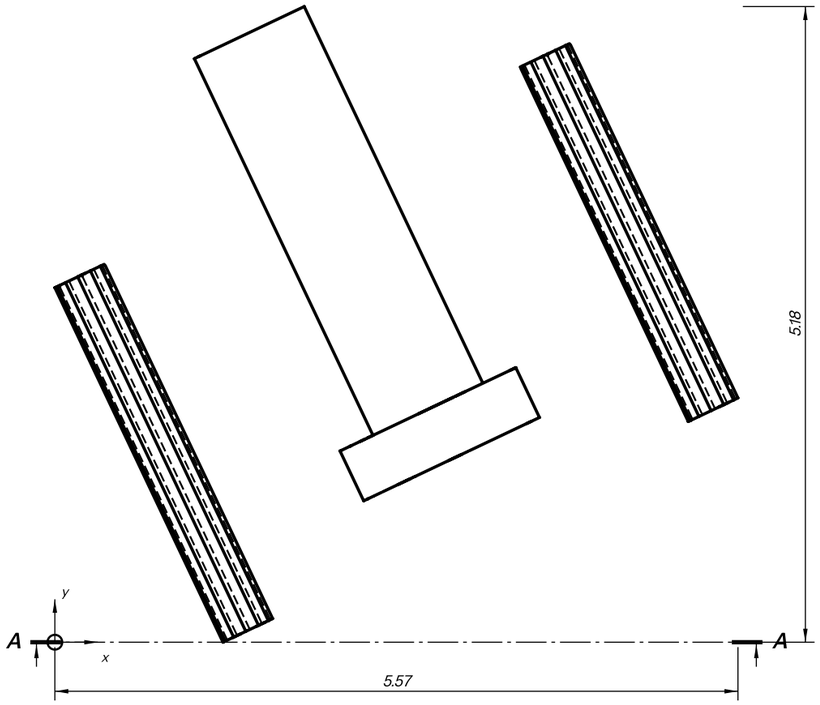
Contour in the plan

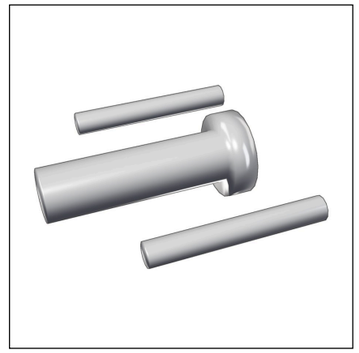
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.50, 0.54	0.50, 0.69		
Outline	2	line	0.50, 0.69	0.00, 0.71		
Outline	3	line	0.00, 0.71	0.00, 0.86		
Outline	4	line	0.00, 0.86	0.87, 0.83		
Outline	5	line	0.87, 0.83	0.88, 1.23		
Outline	6	line	0.88, 1.23	1.54, 1.22		
Outline	7	line	1.54, 1.22	1.53, 0.82		
Outline	8	line	1.53, 0.82	3.84, 0.76		
Outline	9	line	3.84, 0.76	3.84, 0.69		
Outline	10	line	3.84, 0.69	3.84, 0.76		
Outline	11	line	3.84, 0.76	4.23, 0.74		
Outline	12	line	4.23, 0.74	5.73, 0.61		
Outline	13	arc clockwise	5.73, 0.61	7.73, 0.37	-4.21, -91.71	92.85
Outline	14	line	7.73, 0.37	8.93, 0.23		
Outline	15	arc anticlockwise	8.93, 0.23	13.41, 0.71	10.21, 9.37	9.22
Outline	16	line	13.41, 0.71	15.06, 1.23		
Outline	17	line	15.06, 1.23	15.10, 1.09		
Outline	18	line	15.10, 1.09	13.00, 0.43		
Outline	19	arc clockwise	13.00, 0.43	8.78, 0.10	10.20, 9.26	9.26
Outline	20	line	8.78, 0.10	5.98, 0.43		
Outline	21	line	5.98, 0.43	4.92, 0.53		
Outline	22	arc anticlockwise	4.92, 0.53	3.86, 0.61	3.11, -16.15	16.78

Outline	23	line	3.86, 0.61	1.90, 0.66		
Outline	24	line	1.90, 0.66	1.89, 0.51		
Outline	25	line	1.89, 0.51	1.90, 0.51		
Outline	26	line	1.90, 0.51	1.90, 0.35		
Outline	27	arc clockwise	1.90, 0.35	0.48, 0.38	1.25, 2.79	2.53
Outline	28	line	0.48, 0.38	0.49, 0.54		
Outline	29	line	0.49, 0.54	0.50, 0.54		







Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	Zeichnung Nr. / Drawing no. **WN-205**
Behandlung / Treatment			Teil / Part *detent.foot*
		Benennung / Name ***Detent foot screw and pins***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Detent foot screw and pins — sheet 2

The screw and two steady pins that hold the detent's foot to the block.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-205	Stellstift (WN-205 by the coloured section's label), the detent's foot pins	2	not on the drawing		in 050401-1 (embed): 0.636 mm, least gap -0.2222; in 050401-4 (embed): 1.808 mm, least gap -0.2222
WN-13	Detent foot screw, M1 (WN-13 by the coloured section's label, likely)	1	not on the drawing		in 050401-1 (embed): 0.589 mm, least gap -0.4966; in 050401-4 (embed): 2.357 mm, least gap -0.4966

What it does (WN-205). Place the detent's foot on the block.

What it does (WN-13). Clamps the detent's foot to the block's end.

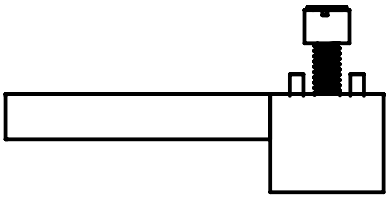
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.59.

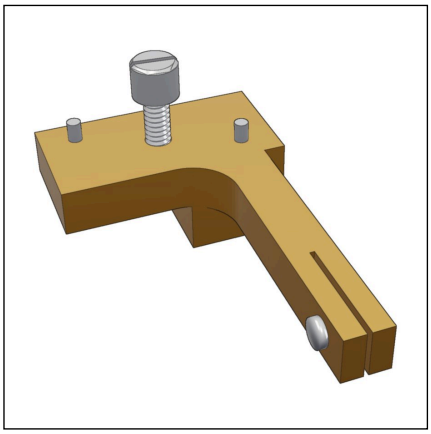
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	3.93, 4.75	4.20, 4.88		
Outline	2	line	4.20, 4.88	5.57, 1.99		
Outline	3	line	5.57, 1.99	5.17, 1.80		
Outline	4	line	5.17, 1.80	3.79, 4.69		
Outline	5	line	3.79, 4.69	3.93, 4.75		
Outline	6	line	1.14, 4.76	2.03, 5.18		
Outline	7	line	2.03, 5.18	3.49, 2.11		
Outline	8	line	3.49, 2.11	3.76, 2.24		
Outline	9	line	3.76, 2.24	3.95, 1.83		
Outline	10	line	3.95, 1.83	2.52, 1.15		
Outline	11	line	2.52, 1.15	2.33, 1.56		
Outline	12	line	2.33, 1.56	2.59, 1.69		
Outline	13	line	2.59, 1.69	1.14, 4.76		
Outline	14	line	1.14, 4.76	1.14, 4.76		
Outline	15	line	0.29, 3.03	0.41, 3.08		
Outline	16	line	0.41, 3.08	1.78, 0.19		
Outline	17	line	1.78, 0.19	1.37, 0.00		
Outline	18	line	1.37, 0.00	0.00, 2.89		
Outline	19	line	0.00, 2.89	0.29, 3.03		
Cut-out 1	20	line	0.65, 2.08	0.65, 2.08		
Cut-out 1	21	arc anticlockwise	0.65, 2.08	0.65, 2.08	1.77, 2.35	1.15

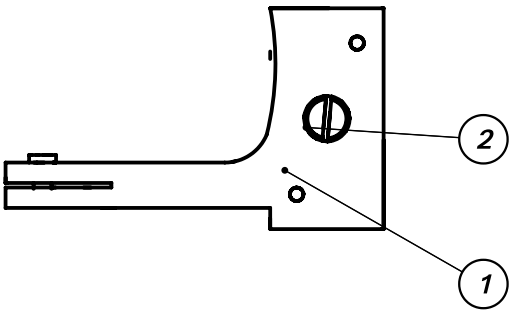
Front view





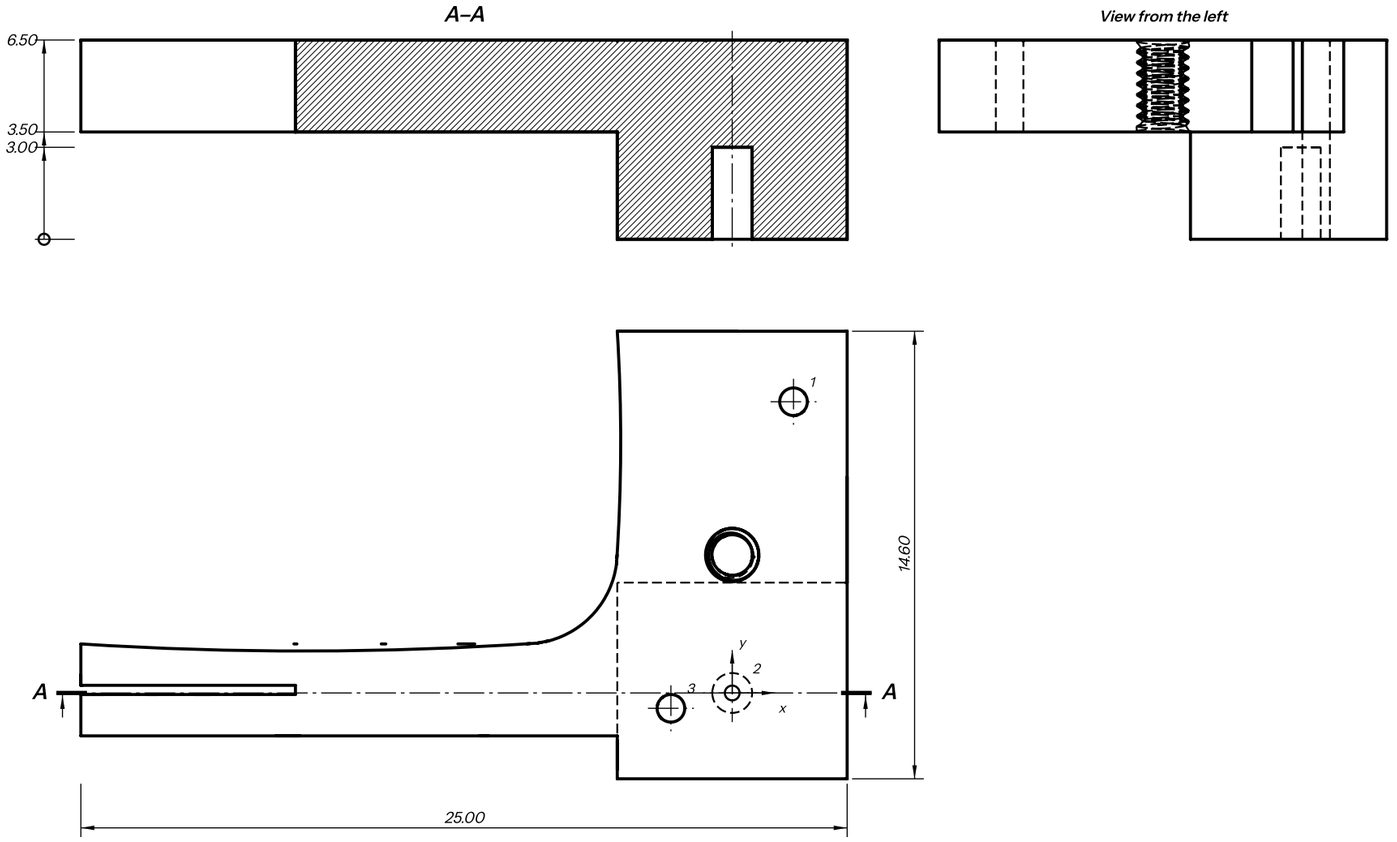
Isometric view, not to scale

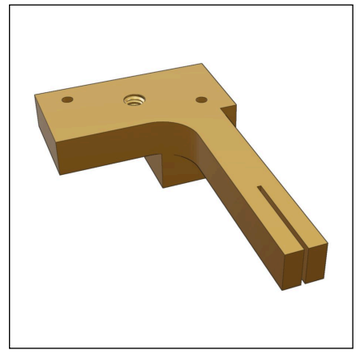
Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	2	Block screws and pins	WN-201	not on the drawing
1	1	Detent block	05 04 01-4	not on the drawing

Werkstoff / Material not on the drawing	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 04 01-4** Teil / Part detentBlock
Behandlung / Treatment		Benennung / Name **Detent block (Gangfederkloben)**	
Blatt / Sheet **1 / 1**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 01-4** *Teil / Part* *detentBlock.block*
Behandlung / Treatment		*Benennung / Name* ***Detent block***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Detent block — sheet 2

The brass L under the top plate that carries the detent: screwed up from the plate's outer face through hole 13, placed by two steady pins. The detent's foot is clamped to the end of its base; a screw crosses the slit end of its upright.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 01-4	Gangfederkloben (detent block)	1	not on the drawing		on 050901: 0 mm

Test on the bench (05 04 01-4). The escapement's figures and the banking, the detent, its gold spring and the balance in; the train's power let down: 05 04_1's layout and the model: the impulse stone's face 28°10' before the line of centres at rest, the locking stone 37.0° on in the wheel's turn and 7.14 from its centre; the gold spring met at -8.8° of the balance (05 04_1's −12°), the wheel released at 4.0° (lock 12.8°), the gold spring off the stone at 6.9° (let-off 2.8°), the return swing past it at -21.6° (overall 28.5°); roller shake 0.011 mm, the teeth 0.39 mm into the roller. Set the banking (the block's cross screw, the 1954 instruction's п. 11) for a lock 0.187 mm deep at the stone, the middle of its window (the drawing's 0.235 lies 146 µm over its floor): never shallower than 0.089 (a held tooth's stone rings more than half the depth: wempe_tripping.js) nor deeper than 0.285 (the discharge stone can't lift the stone further: it doesn't unlock). A wheel at the top of its 14 +0.03 locks 0.250 deep where the drawn wheel locks 0.235: set the banking to the wheel as made. The preload at the stone: a 0.64 g rider must leave the detent on its banking and a 0.80 g one part it; aim near 0.68 g (the model's 0.8 g is an estimate: wempe_tripping.js finds a held tooth's stone ringing 44 % of the depth of lock there, under half from 0.40 g, teeth getting past at a fifth of it and under; and a 3 ms blow of 20 g across the plate lets a tooth go under 0.64 g, the floor taken here, the higher; each gram more costs about 10.4° of swing; never over 0.80 g: above it the stone, falling back after the release, lands again on the escaping tooth's back during the impulse, where the models part: dynamics.js with brass's restitution). The return pass: turn the balance back slowly by hand through the discharge with the train's power on and watch the locking stone under a loupe: the discharge stone must bend the gold spring aside without lifting the detent (wempe_tripping.js: as the model takes the stone's contact on the gold spring no tooth gets past over 333 gold's whole range of modulus, 83-110 GPa, but at 4 times that contact's stiffness 6 in 12 do, whatever the alloy: only the made spring can show it); its tip at rest 0.10-0.15 short of the discharge stone and 0.35-0.40 past the detent's end (the 1958 Navy repair book's checks, p. 60).

What it does (05 04 01-4). Under the top plate, carries the detent. As first drawn (05 04_1, Wempe 3955), a base with a long arm and its banking screw in place of the upright.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 6.50.

Holes

Hole	x	y	Size
1	2.00	9.50	Ø 0.90 through (3.00 thick at 3.50-6.50)
2	0.00	0.00	Ø 1.30 × 3.00 deep from the bottom
3	-2.00	-0.50	Ø 0.90 through

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-14.25, -1.40	-21.25, -1.40		
Outline	2	line	-21.25, -1.40	-21.25, -0.05		
Outline	3	line	-21.25, -0.05	-14.25, -0.05		
Outline	4	line	-14.25, -0.05	-14.25, 0.25		
Outline	5	line	-14.25, 0.25	-21.25, 0.25		
Outline	6	line	-21.25, 0.25	-21.25, 1.60		
Outline	7	arc anticlockwise	-21.25, 1.60	-6.36, 1.63	-14.00, 115.21	113.84
Outline	8	arc anticlockwise	-6.36, 1.63	-3.75, 4.60	-6.75, 4.60	3.00

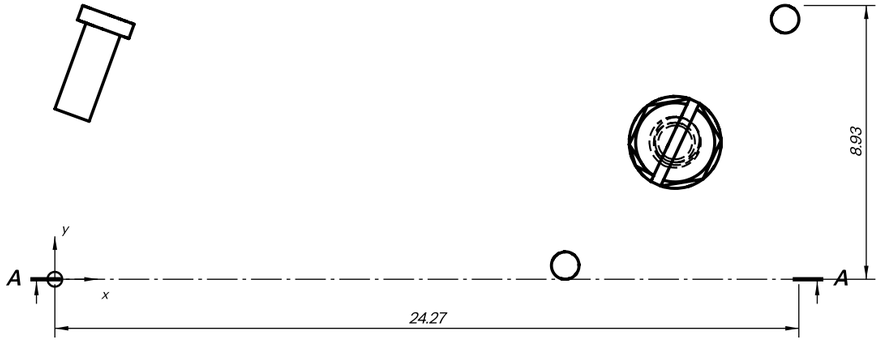
Outline	9	line	-3.75, 4.60	-3.75, 11.80		
Outline	10	line	-3.75, 11.80	3.75, 11.80		
Outline	11	line	3.75, 11.80	3.75, -2.80		
Outline	12	line	3.75, -2.80	-3.75, -2.80		
Outline	13	line	-3.75, -2.80	-3.75, -1.40		
Outline	14	line	-3.75, -1.40	-14.25, -1.40		

Levels

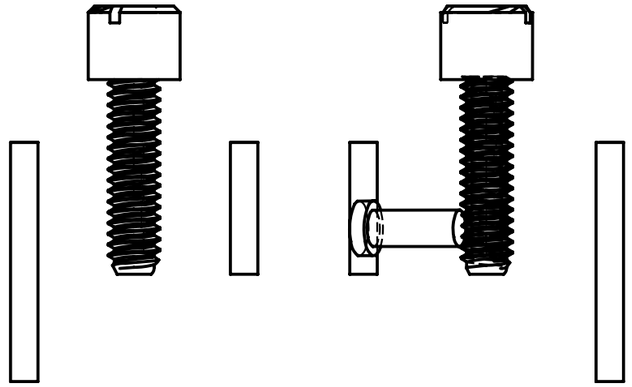
Faces in section A-A, from the bottom face: 3.00, 3.50, 6.50.

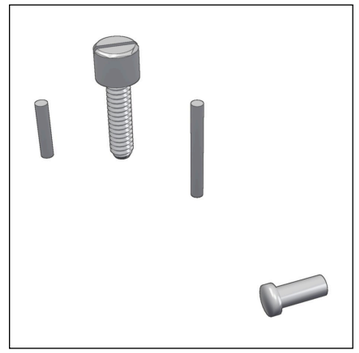
A–A





View from the left





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	Zeichnung Nr. / Drawing no. ***WN-201***
Behandlung / Treatment			Teil / Part *detentBlock.screws*
Blatt / Sheet **1 / 2**		Benennung / Name ***Block screws and pins***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Block screws and pins — sheet 2

The fixing screw from the top plate's outer face, the screw across the upright's slit end, and the two steady pins.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-201	Stellstift (WN-201, "Stellstift 2" by the coloured section's label), the detent block's steady pins Ø 0.9	2	not on the drawing		in 050401-4 (press): 6.521 mm, least gap 0.0045; in 050901 (clear): 1.284 mm, least gap 0.0185
WN-15	Detent block screw, M1.7 (WN-15 by the coloured section's label, likely)	1	not on the drawing		in 050901 (clear): 3.367 mm, least gap 0.0521; in 050401-4 (tap): 2.986 mm, least gap 0.0275
WN-gkA	Screw across the block's slit upright, M1.2 (its WN number not read)	[0, 1]	not on the drawing		in 050401-4 (embed): 2.718 mm, least gap -0.5959

What it does (WN-201). Place the block on the top plate.

What it does (WN-15). From the top plate's outer face through hole 13 into the block. The Soviet 6MX's: a slotted stud flush in the plate's outer face, 2.2 from hole 13 (Kirov No. 247), through the adjusting slide into the block.

What it does (WN-gkA). Across the slit end of the upright.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 12.25.

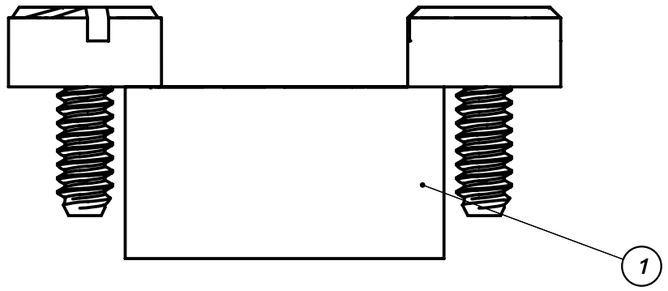
Contour in the plan

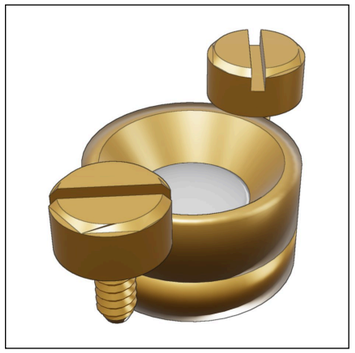
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			23.82, 8.48	0.45
Outline	2	circle			20.24, 4.47	1.50
Cut-out 1	3	line	19.12, 3.83	19.44, 4.23		
Cut-out 1	4	line	19.44, 4.23	19.12, 3.83		
Outline	5	circle			16.65, 0.45	0.45
Outline	6	line	1.02, 8.36	0.74, 8.46		
Outline	7	line	0.74, 8.46	0.91, 8.93		
Outline	8	line	0.91, 8.93	2.59, 8.32		
Outline	9	line	2.59, 8.32	2.42, 7.85		
Outline	10	line	2.42, 7.85	2.14, 7.95		
Outline	11	line	2.14, 7.95	1.12, 5.15		
Outline	12	line	1.12, 5.15	0.00, 5.56		
Outline	13	line	0.00, 5.56	1.02, 8.36		
Cut-out 1	14	line	2.56, 8.28	2.56, 8.28		
Cut-out 1	15	arc anticlockwise	2.56, 8.28	2.56, 8.28	7.47, 6.54	5.20
Cut-out 1	16	line	2.56, 8.28	2.56, 8.28		
Cut-out 2	17	line	0.82, 7.65	0.82, 7.65		
Cut-out 2	18	arc anticlockwise	0.82, 7.65	0.82, 7.65	6.40, 5.44	6.00

Cut-out 2	19	line	0.82, 7.65	0.82, 7.65		

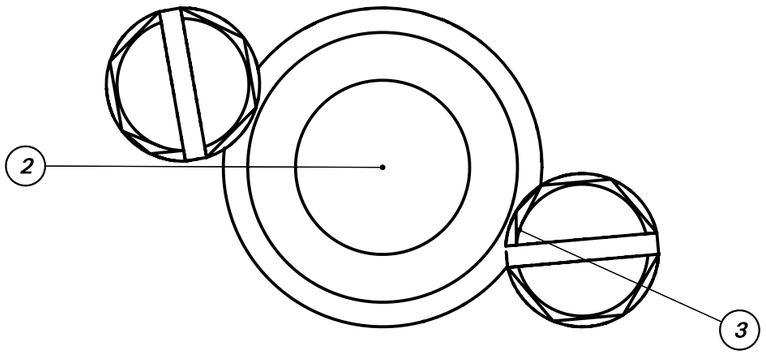
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	2	Setting screws	WN-55.ck	not on the drawing
2	1	Upper endstone	05 04 08-1	not on the drawing
1	1	Upper hole jewel	05 04 06-2	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: 10:1

Behandlung / Treatment:

Blatt / Sheet: 1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

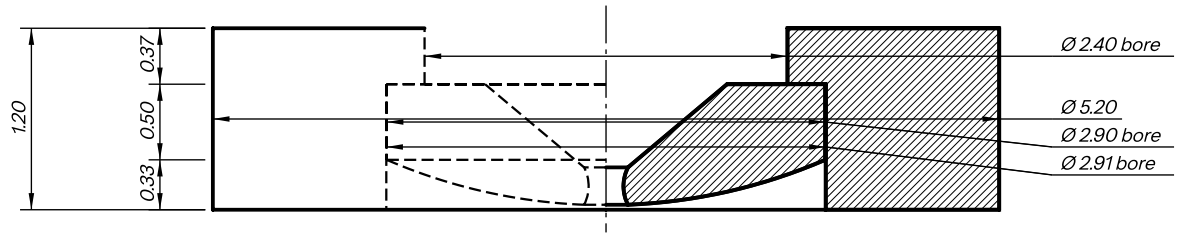
Model v3.35.00 · 2026-10-26

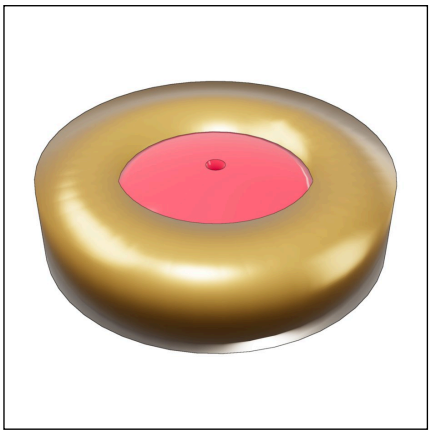
Zeichnung Nr. / Drawing no.: 05 04 06-2

Teil / Part: balanceJewelsUp

Benennung / Name: **Balance jewels, upper**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 06-2** *Teil / Part* *balanceJewelsUp.jewel*
Behandlung / Treatment		*Benennung / Name* ***Upper hole jewel***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Upper hole jewel — sheet 2

The hole jewel and its setting.

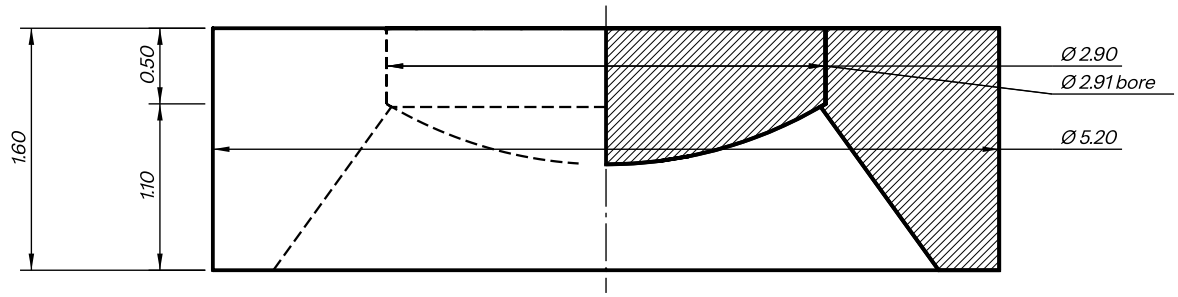
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 06-2	Fassung (jewel setting), balance, upper	1	not on the drawing		in 050803-1 (press): 1.2 mm, least gap 0.0064
05 04 07-1	Lagerstein (hole jewel), balance, upper	1	not on the drawing		takes Unruhwelle (balance staff): in it (run): 0.321 mm, least gap 0.0163

What it does (05 04 06-2). Holds the upper hole jewel in the cock, its flange in the sink below.

What it does (05 04 07-1). The staff's upper pivot runs in it.

Turned about its axis; length 1.20, largest Ø 5.20. Diameters: Ø 5.20; bores Ø 2.40, Ø 2.90, Ø 2.91. Shoulders from the lower end: 0.00, 0.33, 0.83, 1.20.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 08-1** *Teil / Part* balanceJewelsUp.end
Behandlung / Treatment		*Benennung / Name* ***Upper endstone***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Upper endstone — sheet 2

The endstone and its setting.

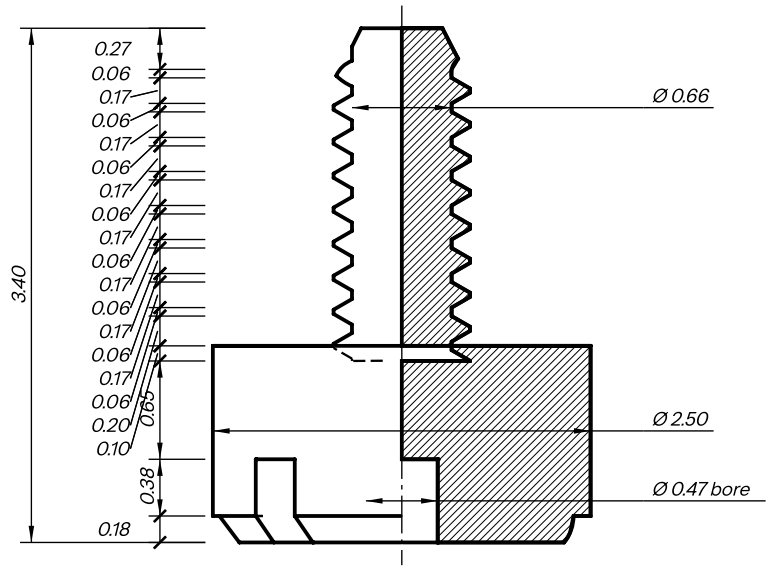
Parts list

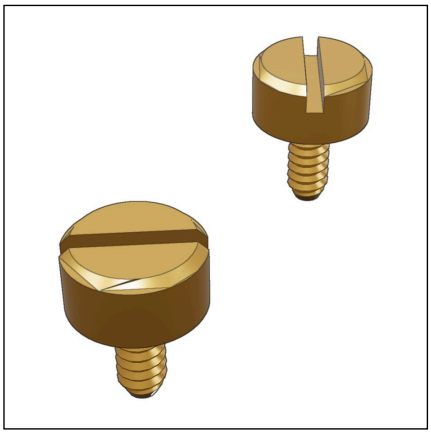
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 08-1	Deckstein (endstone), balance, upper	1	not on the drawing		not measured on the model: see its function, Stops the staff's upper pivot end.
05 04 08-2	Fassung (endstone setting), balance, upper	1	not on the drawing		in 050803-1 (press): 1.507 mm, least gap 0.0064

What it does (05 04 08-1). Stops the staff's upper pivot end.

What it does (05 04 08-2). Holds the upper endstone in the cock.

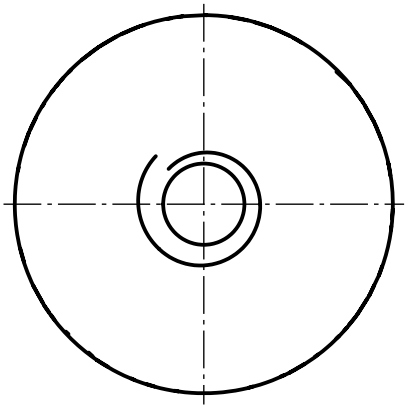
Turned about its axis; length 1.60, largest Ø 5.20. Diameters: Ø 2.90, Ø 5.20; bores Ø 2.91. Shoulders from the lower end: 0.00, 1.10, 1.60.





Isometric view, not to scale

View from above



2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-55.ck** *Teil / Part* *balanceJewelsUp.screws*
Behandlung / Treatment		*Benennung / Name* ***Setting screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Setting screws — sheet 2

Two screws holding the endstone's setting.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

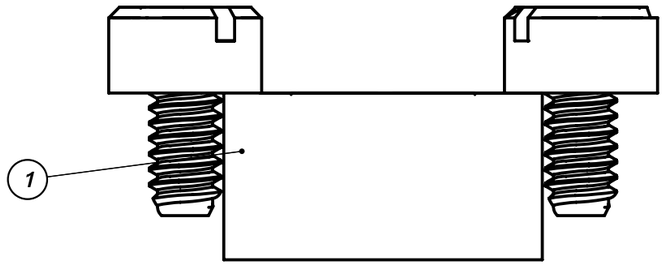
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-55.ck	Schraube (WN-55 by the coloured section), the upper endstone's screws, M1.2	2	not on the drawing		in 050803-1 (tap): 2.021 mm, least gap 0.0145; on 050408-2.bu: 0 mm

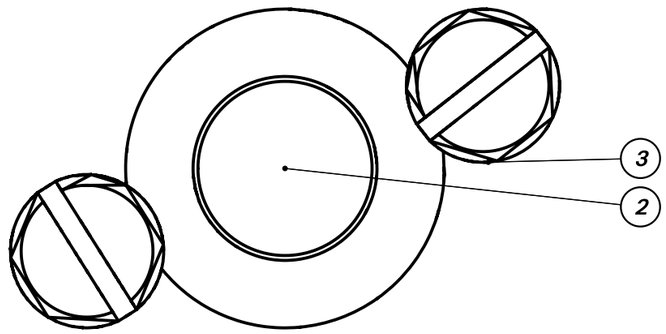
What it does (WN-55.ck). Hold the endstone's setting, their heads over its edge.

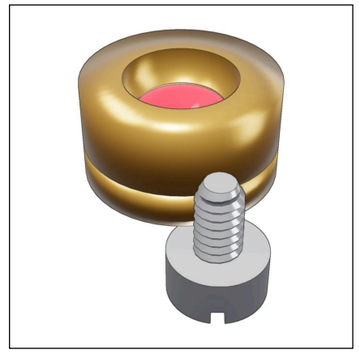
Turned about its axis; length 3.40, largest Ø 2.50. Diameters: Ø 0.66, Ø 2.50; bores Ø 0.47. Shoulders from the lower end: 0.00, 0.18, 0.55, 1.20, 1.30, 1.50, 1.55, 1.72, 1.78, 1.95, 2.00, 2.17, 2.23, 2.40, 2.45, 2.62, 2.68, 2.85, 2.90, 3.07, 3.13, 3.40.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	2	Setting screws	WN-10	not on the drawing
2	1	Lower endstone	05 04 08-1	not on the drawing
1	1	Lower hole jewel	05 04 06-2	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: **10:1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **05 04 06-2**

Teil / Part: balanceJewelsLow

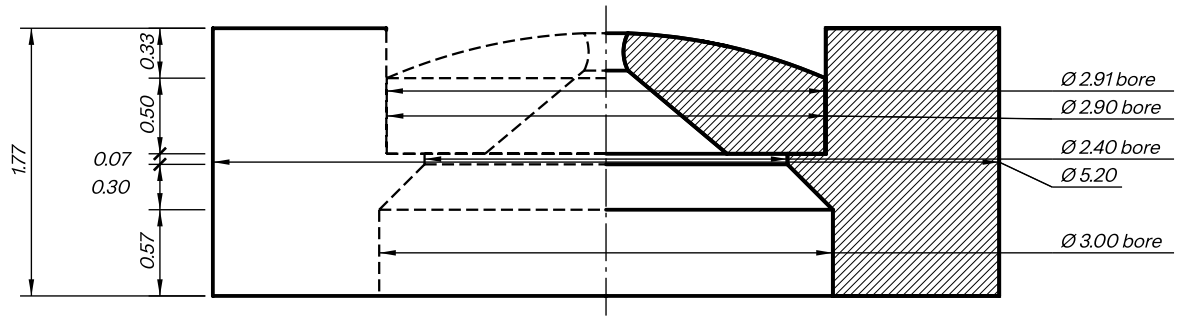
Behandlung / Treatment:

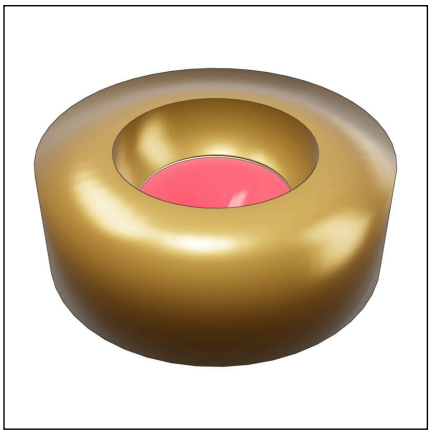
Benennung / Name: **Balance jewels, lower**

Blatt / Sheet: **1 / 1**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 06-2**
Behandlung / Treatment			*Teil / Part* *balanceJewelsLow.jewel*
		Benennung / Name ***Lower hole jewel***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Lower hole jewel — sheet 2

The hole jewel and its setting.

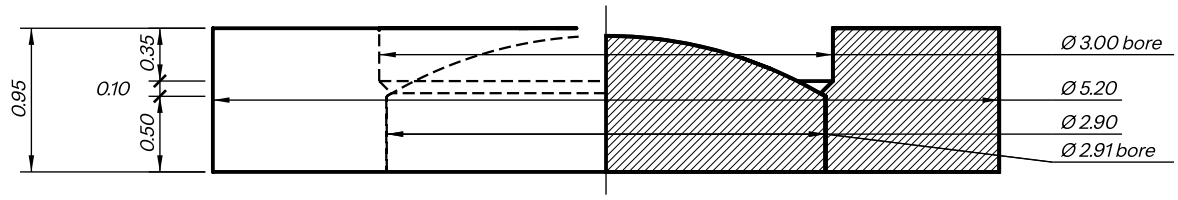
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 06-2	Fassung (jewel setting), balance, lower	1	not on the drawing		in 050800-2 (press): 1.769 mm, least gap 0.0064
05 04 07-1	Lagerstein (hole jewel), balance, lower	1	not on the drawing		takes Unruhwelle (balance staff): in it (run): 0.321 mm, least gap 0.0163

What it does (05 04 06-2). Holds the lower hole jewel in the lug of the lower cock.

What it does (05 04 07-1). The staff's lower pivot runs in it.

Turned about its axis; length 1.77, largest Ø 5.20. Diameters: Ø 5.20; bores Ø 2.40, Ø 2.90, Ø 2.91, Ø 3.00. Shoulders from the lower end: 0.00, 0.57, 0.87, 0.94, 1.44, 1.77.





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **20:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 04 08-1** Teil / Part balanceJewelsLow.end
Behandlung / Treatment		Benennung / Name **Lower endstone**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Lower endstone — sheet 2

The endstone and its setting.

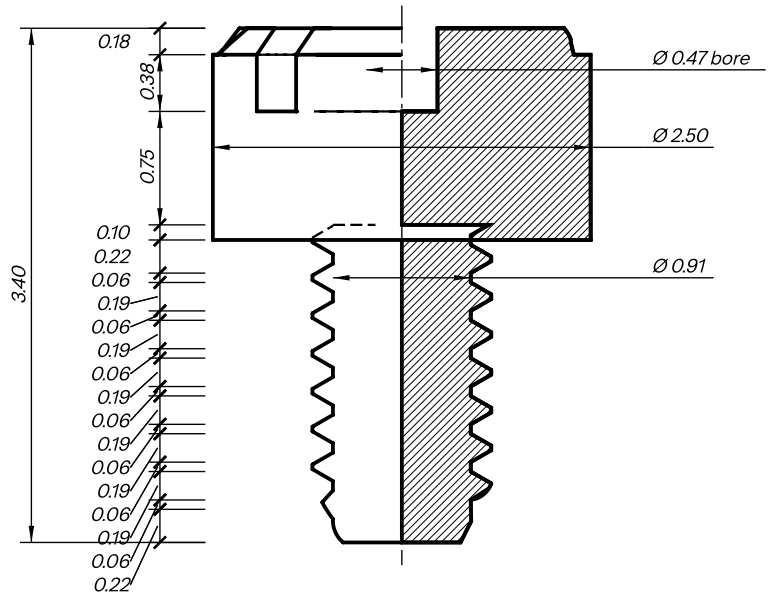
Parts list

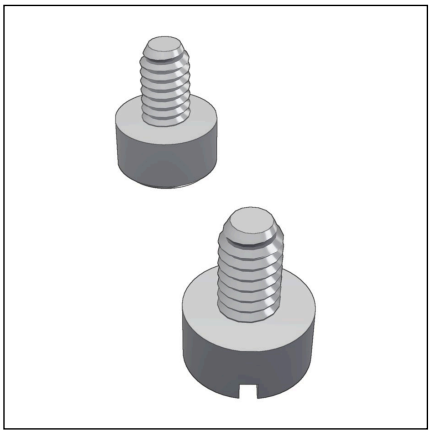
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 08-1	Deckstein (endstone), balance, lower	1	not on the drawing		not measured on the model: see its function, Stops the staff's lower pivot end.
05 04 08-2	Fassung (endstone setting), balance, lower	1	not on the drawing		in 050800-2 (press): 0.95 mm, least gap 0.0064

What it does (05 04 08-1). Stops the staff's lower pivot end.

What it does (05 04 08-2). Holds the lower endstone.

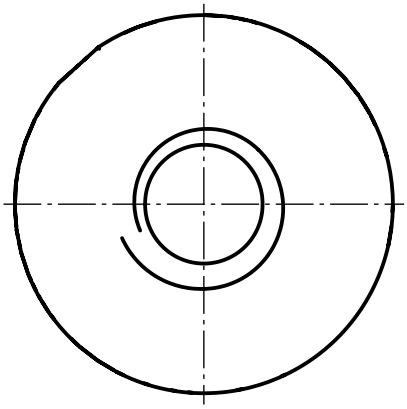
Turned about its axis; length 0.95, largest Ø 5.20. Diameters: Ø 2.90, Ø 5.20; bores Ø 2.91, Ø 3.00. Shoulders from the lower end: 0.00, 0.50, 0.60, 0.95.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-10** *Teil / Part* balanceJewelsLow.screws
Behandlung / Treatment		*Benennung / Name* ***Setting screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Setting screws — sheet 2

Two screws holding the settings in the lug.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

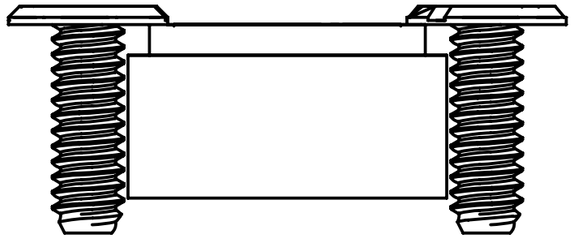
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-10	Schraube (WN-10 by the coloured section's label, its last digit 0 or 8: likely 10), the lower settings' screws, M1.2	2	not on the drawing		in 050800-2 (tap): 2.021 mm, least gap 0.0205; on 050408-2.bl: 0 mm

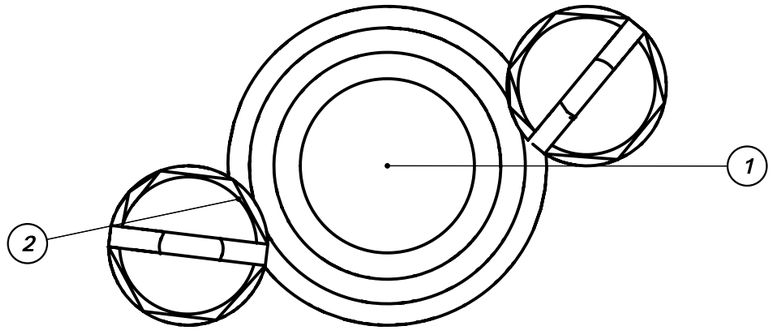
What it does (WN-10). Hold the lower settings in the lug.

Turned about its axis; length 3.40, largest Ø 2.50. Diameters: Ø 0.91, Ø 2.50; bores Ø 0.47. Shoulders from the lower end: 0.00, 0.22, 0.28, 0.47, 0.53, 0.72, 0.78, 0.97, 1.03, 1.22, 1.28, 1.47, 1.53, 1.72, 1.78, 2.00, 2.10, 2.85, 3.23, 3.40.

Front view



Plan

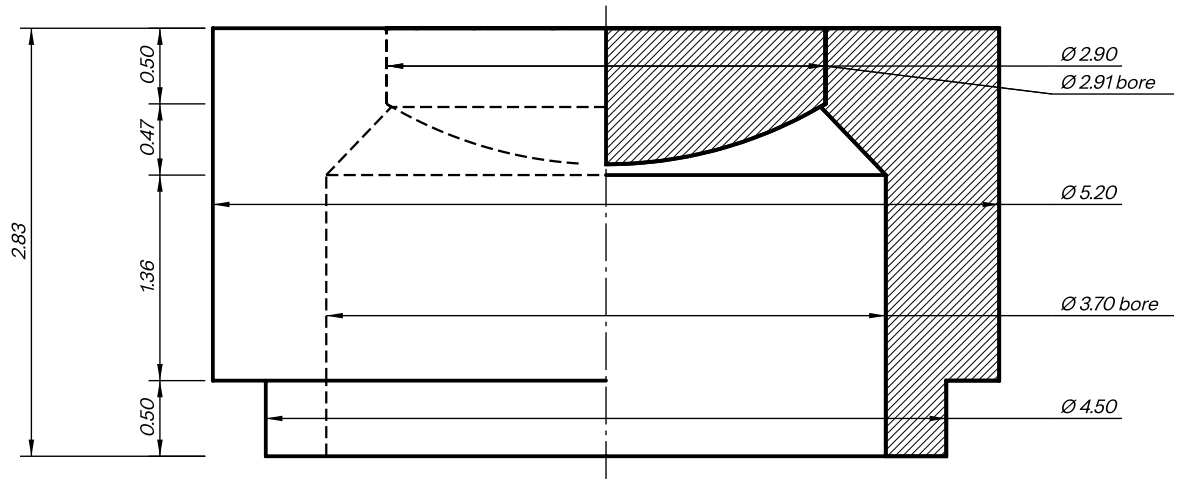


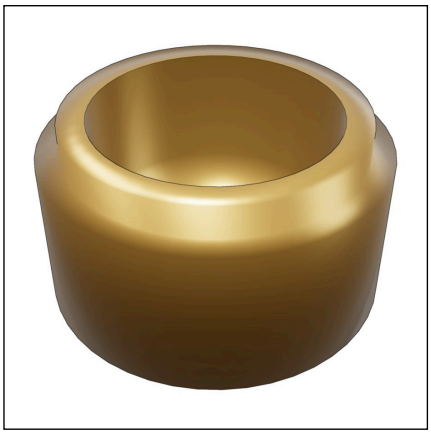


Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	2	Setting screws	WN-es	not on the drawing
1	1	Endstone and setting	05 04 08-1	not on the drawing

Werkstoff / Material: not on the drawing	Maßstab / Scale: **10:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no.: **05 04 08-1** Teil / Part: escEndstoneUp
Behandlung / Treatment		Benennung / Name: **Escape wheel endstone, upper**	
Blatt / Sheet: **1 / 1**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 08-1** *Teil / Part* *escEndstoneUp.setting*
Behandlung / Treatment		*Benennung / Name* ***Endstone and setting***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Endstone and setting — sheet 2

The endstone over the escape wheel arbor's upper pivot: its gilt setting goes up into the top plate's seat from inside, its spigot through the lip flush with the plate's face, the whole domed ruby showing in its bevelled mouth, and two brass screws put in from outside hold it, their heads at its edge.

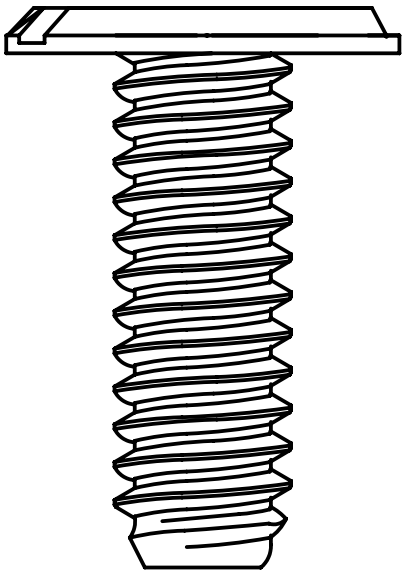
Parts list

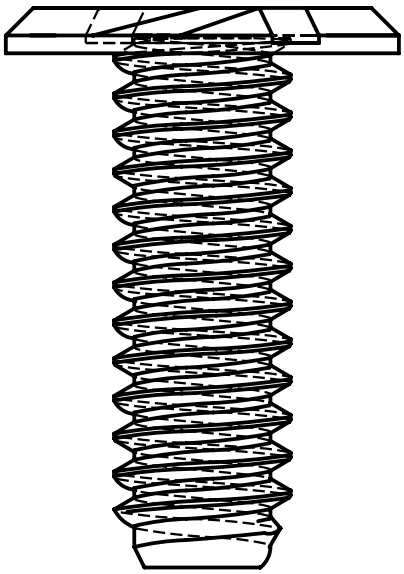
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 08-1	Deckstein (endstone), escape wheel, upper	1	not on the drawing		not measured on the model: see its function, Stops the escape arbor's upper pivot end.
05 04 08-2	Fassung (endstone setting), escape wheel, upper	1	not on the drawing		in 050901 (press): 2.83 mm, least gap 0.0058

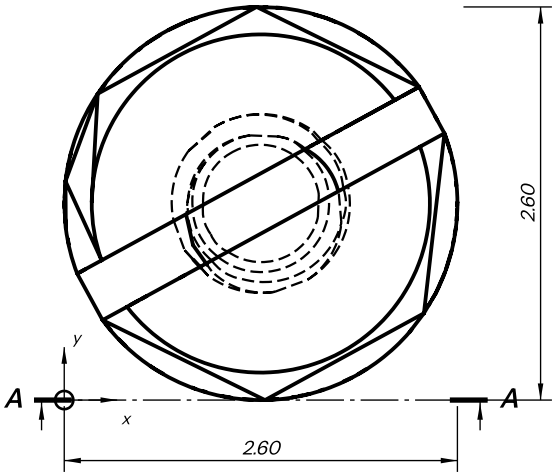
What it does (05 04 08-1). Stops the escape arbor's upper pivot end.

What it does (05 04 08-2). Holds the upper endstone in the top plate's seat.

Turned about its axis; length 2.83, largest Ø 5.20. Diameters: Ø 2.90, Ø 4.50, Ø 5.20; bores Ø 2.91, Ø 3.70. Shoulders from the lower end: 0.00, 0.50, 1.86, 2.33, 2.83.









Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-es** *Teil / Part* *escEndstoneUp.screws*
Behandlung / Treatment		*Benennung / Name* ***Setting screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Setting screws — sheet 2

Two brass screws put in from outside, their heads at the setting's edge.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-es	Escape endstone screws, upper, M1.2 (their WN number not read)	2	not on the drawing		in 050901 (tap): 3.36 mm, least gap 0.0205; on 050901: 0 mm

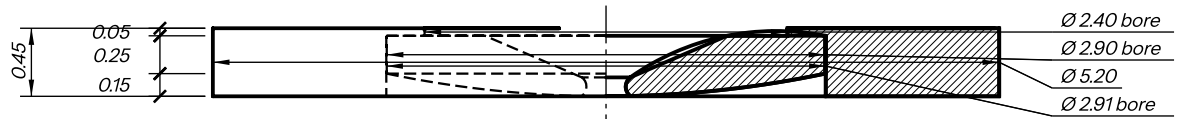
What it does (WN-es). Put in from outside: hold the upper endstone's setting, their heads over its spigot's edge (Wempe 3955).

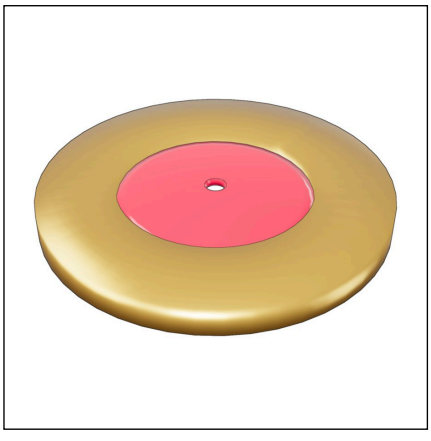
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.70.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			1.30, 1.30	1.30





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 06-1** *Teil / Part* *escJewelUp*
Behandlung / Treatment		*Benennung / Name* ***Escape wheel jewel, upper***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Escape wheel jewel, upper — sheet 2

The escape wheel arbor's upper hole jewel, in its setting flush with the top plate's inner face, held by two screws whose heads lie over its edge. It is domed toward the endstone, with an oil sink opening toward the wheel.

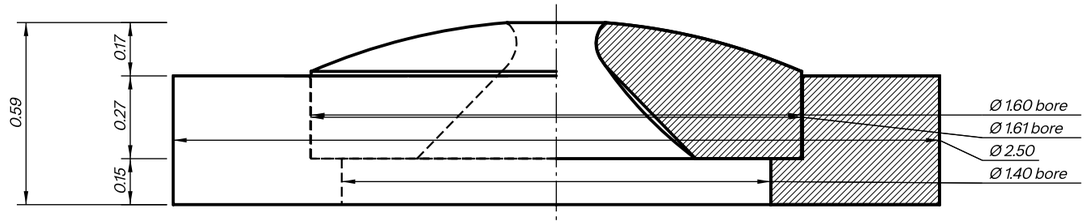
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 06-1	Lagerstein (hole jewel), escape wheel, upper	1	not on the drawing		takes Gangtrieb (escape pinion, z 10, m 0.263, and arbor): in it (run): 0.141 mm, least gap 0.0189
05 04 06-2	Fassung (jewel setting), escape wheel, upper	1	not on the drawing		in 050901 (press): 0.45 mm, least gap 0.0064

What it does (05 04 06-1). The escape arbor's upper pivot (Ø 0.24) runs in it.

What it does (05 04 06-2). Holds the upper hole jewel, flush with the top plate's inner face.

Turned about its axis; length 0.45, largest Ø 5.20. Diameters: Ø 5.20; bores Ø 2.40, Ø 2.90, Ø 2.91. Shoulders from the lower end: 0.00, 0.15, 0.40, 0.45.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **50:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 06-1**
Behandlung / Treatment			*Teil / Part* *escJewelLow*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Escape wheel jewel, lower***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Escape wheel jewel, lower — sheet 2

The escape wheel arbor's lower hole jewel, in a bush pressed into the pillar plate, domed toward the endstone under it, with an oil sink opening toward the wheel.

Parts list

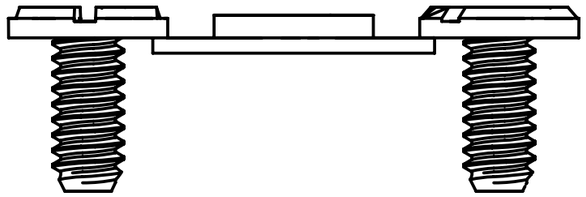
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 06-1	Lagerstein (hole jewel), escape wheel, lower	1	not on the drawing		takes Gangtrieb (escape pinion, z 10, m 0.263, and arbor): in it (run): 0.141 mm, least gap 0.0121
05 04 06-2	Fassung (jewel bush), escape wheel, lower	1	not on the drawing		in 050802-1 (press): 0.42 mm, least gap 0.0009

What it does (05 04 06-1). The escape arbor's lower pivot runs in it.

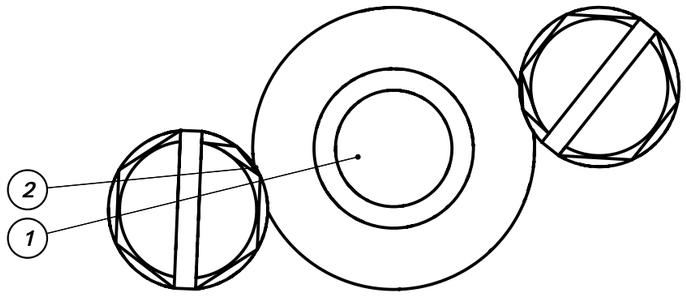
What it does (05 04 06-2). Pressed into the pillar plate's hole 26, holding the lower hole jewel.

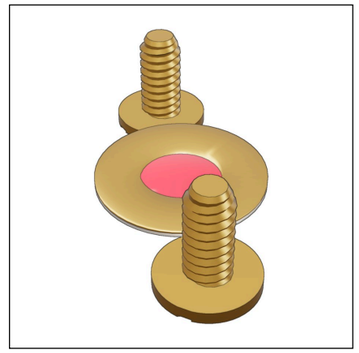
Turned about its axis; length 0.59, largest Ø 2.50. Diameters: Ø 2.50; bores Ø 1.40, Ø 1.60, Ø 1.61. Shoulders from the lower end: 0.00, 0.15, 0.42, 0.59.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	2	Setting screws	WN-ec	not on the drawing
1	1	Endstone and setting	05 04 08-1	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: **10:1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **05 04 08-1**

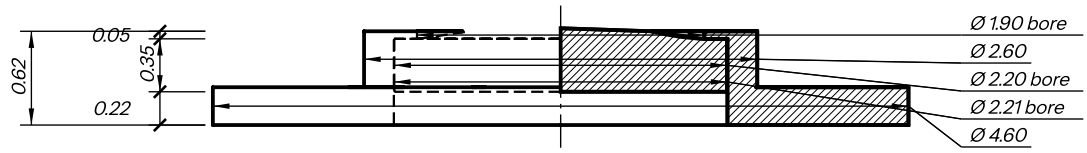
Teil / Part: escEndstoneLow

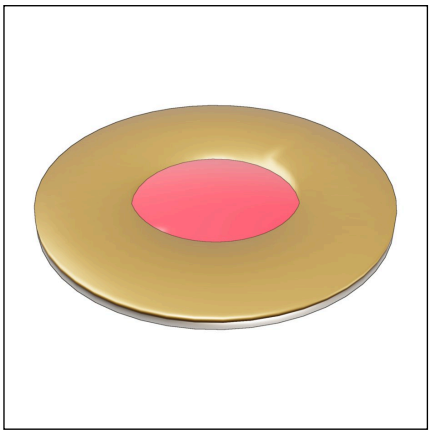
Behandlung / Treatment:

Benennung / Name: **Escape wheel endstone, lower**

Blatt / Sheet: **1 / 1**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 04 08-1** *Teil / Part* *escEndstoneLow.cap*
Behandlung / Treatment		*Benennung / Name* ***Endstone and setting***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Endstone and setting — sheet 2

The endstone under the escape wheel arbor's lower pivot: a round gilt setting flush in a seat in the pillar plate's dial side, the ruby showing, held by two screws whose heads lap its rim.

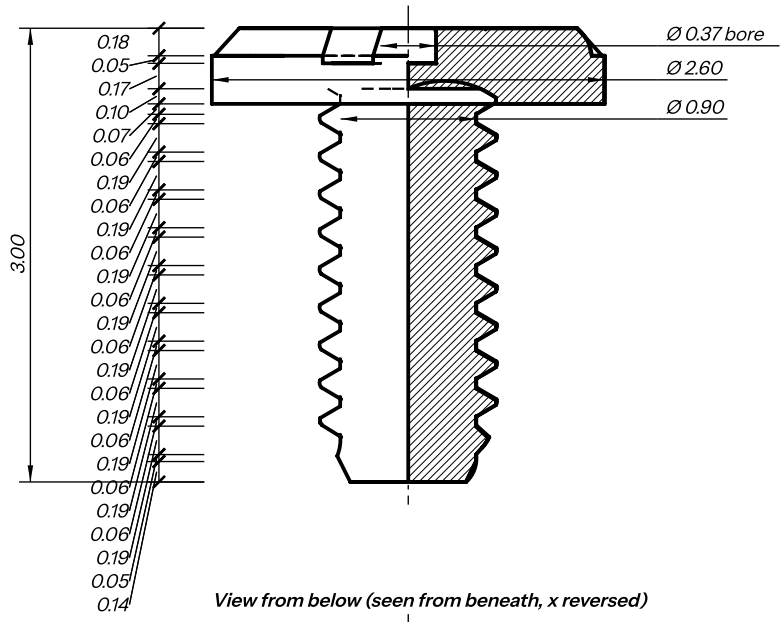
Parts list

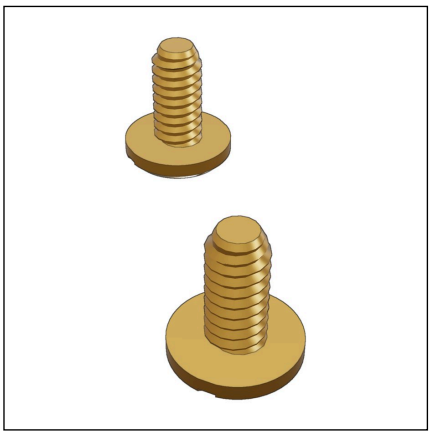
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 04 08-1	Deckstein (endstone), escape wheel, lower	1	not on the drawing		not measured on the model: see its function, Stops the escape arbor's lower pivot end.
05 04 08-2	Fassung (endstone cap), escape wheel, lower	1	not on the drawing		on 050802-1: 0 mm

What it does (05 04 08-1). Stops the escape arbor's lower pivot end.

What it does (05 04 08-2). On the pillar plate's dial side, holding the lower endstone.

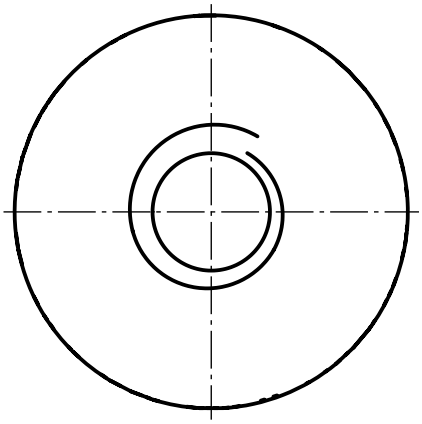
Turned about its axis; length 0.64, largest Ø 4.60. Diameters: Ø 2.60, Ø 4.60; bores Ø 1.90, Ø 2.20, Ø 2.21. Shoulders from the lower end: 0.00, 0.22, 0.57, 0.62.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-ec** *Teil / Part* escEndstoneLow.screws
Behandlung / Treatment		*Benennung / Name* ***Setting screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Setting screws — sheet 2

Two brass screws into the pillar plate, their heads lapping the setting's rim.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

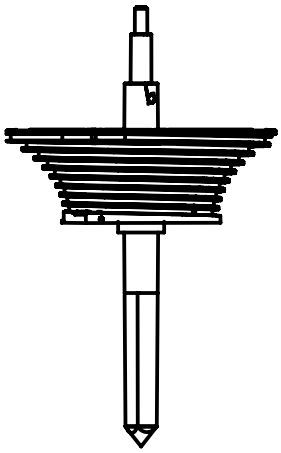
Parts list

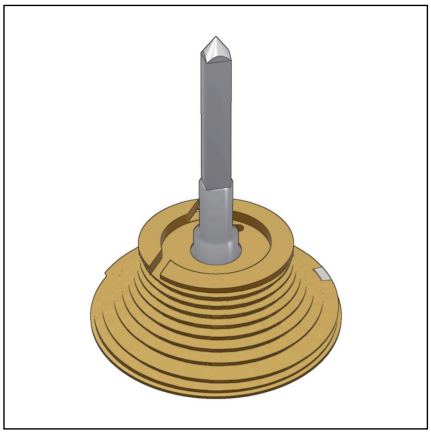
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-ec	Escape endstone screws, lower, M1.2 (their WN number not read)	2	not on the drawing		in 050802-1 (tap): 2.517 mm, least gap 0.0205; on 050802-1: 0 mm

What it does (WN-ec). Hold the lower endstone's setting in its seat, their heads over its rim (Wempe 3955).

Turned about its axis; length 3.00, largest Ø 2.60. Diameters: Ø 0.90, Ø 2.60; bores Ø 0.37. Shoulders from the lower end: 0.00, 0.13, 0.18, 0.37, 0.43, 0.62, 0.68, 0.87, 0.93, 1.12, 1.18, 1.37, 1.43, 1.62, 1.68, 1.87, 1.93, 2.12, 2.18, 2.37, 2.43, 2.50, 2.60, 2.77, 2.82, 3.00.

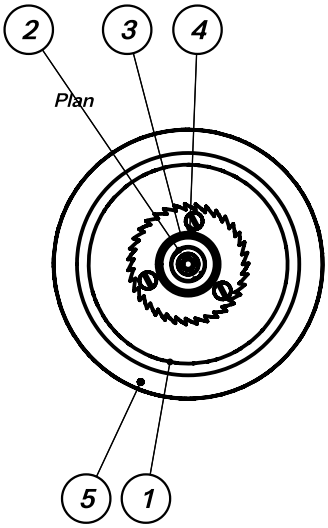
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
5	1	Chain pin (Kettenstift)	05 06 00-6	not on the drawing
4	3	Click wheel screws (WN-19)	WN-19	not on the drawing
3	1	Click wheel (Schneckensperrrad)	05 06 04-3	„Ms Blech 2mm DIN 17.51 Ms 58
2	1	Fusee arbor (Schneckenwelle)	05 06 04-1	„Präz. Rundstahl 7,5 Din 175“
1	1	Fusee body (Schneckenkörper)	05 06 04-2	„Rundmessing 36 Din 1756(?) Ms

Werkstoff / Material
not on the drawing

Maßstab / Scale
1:1

Behandlung / Treatment

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

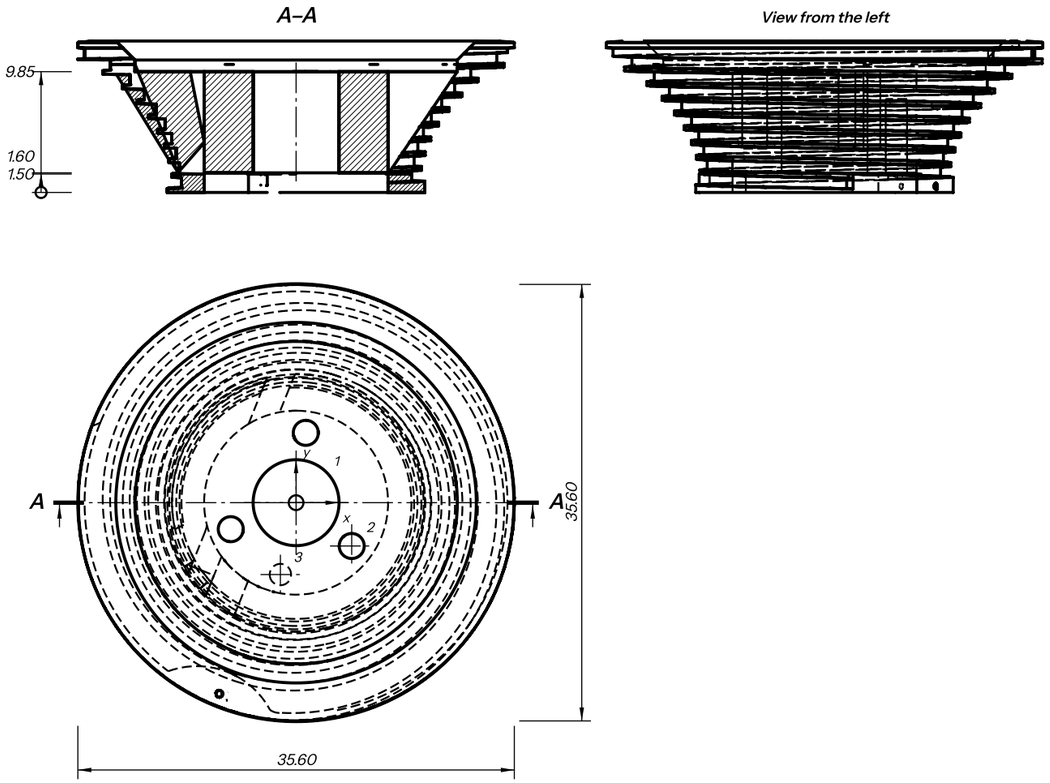
Model v3.35.00 · 2026-10-26

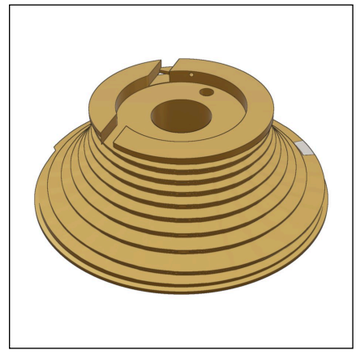
Zeichnung Nr. / Drawing no.
05 06 00-6

Teil / Part
fusee

Benennung / Name
Fusee (Schnecke)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Rundmessing 36 Din 1756(?) Ms 58" ("Rundmessing 36", "Ms 58"; the DIN number reads 1756 or 1758)*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 04-2*** *Teil / Part* *fusee.body*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0.05*			
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Fusee body (Schneckenkörper)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Fusee body (Schneckenkörper) — sheet 2

The brass cone with its nine-turn groove, square-cut: each turn a flat land with a rib 0.33 thick on its small-end side, 0.6 high. The chain's end is pinned in the slot between the flange's two ribs and leaves it over a flat milled across the flange. A recess in its large end holds the click wheel and the maintaining ratchet's cup; across its small end runs the stop bar's slot.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 04-2	Schneckenkörper (fusee body)	1	„Rundmessing 36 Din 1756(?) Ms 58“ ("Rundmessing 36", "Ms 58"; the DIN number reads 1756 or 1758)	surface ∇∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished); untoleranced sizes ±0,05	round 050604-1 (press): 7.942 mm, least gap -0.0103

Test on the bench (05 06 04-2). The pull on the fusee arbor, against 05 06_20, the chain on, the barrel set up ¾ turn, before the train goes in: Each reading within ×0.684–1.253 of the sheet's at the same turn (its readings' mean 132.7 g: 90.8–166.4 g; fully wound 138.4 g), and the run's shape the sheet's: its readings -4.7 to +4.3 % about their mean, the hump at turns 1–1.5, the dip at 4–5.75 and the rise to full wind (the solved spring through the drawn fusee fits it to 2.10 % rms). Low everywhere: the strip is thin (set up a tooth more, or a stronger strip: wempe_tolerances.js); out of shape at one end: the spring's length or set against the fusee. The sources' band, 225–270°, would want 134.3 g mean (×1.012): open (README, Unresolved).

What it does (05 06 04-2). The cone with its nine-turn groove, pressed on the arbor's collar.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 12.35.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 7.01 through (8.25 thick at 1.60–9.85)
2	4.54	-3.53	Ø 2.04 × 6.25 deep from the face at 9.85
3	-1.28	-5.86	Ø 1.74 × 6.00 deep from the face at 1.60

Contour in the plan

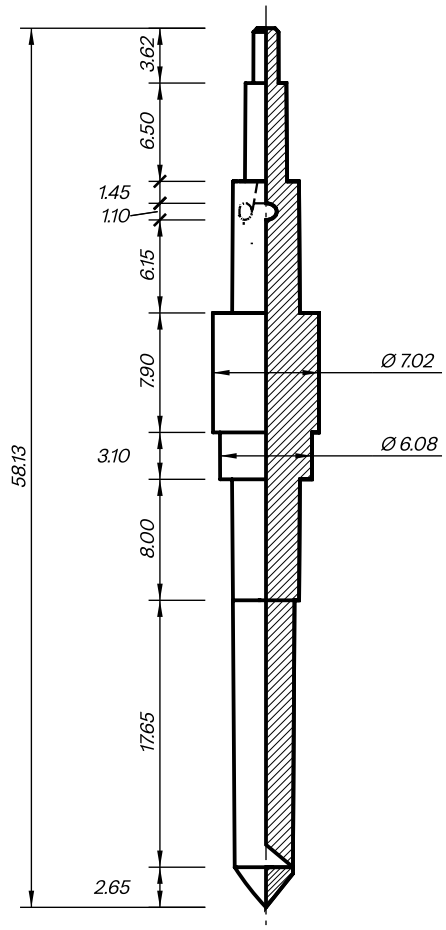
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

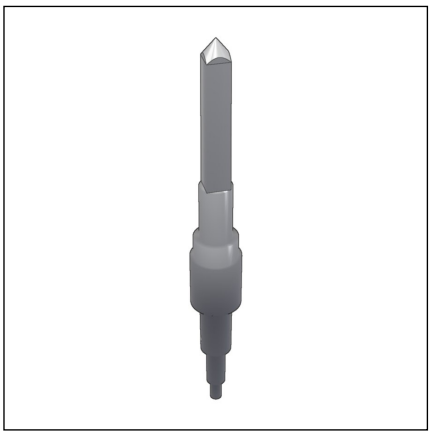
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			0.00, 0.00	17.80
Cut-out 1	2	line	8.41, 15.28	8.41, 15.28		
Cut-out 1	3	line	8.41, 15.28	8.41, 15.28		
Cut-out 1	4	line	8.41, 15.28	8.41, 15.28		
Cut-out 1	5	line	8.41, 15.28	8.41, 15.28		
Cut-out 3	6	arc anticlockwise	2.30, 7.09	2.30, 7.09	1.27, 3.49	3.74
Cut-out 3	7	arc anticlockwise	2.30, 7.09	2.30, 7.09	1.42, 3.44	3.75
Cut-out 4	8	arc anticlockwise	-6.54, -15.54	-6.53, -15.69	-6.32, -15.59	0.24
Cut-out 4	9	line	-6.53, -15.69	-6.48, -15.77		
Cut-out 4	10	arc anticlockwise	-6.48, -15.77	-6.13, -15.82	-6.27, -15.58	0.28
Cut-out 4	11	line	-6.13, -15.82	-6.08, -15.79		
Cut-out 4	12	line	-6.08, -15.79	-6.04, -15.75		
Cut-out 4	13	line	-6.04, -15.75	-6.01, -15.68		
Cut-out 4	14	line	-6.01, -15.68	-5.99, -15.61		
Cut-out 4	15	line	-5.99, -15.61	-6.00, -15.53		

Cut-out 4	16	line	-6.00, -15.53	-6.02, -15.47		
Cut-out 4	17	line	-6.02, -15.47	-6.07, -15.39		
Cut-out 4	18	line	-6.07, -15.39	-6.11, -15.36		
Cut-out 4	19	line	-6.11, -15.36	-6.19, -15.32		
Cut-out 4	20	line	-6.19, -15.32	-6.25, -15.31		
Cut-out 4	21	line	-6.25, -15.31	-6.32, -15.31		
Cut-out 4	22	line	-6.32, -15.31	-6.40, -15.34		
Cut-out 4	23	line	-6.40, -15.34	-6.46, -15.39		
Cut-out 4	24	line	-6.46, -15.39	-6.50, -15.43		
Cut-out 4	25	arc anticlockwise	-6.50, -15.43	-6.54, -15.52	-6.35, -15.55	0.19
Cut-out 4	26	line	-6.54, -15.52	-6.54, -15.54		

Levels

Faces in section A-A, from the bottom face: 1.50, 1.60, 9.85.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



Werkstoff / Material „Präz. Rundstahl 7,5 Din 175" (precision round steel 7,5 dia, DIN 175)	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 04-1*** *Teil / Part* *fusee.arbor*
Behandlung / Treatment untoleranced sizes ±0,1		*Benennung / Name* ***Fusee arbor (Schneckenwelle)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Fusee arbor (Schneckenwelle) — sheet 2

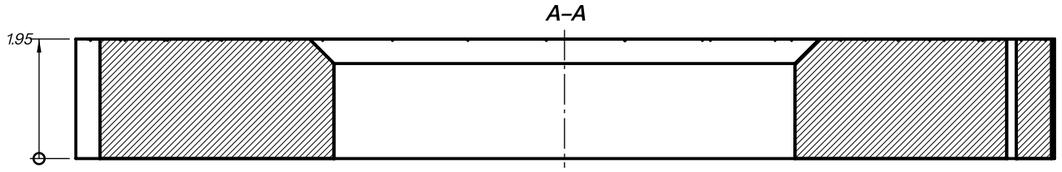
The steel arbor: the dial-side pivot in the pillar plate, the Vorreiber's seat, the collar the body is pressed on, the step for the cover plate, the journal in the fusee bush and the winding square. Its stub on the dial side carries the up-and-down pinion (the motion work's).

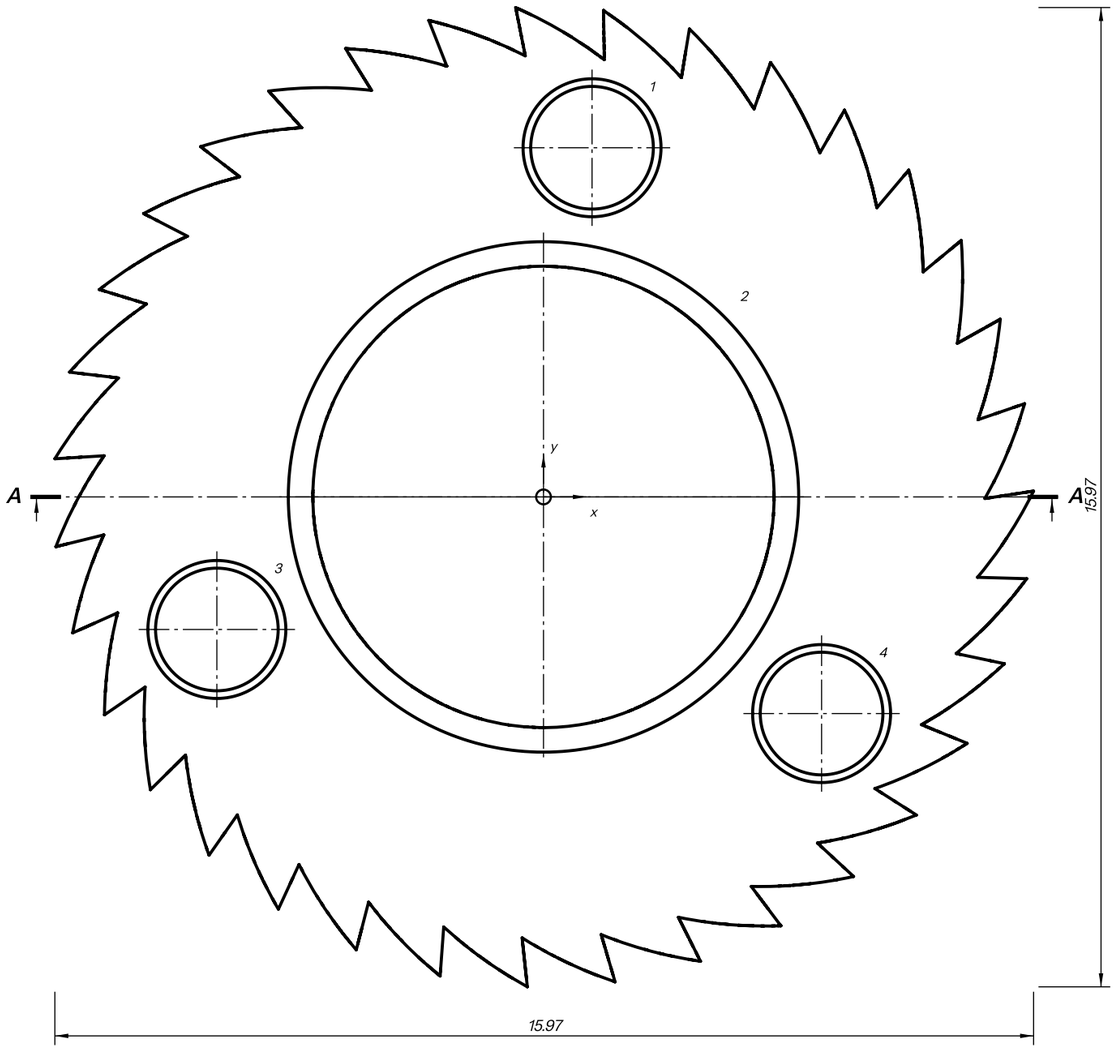
Parts list

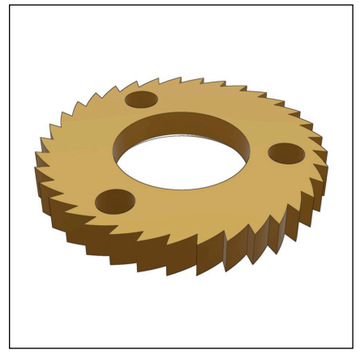
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 04-1	Schneckenwelle (fusee arbor)	1	„Präz. Rundstahl 7,5 Din 175“ (precision round steel 7,5 dia, DIN 175)	untoleranced sizes ±0,1	in 050802-1 (run): 3.249 mm, least gap 0.0133; in 050801-2 (run): 5.776 mm, least gap 0.0163

What it does (05 06 04-1). Runs in the pillar plate (its dial-side pivot) and the fusee bush (its journal); its square takes the key.

Turned about its axis; length 58.13, largest Ø 7.03. Diameters: Ø 6.08, Ø 7.02. Shoulders from the lower end: 0.00, 2.65, 20.30, 28.30, 31.40, 39.30, 45.45, 46.55, 48.00, 54.50, 58.13.







Isometric view, not to scale

Werkstoff / Material *„Ms Blech 2mm DIN 1751 Ms 58 federhart" (brass sheet 2 mm, DIN 1751, Ms 58 spring-hard)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 04-3***
Behandlung / Treatment *surface ▽▽ ((DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*			*Teil / Part* *fusee.clickWheel*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Click wheel (Schneckensperrrad)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Click wheel (Schneckensperrrad) — sheet 2

A 35-tooth brass ratchet screwed to the body's large end: running, its undercut faces push the maintaining clicks forward; winding, they ride over its backs.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 04-3	Schneckensperrrad (click wheel)	1	„Ms Blech 2mm DIN 17.51 Ms 58 federhart“ (brass sheet 2 mm, DIN 1751, Ms 58 spring-hard)	surface ▽▽ ((DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	on 050604-2: 0 mm

What it does (05 06 04-3). On the body's large end, in its recess: drives the maintaining clicks.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.95.

Holes

Hole	x	y	Size
1	0.79	5.70	Ø 2.00 through (0.28 thick at 0.00–0.28); sink Ø 2.25 × 0.19 from the top
2	0.00	0.00	Ø 7.53 through (1.57 thick at 0.00–1.57); countersunk to Ø 8.32
3	-5.33	-2.16	Ø 2.00 through (0.28 thick at 0.00–0.28); sink Ø 2.25 × 0.19 from the top
4	4.54	-3.53	Ø 2.00 through (0.28 thick at 0.00–0.28); sink Ø 2.25 × 0.19 from the top

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

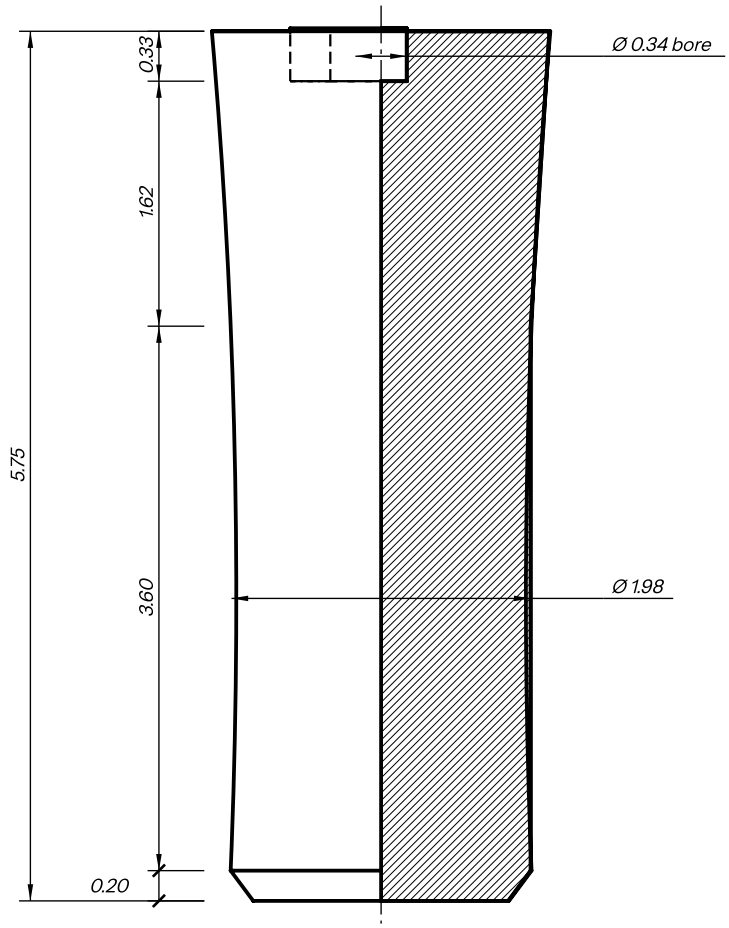
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-6.50, -3.10	-7.16, -3.55		
Outline	2	arc clockwise	-7.16, -3.55	-6.95, -1.89	-0.11, -3.61	7.05
Outline	3	line	-6.95, -1.89	-7.68, -2.22		
Outline	4	arc clockwise	-7.68, -2.22	-7.17, -0.62	-0.76, -3.54	7.05
Outline	5	line	-7.17, -0.62	-7.95, -0.81		
Outline	6	arc clockwise	-7.95, -0.81	-7.17, 0.67	-1.38, -3.34	7.05
Outline	7	line	-7.17, 0.67	-7.97, 0.62		
Outline	8	arc clockwise	-7.97, 0.62	-6.93, 1.94	-1.95, -3.04	7.05
Outline	9	line	-6.93, 1.94	-7.73, 2.04		
Outline	10	arc clockwise	-7.73, 2.04	-6.48, 3.15	-2.46, -2.65	7.05
Outline	11	line	-6.48, 3.15	-7.24, 3.38		
Outline	12	arc clockwise	-7.24, 3.38	-5.81, 4.25	-2.90, -2.17	7.05
Outline	13	line	-5.81, 4.25	-6.52, 4.62		
Outline	14	arc clockwise	-6.52, 4.62	-4.96, 5.22	-3.24, -1.61	7.05
Outline	15	line	-4.96, 5.22	-5.59, 5.71		
Outline	16	arc clockwise	-5.59, 5.71	-3.94, 6.02	-3.47, -1.01	7.05
Outline	17	line	-3.94, 6.02	-4.48, 6.62		
Outline	18	arc clockwise	-4.48, 6.62	-2.81, 6.63	-3.60, -0.37	7.05
Outline	19	line	-2.81, 6.63	-3.23, 7.31		
Outline	20	arc clockwise	-3.23, 7.31	-1.58, 7.03	-3.61, 0.27	7.05

Outline	21	line	-1.58, 7.03	-1.87, 7.77		
Outline	22	arc clockwise	-1.87, 7.77	-0.30, 7.19	-3.50, 0.91	7.05
Outline	23	line	-0.30, 7.19	-0.45, 7.98		
Outline	24	arc clockwise	-0.45, 7.98	0.99, 7.13	-3.28, 1.52	7.05
Outline	25	line	0.99, 7.13	0.98, 7.93		
Outline	26	arc clockwise	0.98, 7.93	2.25, 6.84	-2.95, 2.08	7.05
Outline	27	line	2.25, 6.84	2.38, 7.63		
Outline	28	arc clockwise	2.38, 7.63	3.43, 6.33	-2.53, 2.58	7.05
Outline	29	line	3.43, 6.33	3.70, 7.08		
Outline	30	arc clockwise	3.70, 7.08	4.51, 5.61	-2.03, 2.99	7.05
Outline	31	line	4.51, 5.61	4.91, 6.31		
Outline	32	arc clockwise	4.91, 6.31	5.44, 4.72	-1.47, 3.30	7.05
Outline	33	line	5.44, 4.72	5.96, 5.33		
Outline	34	arc clockwise	5.96, 5.33	6.19, 3.67	-0.85, 3.51	7.05
Outline	35	line	6.19, 3.67	6.81, 4.18		
Outline	36	arc clockwise	6.81, 4.18	6.75, 2.51	-0.21, 3.61	7.05
Outline	37	line	6.75, 2.51	7.45, 2.90		
Outline	38	arc clockwise	7.45, 2.90	7.09, 1.26	0.44, 3.59	7.05
Outline	39	line	7.09, 1.26	7.85, 1.52		
Outline	40	arc clockwise	7.85, 1.52	7.20, -0.03	1.07, 3.45	7.05
Outline	41	line	7.20, -0.03	7.99, 0.10		
Outline	42	arc clockwise	7.99, 0.10	7.08, -1.31	1.67, 3.21	7.05
Outline	43	line	7.08, -1.31	7.88, -1.33		
Outline	44	arc clockwise	7.88, -1.33	6.73, -2.55	2.22, 2.86	7.05
Outline	45	line	6.73, -2.55	7.52, -2.72		
Outline	46	arc clockwise	7.52, -2.72	6.17, -3.72	2.69, 2.42	7.05
Outline	47	line	6.17, -3.72	6.91, -4.02		
Outline	48	arc clockwise	6.91, -4.02	5.40, -4.76	3.08, 1.90	7.05
Outline	49	line	5.40, -4.76	6.08, -5.19		
Outline	50	arc clockwise	6.08, -5.19	4.47, -5.65	3.37, 1.32	7.05
Outline	51	line	4.47, -5.65	5.06, -6.19		
Outline	52	arc clockwise	5.06, -6.19	3.39, -6.35	3.55, 0.69	7.05
Outline	53	line	3.39, -6.35	3.87, -6.99		
Outline	54	arc clockwise	3.87, -6.99	2.20, -6.86	3.62, 0.05	7.05
Outline	55	line	2.20, -6.86	2.56, -7.57		
Outline	56	arc clockwise	2.56, -7.57	0.94, -7.14	3.57, -0.60	7.05
Outline	57	line	0.94, -7.14	1.17, -7.91		
Outline	58	arc clockwise	1.17, -7.91	-0.35, -7.19	3.40, -1.22	7.05
Outline	59	line	-0.35, -7.19	-0.26, -7.99		
Outline	60	arc clockwise	-0.26, -7.99	-1.63, -7.01	3.13, -1.81	7.05
Outline	61	line	-1.63, -7.01	-1.69, -7.82		
Outline	62	arc clockwise	-1.69, -7.82	-2.85, -6.61	2.76, -2.34	7.05
Outline	63	line	-2.85, -6.61	-3.05, -7.39		
Outline	64	arc clockwise	-3.05, -7.39	-3.99, -5.99	2.29, -2.80	7.05
Outline	65	line	-3.99, -5.99	-4.32, -6.72		
Outline	66	arc clockwise	-4.32, -6.72	-4.99, -5.19	1.76, -3.16	7.05
Outline	67	line	-4.99, -5.19	-5.46, -5.84		
Outline	68	arc clockwise	-5.46, -5.84	-5.84, -4.21	1.16, -3.42	7.05

Outline	69	line	-5.84, -4.21	-6.41, -4.78		
Outline	70	arc clockwise	-6.41, -4.78	-6.50, -3.10	0.53, -3.58	7.05

Levels

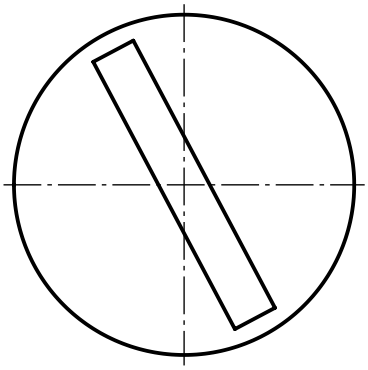
Faces in section A-A, from the bottom face: 1.95.





Isometric view, not to scale

View from above



3 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-19*** *Teil / Part* *fusee.screws*
Behandlung / Treatment		*Benennung / Name* ***Click wheel screws (WN-19)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Click wheel screws (WN-19) — sheet 2

Three screws holding the click wheel to the body, their heads sunk flush in its 1:8 holes.

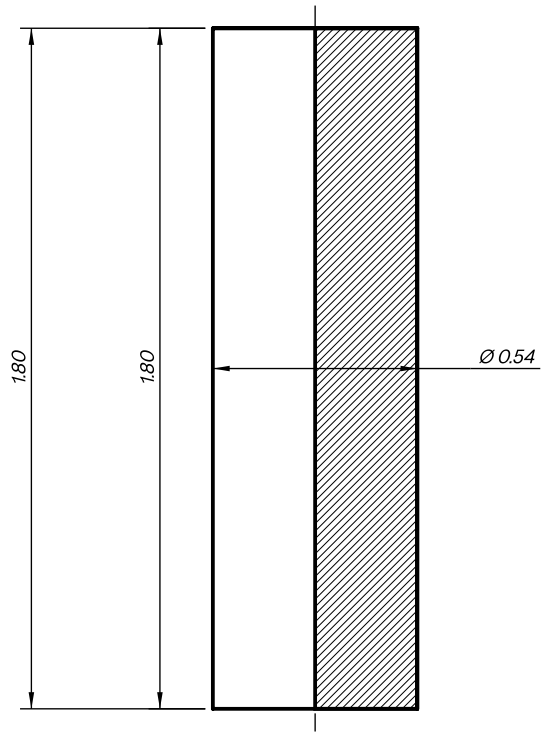
3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

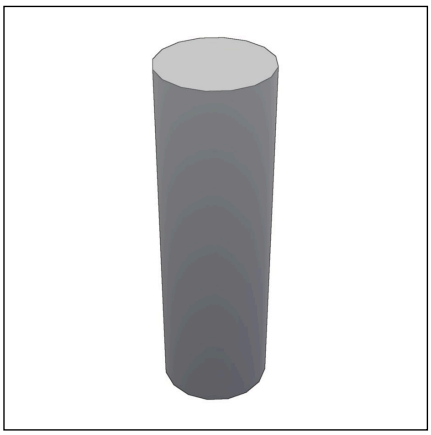
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-19	Schraube WN-19 (click wheel screw)	3	not on the drawing		in 050604-3 (press): 1.627 mm, least gap 0.0025; in 050604-2 (tap): 3.796 mm, least gap 0.0225

What it does (WN-19). Holds the click wheel to the body; its head flush in the wheel's 1:8 hole, its thread in the body's M2, 6 +0.5 deep from the large end's recess floor (05 06 04-2).

Turned about its axis; length 5.77, largest Ø 2.24. Diameters: Ø 1.98; bores Ø 0.34. Shoulders from the lower end: 0.00, 0.20, 3.80, 5.42, 5.75.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **50:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 00-6*** *Teil / Part* *fusee.pin*
Behandlung / Treatment		*Benennung / Name* ***Chain pin (Kettenstift)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Chain pin (Kettenstift) — sheet 2

The pin across the flange’s slot that the chain’s (or band’s) top hook takes.

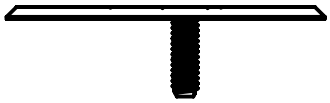
Parts list

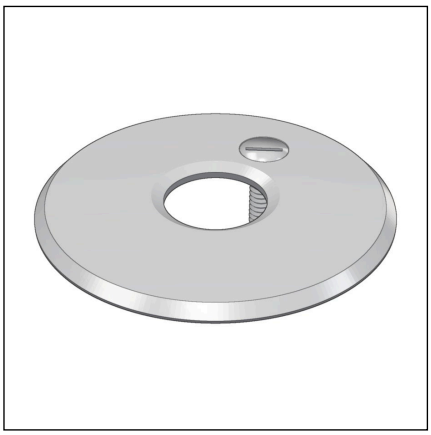
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-6	Kettenstift (chain pin)	1	not on the drawing		in 050604-2 (press): 0.826 mm, least gap 0.0059

What it does (05 06 00-6). Across the flange's slot in its hole through both ribs ("0,55Ø +0,01", 05 06 04-2): the chain's (or band's) top hook takes it.

Turned about its axis; length 1.80, largest Ø 0.54. Diameters: Ø 0.54. Shoulders from the lower end: 0.00, 1.80.

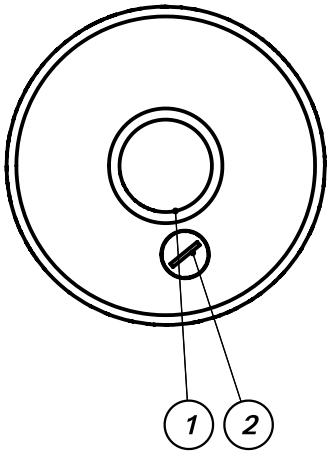
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Cover plate screw (WN-20)	WN-20	not on the drawing
1	1	Cover plate	05 06 00-4	„Bandstahl 24×1 Din 1016“ (str

Werkstoff / Material: „Bandstahl 24×1 Din 1016“ (strip steel 24 × 1, DIN 1016)

Maßstab / Scale: **2:1**

Behandlung / Treatment: surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)

Blatt / Sheet: **1 / 1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

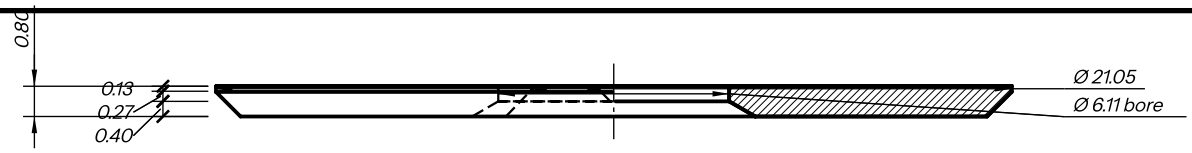
Zeichnung Nr. / Drawing no.: **05 06 00-4**

Teil / Part: coverPlate

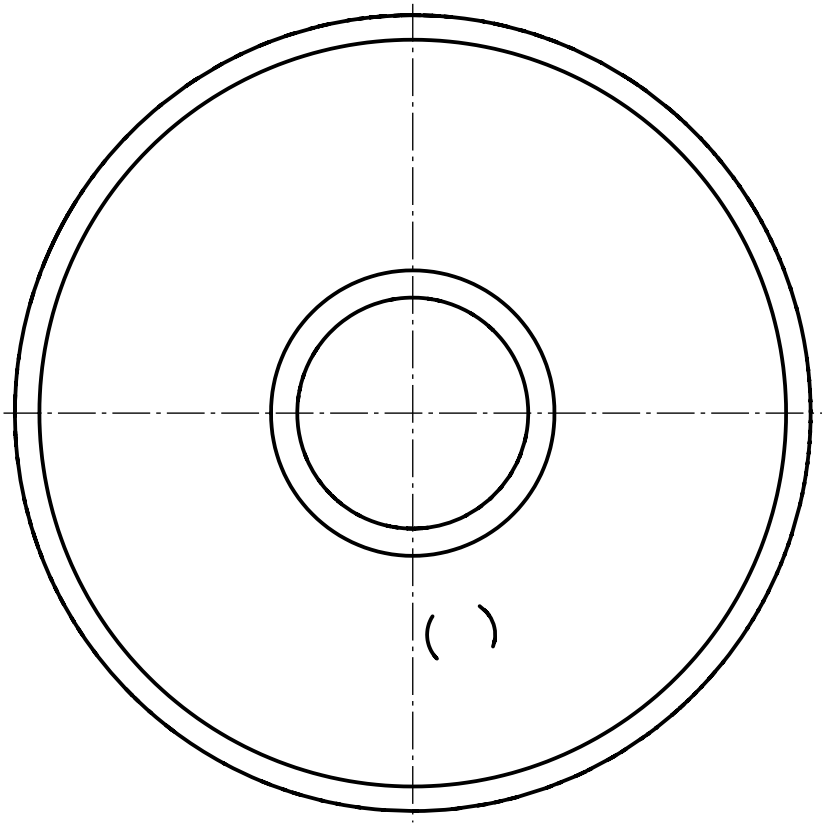
Benennung / Name: **Cover plate (Deckplatte)**

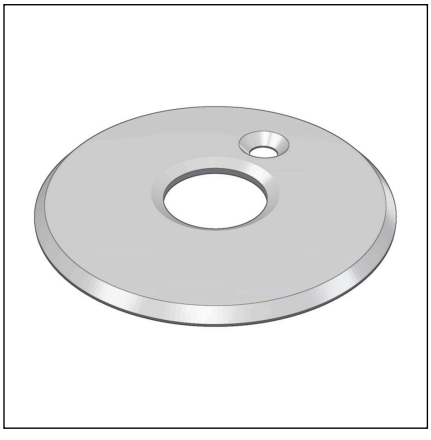
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).



View from below (seen from beneath, x reversed)





Isometric view, not to scale

Werkstoff / Material „Bandstahl 24×1 Din 1016" (strip steel 24 × 1, DIN 1016)	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 00-4*** *Teil / Part* *coverPlate.plate*
Behandlung / Treatment surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)		*Benennung / Name* ***Cover plate***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Cover plate — sheet 2

A steel disc on the fusee's small end, screwed to it, over the stop bar's slot: it keeps the bar in. Wempe 3955's fusee shows it bright, the bar's end standing out from under it.

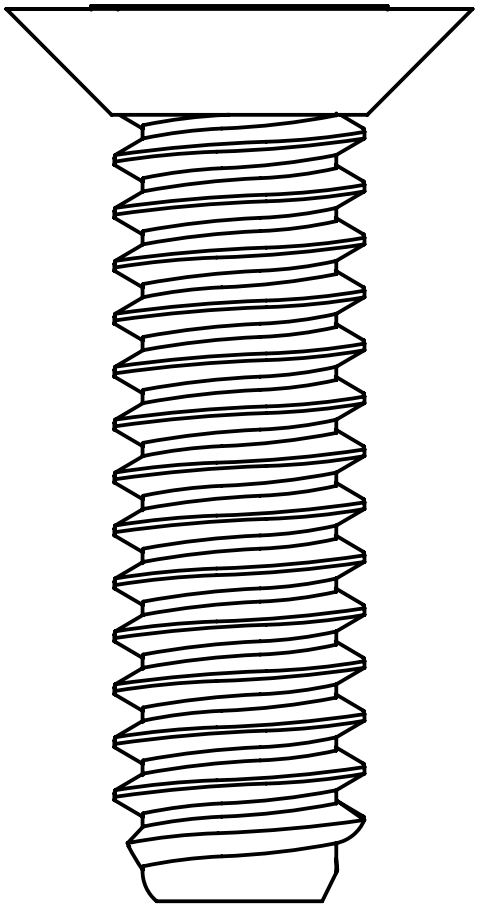
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-4	Deckplatte (cover plate)	1	„Bandstahl 24×1 Din 1016“ (strip steel 24 × 1, DIN 1016)	surface ∇∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	round 050604-1 (run): 0.361 mm, least gap 0.0092; on 050604-2: 0 mm

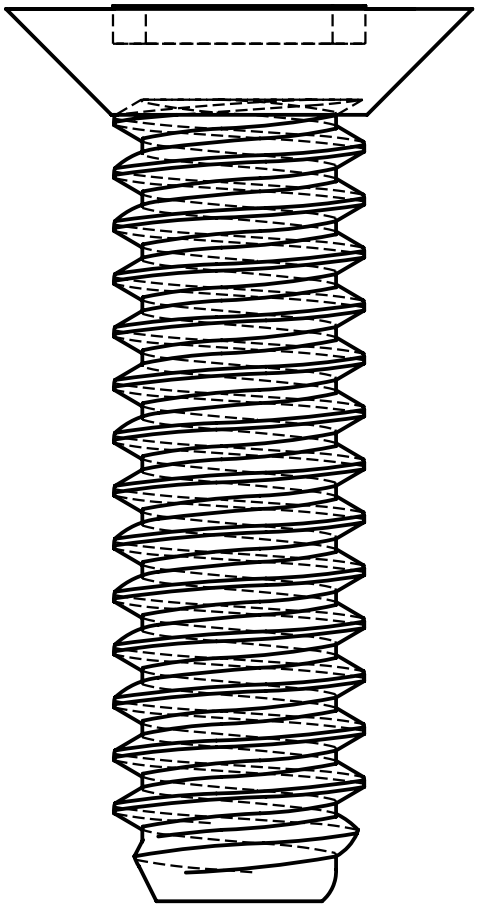
What it does (05 06 00-4). On the arbor's step against the body's small end, over the stop bar.

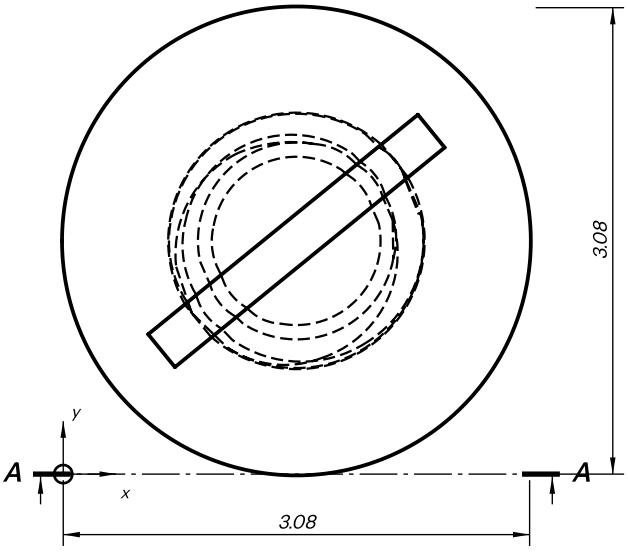
Turned about its axis; length 0.80, largest Ø 21.05. Diameters: Ø 21.05; bores Ø 6.11. Shoulders from the lower end: 0.00, 0.40, 0.67, 0.80.

A–A



View from the left







Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale 20:1	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-20** Teil / Part coverPlate.screw
Behandlung / Treatment		Benennung / Name **Cover plate screw (WN-20)**	
Blatt / Sheet 1 / 2		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Cover plate screw (WN-20) — sheet 2

The countersunk screw holding the cover plate to the fusee's small end.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-20	Schraube WN-20 (cover plate screw)	1	not on the drawing		in 050600-4 (press): 0.493 mm, least gap 0.0143; in 050604-2 (tap): 3.5 mm, least gap 0.0207

What it does (WN-20). Countersunk in the cover plate, into the body's M1.7, 6 deep from the small end's recess floor (05 06 04-2).

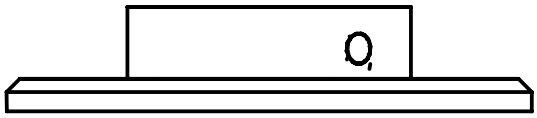
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 5.92.

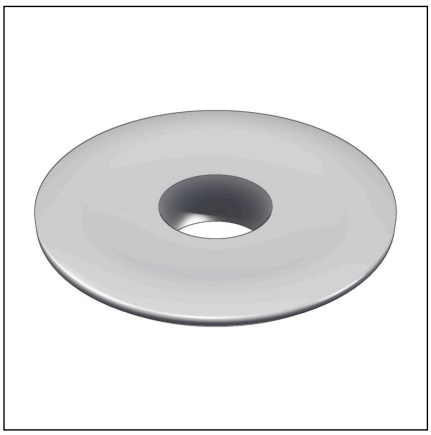
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			1.54, 1.54	1.55
Cut-out 1	2	line	0.91, 2.11	0.91, 2.11		
Cut-out 1	3	line	0.91, 2.11	0.91, 2.11		
Cut-out 1	4	line	0.91, 2.11	0.91, 2.11		
Cut-out 1	5	line	0.91, 2.11	0.91, 2.11		

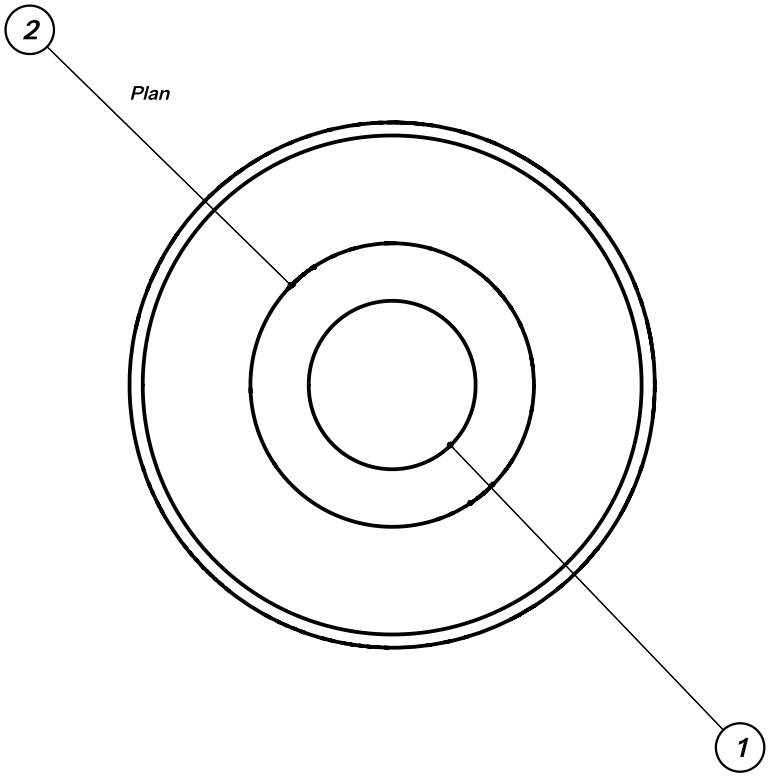
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Vorreiber pin (Vorreiberstift)	05 06 00-5	not on the drawing
1	1	Vorreiber	05 06 00-2	„Präz. Rundstahl 14 Din 175“ (

Werkstoff / Material
„Präz. Rundstahl 14 Din 175“ (precision round steel bar 14 dia, DIN 175; "Rundstahl" cut off at the box edge, likely)

Maßstab / Scale
5:1

Behandlung / Treatment
surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

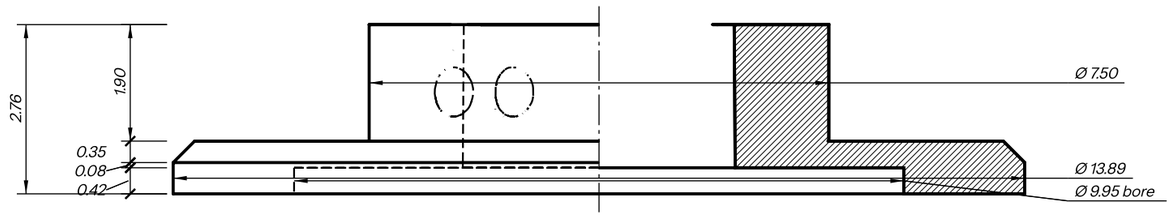
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 06 00-2

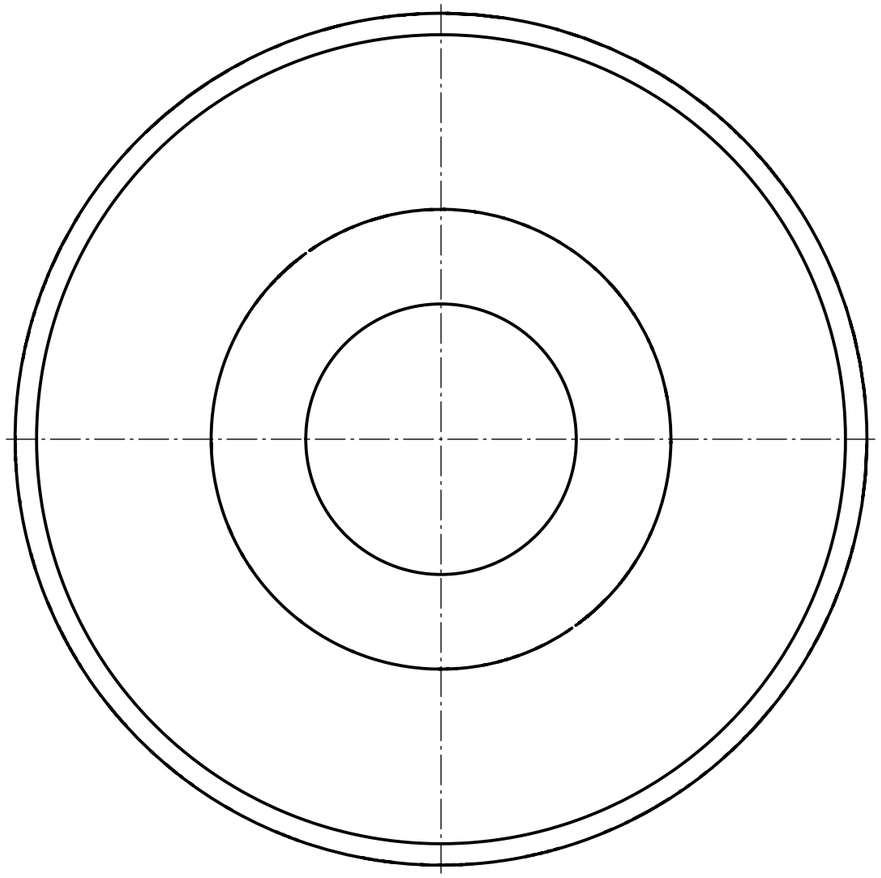
Teil / Part
vorreiber

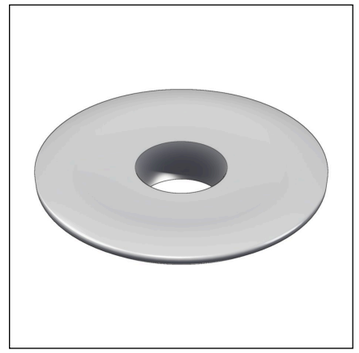
Benennung / Name
Vorreiber

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).



View from above





Isometric view, not to scale

Werkstoff / Material *„Präz. Rundstahl 14 Din 175" (precision round steel bar 14 dia, DIN 175; "Rundstahl" cut off at the box edge, likely)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 00-2***
Behandlung / Treatment *surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*			*Teil / Part* *vorreiber.collar*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Vorreiber***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Vorreiber — sheet 2

A steel collar pinned to the fusee arbor under the fusee wheel, its flange in the wheel's underside recess: it keeps the wheel and the maintaining work on the arbor. Its face under the wheel is re-turned at the fusee's assembly ("bei Schneckenmontage … Fläche a nachdrehen"), which sets the endshake of the stack.

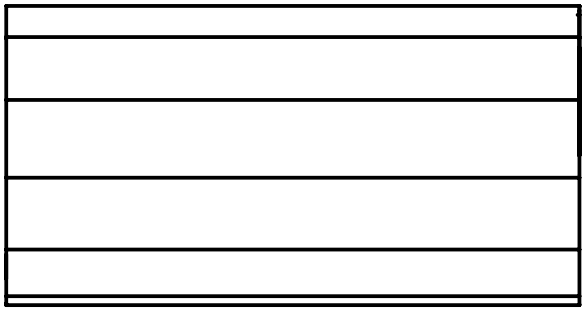
Parts list

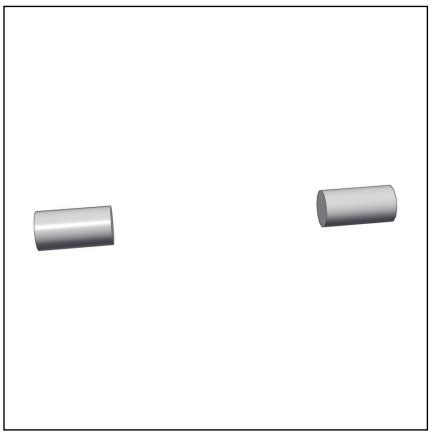
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-2	Vorreiber	1	„Präz. Rundstahl 14 Din 175“ (precision round steel bar 14 dia, DIN 175; "Rundstahl" cut off at the box edge, likely)	surface ∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	round 050604-1 (press): 2.166 mm, least gap 0.0134

What it does (05 06 00-2). Pinned on the arbor's taper under the fusee wheel, its flange in the wheel's underside recess.

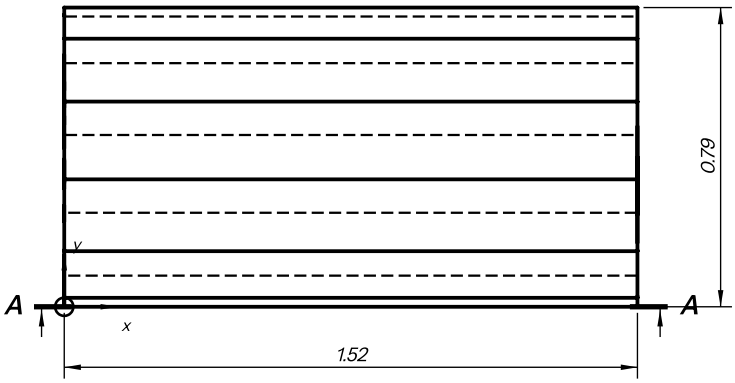
Turned about its axis; length 2.76, largest Ø 13.89. Diameters: Ø 7.50, Ø 13.89; bores Ø 9.95. Shoulders from the lower end: 0.00, 0.42, 0.51, 0.86, 2.76.

A–A





Isometric view, not to scale



2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***50:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 00-5***
Behandlung / Treatment			*Teil / Part* *vorreiber.pin*
		Benennung / Name ***Vorreiber pin (Vorreiberstift)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Vorreiber pin (Vorreiberstift) — sheet 2

The taper pin through the Vorreiber's hub and the arbor's cross hole.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-5	Vorreiberstift (Vorreiber pin)	1	not on the drawing		in 050600-2 (press): 3.091 mm, least gap -0.0004

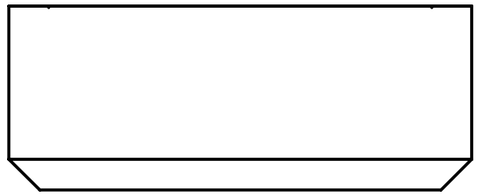
What it does (05 06 00-5). Through the Vorreiber's hub (its Ø 0.8 hole, 05 06 00-2) and the arbor's cross hole (Ø 1.1, 05 06 04-1): drawn as its two ends in the hub, so the Exploded view can part the arbor from the Vorreiber.

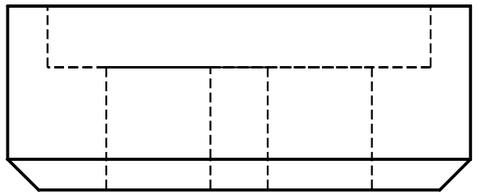
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 0.79.

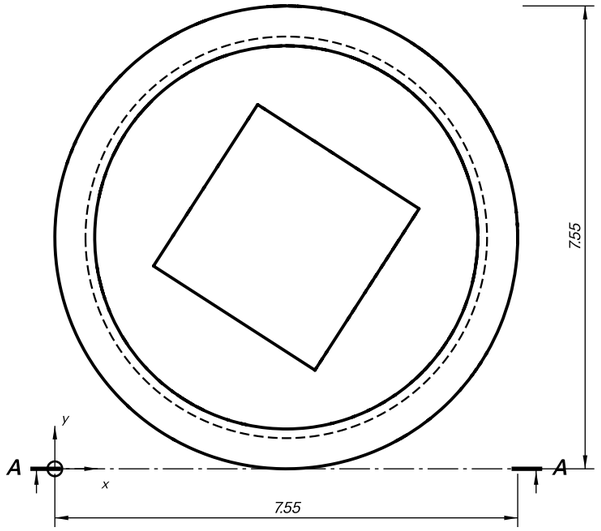
Contour in the plan

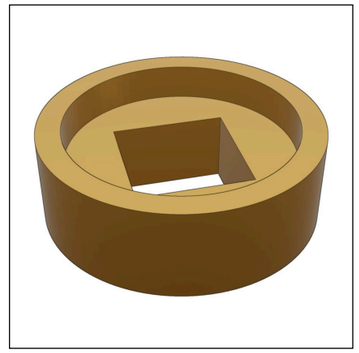
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.25	0.00, 0.79		
Outline	2	line	0.00, 0.79	1.52, 0.79		
Outline	3	line	1.52, 0.79	1.52, 0.00		
Outline	4	line	1.52, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 0.25		









Isometric view, not to scale

Werkstoff / Material *„Rundmessing 8 Din 1758 Ms58" (round brass bar 8 dia, DIN 1758, Ms 58)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 00-1*** *Teil / Part* *dustPlate*
Behandlung / Treatment *none on the drawing*		*Benennung / Name* ***Dust plate (Staubplatte)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Dust plate (Staubplatte) — sheet 2

A brass disc on the winding square, turning with it inside the dust bush, which keeps dust from the winding hole out of the fusee bush.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-1	Staubplatte (dust plate)	1	„Rundmessing 8 Din 1758 Ms58“ (round brass bar 8 dia, DIN 1758, Ms 58)	none on the drawing	round 050604-1 (press): 1.805 mm, least gap 0.0034

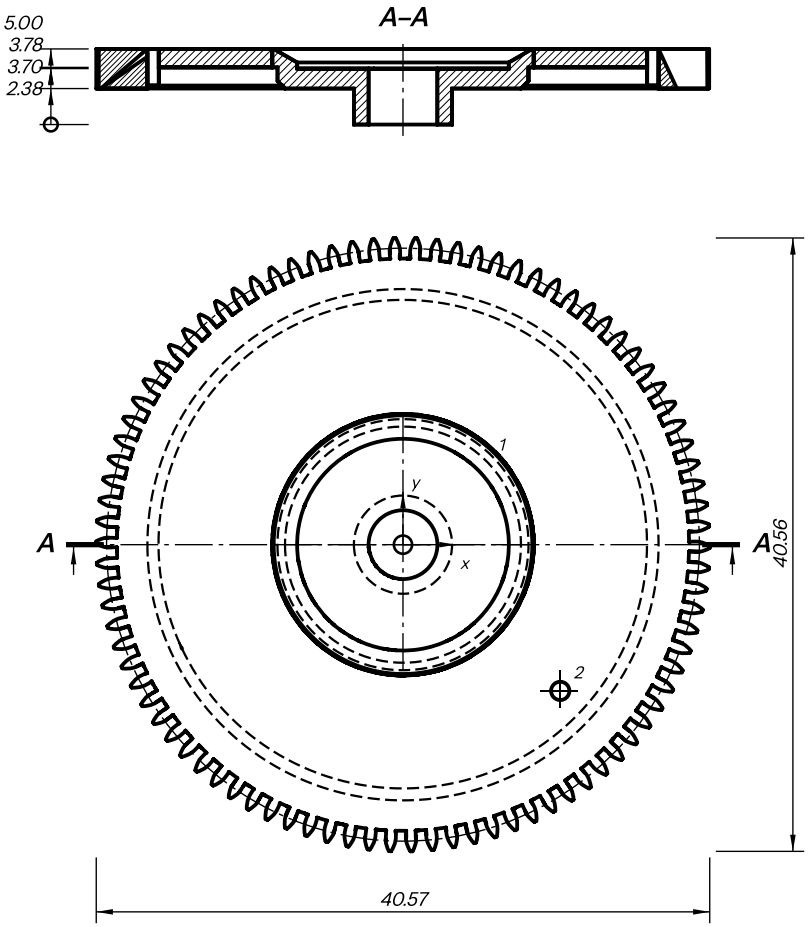
What it does (05 06 00-1). On the winding square, inside the dust bush.

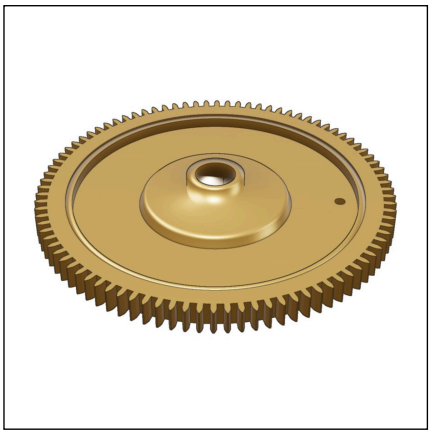
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.00.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			3.77, 3.77	3.77
Cut-out 1	2	line	1.61, 3.31	4.24, 1.61		
Cut-out 1	3	line	4.24, 1.61	5.94, 4.24		
Cut-out 1	4	line	5.94, 4.24	3.31, 5.94		
Cut-out 1	5	line	3.31, 5.94	1.61, 3.31		





Isometric view, not to scale

Werkstoff / Material „Messingblech 5,5 Din 1751 Ms 58 federhart" (brass sheet 5,5 thick, DIN 1751, Ms 58, spring-hard)	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 00-3*** *Teil / Part* *fuseeWheel*
Behandlung / Treatment surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)		*Benennung / Name* ***Fusee wheel (Schneckenrad)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Fusee wheel (Schneckenrad) — sheet 2

The great wheel: it drives the centre pinion, so the whole train. It turns loose on the fusee arbor, driven by the maintaining spring, which lies in the recess in its face; the maintaining ratchet turns on the tube on its hub.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-3	Schneckenrad (fusee wheel)	1	„Messingblech 5,5 Din 1751 Ms 58 federhart“ (brass sheet 5,5 thick, DIN 1751, Ms 58, spring-hard)	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	-6.428571

What it does (05 06 00-3). Loose on the arbor's seat, driven by the maintaining spring; drives the centre pinion.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 5.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 4.55 through (3.70 thick at 0.00–3.70); countersunk to Ø 15.60
2	10.40	-9.67	Ø 1.21 through (1.22 thick at 3.78–5.00)

The contour has 757 elements (its teeth): the drawing and the tooth data define it.

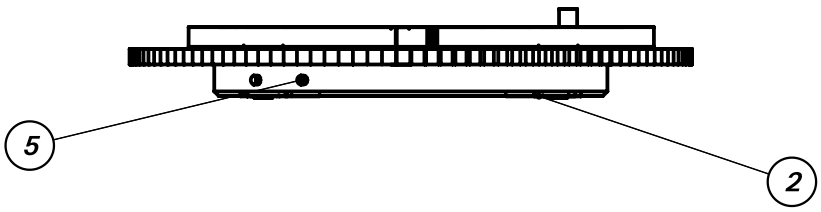
Levels

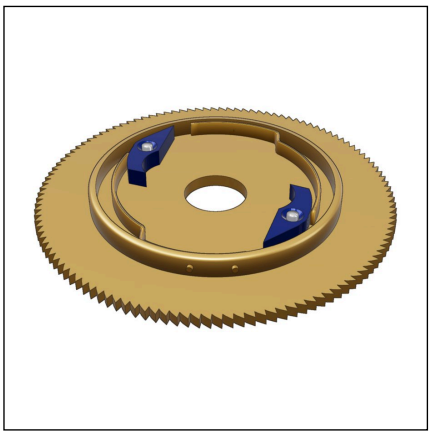
Faces in section A-A, from the bottom face: 2.38, 3.70, 3.78, 5.00.

Teeth

Number	Module	Tip Ø	Root Ø
90	0.4356	40.57	37.84

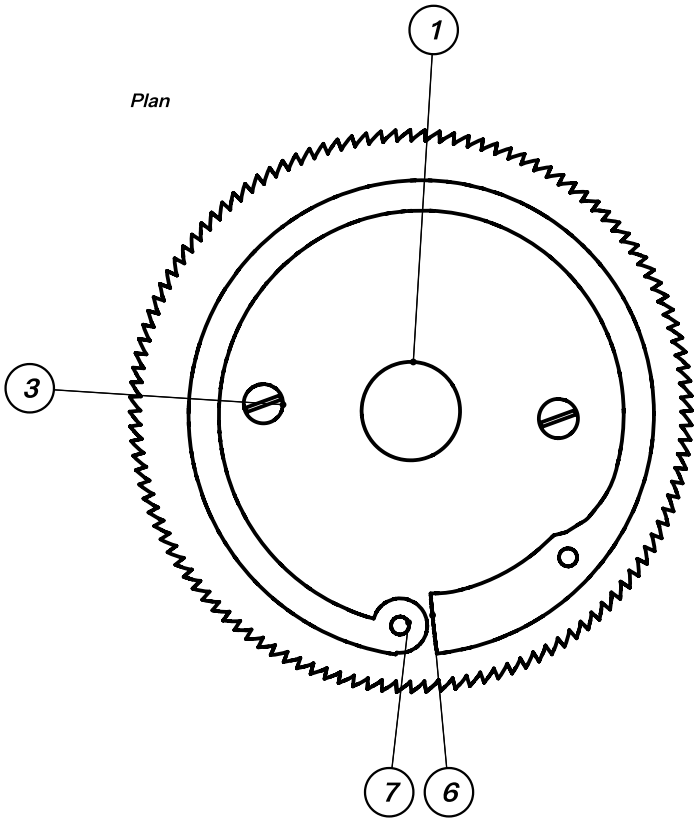
Front view





Isometric view, not to scale

Plan



Not seen in either view (inside the assembly): Pos. 4.

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
7	2	Maintaining spring's pins (Stellstifte)	05 06 00-8	not on the drawing
6	1	Maintaining spring (Gegenfeder)	05 06 01-1	„Bandstahl 35×1,75 Din 1016“ (
5	4	Click spring pins	05 06 02-4	not on the drawing
4	2	Click springs (Sperrfeder)	05 06 02-2	„Bronzeband 1,5×0,4 federhart“
3	2	Click screws (WN-21)	WN-21	not on the drawing
2	2	Clicks (Sperrkegel)	05 06 02-3	„Bandstahl 11×2 Din 1016“ (str
1	1	Maintaining ratchet (Gegensperrrad)	05 06 02-1	„Messingblech 3,5 Din 1751 Ms

Werkstoff / Material: not on the drawing

Maßstab / Scale: **2:1**

Behandlung / Treatment:

Blatt / Sheet: **1 / 1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **05 06 00-8**

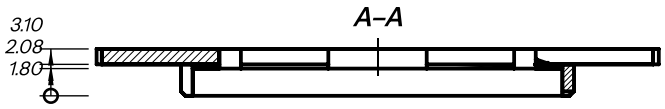
Teil / Part: maintaining

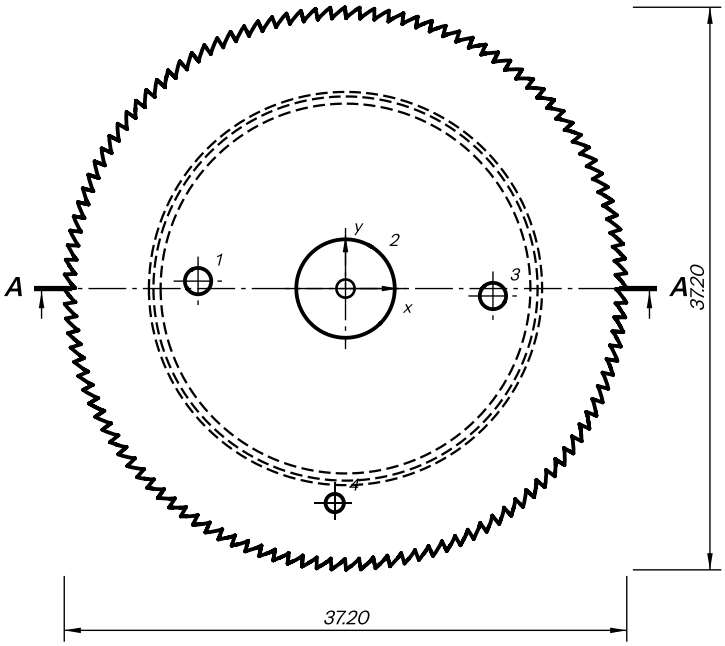
Benennung / Name

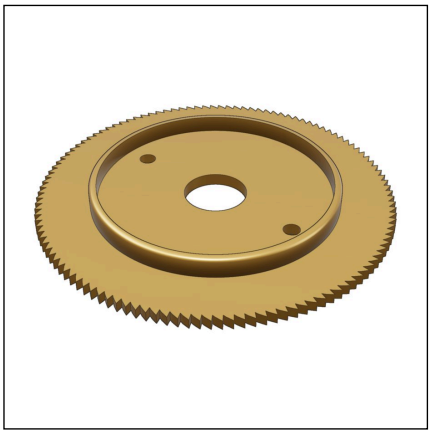
Maintaining work (Gegensperrrad, Gegenfeder)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).







Isometric view, not to scale

Werkstoff / Material „Messingblech 3,5 Din 1751 Ms 58 federhart" (brass sheet 3,5, DIN 1751, Ms 58 spring-hard)	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 02-1*** *Teil / Part* *maintaining.ratchet*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*		*Benennung / Name* ***Maintaining ratchet (Gegensperrrad)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Maintaining ratchet (Gegensperrrad) — sheet 2

A brass ratchet of 120 teeth turning on the fusee wheel's tube, its cup holding the two clicks; the maintaining detent rides on its teeth.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 02-1	Gegensperrrad (maintaining ratchet)	1	„Messingblech 3,5 Din 1751 Ms 58 federhart“ (brass sheet 3,5, DIN 1751, Ms 58 spring-hard)	surface ∇∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	round 050600-3 (free): 1.323 mm, least gap 0.0088

What it does (05 06 02-1). Turns on the fusee wheel's tube; the maintaining detent rides on its teeth.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.10.

Holes

Hole	x	y	Size
1	-9.75	0.49	Ø 1.74 through (1.02 thick at 2.08-3.10)
2	0.00	0.00	Ø 6.51 through (1.02 thick at 2.08-3.10)
3	9.75	-0.49	Ø 1.74 through (1.02 thick at 2.08-3.10)
4	-0.71	-14.18	Ø 1.21 through (1.02 thick at 2.08-3.10)

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-15.72, -9.49	-16.09, -9.33		
Outline	2	line	-16.09, -9.33	-15.50, -8.91		
Outline	3	arc clockwise	-15.50, -8.91	-16.19, -8.65	-15.00, -6.49	2.47
Outline	4	line	-16.19, -8.65	-16.56, -8.48		
Outline	5	line	-16.56, -8.48	-15.94, -8.08		
Outline	6	arc clockwise	-15.94, -8.08	-16.62, -7.79	-15.32, -5.70	2.47
Outline	7	line	-16.62, -7.79	-16.98, -7.60		
Outline	8	line	-16.98, -7.60	-16.34, -7.24		
Outline	9	arc clockwise	-16.34, -7.24	-17.01, -6.91	-15.60, -4.89	2.47
Outline	10	line	-17.01, -6.91	-17.35, -6.70		
Outline	11	line	-17.35, -6.70	-16.70, -6.37		
Outline	12	arc clockwise	-16.70, -6.37	-17.35, -6.01	-15.83, -4.06	2.47
Outline	13	line	-17.35, -6.01	-17.68, -5.78		
Outline	14	line	-17.68, -5.78	-17.01, -5.49		
Outline	15	arc clockwise	-17.01, -5.49	-17.64, -5.10	-16.03, -3.23	2.47
Outline	16	line	-17.64, -5.10	-17.96, -4.85		
Outline	17	line	-17.96, -4.85	-17.28, -4.59		
Outline	18	arc clockwise	-17.28, -4.59	-17.88, -4.17	-16.17, -2.39	2.47
Outline	19	line	-17.88, -4.17	-18.19, -3.90		
Outline	20	line	-18.19, -3.90	-17.49, -3.68		
Outline	21	arc clockwise	-17.49, -3.68	-18.07, -3.22	-16.27, -1.54	2.47

Outline	22	line	-18.07, -3.22	-18.37, -2.95		
Outline	23	line	-18.37, -2.95	-17.66, -2.76		
Outline	24	arc clockwise	-17.66, -2.76	-18.22, -2.27	-16.33, -0.68	2.47
Outline	25	line	-18.22, -2.27	-18.49, -1.98		
Outline	26	line	-18.49, -1.98	-17.78, -1.83		
Outline	27	arc clockwise	-17.78, -1.83	-18.31, -1.32	-16.35, 0.17	2.47
Outline	28	line	-18.31, -1.32	-18.57, -1.01		
Outline	29	line	-18.57, -1.01	-17.85, -0.90		
Outline	30	arc clockwise	-17.85, -0.90	-18.35, -0.36	-16.32, 1.03	2.47
Outline	31	line	-18.35, -0.36	-18.60, -0.04		
Outline	32	line	-18.60, -0.04	-17.87, 0.04		
Outline	33	arc clockwise	-17.87, 0.04	-18.35, 0.60	-16.24, 1.88	2.47
Outline	34	line	-18.35, 0.60	-18.58, 0.94		
Outline	35	line	-18.58, 0.94	-17.85, 0.97		
Outline	36	arc clockwise	-17.85, 0.97	-18.29, 1.56	-16.12, 2.73	2.47
Outline	37	line	-18.29, 1.56	-18.50, 1.91		
Outline	38	line	-18.50, 1.91	-17.77, 1.90		
Outline	39	arc clockwise	-17.77, 1.90	-18.18, 2.52	-15.95, 3.57	2.47
Outline	40	line	-18.18, 2.52	-18.38, 2.87		
Outline	41	line	-18.38, 2.87	-17.65, 2.83		
Outline	42	arc clockwise	-17.65, 2.83	-18.03, 3.47	-15.74, 4.40	2.47
Outline	43	line	-18.03, 3.47	-18.20, 3.83		
Outline	44	line	-18.20, 3.83	-17.48, 3.75		
Outline	45	arc clockwise	-17.48, 3.75	-17.82, 4.41	-15.49, 5.21	2.47
Outline	46	line	-17.82, 4.41	-17.98, 4.78		
Outline	47	line	-17.98, 4.78	-17.26, 4.66		
Outline	48	arc clockwise	-17.26, 4.66	-17.57, 5.33	-15.20, 6.02	2.47
Outline	49	line	-17.57, 5.33	-17.70, 5.71		
Outline	50	line	-17.70, 5.71	-16.99, 5.56		
Outline	51	arc clockwise	-16.99, 5.56	-17.26, 6.24	-14.86, 6.81	2.47
Outline	52	line	-17.26, 6.24	-17.38, 6.63		
Outline	53	line	-17.38, 6.63	-16.68, 6.44		
Outline	54	arc clockwise	-16.68, 6.44	-16.91, 7.14	-14.49, 7.57	2.47
Outline	55	line	-16.91, 7.14	-17.01, 7.53		
Outline	56	line	-17.01, 7.53	-16.32, 7.30		
Outline	57	arc clockwise	-16.32, 7.30	-16.52, 8.01	-14.07, 8.32	2.47
Outline	58	line	-16.52, 8.01	-16.59, 8.41		
Outline	59	line	-16.59, 8.41	-15.91, 8.15		
Outline	60	arc clockwise	-15.91, 8.15	-16.07, 8.87	-13.62, 9.05	2.47
Outline	61	line	-16.07, 8.87	-16.13, 9.27		
Outline	62	line	-16.13, 9.27	-15.46, 8.97		
Outline	63	arc clockwise	-15.46, 8.97	-15.59, 9.70	-13.12, 9.75	2.47
Outline	64	line	-15.59, 9.70	-15.62, 10.10		
Outline	65	line	-15.62, 10.10	-14.97, 9.76		
Outline	66	arc clockwise	-14.97, 9.76	-15.06, 10.50	-12.60, 10.42	2.47
Outline	67	line	-15.06, 10.50	-15.07, 10.90		
Outline	68	line	-15.07, 10.90	-14.44, 10.54		
Outline	69	arc clockwise	-14.44, 10.54	-14.49, 11.27	-12.03, 11.07	2.47

Outline	70	line	-14.49, 11.27	-14.48, 11.68		
Outline	71	line	-14.48, 11.68	-13.87, 11.28		
Outline	72	arc clockwise	-13.87, 11.28	-13.88, 12.02	-11.44, 11.68	2.47
Outline	73	line	-13.88, 12.02	-13.85, 12.42		
Outline	74	line	-13.85, 12.42	-13.26, 11.99		
Outline	75	arc clockwise	-13.26, 11.99	-13.23, 12.73	-10.81, 12.26	2.47
Outline	76	line	-13.23, 12.73	-13.18, 13.13		
Outline	77	line	-13.18, 13.13	-12.61, 12.66		
Outline	78	arc clockwise	-12.61, 12.66	-12.55, 13.40	-10.15, 12.81	2.47
Outline	79	line	-12.55, 13.40	-12.47, 13.80		
Outline	80	line	-12.47, 13.80	-11.93, 13.31		
Outline	81	arc clockwise	-11.93, 13.31	-11.83, 14.04	-9.47, 13.33	2.47
Outline	82	line	-11.83, 14.04	-11.73, 14.43		
Outline	83	line	-11.73, 14.43	-11.22, 13.91		
Outline	84	arc clockwise	-11.22, 13.91	-11.08, 14.64	-8.76, 13.80	2.47
Outline	85	line	-11.08, 14.64	-10.96, 15.03		
Outline	86	line	-10.96, 15.03	-10.48, 14.48		
Outline	87	arc clockwise	-10.48, 14.48	-10.30, 15.20	-8.02, 14.24	2.47
Outline	88	line	-10.30, 15.20	-10.16, 15.58		
Outline	89	line	-10.16, 15.58	-9.71, 15.01		
Outline	90	arc clockwise	-9.71, 15.01	-9.49, 15.72	-7.27, 14.64	2.47
Outline	91	line	-9.49, 15.72	-9.33, 16.09		
Outline	92	line	-9.33, 16.09	-8.91, 15.50		
Outline	93	arc clockwise	-8.91, 15.50	-8.65, 16.19	-6.49, 15.00	2.47
Outline	94	line	-8.65, 16.19	-8.48, 16.56		
Outline	95	line	-8.48, 16.56	-8.08, 15.94		
Outline	96	arc clockwise	-8.08, 15.94	-7.79, 16.62	-5.70, 15.32	2.47
Outline	97	line	-7.79, 16.62	-7.60, 16.98		
Outline	98	line	-7.60, 16.98	-7.24, 16.34		
Outline	99	arc clockwise	-7.24, 16.34	-6.91, 17.01	-4.89, 15.60	2.47
Outline	100	line	-6.91, 17.01	-6.70, 17.35		
Outline	101	line	-6.70, 17.35	-6.37, 16.70		
Outline	102	arc clockwise	-6.37, 16.70	-6.01, 17.35	-4.06, 15.83	2.47
Outline	103	line	-6.01, 17.35	-5.78, 17.68		
Outline	104	line	-5.78, 17.68	-5.49, 17.01		
Outline	105	arc clockwise	-5.49, 17.01	-5.10, 17.64	-3.23, 16.03	2.47
Outline	106	line	-5.10, 17.64	-4.85, 17.96		
Outline	107	line	-4.85, 17.96	-4.59, 17.28		
Outline	108	arc clockwise	-4.59, 17.28	-4.17, 17.88	-2.39, 16.17	2.47
Outline	109	line	-4.17, 17.88	-3.90, 18.19		
Outline	110	line	-3.90, 18.19	-3.68, 17.49		
Outline	111	arc clockwise	-3.68, 17.49	-3.22, 18.07	-1.54, 16.27	2.47
Outline	112	line	-3.22, 18.07	-2.95, 18.37		
Outline	113	line	-2.95, 18.37	-2.76, 17.66		
Outline	114	arc clockwise	-2.76, 17.66	-2.27, 18.22	-0.68, 16.33	2.47
Outline	115	line	-2.27, 18.22	-1.98, 18.49		
Outline	116	line	-1.98, 18.49	-1.83, 17.78		
Outline	117	arc clockwise	-1.83, 17.78	-1.32, 18.31	0.17, 16.35	2.47

Outline	118	line	-1.32, 18.31	-1.01, 18.57		
Outline	119	line	-1.01, 18.57	-0.90, 17.85		
Outline	120	arc clockwise	-0.90, 17.85	-0.36, 18.35	1.03, 16.32	2.47
Outline	121	line	-0.36, 18.35	-0.04, 18.60		
Outline	122	line	-0.04, 18.60	0.04, 17.87		
Outline	123	arc clockwise	0.04, 17.87	0.60, 18.35	1.88, 16.24	2.47
Outline	124	line	0.60, 18.35	0.94, 18.58		
Outline	125	line	0.94, 18.58	0.97, 17.85		
Outline	126	arc clockwise	0.97, 17.85	1.56, 18.29	2.73, 16.12	2.47
Outline	127	line	1.56, 18.29	1.91, 18.50		
Outline	128	line	1.91, 18.50	1.90, 17.77		
Outline	129	arc clockwise	1.90, 17.77	2.52, 18.18	3.57, 15.95	2.47
Outline	130	line	2.52, 18.18	2.87, 18.38		
Outline	131	line	2.87, 18.38	2.83, 17.65		
Outline	132	arc clockwise	2.83, 17.65	3.47, 18.03	4.40, 15.75	2.47
Outline	133	line	3.47, 18.03	3.83, 18.20		
Outline	134	line	3.83, 18.20	3.75, 17.48		
Outline	135	arc clockwise	3.75, 17.48	4.41, 17.82	5.21, 15.49	2.47
Outline	136	line	4.41, 17.82	4.78, 17.98		
Outline	137	line	4.78, 17.98	4.66, 17.26		
Outline	138	arc clockwise	4.66, 17.26	5.33, 17.57	6.02, 15.20	2.47
Outline	139	line	5.33, 17.57	5.71, 17.70		
Outline	140	line	5.71, 17.70	5.56, 16.99		
Outline	141	arc clockwise	5.56, 16.99	6.24, 17.26	6.81, 14.86	2.47
Outline	142	line	6.24, 17.26	6.63, 17.38		
Outline	143	line	6.63, 17.38	6.44, 16.68		
Outline	144	arc clockwise	6.44, 16.68	7.14, 16.91	7.57, 14.49	2.47
Outline	145	line	7.14, 16.91	7.53, 17.01		
Outline	146	line	7.53, 17.01	7.30, 16.32		
Outline	147	arc clockwise	7.30, 16.32	8.01, 16.52	8.32, 14.07	2.47
Outline	148	line	8.01, 16.52	8.41, 16.59		
Outline	149	line	8.41, 16.59	8.15, 15.91		
Outline	150	arc clockwise	8.15, 15.91	8.87, 16.07	9.05, 13.62	2.47
Outline	151	line	8.87, 16.07	9.27, 16.13		
Outline	152	line	9.27, 16.13	8.97, 15.46		
Outline	153	arc clockwise	8.97, 15.46	9.70, 15.59	9.75, 13.12	2.47
Outline	154	line	9.70, 15.59	10.10, 15.62		
Outline	155	line	10.10, 15.62	9.76, 14.97		
Outline	156	arc clockwise	9.76, 14.97	10.50, 15.06	10.42, 12.60	2.47
Outline	157	line	10.50, 15.06	10.90, 15.07		
Outline	158	line	10.90, 15.07	10.54, 14.44		
Outline	159	arc clockwise	10.54, 14.44	11.27, 14.49	11.07, 12.03	2.47
Outline	160	line	11.27, 14.49	11.68, 14.48		
Outline	161	line	11.68, 14.48	11.28, 13.87		
Outline	162	arc clockwise	11.28, 13.87	12.02, 13.88	11.68, 11.44	2.47
Outline	163	line	12.02, 13.88	12.42, 13.85		
Outline	164	line	12.42, 13.85	11.99, 13.26		
Outline	165	arc clockwise	11.99, 13.26	12.73, 13.23	12.26, 10.81	2.47

Outline	166	line	12.73, 13.23	13.13, 13.18		
Outline	167	line	13.13, 13.18	12.66, 12.61		
Outline	168	arc clockwise	12.66, 12.61	13.40, 12.55	12.81, 10.15	2.47
Outline	169	line	13.40, 12.55	13.80, 12.47		
Outline	170	line	13.80, 12.47	13.31, 11.93		
Outline	171	arc clockwise	13.31, 11.93	14.04, 11.83	13.33, 9.47	2.47
Outline	172	line	14.04, 11.83	14.43, 11.73		
Outline	173	line	14.43, 11.73	13.91, 11.22		
Outline	174	arc clockwise	13.91, 11.22	14.64, 11.08	13.80, 8.76	2.47
Outline	175	line	14.64, 11.08	15.03, 10.96		
Outline	176	line	15.03, 10.96	14.48, 10.48		
Outline	177	arc clockwise	14.48, 10.48	15.20, 10.30	14.24, 8.02	2.47
Outline	178	line	15.20, 10.30	15.58, 10.16		
Outline	179	line	15.58, 10.16	15.01, 9.71		
Outline	180	arc clockwise	15.01, 9.71	15.72, 9.49	14.64, 7.27	2.47
Outline	181	line	15.72, 9.49	16.09, 9.33		
Outline	182	line	16.09, 9.33	15.50, 8.91		
Outline	183	arc clockwise	15.50, 8.91	16.19, 8.65	15.00, 6.49	2.47
Outline	184	line	16.19, 8.65	16.56, 8.48		
Outline	185	line	16.56, 8.48	15.94, 8.08		
Outline	186	arc clockwise	15.94, 8.08	16.62, 7.79	15.32, 5.70	2.47
Outline	187	line	16.62, 7.79	16.98, 7.60		
Outline	188	line	16.98, 7.60	16.34, 7.24		
Outline	189	arc clockwise	16.34, 7.24	17.01, 6.91	15.60, 4.89	2.47
Outline	190	line	17.01, 6.91	17.35, 6.70		
Outline	191	line	17.35, 6.70	16.70, 6.37		
Outline	192	arc clockwise	16.70, 6.37	17.35, 6.01	15.83, 4.06	2.47
Outline	193	line	17.35, 6.01	17.68, 5.78		
Outline	194	line	17.68, 5.78	17.01, 5.49		
Outline	195	arc clockwise	17.01, 5.49	17.64, 5.10	16.03, 3.23	2.47
Outline	196	line	17.64, 5.10	17.96, 4.85		
Outline	197	line	17.96, 4.85	17.28, 4.59		
Outline	198	arc clockwise	17.28, 4.59	17.88, 4.17	16.17, 2.39	2.47
Outline	199	line	17.88, 4.17	18.19, 3.90		
Outline	200	line	18.19, 3.90	17.49, 3.68		
Outline	201	arc clockwise	17.49, 3.68	18.07, 3.22	16.27, 1.54	2.47
Outline	202	line	18.07, 3.22	18.37, 2.95		
Outline	203	line	18.37, 2.95	17.66, 2.76		
Outline	204	arc clockwise	17.66, 2.76	18.22, 2.27	16.33, 0.68	2.47
Outline	205	line	18.22, 2.27	18.49, 1.98		
Outline	206	line	18.49, 1.98	17.78, 1.83		
Outline	207	arc clockwise	17.78, 1.83	18.31, 1.32	16.35, -0.17	2.47
Outline	208	line	18.31, 1.32	18.57, 1.01		
Outline	209	line	18.57, 1.01	17.85, 0.90		
Outline	210	arc clockwise	17.85, 0.90	18.35, 0.36	16.32, -1.03	2.47
Outline	211	line	18.35, 0.36	18.60, 0.04		
Outline	212	line	18.60, 0.04	17.87, -0.04		
Outline	213	arc clockwise	17.87, -0.04	18.35, -0.60	16.24, -1.88	2.47

Outline	214	line	18.35, -0.60	18.58, -0.94		
Outline	215	line	18.58, -0.94	17.85, -0.97		
Outline	216	arc clockwise	17.85, -0.97	18.29, -1.56	16.12, -2.73	2.47
Outline	217	line	18.29, -1.56	18.50, -1.91		
Outline	218	line	18.50, -1.91	17.77, -1.90		
Outline	219	arc clockwise	17.77, -1.90	18.18, -2.52	15.95, -3.57	2.47
Outline	220	line	18.18, -2.52	18.38, -2.87		
Outline	221	line	18.38, -2.87	17.65, -2.83		
Outline	222	arc clockwise	17.65, -2.83	18.03, -3.47	15.75, -4.40	2.47
Outline	223	line	18.03, -3.47	18.20, -3.83		
Outline	224	line	18.20, -3.83	17.48, -3.75		
Outline	225	arc clockwise	17.48, -3.75	17.82, -4.41	15.49, -5.21	2.47
Outline	226	line	17.82, -4.41	17.98, -4.78		
Outline	227	line	17.98, -4.78	17.26, -4.66		
Outline	228	arc clockwise	17.26, -4.66	17.57, -5.33	15.20, -6.02	2.47
Outline	229	line	17.57, -5.33	17.70, -5.71		
Outline	230	line	17.70, -5.71	16.99, -5.56		
Outline	231	arc clockwise	16.99, -5.56	17.26, -6.24	14.86, -6.81	2.47
Outline	232	line	17.26, -6.24	17.38, -6.63		
Outline	233	line	17.38, -6.63	16.68, -6.44		
Outline	234	arc clockwise	16.68, -6.44	16.91, -7.14	14.49, -7.57	2.47
Outline	235	line	16.91, -7.14	17.01, -7.53		
Outline	236	line	17.01, -7.53	16.32, -7.30		
Outline	237	arc clockwise	16.32, -7.30	16.52, -8.01	14.07, -8.32	2.47
Outline	238	line	16.52, -8.01	16.59, -8.41		
Outline	239	line	16.59, -8.41	15.91, -8.15		
Outline	240	arc clockwise	15.91, -8.15	16.07, -8.87	13.62, -9.05	2.47
Outline	241	line	16.07, -8.87	16.13, -9.27		
Outline	242	line	16.13, -9.27	15.46, -8.97		
Outline	243	arc clockwise	15.46, -8.97	15.59, -9.70	13.12, -9.75	2.47
Outline	244	line	15.59, -9.70	15.62, -10.10		
Outline	245	line	15.62, -10.10	14.97, -9.76		
Outline	246	arc clockwise	14.97, -9.76	15.06, -10.50	12.60, -10.42	2.47
Outline	247	line	15.06, -10.50	15.07, -10.90		
Outline	248	line	15.07, -10.90	14.44, -10.54		
Outline	249	arc clockwise	14.44, -10.54	14.49, -11.27	12.03, -11.07	2.47
Outline	250	line	14.49, -11.27	14.48, -11.68		
Outline	251	line	14.48, -11.68	13.87, -11.28		
Outline	252	arc clockwise	13.87, -11.28	13.88, -12.02	11.44, -11.68	2.47
Outline	253	line	13.88, -12.02	13.85, -12.42		
Outline	254	line	13.85, -12.42	13.26, -11.99		
Outline	255	arc clockwise	13.26, -11.99	13.23, -12.73	10.81, -12.26	2.47
Outline	256	line	13.23, -12.73	13.18, -13.13		
Outline	257	line	13.18, -13.13	12.61, -12.66		
Outline	258	arc clockwise	12.61, -12.66	12.55, -13.40	10.15, -12.81	2.47
Outline	259	line	12.55, -13.40	12.47, -13.80		
Outline	260	line	12.47, -13.80	11.93, -13.31		
Outline	261	arc clockwise	11.93, -13.31	11.83, -14.04	9.47, -13.33	2.47

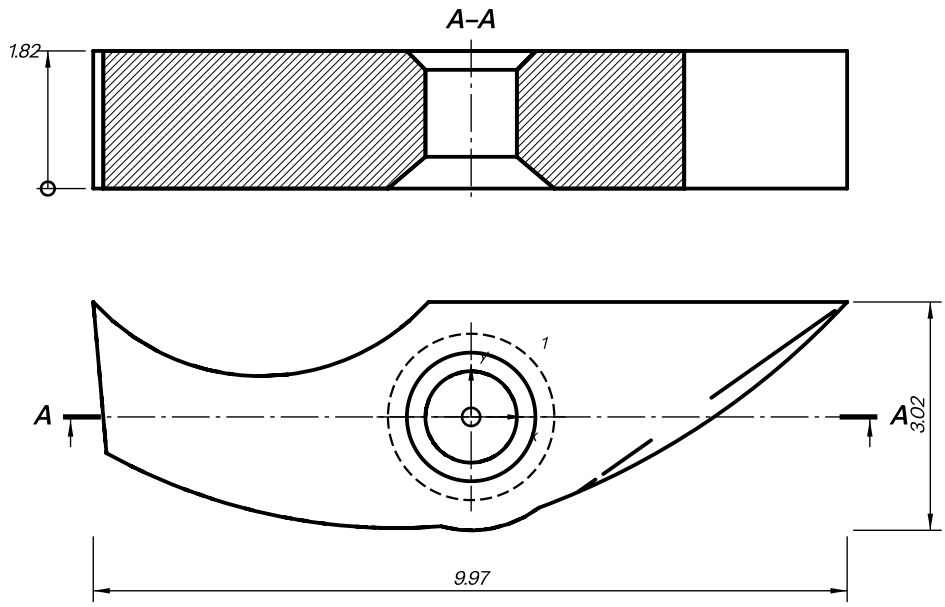
Outline	262	line	11.83, -14.04	11.73, -14.43		
Outline	263	line	11.73, -14.43	11.22, -13.91		
Outline	264	arc clockwise	11.22, -13.91	11.08, -14.64	8.76, -13.80	2.47
Outline	265	line	11.08, -14.64	10.96, -15.03		
Outline	266	line	10.96, -15.03	10.48, -14.48		
Outline	267	arc clockwise	10.48, -14.48	10.30, -15.20	8.02, -14.24	2.47
Outline	268	line	10.30, -15.20	10.16, -15.58		
Outline	269	line	10.16, -15.58	9.71, -15.01		
Outline	270	arc clockwise	9.71, -15.01	9.49, -15.72	7.27, -14.64	2.47
Outline	271	line	9.49, -15.72	9.33, -16.09		
Outline	272	line	9.33, -16.09	8.91, -15.50		
Outline	273	arc clockwise	8.91, -15.50	8.65, -16.19	6.49, -15.00	2.47
Outline	274	line	8.65, -16.19	8.48, -16.56		
Outline	275	line	8.48, -16.56	8.08, -15.94		
Outline	276	arc clockwise	8.08, -15.94	7.79, -16.62	5.70, -15.32	2.47
Outline	277	line	7.79, -16.62	7.60, -16.98		
Outline	278	line	7.60, -16.98	7.24, -16.34		
Outline	279	arc clockwise	7.24, -16.34	6.91, -17.01	4.89, -15.60	2.47
Outline	280	line	6.91, -17.01	6.70, -17.35		
Outline	281	line	6.70, -17.35	6.37, -16.70		
Outline	282	arc clockwise	6.37, -16.70	6.01, -17.35	4.06, -15.83	2.47
Outline	283	line	6.01, -17.35	5.78, -17.68		
Outline	284	line	5.78, -17.68	5.49, -17.01		
Outline	285	arc clockwise	5.49, -17.01	5.10, -17.64	3.23, -16.03	2.47
Outline	286	line	5.10, -17.64	4.85, -17.96		
Outline	287	line	4.85, -17.96	4.59, -17.28		
Outline	288	arc clockwise	4.59, -17.28	4.17, -17.88	2.39, -16.17	2.47
Outline	289	line	4.17, -17.88	3.90, -18.19		
Outline	290	line	3.90, -18.19	3.68, -17.49		
Outline	291	arc clockwise	3.68, -17.49	3.22, -18.07	1.54, -16.27	2.47
Outline	292	line	3.22, -18.07	2.95, -18.37		
Outline	293	line	2.95, -18.37	2.76, -17.66		
Outline	294	arc clockwise	2.76, -17.66	2.27, -18.22	0.68, -16.33	2.47
Outline	295	line	2.27, -18.22	1.98, -18.49		
Outline	296	line	1.98, -18.49	1.83, -17.78		
Outline	297	arc clockwise	1.83, -17.78	1.32, -18.31	-0.17, -16.35	2.47
Outline	298	line	1.32, -18.31	1.01, -18.57		
Outline	299	line	1.01, -18.57	0.90, -17.85		
Outline	300	arc clockwise	0.90, -17.85	0.36, -18.35	-1.03, -16.32	2.47
Outline	301	line	0.36, -18.35	0.04, -18.60		
Outline	302	line	0.04, -18.60	-0.04, -17.87		
Outline	303	arc clockwise	-0.04, -17.87	-0.60, -18.35	-1.88, -16.24	2.47
Outline	304	line	-0.60, -18.35	-0.94, -18.58		
Outline	305	line	-0.94, -18.58	-0.97, -17.85		
Outline	306	arc clockwise	-0.97, -17.85	-1.56, -18.29	-2.73, -16.12	2.47
Outline	307	line	-1.56, -18.29	-1.91, -18.50		
Outline	308	line	-1.91, -18.50	-1.90, -17.77		
Outline	309	arc clockwise	-1.90, -17.77	-2.52, -18.18	-3.57, -15.95	2.47

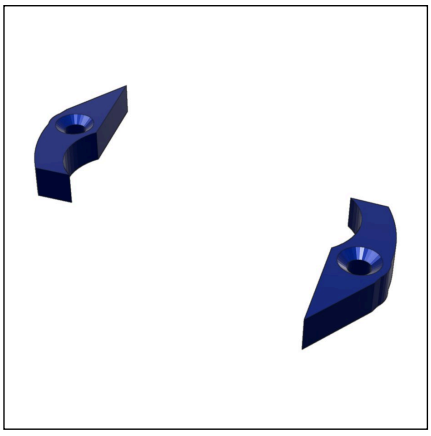
Outline	310	line	-2.52, -18.18	-2.87, -18.38		
Outline	311	line	-2.87, -18.38	-2.83, -17.65		
Outline	312	arc clockwise	-2.83, -17.65	-3.47, -18.03	-4.40, -15.75	2.47
Outline	313	line	-3.47, -18.03	-3.83, -18.20		
Outline	314	line	-3.83, -18.20	-3.75, -17.48		
Outline	315	arc clockwise	-3.75, -17.48	-4.41, -17.82	-5.21, -15.49	2.47
Outline	316	line	-4.41, -17.82	-4.78, -17.98		
Outline	317	line	-4.78, -17.98	-4.66, -17.26		
Outline	318	arc clockwise	-4.66, -17.26	-5.33, -17.57	-6.02, -15.20	2.47
Outline	319	line	-5.33, -17.57	-5.71, -17.70		
Outline	320	line	-5.71, -17.70	-5.56, -16.99		
Outline	321	arc clockwise	-5.56, -16.99	-6.24, -17.26	-6.81, -14.86	2.47
Outline	322	line	-6.24, -17.26	-6.63, -17.38		
Outline	323	line	-6.63, -17.38	-6.44, -16.68		
Outline	324	arc clockwise	-6.44, -16.68	-7.14, -16.91	-7.57, -14.49	2.47
Outline	325	line	-7.14, -16.91	-7.53, -17.01		
Outline	326	line	-7.53, -17.01	-7.30, -16.32		
Outline	327	arc clockwise	-7.30, -16.32	-8.01, -16.52	-8.32, -14.07	2.47
Outline	328	line	-8.01, -16.52	-8.41, -16.59		
Outline	329	line	-8.41, -16.59	-8.15, -15.91		
Outline	330	arc clockwise	-8.15, -15.91	-8.87, -16.07	-9.05, -13.62	2.47
Outline	331	line	-8.87, -16.07	-9.27, -16.13		
Outline	332	line	-9.27, -16.13	-8.97, -15.46		
Outline	333	arc clockwise	-8.97, -15.46	-9.70, -15.59	-9.75, -13.12	2.47
Outline	334	line	-9.70, -15.59	-10.10, -15.62		
Outline	335	line	-10.10, -15.62	-9.76, -14.97		
Outline	336	arc clockwise	-9.76, -14.97	-10.50, -15.06	-10.42, -12.60	2.47
Outline	337	line	-10.50, -15.06	-10.90, -15.07		
Outline	338	line	-10.90, -15.07	-10.54, -14.44		
Outline	339	arc clockwise	-10.54, -14.44	-11.27, -14.49	-11.07, -12.03	2.47
Outline	340	line	-11.27, -14.49	-11.68, -14.48		
Outline	341	line	-11.68, -14.48	-11.28, -13.87		
Outline	342	arc clockwise	-11.28, -13.87	-12.02, -13.88	-11.68, -11.44	2.47
Outline	343	line	-12.02, -13.88	-12.42, -13.85		
Outline	344	line	-12.42, -13.85	-11.99, -13.26		
Outline	345	arc clockwise	-11.99, -13.26	-12.73, -13.23	-12.26, -10.81	2.47
Outline	346	line	-12.73, -13.23	-13.13, -13.18		
Outline	347	line	-13.13, -13.18	-12.66, -12.61		
Outline	348	arc clockwise	-12.66, -12.61	-13.40, -12.55	-12.81, -10.15	2.47
Outline	349	line	-13.40, -12.55	-13.80, -12.47		
Outline	350	line	-13.80, -12.47	-13.31, -11.93		
Outline	351	arc clockwise	-13.31, -11.93	-14.04, -11.83	-13.33, -9.47	2.47
Outline	352	line	-14.04, -11.83	-14.43, -11.73		
Outline	353	line	-14.43, -11.73	-13.91, -11.22		
Outline	354	arc clockwise	-13.91, -11.22	-14.64, -11.08	-13.80, -8.76	2.47
Outline	355	line	-14.64, -11.08	-15.03, -10.96		
Outline	356	line	-15.03, -10.96	-14.48, -10.48		
Outline	357	arc clockwise	-14.48, -10.48	-15.20, -10.30	-14.24, -8.02	2.47

Outline	358	line	-15.20, -10.30	-15.58, -10.16		
Outline	359	line	-15.58, -10.16	-15.01, -9.71		
Outline	360	arc clockwise	-15.01, -9.71	-15.72, -9.49	-14.64, -7.27	2.47

Levels

Faces in section A-A, from the bottom face: 1.80, 2.08, 3.10.





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material „Bandstahl 11×2 Din 1016" (strip steel 11 × 2, DIN 1016)	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 02-3*** *Teil / Part* *maintaining.clicks*
Behandlung / Treatment surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,05		*Benennung / Name* ***Clicks (Sperrkegel)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Clicks (Sperrkegel) — sheet 2

Two blued steel clicks on screws in the ratchet's cup, pressed onto the click wheel by their springs.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 02-3	Sperrkegel (click)	2	„Bandstahl 11×2 Din 1016“ (strip steel 11 × 2, DIN 1016)	surface ∇∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,05	round WN-21 (free): 1.624 mm, least gap 0.0092

What it does (05 06 02-3). In the ratchet's cup, on its screw's shoulder; the click wheel drives it.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.82.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.21 through (1.18 thick at 0.41–1.59); countersunk to Ø 1.70; sink Ø 1.70 × 0.03 from the face at 0.20; countersunk to Ø 2.20

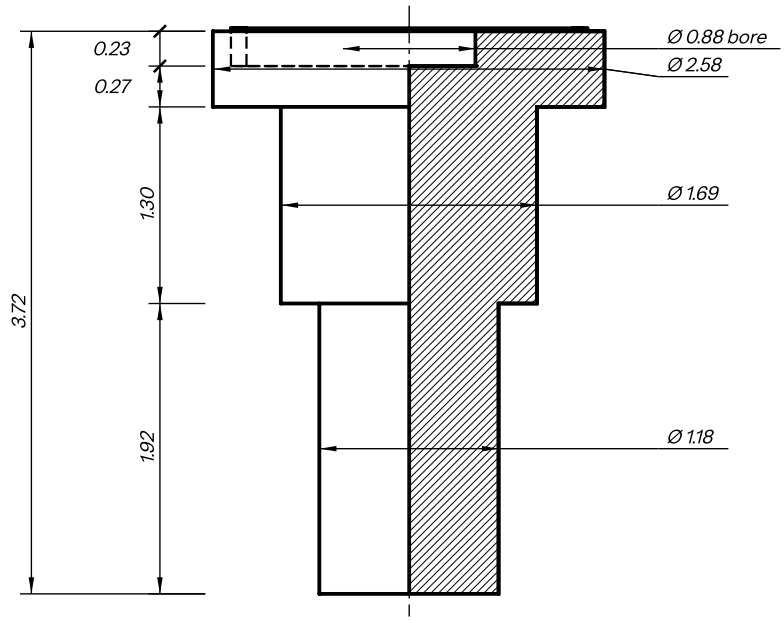
Contour in the plan

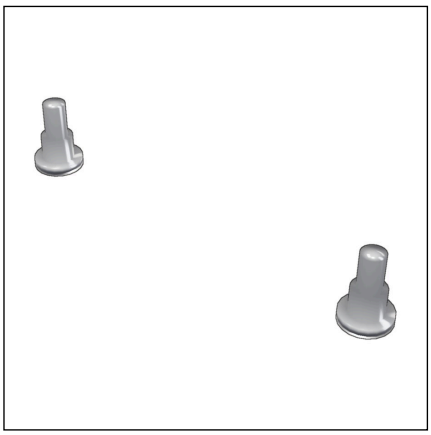
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-4.48, -0.66	-4.82, -0.48		
Outline	2	line	-4.82, -0.48	-5.00, 1.52		
Outline	3	arc anticlockwise	-5.00, 1.52	-0.56, 1.52	-2.78, 3.54	3.00
Outline	4	line	-0.56, 1.52	4.97, 1.52		
Outline	5	arc clockwise	4.97, 1.52	0.90, -1.20	-2.77, 8.71	10.57
Outline	6	arc clockwise	0.90, -1.20	-0.40, -1.45	0.00, 0.00	1.50
Outline	7	line	-0.40, -1.45	-0.78, -1.47		
Outline	8	arc clockwise	-0.78, -1.47	-4.48, -0.66	-0.97, 6.53	8.00

Levels

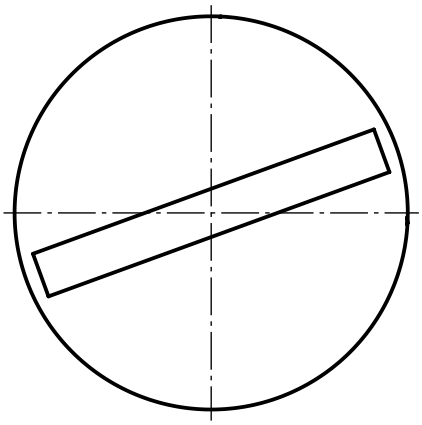
Faces in section A–A, from the bottom face: 1.82.





Isometric view, not to scale

View from above



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-21*** *Teil / Part* *maintaining.clickScrews*
Behandlung / Treatment		*Benennung / Name* ***Click screws (WN-21)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Click screws (WN-21) — sheet 2

The shouldered screws the clicks turn on, put in from under the ratchet.

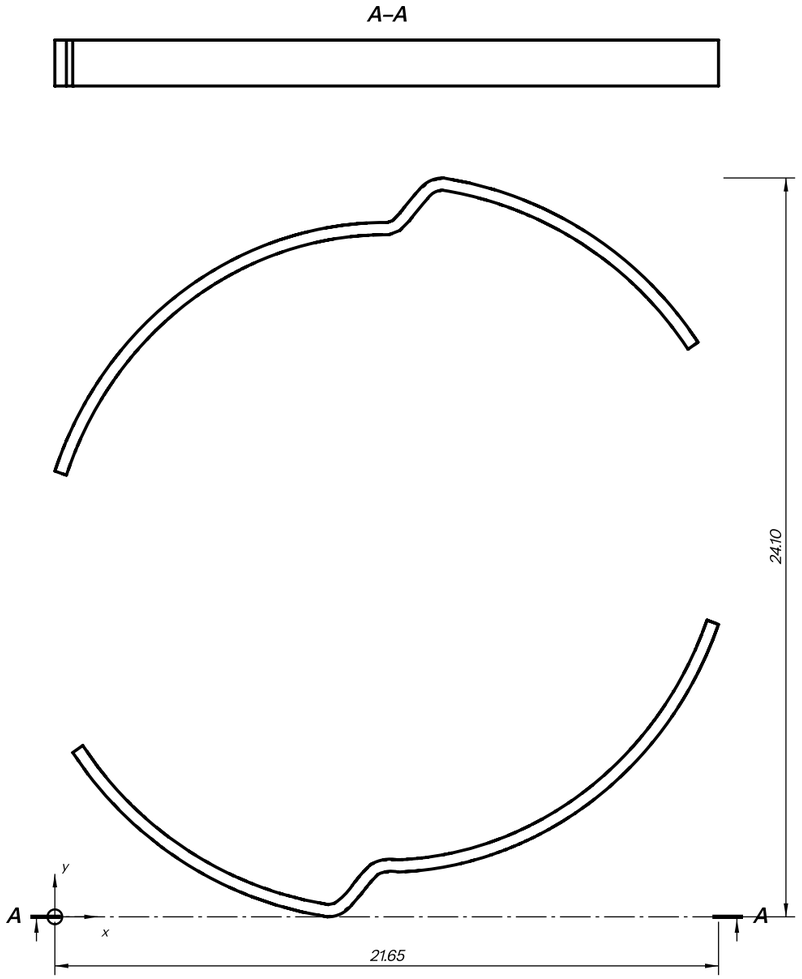
2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-21	Schraube WN-21 (click screw)	2	not on the drawing		in 050602-1 (tap): 1.279 mm, least gap 0.02

What it does (WN-21). Put in from under the ratchet, threaded in its M1.7; the click turns on its shoulder.

Turned about its axis; length 3.74, largest Ø 2.60. Diameters: Ø 1.18, Ø 1.69, Ø 2.58; bores Ø 0.88. Shoulders from the lower end: 0.00, 1.92, 3.22, 3.49, 3.72.





Isometric view, not to scale

Werkstoff / Material *„Bronzeband 1,5×0,4 federhart" (bronze strip 1,5 × 0,4, spring-hard)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 02-2***
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ± 0,1*			*Teil / Part* *maintaining.springs*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Click springs (Sperrfeder)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Click springs (Sperrfeder) — sheet 2

Two bronze strips pinned inside the cup's wall, each pressing a click onto the click wheel.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 02-2	Sperrfeder (click spring)	2	„Bronzeband 1,5×0,4 federhart“ (bronze strip 1,5 × 0,4, spring-hard)	surface ∇∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished); untoleranced sizes ±0,1	not measured on the model: see its function, Pinned inside the cup's wall, pressing its click onto the click wheel.

What it does (05 06 02-2). Pinned inside the cup's wall, pressing its click onto the click wheel.

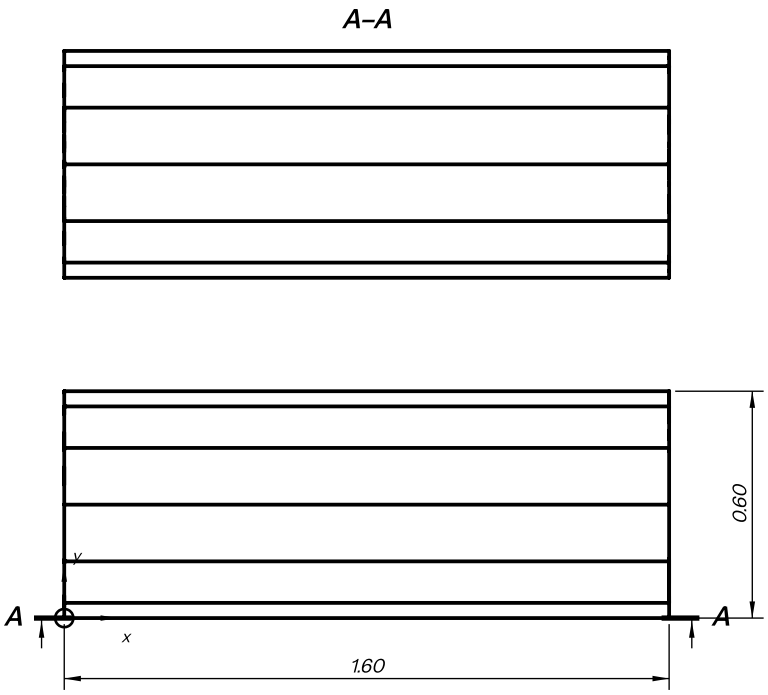
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.50.

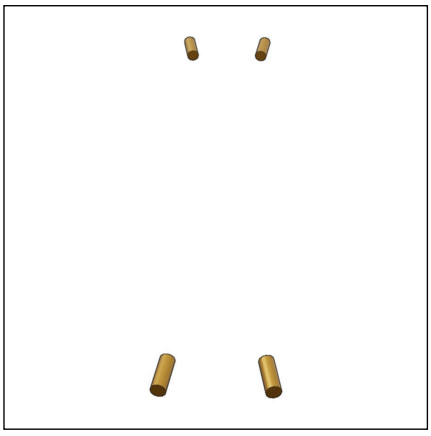
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.58, 5.36	0.91, 5.59		
Outline	2	arc anticlockwise	0.91, 5.59	8.92, 0.40	10.78, 12.05	11.80
Outline	3	arc anticlockwise	8.92, 0.40	9.39, 0.66	8.90, 1.00	0.60
Outline	4	arc anticlockwise	9.39, 0.66	10.00, 1.40	0.33, 8.72	12.13
Outline	5	arc clockwise	10.00, 1.40	10.66, 1.85	10.84, 0.87	1.00
Outline	6	arc anticlockwise	10.66, 1.85	19.61, 6.49	10.63, 12.81	10.97
Outline	7	arc anticlockwise	19.61, 6.49	21.28, 9.68	9.59, 13.74	12.37
Outline	8	line	21.28, 9.68	21.65, 9.54		
Outline	9	arc clockwise	21.65, 9.54	14.70, 2.19	10.20, 13.37	12.06
Outline	10	arc clockwise	14.70, 2.19	10.72, 1.45	10.78, 12.26	10.81
Outline	11	arc anticlockwise	10.72, 1.45	10.30, 1.14	10.85, 0.85	0.62
Outline	12	arc clockwise	10.30, 1.14	9.67, 0.38	1.19, 8.02	11.42
Outline	13	arc clockwise	9.67, 0.38	8.89, 0.00	8.90, 0.98	0.98
Outline	14	arc clockwise	8.89, 0.00	0.58, 5.36	10.78, 12.05	12.20
Outline	15	arc clockwise	0.00, 14.54	7.35, 22.09	11.23, 10.99	11.77
Outline	16	arc clockwise	7.35, 22.09	10.85, 22.65	10.78, 11.94	10.71
Outline	17	arc anticlockwise	10.85, 22.65	11.26, 22.96	10.71, 23.26	0.62
Outline	18	arc clockwise	11.26, 22.96	11.89, 23.73	20.38, 16.08	11.42
Outline	19	arc clockwise	11.89, 23.73	12.68, 24.10	12.66, 23.13	0.98
Outline	20	arc clockwise	12.68, 24.10	20.98, 18.74	10.78, 12.05	12.20
Outline	21	line	20.98, 18.74	20.66, 18.51		
Outline	22	arc anticlockwise	20.66, 18.51	12.64, 23.70	10.78, 12.05	11.80
Outline	23	arc anticlockwise	12.64, 23.70	12.17, 23.44	12.67, 23.10	0.60
Outline	24	arc anticlockwise	12.17, 23.44	11.56, 22.70	21.23, 15.38	12.13
Outline	25	arc clockwise	11.56, 22.70	10.90, 22.25	10.72, 23.23	1.00
Outline	26	arc anticlockwise	10.90, 22.25	1.70, 17.18	10.92, 11.38	10.88
Outline	27	arc anticlockwise	1.70, 17.18	0.38, 14.41	11.82, 10.64	12.04

Outline	28	line	0.38, 14.41	0.00, 14.54		





Isometric view, not to scale

4 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* **50:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 02-4***
Behandlung / Treatment			*Teil / Part* *maintaining.springPins*
		Benennung / Name ***Click spring pins***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Click spring pins — sheet 2

The pins through the cup's wall that hold the click springs.

4 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 02-4	Stift (click spring pin)	4	not on the drawing		in 050602-1 (embed): 0.6 mm, least gap -0.2093

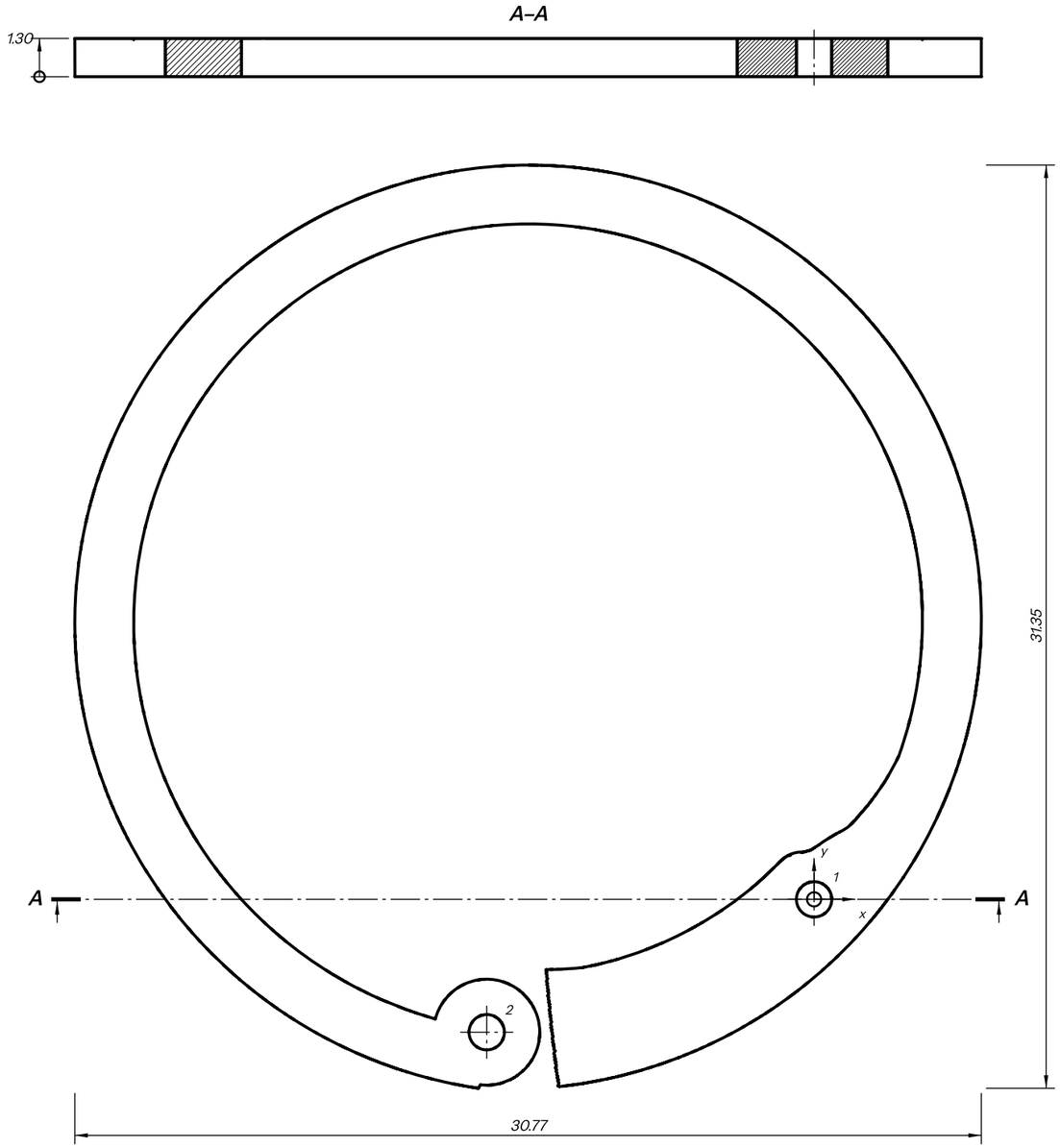
What it does (05 06 02-4). Through the cup's wall and the spring's hole (not cut in the turned cup).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 0.60.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.30	0.00, 0.60		
Outline	2	line	0.00, 0.60	1.60, 0.60		
Outline	3	line	1.60, 0.60	1.60, 0.00		
Outline	4	line	1.60, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 0.30		





Isometric view, not to scale

Werkstoff / Material *„Bandstahl 35×1,75 Din 1016" (strip steel 35 × 1.75, DIN 1016)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 01-1***
Behandlung / Treatment *„400° anlassen"; „härten in Öl"; „310° blau anlassen"; surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0.05*			*Teil / Part* *maintaining.spring*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Maintaining spring (Gegenfeder)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Maintaining spring (Gegenfeder) — sheet 2

An open steel ring, its eye pinned to the fusee wheel and its other end to the ratchet: it carries the whole drive, and drives the train alone while the key turns.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 01-1	Gegenfeder (maintaining spring)	1	„Bandstahl 35×1,75 Din 1016“ (strip steel 35 × 1,75, DIN 1016)	„400° anlassen“; „härten in Öl“; „310° blau anlassen“; surface ∇∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished); untoleranced sizes ±0,05	on 050600-3: 0 mm

Test on the bench (05 06 01-1). The maintaining spring drives the train, the movement running, part wound: Running, the ring closed 17.8–19.7° by the drive, its ends -0.74–-0.28 mm apart at the eye (they would meet after 16.7°; wempe_sustaining.js); the key held, the balance keeps going 11.4–14.6 minutes (the detent catching a whole tooth late or at once), its swing falling; the 1954 instruction gives 1–2 minutes for chronometers in general; a wind takes 10–30 s. Under 5 minutes (estimated): the ring is open (the pins the wrong way round: it bears on the recess's wall and barely gives), weak, or has taken a set; its ends touching in running: weak or set.

What it does (05 06 01-1). An open ring between the fusee wheel and the ratchet, in the wheel's recess on its web: carries the drive.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.30.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.20 through
2	-11.11	-4.51	Ø 1.20 through

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

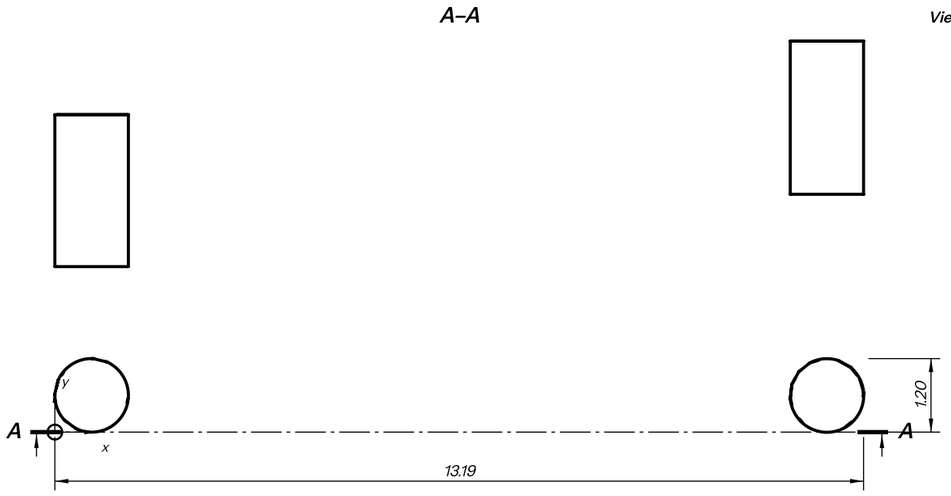
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-24.49, 5.03	-14.36, 24.20	-9.53, 9.41	15.57
Outline	2	arc clockwise	-14.36, 24.20	5.66, 10.26	-9.70, 9.57	15.36
Outline	3	arc clockwise	5.66, 10.26	-6.52, -5.98	-10.32, 9.57	16.00
Outline	4	arc clockwise	-6.52, -5.98	-8.65, -6.36	-10.40, 9.67	16.12
Outline	5	line	-8.65, -6.36	-8.69, -6.33		
Outline	6	line	-8.69, -6.33	-8.66, -6.24		
Outline	7	line	-8.66, -6.24	-8.70, -6.17		
Outline	8	line	-8.70, -6.17	-8.68, -6.12		
Outline	9	line	-8.68, -6.12	-8.72, -6.06		
Outline	10	line	-8.72, -6.06	-8.70, -6.01		
Outline	11	line	-8.70, -6.01	-8.74, -5.95		
Outline	12	line	-8.74, -5.95	-8.72, -5.90		
Outline	13	line	-8.72, -5.90	-8.76, -5.84		
Outline	14	line	-8.76, -5.84	-8.74, -5.79		
Outline	15	line	-8.74, -5.79	-8.78, -5.73		
Outline	16	line	-8.78, -5.73	-8.75, -5.63		

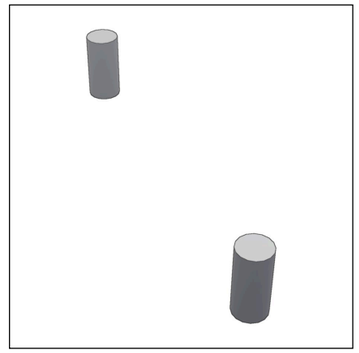
Outline	17	line	-8.75, -5.63	-8.78, -5.57		
Outline	18	line	-8.78, -5.57	-8.77, -5.52		
Outline	19	line	-8.77, -5.52	-8.80, -5.46		
Outline	20	line	-8.80, -5.46	-8.78, -5.41		
Outline	21	line	-8.78, -5.41	-8.82, -5.35		
Outline	22	line	-8.82, -5.35	-8.80, -5.30		
Outline	23	line	-8.80, -5.30	-8.84, -5.23		
Outline	24	arc anticlockwise	-8.84, -5.23	-8.79, -5.14	-8.96, -5.11	0.18
Outline	25	arc clockwise	-8.79, -5.14	-8.84, -5.08	-8.69, -5.02	0.16
Outline	26	line	-8.84, -5.08	-8.82, -5.03		
Outline	27	line	-8.82, -5.03	-8.86, -4.97		
Outline	28	line	-8.86, -4.97	-8.84, -4.92		
Outline	29	line	-8.84, -4.92	-8.87, -4.85		
Outline	30	line	-8.87, -4.85	-8.86, -4.81		
Outline	31	line	-8.86, -4.81	-8.89, -4.74		
Outline	32	line	-8.89, -4.74	-8.86, -4.65		
Outline	33	line	-8.86, -4.65	-8.89, -4.58		
Outline	34	line	-8.89, -4.58	-8.88, -4.54		
Outline	35	line	-8.88, -4.54	-8.91, -4.47		
Outline	36	line	-8.91, -4.47	-8.90, -4.43		
Outline	37	line	-8.90, -4.43	-8.93, -4.36		
Outline	38	line	-8.93, -4.36	-8.91, -4.32		
Outline	39	line	-8.91, -4.32	-8.94, -4.25		
Outline	40	line	-8.94, -4.25	-8.91, -4.16		
Outline	41	line	-8.91, -4.16	-8.95, -4.10		
Outline	42	line	-8.95, -4.10	-8.93, -4.05		
Outline	43	line	-8.93, -4.05	-8.96, -3.99		
Outline	44	line	-8.96, -3.99	-8.95, -3.94		
Outline	45	line	-8.95, -3.94	-8.98, -3.87		
Outline	46	line	-8.98, -3.87	-8.96, -3.83		
Outline	47	line	-8.96, -3.83	-8.99, -3.76		
Outline	48	line	-8.99, -3.76	-8.96, -3.67		
Outline	49	line	-8.96, -3.67	-8.99, -3.61		
Outline	50	line	-8.99, -3.61	-8.98, -3.56		
Outline	51	line	-8.98, -3.56	-9.01, -3.50		
Outline	52	line	-9.01, -3.50	-8.99, -3.45		
Outline	53	line	-8.99, -3.45	-9.02, -3.39		
Outline	54	line	-9.02, -3.39	-9.01, -3.34		
Outline	55	line	-9.01, -3.34	-9.04, -3.28		
Outline	56	line	-9.04, -3.28	-9.01, -3.18		
Outline	57	line	-9.01, -3.18	-9.04, -3.12		
Outline	58	line	-9.04, -3.12	-9.02, -3.07		
Outline	59	line	-9.02, -3.07	-9.05, -3.01		
Outline	60	line	-9.05, -3.01	-9.04, -2.96		
Outline	61	line	-9.04, -2.96	-9.07, -2.90		
Outline	62	line	-9.07, -2.90	-9.05, -2.85		
Outline	63	line	-9.05, -2.85	-9.08, -2.79		
Outline	64	line	-9.08, -2.79	-9.05, -2.70		

Outline	65	line	-9.05, -2.70	-9.08, -2.63		
Outline	66	line	-9.08, -2.63	-9.06, -2.59		
Outline	67	line	-9.06, -2.59	-9.09, -2.52		
Outline	68	line	-9.09, -2.52	-9.08, -2.48		
Outline	69	line	-9.08, -2.48	-9.11, -2.41		
Outline	70	line	-9.11, -2.41	-9.10, -2.39		
Outline	71	arc anticlockwise	-9.10, -2.39	-6.32, -1.92	-8.96, 5.34	7.73
Outline	72	arc anticlockwise	-6.32, -1.92	-1.24, 1.19	-10.22, 10.14	12.67
Outline	73	arc clockwise	-1.24, 1.19	-0.34, 1.61	-0.40, 0.54	1.07
Outline	74	arc anticlockwise	-0.34, 1.61	0.16, 1.84	-0.47, 2.52	0.92
Outline	75	arc clockwise	0.16, 1.84	0.88, 2.29	3.26, -2.37	5.23
Outline	76	arc anticlockwise	0.88, 2.29	1.50, 2.82	0.21, 3.70	1.56
Outline	77	arc anticlockwise	1.50, 2.82	2.87, 4.85	-4.58, 8.40	8.25
Outline	78	arc anticlockwise	2.87, 4.85	-6.06, 22.43	-9.79, 9.49	13.48
Outline	79	arc anticlockwise	-6.06, 22.43	-22.89, 11.76	-9.66, 9.52	13.41
Outline	80	arc anticlockwise	-22.89, 11.76	-12.87, -4.06	-9.11, 9.42	13.99
Outline	81	line	-12.87, -4.06	-12.85, -4.05		
Outline	82	arc clockwise	-12.85, -4.05	-11.33, -6.30	-11.11, -4.51	1.80
Outline	83	line	-11.33, -6.30	-11.40, -6.42		
Outline	84	arc clockwise	-11.40, -6.42	-24.49, 5.03	-9.05, 9.48	16.07

Levels

Faces in section A-A, from the bottom face: 1.30.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 06 00-8**
Behandlung / Treatment			*Teil / Part* *maintaining.springPinsG*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Maintaining spring's pins (Stellstifte)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Maintaining spring’s pins (Stellstifte) — sheet 2

The two pins in the maintaining spring’s holes, one into the fusee wheel, one into the ratchet.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 00-8	Stellstift f. Gegenfeder (maintaining spring's pin)	2	not on the drawing		in 050601-1 (press): 1.312 mm, least gap 0.0031; in 050600-3/050602-1 (run): 1.021 mm, least gap 0.0066

What it does (05 06 00-8). Pressed in the spring's holes: one into the fusee wheel, one into the ratchet.

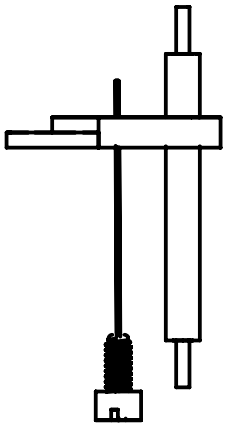
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.68.

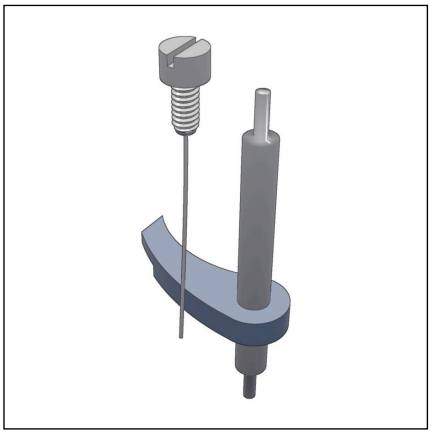
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			12.59, 0.60	0.60
Outline	2	circle			0.60, 0.60	0.60

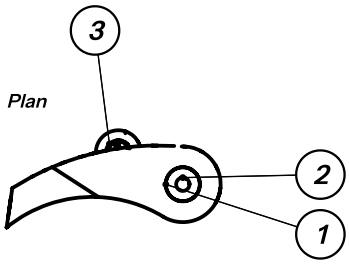
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	1	Detent spring (Feder) and its stud	05 05 06-1	not on the drawing
2	1	Detent arbor	05 05 05-2	not on the drawing
1	1	Detent click	05 05 05-1	„Bandstahl 2,25 × 18 DIN 1016“

Werkstoff / Material
„Bandstahl 2,25 × 18 DIN 1016“ (steel strip 2.25 × 18)

Maßstab / Scale
2:1

Behandlung / Treatment
„härten in Öl, 310° hell-blau anlassen“ (harden in oil, temper light blue at 310°); „um die Bohrung 400° anlassen“ (temper round the hole at 400°); surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished);…

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

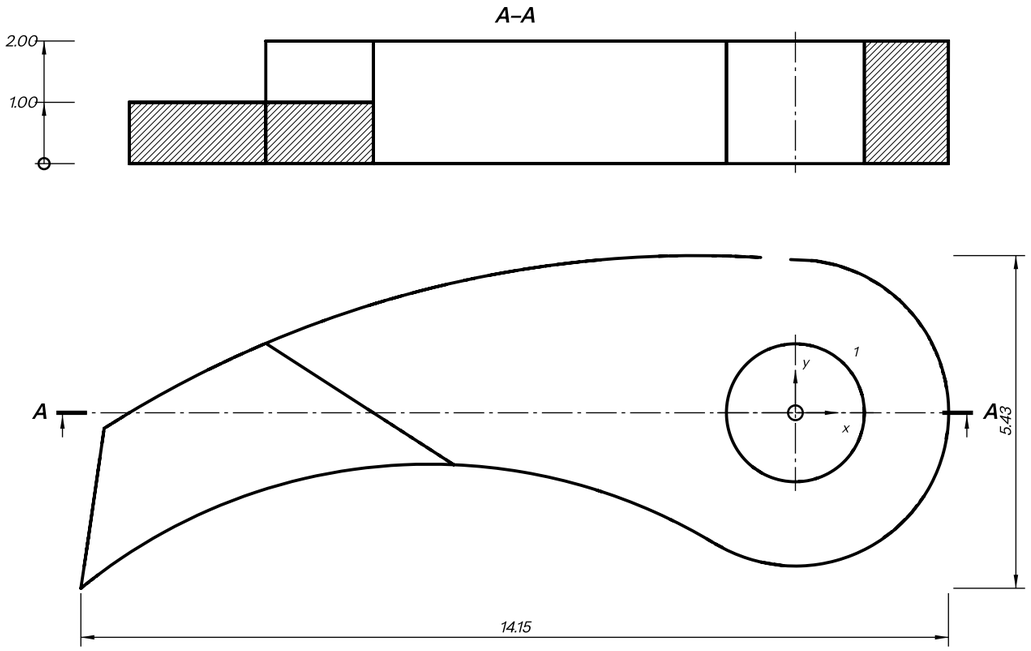
Zeichnung Nr. / Drawing no.
05 05 05-1

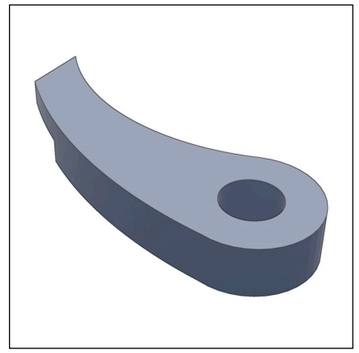
Teil / Part
maintDetent

Benennung / Name
Maintaining detent (Gegensperrkegel)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Bandstahl 2,25 × 18 DIN 1016" (steel strip 2.25 × 18)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 05-1*** *Teil / Part* *maintDetent.click*
Behandlung / Treatment *„härten in Öl, 310° hell-blau anlassen" (harden in oil, temper light blue at 310°); „um die Bohrung 400° anlassen" (temper round the hole at 400°); surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished);…*		*Benennung / Name* ***Detent click***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Detent click — sheet 2

A blued steel click staked on its own arbor between the plates, at hole 4, riding on the maintaining ratchet's teeth: it stops the ratchet turning back while the chronometer is wound, so the maintaining spring drives the train. A straight wire spring hanging from a stud in the top plate presses it in (the coloured section's Feder 05 05 06-1).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 05-1	Gegensperrkegel (maintaining detent)	1	„Bandstahl 2,25 × 18 DIN 1016“ (steel strip 2.25 × 18)	„härten in Öl, 310° hell-blau anlassen“ (harden in oil, temper light blue at 310°); „um die Bohrung 400° anlassen“ (temper round the hole at 400°); surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	round 050505-2 (press): 2.031 mm, least gap 0.0025

What it does (05 05 05-1). Rides on the maintaining ratchet's teeth; holds it while the chronometer is wound.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 2.25 through

Contour in the plan

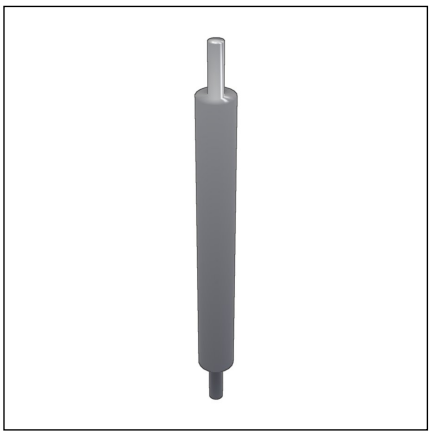
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-11.29, -2.58	-11.66, -2.86		
Outline	2	line	-11.66, -2.86	-11.28, -0.25		
Outline	3	arc clockwise	-11.28, -0.25	-8.64, 1.13	-1.64, -15.39	17.94
Outline	4	arc clockwise	-8.64, 1.13	-0.08, 2.50	-1.61, -15.43	17.99
Outline	5	arc clockwise	-0.08, 2.50	-1.30, -2.14	0.00, 0.00	2.50
Outline	6	arc anticlockwise	-1.30, -2.14	-5.57, -0.85	-5.96, -9.81	8.97
Outline	7	arc anticlockwise	-5.57, -0.85	-11.29, -2.58	-5.97, -9.83	8.99

Levels

Faces in section A-A, from the bottom face: 1.00, 2.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 05-2***
Behandlung / Treatment			*Teil / Part* *maintDetent.arbor*
		Benennung / Name ***Detent arbor***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Detent arbor — sheet 2

The detent's arbor, its pivots in the two plates' holes 4.

Parts list

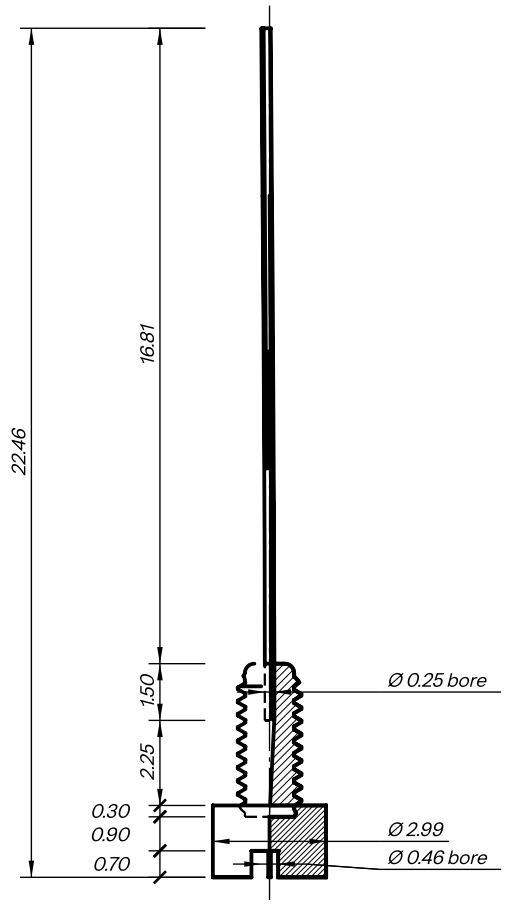
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 05-2	Gegensperrwelle (its arbor)	1	not on the drawing		in 050802-1 (run): 3.124 mm, least gap 0.009; in 050901 (run): 2.499 mm, least gap 0.0111; endshake 0.050 mm (050802-1 / 050901)

What it does (05 05 05-2). Runs in the plates' holes 4 (drilled 0.9).

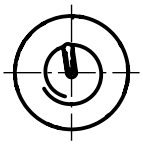
Turned about its axis; length 25.15, largest Ø 2.25. Diameters: Ø 0.88, Ø 2.25. Shoulders from the lower end: 0.00, 3.10, 22.05, 25.15.



Isometric view, not to scale



View from above



Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 06-1*** *Teil / Part* *maintDetent.spring*
Behandlung / Treatment		*Benennung / Name* ***Detent spring (Feder) and its stud***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Detent spring (Feder) and its stud — sheet 2

A straight steel wire held in the end of a slotted stud screwed into the top plate's hole 5. It hangs past the detent's back, which bends it out, so it presses the detent onto the maintaining ratchet's teeth.

Parts list

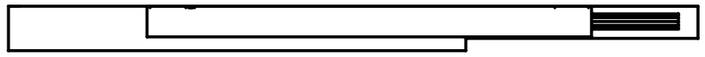
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 06-1	Feder (the maintaining detent's wire spring)	1	not on the drawing		on wireStud: 0 mm; on 050505-1: 0.0015 mm
WN-27	Schraube (WN-27 by the coloured section's label, likely: the maintaining detent's wire spring's stud)	1	not on the drawing		in 050901 (tap): 3.373 mm, least gap 0.0215

What it does (05 05 06-1). A straight steel wire hanging from its stud past the detent's back: bent out by the detent, it presses the detent onto the maintaining ratchet's teeth (no sheet in the set; the coloured section names it, 05 06_4 marks its "Drahtfedermitte").

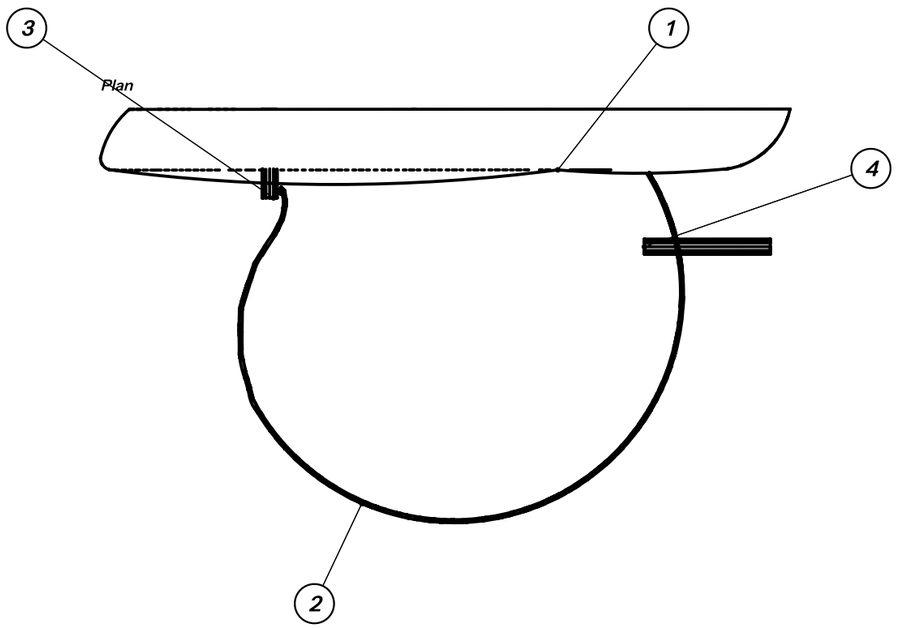
What it does (WN-27). A slotted cheese-head stud screwed into the top plate's hole 5 ("Drahtfeder", M1.7) from its outer face; the wire spring is held in its end under the plate (no sheet; the coloured section draws it, the north-side photographs show it).

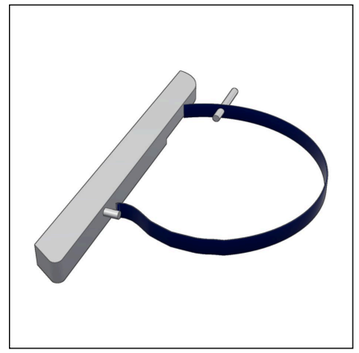
Turned about its axis; length 22.46, largest Ø 3.00. Diameters: Ø 2.99; bores Ø 0.25, Ø 0.46. Shoulders from the lower end: 0.00, 0.70, 1.60, 1.90, 4.15, 5.65, 22.46.

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Stop bar spring's pin (Stift)	05 06 04-4	not on the drawing
3	1	Stop bar's pin (Stift)	05 06 03-2	not on the drawing
2	1	Stop bar spring (Stegfeder)	05 06 03-3	not on the drawing
1	1	Stop bar (Stellungssteg)	05 06 03-1	not on the drawing

Werkstoff / Material
not on the drawing

Maßstab / Scale
5:1

Wempe Einheits-Chronometer
The Marine Chronometer working model
drawn from its solids by tools/drawbook.py
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 06 03-1

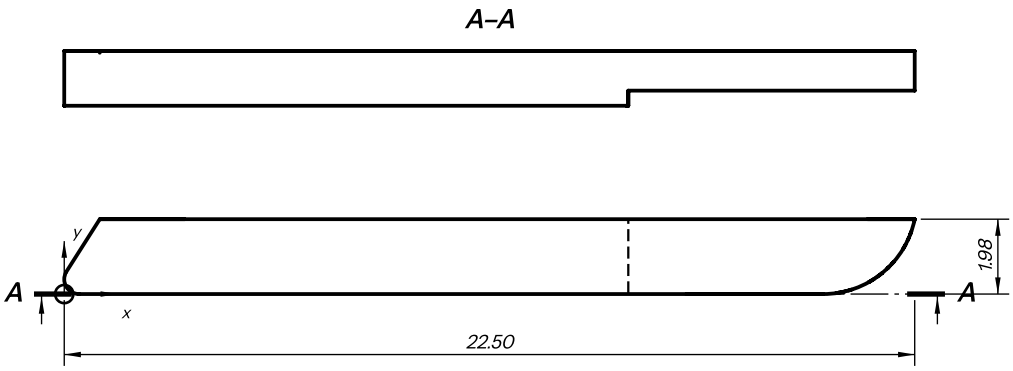
Teil / Part
stopWork

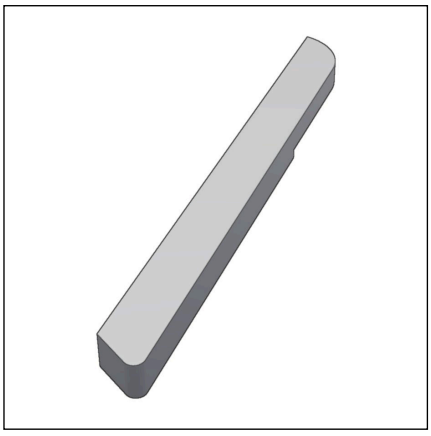
Behandlung / Treatment

Benennung / Name
Stop work (Stellungssteg)

Blatt / Sheet
1 / 1

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 06 03-1***
Behandlung / Treatment			*Teil / Part* *stopWork.bar*
		Benennung / Name ***Stop bar (Stellungssteg)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Stop bar (Stellungssteg) — sheet 2

The steel bar in the slot across the fusee's small end: its lower end, cut obliquely, stands into the top turn's groove until the chain presses it in; its upper end, thinner and rounded on the axis's side, then stands out to meet the stop pin.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 03-1	Stellungssteg (stop bar)	1	not on the drawing		on 050600-4: 0 mm

Test on the bench (05 06 03-1). The stop work: the bar springs back, the stop holds full wind, the bar, its spring and pin in the fusee's small end, the cover plate off; then the fusee in its plates, the chain on and the barrel set up, the train out: The bar springs fully out at once every time, in both positions: its spring (05 06 03-3; no source draws its section: any height 0.9–1.5, its pin's hole to the recess's floor, 0.10 thick; wempe_strength.js) pushes it 0.06–0.08 N half way at the model's 1.0 high, 41 times or more the 0.5 g bar's sliding on the cover plate (μ 0.3, estimated). Assembled: the stop pin (hole 7) stops the fusee with the chain 8.638 turns on it (8.70 from run-down), 0.112 turn short of the groove's end N, and the bar never touches the pin earlier (0.69 off a turn before: wempe_stopwork.py). Sluggish: clean the slot and the bar, and look for a burr at the slot's ends; still slow, bend the spring's free span further in, keeping it under its steel's yield (wempe_strength.js). Stopping short of 8.64 turns: the bar is caught early (its spring too weak to hold it out, or the bar's upper end too long).

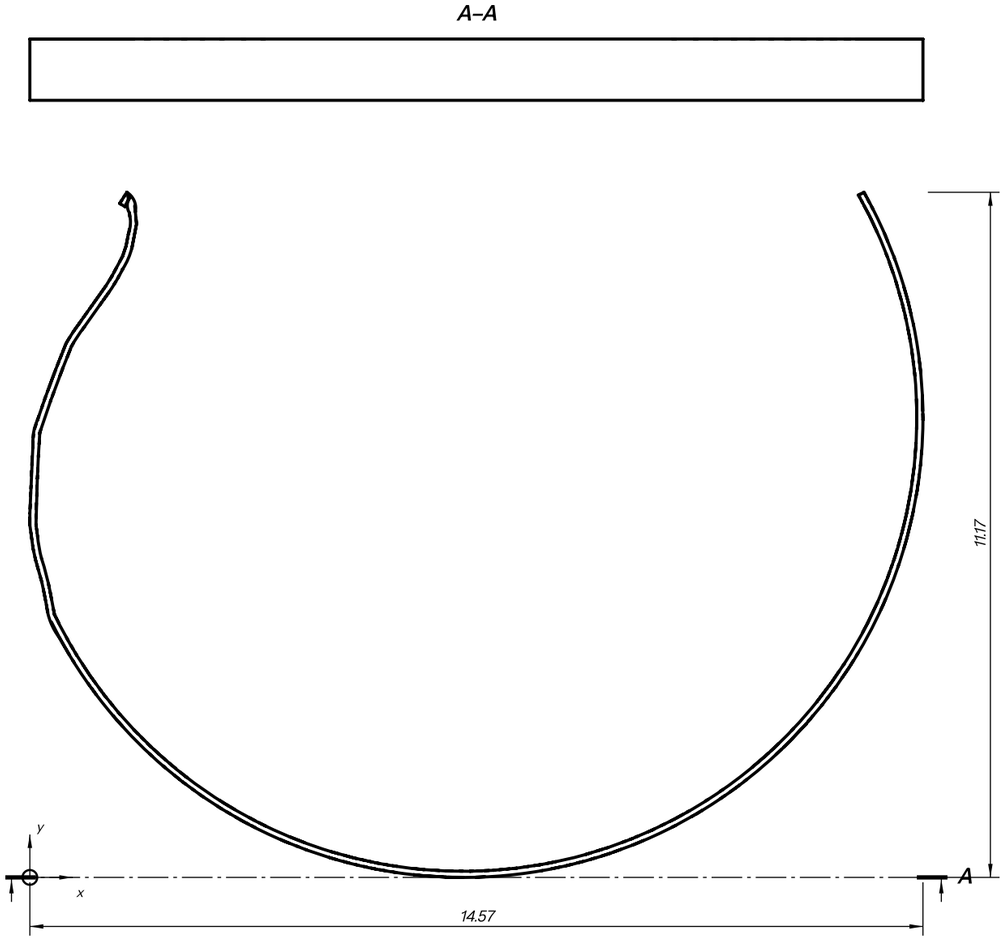
What it does (05 06 03-1). Slides in the slot across the fusee's small end (cut in the body: 1.5 deep, 1.1 toward its upper end, which is thinner), on the cover plate; its spring holds it out, the chain on the top turn presses it in, and its upper end meets the stop pin at full wind.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.45.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.11, 0.11	0.05, 0.59	0.39, 0.39	0.39
Outline	2	line	0.05, 0.59	0.94, 1.98		
Outline	3	line	0.94, 1.98	22.50, 1.98		
Outline	4	arc clockwise	22.50, 1.98	20.08, 0.00	20.08, 2.47	2.47
Outline	5	line	20.08, 0.00	0.39, 0.00		
Outline	6	arc clockwise	0.39, 0.00	0.11, 0.11	0.39, 0.39	0.39





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 06 03-3**
Behandlung / Treatment			*Teil / Part* *stopWork.spring*
		Benennung / Name ***Stop bar spring (Stegfeder)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Stop bar spring (Stegfeder) — sheet 2

A steel strip lying round the wall of the small end's recess, pinned near the slot, its free end in a hole across the bar near its lower end: it holds the bar out toward that end.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 03-3	Stegfeder (the stop bar's spring)	1	not on the drawing		on 050600-4: 0 mm

What it does (05 06 03-3). A strip in the small end's recess, round its wall to the hole across the bar near its lower end, its end in that hole (not cut in the bar): holds the bar out toward that end (no sheet: drawn as 05 06_4's C-D draws it).

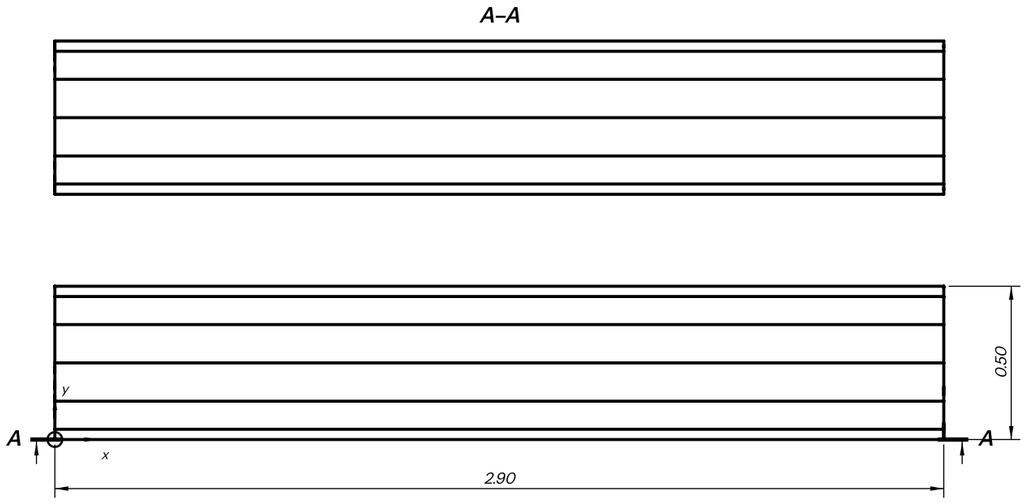
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.00.

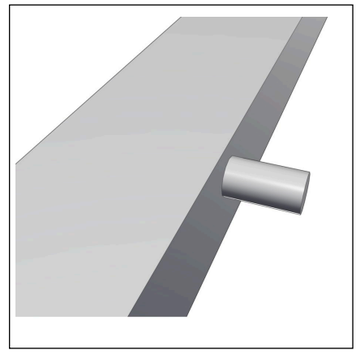
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 5.75	0.01, 6.13		
Outline	2	line	0.01, 6.13	0.04, 6.89		
Outline	3	line	0.04, 6.89	0.06, 7.27		
Outline	4	arc clockwise	0.06, 7.27	0.31, 8.00	14.99, 2.51	15.67
Outline	5	line	0.31, 8.00	0.59, 8.71		
Outline	6	arc clockwise	0.59, 8.71	1.24, 9.65	11.09, 2.08	12.42
Outline	7	arc anticlockwise	1.24, 9.65	1.43, 9.97	-1.03, 11.18	2.74
Outline	8	arc anticlockwise	1.43, 9.97	1.61, 10.48	0.40, 10.61	1.21
Outline	9	arc anticlockwise	1.61, 10.48	1.62, 10.84	1.03, 10.69	0.61
Outline	10	line	1.62, 10.84	1.60, 11.03		
Outline	11	line	1.60, 11.03	1.55, 10.94		
Outline	12	line	1.55, 10.94	1.47, 10.99		
Outline	13	line	1.47, 10.99	1.58, 11.17		
Outline	14	arc clockwise	1.58, 11.17	1.72, 10.85	1.42, 10.91	0.31
Outline	15	arc clockwise	1.72, 10.85	1.70, 10.47	1.05, 10.69	0.69
Outline	16	arc clockwise	1.70, 10.47	1.51, 9.92	0.41, 10.61	1.30
Outline	17	arc clockwise	1.51, 9.92	1.10, 9.28	-2.03, 11.76	4.00
Outline	18	line	1.10, 9.28	0.77, 8.82		
Outline	19	arc anticlockwise	0.77, 8.82	0.61, 8.49	1.65, 8.18	1.09
Outline	20	line	0.61, 8.49	0.34, 7.78		
Outline	21	line	0.34, 7.78	0.16, 7.25		
Outline	22	line	0.16, 7.25	0.12, 6.51		
Outline	23	line	0.12, 6.51	0.10, 5.94		
Outline	24	arc anticlockwise	0.10, 5.94	0.13, 5.57	1.44, 5.84	1.34
Outline	25	arc anticlockwise	0.13, 5.57	0.24, 5.02	2.78, 5.85	2.67
Outline	26	arc clockwise	0.24, 5.02	0.40, 4.29	-5.40, 3.43	5.86
Outline	27	arc anticlockwise	0.40, 4.29	13.52, 11.13	7.07, 7.50	7.40
Outline	28	line	13.52, 11.13	13.61, 11.17		

Outline	29	arc clockwise	13.61, 11.17	0.31, 4.25	7.07, 7.50	7.50
Outline	30	line	0.31, 4.25	0.20, 4.82		
Outline	31	arc clockwise	0.20, 4.82	0.00, 5.75	4.09, 6.12	4.11
Cut-out 1	32	line	2.08, 1.97	2.08, 1.97		
Cut-out 1	33	line	2.08, 1.97	2.08, 1.97		
Cut-out 1	34	line	2.08, 1.97	2.08, 1.97		
Cut-out 2	35	line	1.05, 3.12	1.05, 3.12		
Cut-out 2	36	line	1.05, 3.12	1.05, 3.12		
Cut-out 2	37	line	1.05, 3.12	1.05, 3.12		
Cut-out 2	38	line	1.05, 3.12	1.05, 3.12		





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **50:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 06 03-2**
Behandlung / Treatment			*Teil / Part* *stopWork.barpin*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Stop bar's pin (Stift)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Stop bar’s pin (Stift) — sheet 2

The pin pressed across the stop bar near its lower end; the bar’s spring bears on the end standing out toward the arbor.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 03-2	Stift (05 06 03-2, the coloured section's label), the stop bar's pin, on which its spring's end bears	1	not on the drawing		in 050603-1 (embed): 0.5 mm, least gap -1.3743

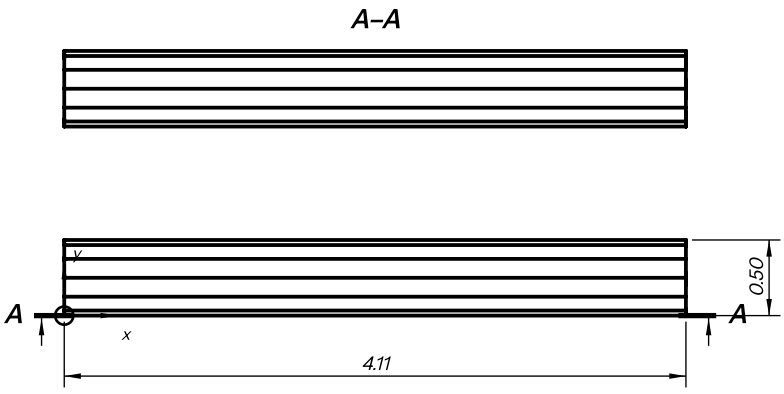
What it does (05 06 03-2). Pressed across the bar near its lower end, standing 0.9 out of its near side, where the bar's spring bears (05 06_4's C-D).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 0.50.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.00	0.00, 0.50		
Outline	2	line	0.00, 0.50	2.90, 0.50		
Outline	3	line	2.90, 0.50	2.90, 0.00		
Outline	4	line	2.90, 0.00	0.00, 0.00		





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 06 04-4** *Teil / Part* *stopWork.pin*
Behandlung / Treatment		*Benennung / Name* ***Stop bar spring's pin (Stift)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Stop bar spring’s pin (Stift) — sheet 2

The pin along the slot in the body’s small end that holds the stop bar’s spring.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 06 04-4	Stift (05 06 04-4: the coloured section's label and 05 06 04's parts list, likely), the stop bar's spring's pin	1	not on the drawing		in 050604-2 (press): 2.94 mm, least gap 0.0063

What it does (05 06 04-4). Along the slot in the body's hole 0.5 H7 (05 06 04-2), from the recess out through the end, standing into the recess through the spring.

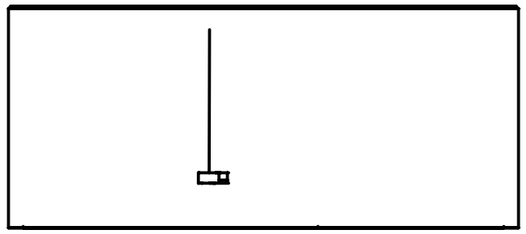
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 0.50.

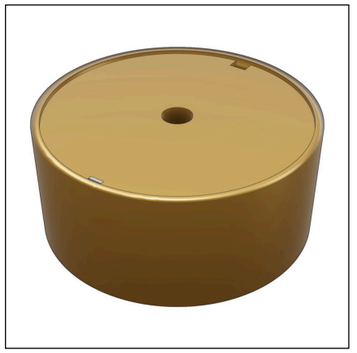
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.25	0.00, 0.50		
Outline	2	line	0.00, 0.50	4.11, 0.50		
Outline	3	line	4.11, 0.50	4.11, 0.00		
Outline	4	line	4.11, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 0.25		

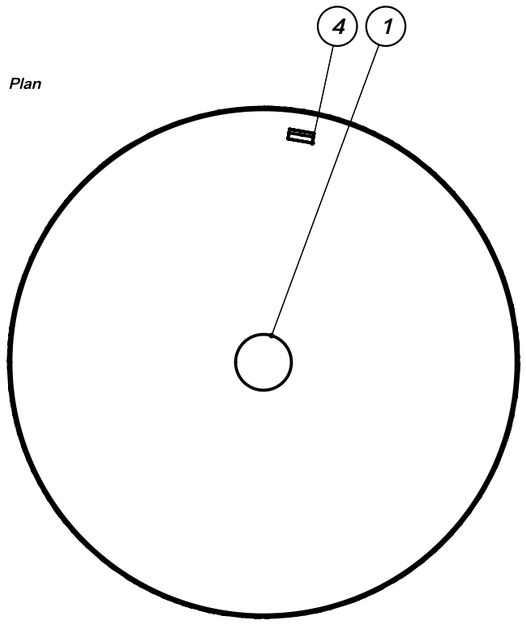
Front view





Isometric view, not to scale

Plan



Not seen in either view (inside the assembly): Pos. 2, 3.

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Steg (spring brace)	05 07 00-1	not on the drawing
3	1	Spring hook block (Federhaushaken)	05 07 01-3	not on the drawing
2	1	Barrel lid (Federhausdeckel)	05 07 01-2	„Ms 58 DIN 1751"
1	1	Barrel drum (Federhaustrommel)	05 07 01-1	„Ms. 63 3" (brass Ms 63, 3 mm,

Werkstoff / Material: not on the drawing

Maßstab / Scale: **2:1**

Behandlung / Treatment:

Blatt / Sheet: **1 / 1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

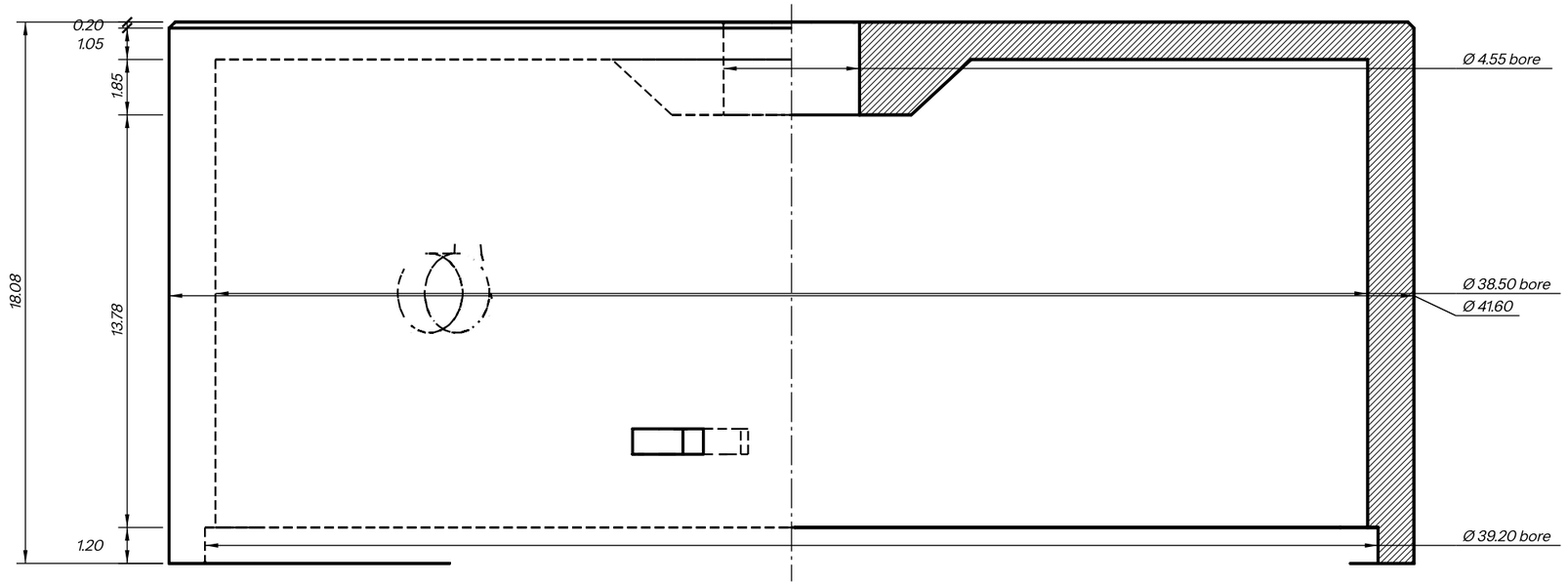
Zeichnung Nr. / Drawing no.: **05 07 00-1**

Teil / Part: barrel

Benennung / Name: **Barrel (Federhaus)**

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Ms. 63 3" (brass Ms 63, 3 mm, i.e. 3 mm sheet)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 01-1*** *Teil / Part* *barrel.drum*
Behandlung / Treatment *surface ∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished); untoleranced sizes ±0,1*			
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Barrel drum (Federhaustrommel)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Barrel drum (Federhaustrommel) — sheet 2

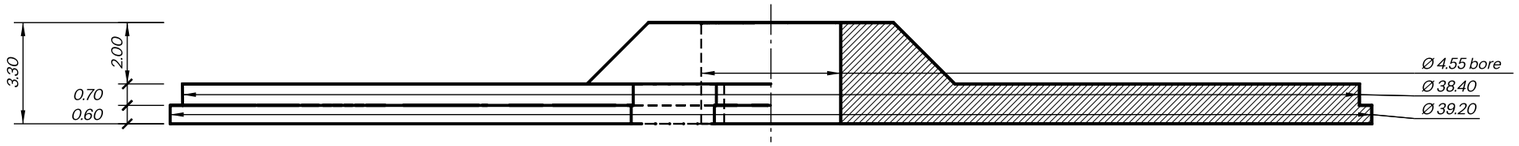
The drum the mainspring is coiled in: its outer end is hooked to the drum's wall, its inner end to the arbor, which the set-up ratchet holds. Running, the spring turns the drum, which winds the chain onto itself off the fusee, about five turns over a wind. Its lid, sprung in, is on the barrel bridge's side.

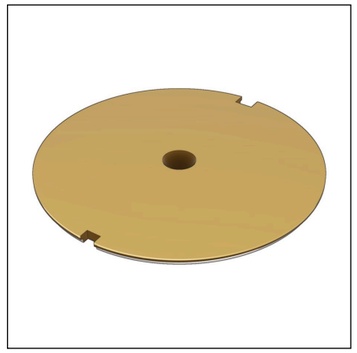
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 01-1	Federhaustrommel (barrel drum)	1	„Ms. 63 3“ (brass Ms 63, 3 mm, i.e. 3 mm sheet)	surface ∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	round 050702-1 (free): 3.03 mm, least gap 0.0396

What it does (05 07 01-1). Turns on the arbor's journals; the chain winds onto it.

Turned about its axis; length 18.08, largest Ø 41.60. Diameters: Ø 41.60; bores Ø 4.55, Ø 38.50, Ø 39.20. Shoulders from the lower end: 0.00, 1.20, 14.98, 16.83, 17.88, 18.08.





Isometric view, not to scale

Werkstoff / Material *„Ms 58 DIN 1751"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 01-2***
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*			*Teil / Part* *barrel.lid*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Barrel lid (Federhausdeckel)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Barrel lid (Federhausdeckel) — sheet 2

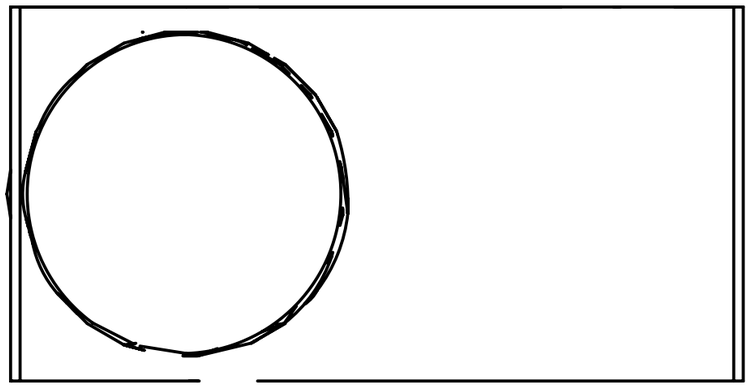
The lid, sprung into the drum's seat on the barrel bridge's side, its boss bearing on the arbor.

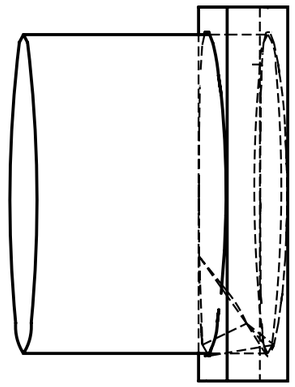
Parts list

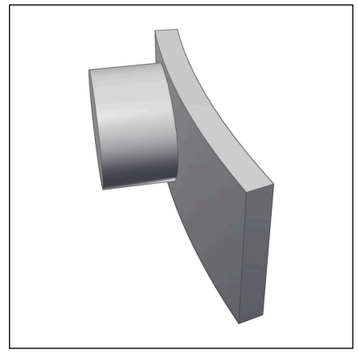
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 01-2	Federhausdeckel (barrel lid)	1	„Ms 58 DIN 1751“	surface ∇∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	round 050702-1 (free): 3.283 mm, least gap 0.0396

What it does (05 07 01-2). Sprung into the drum's seat on the bridge's side; turns on the arbor's upper journal.

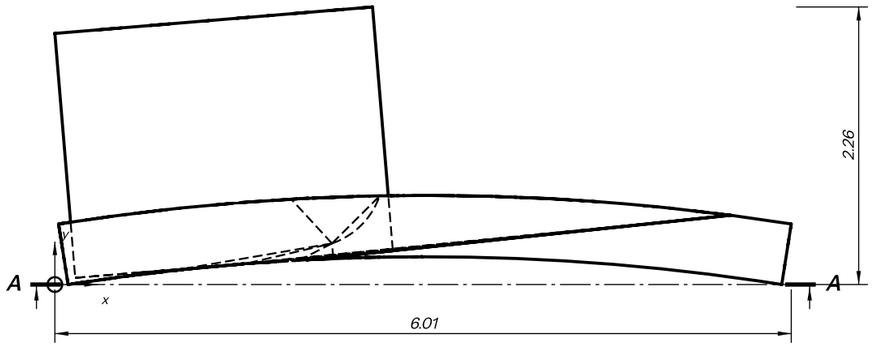
Turned about its axis; length 3.30, largest Ø 39.20. Diameters: Ø 38.40, Ø 39.20; bores Ø 4.55. Shoulders from the lower end: 0.00, 0.60, 1.30, 3.30.







Isometric view, not to scale



Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 07 01-3**
Behandlung / Treatment			*Teil / Part* *barrel.hook*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Spring hook block (Federhaushaken)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Spring hook block (Federhaushaken) — sheet 2

The block riveted inside the drum's wall that takes the mainspring's outer end: the strip's last 5.7 mm lie on it.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 01-3	Federhaushaken (spring hook block)	1	not on the drawing		on 050701-1: 0 mm

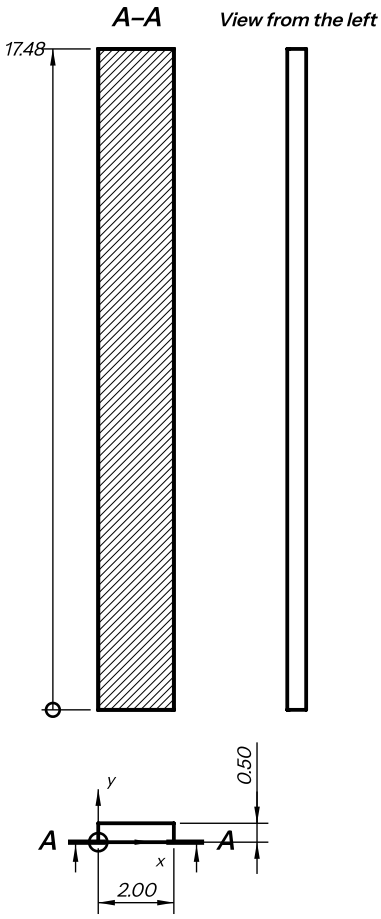
What it does (05 07 01-3). Riveted inside the drum's wall: takes the mainspring's outer end.

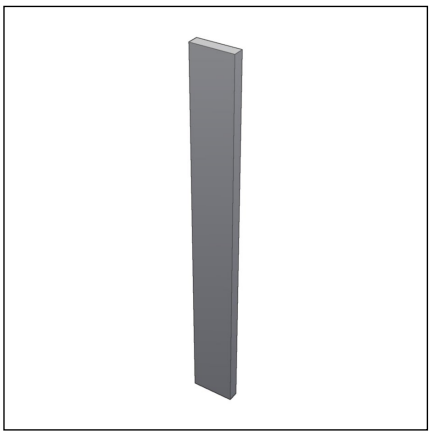
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.05.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.03, 0.49	0.13, 0.51		
Outline	2	line	0.13, 0.51	0.00, 2.05		
Outline	3	line	0.00, 2.05	2.59, 2.26		
Outline	4	line	2.59, 2.26	2.72, 0.72		
Outline	5	line	2.72, 0.72	3.02, 0.73		
Outline	6	arc clockwise	3.02, 0.73	6.01, 0.49	3.02, -18.52	19.25
Outline	7	line	6.01, 0.49	5.93, 0.00		
Outline	8	arc anticlockwise	5.93, 0.00	3.00, 0.23	3.02, -18.50	18.72
Outline	9	line	3.00, 0.23	2.38, 0.21		
Outline	10	line	2.38, 0.21	1.80, 0.19		
Outline	11	line	1.80, 0.19	1.18, 0.14		
Outline	12	line	1.18, 0.14	0.45, 0.05		
Outline	13	line	0.45, 0.05	0.11, 0.00		
Outline	14	line	0.11, 0.00	0.03, 0.49		
Cut-out 1	15	line	2.54, 0.22	2.54, 0.22		
Cut-out 1	16	line	2.54, 0.22	2.54, 0.22		
Cut-out 2	17	line	0.15, 0.01	0.15, 0.01		
Cut-out 2	18	line	0.15, 0.01	0.15, 0.01		
Cut-out 3	19	line	0.11, 0.00	0.15, 0.05		
Cut-out 3	20	line	0.15, 0.05	0.11, 0.00		





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 00-1*** *Teil / Part* *barrel.steg*
Behandlung / Treatment		*Benennung / Name* ***Steg (spring brace)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Steg (spring brace) — sheet 2

A steel bar standing inside the drum's wall from its bottom to its lid, 45° back from the hook block, its ends through a slot in the bottom and the lid's notch. The mainspring's outer coil is threaded between it and the wall, so the strip's last 45° lies on the wall to the hook.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 00-1	Steg (spring brace)	1	not on the drawing		on 050701-2: 0 mm; on 050701-1: 0 mm

What it does (05 07 00-1). Stands inside the drum's wall from its bottom to its lid, its ends through the bottom's slot and the lid's notch: the mainspring's outer coil threaded between it and the wall.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 17.48.

Contour in the plan

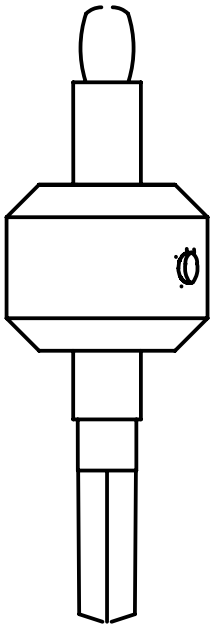
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

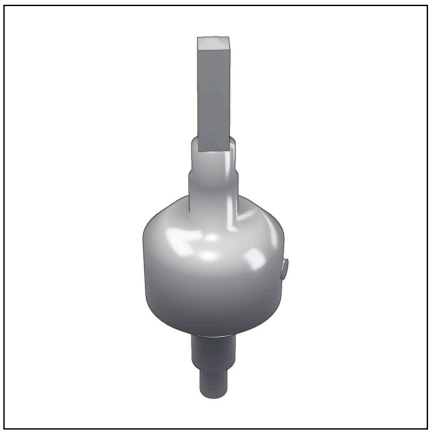
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.50	2.00, 0.50		
Outline	2	line	2.00, 0.50	2.00, 0.00		
Outline	3	line	2.00, 0.00	0.00, 0.00		
Outline	4	line	0.00, 0.00	0.00, 0.50		

Levels

Faces in section A-A, from the bottom face: 17.48.

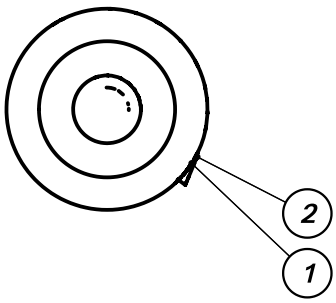
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Arbor hook (Federhaken)	05 07 02-2	not on the drawing
1	1	Barrel arbor (Federkern)	05 07 02-1	„Präz. Rundstahl 14 Din 175“ (

Werkstoff / Material

„Präz. Rundstahl 14 Din 175“ (precision round steel 14 dia, DIN 175)

Maßstab / Scale

2:1

Behandlung / Treatment

„härten bei 800(?)“ (harden at 800?); „anlassen 270“ (temper at 270); „Kegel 1:70 poliert“; „härten bei 800 / anlassen 270“; surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); blank (Rohmaß) 14 × 45…

Blatt / Sheet

1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.

05 07 02-1

Teil / Part

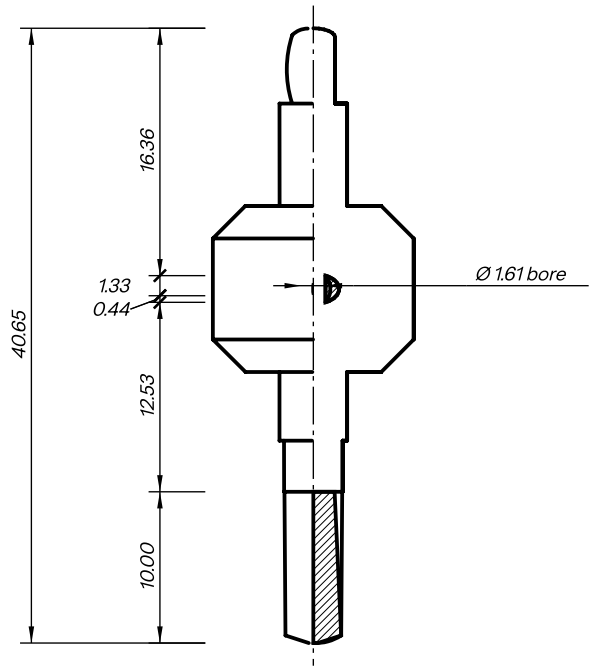
barrelArbor

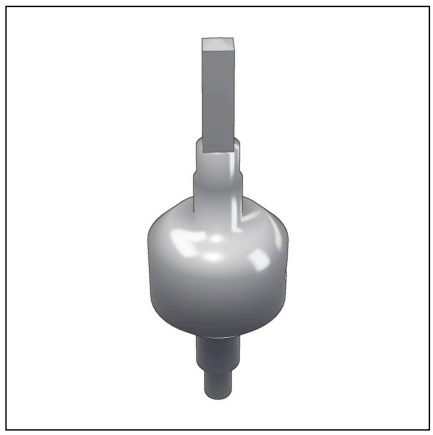
Benennung / Name

Barrel arbor (Federkern)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

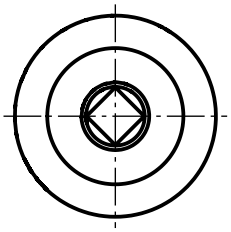
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

View from below (seen from beneath, x reversed)



Werkstoff / Material „Präz. Rundstahl 14 Din 175“ (precision round steel 14 dia, DIN 175)	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 02-1***
Behandlung / Treatment „härten bei 800(?)“ (harden at 800?); „anlassen 270“ (temper at 270); „Kegel 1:70 poliert“; „härten bei 800 / anlassen 270“; surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); blank (Rohmaß) 14 × 45…			*Teil / Part* *barrelArbor.arbor*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Barrel arbor (Federkern)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Barrel arbor (Federkern) — sheet 2

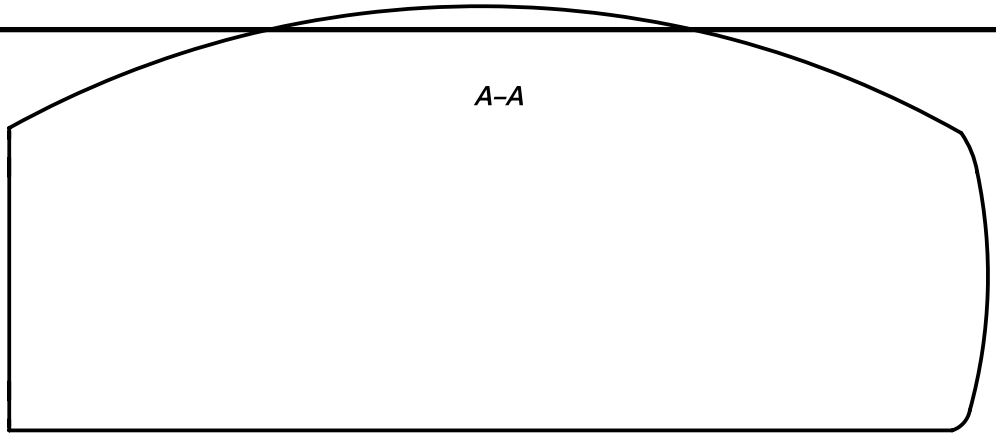
The arbor the mainspring's inner end is hooked to; it doesn't turn in use: the set-up ratchet on its square, held by the barrel bridge's click, keeps the spring set up. The drum turns on its two journals.

Parts list

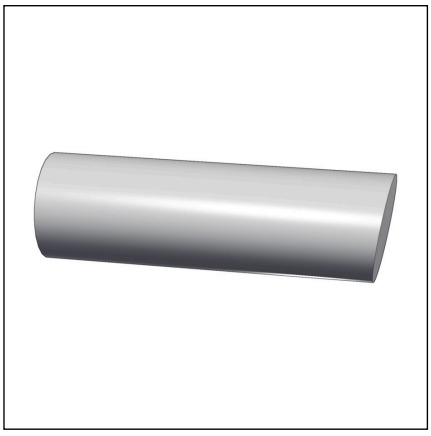
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 02-1	Federkern (barrel arbor)	1	„Präz. Rundstahl 14 Din 175“ (precision round steel 14 dia, DIN 175)	„härten bei 800(?)“ (harden at 800?); „anlassen 270“ (temper at 270); „Kegel 1:70 poliert“; „härten bei 800 / anlassen 270“; surface ∇∇∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); blank (Rohmaß) 14 × 45; untoleranced sizes ±0,05	in 050802-1 (run): 3.535 mm, least gap 0.0124; in 050805-1 (run): 3.283 mm, least gap 0.0135; endshake 0.060 mm (050802-1 / 050805-1)

What it does (05 07 02-1). Runs in the pillar plate and the barrel bridge; held by the set-up ratchet and click.

Turned about its axis; length 40.65, largest Ø 13.29. Diameters: ; bores Ø 1.61. Shoulders from the lower end: 0.00, 10.00, 22.53, 22.96, 24.29, 40.65.







Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 02-2***
Behandlung / Treatment			*Teil / Part* *barrelArbor.hook*
		Benennung / Name ***Arbor hook (Federhaken)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Arbor hook (Federhaken) — sheet 2

The pin in the arbor's core that takes the mainspring's inner end.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 02-2	Federhaken (arbor hook)	1	not on the drawing		in 050702-1 (press): 5.928 mm, least gap -0.0097

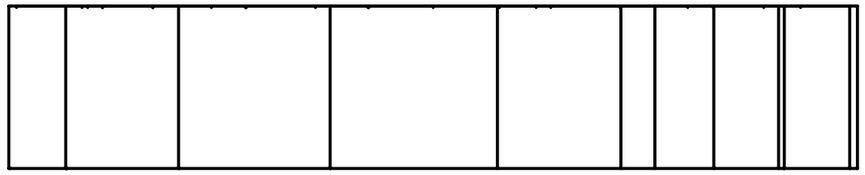
What it does (05 07 02-2). Pressed in the core's hole (Ø 2 × 6, 0207): takes the mainspring's inner end.

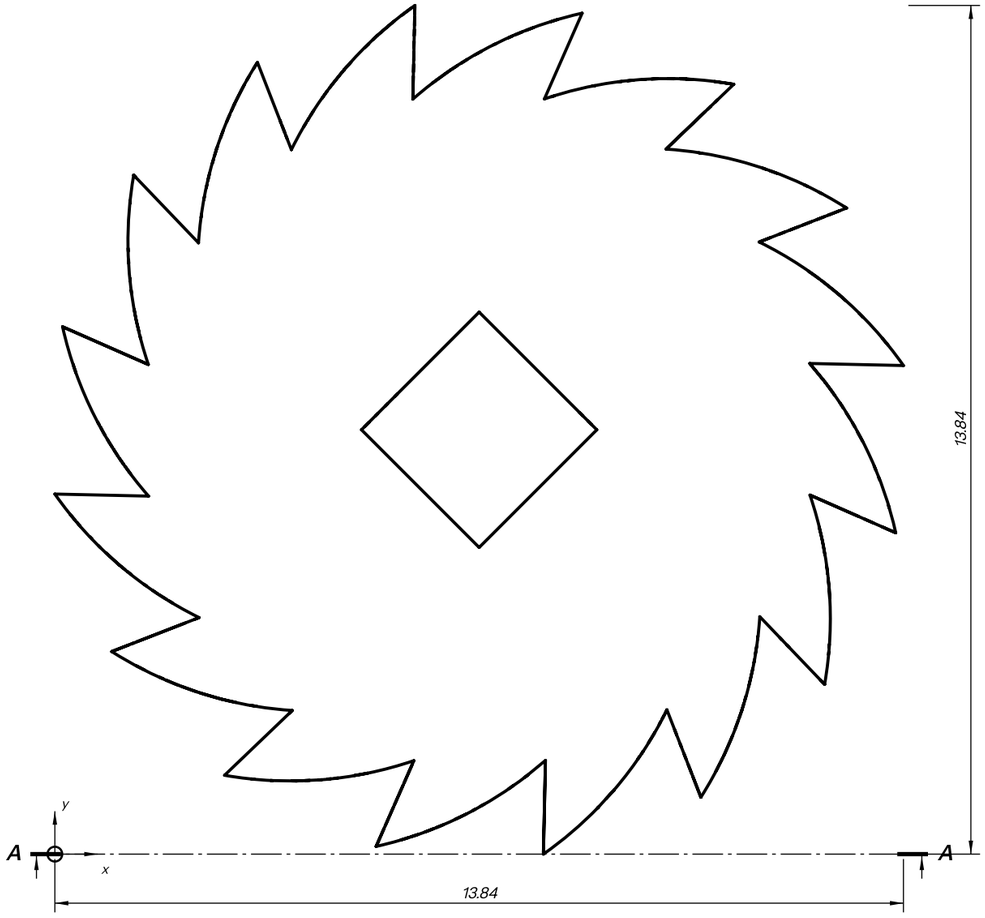
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.00.

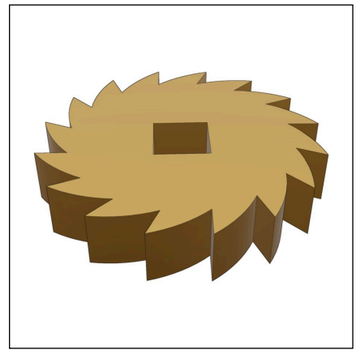
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.50	0.00, 2.00		
Outline	2	line	0.00, 2.00	6.00, 2.00		
Outline	3	line	6.00, 2.00	6.47, 0.00		
Outline	4	line	6.47, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 0.50		







Isometric view, not to scale

Werkstoff / Material *„Messingblech 3 Din 1751 Ms58" (brass sheet 3, DIN 1751, Ms 58)*	*Maßstab / Scale* ***10:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 00-3*** *Teil / Part* *setupRatchet*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*		*Benennung / Name* ***Set-up ratchet (Sperrad)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Set-up ratchet (Sperrad) — sheet 2

The ratchet on the barrel arbor's square above the barrel bridge: the bridge's click holds it, which keeps the mainspring set up ¾ turn at run down.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 00-3	Sperrad (set-up ratchet)	1	„Messingblech 3 Din 1751 Ms58“ (brass sheet 3, DIN 1751, Ms 58)	surface ∇∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished)	round 050702-1 (press): 2.777 mm, least gap 0.0075

What it does (05 07 00-3). On the arbor's square above the barrel bridge; the bridge's click holds it.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.65.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

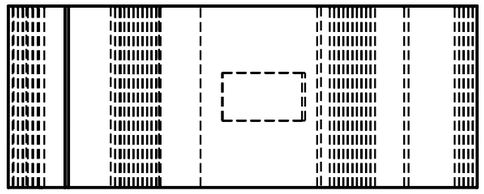
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.93, 3.30	2.35, 3.86		
Outline	2	arc clockwise	2.35, 3.86	0.00, 5.87	5.11, 9.46	6.25
Outline	3	line	0.00, 5.87	1.53, 5.84		
Outline	4	arc clockwise	1.53, 5.84	0.13, 8.60	6.23, 9.96	6.25
Outline	5	line	0.13, 8.60	1.53, 7.99		
Outline	6	arc clockwise	1.53, 7.99	1.29, 11.07	7.44, 10.00	6.25
Outline	7	line	1.29, 11.07	2.34, 9.97		
Outline	8	arc clockwise	2.34, 9.97	3.30, 12.91	8.58, 9.56	6.25
Outline	9	line	3.30, 12.91	3.86, 11.49		
Outline	10	arc clockwise	3.86, 11.49	5.87, 13.84	9.46, 8.73	6.25
Outline	11	line	5.87, 13.84	5.84, 12.31		
Outline	12	arc clockwise	5.84, 12.31	8.60, 13.72	9.96, 7.62	6.25
Outline	13	line	8.60, 13.72	7.99, 12.32		
Outline	14	arc clockwise	7.99, 12.32	11.07, 12.56	10.00, 6.40	6.25
Outline	15	line	11.07, 12.56	9.97, 11.50		
Outline	16	arc clockwise	9.97, 11.50	12.91, 10.54	9.56, 5.26	6.25
Outline	17	line	12.91, 10.54	11.49, 9.98		
Outline	18	arc clockwise	11.49, 9.98	13.84, 7.97	8.73, 4.38	6.25
Outline	19	line	13.84, 7.97	12.31, 8.00		
Outline	20	arc clockwise	12.31, 8.00	13.72, 5.24	7.62, 3.88	6.25
Outline	21	line	13.72, 5.24	12.32, 5.86		
Outline	22	arc clockwise	12.32, 5.86	12.56, 2.77	6.40, 3.85	6.25
Outline	23	line	12.56, 2.77	11.50, 3.87		
Outline	24	arc clockwise	11.50, 3.87	10.54, 0.93	5.26, 4.28	6.25
Outline	25	line	10.54, 0.93	9.98, 2.35		
Outline	26	arc clockwise	9.98, 2.35	7.97, 0.00	4.38, 5.11	6.25
Outline	27	line	7.97, 0.00	8.00, 1.53		
Outline	28	arc clockwise	8.00, 1.53	5.24, 0.13	3.88, 6.23	6.25

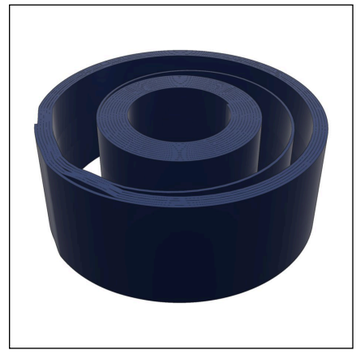
Outline	29	line	5.24, 0.13	5.86, 1.53		
Outline	30	arc clockwise	5.86, 1.53	2.77, 1.29	3.85, 7.44	6.25
Outline	31	line	2.77, 1.29	3.87, 2.34		
Outline	32	arc clockwise	3.87, 2.34	0.93, 3.30	4.28, 8.58	6.25
Cut-out 1	33	line	5.00, 6.92	6.92, 5.00		
Cut-out 1	34	line	6.92, 5.00	8.84, 6.92		
Cut-out 1	35	line	8.84, 6.92	6.92, 8.84		
Cut-out 1	36	line	6.92, 8.84	5.00, 6.92		

A–A

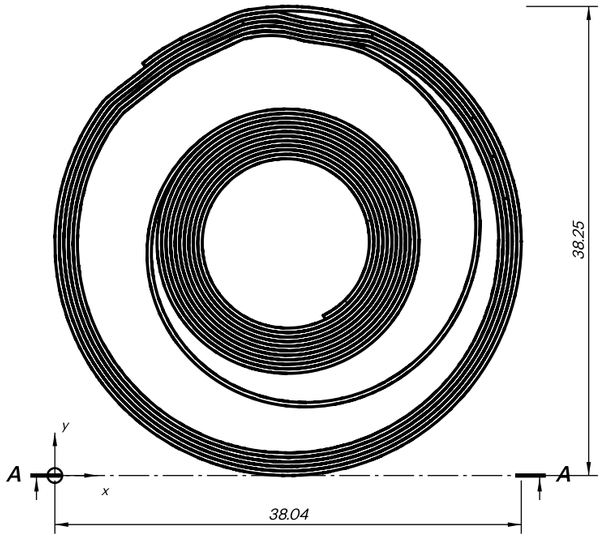


View from the left





Isometric view, not to scale



Werkstoff / Material not on the drawing	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 07 00-6**
Behandlung / Treatment			Teil / Part mainspring
Blatt / Sheet **1 / 2**		Benennung / Name **Mainspring (Zugfeder)**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Mainspring (Zugfeder) — sheet 2

The steel spring that drives the chronometer, 15 mm high and 0.37 thick, coiled in the barrel: wound, its coils draw in round the arbor; run down, they lie out against the wall. Its length isn't on any sheet: 1.39 m, half the room in the barrel, lets it work the barrel's five turns with the ¾ turn of set-up and room to spare.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 00-6	Zugfeder (mainspring)	1	not on the drawing		not measured on the model: see its function, In the drum: inner end on the arbor's hook, outer end on the drum's block. Its length isn't drawn: estimated.

Tolerance (05 07 00-6). The strip 0.393 mm as the build takes it: 0.346–0.424 mm (its torque ×0.68–1.25) keeps the swing in the guard band, 180–270°, over the whole wind: on a 262 mm lever on the fusee arbor 91–166 g, as 05 06_20 weighs it.

Test on the bench (05 07 00-6). The mainspring's torque, before it goes into the barrel, and again in its barrel: Set up ¾ turn past slack (05 06_20's "3/4 Umgang gespannt") and wound the drawn fusee's 5.08 barrel turns, the model's spring runs 546 to 1074 mN·m at the level 05 06_20 measures (the solved shape, wempe_physics.js); every reading within ×0.684–1.253 of it keeps the swing in the guard band over the wind and the strip under a mainspring's stress limit (wempe_tolerances.js). From slack to coil-bound it must have the set-up, the fusee's 5.08 turns and half a turn to spare (the model's 0.93 past full wind). Unwinding reads less than winding: the coils' friction gives 1–3 % on the solved spring, mainsprings 10–20 % (ICMFF9); over 22 %, clean and grease it. After the week, still in the band.

What it does (05 07 00-6). In the drum: inner end on the arbor's hook, outer end on the drum's block. Its length isn't drawn: estimated.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 14.80.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.22, 16.11	3.93, 30.50	18.91, 18.96	18.91
Outline	2	arc clockwise	3.93, 30.50	5.84, 32.10	14.34, 19.83	14.92
Outline	3	arc anticlockwise	5.84, 32.10	7.33, 33.28	-18.43, 64.42	40.42
Outline	4	arc clockwise	7.33, 33.28	12.23, 36.13	18.95, 18.98	18.42
Outline	5	arc anticlockwise	12.23, 36.13	13.59, 36.78	2.48, 58.49	24.39
Outline	6	arc clockwise	13.59, 36.78	15.62, 37.57	17.58, 29.77	8.05
Outline	7	arc clockwise	15.62, 37.57	18.10, 37.82	18.30, 23.54	14.29
Outline	8	arc clockwise	18.10, 37.82	19.04, 37.59	12.52, 12.58	25.85
Outline	9	arc anticlockwise	19.04, 37.59	21.28, 37.16	22.67, 50.15	13.07
Outline	10	arc clockwise	21.28, 37.16	22.85, 36.88	18.93, 19.03	18.28
Outline	11	arc anticlockwise	22.85, 36.88	24.83, 36.63	25.49, 50.06	13.44
Outline	12	line	24.83, 36.63	25.68, 36.55		
Outline	13	arc clockwise	25.68, 36.55	29.68, 34.44	19.47, 20.09	17.61
Outline	14	arc clockwise	29.68, 34.44	14.14, 1.01	18.96, 19.08	18.72
Outline	15	arc clockwise	14.14, 1.01	4.22, 30.28	18.95, 18.96	18.58
Outline	16	arc clockwise	4.22, 30.28	6.10, 31.84	14.22, 20.00	14.34
Outline	17	line	6.10, 31.84	7.03, 32.55		
Outline	18	line	7.03, 32.55	7.63, 33.04		
Outline	19	arc clockwise	7.63, 33.04	12.37, 35.79	18.96, 18.97	18.06
Outline	20	arc anticlockwise	12.37, 35.79	13.62, 36.38	3.97, 55.08	21.04

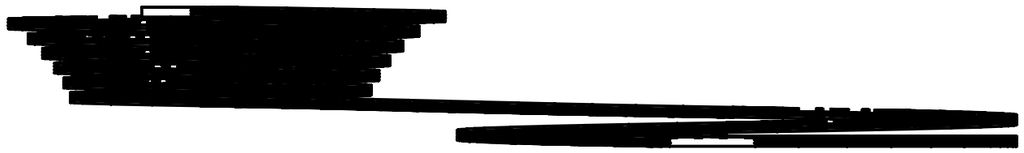
Outline	21	arc clockwise	13.62, 36.38	15.69, 37.21	17.64, 29.51	7.95
Outline	22	arc clockwise	15.69, 37.21	18.11, 37.45	18.31, 23.44	14.02
Outline	23	arc anticlockwise	18.11, 37.45	15.32, 37.14	18.17, 24.53	12.93
Outline	24	arc anticlockwise	15.32, 37.14	13.47, 36.31	20.87, 22.25	15.89
Outline	25	arc clockwise	13.47, 36.31	12.10, 35.69	5.50, 51.97	17.57
Outline	26	arc anticlockwise	12.10, 35.69	7.32, 32.80	18.93, 19.01	18.03
Outline	27	arc clockwise	7.32, 32.80	6.06, 31.81	-20.33, 66.85	43.86
Outline	28	arc anticlockwise	6.06, 31.81	4.20, 30.27	14.62, 19.48	15.00
Outline	29	arc anticlockwise	4.20, 30.27	29.34, 3.49	18.98, 18.97	18.61
Outline	30	arc anticlockwise	29.34, 3.49	27.22, 35.92	18.92, 19.09	18.76
Outline	31	arc anticlockwise	27.22, 35.92	16.03, 37.65	19.06, 20.03	17.87
Outline	32	arc anticlockwise	16.03, 37.65	14.10, 37.03	16.71, 32.37	5.32
Outline	33	line	14.10, 37.03	12.61, 36.31		
Outline	34	line	12.61, 36.31	12.05, 36.07		
Outline	35	arc anticlockwise	12.05, 36.07	7.39, 33.34	18.93, 19.02	18.39
Outline	36	line	7.39, 33.34	7.17, 33.62		
Outline	37	arc clockwise	7.17, 33.62	11.94, 36.41	18.93, 19.02	18.74
Outline	38	arc anticlockwise	11.94, 36.41	13.57, 37.17	2.28, 59.24	24.78
Outline	39	arc clockwise	13.57, 37.17	15.56, 37.94	17.54, 30.00	8.19
Outline	40	arc clockwise	15.56, 37.94	26.25, 36.78	19.04, 19.85	18.41
Outline	41	arc clockwise	26.25, 36.78	36.75, 25.98	18.79, 18.99	19.28
Outline	42	arc clockwise	36.75, 25.98	4.69, 6.49	18.99, 19.06	19.05
Outline	43	arc clockwise	4.69, 6.49	0.22, 16.11	18.95, 18.97	18.95
Cut-out 1	44	arc clockwise	18.99, 0.74	4.52, 30.05	18.96, 18.96	18.21
Cut-out 1	45	arc clockwise	4.52, 30.05	6.37, 31.57	14.44, 19.76	14.30
Cut-out 1	46	arc anticlockwise	6.37, 31.57	7.67, 32.60	-13.64, 58.55	33.59
Cut-out 1	47	arc clockwise	7.67, 32.60	12.33, 35.38	19.11, 18.75	17.96
Cut-out 1	48	arc anticlockwise	12.33, 35.38	7.56, 32.52	18.94, 19.00	17.66
Cut-out 1	49	arc clockwise	7.56, 32.52	6.34, 31.57	-17.41, 63.28	39.63
Cut-out 1	50	line	6.34, 31.57	5.38, 30.82		
Cut-out 1	51	arc anticlockwise	5.38, 30.82	3.82, 29.11	8.49, 26.46	5.36
Cut-out 1	52	arc anticlockwise	3.82, 29.11	29.79, 4.26	18.98, 18.97	18.24
Cut-out 1	53	arc anticlockwise	29.79, 4.26	25.97, 36.08	18.92, 19.09	18.39
Cut-out 1	54	arc anticlockwise	25.97, 36.08	25.55, 36.21	25.50, 35.31	0.90
Cut-out 1	55	arc clockwise	25.55, 36.21	29.47, 34.13	19.45, 20.06	17.27
Cut-out 1	56	arc clockwise	29.47, 34.13	18.99, 0.74	18.95, 19.09	18.35
Cut-out 2	57	arc clockwise	17.31, 1.19	4.05, 28.76	18.96, 18.96	17.84
Cut-out 2	58	arc clockwise	4.05, 28.76	6.51, 31.23	10.29, 25.00	7.29
Cut-out 2	59	line	6.51, 31.23	6.89, 31.52		
Cut-out 2	60	line	6.89, 31.52	5.65, 30.58		
Cut-out 2	61	arc anticlockwise	5.65, 30.58	4.79, 29.82	9.34, 25.47	6.29
Cut-out 2	62	arc anticlockwise	4.79, 29.82	28.60, 3.90	18.98, 18.97	17.87
Cut-out 2	63	arc anticlockwise	28.60, 3.90	25.83, 35.73	18.92, 19.09	18.02
Cut-out 2	64	line	25.83, 35.73	25.47, 35.86		
Cut-out 2	65	line	25.47, 35.86	23.94, 35.98		
Cut-out 2	66	arc clockwise	23.94, 35.98	27.35, 35.00	24.13, 30.21	5.78
Cut-out 2	67	arc clockwise	27.35, 35.00	17.31, 1.19	18.95, 19.09	17.99
Cut-out 3	68	arc anticlockwise	17.04, 36.29	13.84, 35.28	17.40, 29.54	6.76

Cut-out 3	69	arc anticlockwise	13.84, 35.28	10.81, 33.85	49.95, -44.95	87.99
Cut-out 3	70	arc anticlockwise	10.81, 33.85	8.07, 31.98	18.97, 18.96	16.98
Cut-out 3	71	arc anticlockwise	8.07, 31.98	5.39, 29.89	207.17, -226.74	326.46
Cut-out 3	72	arc anticlockwise	5.39, 29.89	2.76, 25.54	16.69, 20.11	14.95
Cut-out 3	73	arc anticlockwise	2.76, 25.54	32.62, 7.95	18.99, 18.98	17.52
Cut-out 3	74	arc anticlockwise	32.62, 7.95	25.63, 35.41	18.91, 19.08	17.66
Cut-out 3	75	line	25.63, 35.41	25.51, 35.43		
Cut-out 3	76	arc clockwise	25.51, 35.43	36.25, 15.78	18.93, 19.09	17.63
Cut-out 3	77	arc clockwise	36.25, 15.78	1.58, 17.07	19.00, 19.02	17.54
Cut-out 3	78	arc clockwise	1.58, 17.07	4.31, 28.49	18.91, 18.96	17.43
Cut-out 3	79	arc clockwise	4.31, 28.49	6.71, 30.92	10.50, 24.79	7.21
Cut-out 3	80	arc clockwise	6.71, 30.92	9.34, 32.93	121.85, -117.10	187.53
Cut-out 3	81	arc clockwise	9.34, 32.93	12.23, 34.53	18.92, 19.03	16.88
Cut-out 3	82	line	12.23, 34.53	15.28, 35.94		
Cut-out 3	83	arc clockwise	15.28, 35.94	20.35, 35.81	17.60, 27.20	9.04
Cut-out 3	84	line	20.35, 35.81	18.72, 36.18		
Cut-out 3	85	line	18.72, 36.18	17.04, 36.29		
Cut-out 4	86	arc anticlockwise	17.01, 35.86	13.88, 34.90	17.14, 29.82	6.03
Cut-out 4	87	arc anticlockwise	13.88, 34.90	10.92, 33.49	45.76, -35.65	77.42
Cut-out 4	88	arc anticlockwise	10.92, 33.49	8.24, 31.65	19.38, 18.30	17.39
Cut-out 4	89	arc anticlockwise	8.24, 31.65	5.62, 29.60	75.20, -56.58	110.76
Cut-out 4	90	arc anticlockwise	5.62, 29.60	3.07, 25.33	16.79, 20.05	14.70
Cut-out 4	91	arc anticlockwise	3.07, 25.33	32.38, 8.25	18.99, 18.98	17.15
Cut-out 4	92	arc anticlockwise	32.38, 8.25	25.41, 35.08	18.93, 19.08	17.27
Cut-out 4	93	arc clockwise	25.41, 35.08	22.04, 35.55	29.82, 79.34	44.48
Cut-out 4	94	line	22.04, 35.55	20.45, 35.80		
Cut-out 4	95	line	20.45, 35.80	23.59, 35.30		
Cut-out 4	96	arc clockwise	23.59, 35.30	26.69, 34.08	22.06, 26.88	8.56
Cut-out 4	97	arc clockwise	26.69, 34.08	34.46, 17.73	18.85, 20.31	15.83
Cut-out 4	98	arc clockwise	34.46, 17.73	24.47, 6.21	20.05, 20.10	14.59
Cut-out 4	99	arc clockwise	24.47, 6.21	11.23, 9.32	20.45, 18.87	13.28
Cut-out 4	100	arc clockwise	11.23, 9.32	9.09, 24.05	19.70, 18.09	12.18
Cut-out 4	101	arc clockwise	9.09, 24.05	17.65, 29.82	19.08, 18.50	11.43
Cut-out 4	102	arc clockwise	17.65, 29.82	23.24, 9.21	18.91, 19.09	10.80
Cut-out 4	103	arc clockwise	23.24, 9.21	12.05, 27.01	18.97, 18.96	10.63
Cut-out 4	104	arc clockwise	12.05, 27.01	28.21, 14.36	18.87, 19.06	10.46
Cut-out 4	105	arc clockwise	28.21, 14.36	8.86, 20.73	19.00, 19.00	10.30
Cut-out 4	106	arc clockwise	8.86, 20.73	28.89, 20.31	18.85, 19.02	10.13
Cut-out 4	107	arc clockwise	28.89, 20.31	9.93, 14.95	19.00, 19.04	9.96
Cut-out 4	108	arc clockwise	9.93, 14.95	26.22, 25.44	18.87, 18.98	9.80
Cut-out 4	109	arc clockwise	26.22, 25.44	13.92, 10.90	18.97, 19.08	9.63
Cut-out 4	110	arc clockwise	13.92, 10.90	21.30, 28.09	18.90, 18.95	9.46
Cut-out 4	111	arc clockwise	21.30, 28.09	19.30, 9.82	18.94, 19.09	9.29
Cut-out 4	112	arc clockwise	19.30, 9.82	15.93, 27.55	18.94, 18.95	9.13
Cut-out 4	113	arc clockwise	15.93, 27.55	24.19, 11.87	18.89, 19.08	8.96
Cut-out 4	114	arc clockwise	24.19, 11.87	11.94, 24.22	18.98, 18.97	8.79
Cut-out 4	115	arc clockwise	11.94, 24.22	26.97, 16.15	18.86, 19.05	8.62
Cut-out 4	116	arc clockwise	26.97, 16.15	10.56, 19.41	19.00, 19.01	8.46

Cut-out 4	117	arc clockwise	10.56, 19.41	26.88, 21.05	18.86, 19.01	8.29
Cut-out 4	118	arc clockwise	26.88, 21.05	12.06, 14.84	18.99, 19.05	8.12
Cut-out 4	119	arc clockwise	12.06, 14.84	24.16, 24.90	18.87, 18.97	7.95
Cut-out 4	120	arc clockwise	24.16, 24.90	15.71, 12.02	18.97, 19.08	7.79
Cut-out 4	121	arc clockwise	15.71, 12.02	19.94, 26.49	18.91, 18.95	7.62
Cut-out 4	122	arc clockwise	19.94, 26.49	20.12, 11.75	18.93, 19.09	7.45
Cut-out 4	123	arc clockwise	20.12, 11.75	15.74, 25.47	18.95, 18.95	7.28
Cut-out 4	124	arc clockwise	15.74, 25.47	23.89, 14.04	18.89, 19.08	7.12
Cut-out 4	125	arc clockwise	23.89, 14.04	22.06, 12.74	18.98, 19.04	7.01
Cut-out 4	126	line	22.06, 12.74	21.89, 13.06		
Cut-out 4	127	arc anticlockwise	21.89, 13.06	17.81, 25.75	18.91, 19.09	6.74
Cut-out 4	128	arc anticlockwise	17.81, 25.75	18.01, 12.09	18.93, 18.95	6.91
Cut-out 4	129	arc anticlockwise	18.01, 12.09	21.90, 25.53	18.95, 19.09	7.07
Cut-out 4	130	arc anticlockwise	21.90, 25.53	14.07, 13.54	18.89, 18.96	7.24
Cut-out 4	131	arc anticlockwise	14.07, 13.54	25.32, 22.93	18.98, 19.07	7.41
Cut-out 4	132	arc anticlockwise	25.32, 22.93	11.51, 17.11	18.86, 18.99	7.58
Cut-out 4	133	arc anticlockwise	11.51, 17.11	26.75, 18.68	19.00, 19.03	7.74
Cut-out 4	134	arc anticlockwise	26.75, 18.68	11.39, 21.69	18.86, 19.03	7.91
Cut-out 4	135	arc anticlockwise	11.39, 21.69	25.49, 14.16	18.99, 18.99	8.08
Cut-out 4	136	arc anticlockwise	25.49, 14.16	13.97, 25.72	18.88, 19.07	8.25
Cut-out 4	137	arc anticlockwise	13.97, 25.72	20.96, 10.78	18.96, 18.95	8.41
Cut-out 4	138	arc anticlockwise	20.96, 10.78	19.40, 27.67	18.92, 19.09	8.58
Cut-out 4	139	arc anticlockwise	19.40, 27.67	15.89, 10.73	18.91, 18.95	8.74
Cut-out 4	140	arc anticlockwise	15.89, 10.73	24.36, 26.19	18.96, 19.08	8.91
Cut-out 4	141	arc anticlockwise	24.36, 26.19	11.50, 13.65	18.88, 18.97	9.08
Cut-out 4	142	arc anticlockwise	11.50, 13.65	27.76, 22.04	18.99, 19.05	9.25
Cut-out 4	143	arc anticlockwise	27.76, 22.04	9.44, 18.71	18.86, 19.01	9.41
Cut-out 4	144	arc anticlockwise	9.44, 18.71	28.26, 16.50	19.00, 19.01	9.58
Cut-out 4	145	arc anticlockwise	28.26, 16.50	10.60, 24.25	18.86, 19.05	9.75
Cut-out 4	146	arc anticlockwise	10.60, 24.25	25.47, 11.46	18.98, 18.97	9.92
Cut-out 4	147	arc anticlockwise	25.47, 11.46	14.81, 28.32	18.89, 19.08	10.08
Cut-out 4	148	arc anticlockwise	14.81, 28.32	20.17, 8.76	18.95, 18.95	10.25
Cut-out 4	149	arc anticlockwise	20.17, 8.76	20.77, 29.36	18.93, 19.09	10.42
Cut-out 4	150	arc anticlockwise	20.77, 29.36	11.30, 26.35	18.88, 18.92	10.61
Cut-out 4	151	arc anticlockwise	11.30, 26.35	9.98, 11.38	19.57, 18.09	11.70
Cut-out 4	152	arc anticlockwise	9.98, 11.38	23.07, 6.22	20.40, 18.71	12.76
Cut-out 4	153	arc anticlockwise	23.07, 6.22	33.77, 16.33	20.14, 20.02	14.11
Cut-out 4	154	arc anticlockwise	33.77, 16.33	27.87, 32.88	18.94, 20.35	15.38
Cut-out 4	155	arc anticlockwise	27.87, 32.88	25.07, 34.51	18.52, 19.97	15.94
Cut-out 4	156	arc anticlockwise	25.07, 34.51	21.90, 35.20	20.65, 21.90	13.36
Cut-out 4	157	arc clockwise	21.90, 35.20	18.65, 35.82	25.17, 61.11	26.11
Cut-out 4	158	line	18.65, 35.82	17.01, 35.86		
Cut-out 5	159	line	25.41, 35.08	25.50, 35.43		
Cut-out 5	160	line	25.50, 35.43	25.41, 35.08		
Cut-out 6	161	line	24.81, 23.94	24.81, 23.94		
Cut-out 6	162	line	24.81, 23.94	24.81, 23.94		
Cut-out 6	163	line	24.81, 23.94	24.81, 23.94		
Cut-out 7	164	arc anticlockwise	19.24, 37.17	21.24, 36.79	22.42, 48.37	11.64

Cut-out 7	165	arc clockwise	21.24, 36.79	22.77, 36.52	18.93, 19.03	17.91
Cut-out 7	166	arc anticlockwise	22.77, 36.52	24.12, 36.33	24.78, 45.77	9.46
Cut-out 7	167	line	24.12, 36.33	24.28, 36.32		
Cut-out 7	168	arc clockwise	24.28, 36.32	22.49, 36.59	25.14, 47.83	11.54
Cut-out 7	169	arc anticlockwise	22.49, 36.59	20.85, 36.85	18.91, 18.90	18.05
Cut-out 7	170	arc clockwise	20.85, 36.85	19.24, 37.17	22.54, 49.34	12.61
Cut-out 8	171	line	19.33, 37.15	19.25, 36.81		
Cut-out 8	172	line	19.25, 36.81	19.33, 37.15		
Cut-out 9	173	line	22.10, 35.91	23.84, 35.63		
Cut-out 9	174	arc anticlockwise	23.84, 35.63	20.70, 36.12	-1.00, -114.03	151.71
Cut-out 9	175	line	20.70, 36.12	22.10, 35.91		
Cut-out 10	176	line	22.25, 36.25	22.49, 36.22		
Cut-out 10	177	arc anticlockwise	22.49, 36.22	20.81, 36.48	18.96, 19.27	17.31
Cut-out 10	178	line	20.81, 36.48	20.45, 36.53		
Cut-out 10	179	arc anticlockwise	20.45, 36.53	22.25, 36.25	23.11, 47.74	11.52
Cut-out 11	180	arc anticlockwise	17.08, 36.67	15.28, 36.34	16.80, 33.12	3.56
Cut-out 11	181	arc clockwise	15.28, 36.34	18.78, 36.54	17.45, 29.27	7.39
Cut-out 11	182	line	18.78, 36.54	20.49, 36.15		
Cut-out 11	183	arc anticlockwise	20.49, 36.15	17.08, 36.67	17.17, 25.66	11.01
Cut-out 12	184	line	18.80, 36.54	18.86, 36.90		
Cut-out 12	185	line	18.86, 36.90	18.80, 36.54		
Cut-out 13	186	arc clockwise	13.66, 35.99	18.85, 36.90	17.42, 29.74	7.30
Cut-out 13	187	line	18.85, 36.90	18.08, 37.09		
Cut-out 13	188	arc anticlockwise	18.08, 37.09	15.39, 36.77	18.33, 23.51	13.59
Cut-out 13	189	arc anticlockwise	15.39, 36.77	13.58, 35.96	20.86, 22.24	15.53
Cut-out 13	190	line	13.58, 35.96	13.66, 35.99		
Cut-out 14	191	line	8.09, 32.47	7.47, 31.97		
Cut-out 14	192	arc clockwise	7.47, 31.97	12.16, 34.90	19.34, 18.28	18.11
Cut-out 14	193	arc clockwise	12.16, 34.90	15.24, 36.31	141.92, -244.13	307.73
Cut-out 14	194	arc anticlockwise	15.24, 36.31	10.71, 34.21	72.79, -93.90	142.37
Cut-out 14	195	arc anticlockwise	10.71, 34.21	8.09, 32.47	18.93, 19.01	17.29

Front view

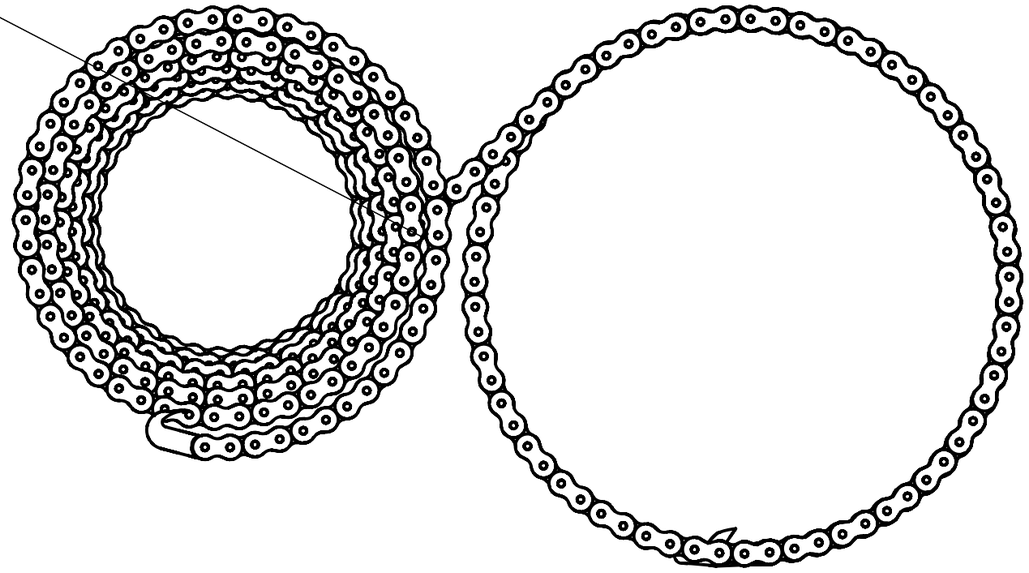




Isometric view, not to scale

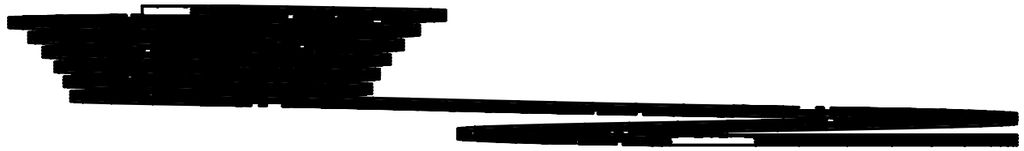
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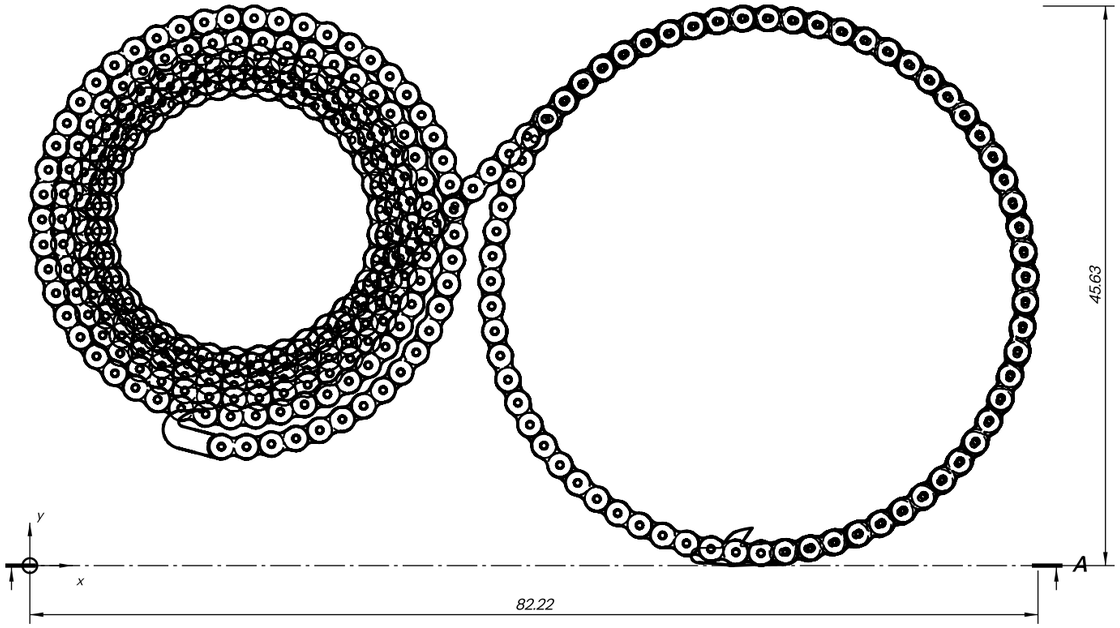
Plan

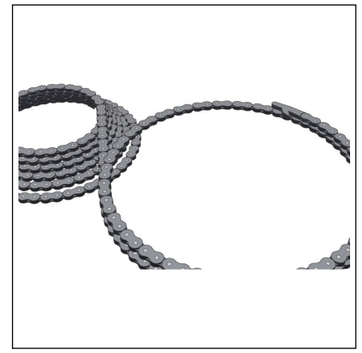


1	[0, 1]	Chain links	05 07 04	„Bandstahl 5×0,3; 10×0,9; Präz
Pos.	**Units**	**Benennung / Name**	**Zeichnung / Drawing**	**Werkstoff / Material**

Werkstoff / Material „Bandstahl 5×0,3; 10×0,9; Präz. Rundstahl 0,6" (strip steel 5 × 0,3 [the link plates, inferred]; 10 × 0,9 [the end hooks, inferred]; precision round steel 0,6 [the…	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26
Behandlung / Treatment „leicht polieren" (polish lightly); „▽▽▽ härten, 310° anlassen"; untoleranced sizes ±0,05		Zeichnung Nr. / Drawing no. **05 07 04** Teil / Part chain
Blatt / Sheet **1 / 1**		Benennung / Name **Fusee chain (Kette)** Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).







Isometric view, not to scale

Werkstoff / Material *„Bandstahl 5×0,3; 10×0,9; Präz. Rundstahl 0,6" (strip steel 5 × 0,3 [the link plates, inferred]; 10 × 0,9 [the end hooks, inferred]; precision round steel 0,6 [the…*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 04***
Behandlung / Treatment *„leicht polieren" (polish lightly); „▽▽▽ härten, 310° anlassen"; untoleranced sizes ±0,05*			*Teil / Part* *chain.links*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Chain links***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Chain links — sheet 2

The steel chain between the barrel and the fusee: 374 links of figure-of-eight plates three deep, riveted to move: an outer link two plates, an inner link one between them, each joint one steel rivet through all three. Its top hook takes the pin in the fusee's flange, its bottom hook a slot in the barrel's wall. Fully wound it lies in the fusee's groove up to the top turn, with a quarter turn left on the barrel; run down it is all on the barrel. Wempe 3955 and Lange 5177 run with a chain.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 04	Kette (fusee chain)	[0, 1]	„Bandstahl 5×0,3; 10×0,9; Präz. Rundstahl 0,6“ (strip steel 5 × 0,3 [the link plates, inferred]; 10 × 0,9 [the end hooks, inferred]; precision round steel 0,6 [the rivets, inferred])	„leicht polieren“ (polish lightly); „▽▽▽ härten, 310° anlassen“; untoleranced sizes ±0,05	not measured on the model: see its function, Between the barrel and the fusee: its top hook on the fusee's pin, its bottom hook in the drum's slot (the page's default; the band under Variants).

What it does (05 07 04). Between the barrel and the fusee: its top hook on the fusee's pin, its bottom hook in the drum's slot (the page's default; the band under Variants).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 11.34.

The contour has 3612 elements (its teeth): the drawing and the tooth data define it.

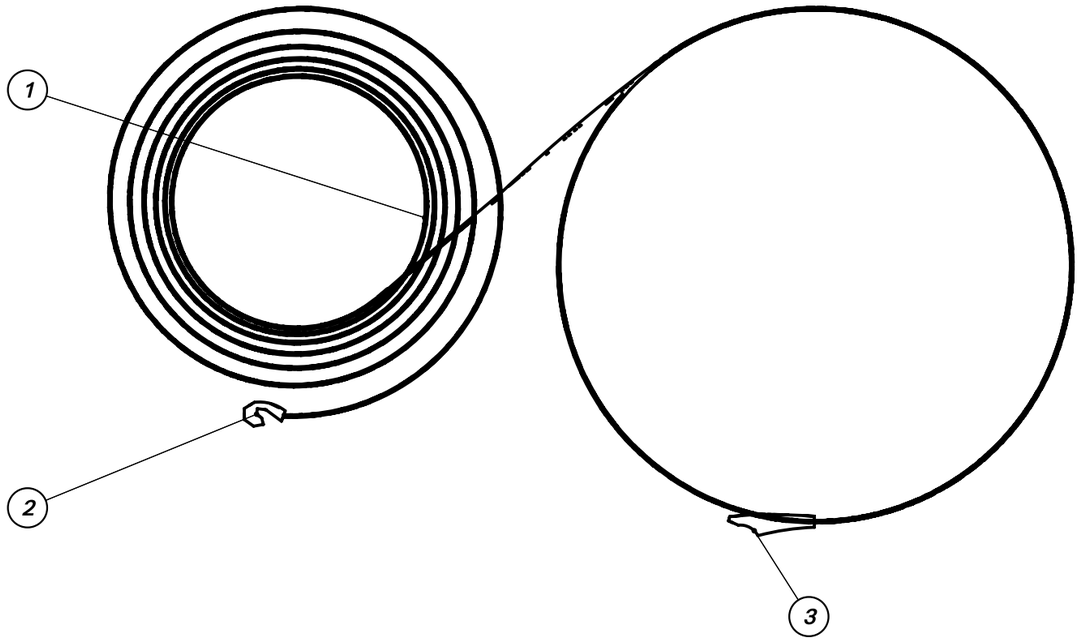
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	[0, 1]	Band's drum hook (Bandhaken f. Trommel)	05 07 03-3	„Bandstahl 10×0,8 Din 1016“ (s
2	[0, 1]	Band's round hook (Bandhaken rund)	05 07 03-2	„Bandstahl 10×0,8 Din 1016“ (s
1	[0, 1]	Band (Schneckenband)	05 07 03-1	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: **2:1**

Behandlung / Treatment:

Blatt / Sheet: 1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **05 07 03-1**

Teil / Part: band

Benennung / Name: **Fusee band (Schneckenband)**

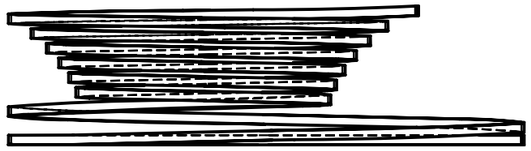
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).

A–A

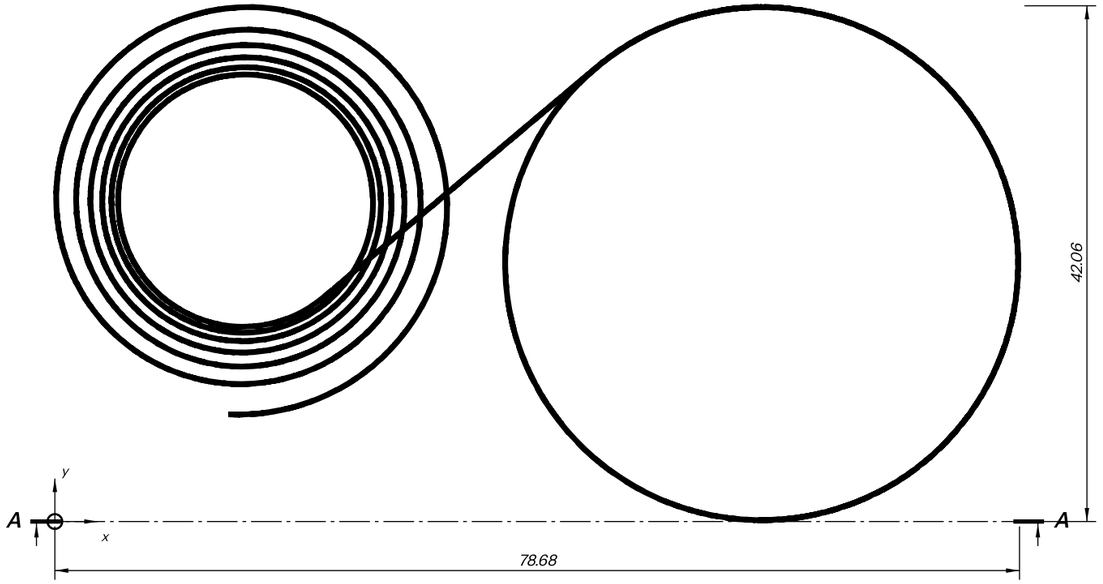


View from the left





Isometric view, not to scale



Werkstoff / Material not on the drawing	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 07 03-1**
Behandlung / Treatment			Teil / Part band.band
Blatt / Sheet **1 / 2**		Benennung / Name **Band (Schneckenband)**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Band (Schneckenband) — sheet 2

The steel band drawn in 1943 in place of the chain: a strip 0.22 thick with a round hook brazed to one end, for the fusee's pin, and a barbed hook to the other, for the barrel. Greenwich's wartime Wempe has one; Wempe 3955 came with both. Chosen under Variants.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 03-1	Band (fusee band)	[0, 1]	not on the drawing		not measured on the model: see its function, The strip, 0.22 thick, between the barrel and the fusee.

What it does (05 07 03-1). The strip, 0.22 thick, between the barrel and the fusee.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 11.32.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.22, 23.75	14.41, 42.00	15.68, 26.36	15.68
Outline	2	arc clockwise	14.41, 42.00	31.95, 27.85	15.78, 25.77	16.29
Outline	3	arc clockwise	31.95, 27.85	32.04, 26.96	21.84, 26.35	10.22
Outline	4	arc clockwise	32.04, 26.96	44.26, 37.25	274.74, -248.96	367.47
Outline	5	arc clockwise	44.26, 37.25	44.62, 37.53	47.74, 33.30	5.26
Outline	6	arc clockwise	44.62, 37.53	44.98, 37.81	48.01, 33.52	5.26
Outline	7	arc clockwise	44.98, 37.81	45.35, 38.08	48.28, 33.72	5.26
Outline	8	arc clockwise	45.35, 38.08	45.72, 38.35	48.57, 33.92	5.26
Outline	9	arc clockwise	45.72, 38.35	46.10, 38.60	48.84, 34.11	5.26
Outline	10	arc clockwise	46.10, 38.60	46.49, 38.85	49.14, 34.31	5.26
Outline	11	arc clockwise	46.49, 38.85	46.88, 39.09	49.42, 34.48	5.26
Outline	12	arc clockwise	46.88, 39.09	47.27, 39.32	49.72, 34.66	5.26
Outline	13	arc clockwise	47.27, 39.32	47.67, 39.54	50.01, 34.83	5.26
Outline	14	arc clockwise	47.67, 39.54	48.08, 39.75	50.32, 35.00	5.26
Outline	15	arc clockwise	48.08, 39.75	48.48, 39.95	50.62, 35.15	5.26
Outline	16	arc clockwise	48.48, 39.95	48.90, 40.15	50.93, 35.30	5.26
Outline	17	arc clockwise	48.90, 40.15	49.32, 40.33	51.24, 35.44	5.26
Outline	18	arc clockwise	49.32, 40.33	49.74, 40.51	51.56, 35.58	5.26
Outline	19	arc clockwise	49.74, 40.51	50.16, 40.68	51.87, 35.70	5.26
Outline	20	arc clockwise	50.16, 40.68	50.59, 40.84	52.20, 35.83	5.26
Outline	21	arc clockwise	50.59, 40.84	51.02, 40.99	52.51, 35.94	5.26
Outline	22	arc clockwise	51.02, 40.99	51.46, 41.12	52.85, 36.05	5.26
Outline	23	arc clockwise	51.46, 41.12	51.90, 41.25	53.16, 36.15	5.26
Outline	24	arc clockwise	51.90, 41.25	52.34, 41.37	53.50, 36.25	5.26
Outline	25	arc clockwise	52.34, 41.37	52.78, 41.48	53.83, 36.33	5.26
Outline	26	arc clockwise	52.78, 41.48	53.23, 41.59	54.17, 36.41	5.26
Outline	27	arc clockwise	53.23, 41.59	53.67, 41.68	54.49, 36.48	5.26
Outline	28	arc clockwise	53.67, 41.68	54.12, 41.76	54.84, 36.55	5.26

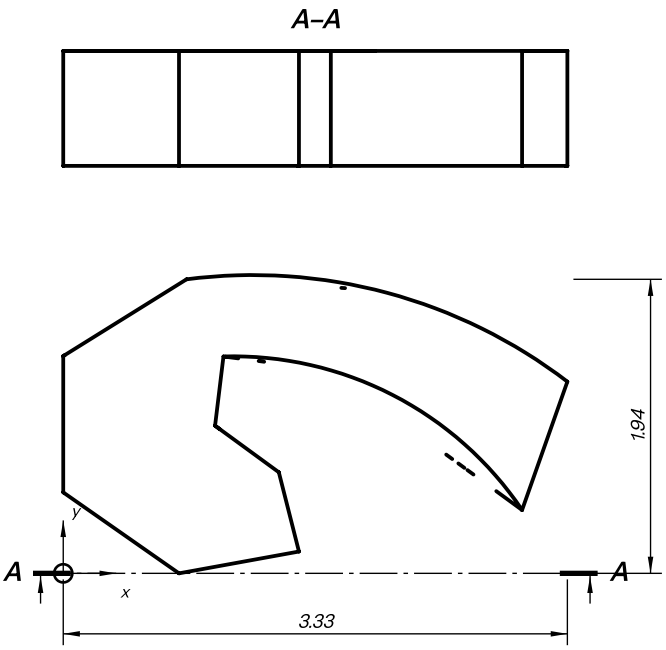
Outline	29	arc clockwise	54.12, 41.76	54.58, 41.83	55.17, 36.61	5.26
Outline	30	arc clockwise	54.58, 41.83	55.03, 41.89	55.52, 36.66	5.26
Outline	31	arc clockwise	55.03, 41.89	55.48, 41.94	55.85, 36.70	5.26
Outline	32	arc clockwise	55.48, 41.94	55.94, 41.99	56.20, 36.74	5.26
Outline	33	arc clockwise	55.94, 41.99	56.39, 42.02	56.53, 36.76	5.26
Outline	34	arc clockwise	56.39, 42.02	56.85, 42.04	56.88, 36.78	5.26
Outline	35	arc clockwise	56.85, 42.04	57.31, 42.05	57.22, 36.79	5.26
Outline	36	arc clockwise	57.31, 42.05	57.76, 42.06	57.57, 36.80	5.26
Outline	37	arc clockwise	57.76, 42.06	58.22, 42.05	57.90, 36.80	5.26
Outline	38	arc clockwise	58.22, 42.05	58.68, 42.03	58.25, 36.79	5.26
Outline	39	arc clockwise	58.68, 42.03	59.13, 42.00	58.59, 36.77	5.26
Outline	40	arc clockwise	59.13, 42.00	59.59, 41.97	58.94, 36.75	5.26
Outline	41	arc clockwise	59.59, 41.97	60.04, 41.92	59.27, 36.72	5.26
Outline	42	arc clockwise	60.04, 41.92	60.50, 41.86	59.62, 36.68	5.26
Outline	43	arc clockwise	60.50, 41.86	60.95, 41.80	59.95, 36.63	5.26
Outline	44	arc clockwise	60.95, 41.80	61.40, 41.72	60.30, 36.58	5.26
Outline	45	arc clockwise	61.40, 41.72	61.85, 41.63	60.63, 36.52	5.26
Outline	46	arc clockwise	61.85, 41.63	62.30, 41.54	60.97, 36.45	5.26
Outline	47	arc clockwise	62.30, 41.54	62.74, 41.43	61.30, 36.37	5.26
Outline	48	arc clockwise	62.74, 41.43	63.18, 41.32	61.64, 36.29	5.26
Outline	49	arc clockwise	63.18, 41.32	63.62, 41.19	61.96, 36.20	5.26
Outline	50	arc clockwise	63.62, 41.19	64.06, 41.06	62.30, 36.10	5.26
Outline	51	arc clockwise	64.06, 41.06	64.49, 40.91	62.61, 36.00	5.26
Outline	52	arc clockwise	64.49, 40.91	64.92, 40.76	62.95, 35.88	5.26
Outline	53	arc clockwise	64.92, 40.76	65.35, 40.59	63.26, 35.77	5.26
Outline	54	arc clockwise	65.35, 40.59	65.77, 40.42	63.59, 35.64	5.26
Outline	55	arc clockwise	65.77, 40.42	66.19, 40.24	63.90, 35.51	5.26
Outline	56	arc clockwise	66.19, 40.24	66.61, 40.05	64.22, 35.37	5.26
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Outline	60	arc clockwise	67.83, 39.43	68.23, 39.20	65.44, 34.74	5.26
Outline	61	arc clockwise	68.23, 39.20	68.62, 38.97	65.73, 34.58	5.26
Outline	62	arc clockwise	68.62, 38.97	69.01, 38.73	66.03, 34.39	5.26
Outline	63	arc clockwise	69.01, 38.73	69.39, 38.47	66.31, 34.21	5.26
Outline	64	arc clockwise	69.39, 38.47	69.76, 38.22	66.60, 34.02	5.26
Outline	65	arc clockwise	69.76, 38.22	70.13, 37.95	66.87, 33.82	5.26
Outline	66	arc clockwise	70.13, 37.95	70.50, 37.67	67.16, 33.61	5.26
Outline	67	arc clockwise	70.50, 37.67	70.86, 37.39	67.42, 33.41	5.26
Outline	68	arc clockwise	70.86, 37.39	71.21, 37.10	67.69, 33.19	5.26
Outline	69	arc clockwise	71.21, 37.10	71.56, 36.80	67.95, 32.97	5.26
Outline	70	arc clockwise	71.56, 36.80	71.90, 36.49	68.21, 32.74	5.26
Outline	71	arc clockwise	71.90, 36.49	72.23, 36.18	68.46, 32.52	5.26
Outline	72	arc clockwise	72.23, 36.18	72.55, 35.86	68.71, 32.27	5.26
Outline	73	arc clockwise	72.55, 35.86	72.87, 35.53	68.95, 32.03	5.26
Outline	74	arc clockwise	72.87, 35.53	73.18, 35.20	69.19, 31.78	5.26
Outline	75	arc clockwise	73.18, 35.20	73.49, 34.86	69.41, 31.53	5.26
Outline	76	arc clockwise	73.49, 34.86	73.79, 34.51	69.65, 31.27	5.26

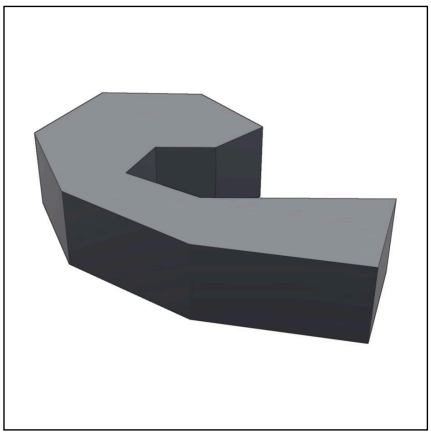
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Outline	78	arc clockwise	74.07, 34.16	74.36, 33.80	70.08, 30.74	5.26
Outline	79	arc clockwise	74.36, 33.80	74.63, 33.43	70.28, 30.47	5.26
Outline	80	arc clockwise	74.63, 33.43	74.89, 33.06	70.49, 30.19	5.26
Outline	81	arc clockwise	74.89, 33.06	75.15, 32.68	70.68, 29.92	5.26
Outline	82	arc clockwise	75.15, 32.68	75.40, 32.30	70.87, 29.62	5.26
Outline	83	arc clockwise	75.40, 32.30	75.64, 31.91	71.05, 29.34	5.26
Outline	84	arc clockwise	75.64, 31.91	75.87, 31.51	71.24, 29.04	5.26
Outline	85	arc clockwise	75.87, 31.51	76.10, 31.12	71.40, 28.75	5.26
Outline	86	arc clockwise	76.10, 31.12	76.31, 30.71	71.57, 28.44	5.26
Outline	87	arc clockwise	76.31, 30.71	76.52, 30.30	71.72, 28.14	5.26
Outline	88	arc clockwise	76.52, 30.30	76.72, 29.89	71.88, 27.83	5.26
Outline	89	arc clockwise	76.72, 29.89	76.90, 29.48	72.02, 27.53	5.26
Outline	90	arc clockwise	76.90, 29.48	77.08, 29.06	72.16, 27.20	5.26
Outline	91	arc clockwise	77.08, 29.06	77.25, 28.63	72.29, 26.90	5.26
Outline	92	arc clockwise	77.25, 28.63	77.41, 28.20	72.42, 26.57	5.26
Outline	93	arc clockwise	77.41, 28.20	77.56, 27.77	72.53, 26.25	5.26
Outline	94	arc clockwise	77.56, 27.77	77.71, 27.34	72.64, 25.92	5.26
Outline	95	arc clockwise	77.71, 27.34	77.84, 26.90	72.74, 25.60	5.26
Outline	96	arc clockwise	77.84, 26.90	77.96, 26.46	72.84, 25.26	5.26
Outline	97	arc clockwise	77.96, 26.46	78.07, 26.02	72.93, 24.94	5.26
Outline	98	arc clockwise	78.07, 26.02	78.18, 25.57	73.01, 24.60	5.26
Outline	99	arc clockwise	78.18, 25.57	78.27, 25.13	73.08, 24.27	5.26
Outline	100	arc clockwise	78.27, 25.13	78.36, 24.68	73.15, 23.93	5.26
Outline	101	arc clockwise	78.36, 24.68	78.43, 24.22	73.21, 23.60	5.26
Outline	102	arc clockwise	78.43, 24.22	78.50, 23.77	73.27, 23.25	5.26
Outline	103	arc clockwise	78.50, 23.77	78.55, 23.32	73.31, 22.92	5.26
Outline	104	arc clockwise	78.55, 23.32	78.59, 22.86	73.34, 22.57	5.26
Outline	105	arc clockwise	78.59, 22.86	78.63, 22.41	73.37, 22.24	5.26
Outline	106	arc clockwise	78.63, 22.41	78.65, 21.95	73.40, 21.89	5.26
Outline	107	arc clockwise	78.65, 21.95	78.67, 21.50	73.41, 21.56	5.26
Outline	108	arc clockwise	78.67, 21.50	78.67, 21.04	73.42, 21.20	5.26
Outline	109	arc clockwise	78.67, 21.04	78.67, 20.58	73.42, 20.87	5.26
Outline	110	arc clockwise	78.67, 20.58	78.66, 20.12	73.41, 20.52	5.26
Outline	111	arc clockwise	78.66, 20.12	78.63, 19.67	73.40, 20.18	5.26
Outline	112	arc clockwise	78.63, 19.67	78.60, 19.21	73.37, 19.83	5.26
Outline	113	arc clockwise	78.60, 19.21	78.55, 18.76	73.35, 19.50	5.26
Outline	114	arc clockwise	78.55, 18.76	78.50, 18.30	73.31, 19.15	5.26
Outline	115	arc clockwise	78.50, 18.30	78.43, 17.85	73.26, 18.82	5.26
Outline	116	arc clockwise	78.43, 17.85	78.36, 17.40	73.21, 18.47	5.26
Outline	117	arc clockwise	78.36, 17.40	78.28, 16.95	73.15, 18.14	5.26
Outline	118	arc clockwise	78.28, 16.95	78.18, 16.50	73.09, 17.80	5.26
Outline	119	arc clockwise	78.18, 16.50	78.08, 16.06	73.01, 17.47	5.26
Outline	120	arc clockwise	78.08, 16.06	77.97, 15.62	72.93, 17.13	5.26
Outline	121	arc clockwise	77.97, 15.62	77.84, 15.18	72.84, 16.81	5.26
Outline	122	arc clockwise	77.84, 15.18	77.71, 14.74	72.75, 16.47	5.26
Outline	123	arc clockwise	77.71, 14.74	77.57, 14.30	72.65, 16.15	5.26
Outline	124	arc clockwise	77.57, 14.30	77.42, 13.87	72.53, 15.82	5.26

Outline	125	arc clockwise	77.42, 13.87	77.26, 13.44	72.42, 15.50	5.26
Outline	126	arc clockwise	77.26, 13.44	77.09, 13.02	72.29, 15.18	5.26
Outline	127	arc clockwise	77.09, 13.02	76.91, 12.60	72.17, 14.87	5.26
Outline	128	arc clockwise	76.91, 12.60	76.72, 12.18	72.03, 14.55	5.26
Outline	129	arc clockwise	76.72, 12.18	76.53, 11.77	71.88, 14.24	5.26
Outline	130	arc clockwise	76.53, 11.77	76.32, 11.36	71.73, 13.93	5.26
Outline	131	arc clockwise	76.32, 11.36	76.11, 10.96	71.58, 13.63	5.26
Outline	132	arc clockwise	76.11, 10.96	75.88, 10.56	71.41, 13.32	5.26
Outline	133	arc clockwise	75.88, 10.56	75.65, 10.17	71.24, 13.03	5.26
Outline	134	arc clockwise	75.65, 10.17	75.41, 9.78	71.06, 12.73	5.26
Outline	135	arc clockwise	75.41, 9.78	75.16, 9.39	70.88, 12.45	5.26
Outline	136	arc clockwise	75.16, 9.39	74.90, 9.02	70.69, 12.16	5.26
Outline	137	arc clockwise	74.90, 9.02	74.64, 8.64	70.50, 11.88	5.26
Outline	138	arc clockwise	74.64, 8.64	74.37, 8.28	70.29, 11.60	5.26
Outline	139	arc clockwise	74.37, 8.28	74.09, 7.92	70.09, 11.33	5.26
Outline	140	arc clockwise	74.09, 7.92	73.80, 7.56	69.87, 11.06	5.26
Outline	141	arc clockwise	73.80, 7.56	73.50, 7.22	69.65, 10.80	5.26
Outline	142	arc clockwise	73.50, 7.22	73.20, 6.87	69.42, 10.53	5.26
Outline	143	arc clockwise	73.20, 6.87	72.88, 6.54	69.20, 10.29	5.26
Outline	144	arc clockwise	72.88, 6.54	72.57, 6.21	68.95, 10.03	5.26
Outline	145	arc clockwise	72.57, 6.21	72.24, 5.89	68.72, 9.80	5.26
Outline	146	arc clockwise	72.24, 5.89	71.91, 5.58	68.47, 9.55	5.26
Outline	147	arc clockwise	71.91, 5.58	71.57, 5.27	68.22, 9.33	5.26
Outline	148	arc clockwise	71.57, 5.27	71.22, 4.97	67.96, 9.09	5.26
Outline	149	arc clockwise	71.22, 4.97	70.87, 4.68	67.70, 8.88	5.26
Outline	150	arc clockwise	70.87, 4.68	70.51, 4.40	67.43, 8.66	5.26
Outline	151	arc clockwise	70.51, 4.40	70.15, 4.12	67.17, 8.45	5.26
Outline	152	arc clockwise	70.15, 4.12	69.78, 3.85	66.88, 8.24	5.26
Outline	153	arc clockwise	69.78, 3.85	69.40, 3.60	66.61, 8.05	5.26
Outline	154	arc clockwise	69.40, 3.60	69.02, 3.34	66.32, 7.85	5.26
Outline	155	arc clockwise	69.02, 3.34	68.63, 3.10	66.04, 7.67	5.26
Outline	156	arc clockwise	68.63, 3.10	68.24, 2.87	65.74, 7.49	5.26
Outline	157	arc clockwise	68.24, 2.87	67.84, 2.64	65.45, 7.32	5.26
Outline	158	arc clockwise	67.84, 2.64	67.44, 2.42	65.14, 7.15	5.26
Outline	159	arc clockwise	67.44, 2.42	67.03, 2.22	64.85, 7.00	5.26
Outline	160	arc clockwise	67.03, 2.22	66.62, 2.02	64.53, 6.84	5.26
Outline	161	arc clockwise	66.62, 2.02	66.21, 1.82	64.23, 6.70	5.26
Outline	162	arc clockwise	66.21, 1.82	65.79, 1.64	63.91, 6.55	5.26
Outline	163	arc clockwise	65.79, 1.64	65.37, 1.47	63.60, 6.42	5.26
Outline	164	arc clockwise	65.37, 1.47	64.94, 1.31	63.27, 6.30	5.26
Outline	165	arc clockwise	64.94, 1.31	64.51, 1.15	62.96, 6.18	5.26
Outline	166	arc clockwise	64.51, 1.15	64.07, 1.01	62.63, 6.06	5.26
Outline	167	arc clockwise	64.07, 1.01	63.64, 0.87	62.31, 5.96	5.26
Outline	168	arc clockwise	63.64, 0.87	63.20, 0.75	61.97, 5.86	5.26
Outline	169	arc clockwise	63.20, 0.75	62.76, 0.63	61.65, 5.77	5.26
Outline	170	arc clockwise	62.76, 0.63	62.31, 0.53	61.31, 5.69	5.26
Outline	171	arc clockwise	62.31, 0.53	61.87, 0.43	60.98, 5.61	5.26
Outline	172	arc clockwise	61.87, 0.43	61.42, 0.34	60.64, 5.54	5.26

Outline	173	arc clockwise	61.42, 0.34	60.97, 0.27	60.31, 5.48	5.26
Outline	174	arc clockwise	60.97, 0.27	60.51, 0.20	59.96, 5.43	5.26
Outline	175	arc clockwise	60.51, 0.20	60.06, 0.14	59.63, 5.38	5.26
Outline	176	arc clockwise	60.06, 0.14	59.61, 0.09	59.28, 5.34	5.26
Outline	177	arc clockwise	59.61, 0.09	59.15, 0.06	58.95, 5.31	5.26
Outline	178	arc clockwise	59.15, 0.06	58.69, 0.03	58.60, 5.29	5.26
Outline	179	arc clockwise	58.69, 0.03	58.24, 0.01	58.27, 5.28	5.27
Outline	180	arc clockwise	58.24, 0.01	57.78, 0.00	57.91, 5.25	5.24
Outline	181	line	57.78, 0.00	57.57, 0.00		
Outline	182	line	57.57, 0.00	57.57, 0.00		
Outline	183	arc clockwise	57.57, 0.00	41.94, 35.01	57.65, 21.03	21.03
Outline	184	line	41.94, 35.01	32.06, 26.69		
Outline	185	arc clockwise	32.06, 26.69	18.04, 8.89	15.10, 25.65	17.00
Outline	186	arc clockwise	18.04, 8.89	14.26, 8.62	14.93, 26.03	17.42
Outline	187	line	14.26, 8.62	14.26, 8.84		
Outline	188	arc anticlockwise	14.26, 8.84	31.64, 22.81	15.02, 25.73	16.89
Outline	189	arc anticlockwise	31.64, 22.81	31.85, 26.51	15.03, 25.61	16.85
Outline	190	line	31.85, 26.51	29.92, 24.88		
Outline	191	arc clockwise	29.92, 24.88	13.77, 11.16	15.20, 25.87	14.76
Outline	192	arc clockwise	13.77, 11.16	0.22, 23.75	15.20, 26.28	15.18
Cut-out 1	193	line	44.10, 37.12	44.10, 37.11		
Cut-out 1	194	line	44.10, 37.11	44.10, 37.12		
Cut-out 3	195	arc anticlockwise	19.90, 41.31	2.28, 34.08	15.80, 26.18	15.67
Cut-out 3	196	arc anticlockwise	2.28, 34.08	6.05, 14.43	15.43, 26.40	15.22
Cut-out 3	197	arc anticlockwise	6.05, 14.43	27.44, 18.01	15.12, 26.03	14.71
Cut-out 3	198	arc anticlockwise	27.44, 18.01	29.68, 24.68	15.37, 25.78	14.36
Cut-out 3	199	line	29.68, 24.68	28.43, 23.63		
Cut-out 3	200	arc clockwise	28.43, 23.63	7.34, 15.07	15.23, 25.94	13.42
Cut-out 3	201	arc clockwise	7.34, 15.07	5.78, 36.18	15.47, 26.27	13.85
Cut-out 3	202	arc clockwise	5.78, 36.18	28.07, 32.97	15.67, 26.02	14.20
Cut-out 3	203	arc clockwise	28.07, 32.97	29.93, 25.18	15.50, 25.84	14.45
Cut-out 3	204	line	29.93, 25.18	31.84, 26.78		
Cut-out 3	205	arc anticlockwise	31.84, 26.78	19.90, 41.31	15.72, 25.72	16.15
Cut-out 4	206	arc anticlockwise	19.96, 39.32	1.91, 27.57	15.63, 26.19	13.81
Cut-out 4	207	arc anticlockwise	1.91, 27.57	16.27, 12.79	15.30, 26.19	13.45
Cut-out 4	208	arc anticlockwise	16.27, 12.79	28.17, 23.41	15.26, 25.91	13.15
Cut-out 4	209	arc anticlockwise	28.17, 23.41	26.96, 22.13	28.55, 21.84	1.61
Cut-out 4	210	arc clockwise	26.96, 22.13	6.37, 17.43	15.25, 26.02	12.35
Cut-out 4	211	arc clockwise	6.37, 17.43	11.85, 38.41	15.49, 26.24	12.69
Cut-out 4	212	arc clockwise	11.85, 38.41	28.60, 25.18	15.58, 25.94	13.03
Cut-out 4	213	arc clockwise	28.60, 25.18	28.48, 23.96	15.96, 25.80	12.66
Cut-out 4	214	line	28.48, 23.96	29.70, 24.99		
Cut-out 4	215	arc anticlockwise	29.70, 24.99	19.96, 39.32	15.57, 25.86	14.16
Cut-out 5	216	arc anticlockwise	20.07, 37.94	3.01, 26.36	15.59, 26.17	12.59
Cut-out 5	217	arc anticlockwise	3.01, 26.36	20.28, 14.98	15.28, 26.16	12.26
Cut-out 5	218	arc anticlockwise	20.28, 14.98	26.76, 22.22	15.32, 25.94	12.03
Cut-out 5	219	arc anticlockwise	26.76, 22.22	25.58, 21.04	27.42, 20.39	1.95
Cut-out 5	220	arc clockwise	25.58, 21.04	5.50, 20.06	15.29, 26.08	11.48

Cut-out 5	221	arc clockwise	5.50, 20.06	16.23, 37.94	15.51, 26.20	11.75
Cut-out 5	222	arc clockwise	16.23, 37.94	27.17, 22.86	15.52, 25.93	12.04
Cut-out 5	223	line	27.17, 22.86	28.23, 23.74		
Cut-out 5	224	arc anticlockwise	28.23, 23.74	20.07, 37.94	15.52, 25.89	12.88
Cut-out 6	225	arc anticlockwise	4.79, 21.95	24.67, 19.87	15.29, 26.10	11.27
Cut-out 6	226	arc anticlockwise	24.67, 19.87	25.32, 21.01	15.95, 25.61	10.44
Cut-out 6	227	arc anticlockwise	25.32, 21.01	24.30, 20.11	26.21, 18.97	2.23
Cut-out 6	228	arc clockwise	24.30, 20.11	4.64, 24.42	15.32, 26.09	10.80
Cut-out 6	229	arc clockwise	4.64, 24.42	20.37, 36.05	15.53, 26.13	11.03
Cut-out 6	230	arc clockwise	20.37, 36.05	25.91, 21.80	15.48, 25.95	11.23
Cut-out 6	231	arc anticlockwise	25.91, 21.80	26.90, 22.70	25.72, 23.01	1.23
Cut-out 6	232	arc anticlockwise	26.90, 22.70	13.18, 37.49	15.54, 25.94	11.80
Cut-out 6	233	arc anticlockwise	13.18, 37.49	4.79, 21.95	15.50, 26.20	11.53
Cut-out 7	234	arc anticlockwise	19.84, 36.05	4.94, 24.00	15.52, 26.13	10.80
Cut-out 7	235	arc anticlockwise	4.94, 24.00	23.69, 19.63	15.32, 26.09	10.58
Cut-out 7	236	arc clockwise	23.69, 19.63	20.73, 17.29	7.83, 36.53	23.17
Cut-out 7	237	arc clockwise	20.73, 17.29	5.33, 28.55	15.37, 26.10	10.32
Cut-out 7	238	arc clockwise	5.33, 28.55	25.31, 29.89	15.52, 26.07	10.50
Cut-out 7	239	arc clockwise	25.31, 29.89	24.70, 20.77	15.41, 25.98	10.65
Cut-out 7	240	arc anticlockwise	24.70, 20.77	25.58, 21.57	24.30, 22.11	1.39
Cut-out 7	241	arc anticlockwise	25.58, 21.57	19.84, 36.05	15.48, 25.95	11.01
Cut-out 8	242	arc anticlockwise	20.01, 35.31	7.52, 19.74	15.49, 26.13	10.22
Cut-out 8	243	arc anticlockwise	7.52, 19.74	21.88, 18.40	15.35, 26.04	10.05
Cut-out 8	244	arc anticlockwise	21.88, 18.40	24.98, 21.84	19.07, 24.03	6.30
Cut-out 8	245	arc anticlockwise	24.98, 21.84	20.01, 35.31	15.45, 25.98	10.39
Cut-out 9	246	line	22.27, 17.11	22.27, 17.11		
Cut-out 9	247	line	22.27, 17.11	22.27, 17.11		
Cut-out 9	248	line	22.27, 17.11	22.27, 17.11		
Cut-out 10	249	line	22.24, 40.61	22.24, 40.61		
Cut-out 10	250	line	22.24, 40.61	22.24, 40.61		
Cut-out 10	251	line	22.24, 40.61	22.24, 40.61		
Cut-out 10	252	line	22.24, 40.61	22.24, 40.61		





Isometric view, not to scale

Werkstoff / Material „Bandstahl 10×0,8 Din 1016" (strip steel 10 × 0,8, DIN 1016)	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 03-2*** *Teil / Part* *band.hookF*
Behandlung / Treatment surface∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,05		*Benennung / Name* ***Band's round hook (Bandhaken rund)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Band’s round hook (Bandhaken rund) — sheet 2

Brazed to the band’s fusee end; it takes the fusee’s pin.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 03-2	Bandhaken rund (band's round hook)	[0, 1]	„Bandstahl 10×0,8 Din 1016“ (strip steel 10 × 0,8, DIN 1016)	surface ∇ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,05	not measured on the model: see its function, On the fusee's pin.

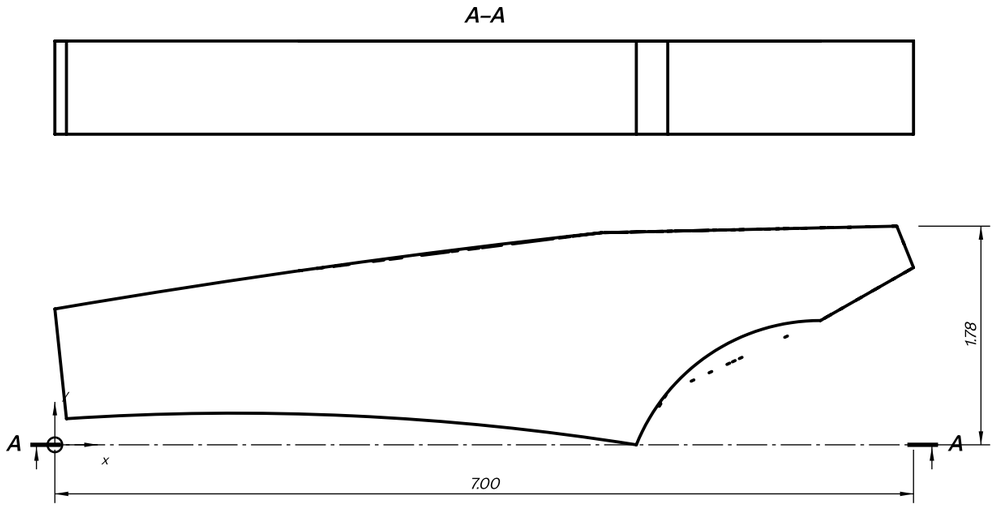
What it does (05 07 03-2). On the fusee's pin.

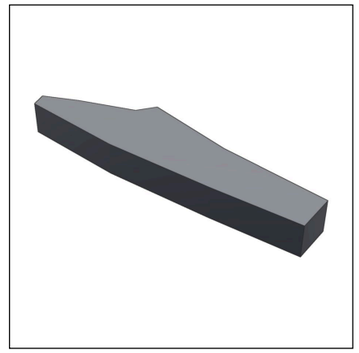
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 0.76.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.54	0.00, 1.44		
Outline	2	line	0.00, 1.44	0.82, 1.94		
Outline	3	arc clockwise	0.82, 1.94	3.33, 1.27	1.25, -1.45	3.43
Outline	4	line	3.33, 1.27	3.03, 0.42		
Outline	5	arc anticlockwise	3.03, 0.42	1.06, 1.43	1.12, -0.87	2.30
Outline	6	line	1.06, 1.43	1.00, 0.98		
Outline	7	line	1.00, 0.98	1.43, 0.67		
Outline	8	line	1.43, 0.67	1.56, 0.14		
Outline	9	line	1.56, 0.14	0.77, 0.00		
Outline	10	line	0.77, 0.00	0.00, 0.54		





Isometric view, not to scale

Werkstoff / Material *„Bandstahl 10×0,8 Din 1016" (strip steel 10 × 0.8, DIN 1016)*	*Maßstab / Scale* ***20:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 07 03-3*** *Teil / Part* *band.hookB*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0.05*		*Benennung / Name* ***Band's drum hook (Bandhaken f. Trommel)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Band’s drum hook (Bandhaken f. Trommel) — sheet 2

Brazed to the band’s barrel end; its barb takes the slot in the drum.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 07 03-3	Bandhaken f. Trommel (band's drum hook)	[0, 1]	„Bandstahl 10×0,8 Din 1016“ (strip steel 10 × 0,8, DIN 1016)	surface ∇∇∇ (DIN 140: ∇ roughed, ∇∇ smoothed, ∇∇∇ fine, ∇∇∇∇ polished); untoleranced sizes ±0,05	not measured on the model: see its function, In the drum's slot.

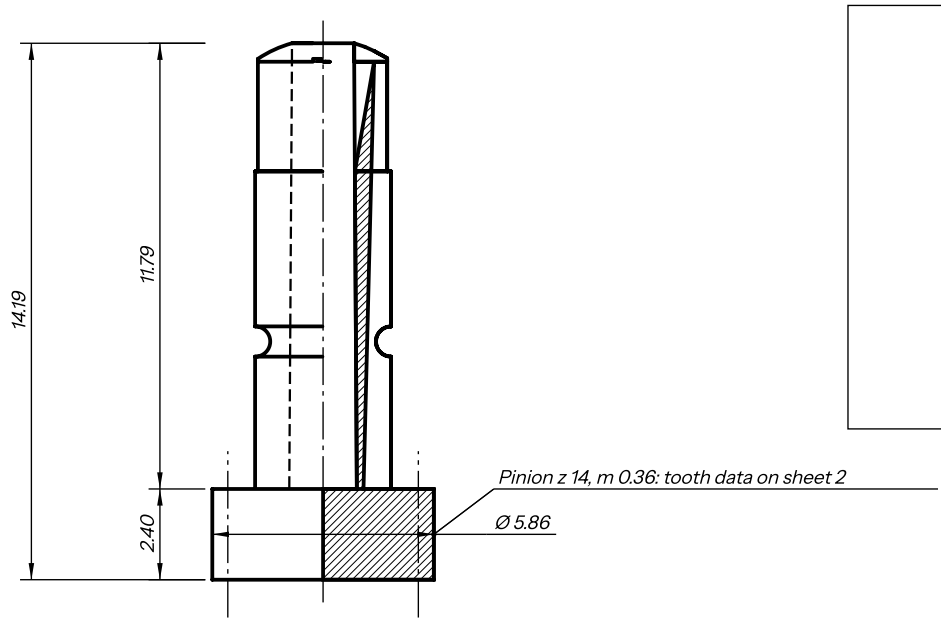
What it does (05 07 03-3). In the drum's slot.

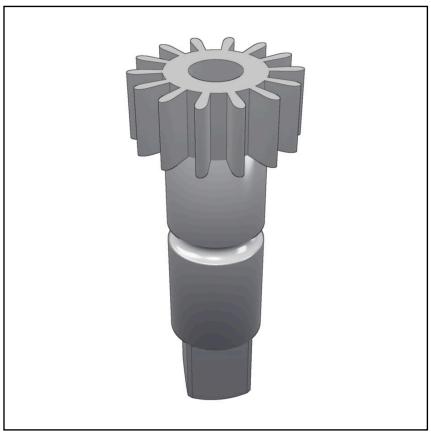
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 0.76.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

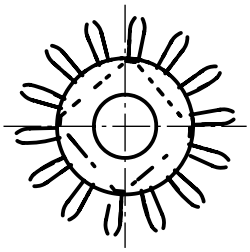
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	2.61, 0.23	0.09, 0.21		
Outline	2	line	0.09, 0.21	0.00, 1.11		
Outline	3	arc clockwise	0.00, 1.11	4.46, 1.73	12.60, -72.98	75.15
Outline	4	line	4.46, 1.73	6.87, 1.78		
Outline	5	line	6.87, 1.78	7.00, 1.45		
Outline	6	arc clockwise	7.00, 1.45	5.00, 0.43	-0.40, 13.57	14.21
Outline	7	line	5.00, 0.43	4.74, 0.00		
Outline	8	line	4.74, 0.00	2.61, 0.23		





Isometric view, not to scale

View from above



Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 05 03-3*** *Teil / Part* *cannon*
Behandlung / Treatment		*Benennung / Name* ***Cannon tube (Minutenrohr)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Cannon tube (Minutenrohr) — sheet 2

A steel tube pressed on the centre arbor's tapered shank, held there by centre-punch marks in its groove, so it turns with the centre arbor, once an hour. Its foot is the cannon pinion, which drives the change wheel; its square top carries the minute hand.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 05 03-3	Minutenrohr (cannon tube)	1	not on the drawing		round 050503-2 (press): 13.777 mm, least gap 0.0104

What it does (05 05 03-3). Pressed on the centre arbor's 1:80 shank, held by centre-punch marks in its groove: its foot is the cannon pinion (14 leaves, m 0.36), which drives the change wheel; its square carries the minute hand (no sheet of its own: 05 05 03 gives it).

Turned about its axis; length 14.19, largest Ø 5.83. Diameters: Ø 5.86. Shoulders from the lower end: 0.00, 2.40, 14.19.

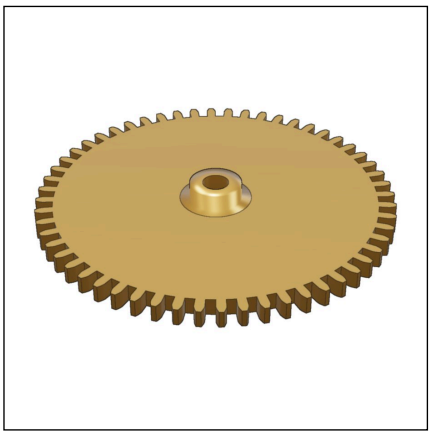
Leaves 0.00–2.40 from the lower end, pitch Ø 5.04; their form is the view from the end, drawn at the sheet's scale.

Teeth

Number	Module	Tip Ø	Root Ø
14	0.36	5.86	

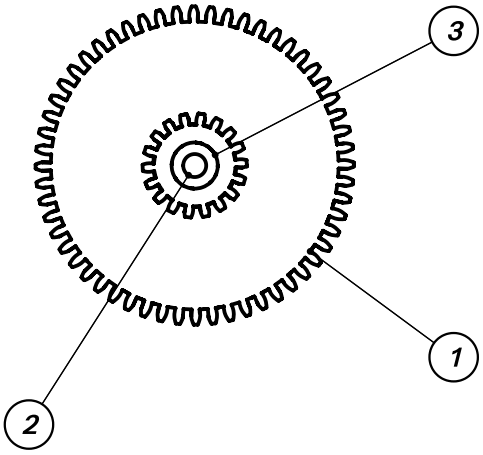
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
3	1	Change pinion (Wechseltrieb)	05 02 02-3	not on the drawing
2	1	Change wheel pipe (Wechselbutzen)	05 02 02-2	not on the drawing
1	1	Change wheel	05 02 02-1	„Messingblech 1,2 Din 1751 Ms

Werkstoff / Material
„Messingblech 1,2 Din 1751 Ms 58 federhart"

Maßstab / Scale
2:1

Behandlung / Treatment
surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); blank (Rohmaß) 1,2 × 22,5 × 22,5; untoleranced sizes ±0,1

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

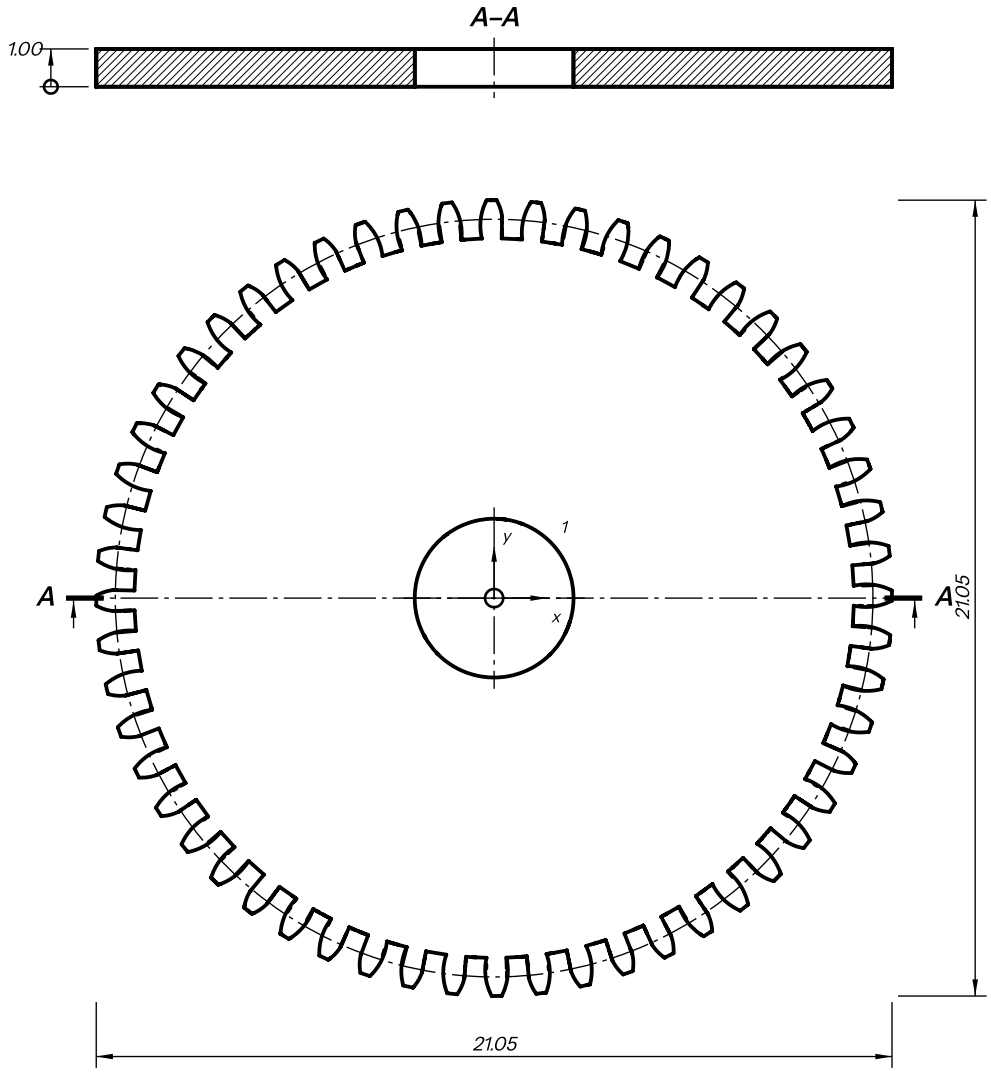
Model v3.35.00 · 2026-10-26

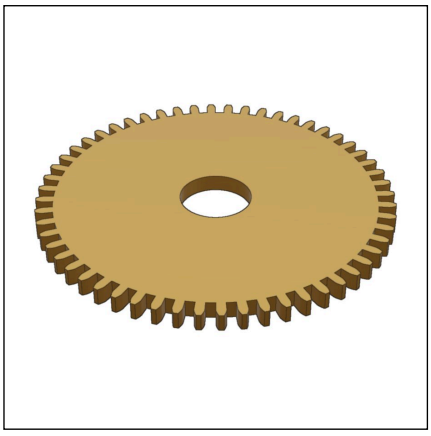
Zeichnung Nr. / Drawing no.
05 02 02-1

Teil / Part
changeWheel

Benennung / Name
Change wheel (Wechselrad)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material „Messingblech 1,2 Din 1751 Ms 58 federhart"	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 02 02-1** *Teil / Part* *changeWheel.wheel*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); blank (Rohmaß) 1,2 × 22,5 × 22,5; untoleranced sizes ±0,1*		*Benennung / Name* ***Change wheel***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Change wheel — sheet 2

The wheel the cannon pinion drives: a plain spring-hard brass disc, riveted on its pipe.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 02-1	Wechselrad (change wheel)	1	„Messingblech 1,2 Din 1751 Ms 58 federhart“	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); blank (Rohmaß) 1,2 × 22,5 × 22,5; untoleranced sizes ±0,1	-4

What it does (05 02 02-1). Riveted on its pipe; the cannon pinion drives it.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 4.20 through

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-10.11, -2.15	-9.32, -1.92	-9.29, -3.50	1.58
Outline	2	line	-9.32, -1.92	-9.41, -1.39		
Outline	3	arc clockwise	-9.41, -1.39	-10.43, -1.38	-9.92, -0.01	1.46
Outline	4	line	-10.43, -1.38	-10.47, -1.12		
Outline	5	arc clockwise	-10.47, -1.12	-9.48, -0.87	-9.64, -2.32	1.46
Outline	6	line	-9.48, -0.87	-9.51, -0.32		
Outline	7	arc clockwise	-9.51, -0.32	-10.52, -0.20	-9.85, 1.10	1.46
Outline	8	line	-10.52, -0.20	-10.52, 0.06		
Outline	9	arc clockwise	-10.52, 0.06	-9.51, 0.20	-9.84, -1.23	1.46
Outline	10	line	-9.51, 0.20	-9.49, 0.74		
Outline	11	arc clockwise	-9.49, 0.74	-10.48, 0.98	-9.67, 2.20	1.46
Outline	12	line	-10.48, 0.98	-10.45, 1.24		
Outline	13	arc clockwise	-10.45, 1.24	-9.43, 1.26	-9.92, -0.12	1.46
Outline	14	line	-9.43, 1.26	-9.34, 1.80		
Outline	15	arc clockwise	-9.34, 1.80	-10.30, 2.15	-9.36, 3.26	1.46
Outline	16	line	-10.30, 2.15	-10.25, 2.40		
Outline	17	arc clockwise	-10.25, 2.40	-9.23, 2.31	-9.87, 0.99	1.46
Outline	18	line	-9.23, 2.31	-9.08, 2.84		
Outline	19	arc clockwise	-9.08, 2.84	-10.00, 3.29	-8.94, 4.29	1.46
Outline	20	line	-10.00, 3.29	-9.91, 3.54		
Outline	21	arc clockwise	-9.91, 3.54	-8.91, 3.33	-9.69, 2.09	1.46
Outline	22	line	-8.91, 3.33	-8.71, 3.83		
Outline	23	arc clockwise	-8.71, 3.83	-9.57, 4.39	-8.40, 5.27	1.46
Outline	24	line	-9.57, 4.39	-9.45, 4.62		

Outline	25	arc clockwise	-9.45, 4.62	-8.48, 4.31	-9.40, 3.16	1.46
Outline	26	line	-8.48, 4.31	-8.22, 4.79		
Outline	27	arc clockwise	-8.22, 4.79	-9.02, 5.43	-7.76, 6.17	1.46
Outline	28	line	-9.02, 5.43	-8.88, 5.65		
Outline	29	arc clockwise	-8.88, 5.65	-7.95, 5.23	-8.98, 4.20	1.46
Outline	30	line	-7.95, 5.23	-7.64, 5.68		
Outline	31	arc clockwise	-7.64, 5.68	-8.35, 6.40	-7.02, 7.00	1.46
Outline	32	line	-8.35, 6.40	-8.19, 6.61		
Outline	33	arc clockwise	-8.19, 6.61	-7.31, 6.09	-8.46, 5.18	1.46
Outline	34	line	-7.31, 6.09	-6.95, 6.50		
Outline	35	arc clockwise	-6.95, 6.50	-7.58, 7.30	-6.19, 7.75	1.46
Outline	36	line	-7.58, 7.30	-7.40, 7.49		
Outline	37	arc clockwise	-7.40, 7.49	-6.59, 6.87	-7.82, 6.09	1.46
Outline	38	line	-6.59, 6.87	-6.18, 7.23		
Outline	39	arc clockwise	-6.18, 7.23	-6.72, 8.10	-5.29, 8.39	1.46
Outline	40	line	-6.72, 8.10	-6.51, 8.27		
Outline	41	arc clockwise	-6.51, 8.27	-5.77, 7.56	-7.09, 6.93	1.46
Outline	42	line	-5.77, 7.56	-5.33, 7.88		
Outline	43	arc clockwise	-5.33, 7.88	-5.77, 8.80	-4.31, 8.93	1.46
Outline	44	line	-5.77, 8.80	-5.55, 8.95		
Outline	45	arc clockwise	-5.55, 8.95	-4.89, 8.16	-6.27, 7.68	1.46
Outline	46	line	-4.89, 8.16	-4.42, 8.43		
Outline	47	arc clockwise	-4.42, 8.43	-4.75, 9.39	-3.29, 9.36	1.46
Outline	48	line	-4.75, 9.39	-4.51, 9.51		
Outline	49	arc clockwise	-4.51, 9.51	-3.95, 8.66	-5.37, 8.33	1.46
Outline	50	line	-3.95, 8.66	-3.45, 8.87		
Outline	51	arc clockwise	-3.45, 8.87	-3.66, 9.87	-2.22, 9.66	1.46
Outline	52	line	-3.66, 9.87	-3.42, 9.95		
Outline	53	arc clockwise	-3.42, 9.95	-2.95, 9.05	-4.41, 8.88	1.46
Outline	54	line	-2.95, 9.05	-2.43, 9.20		
Outline	55	arc clockwise	-2.43, 9.20	-2.54, 10.21	-1.12, 9.85	1.46
Outline	56	line	-2.54, 10.21	-2.28, 10.27		
Outline	57	arc clockwise	-2.28, 10.27	-1.92, 9.32	-3.39, 9.32	1.46
Outline	58	line	-1.92, 9.32	-1.39, 9.41		
Outline	59	arc clockwise	-1.39, 9.41	-1.38, 10.43	-0.01, 9.92	1.46
Outline	60	line	-1.38, 10.43	-1.12, 10.47		
Outline	61	arc clockwise	-1.12, 10.47	-0.87, 9.48	-2.32, 9.64	1.46
Outline	62	line	-0.87, 9.48	-0.32, 9.51		
Outline	63	arc clockwise	-0.32, 9.51	-0.20, 10.52	1.10, 9.85	1.46
Outline	64	line	-0.20, 10.52	0.06, 10.52		
Outline	65	arc clockwise	0.06, 10.52	0.20, 9.51	-1.23, 9.84	1.46
Outline	66	line	0.20, 9.51	0.74, 9.49		
Outline	67	arc clockwise	0.74, 9.49	0.98, 10.48	2.20, 9.67	1.46
Outline	68	line	0.98, 10.48	1.24, 10.45		
Outline	69	arc clockwise	1.24, 10.45	1.26, 9.43	-0.12, 9.92	1.46
Outline	70	line	1.26, 9.43	1.80, 9.34		
Outline	71	arc clockwise	1.80, 9.34	2.15, 10.30	3.26, 9.36	1.46
Outline	72	line	2.15, 10.30	2.40, 10.25		

Outline	73	arc clockwise	2.40, 10.25	2.31, 9.23	0.99, 9.87	1.46
Outline	74	line	2.31, 9.23	2.84, 9.08		
Outline	75	arc clockwise	2.84, 9.08	3.29, 10.00	4.29, 8.94	1.46
Outline	76	line	3.29, 10.00	3.54, 9.91		
Outline	77	arc clockwise	3.54, 9.91	3.33, 8.91	2.09, 9.69	1.46
Outline	78	line	3.33, 8.91	3.83, 8.71		
Outline	79	arc clockwise	3.83, 8.71	4.39, 9.57	5.27, 8.40	1.46
Outline	80	line	4.39, 9.57	4.62, 9.45		
Outline	81	arc clockwise	4.62, 9.45	4.31, 8.48	3.16, 9.40	1.46
Outline	82	line	4.31, 8.48	4.79, 8.22		
Outline	83	arc clockwise	4.79, 8.22	5.43, 9.02	6.17, 7.76	1.46
Outline	84	line	5.43, 9.02	5.65, 8.88		
Outline	85	arc clockwise	5.65, 8.88	5.23, 7.95	4.20, 8.98	1.46
Outline	86	line	5.23, 7.95	5.68, 7.64		
Outline	87	arc clockwise	5.68, 7.64	6.40, 8.35	7.00, 7.02	1.46
Outline	88	line	6.40, 8.35	6.61, 8.19		
Outline	89	arc clockwise	6.61, 8.19	6.09, 7.31	5.18, 8.46	1.46
Outline	90	line	6.09, 7.31	6.50, 6.95		
Outline	91	arc clockwise	6.50, 6.95	7.30, 7.58	7.75, 6.19	1.46
Outline	92	line	7.30, 7.58	7.49, 7.40		
Outline	93	arc clockwise	7.49, 7.40	6.87, 6.59	6.09, 7.82	1.46
Outline	94	line	6.87, 6.59	7.23, 6.18		
Outline	95	arc clockwise	7.23, 6.18	8.10, 6.72	8.39, 5.29	1.46
Outline	96	line	8.10, 6.72	8.27, 6.51		
Outline	97	arc clockwise	8.27, 6.51	7.56, 5.77	6.93, 7.09	1.46
Outline	98	line	7.56, 5.77	7.88, 5.33		
Outline	99	arc clockwise	7.88, 5.33	8.80, 5.77	8.93, 4.31	1.46
Outline	100	line	8.80, 5.77	8.95, 5.55		
Outline	101	arc clockwise	8.95, 5.55	8.16, 4.89	7.68, 6.27	1.46
Outline	102	line	8.16, 4.89	8.43, 4.42		
Outline	103	arc clockwise	8.43, 4.42	9.39, 4.75	9.36, 3.29	1.46
Outline	104	line	9.39, 4.75	9.51, 4.51		
Outline	105	arc clockwise	9.51, 4.51	8.66, 3.95	8.33, 5.37	1.46
Outline	106	line	8.66, 3.95	8.87, 3.45		
Outline	107	arc clockwise	8.87, 3.45	9.87, 3.66	9.66, 2.22	1.46
Outline	108	line	9.87, 3.66	9.95, 3.42		
Outline	109	arc clockwise	9.95, 3.42	9.05, 2.95	8.88, 4.41	1.46
Outline	110	line	9.05, 2.95	9.20, 2.43		
Outline	111	arc clockwise	9.20, 2.43	10.21, 2.54	9.85, 1.12	1.46
Outline	112	line	10.21, 2.54	10.27, 2.28		
Outline	113	arc clockwise	10.27, 2.28	9.32, 1.92	9.32, 3.39	1.46
Outline	114	line	9.32, 1.92	9.41, 1.39		
Outline	115	arc clockwise	9.41, 1.39	10.43, 1.38	9.92, 0.01	1.46
Outline	116	line	10.43, 1.38	10.47, 1.12		
Outline	117	arc clockwise	10.47, 1.12	9.48, 0.87	9.64, 2.32	1.46
Outline	118	line	9.48, 0.87	9.51, 0.32		
Outline	119	arc clockwise	9.51, 0.32	10.52, 0.20	9.85, -1.10	1.46
Outline	120	line	10.52, 0.20	10.52, -0.06		

Outline	121	arc clockwise	10.52, -0.06	9.51, -0.20	9.84, 1.23	1.46
Outline	122	line	9.51, -0.20	9.49, -0.74		
Outline	123	arc clockwise	9.49, -0.74	10.48, -0.98	9.67, -2.20	1.46
Outline	124	line	10.48, -0.98	10.45, -1.24		
Outline	125	arc clockwise	10.45, -1.24	9.43, -1.26	9.92, 0.12	1.46
Outline	126	line	9.43, -1.26	9.34, -1.80		
Outline	127	arc clockwise	9.34, -1.80	10.30, -2.15	9.36, -3.26	1.46
Outline	128	line	10.30, -2.15	10.25, -2.40		
Outline	129	arc clockwise	10.25, -2.40	9.23, -2.31	9.87, -0.99	1.46
Outline	130	line	9.23, -2.31	9.08, -2.84		
Outline	131	arc clockwise	9.08, -2.84	10.00, -3.29	8.94, -4.29	1.46
Outline	132	line	10.00, -3.29	9.91, -3.54		
Outline	133	arc clockwise	9.91, -3.54	8.91, -3.33	9.69, -2.09	1.46
Outline	134	line	8.91, -3.33	8.71, -3.83		
Outline	135	arc clockwise	8.71, -3.83	9.57, -4.39	8.40, -5.27	1.46
Outline	136	line	9.57, -4.39	9.45, -4.62		
Outline	137	arc clockwise	9.45, -4.62	8.48, -4.31	9.40, -3.16	1.46
Outline	138	line	8.48, -4.31	8.22, -4.79		
Outline	139	arc clockwise	8.22, -4.79	9.02, -5.43	7.76, -6.17	1.46
Outline	140	line	9.02, -5.43	8.88, -5.65		
Outline	141	arc clockwise	8.88, -5.65	7.95, -5.23	8.98, -4.20	1.46
Outline	142	line	7.95, -5.23	7.64, -5.68		
Outline	143	arc clockwise	7.64, -5.68	8.35, -6.40	7.02, -7.00	1.46
Outline	144	line	8.35, -6.40	8.19, -6.61		
Outline	145	arc clockwise	8.19, -6.61	7.31, -6.09	8.46, -5.18	1.46
Outline	146	line	7.31, -6.09	6.95, -6.50		
Outline	147	arc clockwise	6.95, -6.50	7.58, -7.30	6.19, -7.75	1.46
Outline	148	line	7.58, -7.30	7.40, -7.49		
Outline	149	arc clockwise	7.40, -7.49	6.59, -6.87	7.82, -6.09	1.46
Outline	150	line	6.59, -6.87	6.18, -7.23		
Outline	151	arc clockwise	6.18, -7.23	6.72, -8.10	5.29, -8.39	1.46
Outline	152	line	6.72, -8.10	6.51, -8.27		
Outline	153	arc clockwise	6.51, -8.27	5.77, -7.56	7.09, -6.93	1.46
Outline	154	line	5.77, -7.56	5.33, -7.88		
Outline	155	arc clockwise	5.33, -7.88	5.77, -8.80	4.31, -8.93	1.46
Outline	156	line	5.77, -8.80	5.55, -8.95		
Outline	157	arc clockwise	5.55, -8.95	4.89, -8.16	6.27, -7.68	1.46
Outline	158	line	4.89, -8.16	4.42, -8.43		
Outline	159	arc clockwise	4.42, -8.43	4.75, -9.39	3.29, -9.36	1.46
Outline	160	line	4.75, -9.39	4.51, -9.51		
Outline	161	arc clockwise	4.51, -9.51	3.95, -8.66	5.37, -8.33	1.46
Outline	162	line	3.95, -8.66	3.45, -8.87		
Outline	163	arc clockwise	3.45, -8.87	3.66, -9.87	2.22, -9.66	1.46
Outline	164	line	3.66, -9.87	3.42, -9.95		
Outline	165	arc clockwise	3.42, -9.95	2.95, -9.05	4.41, -8.88	1.46
Outline	166	line	2.95, -9.05	2.43, -9.20		
Outline	167	arc clockwise	2.43, -9.20	2.54, -10.21	1.12, -9.85	1.46
Outline	168	line	2.54, -10.21	2.28, -10.27		

Outline	169	arc clockwise	2.28, -10.27	1.92, -9.32	3.39, -9.32	1.46
Outline	170	line	1.92, -9.32	1.39, -9.41		
Outline	171	arc clockwise	1.39, -9.41	1.38, -10.43	0.01, -9.92	1.46
Outline	172	line	1.38, -10.43	1.12, -10.47		
Outline	173	arc clockwise	1.12, -10.47	0.87, -9.48	2.32, -9.64	1.46
Outline	174	line	0.87, -9.48	0.32, -9.51		
Outline	175	arc clockwise	0.32, -9.51	0.20, -10.52	-1.10, -9.85	1.46
Outline	176	line	0.20, -10.52	-0.06, -10.52		
Outline	177	arc clockwise	-0.06, -10.52	-0.20, -9.51	1.23, -9.84	1.46
Outline	178	line	-0.20, -9.51	-0.74, -9.49		
Outline	179	arc clockwise	-0.74, -9.49	-0.98, -10.48	-2.20, -9.67	1.46
Outline	180	line	-0.98, -10.48	-1.24, -10.45		
Outline	181	arc clockwise	-1.24, -10.45	-1.26, -9.43	0.12, -9.92	1.46
Outline	182	line	-1.26, -9.43	-1.80, -9.34		
Outline	183	arc clockwise	-1.80, -9.34	-2.15, -10.30	-3.26, -9.36	1.46
Outline	184	line	-2.15, -10.30	-2.40, -10.25		
Outline	185	arc clockwise	-2.40, -10.25	-2.31, -9.23	-0.99, -9.87	1.46
Outline	186	line	-2.31, -9.23	-2.84, -9.08		
Outline	187	arc clockwise	-2.84, -9.08	-3.29, -10.00	-4.29, -8.94	1.46
Outline	188	line	-3.29, -10.00	-3.54, -9.91		
Outline	189	arc clockwise	-3.54, -9.91	-3.33, -8.91	-2.09, -9.69	1.46
Outline	190	line	-3.33, -8.91	-3.83, -8.71		
Outline	191	arc clockwise	-3.83, -8.71	-4.39, -9.57	-5.27, -8.40	1.46
Outline	192	line	-4.39, -9.57	-4.62, -9.45		
Outline	193	arc clockwise	-4.62, -9.45	-4.31, -8.48	-3.16, -9.40	1.46
Outline	194	line	-4.31, -8.48	-4.79, -8.22		
Outline	195	arc clockwise	-4.79, -8.22	-5.43, -9.02	-6.17, -7.76	1.46
Outline	196	line	-5.43, -9.02	-5.65, -8.88		
Outline	197	arc clockwise	-5.65, -8.88	-5.23, -7.95	-4.20, -8.98	1.46
Outline	198	line	-5.23, -7.95	-5.68, -7.64		
Outline	199	arc clockwise	-5.68, -7.64	-6.40, -8.35	-7.00, -7.02	1.46
Outline	200	line	-6.40, -8.35	-6.61, -8.19		
Outline	201	arc clockwise	-6.61, -8.19	-6.09, -7.31	-5.18, -8.46	1.46
Outline	202	line	-6.09, -7.31	-6.50, -6.95		
Outline	203	arc clockwise	-6.50, -6.95	-7.30, -7.58	-7.75, -6.19	1.46
Outline	204	line	-7.30, -7.58	-7.49, -7.40		
Outline	205	arc clockwise	-7.49, -7.40	-6.87, -6.59	-6.09, -7.82	1.46
Outline	206	line	-6.87, -6.59	-7.23, -6.18		
Outline	207	arc clockwise	-7.23, -6.18	-8.10, -6.72	-8.39, -5.29	1.46
Outline	208	line	-8.10, -6.72	-8.27, -6.51		
Outline	209	arc clockwise	-8.27, -6.51	-7.56, -5.77	-6.93, -7.09	1.46
Outline	210	line	-7.56, -5.77	-7.88, -5.33		
Outline	211	arc clockwise	-7.88, -5.33	-8.80, -5.77	-8.93, -4.31	1.46
Outline	212	line	-8.80, -5.77	-8.95, -5.55		
Outline	213	arc clockwise	-8.95, -5.55	-8.16, -4.89	-7.68, -6.27	1.46
Outline	214	line	-8.16, -4.89	-8.43, -4.42		
Outline	215	arc clockwise	-8.43, -4.42	-9.39, -4.75	-9.36, -3.29	1.46
Outline	216	line	-9.39, -4.75	-9.51, -4.51		

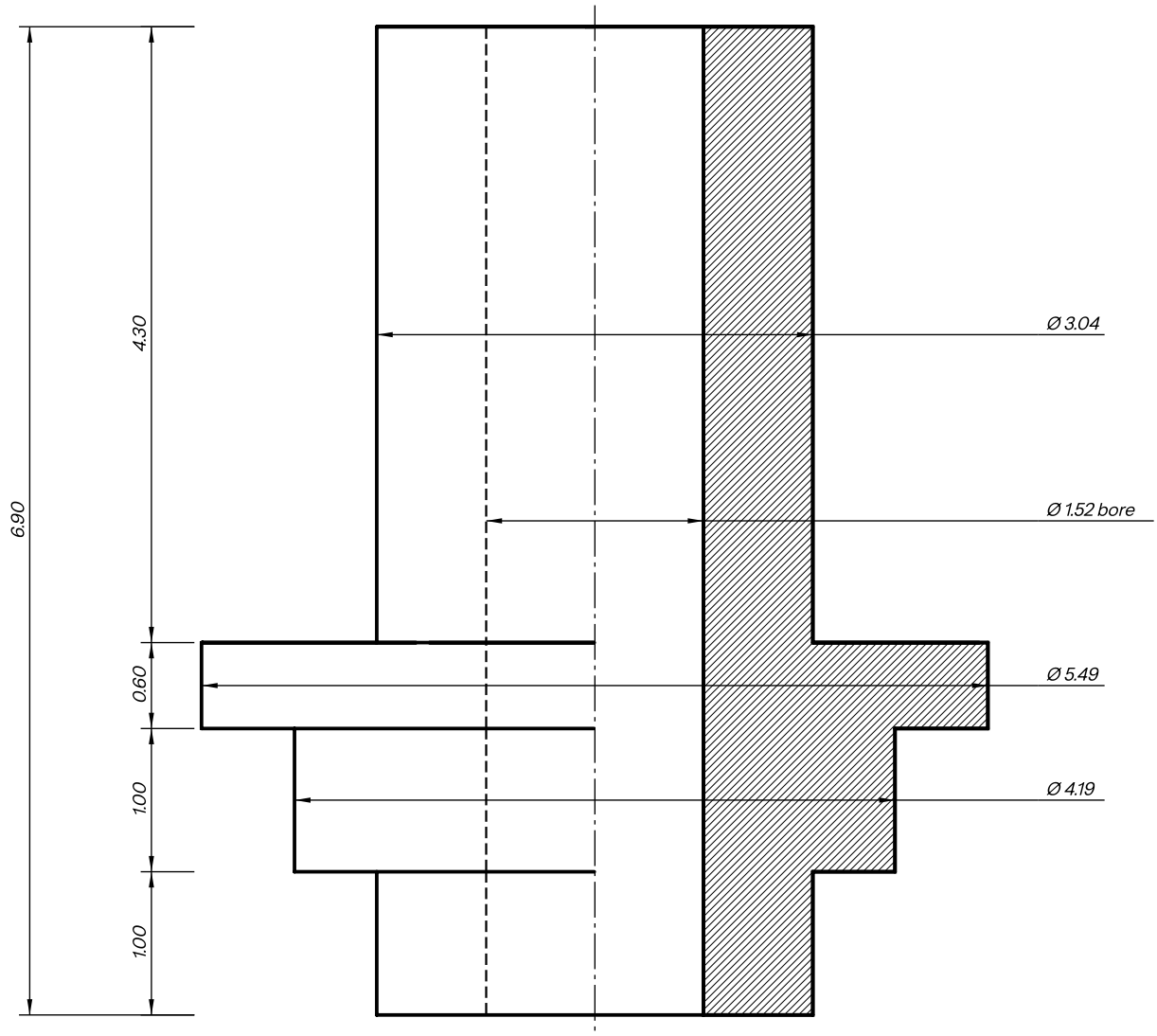
Outline	217	arc clockwise	-9.51, -4.51	-8.66, -3.95	-8.33, -5.37	1.46
Outline	218	line	-8.66, -3.95	-8.87, -3.45		
Outline	219	arc clockwise	-8.87, -3.45	-9.87, -3.66	-9.66, -2.22	1.46
Outline	220	line	-9.87, -3.66	-9.95, -3.42		
Outline	221	arc clockwise	-9.95, -3.42	-9.05, -2.95	-8.88, -4.41	1.46
Outline	222	line	-9.05, -2.95	-9.20, -2.43		
Outline	223	arc clockwise	-9.20, -2.43	-10.21, -2.54	-9.85, -1.12	1.46
Outline	224	line	-10.21, -2.54	-10.27, -2.28		
Outline	225	line	-10.27, -2.28	-10.11, -2.15		

Levels

Faces in section A-A, from the bottom face: 1.00.

Teeth

Number	Module	Tip Ø	Root Ø
56	0.36	21.05	19.03





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **20:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 02 02-2** Teil / Part changeWheel.pipe
Behandlung / Treatment		Benennung / Name **Change wheel pipe (Wechselbutzen)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Change wheel pipe (Wechselbutzen) — sheet 2

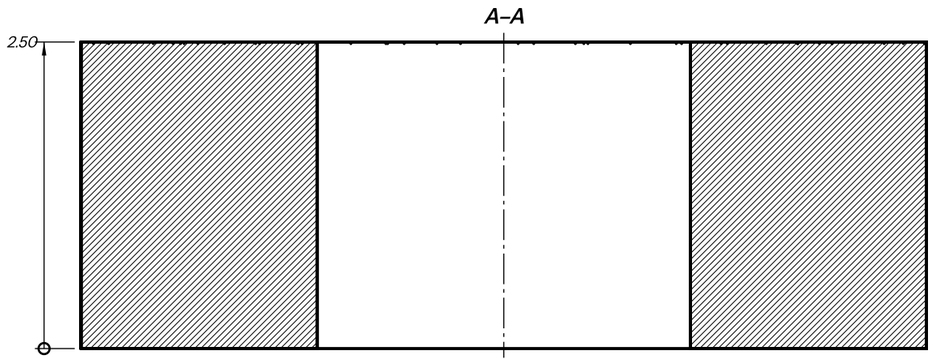
The pipe the wheel and pinion are fixed on, running on the short stud.

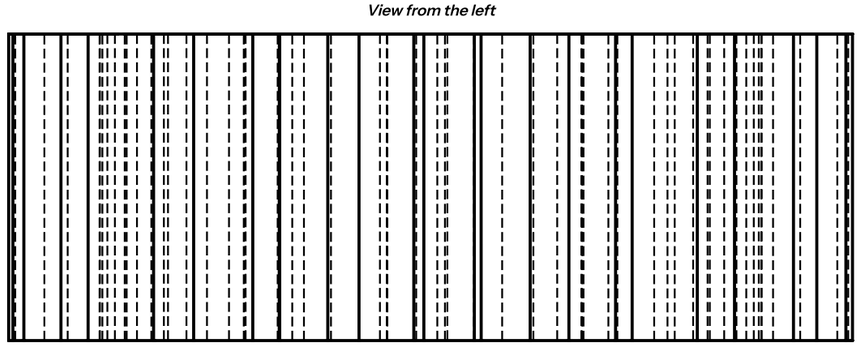
Parts list

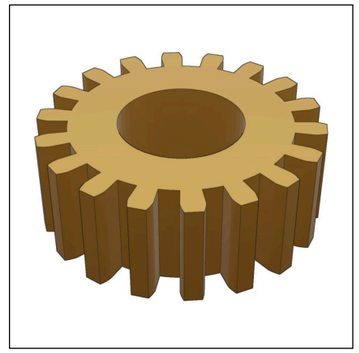
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 02-2	Wechselbutzen (change wheel pipe)	1	not on the drawing		round 050200-7 (run): 6.552 mm, least gap 0.0109; on 050802-2: 0 mm

What it does (05 02 02-2). Runs on the short stud (Laufsitz auf Anrichtstift), its foot on the stud bush.

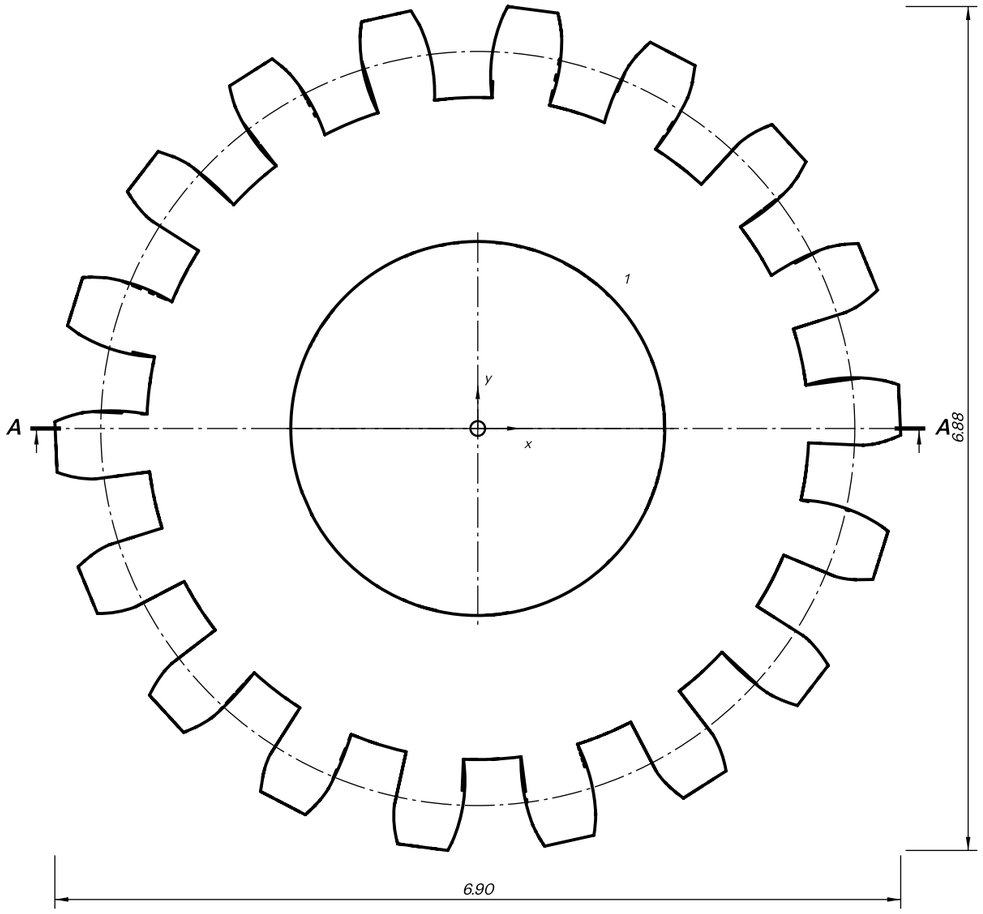
Turned about its axis; length 6.90, largest Ø 5.49. Diameters: Ø 3.04, Ø 4.19, Ø 5.49; bores Ø 1.52. Shoulders from the lower end: 0.00, 1.00, 2.00, 2.60, 6.90.







Isometric view, not to scale



Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 02 02-3**
Behandlung / Treatment			*Teil / Part* *changeWheel.pinion*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Change pinion (Wechseltrieb)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Change pinion (Wechseltrieb) — sheet 2

The pinion that drives the hour wheel, knocked on the pipe to a press fit; brass, as the coloured section colours it.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 02-3	Wechseltrieb (change pinion)	1	not on the drawing		round 050202-2 (press): 2.465 mm, least gap -0.0

What it does (05 02 02-3). Knocked on the pipe to a press fit; drives the hour wheel (no sheet in the set).

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.50.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 3.05 through

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-3.18, -1.06	-2.57, -0.81	-2.42, -2.08	1.28
Outline	2	arc clockwise	-2.57, -0.81	-2.68, -0.35	0.00, 0.00	2.70
Outline	3	arc clockwise	-2.68, -0.35	-3.43, -0.36	-3.06, 0.78	1.20
Outline	4	line	-3.43, -0.36	-3.45, 0.05		
Outline	5	arc clockwise	-3.45, 0.05	-2.70, 0.12	-2.98, -1.05	1.20
Outline	6	arc clockwise	-2.70, 0.12	-2.64, 0.58	0.00, 0.00	2.70
Outline	7	arc clockwise	-2.64, 0.58	-3.35, 0.84	-2.61, 1.78	1.20
Outline	8	line	-3.35, 0.84	-3.22, 1.23		
Outline	9	arc clockwise	-3.22, 1.23	-2.49, 1.03	-3.16, 0.04	1.20
Outline	10	arc clockwise	-2.49, 1.03	-2.28, 1.45	0.00, 0.00	2.70
Outline	11	arc clockwise	-2.28, 1.45	-2.86, 1.93	-1.84, 2.57	1.20
Outline	12	line	-2.86, 1.93	-2.61, 2.26		
Outline	13	arc clockwise	-2.61, 2.26	-1.99, 1.82	-2.96, 1.11	1.20
Outline	14	arc clockwise	-1.99, 1.82	-1.64, 2.14	0.00, 0.00	2.70
Outline	15	arc clockwise	-1.64, 2.14	-2.02, 2.79	-0.85, 3.05	1.20
Outline	16	line	-2.02, 2.79	-1.68, 3.01		
Outline	17	arc clockwise	-1.68, 3.01	-1.25, 2.39	-2.40, 2.06	1.20
Outline	18	arc clockwise	-1.25, 2.39	-0.81, 2.57	0.00, 0.00	2.70
Outline	19	arc clockwise	-0.81, 2.57	-0.95, 3.32	0.24, 3.15	1.20
Outline	20	line	-0.95, 3.32	-0.54, 3.41		
Outline	21	arc clockwise	-0.54, 3.41	-0.35, 2.68	-1.55, 2.76	1.20
Outline	22	arc clockwise	-0.35, 2.68	0.12, 2.70	0.00, 0.00	2.70
Outline	23	arc clockwise	0.12, 2.70	0.24, 3.44	1.30, 2.88	1.20
Outline	24	line	0.24, 3.44	0.65, 3.39		
Outline	25	arc clockwise	0.65, 3.39	0.58, 2.64	-0.51, 3.12	1.20
Outline	26	arc clockwise	0.58, 2.64	1.03, 2.49	0.00, 0.00	2.70

Outline	27	arc clockwise	1.03, 2.49	1.41, 3.15	2.21, 2.26	1.20
Outline	28	line	1.41, 3.15	1.77, 2.96		
Outline	29	arc clockwise	1.77, 2.96	1.45, 2.28	0.58, 3.11	1.20
Outline	30	arc clockwise	1.45, 2.28	1.82, 1.99	0.00, 0.00	2.70
Outline	31	arc clockwise	1.82, 1.99	2.40, 2.48	2.85, 1.37	1.20
Outline	32	line	2.40, 2.48	2.68, 2.18		
Outline	33	arc clockwise	2.68, 2.18	2.14, 1.64	1.61, 2.72	1.20
Outline	34	arc clockwise	2.14, 1.64	2.39, 1.25	0.00, 0.00	2.70
Outline	35	arc clockwise	2.39, 1.25	3.10, 1.51	3.15, 0.31	1.20
Outline	36	line	3.10, 1.51	3.26, 1.13		
Outline	37	arc clockwise	3.26, 1.13	2.57, 0.81	2.44, 2.01	1.20
Outline	38	arc clockwise	2.57, 0.81	2.68, 0.35	0.00, 0.00	2.70
Outline	39	arc clockwise	2.68, 0.35	3.43, 0.36	3.06, -0.78	1.20
Outline	40	line	3.43, 0.36	3.45, -0.05		
Outline	41	arc clockwise	3.45, -0.05	2.70, -0.12	2.98, 1.05	1.20
Outline	42	arc clockwise	2.70, -0.12	2.64, -0.58	0.00, 0.00	2.70
Outline	43	arc clockwise	2.64, -0.58	3.35, -0.84	2.61, -1.78	1.20
Outline	44	line	3.35, -0.84	3.22, -1.23		
Outline	45	arc clockwise	3.22, -1.23	2.49, -1.03	3.16, -0.04	1.20
Outline	46	arc clockwise	2.49, -1.03	2.28, -1.45	0.00, 0.00	2.70
Outline	47	arc clockwise	2.28, -1.45	2.86, -1.93	1.84, -2.57	1.20
Outline	48	line	2.86, -1.93	2.61, -2.26		
Outline	49	arc clockwise	2.61, -2.26	1.99, -1.82	2.96, -1.11	1.20
Outline	50	arc clockwise	1.99, -1.82	1.64, -2.14	0.00, 0.00	2.70
Outline	51	arc clockwise	1.64, -2.14	2.02, -2.79	0.85, -3.05	1.20
Outline	52	line	2.02, -2.79	1.68, -3.01		
Outline	53	arc clockwise	1.68, -3.01	1.25, -2.39	2.40, -2.06	1.20
Outline	54	arc clockwise	1.25, -2.39	0.81, -2.57	0.00, 0.00	2.70
Outline	55	arc clockwise	0.81, -2.57	0.95, -3.32	-0.24, -3.15	1.20
Outline	56	line	0.95, -3.32	0.54, -3.41		
Outline	57	arc clockwise	0.54, -3.41	0.35, -2.68	1.55, -2.76	1.20
Outline	58	arc clockwise	0.35, -2.68	-0.12, -2.70	0.00, 0.00	2.70
Outline	59	arc clockwise	-0.12, -2.70	-0.24, -3.44	-1.30, -2.88	1.20
Outline	60	line	-0.24, -3.44	-0.65, -3.39		
Outline	61	arc clockwise	-0.65, -3.39	-0.58, -2.64	0.51, -3.12	1.20
Outline	62	arc clockwise	-0.58, -2.64	-1.03, -2.49	0.00, 0.00	2.70
Outline	63	arc clockwise	-1.03, -2.49	-1.41, -3.15	-2.21, -2.26	1.20
Outline	64	line	-1.41, -3.15	-1.77, -2.96		
Outline	65	arc clockwise	-1.77, -2.96	-1.45, -2.28	-0.58, -3.11	1.20
Outline	66	arc clockwise	-1.45, -2.28	-1.82, -1.99	0.00, 0.00	2.70
Outline	67	arc clockwise	-1.82, -1.99	-2.40, -2.48	-2.85, -1.37	1.20
Outline	68	line	-2.40, -2.48	-2.68, -2.18		
Outline	69	arc clockwise	-2.68, -2.18	-2.14, -1.64	-1.61, -2.72	1.20
Outline	70	arc clockwise	-2.14, -1.64	-2.39, -1.25	0.00, 0.00	2.70
Outline	71	arc clockwise	-2.39, -1.25	-3.10, -1.51	-3.15, -0.31	1.20
Outline	72	line	-3.10, -1.51	-3.26, -1.13		
Outline	73	line	-3.26, -1.13	-3.18, -1.06		

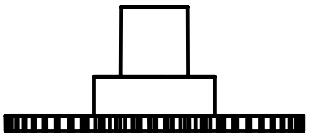
Levels

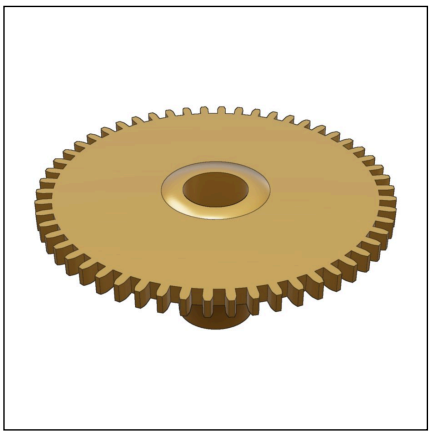
Faces in section A-A, from the bottom face: 2.50.

Teeth

Number	Module	Tip Ø	Root Ø
18	0.35	6.90	5.40

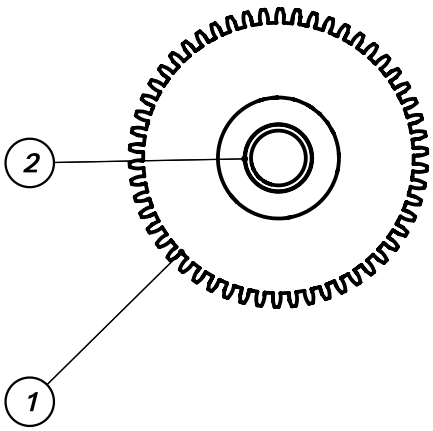
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Hour wheel pipe (Stundenbutzen)	05 02 03-2	not on the drawing
1	1	Hour wheel	05 02 03-1	„Messingblech 1,2 Din 1751 Ms

Werkstoff / Material
„Messingblech 1,2 Din 1751 Ms 58 federhart"

Maßstab / Scale
2:1

Behandlung / Treatment
surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

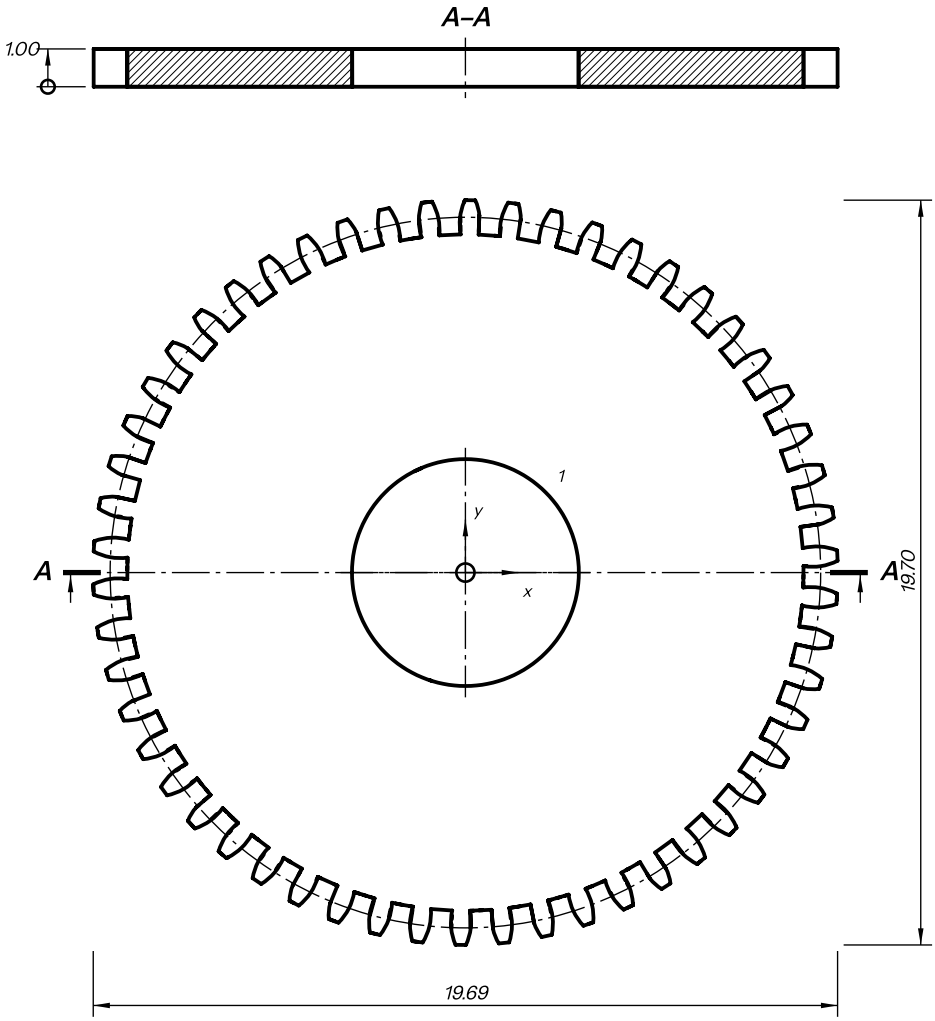
Model v3.35.00 · 2026-10-26

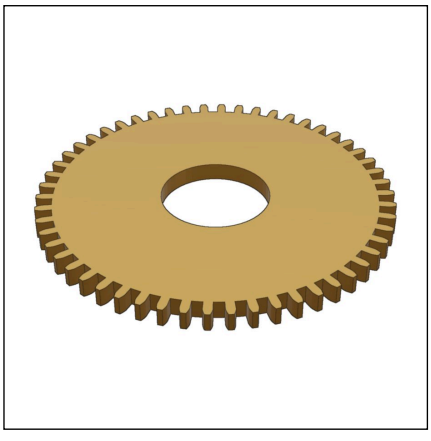
Zeichnung Nr. / Drawing no.
05 02 03-1

Teil / Part
hourWheel

Benennung / Name
Hour wheel (Stundenrad)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material „Messingblech 1,2 Din 1751 Ms 58 federhart"	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 03-1*** *Teil / Part* *hourWheel.wheel*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*		*Benennung / Name* ***Hour wheel***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Hour wheel — sheet 2

The wheel the change pinion drives: a plain spring-hard brass disc, riveted on its pipe.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 03-1	Stundenrad (hour wheel)	1	„Messingblech 1,2 Din 1751 Ms 58 federhart“	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	-3

What it does (05 02 03-1). Riveted on its pipe; the change pinion drives it.

Coordinates from the datum, the movement's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 6.00 through

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-9.43, -2.51	-8.66, -2.23	-8.54, -3.74	1.52
Outline	2	arc clockwise	-8.66, -2.23	-8.79, -1.68	0.00, 0.00	8.94
Outline	3	arc clockwise	-8.79, -1.68	-9.70, -1.72	-9.32, -0.30	1.48
Outline	4	line	-9.70, -1.72	-9.74, -1.45		
Outline	5	arc clockwise	-9.74, -1.45	-8.86, -1.21	-8.93, -2.69	1.48
Outline	6	line	-8.86, -1.21	-8.92, -0.64		
Outline	7	arc clockwise	-8.92, -0.64	-9.83, -0.59	-9.29, 0.79	1.48
Outline	8	line	-9.83, -0.59	-9.84, -0.31		
Outline	9	arc clockwise	-9.84, -0.31	-8.94, -0.17	-9.18, -1.63	1.48
Outline	10	arc clockwise	-8.94, -0.17	-8.94, 0.40	0.00, 0.00	8.94
Outline	11	arc clockwise	-8.94, 0.40	-9.83, 0.56	-9.14, 1.86	1.48
Outline	12	line	-9.83, 0.56	-9.81, 0.83		
Outline	13	arc clockwise	-9.81, 0.83	-8.90, 0.87	-9.31, -0.55	1.48
Outline	14	arc clockwise	-8.90, 0.87	-8.83, 1.43	0.00, 0.00	8.95
Outline	15	arc clockwise	-8.83, 1.43	-9.70, 1.70	-8.86, 2.91	1.48
Outline	16	line	-9.70, 1.70	-9.65, 1.97		
Outline	17	arc clockwise	-9.65, 1.97	-8.74, 1.90	-9.31, 0.53	1.48
Outline	18	arc clockwise	-8.74, 1.90	-8.60, 2.45	0.00, 0.00	8.95
Outline	19	arc clockwise	-8.60, 2.45	-9.44, 2.81	-8.46, 3.92	1.48
Outline	20	line	-9.44, 2.81	-9.36, 3.07		
Outline	21	arc clockwise	-9.36, 3.07	-8.46, 2.90	-9.18, 1.61	1.48
Outline	22	arc clockwise	-8.46, 2.90	-8.26, 3.43	0.00, 0.00	8.94
Outline	23	arc clockwise	-8.26, 3.43	-9.05, 3.89	-7.95, 4.87	1.48
Outline	24	line	-9.05, 3.89	-8.94, 4.14		
Outline	25	arc clockwise	-8.94, 4.14	-8.07, 3.86	-8.94, 2.66	1.48

Outline	26	arc clockwise	-8.07, 3.86	-7.81, 4.36	0.00, 0.00	8.95
Outline	27	arc clockwise	-7.81, 4.36	-8.54, 4.91	-7.33, 5.76	1.48
Outline	28	line	-8.54, 4.91	-8.40, 5.15		
Outline	29	arc clockwise	-8.40, 5.15	-7.57, 4.77	-8.57, 3.68	1.48
Outline	30	arc clockwise	-7.57, 4.77	-7.25, 5.24	0.00, 0.00	8.94
Outline	31	arc clockwise	-7.25, 5.24	-7.91, 5.87	-6.61, 6.58	1.48
Outline	32	line	-7.91, 5.87	-7.74, 6.09		
Outline	33	arc clockwise	-7.74, 6.09	-6.96, 5.62	-8.08, 4.65	1.48
Outline	34	arc clockwise	-6.96, 5.62	-6.59, 6.05	0.00, 0.00	8.94
Outline	35	arc clockwise	-6.59, 6.05	-7.17, 6.75	-5.80, 7.30	1.48
Outline	36	line	-7.17, 6.75	-6.98, 6.95		
Outline	37	arc clockwise	-6.98, 6.95	-6.26, 6.39	-7.49, 5.56	1.48
Outline	38	arc clockwise	-6.26, 6.39	-5.84, 6.77	0.00, 0.00	8.94
Outline	39	arc clockwise	-5.84, 6.77	-6.34, 7.54	-4.92, 7.92	1.48
Outline	40	line	-6.34, 7.54	-6.13, 7.71		
Outline	41	arc clockwise	-6.13, 7.71	-5.48, 7.07	-6.79, 6.39	1.48
Outline	42	arc clockwise	-5.48, 7.07	-5.02, 7.40	0.00, 0.00	8.94
Outline	43	arc clockwise	-5.02, 7.40	-5.42, 8.22	-3.96, 8.44	1.48
Outline	44	line	-5.42, 8.22	-5.19, 8.37		
Outline	45	arc clockwise	-5.19, 8.37	-4.62, 7.66	-6.00, 7.13	1.48
Outline	46	arc clockwise	-4.62, 7.66	-4.13, 7.94	0.00, 0.00	8.94
Outline	47	arc clockwise	-4.13, 7.94	-4.43, 8.80	-2.96, 8.84	1.48
Outline	48	line	-4.43, 8.80	-4.19, 8.92		
Outline	49	arc clockwise	-4.19, 8.92	-3.70, 8.14	-5.13, 7.78	1.48
Outline	50	arc clockwise	-3.70, 8.14	-3.18, 8.36	0.00, 0.00	8.94
Outline	51	arc clockwise	-3.18, 8.36	-3.38, 9.25	-1.91, 9.13	1.48
Outline	52	line	-3.38, 9.25	-3.12, 9.34		
Outline	53	arc clockwise	-3.12, 9.34	-2.73, 8.52	-4.20, 8.33	1.48
Outline	54	arc clockwise	-2.73, 8.52	-2.18, 8.67	0.00, 0.00	8.95
Outline	55	arc clockwise	-2.18, 8.67	-2.28, 9.58	-0.84, 9.29	1.48
Outline	56	line	-2.28, 9.58	-2.02, 9.64		
Outline	57	arc clockwise	-2.02, 9.64	-1.72, 8.78	-3.20, 8.76	1.48
Outline	58	line	-1.72, 8.78	-1.16, 8.87		
Outline	59	arc clockwise	-1.16, 8.87	-1.16, 9.78	0.25, 9.32	1.48
Outline	60	line	-1.16, 9.78	-0.88, 9.81		
Outline	61	arc clockwise	-0.88, 9.81	-0.69, 8.92	-2.16, 9.07	1.48
Outline	62	arc clockwise	-0.69, 8.92	-0.12, 8.94	0.00, 0.00	8.94
Outline	63	arc clockwise	-0.12, 8.94	-0.01, 9.85	1.33, 9.23	1.48
Outline	64	line	-0.01, 9.85	0.26, 9.85		
Outline	65	arc clockwise	0.26, 9.85	0.35, 8.94	-1.09, 9.26	1.48
Outline	66	arc clockwise	0.35, 8.94	0.91, 8.90	0.00, 0.00	8.95
Outline	67	arc clockwise	0.91, 8.90	1.13, 9.78	2.39, 9.01	1.48
Outline	68	line	1.13, 9.78	1.40, 9.75		
Outline	69	arc clockwise	1.40, 9.75	1.38, 8.84	-0.01, 9.32	1.48
Outline	70	arc clockwise	1.38, 8.84	1.94, 8.73	0.00, 0.00	8.94
Outline	71	arc clockwise	1.94, 8.73	2.26, 9.59	3.42, 8.67	1.48
Outline	72	line	2.26, 9.59	2.52, 9.52		
Outline	73	arc clockwise	2.52, 9.52	2.40, 8.62	1.07, 9.26	1.48

Outline	74	arc clockwise	2.40, 8.62	2.94, 8.45	0.00, 0.00	8.94
Outline	75	arc clockwise	2.94, 8.45	3.36, 9.26	4.40, 8.22	1.48
Outline	76	line	3.36, 9.26	3.61, 9.16		
Outline	77	arc clockwise	3.61, 9.16	3.38, 8.28	2.14, 9.08	1.48
Outline	78	arc clockwise	3.38, 8.28	3.90, 8.05	0.00, 0.00	8.94
Outline	79	arc clockwise	3.90, 8.05	4.41, 8.81	5.33, 7.65	1.48
Outline	80	line	4.41, 8.81	4.65, 8.68		
Outline	81	arc clockwise	4.65, 8.68	4.32, 7.83	3.18, 8.77	1.48
Outline	82	arc clockwise	4.32, 7.83	4.81, 7.54	0.00, 0.00	8.94
Outline	83	arc clockwise	4.81, 7.54	5.40, 8.24	6.18, 6.98	1.48
Outline	84	line	5.40, 8.24	5.63, 8.08		
Outline	85	arc clockwise	5.63, 8.08	5.20, 7.28	4.17, 8.34	1.48
Outline	86	arc clockwise	5.20, 7.28	5.65, 6.93	0.00, 0.00	8.94
Outline	87	arc clockwise	5.65, 6.93	6.32, 7.55	6.95, 6.22	1.48
Outline	88	line	6.32, 7.55	6.53, 7.38		
Outline	89	arc clockwise	6.53, 7.38	6.01, 6.62	5.11, 7.80	1.48
Outline	90	arc clockwise	6.01, 6.62	6.42, 6.23	0.00, 0.00	8.95
Outline	91	arc clockwise	6.42, 6.23	7.16, 6.77	7.62, 5.37	1.48
Outline	92	line	7.16, 6.77	7.34, 6.57		
Outline	93	arc clockwise	7.34, 6.57	6.74, 5.88	5.98, 7.15	1.48
Outline	94	arc clockwise	6.74, 5.88	7.10, 5.44	0.00, 0.00	8.94
Outline	95	arc clockwise	7.10, 5.44	7.89, 5.89	8.20, 4.45	1.48
Outline	96	line	7.89, 5.89	8.05, 5.67		
Outline	97	arc clockwise	8.05, 5.67	7.38, 5.06	6.77, 6.41	1.48
Outline	98	arc clockwise	7.38, 5.06	7.68, 4.58	0.00, 0.00	8.94
Outline	99	arc clockwise	7.68, 4.58	8.52, 4.94	8.66, 3.47	1.48
Outline	100	line	8.52, 4.94	8.66, 4.70		
Outline	101	arc clockwise	8.66, 4.70	7.92, 4.17	7.47, 5.58	1.48
Outline	102	line	7.92, 4.17	8.16, 3.66		
Outline	103	arc clockwise	8.16, 3.66	9.04, 3.91	9.00, 2.44	1.48
Outline	104	line	9.04, 3.91	9.14, 3.66		
Outline	105	arc clockwise	9.14, 3.66	8.35, 3.22	8.07, 4.67	1.48
Outline	106	arc clockwise	8.35, 3.22	8.53, 2.68	0.00, 0.00	8.94
Outline	107	arc clockwise	8.53, 2.68	9.43, 2.84	9.22, 1.38	1.48
Outline	108	line	9.43, 2.84	9.51, 2.57		
Outline	109	arc clockwise	9.51, 2.57	8.66, 2.23	8.56, 3.70	1.48
Outline	110	arc clockwise	8.66, 2.23	8.79, 1.68	0.00, 0.00	8.94
Outline	111	arc clockwise	8.79, 1.68	9.70, 1.72	9.32, 0.30	1.48
Outline	112	line	9.70, 1.72	9.74, 1.45		
Outline	113	arc clockwise	9.74, 1.45	8.86, 1.21	8.93, 2.69	1.48
Outline	114	line	8.86, 1.21	8.92, 0.64		
Outline	115	arc clockwise	8.92, 0.64	9.83, 0.59	9.29, -0.79	1.48
Outline	116	line	9.83, 0.59	9.84, 0.31		
Outline	117	arc clockwise	9.84, 0.31	8.94, 0.17	9.18, 1.63	1.48
Outline	118	arc clockwise	8.94, 0.17	8.94, -0.40	0.00, 0.00	8.94
Outline	119	arc clockwise	8.94, -0.40	9.83, -0.56	9.14, -1.86	1.48
Outline	120	line	9.83, -0.56	9.81, -0.83		
Outline	121	arc clockwise	9.81, -0.83	8.90, -0.87	9.31, 0.55	1.48

Outline	122	arc clockwise	8.90, -0.87	8.83, -1.43	0.00, 0.00	8.95
Outline	123	arc clockwise	8.83, -1.43	9.70, -1.70	8.86, -2.91	1.48
Outline	124	line	9.70, -1.70	9.65, -1.97		
Outline	125	arc clockwise	9.65, -1.97	8.74, -1.90	9.31, -0.53	1.48
Outline	126	arc clockwise	8.74, -1.90	8.60, -2.45	0.00, 0.00	8.95
Outline	127	arc clockwise	8.60, -2.45	9.44, -2.81	8.46, -3.92	1.48
Outline	128	line	9.44, -2.81	9.36, -3.07		
Outline	129	arc clockwise	9.36, -3.07	8.46, -2.90	9.18, -1.61	1.48
Outline	130	arc clockwise	8.46, -2.90	8.26, -3.43	0.00, 0.00	8.94
Outline	131	arc clockwise	8.26, -3.43	9.05, -3.89	7.95, -4.87	1.48
Outline	132	line	9.05, -3.89	8.94, -4.14		
Outline	133	arc clockwise	8.94, -4.14	8.07, -3.86	8.94, -2.66	1.48
Outline	134	arc clockwise	8.07, -3.86	7.81, -4.36	0.00, 0.00	8.95
Outline	135	arc clockwise	7.81, -4.36	8.54, -4.91	7.33, -5.76	1.48
Outline	136	line	8.54, -4.91	8.40, -5.15		
Outline	137	arc clockwise	8.40, -5.15	7.57, -4.77	8.57, -3.68	1.48
Outline	138	arc clockwise	7.57, -4.77	7.25, -5.24	0.00, 0.00	8.94
Outline	139	arc clockwise	7.25, -5.24	7.91, -5.87	6.61, -6.58	1.48
Outline	140	line	7.91, -5.87	7.74, -6.09		
Outline	141	arc clockwise	7.74, -6.09	6.96, -5.62	8.08, -4.65	1.48
Outline	142	arc clockwise	6.96, -5.62	6.59, -6.05	0.00, 0.00	8.94
Outline	143	arc clockwise	6.59, -6.05	7.17, -6.75	5.80, -7.30	1.48
Outline	144	line	7.17, -6.75	6.98, -6.95		
Outline	145	arc clockwise	6.98, -6.95	6.26, -6.39	7.49, -5.56	1.48
Outline	146	arc clockwise	6.26, -6.39	5.84, -6.77	0.00, 0.00	8.94
Outline	147	arc clockwise	5.84, -6.77	6.34, -7.54	4.92, -7.92	1.48
Outline	148	line	6.34, -7.54	6.13, -7.71		
Outline	149	arc clockwise	6.13, -7.71	5.48, -7.07	6.79, -6.39	1.48
Outline	150	arc clockwise	5.48, -7.07	5.02, -7.40	0.00, 0.00	8.94
Outline	151	arc clockwise	5.02, -7.40	5.42, -8.22	3.96, -8.44	1.48
Outline	152	line	5.42, -8.22	5.19, -8.37		
Outline	153	arc clockwise	5.19, -8.37	4.62, -7.66	6.00, -7.13	1.48
Outline	154	arc clockwise	4.62, -7.66	4.13, -7.94	0.00, 0.00	8.94
Outline	155	arc clockwise	4.13, -7.94	4.43, -8.80	2.96, -8.84	1.48
Outline	156	line	4.43, -8.80	4.19, -8.92		
Outline	157	arc clockwise	4.19, -8.92	3.70, -8.14	5.13, -7.78	1.48
Outline	158	arc clockwise	3.70, -8.14	3.18, -8.36	0.00, 0.00	8.94
Outline	159	arc clockwise	3.18, -8.36	3.38, -9.25	1.91, -9.13	1.48
Outline	160	line	3.38, -9.25	3.12, -9.34		
Outline	161	arc clockwise	3.12, -9.34	2.73, -8.52	4.20, -8.33	1.48
Outline	162	arc clockwise	2.73, -8.52	2.18, -8.67	0.00, 0.00	8.95
Outline	163	arc clockwise	2.18, -8.67	2.28, -9.58	0.84, -9.29	1.48
Outline	164	line	2.28, -9.58	2.02, -9.64		
Outline	165	arc clockwise	2.02, -9.64	1.72, -8.78	3.20, -8.76	1.48
Outline	166	line	1.72, -8.78	1.16, -8.87		
Outline	167	arc clockwise	1.16, -8.87	1.16, -9.78	-0.25, -9.32	1.48
Outline	168	line	1.16, -9.78	0.88, -9.81		
Outline	169	arc clockwise	0.88, -9.81	0.69, -8.92	2.16, -9.07	1.48

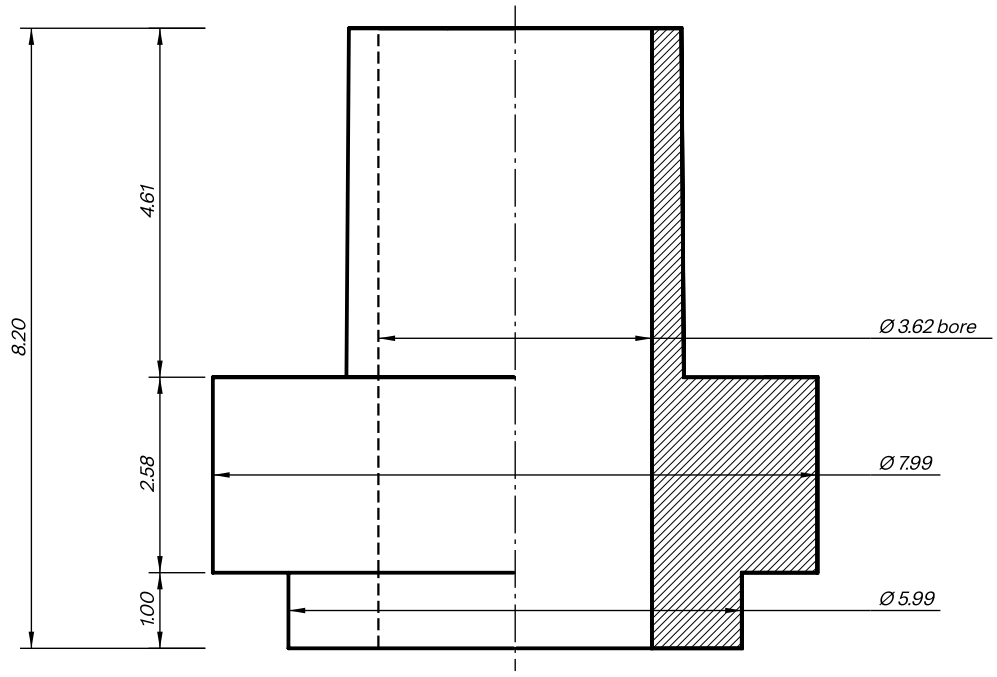
Outline	170	arc clockwise	0.69, -8.92	0.12, -8.94	0.00, 0.00	8.94
Outline	171	arc clockwise	0.12, -8.94	0.01, -9.85	-1.33, -9.23	1.48
Outline	172	line	0.01, -9.85	-0.26, -9.85		
Outline	173	arc clockwise	-0.26, -9.85	-0.35, -8.94	1.09, -9.26	1.48
Outline	174	arc clockwise	-0.35, -8.94	-0.91, -8.90	0.00, 0.00	8.95
Outline	175	arc clockwise	-0.91, -8.90	-1.13, -9.78	-2.39, -9.01	1.48
Outline	176	line	-1.13, -9.78	-1.40, -9.75		
Outline	177	arc clockwise	-1.40, -9.75	-1.38, -8.84	0.01, -9.32	1.48
Outline	178	arc clockwise	-1.38, -8.84	-1.94, -8.73	0.00, 0.00	8.94
Outline	179	arc clockwise	-1.94, -8.73	-2.26, -9.59	-3.42, -8.67	1.48
Outline	180	line	-2.26, -9.59	-2.52, -9.52		
Outline	181	arc clockwise	-2.52, -9.52	-2.40, -8.62	-1.07, -9.26	1.48
Outline	182	arc clockwise	-2.40, -8.62	-2.94, -8.45	0.00, 0.00	8.94
Outline	183	arc clockwise	-2.94, -8.45	-3.36, -9.26	-4.40, -8.22	1.48
Outline	184	line	-3.36, -9.26	-3.61, -9.16		
Outline	185	arc clockwise	-3.61, -9.16	-3.38, -8.28	-2.14, -9.08	1.48
Outline	186	arc clockwise	-3.38, -8.28	-3.90, -8.05	0.00, 0.00	8.94
Outline	187	arc clockwise	-3.90, -8.05	-4.41, -8.81	-5.33, -7.65	1.48
Outline	188	line	-4.41, -8.81	-4.65, -8.68		
Outline	189	arc clockwise	-4.65, -8.68	-4.32, -7.83	-3.18, -8.77	1.48
Outline	190	arc clockwise	-4.32, -7.83	-4.81, -7.54	0.00, 0.00	8.94
Outline	191	arc clockwise	-4.81, -7.54	-5.40, -8.24	-6.18, -6.98	1.48
Outline	192	line	-5.40, -8.24	-5.63, -8.08		
Outline	193	arc clockwise	-5.63, -8.08	-5.20, -7.28	-4.17, -8.34	1.48
Outline	194	arc clockwise	-5.20, -7.28	-5.65, -6.93	0.00, 0.00	8.94
Outline	195	arc clockwise	-5.65, -6.93	-6.32, -7.55	-6.95, -6.22	1.48
Outline	196	line	-6.32, -7.55	-6.53, -7.38		
Outline	197	arc clockwise	-6.53, -7.38	-6.01, -6.62	-5.11, -7.80	1.48
Outline	198	arc clockwise	-6.01, -6.62	-6.42, -6.23	0.00, 0.00	8.95
Outline	199	arc clockwise	-6.42, -6.23	-7.16, -6.77	-7.62, -5.37	1.48
Outline	200	line	-7.16, -6.77	-7.34, -6.57		
Outline	201	arc clockwise	-7.34, -6.57	-6.74, -5.88	-5.98, -7.15	1.48
Outline	202	arc clockwise	-6.74, -5.88	-7.10, -5.44	0.00, 0.00	8.94
Outline	203	arc clockwise	-7.10, -5.44	-7.89, -5.89	-8.20, -4.45	1.48
Outline	204	line	-7.89, -5.89	-8.05, -5.67		
Outline	205	arc clockwise	-8.05, -5.67	-7.38, -5.06	-6.77, -6.41	1.48
Outline	206	arc clockwise	-7.38, -5.06	-7.68, -4.58	0.00, 0.00	8.94
Outline	207	arc clockwise	-7.68, -4.58	-8.52, -4.94	-8.66, -3.47	1.48
Outline	208	line	-8.52, -4.94	-8.66, -4.70		
Outline	209	arc clockwise	-8.66, -4.70	-7.92, -4.17	-7.47, -5.58	1.48
Outline	210	line	-7.92, -4.17	-8.16, -3.66		
Outline	211	arc clockwise	-8.16, -3.66	-9.04, -3.91	-9.00, -2.44	1.48
Outline	212	line	-9.04, -3.91	-9.14, -3.66		
Outline	213	arc clockwise	-9.14, -3.66	-8.35, -3.22	-8.07, -4.67	1.48
Outline	214	arc clockwise	-8.35, -3.22	-8.53, -2.68	0.00, 0.00	8.94
Outline	215	arc clockwise	-8.53, -2.68	-9.43, -2.84	-9.22, -1.38	1.48
Outline	216	line	-9.43, -2.84	-9.51, -2.57		
Outline	217	line	-9.51, -2.57	-9.43, -2.51		

Levels

Faces in section A-A, from the bottom face: 1.00.

Teeth

Number	Module	Tip Ø	Root Ø
54	0.35	19.70	17.89





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **10:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 02 03-2**
Behandlung / Treatment			Teil / Part hourWheel.pipe
		Benennung / Name **Hour wheel pipe (Stundenbutzen)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Hour wheel pipe (Stundenbutzen) — sheet 2

The pipe the wheel is riveted on: it runs on the cannon tube, its collar under the dial, its tapered seat through the dial for the hour hand.

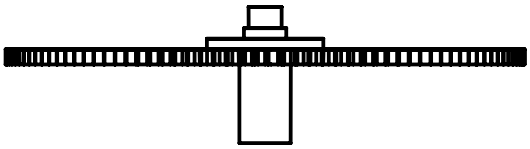
Parts list

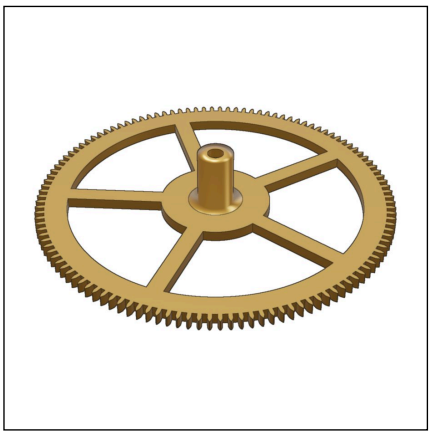
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 03-2	Stundenbutzen (hour wheel pipe)	1	not on the drawing		round 050503-3 (run): 7.4 mm, least gap 0.0096

What it does (05 02 03-2). Runs on the cannon tube, over the cannon pinion; carries the hour hand on its tapered seat (no sheet of its own: 05 02 03 gives it).

Turned about its axis; length 8.20, largest Ø 8.00. Diameters: Ø 5.99, Ø 7.99; bores Ø 3.62. Shoulders from the lower end: 0.00, 1.00, 3.58, 8.20.

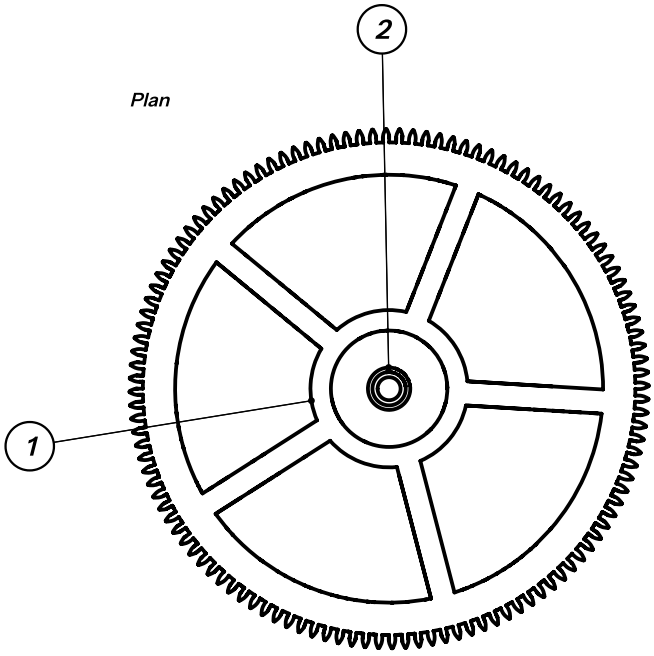
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Up-and-down wheel pipe	05 02 04-2	not on the drawing
1	1	Up-and-down wheel	05 02 04-1	„Messingblech 1,2 Din 1751 Ms

Werkstoff / Material
„Messingblech 1,2 Din 1751 Ms 58 federhart"

Maßstab / Scale
2:1

Behandlung / Treatment
surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,03

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

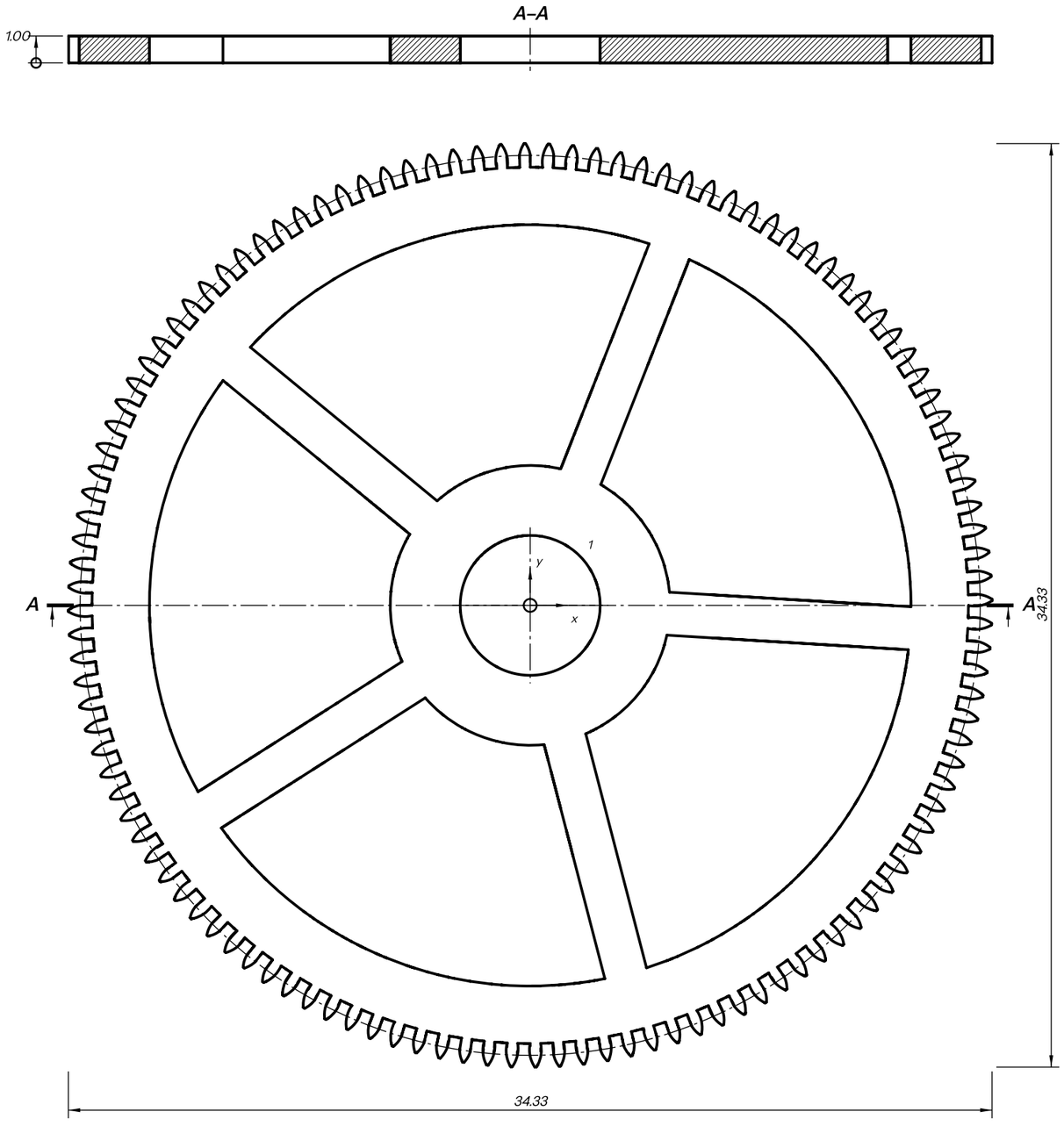
Model v3.35.00 · 2026-10-26

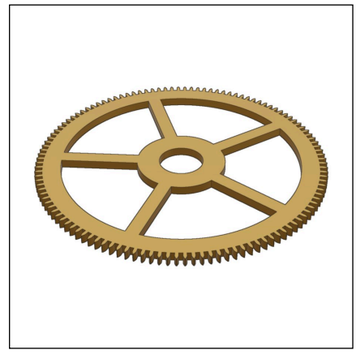
Zeichnung Nr. / Drawing no.
05 02 04-1

Teil / Part
udWheel

Benennung / Name
Up-and-down wheel (Auf- und Abrad)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Messingblech 1,2 Din 1751 Ms 58 federhart"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 04-1*** *Teil / Part* *udWheel.wheel*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,03*		*Benennung / Name* ***Up-and-down wheel***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Up-and-down wheel — sheet 2

The five-spoked wheel the fusee's pinion turns.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 04-1	Auf u. Abrad (up-and-down wheel)	1	„Messingblech 1,2 Din 1751 Ms 58 federhart“	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,03	-10

What it does (05 02 04-1). Riveted on its pipe; the up-and-down pinion on the fusee arbor drives it.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 5.20 through

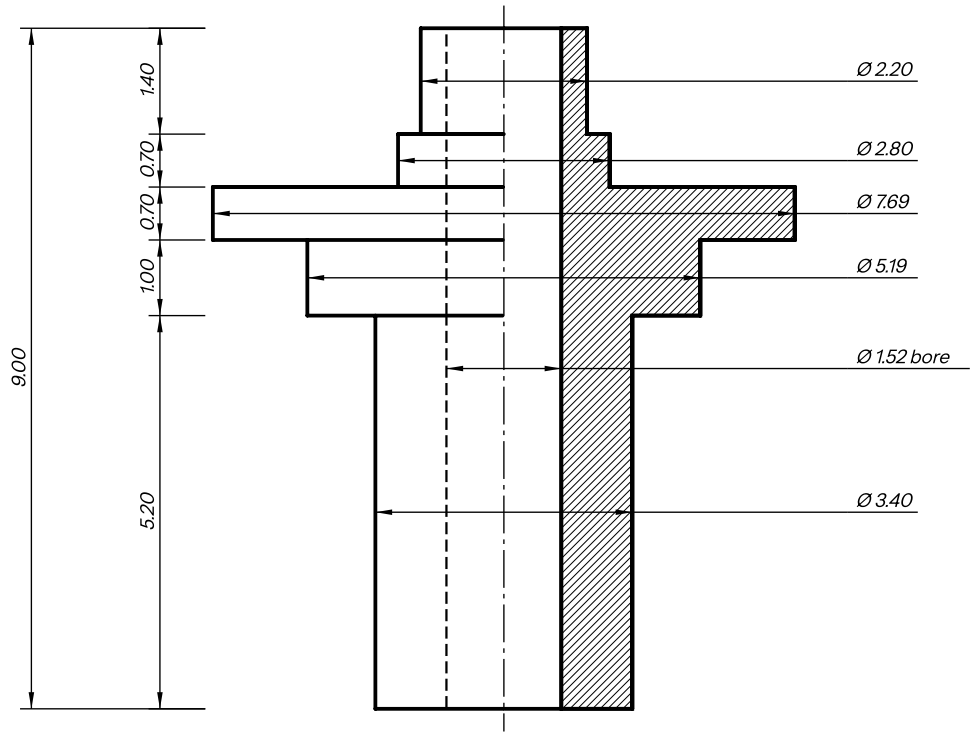
The contour has 506 elements (its teeth): the drawing and the tooth data define it.

Levels

Faces in section A-A, from the bottom face: 1.00.

Teeth

Number	Module	Tip Ø	Root Ø
120	0.2787833333333333	34.33	32.57





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 02 04-2** *Teil / Part* *udWheel.pipe*
Behandlung / Treatment		*Benennung / Name* ***Up-and-down wheel pipe***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Up-and-down wheel pipe — sheet 2

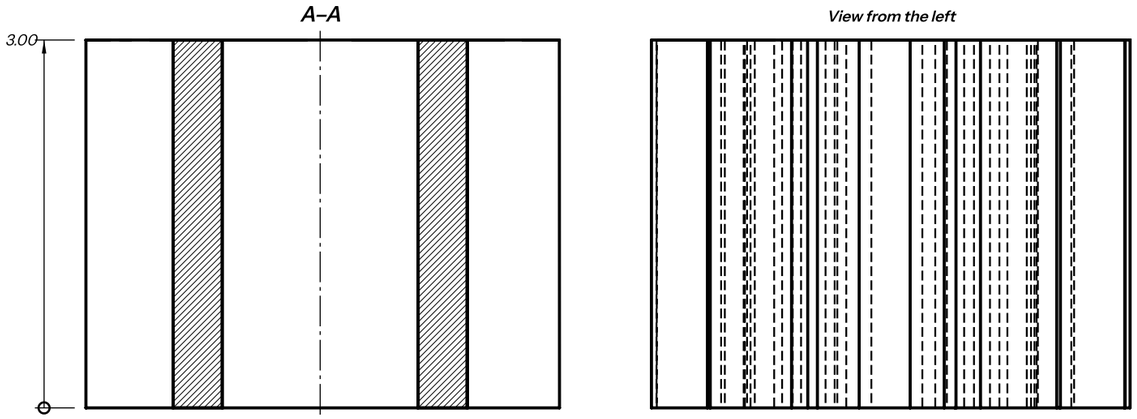
The pipe it is riveted on, running on the long stud, with the hand's seat at its top.

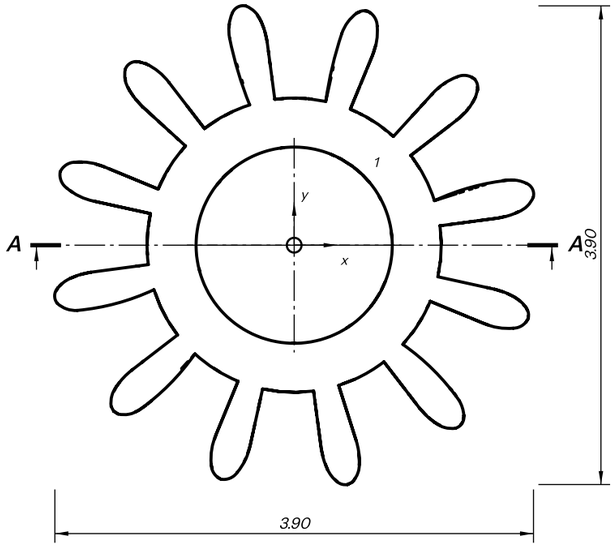
Parts list

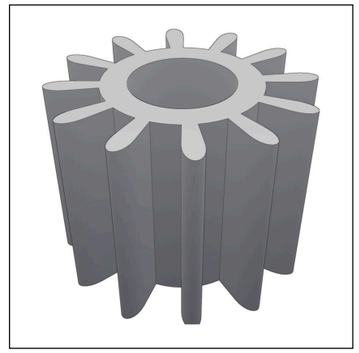
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 04-2	Auf u. Abrad-Butzen (up-and-down wheel pipe)	1	not on the drawing		round 050200-6 (run): 8.71 mm, least gap 0.0122; on 050802-2: 0 mm

What it does (05 02 04-2). Runs on the long stud (Laufsitz auf Anrichtstift), its foot on the stud bush; carries the up-and-down hand.

Turned about its axis; length 9.00, largest Ø 7.70. Diameters: Ø 2.20, Ø 2.80, Ø 3.40, Ø 5.19, Ø 7.69; bores Ø 1.52. Shoulders from the lower end: 0.00, 5.20, 6.20, 6.90, 7.60, 9.00.







Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 02 00-5**
Behandlung / Treatment			*Teil / Part* *udPinion*
		Benennung / Name ***Up-and-down pinion (Auf- und Abtrieb)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Up-and-down pinion (Auf- und Abtrieb) — sheet 2

A small steel pinion pressed on the dial-side end of the fusee arbor, under the dial: it drives the up-and-down wheel.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 00-5	Auf u. Abtrieb (up-and-down pinion)	1	not on the drawing		not measured on the model: see its function, Pressed on the fusee arbor's dial-side end (the fusee group's): drives the up-and-down wheel (no sheet in the set; numbered by the coloured section's label).

What it does (05 02 00-5). Pressed on the fusee arbor's dial-side end (the fusee group's): drives the up-and-down wheel (no sheet in the set; numbered by the coloured section's label).

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 1.60 through

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-1.47, -1.35	-1.45, -1.15	-1.32, -1.27	0.17
Outline	2	arc clockwise	-1.45, -1.15	-0.95, -0.73	0.00, -2.37	1.90
Outline	3	arc clockwise	-0.95, -0.73	-1.14, -0.36	0.00, 0.00	1.20
Outline	4	arc clockwise	-1.14, -0.36	-1.77, -0.53	-1.95, 1.36	1.90
Outline	5	arc clockwise	-1.77, -0.53	-1.95, -0.44	-1.80, -0.36	0.17
Outline	6	arc clockwise	-1.95, -0.44	-1.83, -0.27	-1.78, -0.44	0.17
Outline	7	arc clockwise	-1.83, -0.27	-1.19, -0.16	-1.19, -2.05	1.90
Outline	8	arc clockwise	-1.19, -0.16	-1.17, 0.26	0.00, 0.00	1.20
Outline	9	arc clockwise	-1.17, 0.26	-1.80, 0.42	-1.01, 2.15	1.90
Outline	10	arc clockwise	-1.80, 0.42	-1.91, 0.60	-1.74, 0.59	0.17
Outline	11	arc clockwise	-1.91, 0.60	-1.72, 0.68	-1.76, 0.51	0.17
Outline	12	arc clockwise	-1.72, 0.68	-1.11, 0.46	-2.05, -1.19	1.90
Outline	13	arc clockwise	-1.11, 0.46	-0.89, 0.81	0.00, 0.00	1.20
Outline	14	arc clockwise	-0.89, 0.81	-1.34, 1.27	0.20, 2.36	1.90
Outline	15	arc clockwise	-1.34, 1.27	-1.35, 1.47	-1.21, 1.38	0.17
Outline	16	arc clockwise	-1.35, 1.47	-1.15, 1.45	-1.27, 1.32	0.17
Outline	17	arc clockwise	-1.15, 1.45	-0.73, 0.95	-2.37, 0.00	1.90
Outline	18	arc clockwise	-0.73, 0.95	-0.36, 1.14	0.00, 0.00	1.20
Outline	19	arc clockwise	-0.36, 1.14	-0.53, 1.77	1.36, 1.95	1.90
Outline	20	arc clockwise	-0.53, 1.77	-0.44, 1.95	-0.36, 1.80	0.17
Outline	21	arc clockwise	-0.44, 1.95	-0.27, 1.83	-0.44, 1.78	0.17
Outline	22	arc clockwise	-0.27, 1.83	-0.16, 1.19	-2.05, 1.19	1.90
Outline	23	arc clockwise	-0.16, 1.19	0.26, 1.17	0.00, 0.00	1.20
Outline	24	arc clockwise	0.26, 1.17	0.42, 1.80	2.15, 1.01	1.90

Outline	25	arc clockwise	0.42, 1.80	0.60, 1.91	0.59, 1.74	0.17
Outline	26	arc clockwise	0.60, 1.91	0.68, 1.72	0.51, 1.76	0.17
Outline	27	arc clockwise	0.68, 1.72	0.46, 1.11	-1.19, 2.05	1.90
Outline	28	arc clockwise	0.46, 1.11	0.81, 0.89	0.00, 0.00	1.20
Outline	29	arc clockwise	0.81, 0.89	1.27, 1.34	2.36, -0.20	1.90
Outline	30	arc clockwise	1.27, 1.34	1.47, 1.35	1.38, 1.21	0.17
Outline	31	arc clockwise	1.47, 1.35	1.45, 1.15	1.32, 1.27	0.17
Outline	32	arc clockwise	1.45, 1.15	0.95, 0.73	0.00, 2.37	1.90
Outline	33	arc clockwise	0.95, 0.73	1.14, 0.36	0.00, 0.00	1.20
Outline	34	arc clockwise	1.14, 0.36	1.77, 0.53	1.95, -1.36	1.90
Outline	35	arc clockwise	1.77, 0.53	1.95, 0.44	1.80, 0.36	0.17
Outline	36	arc clockwise	1.95, 0.44	1.83, 0.27	1.78, 0.44	0.17
Outline	37	arc clockwise	1.83, 0.27	1.19, 0.16	1.19, 2.05	1.90
Outline	38	arc clockwise	1.19, 0.16	1.17, -0.26	0.00, 0.00	1.20
Outline	39	arc clockwise	1.17, -0.26	1.80, -0.42	1.01, -2.15	1.90
Outline	40	arc clockwise	1.80, -0.42	1.91, -0.60	1.74, -0.59	0.17
Outline	41	arc clockwise	1.91, -0.60	1.72, -0.68	1.76, -0.51	0.17
Outline	42	arc clockwise	1.72, -0.68	1.11, -0.46	2.05, 1.19	1.90
Outline	43	arc clockwise	1.11, -0.46	0.89, -0.81	0.00, 0.00	1.20
Outline	44	arc clockwise	0.89, -0.81	1.34, -1.27	-0.20, -2.36	1.90
Outline	45	arc clockwise	1.34, -1.27	1.35, -1.47	1.21, -1.38	0.17
Outline	46	arc clockwise	1.35, -1.47	1.15, -1.45	1.27, -1.32	0.17
Outline	47	arc clockwise	1.15, -1.45	0.73, -0.95	2.37, 0.00	1.90
Outline	48	arc clockwise	0.73, -0.95	0.36, -1.14	0.00, 0.00	1.20
Outline	49	arc clockwise	0.36, -1.14	0.53, -1.77	-1.36, -1.95	1.90
Outline	50	arc clockwise	0.53, -1.77	0.44, -1.95	0.36, -1.80	0.17
Outline	51	arc clockwise	0.44, -1.95	0.27, -1.83	0.44, -1.78	0.17
Outline	52	arc clockwise	0.27, -1.83	0.16, -1.19	2.06, -1.19	1.90
Outline	53	arc clockwise	0.16, -1.19	-0.26, -1.17	0.00, 0.00	1.20
Outline	54	arc clockwise	-0.26, -1.17	-0.42, -1.80	-2.15, -1.01	1.90
Outline	55	arc clockwise	-0.42, -1.80	-0.60, -1.91	-0.59, -1.74	0.17
Outline	56	arc clockwise	-0.60, -1.91	-0.68, -1.72	-0.51, -1.76	0.17
Outline	57	arc clockwise	-0.68, -1.72	-0.46, -1.11	1.19, -2.05	1.90
Outline	58	arc clockwise	-0.46, -1.11	-0.81, -0.89	0.00, 0.00	1.20
Outline	59	arc clockwise	-0.81, -0.89	-1.27, -1.34	-2.36, 0.20	1.90
Outline	60	arc clockwise	-1.27, -1.34	-1.47, -1.35	-1.38, -1.21	0.17

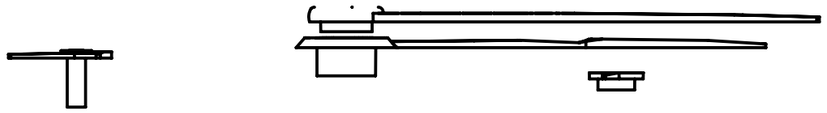
Levels

Faces in section A-A, from the bottom face: 3.00.

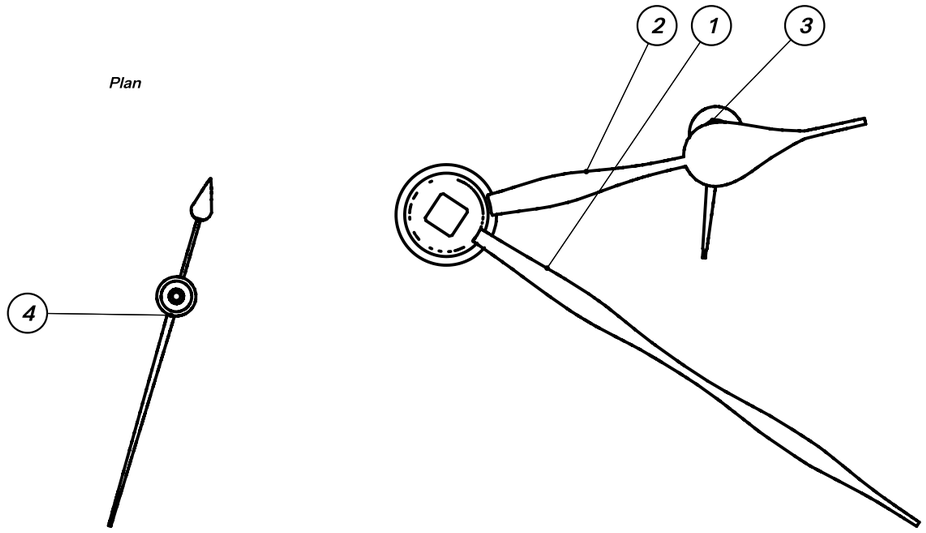
Teeth

Number	Module	Tip Ø	Root Ø
12	0.2787833333333333		

Front view



Plan





Isometric view, not to scale

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Seconds hand (Sekundenzeiger) and collet	05 02 01-1	„Bandstahl 30 × 0,6 Din 1016“
3	1	Up-and-down hand (Auf- und Abzeiger)	05 02 00-3	„Bandstahl 0,6 × 15“
2	1	Hour hand (Stundenzeiger)	05 02 00-2	„Goldin“
1	1	Minute hand (Minutenzeiger)	05 02 00-1	„Goldin“ (a gold-coloured bras

Werkstoff / Material

„Goldin“ (a gold-coloured brass alloy)

Maßstab / Scale

2:1

Behandlung / Treatment

„Oberfläche poliert“ (surface polished); surface ▽▽▽▽ (▽▽▽) (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1

Blatt / Sheet

1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.

05 02 00-1

Teil / Part

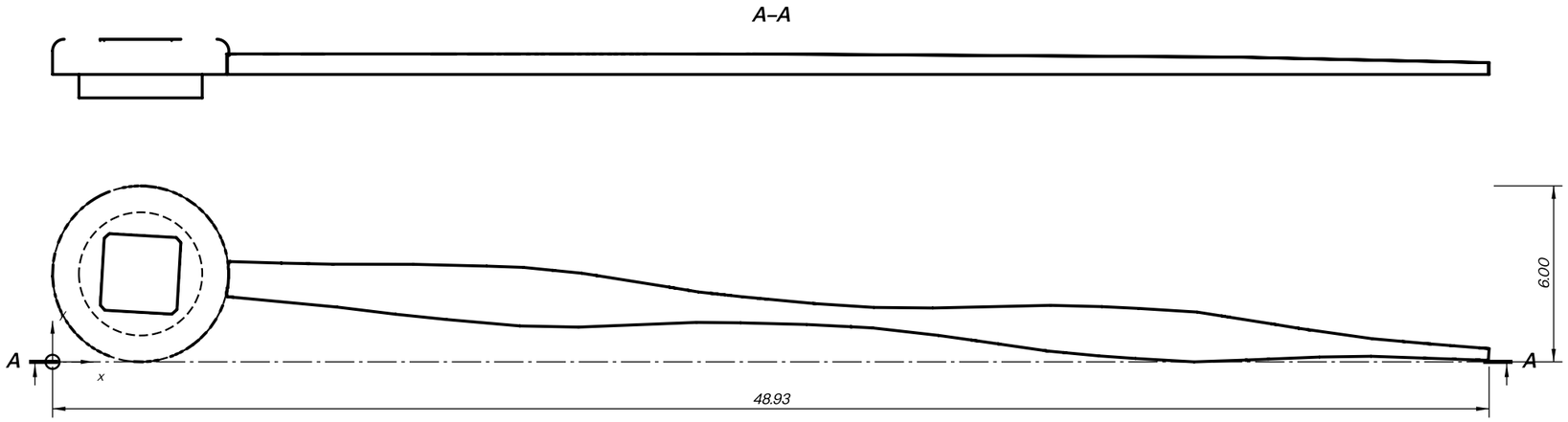
hands

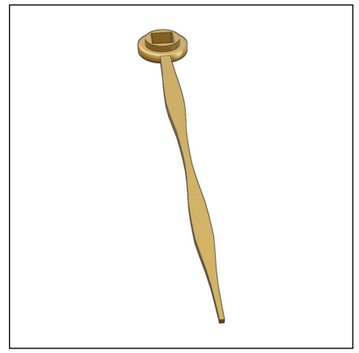
Benennung / Name

Hands

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Goldin" (a gold-coloured brass alloy)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 00-1*** *Teil / Part* *hands.minute*
Behandlung / Treatment *„Oberfläche poliert" (surface polished); surface ▽▽▽▽ (▽▽▽) (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*		*Benennung / Name* ***Minute hand (Minutenzeiger)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Minute hand (Minutenzeiger) — sheet 2

The minute hand: two swellings along its blade, Goldin, polished, on the cannon tube's square.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 00-1	Minutenzeiger (minute hand)	1	„Goldin“ (a gold-coloured brass alloy)	„Oberfläche poliert“ (surface polished); surface ▽▽▽▽ (▽▽▽) (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	not measured on the model: see its function, On the cannon tube's square (no relation here: as the Hamilton's hands, so the centre arbor's endshake isn't sampled through it).

What it does (05 02 00-1). On the cannon tube's square (no relation here: as the Hamilton's hands, so the centre arbor's endshake isn't sampled through it).

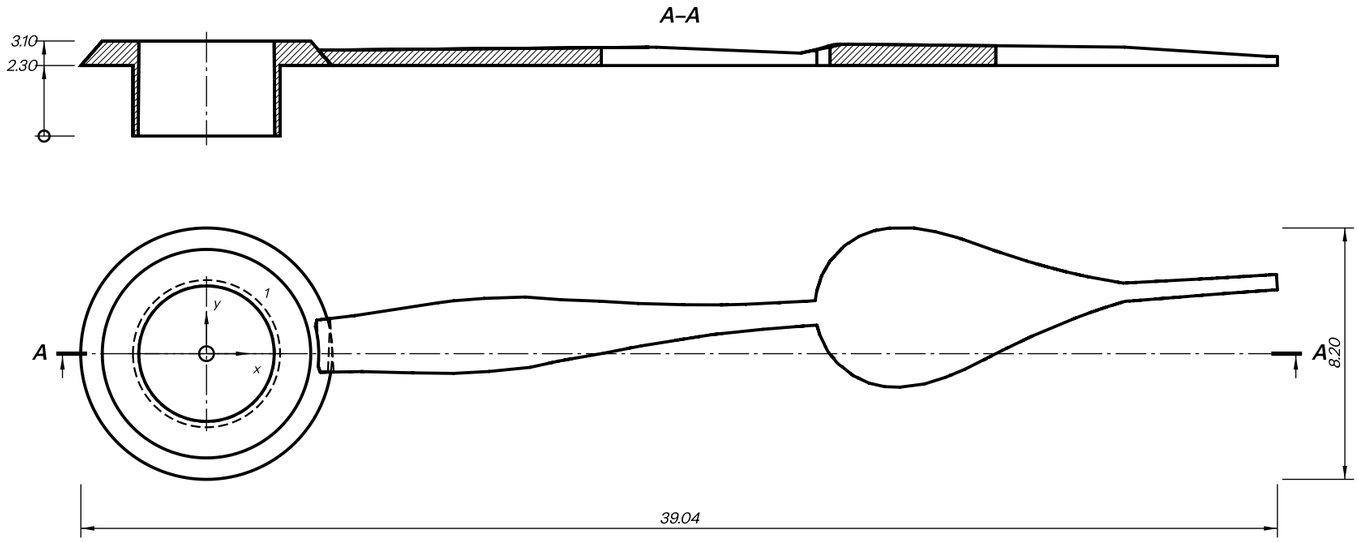
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.00.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.02, 2.66	6.00, 3.09	3.00, 3.00	3.00
Outline	2	line	6.00, 3.09	5.99, 3.27		
Outline	3	line	5.99, 3.27	6.03, 3.42		
Outline	4	line	6.03, 3.42	8.53, 3.35		
Outline	5	line	8.53, 3.35	10.03, 3.32		
Outline	6	line	10.03, 3.32	12.03, 3.33		
Outline	7	line	12.03, 3.33	14.29, 3.28		
Outline	8	line	14.29, 3.28	16.04, 3.22		
Outline	9	line	16.04, 3.22	18.03, 3.02		
Outline	10	line	18.03, 3.02	20.51, 2.63		
Outline	11	line	20.51, 2.63	22.00, 2.38		
Outline	12	line	22.00, 2.38	23.99, 2.16		
Outline	13	line	23.99, 2.16	25.98, 1.98		
Outline	14	line	25.98, 1.98	27.98, 1.85		
Outline	15	line	27.98, 1.85	29.98, 1.84		
Outline	16	line	29.98, 1.84	33.99, 1.93		
Outline	17	line	33.99, 1.93	35.99, 1.88		
Outline	18	line	35.99, 1.88	38.24, 1.75		
Outline	19	line	38.24, 1.75	38.99, 1.70		
Outline	20	line	38.99, 1.70	41.47, 1.31		
Outline	21	line	41.47, 1.31	42.95, 1.06		
Outline	22	line	42.95, 1.06	44.94, 0.79		
Outline	23	line	44.94, 0.79	47.43, 0.57		
Outline	24	line	47.43, 0.57	48.93, 0.45		
Outline	25	line	48.93, 0.45	48.90, 0.05		
Outline	26	line	48.90, 0.05	46.41, 0.15		
Outline	27	line	46.41, 0.15	44.91, 0.19		

Outline	28	line	44.91, 0.19	42.90, 0.16		
Outline	29	line	42.90, 0.16	40.39, 0.05		
Outline	30	line	40.39, 0.05	38.88, 0.00		
Outline	31	line	38.88, 0.00	37.64, 0.07		
Outline	32	line	37.64, 0.07	35.89, 0.18		
Outline	33	line	35.89, 0.18	33.90, 0.37		
Outline	34	line	33.90, 0.37	29.92, 0.94		
Outline	35	line	29.92, 0.94	27.93, 1.16		
Outline	36	line	27.93, 1.16	25.94, 1.27		
Outline	37	line	25.94, 1.27	23.94, 1.33		
Outline	38	line	23.94, 1.33	21.93, 1.34		
Outline	39	line	21.93, 1.34	19.42, 1.24		
Outline	40	line	19.42, 1.24	17.92, 1.18		
Outline	41	line	17.92, 1.18	15.92, 1.22		
Outline	42	line	15.92, 1.22	13.67, 1.42		
Outline	43	line	13.67, 1.42	11.93, 1.59		
Outline	44	line	11.93, 1.59	9.94, 1.84		
Outline	45	line	9.94, 1.84	7.45, 2.09		
Outline	46	line	7.45, 2.09	5.96, 2.22		
Outline	47	arc clockwise	5.96, 2.22	5.96, 2.52	6.37, 2.37	0.43
Outline	48	line	5.96, 2.52	5.92, 2.30		
Outline	49	arc clockwise	5.92, 2.30	0.02, 2.66	3.00, 3.00	3.00
Cut-out 1	50	line	5.97, 2.58	5.99, 2.82		
Cut-out 1	51	line	5.99, 2.82	5.97, 2.58		
Cut-out 2	52	line	5.99, 2.82	6.00, 3.06		
Cut-out 2	53	line	6.00, 3.06	5.99, 2.82		
Cut-out 3	54	arc anticlockwise	1.94, 4.37	1.76, 4.21	2.95, 3.06	1.66
Cut-out 3	55	line	1.76, 4.21	1.62, 1.94		
Cut-out 3	56	arc anticlockwise	1.62, 1.94	1.79, 1.76	2.94, 2.95	1.66
Cut-out 3	57	line	1.79, 1.76	4.06, 1.62		
Cut-out 3	58	arc anticlockwise	4.06, 1.62	4.24, 1.79	3.05, 2.94	1.66
Cut-out 3	59	line	4.24, 1.79	4.37, 4.06		
Cut-out 3	60	arc anticlockwise	4.37, 4.06	4.21, 4.24	3.06, 3.05	1.66
Cut-out 3	61	line	4.21, 4.24	1.94, 4.37		
Cut-out 4	62	line	4.06, 5.45	4.06, 5.45		
Cut-out 4	63	line	4.06, 5.45	4.06, 5.45		
Cut-out 5	64	line	2.68, 5.73	2.68, 5.73		
Cut-out 5	65	line	2.68, 5.73	2.68, 5.73		
Cut-out 5	66	line	2.68, 5.73	2.68, 5.73		
Cut-out 6	67	line	2.10, 0.29	2.10, 0.29		
Cut-out 6	68	line	2.10, 0.29	2.10, 0.29		
Cut-out 7	69	line	0.65, 4.36	0.65, 4.36		
Cut-out 7	70	line	0.65, 4.36	0.65, 4.36		
Cut-out 7	71	line	0.65, 4.36	0.65, 4.36		
Cut-out 7	72	line	0.65, 4.36	0.65, 4.36		





Isometric view, not to scale

Werkstoff / Material *„Goldin"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 00-2***
Behandlung / Treatment *„Oberfläche: hochglanz poliert" (surface high-gloss polished); surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*			*Teil / Part* *hands.hour*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Hour hand (Stundenzeiger)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Hour hand (Stundenzeiger) — sheet 2

The hour hand: a pear-shaped spade on a narrow blade, Goldin, high-gloss polished, on the hour wheel's pipe.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 00-2	Stundenzeiger (hour hand)	1	„Goldin“	„Oberfläche: hochglanz poliert“ (surface high-gloss polished); surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	round 050203-2 (press): 2.698 mm, least gap 0.0111

What it does (05 02 00-2). On the hour wheel's pipe's tapered seat.

Coordinates from the datum, the movement's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.10.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 4.41 through (2.36 thick at 0.74–3.10); sink Ø 4.46 × 1.86 from the bottom

Contour in the plan

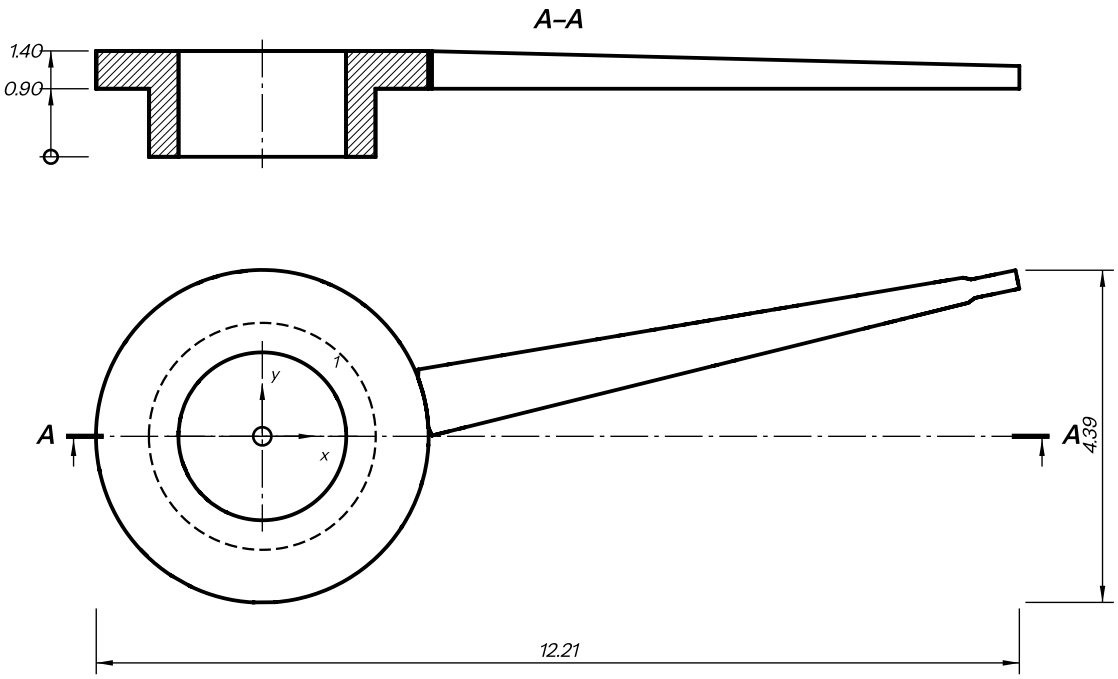
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-4.09, -0.27	3.94, 1.14	0.00, 0.00	4.10
Outline	2	line	3.94, 1.14	4.93, 1.25		
Outline	3	line	4.93, 1.25	6.58, 1.52		
Outline	4	line	6.58, 1.52	7.90, 1.71		
Outline	5	line	7.90, 1.71	8.90, 1.80		
Outline	6	line	8.90, 1.80	10.40, 1.85		
Outline	7	line	10.40, 1.85	11.41, 1.77		
Outline	8	line	11.41, 1.77	12.61, 1.73		
Outline	9	line	12.61, 1.73	13.83, 1.65		
Outline	10	line	13.83, 1.65	15.34, 1.60		
Outline	11	line	15.34, 1.60	16.43, 1.58		
Outline	12	line	16.43, 1.58	18.09, 1.63		
Outline	13	line	18.09, 1.63	19.86, 1.73		
Outline	14	arc clockwise	19.86, 1.73	20.04, 2.46	22.24, 1.52	2.39
Outline	15	arc clockwise	20.04, 2.46	20.26, 2.89	21.17, 2.15	1.17
Outline	16	arc clockwise	20.26, 2.89	20.50, 3.18	20.99, 2.56	0.79
Outline	17	line	20.50, 3.18	20.81, 3.52		
Outline	18	line	20.81, 3.52	21.29, 3.83		
Outline	19	line	21.29, 3.83	21.78, 4.01		
Outline	20	line	21.78, 4.01	22.28, 4.10		
Outline	21	line	22.28, 4.10	23.08, 4.10		
Outline	22	line	23.08, 4.10	23.79, 3.98		
Outline	23	line	23.79, 3.98	24.46, 3.80		
Outline	24	line	24.46, 3.80	25.01, 3.62		
Outline	25	line	25.01, 3.62	25.63, 3.39		

Outline	26	line	25.63, 3.39	26.86, 2.99		
Outline	27	line	26.86, 2.99	27.79, 2.74		
Outline	28	line	27.79, 2.74	28.30, 2.61		
Outline	29	line	28.30, 2.61	28.81, 2.50		
Outline	30	line	28.81, 2.50	29.32, 2.40		
Outline	31	line	29.32, 2.40	29.91, 2.30		
Outline	32	line	29.91, 2.30	33.32, 2.49		
Outline	33	line	33.32, 2.49	34.91, 2.59		
Outline	34	line	34.91, 2.59	34.94, 2.09		
Outline	35	line	34.94, 2.09	32.36, 1.90		
Outline	36	line	32.36, 1.90	29.95, 1.70		
Outline	37	line	29.95, 1.70	29.36, 1.52		
Outline	38	line	29.36, 1.52	28.64, 1.27		
Outline	39	line	28.64, 1.27	28.15, 1.09		
Outline	40	line	28.15, 1.09	27.18, 0.68		
Outline	41	line	27.18, 0.68	26.93, 0.57		
Outline	42	line	26.93, 0.57	25.85, 0.06		
Outline	43	line	25.85, 0.06	25.24, -0.27		
Outline	44	line	25.24, -0.27	24.69, -0.53		
Outline	45	line	24.69, -0.53	24.11, -0.77		
Outline	46	line	24.11, -0.77	23.42, -0.99		
Outline	47	line	23.42, -0.99	22.62, -1.09		
Outline	48	line	22.62, -1.09	22.12, -1.08		
Outline	49	line	22.12, -1.08	21.61, -0.96		
Outline	50	line	21.61, -0.96	21.09, -0.71		
Outline	51	line	21.09, -0.71	20.57, -0.29		
Outline	52	line	20.57, -0.29	20.24, 0.14		
Outline	53	line	20.24, 0.14	20.00, 0.63		
Outline	54	line	20.00, 0.63	19.91, 0.93		
Outline	55	line	19.91, 0.93	17.90, 0.77		
Outline	56	line	17.90, 0.77	16.41, 0.61		
Outline	57	line	16.41, 0.61	14.92, 0.38		
Outline	58	line	14.92, 0.38	13.92, 0.21		
Outline	59	line	13.92, 0.21	12.69, -0.04		
Outline	60	line	12.69, -0.04	11.54, -0.23		
Outline	61	line	11.54, -0.23	10.55, -0.45		
Outline	62	line	10.55, -0.45	9.06, -0.60		
Outline	63	line	9.06, -0.60	8.06, -0.64		
Outline	64	line	8.06, -0.64	6.22, -0.61		
Outline	65	line	6.22, -0.61	4.96, -0.59		
Outline	66	line	4.96, -0.59	4.05, -0.60		
Outline	67	arc clockwise	4.05, -0.60	-4.09, -0.27	0.00, 0.00	4.10
Cut-out 2	68	line	1.23, 3.17	1.23, 3.17		
Cut-out 2	69	line	1.23, 3.17	1.23, 3.17		
Cut-out 2	70	line	1.23, 3.17	1.23, 3.17		
Cut-out 2	71	line	1.23, 3.17	1.23, 3.17		

Levels

Faces in section A-A, from the bottom face: 2.30, 3.10.





Isometric view, not to scale

Werkstoff / Material „Bandstahl 0.6 × 15“	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 00-3*** *Teil / Part* *hands.ud*
Behandlung / Treatment *surface ▽▽▽ (▽▽▽) (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*		*Benennung / Name* ***Up-and-down hand (Auf- und Abzeiger)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Up-and-down hand (Auf- und Abzeiger) — sheet 2

The state-of-wind hand: blued steel, on its wheel's pipe; 0 at full wind, 56 run down.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 00-3	Auf u. Abzeiger (up-and-down hand)	1	„Bandstahl 0,6 × 15“	surface ▽▽▽ (▽▽▽) (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	round 050204-2 (press): 1.397 mm, least gap 0.0089; on 050204-2: 0 mm

What it does (05 02 00-3). On the up-and-down wheel's pipe's seat, down on its shoulder.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 1.40.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 2.22 through

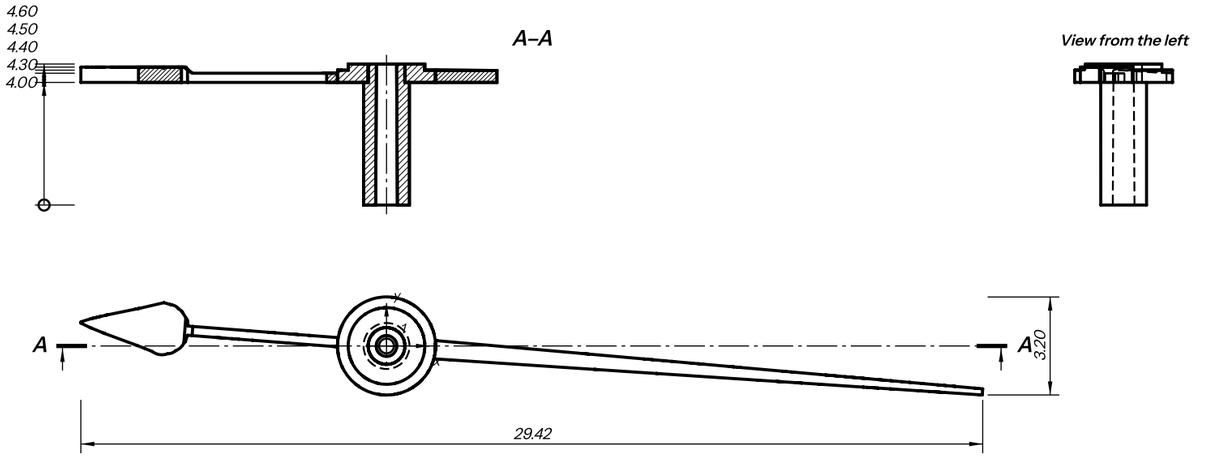
Contour in the plan

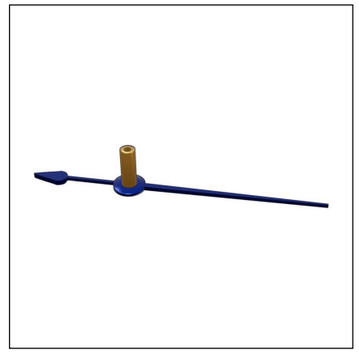
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-2.20, 0.13	2.07, 0.73	0.00, 0.00	2.20
Outline	2	arc clockwise	2.07, 0.73	2.06, 0.88	2.27, 0.82	0.22
Outline	3	line	2.06, 0.88	9.27, 2.10		
Outline	4	arc anticlockwise	9.27, 2.10	9.47, 2.09	9.37, 2.32	0.25
Outline	5	line	9.47, 2.09	9.96, 2.20		
Outline	6	line	9.96, 2.20	10.01, 1.95		
Outline	7	line	10.01, 1.95	9.42, 1.83		
Outline	8	arc clockwise	9.42, 1.83	9.29, 1.75	9.26, 1.96	0.21
Outline	9	line	9.29, 1.75	2.24, 0.01		
Outline	10	arc clockwise	2.24, 0.01	2.19, 0.15	2.41, 0.15	0.22
Outline	11	arc clockwise	2.19, 0.15	-2.20, 0.13	0.00, 0.00	2.20
Cut-out 1	12	line	2.08, 0.71	2.15, 0.45		
Cut-out 1	13	arc anticlockwise	2.15, 0.45	2.08, 0.71	0.20, 0.05	1.99
Cut-out 2	14	line	2.15, 0.45	2.19, 0.23		
Cut-out 2	15	line	2.19, 0.23	2.15, 0.45		

Levels

Faces in section A-A, from the bottom face: 0.90, 1.40.





Isometric view, not to scale

Werkstoff / Material *„Bandstahl 30 × 0,6 Din 1016"*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 01-1*** *Teil / Part* *hands.seconds*
Behandlung / Treatment *surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0.1*		*Benennung / Name* ***Seconds hand (Sekundenzeiger) and collet***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Seconds hand (Sekundenzeiger) and collet — sheet 2

The seconds hand, blued steel with a pointed teardrop counterpoise, on a brass collet (Zeigerfutter) pressed on the seconds arbor's post through the dial.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 01-1	Sekundenzeiger (seconds hand)	1	„Bandstahl 30 × 0,6 Din 1016“	surface ▽▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	round 050201-2 (press): 0.594 mm, least gap 0.0016
05 02 01-2	Zeigerfutter (seconds hand collet)	1	not on the drawing		round 050501-2 (press): 4.504 mm, least gap 0.005

What it does (05 02 01-1). On its collet.

What it does (05 02 01-2). Pressed on the seconds arbor's post through the dial; the hand riveted on its end (no sheet in the set; numbered by the coloured section's label).

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 4.60.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 0.66 × 2.94 deep from the top; sink Ø 0.71 × 2.57 from the bottom

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

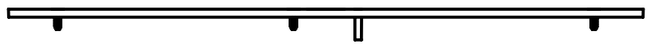
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	-6.49, 0.35	-6.61, 0.09	-6.87, 0.37	0.37
Outline	2	arc clockwise	-6.61, 0.09	-6.88, -0.16	-7.07, 0.32	0.51
Outline	3	arc clockwise	-6.88, -0.16	-7.14, -0.25	-7.14, 0.13	0.38
Outline	4	arc clockwise	-7.14, -0.25	-7.64, -0.18	-7.33, 0.23	0.51
Outline	5	line	-7.64, -0.18	-8.03, -0.02		
Outline	6	line	-8.03, -0.02	-8.61, 0.18		
Outline	7	line	-8.61, 0.18	-9.68, 0.61		
Outline	8	line	-9.68, 0.61	-9.97, 0.75		
Outline	9	line	-9.97, 0.75	-9.97, 0.79		
Outline	10	line	-9.97, 0.79	-9.66, 0.88		
Outline	11	line	-9.66, 0.88	-8.79, 1.08		
Outline	12	arc clockwise	-8.79, 1.08	-8.28, 1.19	-6.87, -7.12	8.42
Outline	13	line	-8.28, 1.19	-7.93, 1.25		
Outline	14	line	-7.93, 1.25	-7.26, 1.41		
Outline	15	line	-7.26, 1.41	-6.92, 1.32		
Outline	16	line	-6.92, 1.32	-6.68, 1.14		
Outline	17	line	-6.68, 1.14	-6.50, 0.85		
Outline	18	line	-6.50, 0.85	-6.47, 0.65		
Outline	19	line	-6.47, 0.65	-1.58, 0.27		
Outline	20	line	-1.58, 0.27	-1.61, -0.03		

Outline	21	line	-1.61, -0.03	-6.49, 0.35		
Outline	22	arc clockwise	-1.60, 0.12	1.60, 0.09	0.00, 0.00	1.60
Outline	23	line	1.60, 0.09	1.62, 0.18		
Outline	24	line	1.62, 0.18	19.45, -1.40		
Outline	25	line	19.45, -1.40	19.43, -1.60		
Outline	26	line	19.43, -1.60	1.57, -0.42		
Outline	27	line	1.57, -0.42	1.56, -0.33		
Outline	28	arc clockwise	1.56, -0.33	-1.60, 0.12	0.00, 0.00	1.60
Cut-out 1	29	line	1.57, -0.32	1.60, -0.12		
Cut-out 1	30	line	1.60, -0.12	1.57, -0.32		
Cut-out 2	31	line	1.60, -0.12	1.60, 0.08		
Cut-out 2	32	line	1.60, 0.08	1.60, -0.12		
Cut-out 4	33	line	-0.51, -0.66	-0.51, -0.66		
Cut-out 4	34	line	-0.51, -0.66	-0.51, -0.66		
Cut-out 4	35	line	-0.51, -0.66	-0.51, -0.66		
Cut-out 4	36	line	-0.51, -0.66	-0.51, -0.66		
Cut-out 5	37	line	-1.18, 0.42	-1.18, 0.42		
Cut-out 5	38	line	-1.18, 0.42	-1.18, 0.42		
Cut-out 5	39	line	-1.18, 0.42	-1.18, 0.42		
Cut-out 5	40	line	-1.18, 0.42	-1.18, 0.42		

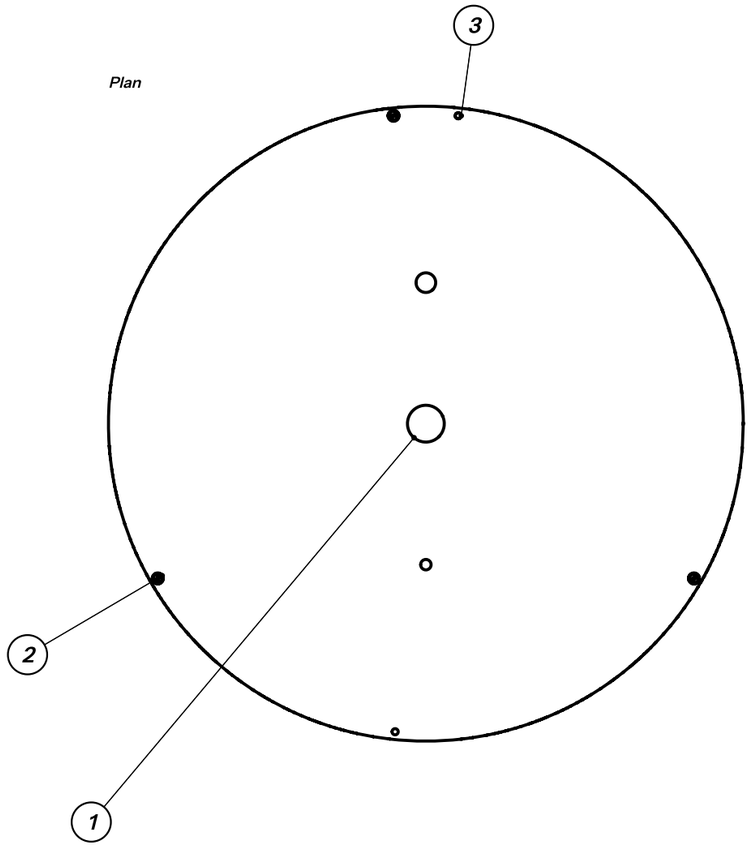
Levels

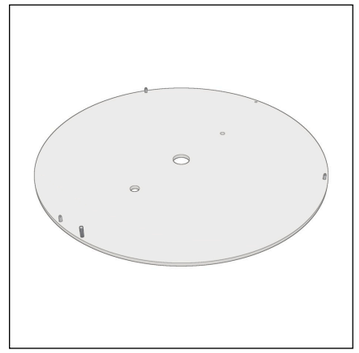
Faces in section A-A, from the bottom face: 4.00, 4.30, 4.40, 4.50, 4.60.

Front view



Plan





Isometric view, not to scale

3	1	Dial pin	dialPin	not on the drawing
2	3	Dial screws	WN-10	not on the drawing
1	1	Dial	05 02 00-4	„Messingblech 1,3 Din 1751 Ms
Pos.	**Units**	**Benennung / Name**	**Zeichnung / Drawing**	**Werkstoff / Material**

Werkstoff / Material

„Messingblech 1,3 Din 1751 Ms 58" (1,3 struck, pencil "1,2")

Maßstab / Scale

1:1

Behandlung / Treatment

„Eisenblech 1,3 gebondert" (iron sheet 1,3, bonderized); „weiß lackiert" (struck through: white lacquered); blank (Rohmaß) 1,3 × 110 × 110

Blatt / Sheet

1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.

05 02 00-4

Teil / Part

dial

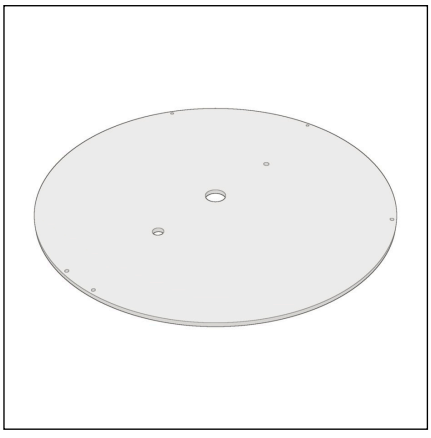
Benennung / Name

Dial (Zifferblatt)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material „Messingblech 1,3 Din 1751 Ms 58" (1,3 struck, pencil "1,2")	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 02 00-4*** *Teil / Part* *dial.dial*
Behandlung / Treatment „Eisenblech 1,3 gebondert" (iron sheet 1.3, bonderized); „weiß lackiert" (struck through: white lacquered); blank (Rohmaß) 1,3 × 110 × 110		*Benennung / Name* ***Dial***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Dial — sheet 2

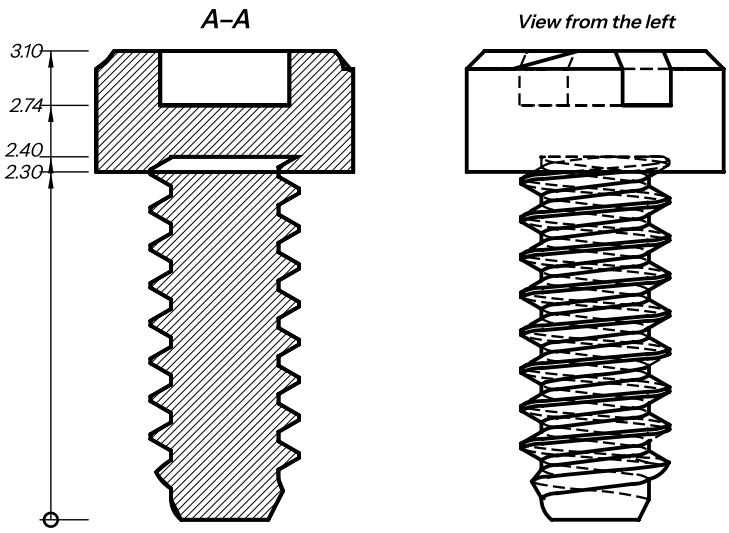
The dial, white with its figures etched and filled black, printed as on Wempe No. 3955: the minute track and the hours, the up-and-down sub-dial at 12 (0 to 56 hours since winding) and the seconds sub-dial at 6 with the dial's number, CHRONOMETERWERKE WEMPE HAMBURG and the maker's anchor. It lies in the recess of the movement ring's flange, placed by a pin and held by three screws at the 20, 40 and 59 minute marks.

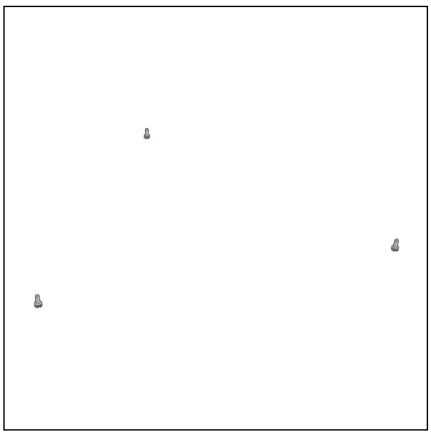
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 02 00-4	Zifferblatt (dial)	1	„Messingblech 1,3 Din 1751 Ms 58“ (1,3 struck, pencil "1,2")	„Eisenblech 1,3 gebondert“ (iron sheet 1.3, bonderized); „weiß lackiert“ (struck through: white lacquered); blank (Rohmaß) 1,3 × 110 × 110	on 050800-3: 0 mm

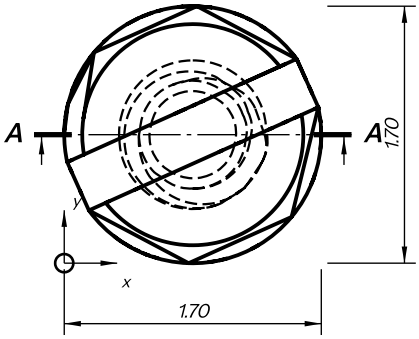
What it does (05 02 00-4). In the movement ring's recess, placed by its pin, held by three screws.

Turned about its axis; length 1.30, largest Ø 103.55. Diameters: Ø 103.55; bores Ø 6.00. Shoulders from the lower end: 0.00, 1.30.





Isometric view, not to scale



3 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-10** *Teil / Part* *dial.screws*
Behandlung / Treatment		*Benennung / Name* ***Dial screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Dial screws — sheet 2

Three small screws through the dial’s sunk holes into the movement ring’s flange, at the 20, 40 and 59 minute marks.

3 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-10	Schraube (dial screw)	3	not on the drawing		in 050800-3 (tap): 2.017 mm, least gap 0.0212; in 050200-4 (clear): 1.082 mm, least gap 0.0454

What it does (WN-10). M1 through the dial's sunk holes into the ring's flange (WN-10 by the coloured section's label, likely).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.10.

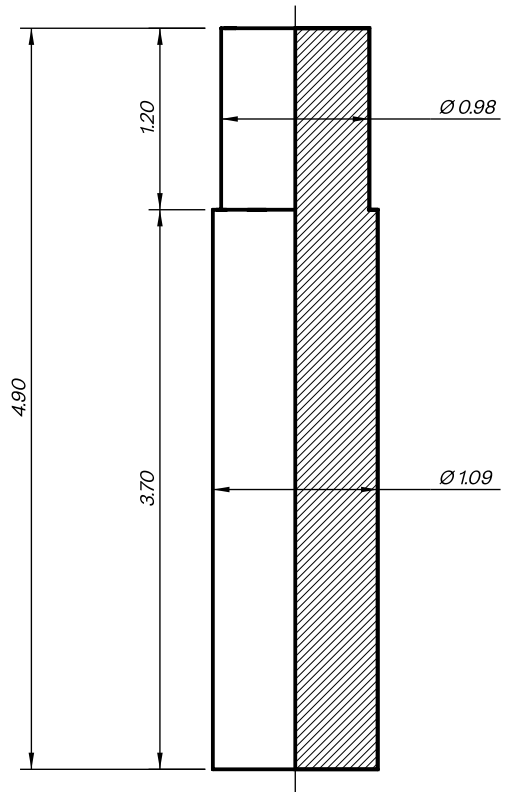
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			0.85, 0.85	0.85

Levels

Faces in section A-A, from the bottom face: 2.30, 2.40, 2.74, 3.10.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***dialPin***
Behandlung / Treatment			*Teil / Part* *dial.pin*
		Benennung / Name ***Dial pin***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Dial pin — sheet 2

The pin at 6° past 12 that places the dial in the ring: in the ring’s hole, stepped to the dial’s.

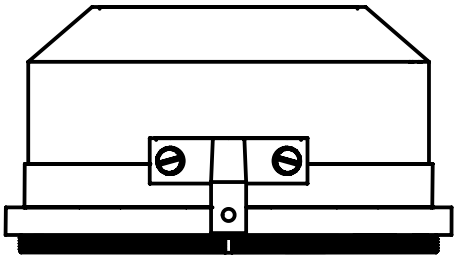
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
dialPin	Stift (dial pin)	1	not on the drawing		in 050800-3 (press): 3.724 mm, least gap 0.0048; in 050200-4 (press): 1.176 mm, least gap 0.0099

What it does (dialPin). In the ring's pin hole at 6 deg past 12 (05 08 00-3), stepped into the dial's (no number read).

Turned about its axis; length 4.90, largest Ø 1.09. Diameters: Ø 0.98, Ø 1.09. Shoulders from the lower end: 0.00, 3.70, 4.90.

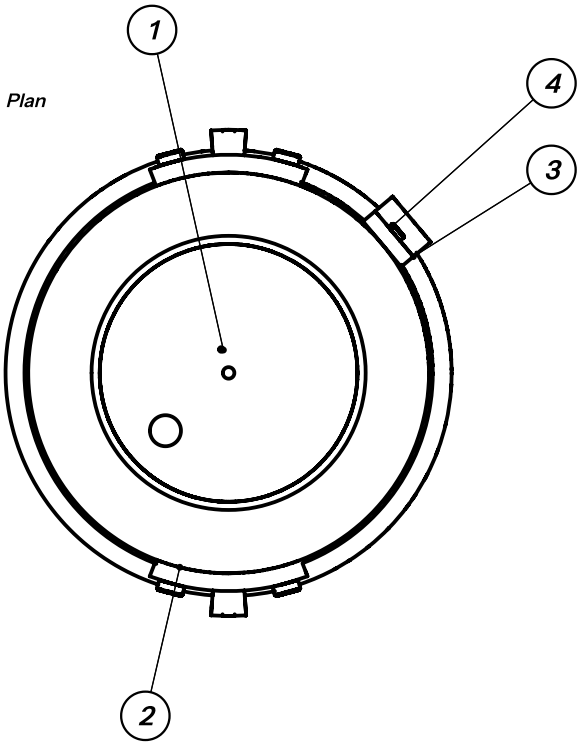
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	5	Bracket screws (WN-78)	WN-78	not on the drawing
3	1	Locking bracket (Feststellkloben)	05 01 00-4	not on the drawing
2	2	Case brackets (Gehäusekloben)	05 01 00-3	not on the drawing
1	1	Bowl	R-05 01 00-1	„Ms-Sandguß (Kokillenguß)“

Werkstoff / Material
„Ms-Sandguß (Kokillenguß)“

Maßstab / Scale
1:2

Behandlung / Treatment
none on the drawing

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

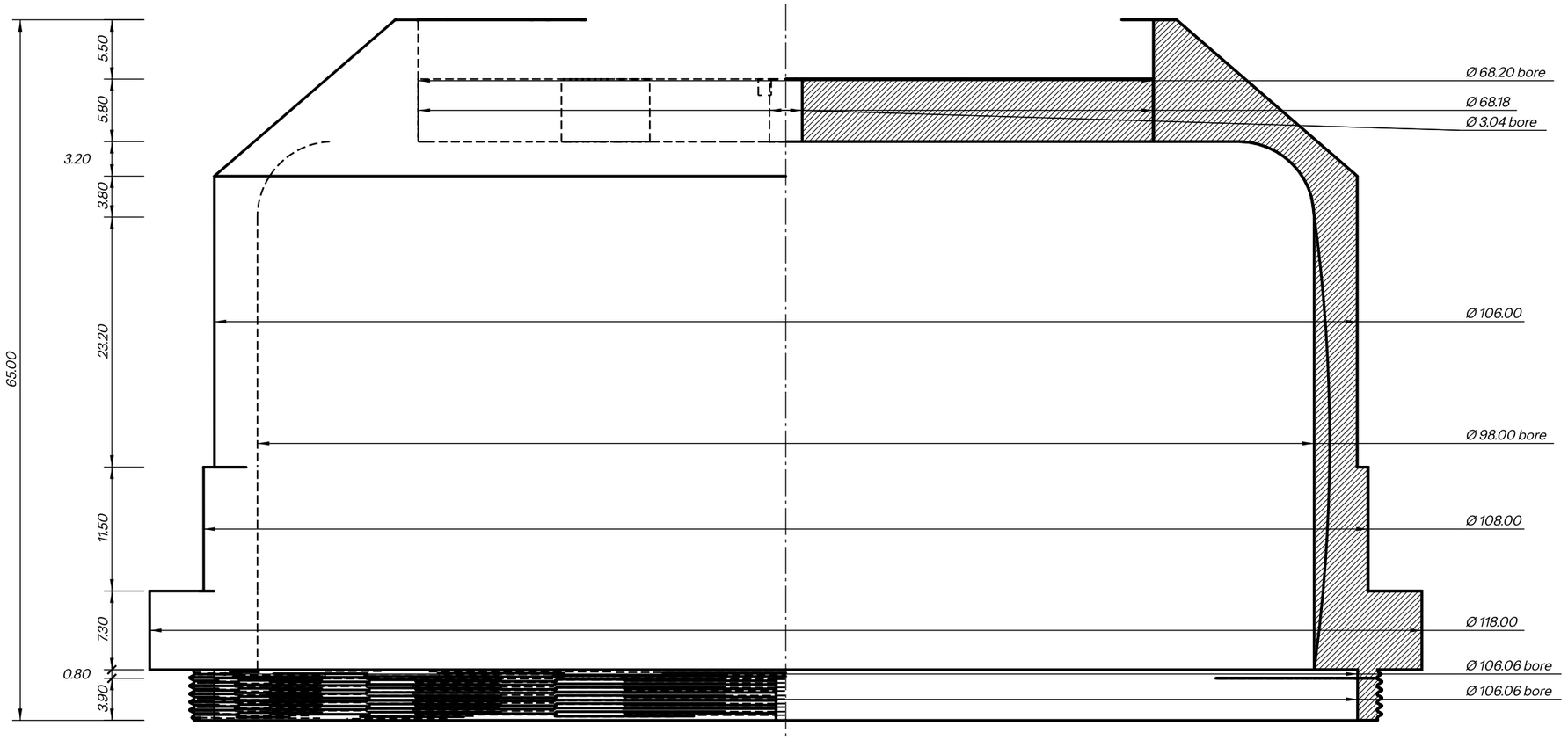
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
R-05 01 00-1

Teil / Part
bowl

Benennung / Name
Bowl (Gehäusetopf)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Ms-Sandguß (Kokillenguß)"*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***R-05 01 00-1***
Behandlung / Treatment *none on the drawing*			*Teil / Part* *bowl.bowl*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Bowl***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Bowl — sheet 2

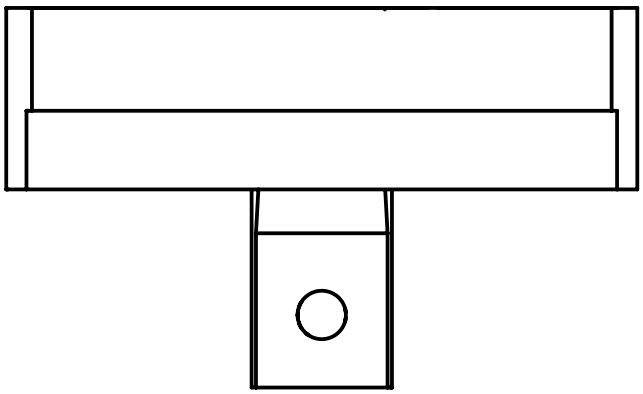
The case the movement hangs in: a cast brass cup, a flange at its top, its body narrowing in a 40° cone to its foot. Its rim is turned down to a fine thread, which the bezel screws onto; the movement ring's flange lies in a rebate inside the rim, the ring in its bore, and a pin in the ring sits in a notch cut through the rim, setting the movement's turn, so the winding hole in its bottom lies over the fusee's square, and holding it while the bezel is screwed down. The thread isn't on the casting's sheet: the bezel's sheet marks its half, and the 6MX's bowls show it. Two brackets on its side carry the inner pivots, slid along slots until it hangs level ("Kloben verstellen bis Gleichgewicht erreicht ist"); a third, cranked, is where the transit lock holds it. Its sheet says a brass casting; the wartime Glashütte bowls were Bakelite, with a heavy weight in the bottom from No. 5100, and the 1942 arrangement names a Buna (synthetic rubber) case; the 6MX's is brass. The sheet's weight, 350 g, is under a third of this bowl's in brass.

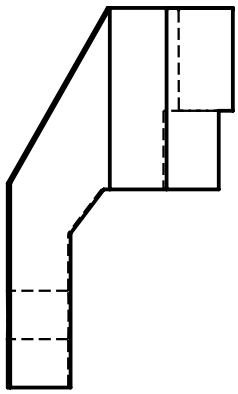
Parts list

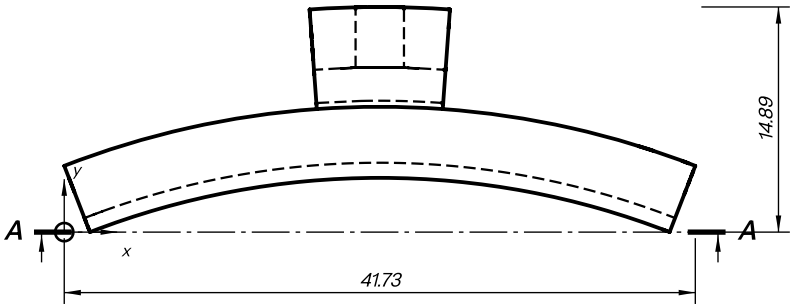
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
R-05 01 00-1	Gehäusetopf (bowl)	1	„Ms-Sandguß (Kokillenguß)“	none on the drawing	takes Schraube WN-71 (7 on 05 01, the number unsure): the closing plate's screw: in it (tap): 5.527 mm, least gap 0.0243; Glasrand (bezel): on it: 0.0124 mm; Gehäusekloben (case bracket): on it: 0 mm; Feststellkloben (locking bracket): on it: 0 mm; Verschlußplatte (closing plate): on it: 0 mm

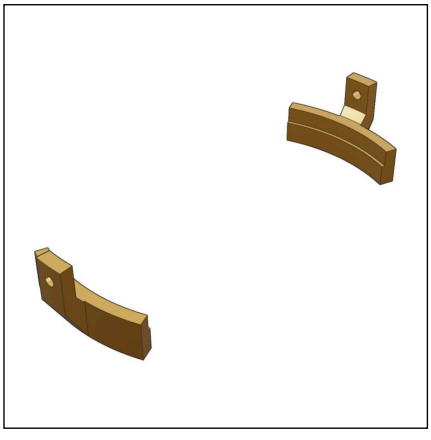
What it does (R-05 01 00-1). The cast brass bowl: the movement ring's flange in a rebate in its rim, the rim threaded outside for the bezel and notched for the ring's pin; the winding hole in its bottom over the fusee's square.

Turned about its axis; length 65.00, largest Ø 118.00. Diameters: Ø 68.18, Ø 106.00, Ø 108.00, Ø 118.00; bores Ø 3.04, Ø 68.20, Ø 98.00, Ø 106.06, Ø 106.06. Shoulders from the lower end: 0.00, 3.90, 4.70, 12.00, 23.50, 46.70, 50.50, 53.70, 59.50, 65.00.









Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 01 00-3** *Teil / Part* bowl.brackets
Behandlung / Treatment		*Benennung / Name* ***Case brackets (Gehäusekloben)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Case brackets (Gehäusekloben) — sheet 2

Two brass brackets on the bowl at 12 and 6 o'clock, each a plate screwed to the bowl through two slots and a block standing out to the ring with the inner pivot's hole: slid along the bowl until it hangs level.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 00-3	Gehäusekloben (case bracket)	2	not on the drawing		on 050100-1: 0 mm

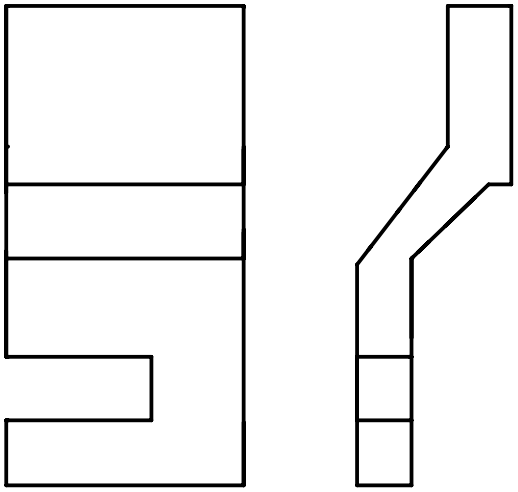
What it does (05 01 00-3). On the bowl at 12 and 6 o'clock, carrying the inner pivots; slid along its slots to level the bowl.

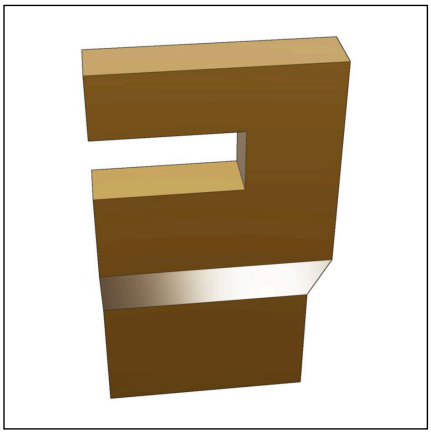
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 25.10.

Contour in the plan

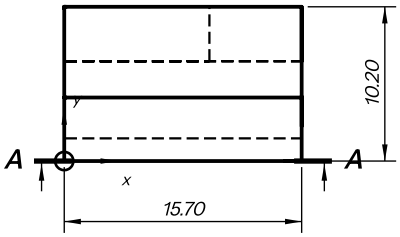
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc anticlockwise	7.96, 1.99	1.70, 0.00	20.87, -49.41	53.00
Outline	2	line	1.70, 0.00	0.00, 4.38		
Outline	3	arc clockwise	0.00, 4.38	16.70, 8.14	20.87, -49.41	57.70
Outline	4	line	16.70, 8.14	16.23, 14.72		
Outline	5	arc clockwise	16.23, 14.72	19.15, 14.86	20.75, -47.42	62.31
Outline	6	line	19.15, 14.86	20.48, 14.89		
Outline	7	line	20.48, 14.89	21.83, 14.88		
Outline	8	line	21.83, 14.88	22.94, 14.85		
Outline	9	arc clockwise	22.94, 14.85	25.51, 14.72	20.87, -49.41	64.30
Outline	10	line	25.51, 14.72	25.03, 8.14		
Outline	11	arc clockwise	25.03, 8.14	41.73, 4.38	20.87, -49.41	57.70
Outline	12	line	41.73, 4.38	40.03, 0.00		
Outline	13	arc anticlockwise	40.03, 0.00	7.96, 1.99	20.87, -49.41	53.00





Isometric view, not to scale



Werkstoff / Material not on the drawing	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 01 00-4** *Teil / Part* *bowl.lockBracket*
Behandlung / Treatment		*Benennung / Name* ***Locking bracket (Feststellkloben)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Locking bracket (Feststellkloben) — sheet 2

A cranked brass bracket on the bowl at 50° from 6 o'clock: a foot screwed to the body, an arm leaning out to an upright, and across the upright at the pivots' height a slot open toward 6 o'clock, which the transit lock's lever turns into until it stops against the slot's end.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 00-4	Feststellkloben (locking bracket)	1	not on the drawing		on 050100-1: 0 mm

What it does (05 01 00-4). On the bowl at 50 deg from 6 o'clock: the transit lock presses the ring onto it.

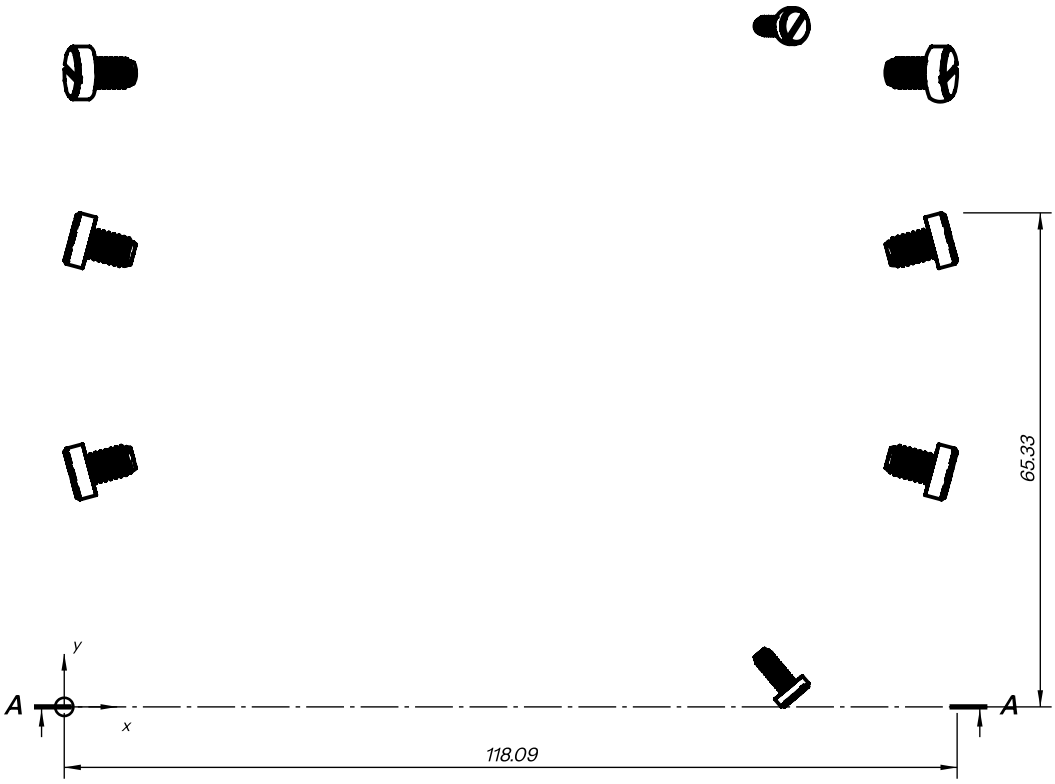
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 31.70.

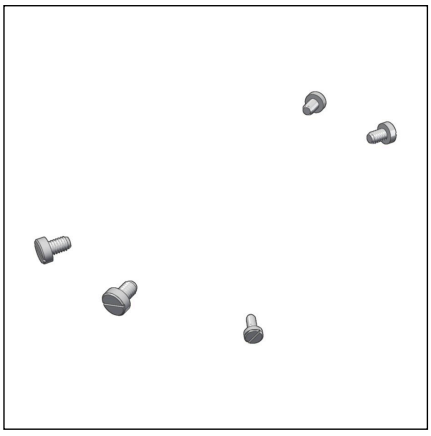
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 10.20	15.70, 10.20		
Outline	2	line	15.70, 10.20	15.70, 0.00		
Outline	3	line	15.70, 0.00	0.00, 0.00		
Outline	4	line	0.00, 0.00	0.00, 10.20		

A–A





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **1:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-78** Teil / Part bowl.screws
Behandlung / Treatment		Benennung / Name **Bracket screws (WN-78)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Bracket screws (WN-78) — sheet 2

The screws holding the brackets to the bowl.

Parts list

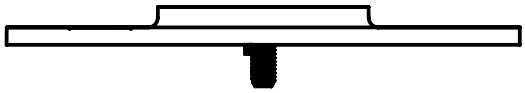
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-78	Schraube WN-78 (13 on 05 01): the brackets' screws	5	not on the drawing		not measured on the model: see its function, Hold the case brackets and the locking bracket to the bowl (their holes across the turned bowl not cut).

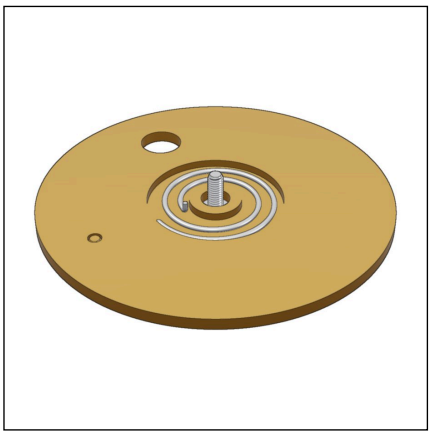
What it does (WN-78). Hold the case brackets and the locking bracket to the bowl (their holes across the turned bowl not cut).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 12.10.

The contour has 417 elements (its teeth): the drawing and the tooth data define it.

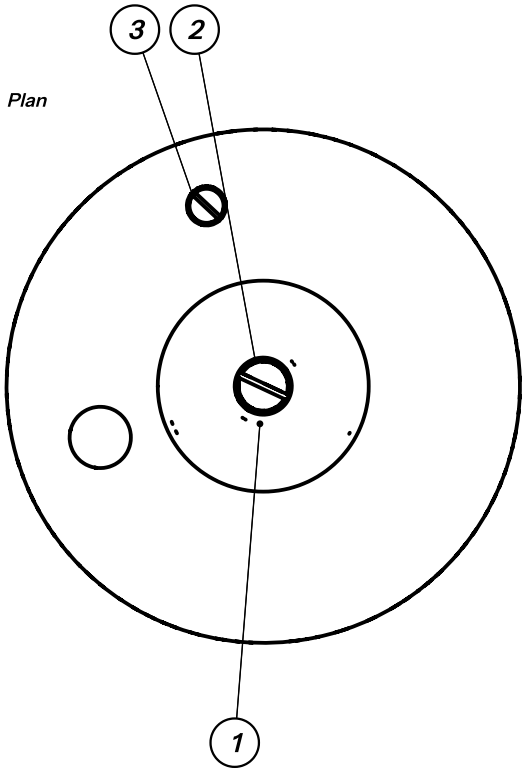
Front view





Isometric view, not to scale

Plan



Not seen in either view (inside the assembly): Pos. 4.

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Closing spring (Verschlußfeder)	05 01 00-7	not on the drawing
3	1	Closing plate knob	WN-72	not on the drawing
2	1	Closing plate screw	WN-71	not on the drawing
1	1	Closing plate	05 01 00-2	„Ms.“ (brass; sure)

Werkstoff / Material
„Ms.“ (brass; sure)

Maßstab / Scale
1:1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
05 01 00-2

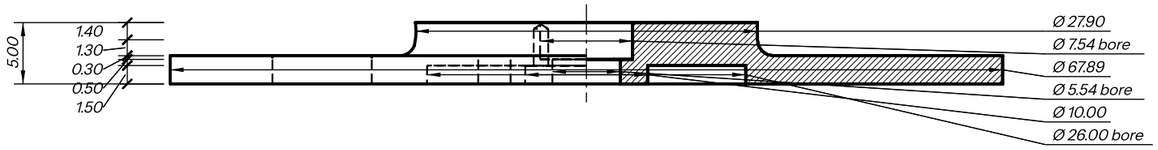
Teil / Part
closingPlate

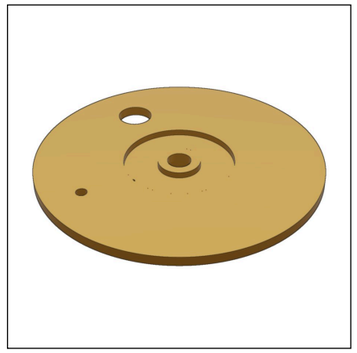
Behandlung / Treatment
surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1

Benennung / Name
Closing plate (Verschlußplatte)

Blatt / Sheet
1 / 1

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale

Werkstoff / Material *„Ms." (brass; sure)*	*Maßstab / Scale* ***2:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 01 00-2***
Behandlung / Treatment *surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1*			*Teil / Part* *closingPlate.plate*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Closing plate***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Closing plate — sheet 2

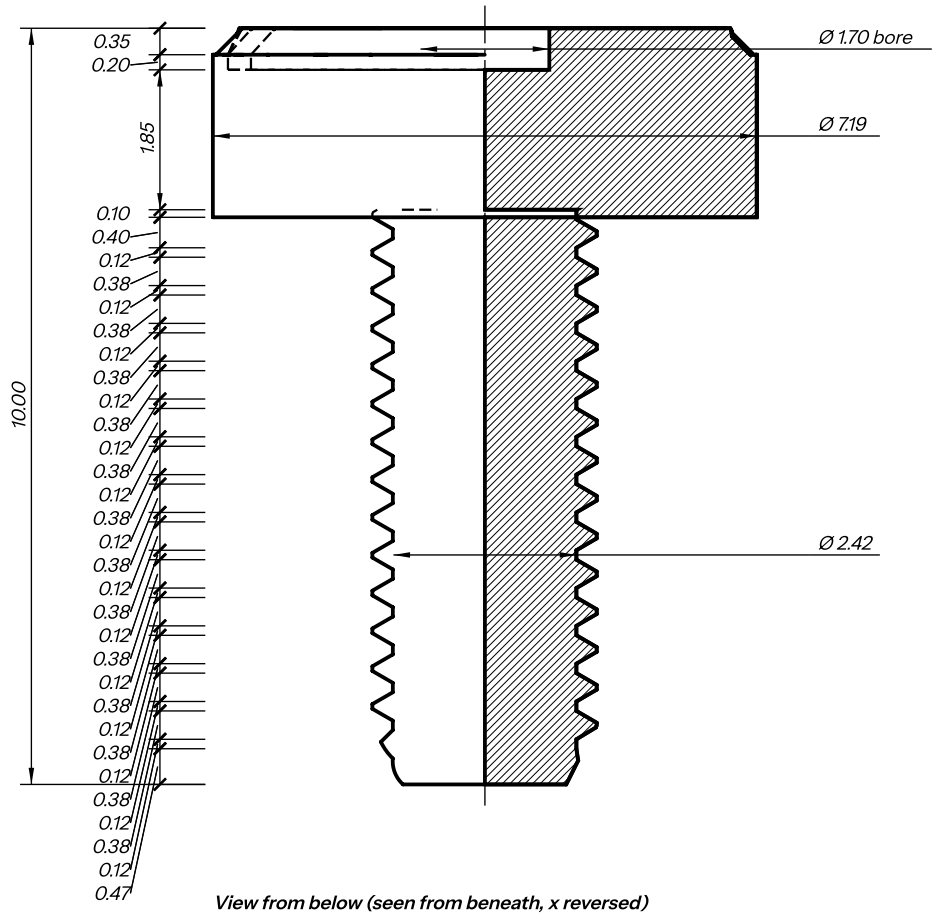
A brass disc in the bowl's foot, turning on a screw in the bottom: turned by its knob, its hole comes over the winding hole for the key, and away from it to shut it: shut at rest, opened when it is wound in its case. A spring of wire, a flat spiral in the groove on its inner face between one end in the plate and the other in the bowl's bottom, turns it back shut when the key comes out (Verschlußfeder 05 01 00-7: no sheet, estimated from the plate's).

Parts list

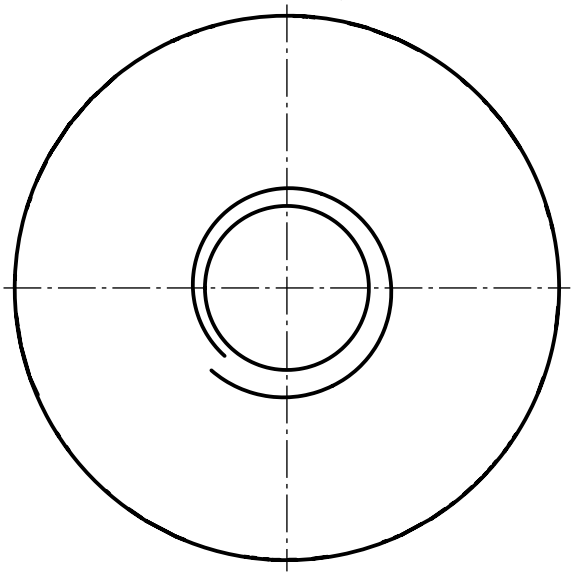
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 00-2	Verschlußplatte (closing plate)	1	„Ms." (brass; sure)	surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished); untoleranced sizes ±0,1	on 050100-1: 0 mm

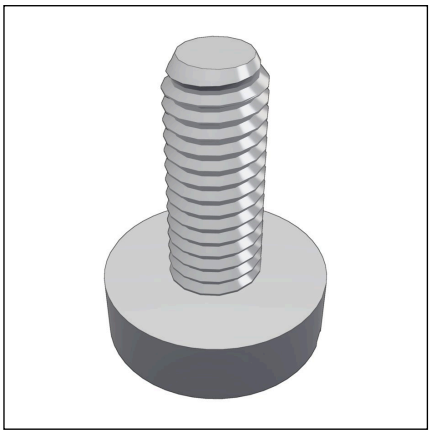
What it does (05 01 00-2). In the bowl's foot, turning on its screw to open and shut the winding hole.

Turned about its axis; length 5.00, largest Ø 67.90. Diameters: Ø 10.00, Ø 27.90, Ø 67.89; bores Ø 5.54, Ø 7.54, Ø 26.00. Shoulders from the lower end: 0.00, 1.50, 2.00, 2.30, 3.60, 5.00.



View from below (seen from beneath, x reversed)





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-71** *Teil / Part* *closingPlate.screw*
Behandlung / Treatment		*Benennung / Name* ***Closing plate screw***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Closing plate screw — sheet 2

The screw in the bowl's bottom the plate turns on, its head in the plate's counterbore.

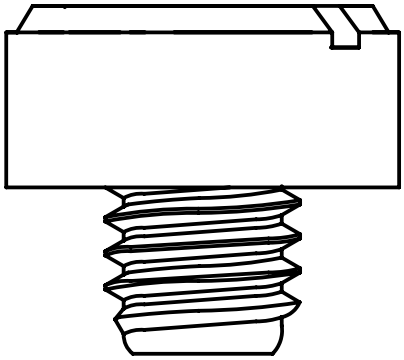
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-71	Schraube WN-71 (7 on 05 01, the number unsure): the closing plate's screw	1	not on the drawing		in 050100-1 (tap): 5.527 mm, least gap 0.0243

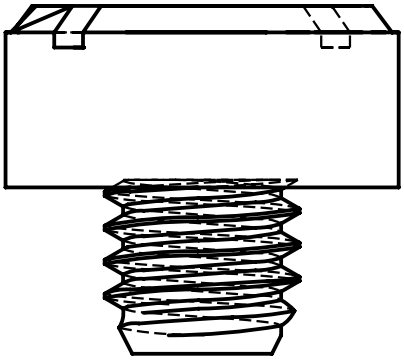
What it does (WN-71). The screw in the bowl's bottom the closing plate turns on.

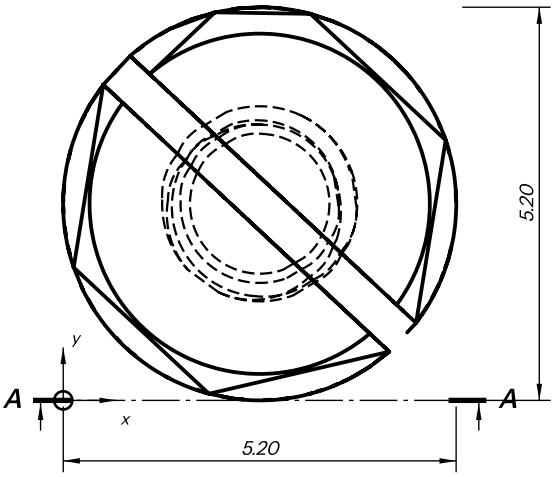
Turned about its axis; length 10.00, largest Ø 7.20. Diameters: Ø 2.42, Ø 7.19; bores Ø 1.70. Shoulders from the lower end: 0.00, 0.47, 0.60, 0.97, 1.10, 1.47, 1.60, 1.97, 2.10, 2.47, 2.60, 2.97, 3.10, 3.47, 3.60, 3.97, 4.10, 4.47, 4.60, 4.97, 5.10, 5.47, 5.60, 5.97, 6.10, 6.47, 6.60, 6.97, 7.10, 7.50, 7.60, 9.45, 9.65, 10.00.

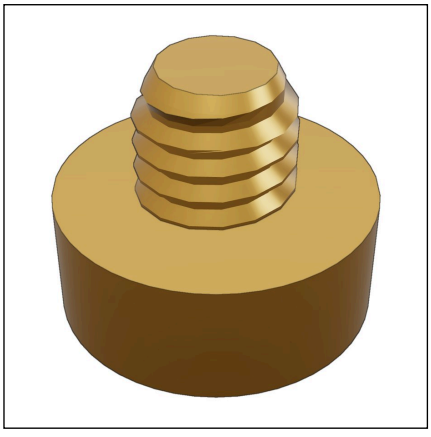
A–A



View from the left







Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **10:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-72** Teil / Part closingPlate.knob
Behandlung / Treatment		Benennung / Name **Closing plate knob**	
Blatt / Sheet 1 / 2		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Closing plate knob — sheet 2

The brass knob in the plate's M2.6, to turn it.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-72	Schraube WN-72: the closing plate's knob	1	not on the drawing		in 050100-2 (tap): 2.2 mm, least gap 0.0193

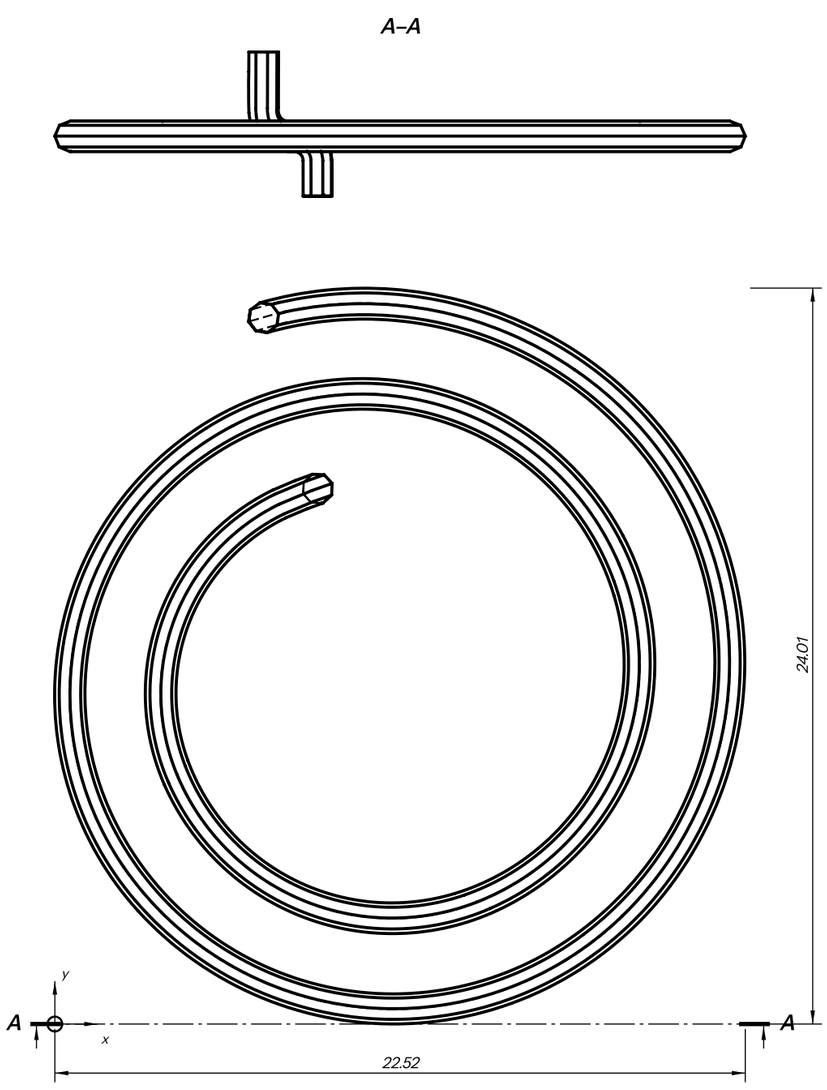
What it does (WN-72). The knob in the closing plate's M2.6, to turn it.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 4.60.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			2.60, 2.60	2.60
Cut-out 1	2	line	3.84, 0.72	3.14, 1.45		
Cut-out 1	3	line	3.14, 1.45	3.84, 0.72		





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 01 00-7**
Behandlung / Treatment			*Teil / Part* *closingPlate.spring*
		Benennung / Name ***Closing spring (Verschlußfeder)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Closing spring (Verschlußfeder) — sheet 2

A steel wire wound as a flat spiral of two turns in the groove on the plate's inner face, closed by the bowl's bottom: its outer end turned down into the plate's Ø 1.2, its inner end up into a hole in the bowl's bottom. Unloaded with the plate shut; turned 25° open it winds up, and turns the plate back when the key comes out.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 00-7	Verschlußfeder (closing spring)	1	not on the drawing		not measured on the model: see its function, Holds the closing plate shut over the winding hole and turns it back when the key is out: a wire spiral in the plate's groove, one end in the plate's Ø 1.2, the other in the bowl's bottom (no sheet: estimated from the plate's; its ends sit free in their holes, so no relation is measured).

What it does (05 01 00-7). Holds the closing plate shut over the winding hole and turns it back when the key is out: a wire spiral in the plate's groove, one end in the plate's Ø 1.2, the other in the bowl's bottom (no sheet: estimated from the plate's; its ends sit free in their holes, so no relation is measured).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 4.70.

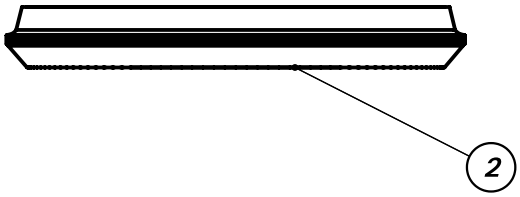
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.10, 9.37	7.39, 20.68	10.25, 10.84	10.25
Outline	2	arc clockwise	7.39, 20.68	18.81, 15.35	10.07, 11.47	9.57
Outline	3	arc clockwise	18.81, 15.35	15.28, 4.15	10.67, 11.74	8.89
Outline	4	arc clockwise	15.28, 4.15	4.68, 5.96	11.01, 11.18	8.22
Outline	5	arc clockwise	4.68, 5.96	4.96, 15.86	10.51, 10.77	7.54
Outline	6	arc clockwise	4.96, 15.86	7.59, 17.62	10.11, 11.02	7.07
Outline	7	line	7.59, 17.62	7.98, 17.79		
Outline	8	line	7.98, 17.79	8.40, 17.93		
Outline	9	line	8.40, 17.93	8.79, 17.92		
Outline	10	line	8.79, 17.92	9.05, 17.64		
Outline	11	line	9.05, 17.64	9.03, 17.25		
Outline	12	line	9.03, 17.25	8.74, 16.99		
Outline	13	arc anticlockwise	8.74, 16.99	8.10, 16.75	10.48, 11.40	5.86
Outline	14	line	8.10, 16.75	7.66, 16.57		
Outline	15	arc anticlockwise	7.66, 16.57	4.25, 8.87	10.32, 10.81	6.36
Outline	16	arc anticlockwise	4.25, 8.87	12.19, 4.06	10.95, 11.00	7.04
Outline	17	arc anticlockwise	12.19, 4.06	18.55, 11.97	10.83, 11.65	7.72
Outline	18	arc anticlockwise	18.55, 11.97	11.01, 19.99	10.18, 11.62	8.39
Outline	19	arc anticlockwise	11.01, 19.99	1.26, 13.02	10.12, 10.97	9.07
Outline	20	arc anticlockwise	1.26, 13.02	7.31, 1.69	10.75, 10.82	9.75
Outline	21	arc anticlockwise	7.31, 1.69	20.16, 6.45	10.99, 11.43	10.43
Outline	22	arc anticlockwise	20.16, 6.45	16.92, 20.77	10.41, 11.75	11.11
Outline	23	arc anticlockwise	16.92, 20.77	7.33, 22.68	10.06, 11.38	11.63
Outline	24	line	7.33, 22.68	6.93, 22.56		

Outline	25	line	6.93, 22.56	6.55, 22.61		
Outline	26	line	6.55, 22.61	6.32, 22.91		
Outline	27	line	6.32, 22.91	6.37, 23.29		
Outline	28	line	6.37, 23.29	6.69, 23.53		
Outline	29	line	6.69, 23.53	7.22, 23.68		
Outline	30	arc clockwise	7.22, 23.68	21.81, 15.83	10.08, 11.50	12.51
Outline	31	arc clockwise	21.81, 15.83	16.23, 1.28	10.70, 11.73	11.83
Outline	32	arc clockwise	16.23, 1.28	1.91, 4.71	11.00, 11.15	11.15
Outline	33	arc clockwise	1.91, 4.71	0.10, 9.37	10.69, 10.82	10.70

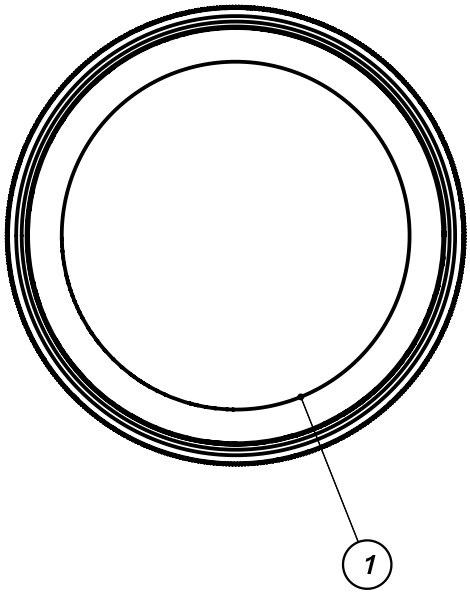
Front view





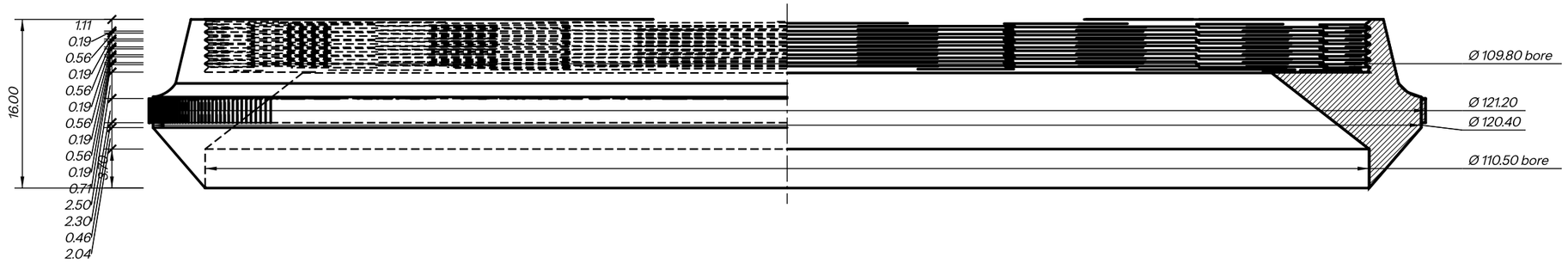
Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
2	1	Glass	05 01 01-1	not on the drawing
1	1	Bezel (Glasrand)	R-05 01 01-2	„Ms.“

Werkstoff / Material not on the drawing	Maßstab / Scale **1:2**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	Zeichnung Nr. / Drawing no. ***05 01 01-1*** Teil / Part *bezel*
Behandlung / Treatment		Benennung / Name ***Bezel and glass (Glasrand)***	
Blatt / Sheet **1 / 1**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	





Isometric view, not to scale

Werkstoff / Material „Ms."	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **R-05 01 01-2**
Behandlung / Treatment none on the drawing			Teil / Part bezel.ring
Blatt / Sheet 1 / 2		Benennung / Name **Bezel (Glasrand)**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Bezel (Glasrand) — sheet 2

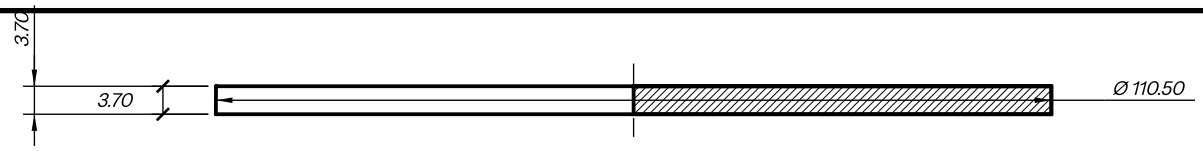
The brass ring that closes the bowl, machined from its casting: a fine thread inside its skirt screws it onto the bowl's threaded rim round the movement ring's flange until its step bears on the dial, holding the movement down, and a cone inside it slopes up and out from the dial to the glass, which sits in its top edge, 16.7 mm over the pillar plate's dial face: the motion work's sheet asks 17.0 (corrected in pencil from 15.8).

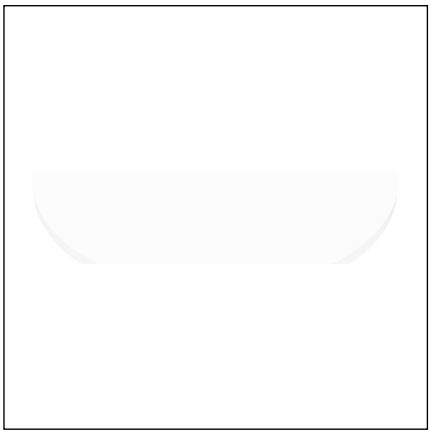
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
R-05 01 01-2	Glasrand (bezel)	1	„Ms.“	none on the drawing	on 050100-1: 0.0124 mm

What it does (R-05 01 01-2). Screws onto the bowl's threaded rim round the movement ring's flange, its step on the dial holding the movement down; the glass in its top.

Turned about its axis; length 16.00, largest Ø 121.20. Diameters: Ø 120.40, Ø 121.20; bores Ø 109.80, Ø 110.50. Shoulders from the lower end: 0.00, 3.70, 5.74, 6.20, 8.50, 11.00, 11.71, 11.89, 12.46, 12.64, 13.21, 13.39, 13.96, 14.14, 14.71, 14.89, 16.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 01 01-1** *Teil / Part* *bezel.glass*
Behandlung / Treatment		*Benennung / Name* ***Glass***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Glass — sheet 2

The glass over the dial, in the seat at the bezel’s top edge, flush with it.

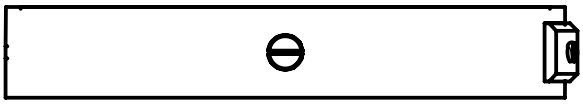
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 01-1	Glas (glass)	1	not on the drawing		on 050101-2: 0 mm

What it does (05 01 01-1). In the bezel's top, flush with it.

Turned about its axis; length 3.70, largest Ø 110.50. Diameters: Ø 110.50. Shoulders from the lower end: 0.00, 3.70.

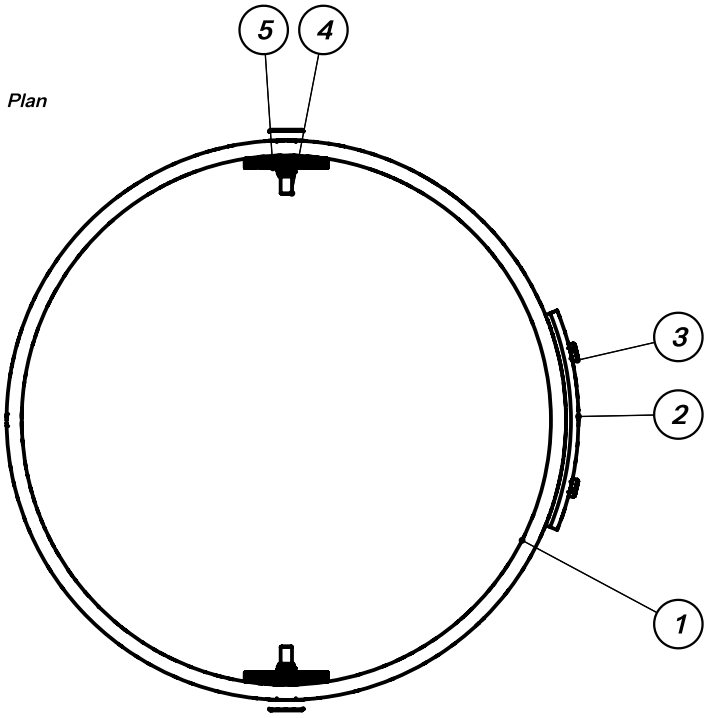
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
5	2	Lock nuts (WN-400)	WN-400	not on the drawing
4	2	Inner pivot screws (WN-79)	WN-79	not on the drawing
3	2	Ring bracket screws	WN-kr	not on the drawing
2	1	Ring bracket (Kardanringkloben)	05 01 00-6	not on the drawing
1	1	Gimbal ring	05 01 00-5	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: **1:2**

Behandlung / Treatment:

Blatt / Sheet: **1 / 1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **05 01 00-5**

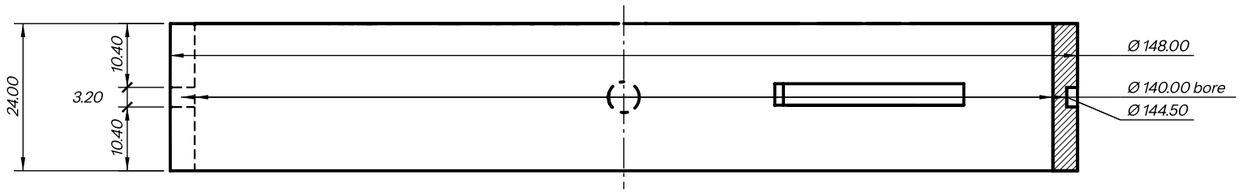
Teil / Part: gimbalRing

Benennung / Name

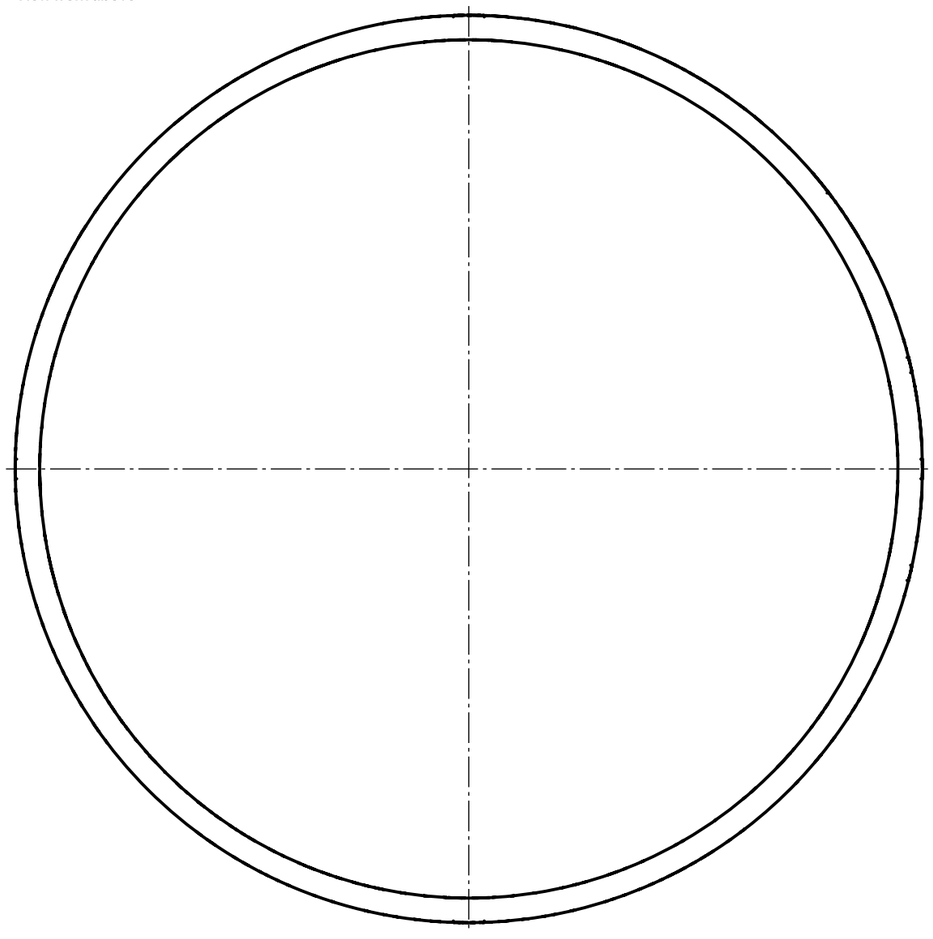
Gimbal ring (Kardanring)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).



View from above





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **05 01 00-5** *Teil / Part* *gimbalRing.ring*
Behandlung / Treatment		*Benennung / Name* ***Gimbal ring***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Gimbal ring — sheet 2

The ring between the bowl and the box: its outer pivots, on the 3-9 line, are screws through the box's walls; its inner pivots, screws on the 12-6 line with lock nuts, run in the bowl's brackets ("Lagerstellen a u. b leicht spielend": the bearings a and b running freely). Both pairs lie in one plane, so the bowl hangs level whatever the ship does. A bracket on the ring at 3 o'clock, slid to level the bowl the other way, carries one outer pivot. A slot through its band, between 20° and 52° from 6 o'clock, lets the transit lock's lever in to the bowl. Wempe 3955's ring is Bakelite; no material is given.

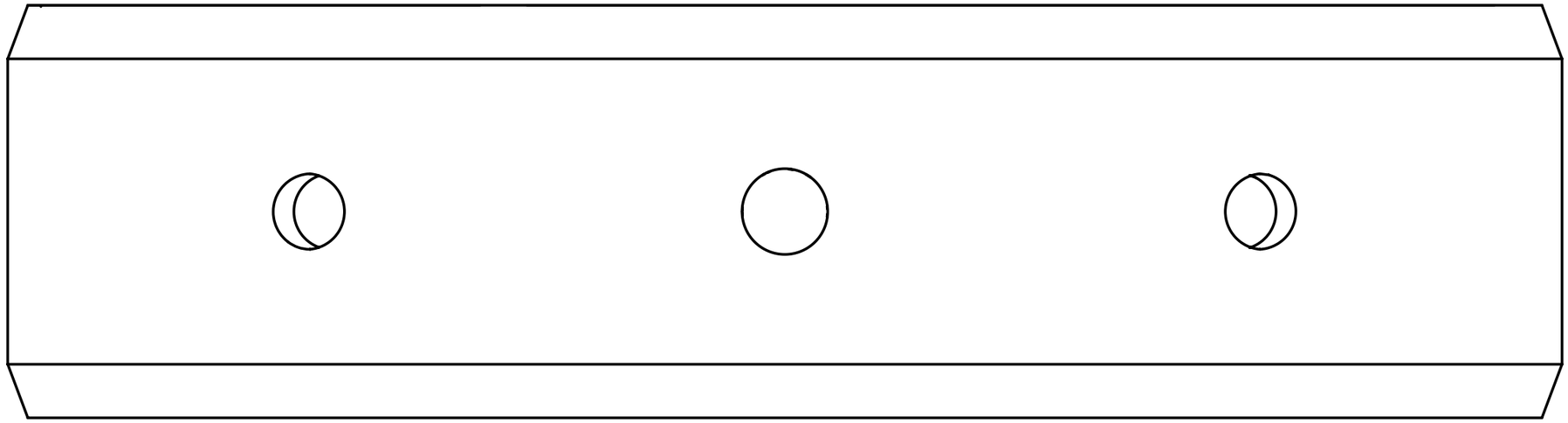
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 00-5	Kardanring (gimbal ring)	1	not on the drawing		takes Kontermutter WN-400 (10 on 05 01), knurled brass: on it: 0 mm; Kardanringkloben (ring bracket): on it: 0 mm

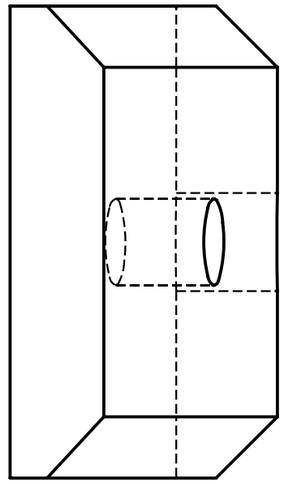
What it does (05 01 00-5). Between the bowl and the box: its outer pivots in the box's walls, its inner pivots in the bowl's brackets.

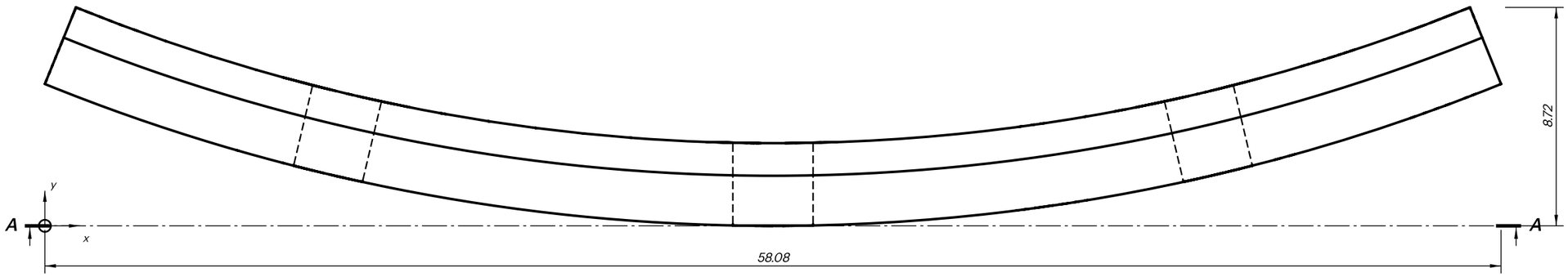
Turned about its axis; length 24.00, largest Ø 148.00. Diameters: Ø 144.50, Ø 148.00; bores Ø 140.00. Shoulders from the lower end: 0.00, 10.40, 13.60, 24.00.

A–A



View from the left







Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **05 01 00-6**
Behandlung / Treatment			Teil / Part gimbalRing.bracket
Blatt / Sheet **1 / 2**		Benennung / Name **Ring bracket (Kardanringkloben)**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Ring bracket (Kardanringkloben) — sheet 2

A curved brass plate on the ring at 3 o'clock, its two screws in slots, carrying that outer pivot: moved along the ring to level the bowl.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 00-6	Kardanringkloben (ring bracket)	1	not on the drawing		on 050100-5: 0 mm

What it does (05 01 00-6). On the ring at 3 o'clock, carrying that outer pivot; slid along its slots to level the bowl.

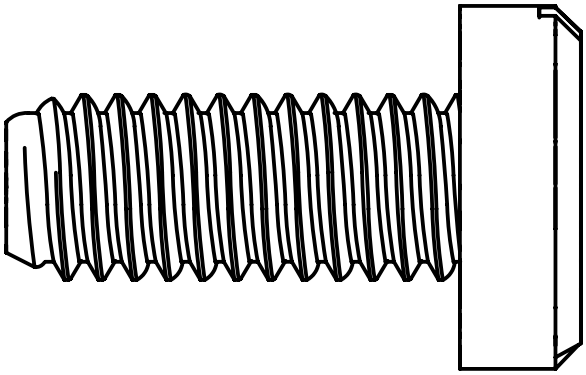
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 15.40.

Contour in the plan

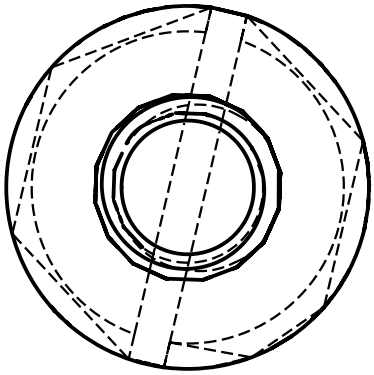
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

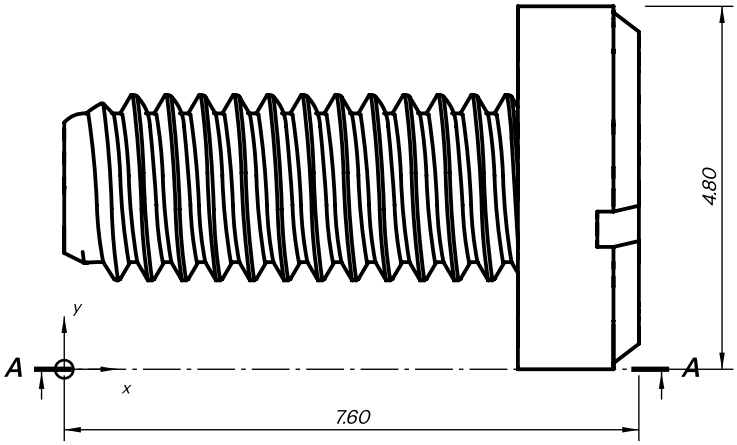
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	7.50, 3.06	0.00, 5.66	29.04, 77.30	77.30
Outline	2	line	0.00, 5.66	1.24, 8.72		
Outline	3	arc anticlockwise	1.24, 8.72	56.84, 8.72	29.04, 77.31	74.01
Outline	4	line	56.84, 8.72	58.08, 5.66		
Outline	5	arc clockwise	58.08, 5.66	48.20, 2.41	29.06, 77.24	77.23
Outline	6	line	48.20, 2.41	46.79, 2.07		
Outline	7	line	46.79, 2.07	45.37, 1.75		
Outline	8	arc clockwise	45.37, 1.75	30.76, 0.02	29.05, 77.23	77.23
Outline	9	line	30.76, 0.02	29.43, 0.00		
Outline	10	arc clockwise	29.43, 0.00	27.70, 0.01	29.04, 77.03	77.03
Outline	11	line	27.70, 0.01	26.33, 0.05		
Outline	12	arc clockwise	26.33, 0.05	12.71, 1.75	29.04, 77.27	77.27
Outline	13	line	12.71, 1.75	11.29, 2.07		
Outline	14	line	11.29, 2.07	9.88, 2.41		
Outline	15	arc clockwise	9.88, 2.41	7.50, 3.06	28.56, 75.58	75.52

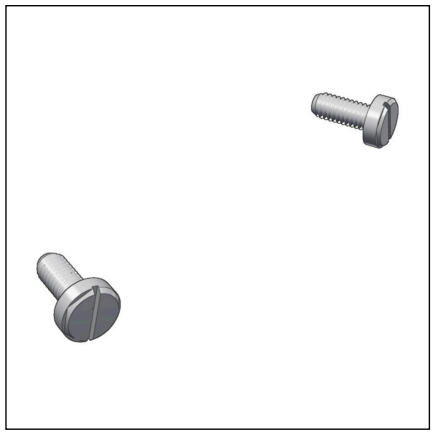
A–A



View from the left







Isometric view, not to scale

2 made; one drawn. Hidden lines are left out of the plan (too many to read); the section and the hole table give the depths.

Werkstoff / Material not on the drawing	Maßstab / Scale 10:1	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-kr** Teil / Part gimbalRing.bracketScrews
Behandlung / Treatment		Benennung / Name **Ring bracket screws**	
Blatt / Sheet 1 / 2		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Ring bracket screws — sheet 2

The bracket's two screws.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-kr	Schraube (the ring bracket's screws, no number)	2	not on the drawing		not measured on the model: see its function, Hold the ring bracket to the ring: through its holes, tapped 2.75 into the ring (cut along the radius since 25 October 2026).

What it does (WN-kr). Hold the ring bracket to the ring: through its holes, tapped 2.75 into the ring (cut along the radius since 25 October 2026).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 4.80.

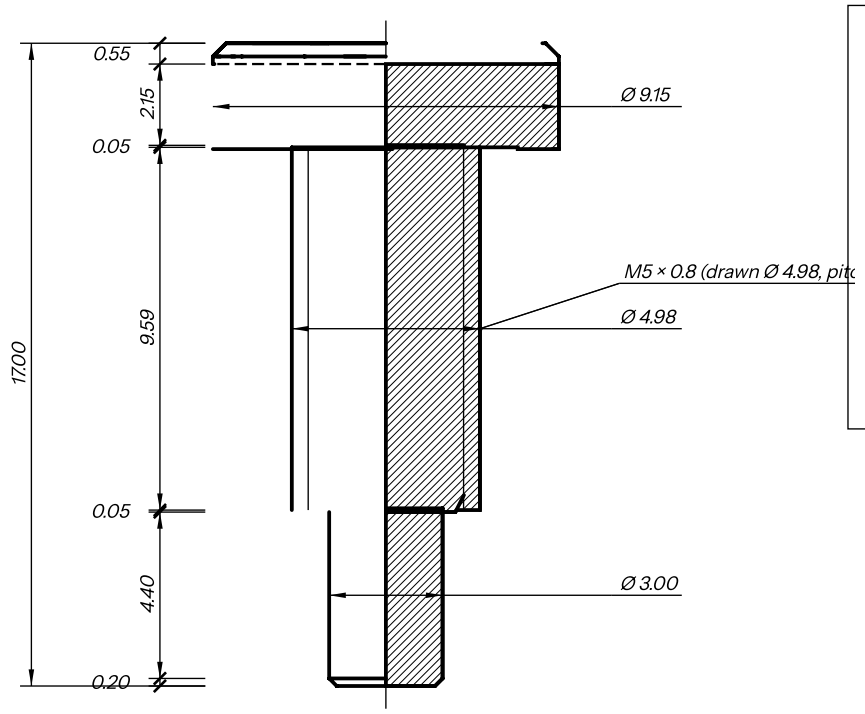
Contour in the plan

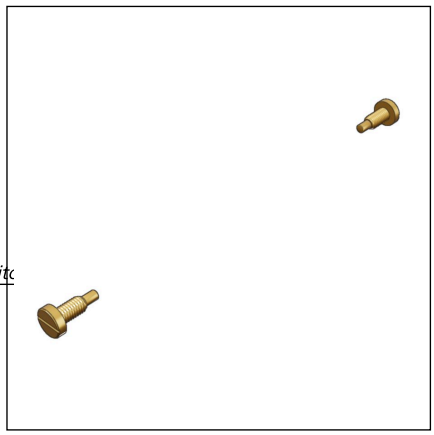
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 1.54	0.00, 3.26		
Outline	2	arc clockwise	0.00, 3.26	0.29, 3.38	0.27, 3.03	0.36
Outline	3	arc clockwise	0.29, 3.38	0.49, 3.51	0.89, 2.69	0.92
Outline	4	line	0.49, 3.51	0.52, 3.51		
Outline	5	line	0.52, 3.51	0.64, 3.38		
Outline	6	line	0.64, 3.38	0.74, 3.38		
Outline	7	line	0.74, 3.38	0.88, 3.62		
Outline	8	arc clockwise	0.88, 3.62	0.97, 3.61	0.91, 3.55	0.08
Outline	9	line	0.97, 3.61	1.10, 3.38		
Outline	10	line	1.10, 3.38	1.19, 3.38		
Outline	11	line	1.19, 3.38	1.33, 3.62		
Outline	12	arc clockwise	1.33, 3.62	1.42, 3.61	1.36, 3.55	0.08
Outline	13	line	1.42, 3.61	1.55, 3.38		
Outline	14	line	1.55, 3.38	1.64, 3.38		
Outline	15	line	1.64, 3.38	1.78, 3.62		
Outline	16	arc clockwise	1.78, 3.62	1.87, 3.61	1.81, 3.55	0.08
Outline	17	line	1.87, 3.61	2.00, 3.38		
Outline	18	line	2.00, 3.38	2.09, 3.38		
Outline	19	line	2.09, 3.38	2.23, 3.62		
Outline	20	arc clockwise	2.23, 3.62	2.32, 3.61	2.26, 3.55	0.08
Outline	21	line	2.32, 3.61	2.45, 3.38		
Outline	22	line	2.45, 3.38	2.54, 3.38		
Outline	23	line	2.54, 3.38	2.68, 3.62		
Outline	24	arc clockwise	2.68, 3.62	2.77, 3.61	2.71, 3.55	0.08
Outline	25	line	2.77, 3.61	2.90, 3.38		
Outline	26	line	2.90, 3.38	2.99, 3.38		
Outline	27	line	2.99, 3.38	3.13, 3.62		

Outline	28	arc clockwise	3.13, 3.62	3.22, 3.61	3.16, 3.55	0.08
Outline	29	line	3.22, 3.61	3.35, 3.38		
Outline	30	line	3.35, 3.38	3.44, 3.38		
Outline	31	line	3.44, 3.38	3.58, 3.62		
Outline	32	arc clockwise	3.58, 3.62	3.67, 3.61	3.61, 3.55	0.08
Outline	33	line	3.67, 3.61	3.80, 3.38		
Outline	34	line	3.80, 3.38	3.89, 3.38		
Outline	35	line	3.89, 3.38	4.03, 3.62		
Outline	36	arc clockwise	4.03, 3.62	4.12, 3.61	4.06, 3.55	0.08
Outline	37	line	4.12, 3.61	4.25, 3.38		
Outline	38	line	4.25, 3.38	4.34, 3.38		
Outline	39	line	4.34, 3.38	4.48, 3.62		
Outline	40	arc clockwise	4.48, 3.62	4.57, 3.61	4.51, 3.55	0.08
Outline	41	line	4.57, 3.61	4.70, 3.38		
Outline	42	line	4.70, 3.38	4.79, 3.38		
Outline	43	line	4.79, 3.38	4.93, 3.62		
Outline	44	arc clockwise	4.93, 3.62	5.02, 3.61	4.96, 3.55	0.08
Outline	45	line	5.02, 3.61	5.15, 3.38		
Outline	46	line	5.15, 3.38	5.24, 3.38		
Outline	47	line	5.24, 3.38	5.38, 3.62		
Outline	48	arc clockwise	5.38, 3.62	5.47, 3.61	5.41, 3.55	0.08
Outline	49	line	5.47, 3.61	5.60, 3.38		
Outline	50	line	5.60, 3.38	5.69, 3.38		
Outline	51	line	5.69, 3.38	5.83, 3.62		
Outline	52	arc clockwise	5.83, 3.62	6.00, 3.46	5.85, 3.47	0.15
Outline	53	line	6.00, 3.46	6.00, 4.80		
Outline	54	line	6.00, 4.80	7.26, 4.80		
Outline	55	line	7.26, 4.80	7.26, 4.72		
Outline	56	line	7.26, 4.72	7.60, 4.46		
Outline	57	line	7.60, 4.46	7.60, 0.34		
Outline	58	line	7.60, 0.34	7.26, 0.08		
Outline	59	line	7.26, 0.08	7.26, 0.00		
Outline	60	line	7.26, 0.00	6.00, 0.00		
Outline	61	line	6.00, 0.00	6.00, 1.27		
Outline	62	line	6.00, 1.27	5.92, 1.42		
Outline	63	line	5.92, 1.42	5.82, 1.42		
Outline	64	arc clockwise	5.82, 1.42	5.66, 1.18	5.52, 1.45	0.31
Outline	65	line	5.66, 1.18	5.61, 1.18		
Outline	66	line	5.61, 1.18	5.47, 1.42		
Outline	67	line	5.47, 1.42	5.37, 1.42		
Outline	68	arc clockwise	5.37, 1.42	5.21, 1.18	5.07, 1.45	0.31
Outline	69	line	5.21, 1.18	5.16, 1.18		
Outline	70	line	5.16, 1.18	5.02, 1.42		
Outline	71	line	5.02, 1.42	4.92, 1.42		
Outline	72	arc clockwise	4.92, 1.42	4.76, 1.18	4.62, 1.45	0.31
Outline	73	line	4.76, 1.18	4.71, 1.18		
Outline	74	line	4.71, 1.18	4.57, 1.42		
Outline	75	line	4.57, 1.42	4.47, 1.42		

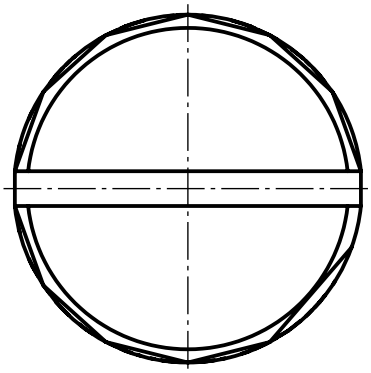
Outline	76	arc clockwise	4.47, 1.42	4.31, 1.18	4.17, 1.45	0.31
Outline	77	line	4.31, 1.18	4.26, 1.18		
Outline	78	line	4.26, 1.18	4.12, 1.42		
Outline	79	line	4.12, 1.42	4.02, 1.42		
Outline	80	arc clockwise	4.02, 1.42	3.86, 1.18	3.72, 1.45	0.31
Outline	81	line	3.86, 1.18	3.81, 1.18		
Outline	82	line	3.81, 1.18	3.67, 1.42		
Outline	83	line	3.67, 1.42	3.57, 1.42		
Outline	84	arc clockwise	3.57, 1.42	3.41, 1.18	3.27, 1.45	0.31
Outline	85	line	3.41, 1.18	3.36, 1.18		
Outline	86	line	3.36, 1.18	3.22, 1.42		
Outline	87	line	3.22, 1.42	3.12, 1.42		
Outline	88	arc clockwise	3.12, 1.42	2.96, 1.18	2.82, 1.45	0.31
Outline	89	line	2.96, 1.18	2.91, 1.18		
Outline	90	line	2.91, 1.18	2.77, 1.42		
Outline	91	line	2.77, 1.42	2.67, 1.42		
Outline	92	arc clockwise	2.67, 1.42	2.51, 1.18	2.37, 1.45	0.31
Outline	93	line	2.51, 1.18	2.46, 1.18		
Outline	94	line	2.46, 1.18	2.32, 1.42		
Outline	95	line	2.32, 1.42	2.22, 1.42		
Outline	96	arc clockwise	2.22, 1.42	2.06, 1.18	1.92, 1.45	0.31
Outline	97	line	2.06, 1.18	2.01, 1.18		
Outline	98	line	2.01, 1.18	1.87, 1.42		
Outline	99	line	1.87, 1.42	1.77, 1.42		
Outline	100	arc clockwise	1.77, 1.42	1.61, 1.18	1.47, 1.45	0.31
Outline	101	line	1.61, 1.18	1.56, 1.18		
Outline	102	line	1.56, 1.18	1.42, 1.42		
Outline	103	line	1.42, 1.42	1.32, 1.42		
Outline	104	arc clockwise	1.32, 1.42	1.16, 1.18	1.02, 1.45	0.31
Outline	105	line	1.16, 1.18	1.11, 1.18		
Outline	106	line	1.11, 1.18	0.97, 1.42		
Outline	107	line	0.97, 1.42	0.87, 1.42		
Outline	108	arc clockwise	0.87, 1.42	0.69, 1.18	0.64, 1.40	0.23
Outline	109	arc clockwise	0.69, 1.18	0.52, 1.42	0.82, 1.45	0.31
Outline	110	arc anticlockwise	0.52, 1.42	0.29, 1.41	0.45, 0.33	1.09
Outline	111	arc clockwise	0.29, 1.41	0.24, 1.42	0.28, 1.46	0.05
Outline	112	line	0.24, 1.42	0.00, 1.54		
Cut-out 1	113	line	1.59, 2.99	1.59, 2.99		
Cut-out 1	114	line	1.59, 2.99	1.59, 2.99		
Cut-out 1	115	line	1.59, 2.99	1.59, 2.99		
Cut-out 1	116	line	1.59, 2.99	1.59, 2.99		





Isometric view, not to scale

View from above



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-79*** *Teil / Part* *gimbalRing.pivots*
Behandlung / Treatment		*Benennung / Name* ***Inner pivot screws (WN-79)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Inner pivot screws (WN-79) — sheet 2

Two screws through the ring at 12 and 6 o’clock, their points the inner pivots in the bowl’s brackets.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-79	Schraube WN-79 (12 on 05 01): the inner pivots	2	not on the drawing		not measured on the model: see its function, Through the ring at 12 and 6 o'clock (M5, tapped in it), their plain tips the inner pivots in the case brackets' holes (05 01 Schnitt C-D).

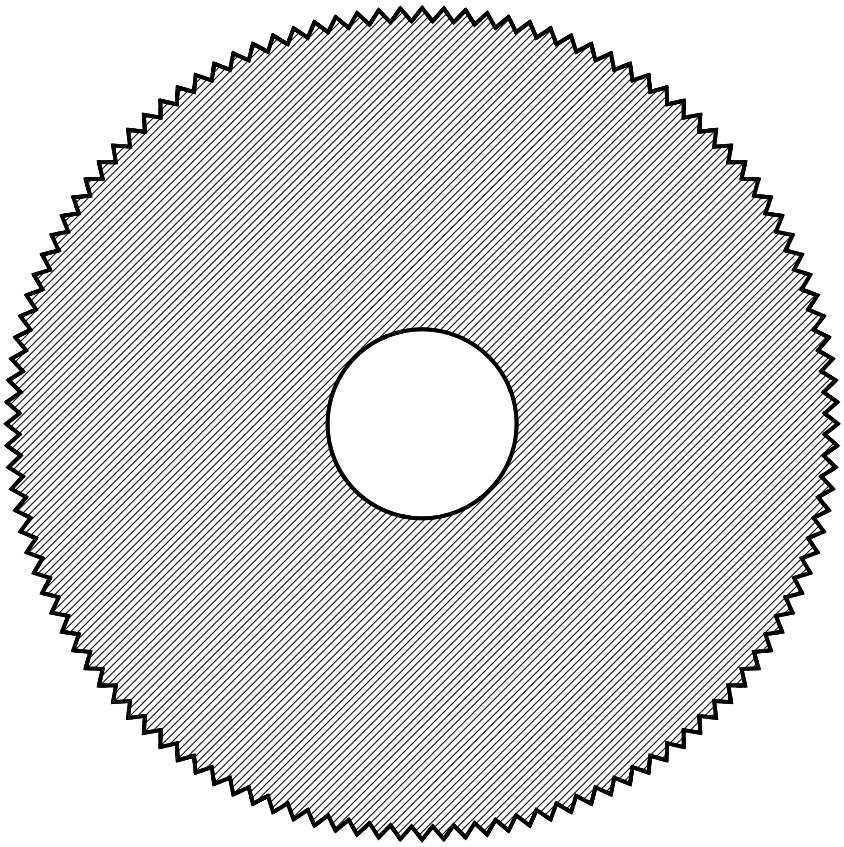
What it does (WN-79). Through the ring at 12 and 6 o'clock (M5, tapped in it), their plain tips the inner pivots in the case brackets' holes (05 01 Schnitt C-D).

Turned about its axis; length 17.00, largest Ø 9.15. Diameters: Ø 3.00, Ø 4.98, Ø 9.15. Shoulders from the lower end: 0.00, 0.20, 4.60, 4.65, 14.25, 14.30, 16.45, 17.00.

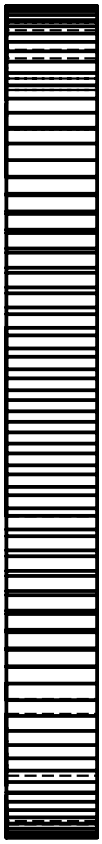
Thread

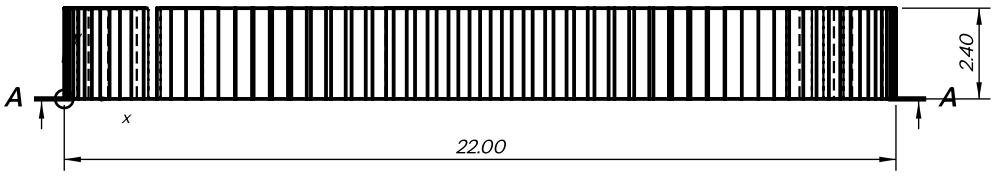
Outside thread 4.65–14.25 from the lower end, 9.59 long: drawn Ø 4.98 over its crests, 4.11 at its roots, pitch 0.80. Cut the ISO 261 coarse thread nearest it, M5 × 0.8, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).

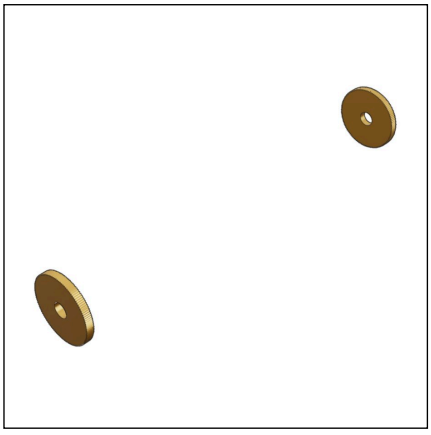
A–A



View from the left







Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-400** Teil / Part gimbalRing.nuts
Behandlung / Treatment		Benennung / Name **Lock nuts (WN-400)**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Lock nuts (WN-400) — sheet 2

The lock nuts that hold the inner pivot screws where they are set: knurled brass discs inside the ring, against its inner face, turned by hand (Wempe 3955).

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-400	Kontermutter WN-400 (10 on 05 01), knurled brass	2	not on the drawing		on 050100-5: 0 mm

What it does (WN-400). Lock the inner pivot screws where they are set: inside the ring, against its inner face (05 01 Schnitt C-D: Ø 22 × 2.4).

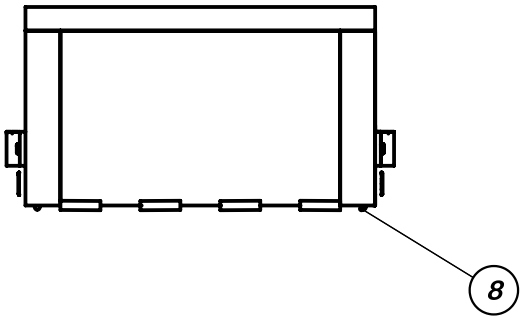
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 22.00.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.38, 2.40	22.00, 2.40		
Outline	2	line	22.00, 2.40	22.00, 0.00		
Outline	3	line	22.00, 0.00	0.00, 0.00		
Outline	4	line	0.00, 0.00	0.00, 2.40		
Outline	5	line	0.00, 2.40	0.38, 2.40		

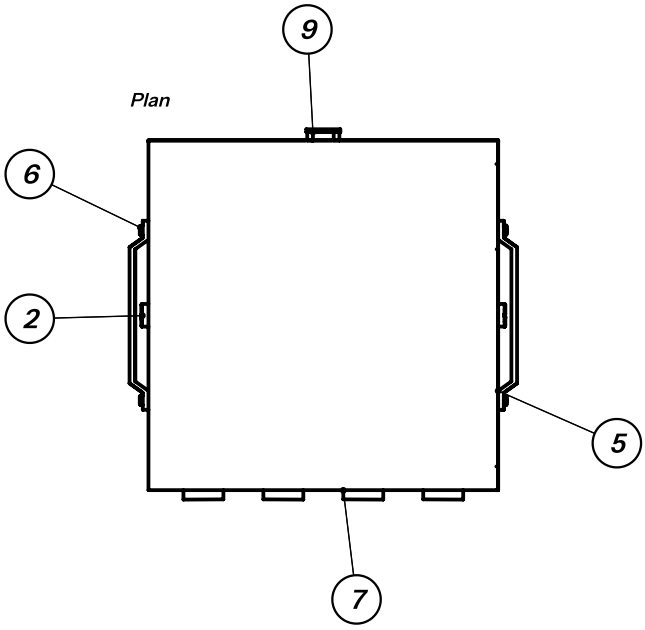
Front view





Isometric view, not to scale

Plan



Not seen in either view (inside the assembly): Pos. 1, 3, 4, 10, 11, 12.

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
12	2	Key holder's screws	WN-kh	not on the drawing
11	1	Key holder's fitting	Schluesselhalter.F	not on the drawing
10	1	Key holder	Schluesselhalter	not on the drawing
9	1	Keeper	Schliessblech	not on the drawing
8	2	Pins over the pivots	Kastenstift	not on the drawing
7	1	Hinge	Scharnier	not on the drawing
6	4	Handle screws	WN-hg	not on the drawing
5	2	Handles (Griff f. Chronometerkasten)	51 00 10	not on the drawing
4	1	Lid stay	Deckelstuetze	not on the drawing
3	2	Bearing discs	Lagerscheibe	not on the drawing
2	2	Outer pivot screws (WN-80)	WN-80	not on the drawing
1	1	Box	127-556.02	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: 1:4

Behandlung / Treatment:

Blatt / Sheet: 1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

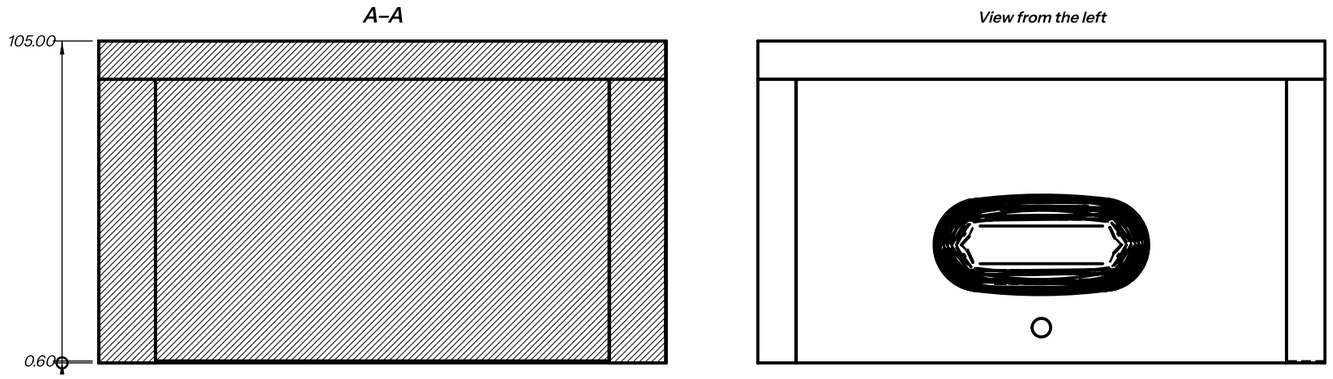
Zeichnung Nr. / Drawing no.: **51 00 10**

Teil / Part: box

Benennung / Name: **Wooden box (Holzkasten)**

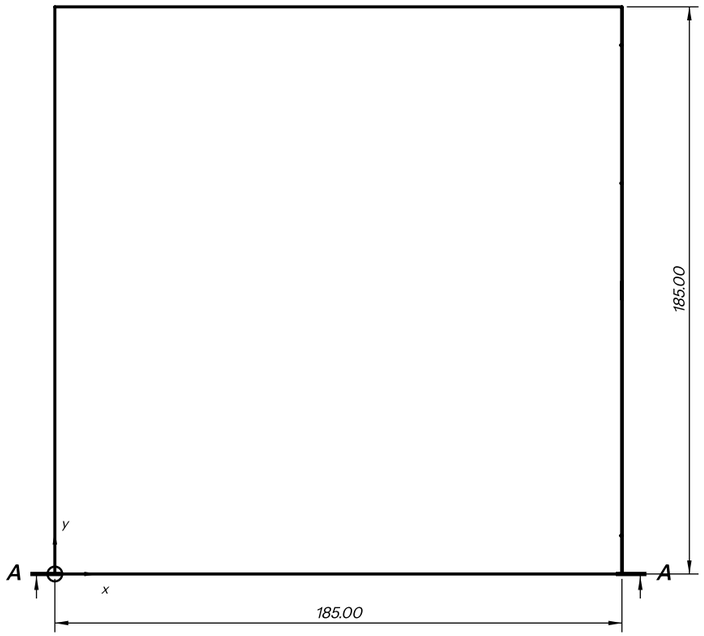
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale



Hidden lines are left out of the plan (too many to read); the section and the hole table give the depths.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:2**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **127-556.02**
Behandlung / Treatment			*Teil / Part* box.body
		Benennung / Name **Box**	
Blatt / Sheet 1 / 2		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Box — sheet 2

The mahogany box the gimbals hang in, 185 mm square and 180 high: the outer pivots are screws through its side walls, below them a pressed handle on each side. Its glazed lid is hinged at the back on one long hinge and held open by a slotted brass stay on the left. For carrying it goes into the padded outer box (Transport box view).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.02	Holzkasten (wooden box)	1	not on the drawing		takes Bearing disc (no number): on it: 0 mm; Schlüsselhalter (key holder, estimated): on it: 0 mm; Schließblech (keeper, estimated): on it: 0 mm; Pin over the outer pivot (no number; inferred a lid-locating pin): in it (embed): 3.291 mm, least gap -1.2458; Griff f. Chronometerkasten (handle): on it: 0 mm

What it does (127-556.02). 185 x 185 x 180 outside: the gimbals hang in it (no sheet in the set).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 105.00.

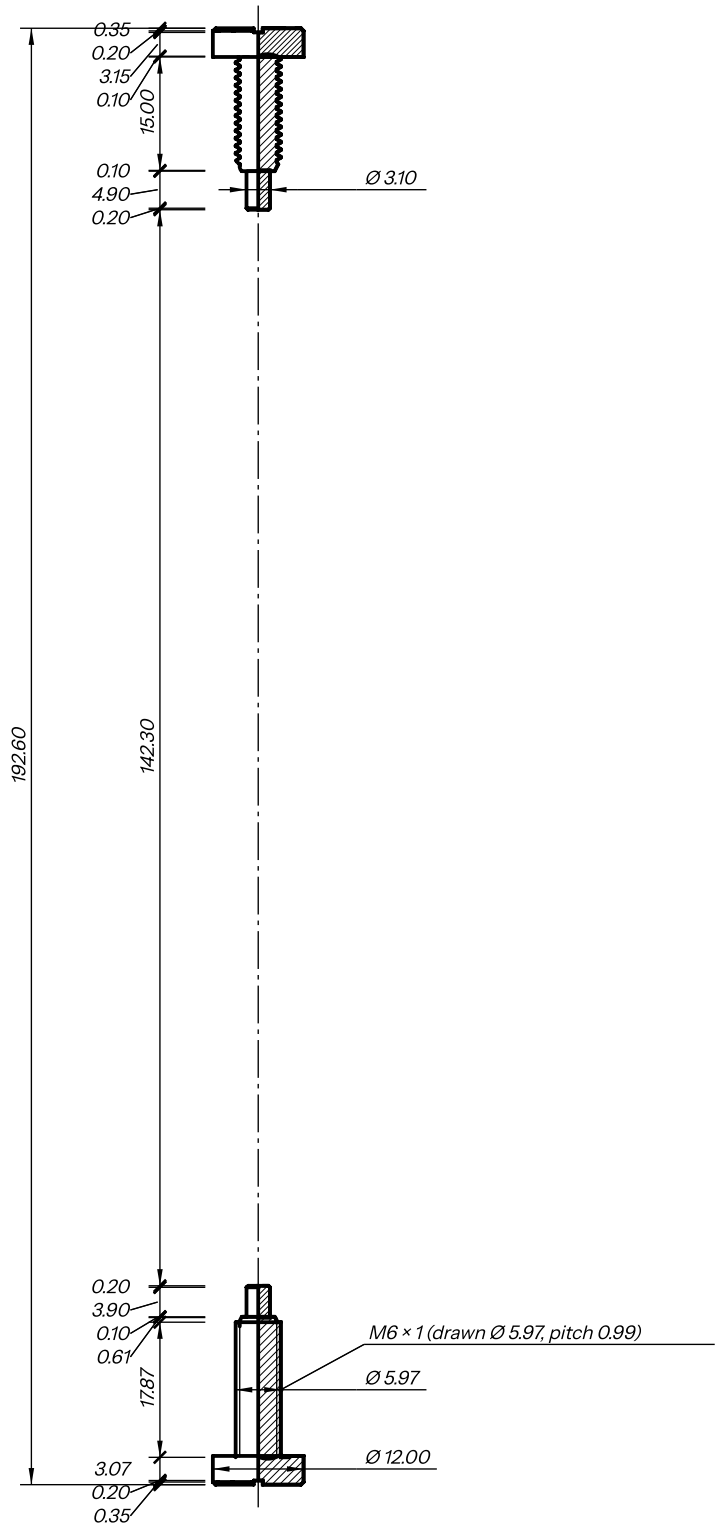
Contour in the plan

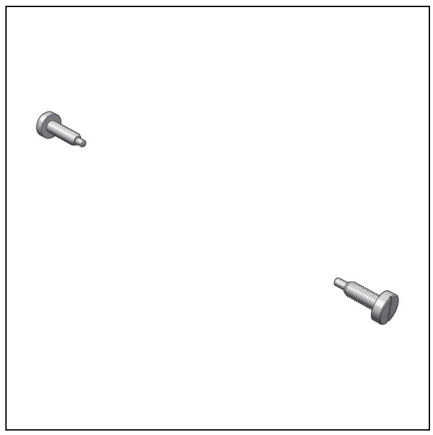
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 63.37	0.00, 185.00		
Outline	2	line	0.00, 185.00	185.00, 185.00		
Outline	3	line	185.00, 185.00	185.00, 0.00		
Outline	4	line	185.00, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 63.37		

Levels

Faces in section A-A, from the bottom face: 0.60, 105.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-80*** *Teil / Part* *box.pivots*
Behandlung / Treatment		*Benennung / Name* ***Outer pivot screws (WN-80)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Outer pivot screws (WN-80) — sheet 2

Two M6 screws through the box’s side walls, their ends the gimbal ring’s pivots.

Parts list

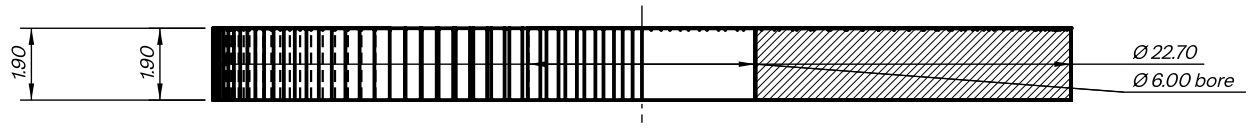
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-80	Schraube WN-80 (11 on 05 01): the outer pivots, M6	2	not on the drawing		not measured on the model: see its function, Through the box's side walls, their tips the gimbal ring's pivots: through the ring at 9 o'clock, through its bracket into the ring at 3 (05 01's elevation).

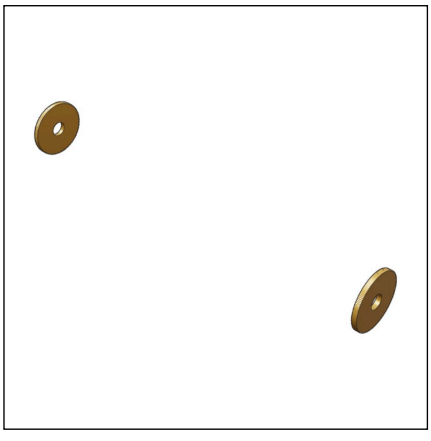
What it does (WN-80). Through the box's side walls, their tips the gimbal ring's pivots: through the ring at 9 o'clock, through its bracket into the ring at 3 (05 01's elevation).

Turned about its axis; length 192.60, largest Ø 12.00. Diameters: Ø 3.10, Ø 5.97, Ø 12.00. Shoulders from the lower end: 0.00, 0.35, 0.55, 3.62, 21.49, 22.10, 22.20, 26.10, 26.30, 168.60, 168.80, 173.70, 173.80, 188.80, 188.90, 192.05, 192.25, 192.60.

Thread

Outside thread 3.62–21.48 from the lower end, 17.87 long: drawn Ø 5.97 over its crests, 4.89 at its roots, pitch 0.99. Cut the ISO 261 coarse thread nearest it, M6 × 1, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Lagerscheibe*** *Teil / Part* *box.discs*
Behandlung / Treatment		*Benennung / Name* ***Bearing discs***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Bearing discs — sheet 2

Brass discs on the side walls' inside round the outer pivots, their edges knurled.

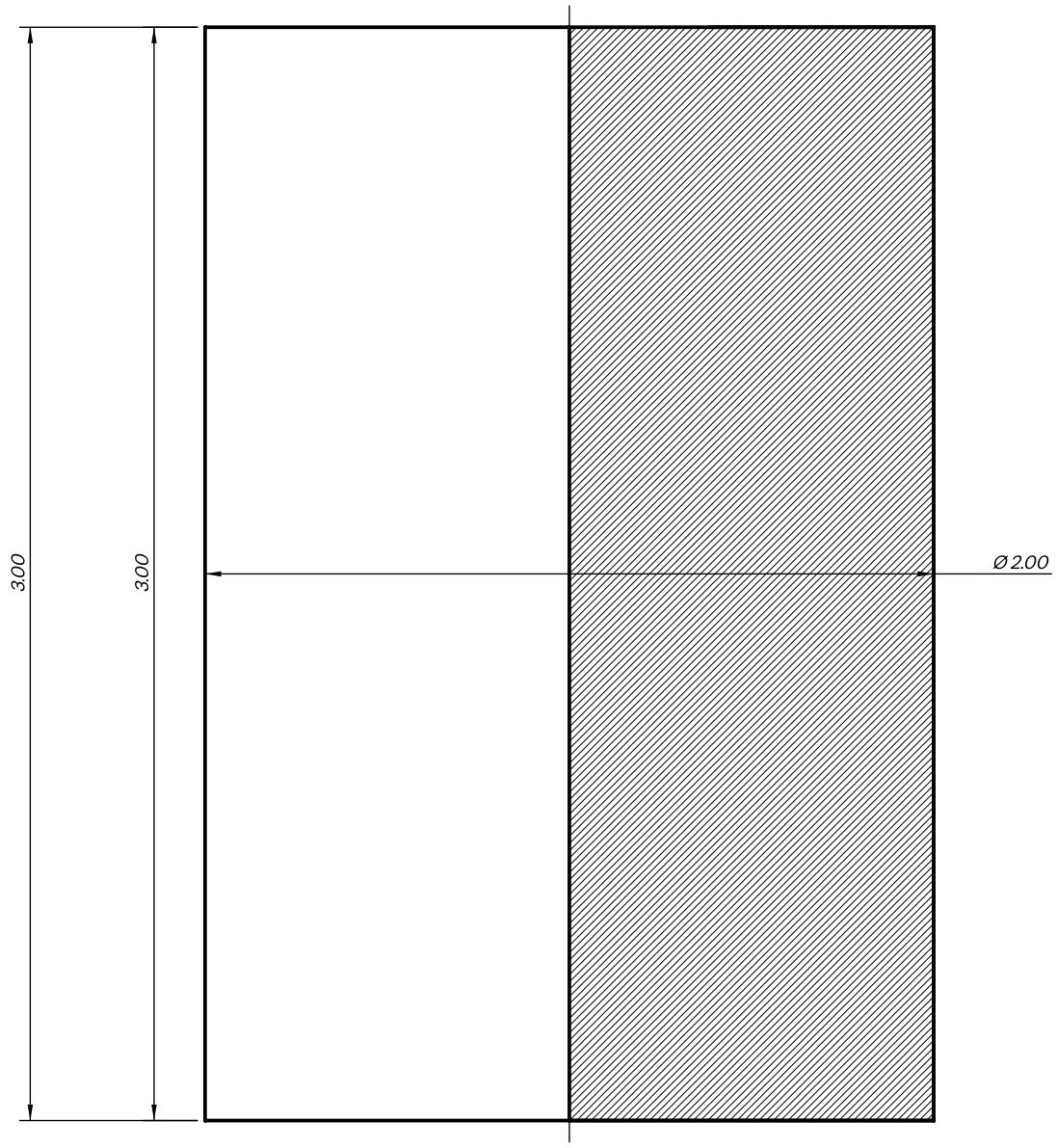
2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

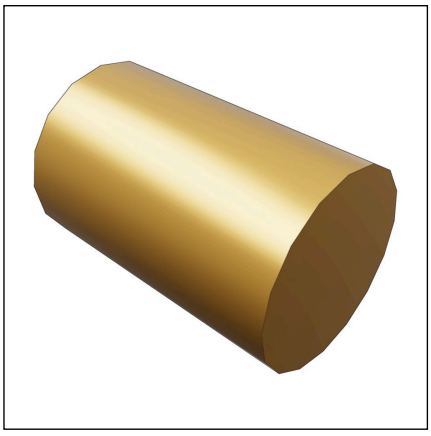
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Lagerscheibe	Bearing disc (no number)	2	not on the drawing		on Holzkasten/Kasten6: 0 mm

What it does (Lagerscheibe). On each side wall's inside round the outer pivot.

Turned about its axis; length 1.90, largest Ø 22.70. Diameters: Ø 22.70; bores Ø 6.00. Shoulders from the lower end: 0.00, 1.90.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **50:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Deckelstuetze*** *Teil / Part* *box.stay*
Behandlung / Treatment		*Benennung / Name* ***Lid stay***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Lid stay — sheet 2

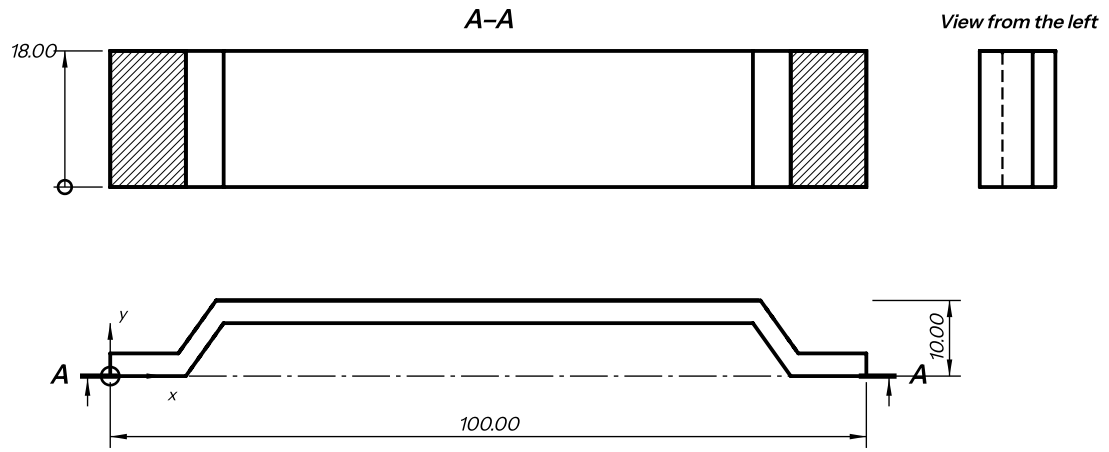
A brass bar with a long slot, pivoted on the lid’s left wall and riding on a pin in the box’s: opened, the pin reaches the slot’s end and holds the lid; shut, the bar lies along the wall.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Deckelstuetze	Deckelstütze (lid stay, estimated)	1	not on the drawing		not measured on the model: see its function, Holds the lid open: a slotted brass bar pivoted on the lid's left wall, riding on a pin in the box's (Wempe 6161).

What it does (Deckelstuetze). Holds the lid open: a slotted brass bar pivoted on the lid's left wall, riding on a pin in the box's (Wempe 6161).

Turned about its axis; length 3.00, largest Ø 2.00. Diameters: Ø 2.00. Shoulders from the lower end: 0.00, 3.00.





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **51 00 10** *Teil / Part* box.handles
Behandlung / Treatment		*Benennung / Name* ***Handles (Griff f. Chronometerkasten)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Handles (Griff f. Chronometerkasten) — sheet 2

A pressed strip handle on each side, under the pivot, its grip an oval standing 10 mm off the side.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
51 00 10	Griff f. Chronometerkasten (handle)	2	not on the drawing		on Holzkasten: 0 mm

What it does (51 00 10). On each side, under the pivot, over the wall's finger recess (an oval 70 × 30, 10.5 deep in the 12.5 wall).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 18.00.

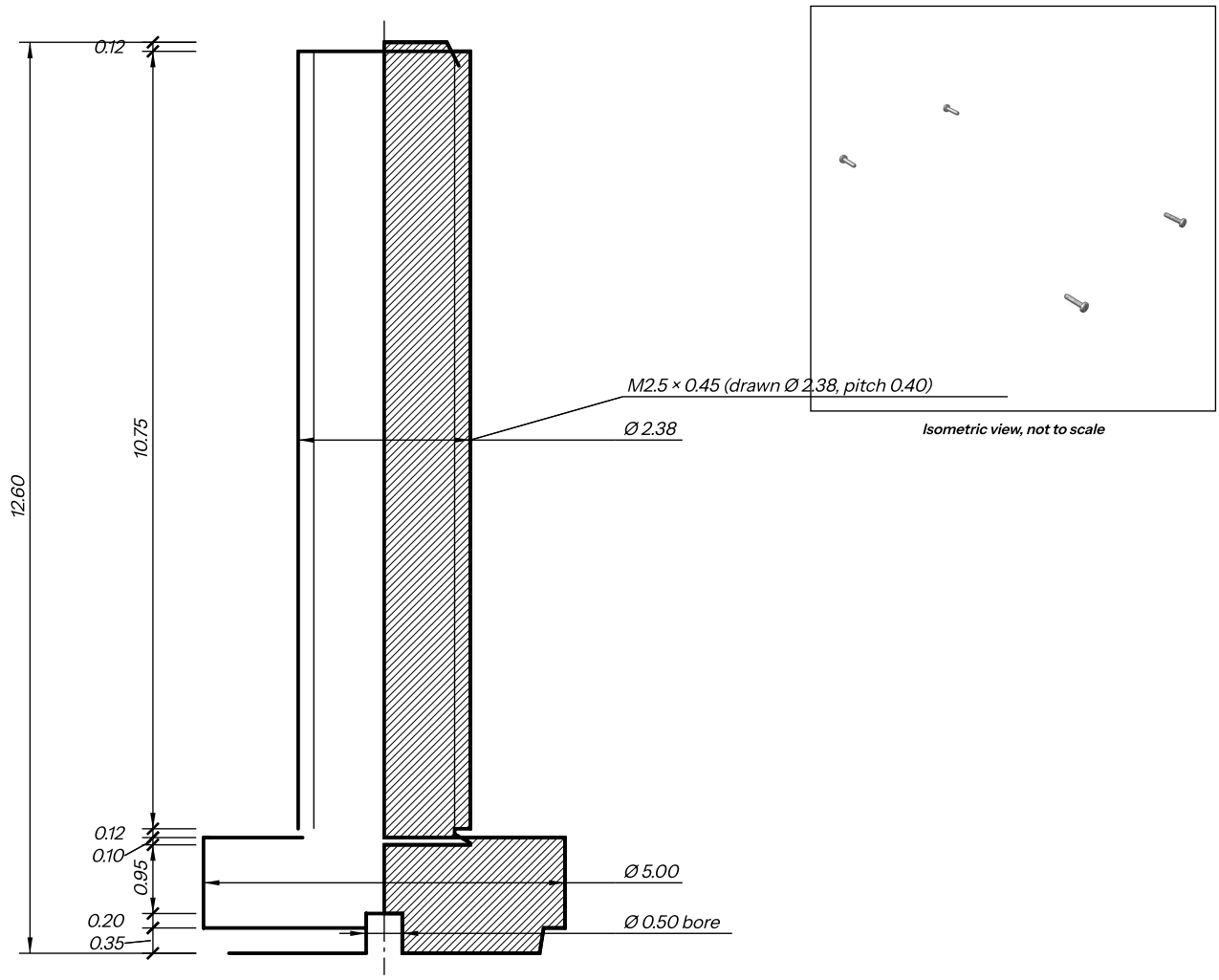
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	14.00, 10.00	86.00, 10.00		
Outline	2	line	86.00, 10.00	91.00, 3.00		
Outline	3	line	91.00, 3.00	100.00, 3.00		
Outline	4	line	100.00, 3.00	100.00, 0.00		
Outline	5	line	100.00, 0.00	90.00, 0.00		
Outline	6	line	90.00, 0.00	85.00, 7.00		
Outline	7	line	85.00, 7.00	15.00, 7.00		
Outline	8	line	15.00, 7.00	10.00, 0.00		
Outline	9	line	10.00, 0.00	0.00, 0.00		
Outline	10	line	0.00, 0.00	0.00, 3.00		
Outline	11	line	0.00, 3.00	9.00, 3.00		
Outline	12	line	9.00, 3.00	14.00, 10.00		

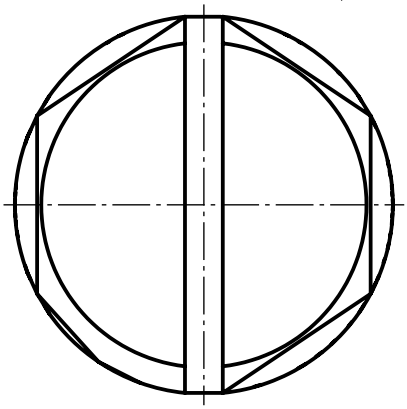
Levels

Faces in section A-A, from the bottom face: 18.00.



Isometric view, not to scale

View from below (seen from beneath, x reversed)



4 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-hg** *Teil / Part* box.handleScrews
Behandlung / Treatment		*Benennung / Name* ***Handle screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Handle screws — sheet 2

The screws through the handles’ ends.

4 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

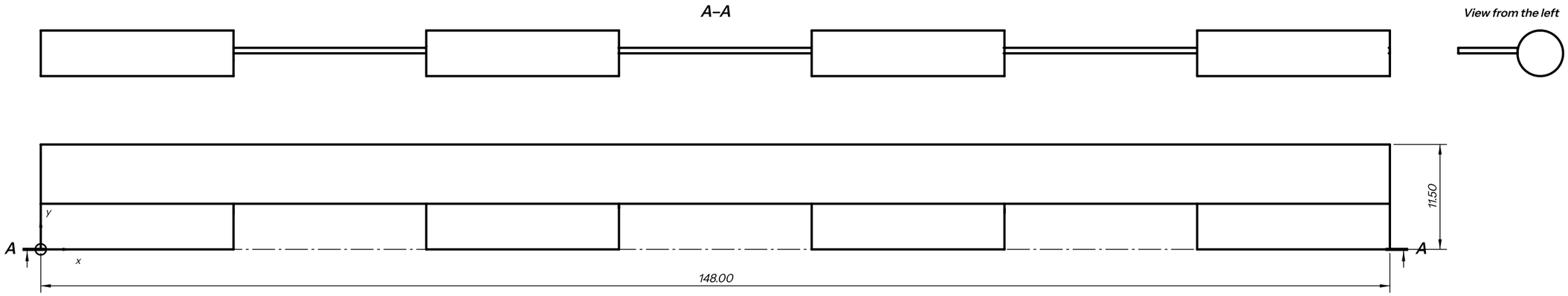
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-hg	Schraube (the handles’ screws, estimated)	4	not on the drawing		not measured on the model: see its function, Through the handles' ends into the side walls.

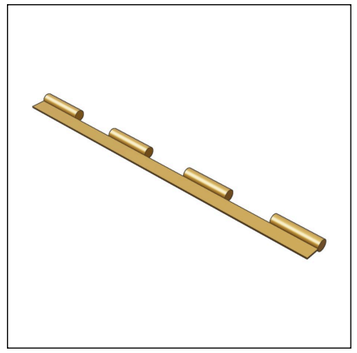
What it does (WN-hg). Through the handles' ends into the side walls.

Turned about its axis; length 12.60, largest Ø 5.00. Diameters: Ø 2.38, Ø 5.00; bores Ø 0.50. Shoulders from the lower end: 0.00, 0.35, 0.55, 1.50, 1.60, 1.72, 12.47, 12.60.

Thread

Outside thread 1.72–12.47 from the lower end, 10.75 long: drawn Ø 2.38 over its crests, 1.95 at its roots, pitch 0.40. Cut the ISO 261 coarse thread nearest it, M2.5 × 0.45, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Scharnier***
Behandlung / Treatment			*Teil / Part* *box.hinges*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Hinge***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Hinge — sheet 2

One brass piano hinge across the back joining the lid to the box, its leaves let into the parting faces.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Scharnier	Scharnier (piano hinge, estimated)	1	not on the drawing		not measured on the model: see its function, Join the lid to the box at the back: one hinge 148 long, its leaves let into the parting faces (as every photographed box).

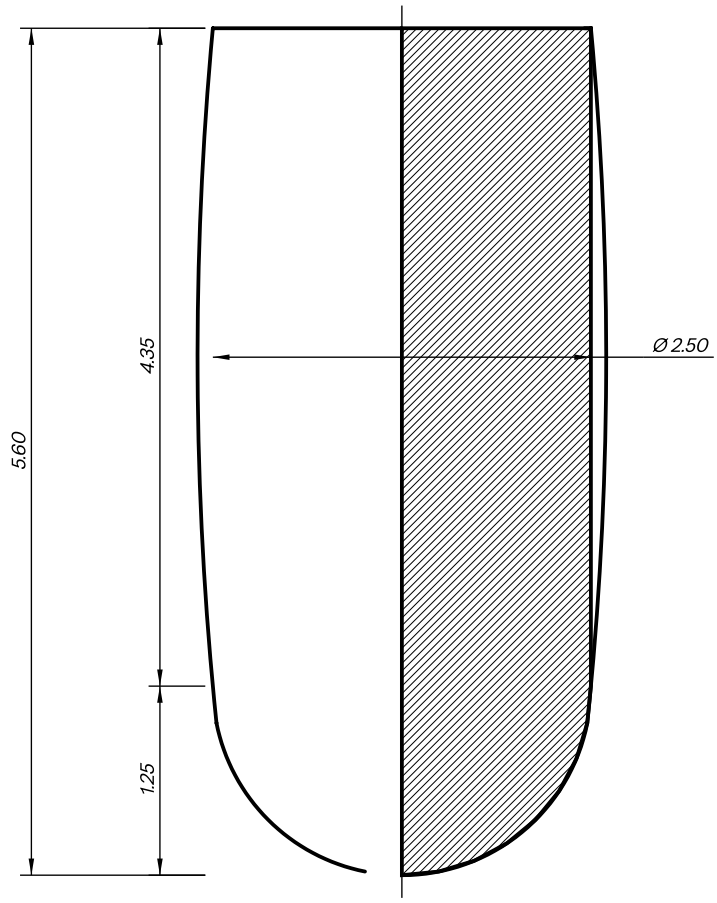
What it does (Scharnier). Join the lid to the box at the back: one hinge 148 long, its leaves let into the parting faces (as every photographed box).

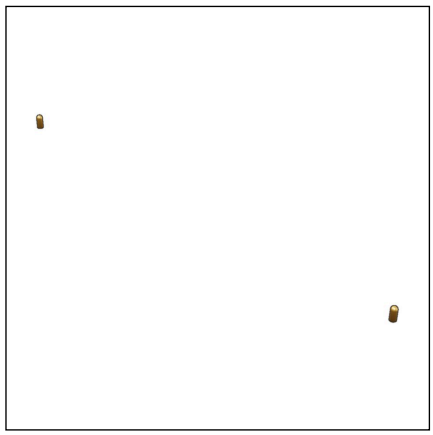
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 5.00.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 1.73	0.00, 11.50		
Outline	2	line	0.00, 11.50	148.00, 11.50		
Outline	3	line	148.00, 11.50	148.00, 0.00		
Outline	4	line	148.00, 0.00	126.86, 0.00		
Outline	5	line	126.86, 0.00	126.86, 5.00		
Outline	6	line	126.86, 5.00	105.71, 5.00		
Outline	7	line	105.71, 5.00	105.71, 0.00		
Outline	8	line	105.71, 0.00	84.57, 0.00		
Outline	9	line	84.57, 0.00	84.57, 5.00		
Outline	10	line	84.57, 5.00	63.43, 5.00		
Outline	11	line	63.43, 5.00	63.43, 0.00		
Outline	12	line	63.43, 0.00	42.29, 0.00		
Outline	13	line	42.29, 0.00	42.29, 5.00		
Outline	14	line	42.29, 5.00	21.14, 5.00		
Outline	15	line	21.14, 5.00	21.14, 0.00		
Outline	16	line	21.14, 0.00	0.00, 0.00		
Outline	17	line	0.00, 0.00	0.00, 1.73		





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***20:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Kastenstift*** *Teil / Part* *box.pins*
Behandlung / Treatment		*Benennung / Name* ***Pins over the pivots***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Pins over the pivots — sheet 2

A small round-headed pin standing out of each side wall’s top face over its outer pivot, into a hole in the closed lid: inferred to locate the lid on the box (05 01 draws it without a number; no photograph shows one).

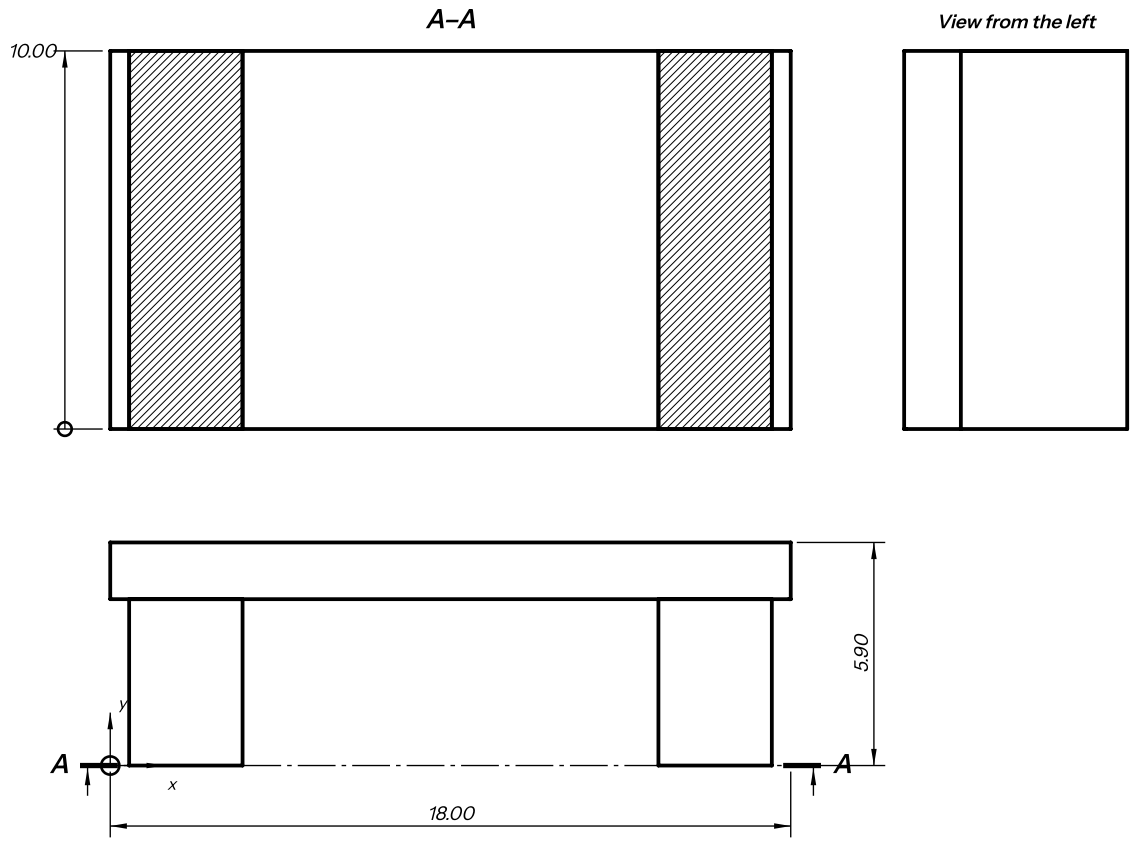
2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

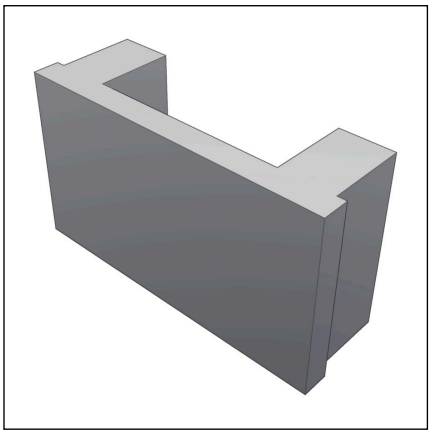
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Kastenstift	Pin over the outer pivot (no number; inferred a lid-locating pin)	2	not on the drawing		in Holzkasten (embed): 3.291 mm, least gap -1.2458

What it does (Kastenstift). Standing 2.3 out of each side wall's parting face over its outer pivot, into a hole in the closed lid's wall (05 01's elevation draws it; its purpose inferred).

Turned about its axis; length 5.60, largest Ø 2.50. Diameters: Ø 2.50. Shoulders from the lower end: 0.00, 1.25, 5.60.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Schliessblech*** *Teil / Part* *box.keeper*
Behandlung / Treatment		*Benennung / Name* ***Keeper***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Keeper — sheet 2

The strap on the box's front that the lid's latch tongue slides under.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Schliessblech	Schließblech (keeper, estimated)	1	not on the drawing		on Holzkasten: 0 mm

What it does (Schliessblech). On the box's front: the latch tongue's end goes under it.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 10.00.

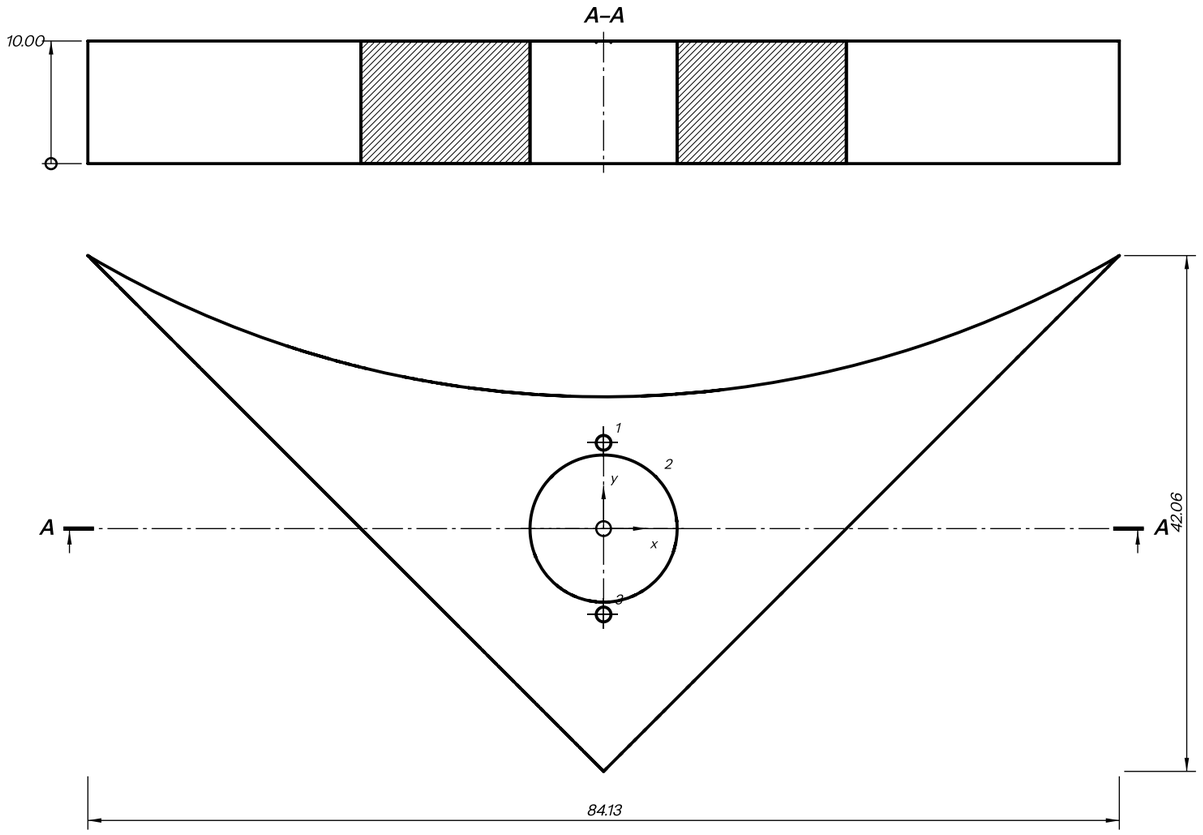
Contour in the plan

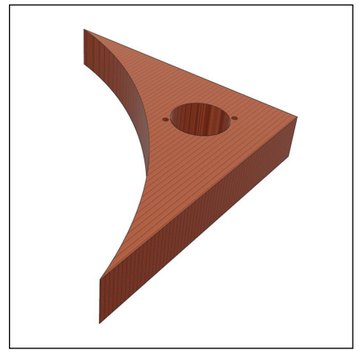
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.50, 0.00	0.50, 4.40		
Outline	2	line	0.50, 4.40	0.00, 4.40		
Outline	3	line	0.00, 4.40	0.00, 5.90		
Outline	4	line	0.00, 5.90	18.00, 5.90		
Outline	5	line	18.00, 5.90	18.00, 4.40		
Outline	6	line	18.00, 4.40	17.50, 4.40		
Outline	7	line	17.50, 4.40	17.50, 0.00		
Outline	8	line	17.50, 0.00	14.50, 0.00		
Outline	9	line	14.50, 0.00	14.50, 4.40		
Outline	10	line	14.50, 4.40	3.50, 4.40		
Outline	11	line	3.50, 4.40	3.50, 0.00		
Outline	12	line	3.50, 0.00	0.50, 0.00		

Levels

Faces in section A-A, from the bottom face: 10.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Schluesselhalter*** *Teil / Part* *box.keyHolder*
Behandlung / Treatment		*Benennung / Name* ***Key holder***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Key holder — sheet 2

A wooden shelf across the box's back right corner, its front edge curved round the gimbal ring (Wempe 6161, 3044; the box's plan, 127-556. A-1, draws it straight): the winding key stands in a round fitting on it, its tube down through the shelf, paddle up, ready to hand, as every photographed box keeps it (a wartime Wempe at Greenwich, Wempe 5377, Lange 2285 and 5510, a 6MX).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Schluesselhalter	Schlüsselhalter (key holder, estimated)	1	not on the drawing		on Holzkasten/Kasten6: 0 mm

What it does (Schluesselhalter). A shelf across the box's back right corner, its front edge curved round the gimbal ring (Wempe 6161, 3044; 51-00_1 draws a triangle), the winding key standing in a round fitting on it, its tube down through the shelf.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 10.00.

Holes

Hole	x	y	Size
1	0.00	7.00	Ø 1.20 through
2	0.00	0.00	Ø 12.00 through
3	0.00	-7.00	Ø 1.20 through

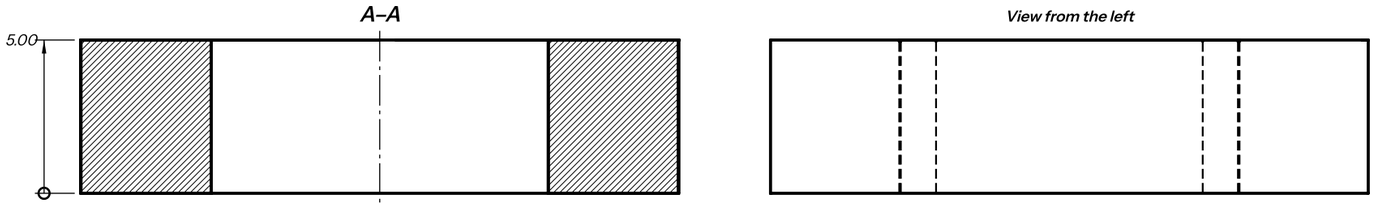
Contour in the plan

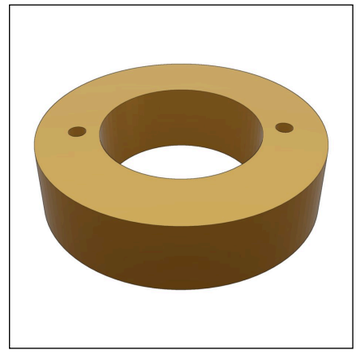
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc anticlockwise	-42.06, 22.26	42.06, 22.26	0.00, 93.34	82.59
Outline	2	line	42.06, 22.26	0.00, -19.80		
Outline	3	line	0.00, -19.80	-42.06, 22.26		

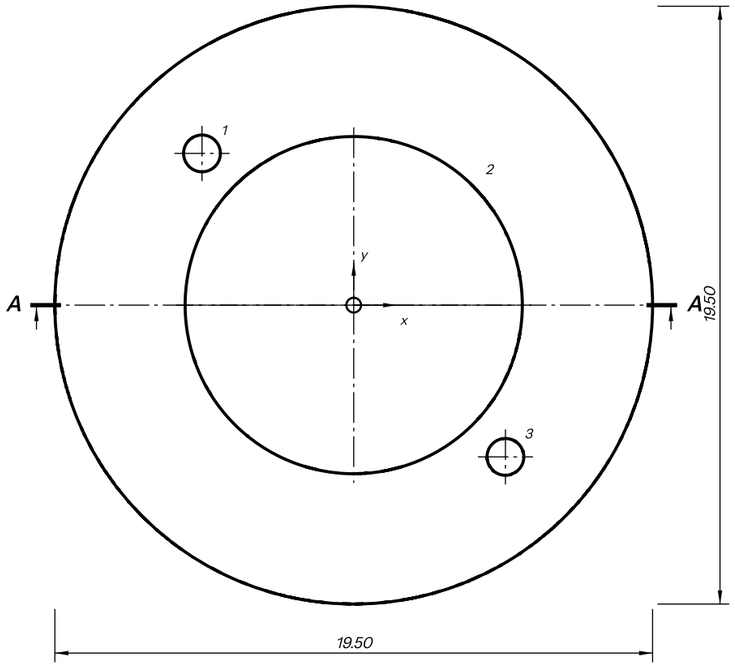
Levels

Faces in section A-A, from the bottom face: 10.00.





Isometric view, not to scale



Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **Schluesselhalter.F** Teil / Part box.keyFitting
Behandlung / Treatment		Benennung / Name **Key holder's fitting**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Key holder's fitting — sheet 2

A round brass cup standing on the shelf, the key's cone seated in its bore (51-00_1 draws the circle on the shelf's edge; Lange VEB 2285's box has one, black; Wempe 6161's key stands in a brass ring, and Wempe 3044's corner shows nothing black).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Schluesselhalter.F	Fassung (the key holder's round fitting, estimated)	1	not on the drawing		on Schluesselhalter: 0 mm

What it does (Schluesselhalter.F). A round brass cup standing on the shelf (51-00_1's circle; Lange VEB 2285's, on its shelf): the key's cone seats on its bore.

Coordinates from the datum, hole 2's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 5.00.

Holes

Hole	x	y	Size
1	-4.95	4.95	Ø 1.20 through
2	0.00	0.00	Ø 11.00 through
3	4.95	-4.95	Ø 1.20 through

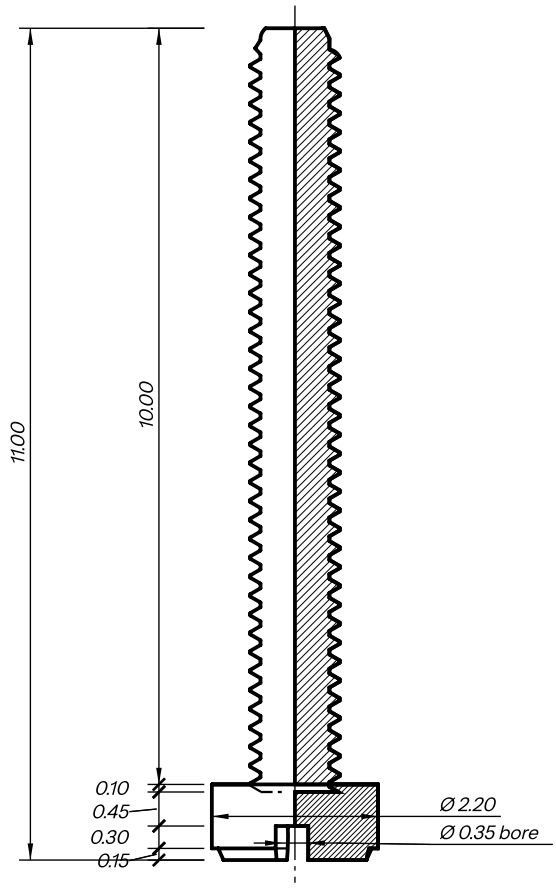
Contour in the plan

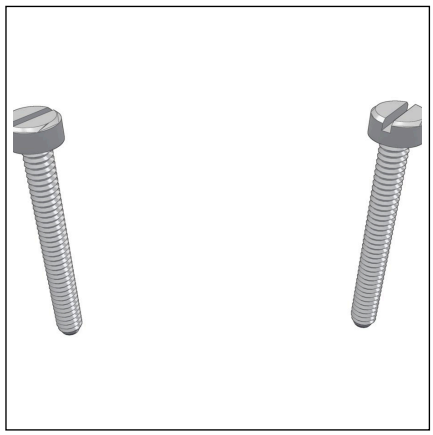
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	circle			0.00, 0.00	9.75

Levels

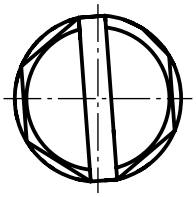
Faces in section A-A, from the bottom face: 5.00.





Isometric view, not to scale

View from below (seen from beneath, x reversed)



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-kh** *Teil / Part* box.keyFittingScrews
Behandlung / Treatment		*Benennung / Name* ***Key holder's screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Key holder’s screws — sheet 2

The fitting’s two screws on the box’s diagonal, either side of the key, into the shelf (51-00_1’s two dots).

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

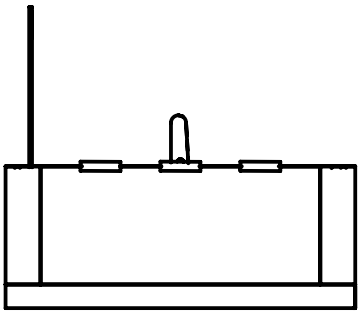
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-kh	Schraube (the key holder's fitting's screws, estimated)	2	not on the drawing		not measured on the model: see its function, Through the fitting into the shelf, either side of the key (51-00_1's two dots).

What it does (WN-kh). Through the fitting into the shelf, either side of the key (51-00_1's two dots).

Turned about its axis; length 11.00, largest Ø 2.20. Diameters: Ø 2.20; bores Ø 0.35. Shoulders from the lower end: 0.00, 0.15, 0.45, 0.90, 1.00, 11.00.

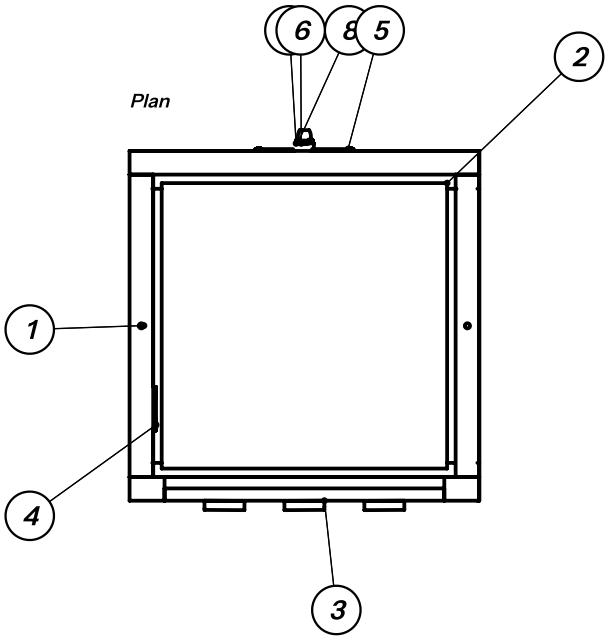
Front view





Isometric view, not to scale

Plan



8	1	Tongue knob	WN-zk	not on the drawing
7	1	Tongue screw	WN-zg	not on the drawing
6	1	Latch tongue (Zunge)	P. 51 00 02-2	„Eisen 10 × 4 blank gezogen“
5	1	Maker's plate (Firmenschild)	00 01 01-9	„Alu 10“
4	1	Lid stay	Deckelstuetze	not on the drawing
3	1	Hinge	Scharnier	not on the drawing
2	1	Lid glass	Deckelglas	not on the drawing
1	1	Lid	127-556.02	not on the drawing
Pos.	**Units**	**Benennung / Name**	**Zeichnung / Drawing**	**Werkstoff / Material**

Werkstoff / Material
„Alu 10“

Maßstab / Scale
1:4

Behandlung / Treatment
none on the drawing

Blatt / Sheet
1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

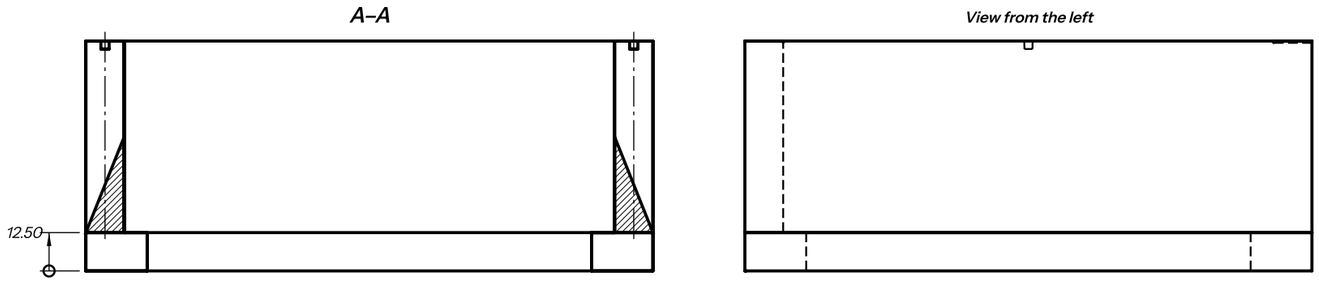
Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.
00 01 01-9

Teil / Part
lid

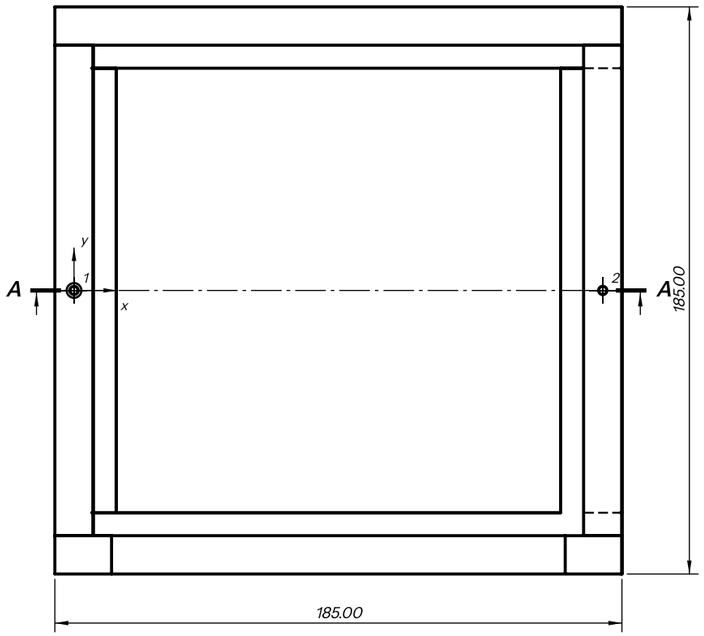
Benennung / Name
Lid (Deckel)

Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.
First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale



Werkstoff / Material not on the drawing	Maßstab / Scale **1:2**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **127-556.02** Teil / Part lid.lid
Behandlung / Treatment		Benennung / Name **Lid**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Lid — sheet 2

The box's lid, with a glass in its top so the dial can be read with it shut, hinged at the back. On its front the maker's plate, "Chronometerwerke Wempe Hamburg, Type 3371" (the number runs on), and the latch tongue.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.02	Deckel (the box's lid)	1	not on the drawing		takes Deckelglas (the lid's glass, estimated): on it: 0 mm; Zunge (latch tongue): on it: 0 mm; Firmenschild (maker's plate), Type 3371: on it: 0 mm

What it does (127-556.02). The glazed lid, hinged at the back.

Coordinates from the datum, hole 1's centre (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 75.00.

Holes

Hole	x	y	Size
1	0.00	0.00	Ø 2.70 × 2.60 deep from the top
2	172.50	0.00	Ø 2.70 × 2.60 deep from the top

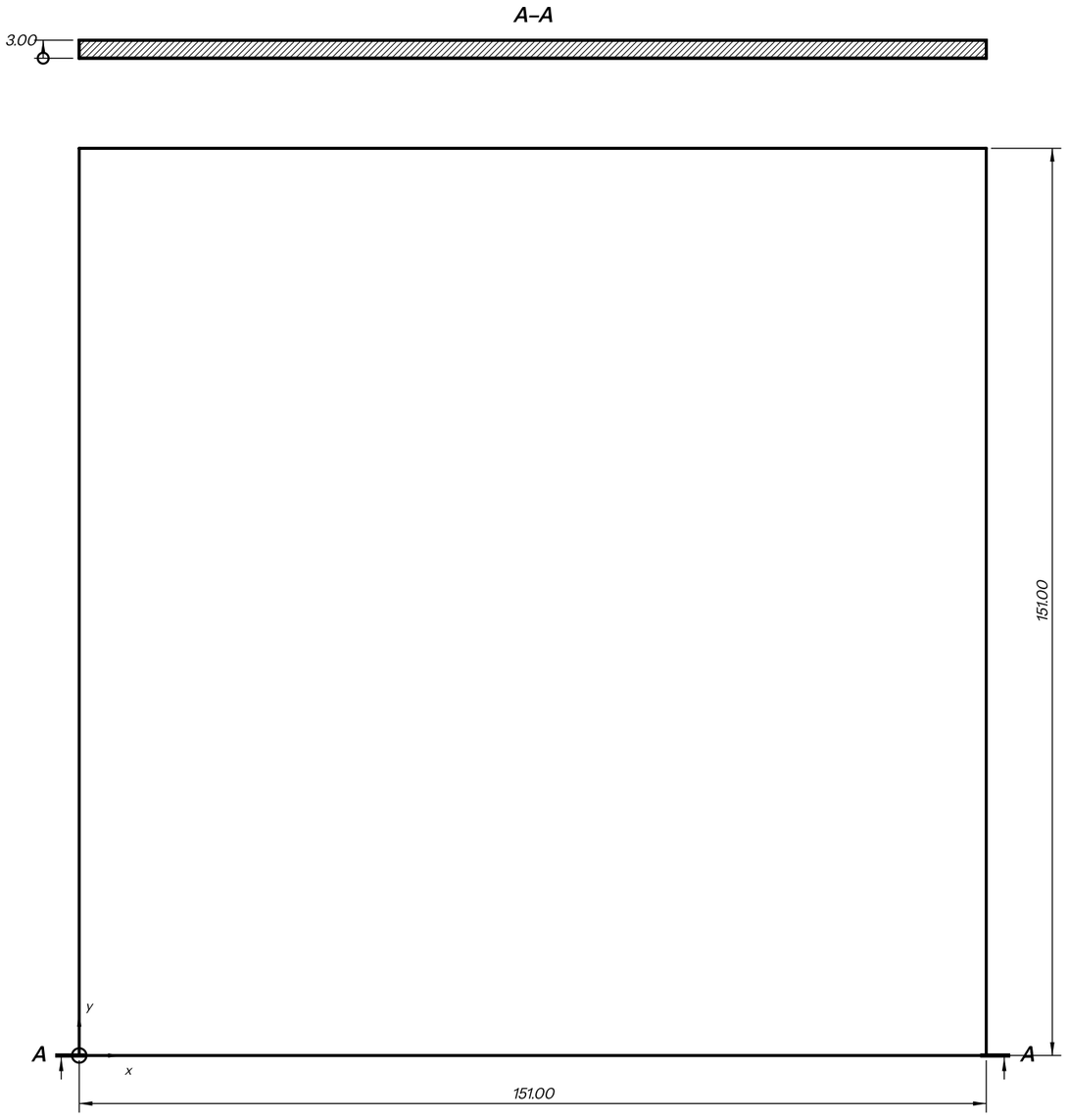
Contour in the plan

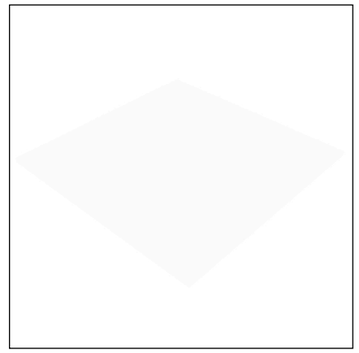
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	-6.25, 72.50	-6.25, 92.50		
Outline	2	line	-6.25, 92.50	178.75, 92.50		
Outline	3	line	178.75, 92.50	178.75, -92.50		
Outline	4	line	178.75, -92.50	-6.25, -92.50		
Outline	5	line	-6.25, -92.50	-6.25, 72.50		
Cut-out 1	6	line	13.75, 72.50	13.75, -72.50		
Cut-out 1	7	line	13.75, -72.50	158.75, -72.50		
Cut-out 1	8	line	158.75, -72.50	158.75, 72.50		
Cut-out 1	9	line	158.75, 72.50	13.75, 72.50		

Levels

Faces in section A-A, from the bottom face: 12.50.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Deckelglas***
Behandlung / Treatment			*Teil / Part* *lid.glass*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Lid glass***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Lid glass — sheet 2

The glass in the lid’s top.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Deckelglas	Deckelglas (the lid’s glass, estimated)	1	not on the drawing		on Holzkasten.D/Kasten6.D: 0 mm

What it does (Deckelglas). In the lid's top, under its frame.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 3.00.

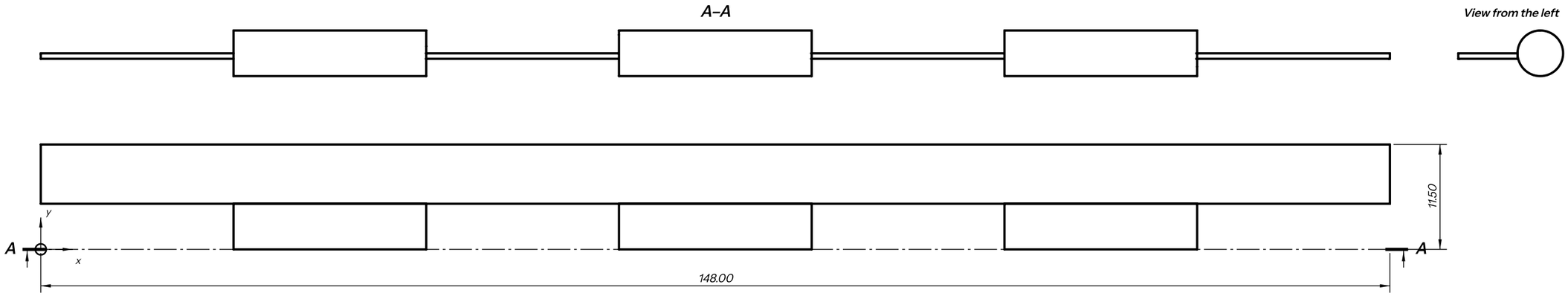
Contour in the plan

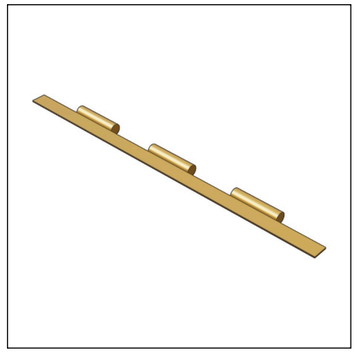
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	151.00, 151.00	151.00, 0.00		
Outline	2	line	151.00, 0.00	0.00, 0.00		
Outline	3	line	0.00, 0.00	0.00, 151.00		
Outline	4	line	0.00, 151.00	151.00, 151.00		

Levels

Faces in section A-A, from the bottom face: 3.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **Scharnier** Teil / Part lid.hinges
Behandlung / Treatment		Benennung / Name **Hinge**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Hinge — sheet 2

The piano hinge's leaf and knuckles on the lid, let into its back wall's underside.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Scharnier	Scharnier (piano hinge, estimated)	1	not on the drawing		not measured on the model: see its function, Join the lid to the box at the back: one hinge 148 long, its leaves let into the parting faces (as every photographed box).

What it does (Scharnier). Join the lid to the box at the back: one hinge 148 long, its leaves let into the parting faces (as every photographed box).

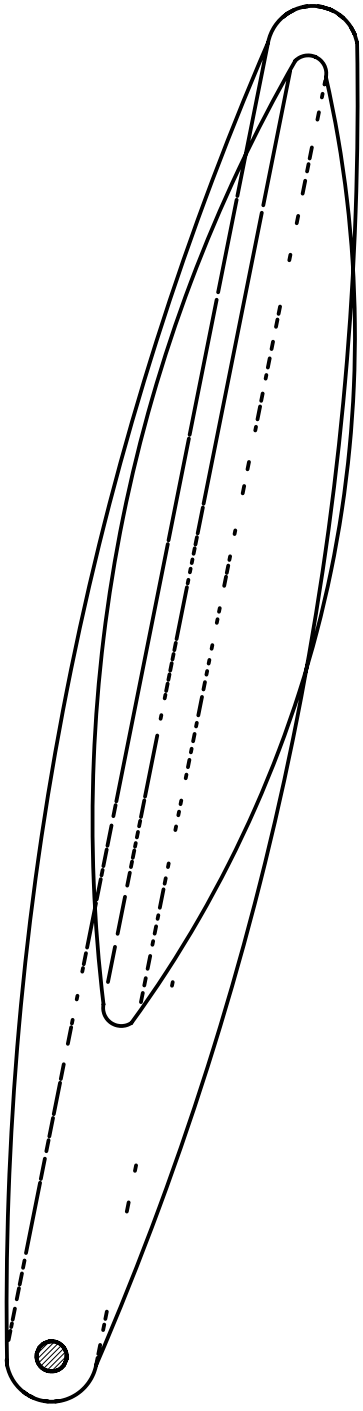
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 5.00.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	21.14, 1.03	21.14, 5.00		
Outline	2	line	21.14, 5.00	0.00, 5.00		
Outline	3	line	0.00, 5.00	0.00, 11.50		
Outline	4	line	0.00, 11.50	148.00, 11.50		
Outline	5	line	148.00, 11.50	148.00, 5.00		
Outline	6	line	148.00, 5.00	126.86, 5.00		
Outline	7	line	126.86, 5.00	126.86, 0.00		
Outline	8	line	126.86, 0.00	105.71, 0.00		
Outline	9	line	105.71, 0.00	105.71, 5.00		
Outline	10	line	105.71, 5.00	84.57, 5.00		
Outline	11	line	84.57, 5.00	84.57, 0.00		
Outline	12	line	84.57, 0.00	63.43, 0.00		
Outline	13	line	63.43, 0.00	63.43, 5.00		
Outline	14	line	63.43, 5.00	42.29, 5.00		
Outline	15	line	42.29, 5.00	42.29, 0.00		
Outline	16	line	42.29, 0.00	21.14, 0.00		
Outline	17	line	21.14, 0.00	21.14, 1.03		

A–A

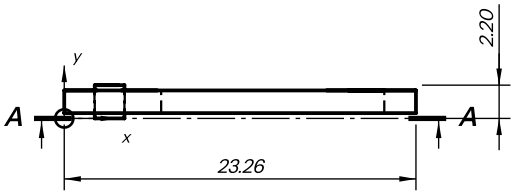


View from the left





Isometric view, not to scale



Werkstoff / Material not on the drawing	Maßstab / Scale **2:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **Deckelstuetze** Teil / Part lid.stay
Behandlung / Treatment		Benennung / Name **Lid stay**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Lid stay — sheet 2

The lid’s stay, a slotted brass bar on a pivot screw in the lid’s left wall; its pin is in the box’s wall.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Deckelstuetze	Deckelstütze (lid stay, estimated)	1	not on the drawing		not measured on the model: see its function, Holds the lid open: a slotted brass bar pivoted on the lid's left wall, riding on a pin in the box's (Wempe 6161).

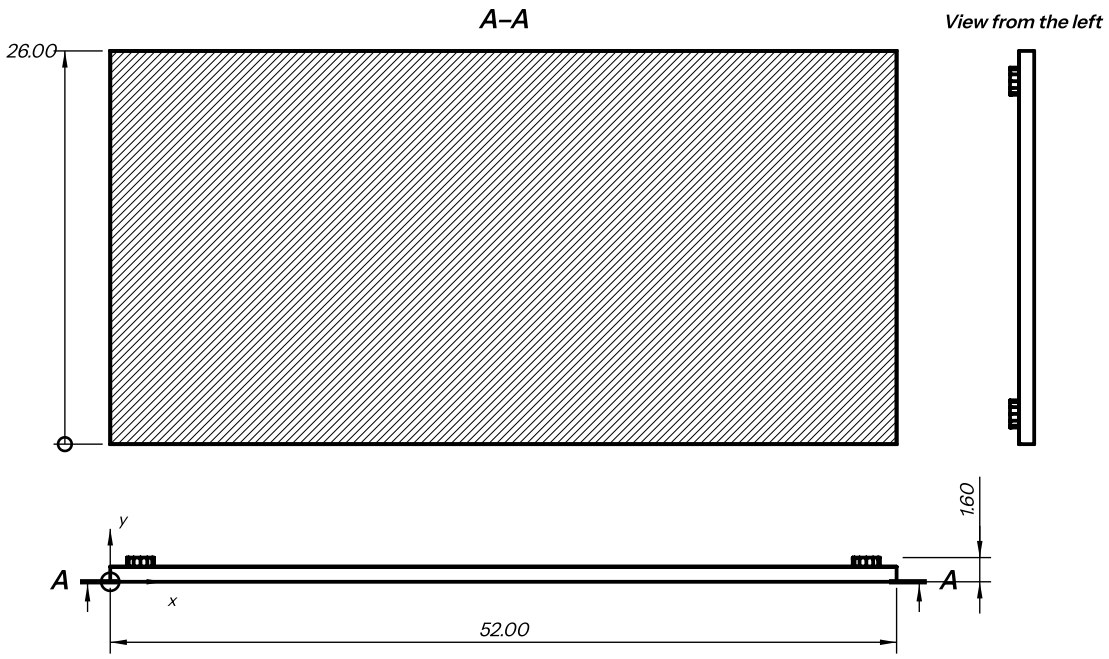
What it does (Deckelstuetze). Holds the lid open: a slotted brass bar pivoted on the lid's left wall, riding on a pin in the box's (Wempe 6161).

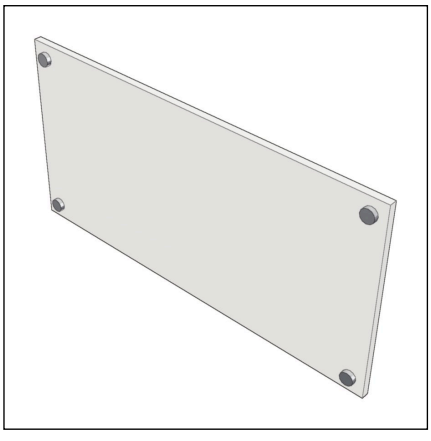
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 92.29.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	1.85, 1.85	2.02, 1.85		
Outline	2	line	2.02, 1.85	2.02, 2.20		
Outline	3	line	2.02, 2.20	3.98, 2.20		
Outline	4	line	3.98, 2.20	3.98, 1.85		
Outline	5	line	3.98, 1.85	23.26, 1.85		
Outline	6	line	23.26, 1.85	23.26, 0.35		
Outline	7	line	23.26, 0.35	3.98, 0.35		
Outline	8	line	3.98, 0.35	3.98, 0.00		
Outline	9	line	3.98, 0.00	2.02, 0.00		
Outline	10	line	2.02, 0.00	2.02, 0.35		
Outline	11	line	2.02, 0.35	0.00, 0.35		
Outline	12	line	0.00, 0.35	0.00, 1.85		
Outline	13	line	0.00, 1.85	1.85, 1.85		





Isometric view, not to scale

Werkstoff / Material *„Alu 10"*	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***00 01 01-9***
Behandlung / Treatment *none on the drawing*			*Teil / Part* *lid.plate*
		Benennung / Name ***Maker's plate (Firmenschild)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Maker’s plate (Firmenschild) — sheet 2

Aluminium, white, its lettering sunk and filled black: CHRONOMETERWERKE, WEMPE, the C-W monogram, HAMBURG, Type 3371 (the number runs on).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
00 01 01-9	Firmenschild (maker’s plate), Type 3371	1	„Alu 10“	none on the drawing	on Holzkasten.D: 0 mm

What it does (00 01 01-9). On the lid’s front.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 26.00.

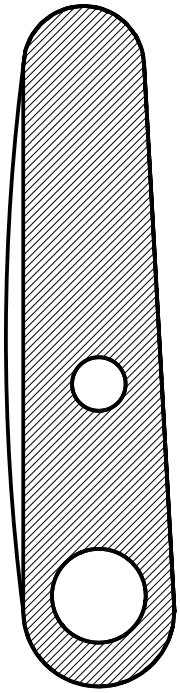
Contour in the plan

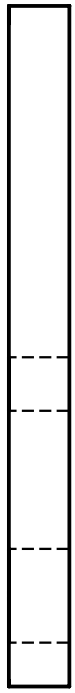
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	2.00, 1.60	2.90, 1.60		
Outline	2	line	2.90, 1.60	2.90, 1.00		
Outline	3	line	2.90, 1.00	49.10, 1.00		
Outline	4	line	49.10, 1.00	49.10, 1.60		
Outline	5	line	49.10, 1.60	50.90, 1.60		
Outline	6	line	50.90, 1.60	50.90, 1.00		
Outline	7	line	50.90, 1.00	52.00, 1.00		
Outline	8	line	52.00, 1.00	52.00, 0.00		
Outline	9	line	52.00, 0.00	0.00, 0.00		
Outline	10	line	0.00, 0.00	0.00, 1.00		
Outline	11	line	0.00, 1.00	1.10, 1.00		
Outline	12	line	1.10, 1.00	1.10, 1.60		
Outline	13	line	1.10, 1.60	2.00, 1.60		

Levels

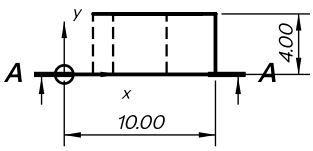
Faces in section A-A, from the bottom face: 26.00.







Isometric view, not to scale



Werkstoff / Material „Eisen 10 × 4 blank gezogen“	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***P. 5100 02-2*** *Teil / Part* *lid.tongue*
Behandlung / Treatment „gebondert, schwarz lackiert“		*Benennung / Name* ***Latch tongue (Zunge)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Latch tongue (Zunge) — sheet 2

An iron tongue, black, turning on a screw on the lid's front: hung down, its end goes under the keeper on the box and holds the lid shut. Its sheet doesn't say where it goes: taken as the lid's latch.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
P. 51 00 02-2	Zunge (latch tongue)	1	„Eisen 10 × 4 blank gezogen“	„gebondert, schwarz lackiert“	on Holzkasten.D: 0 mm

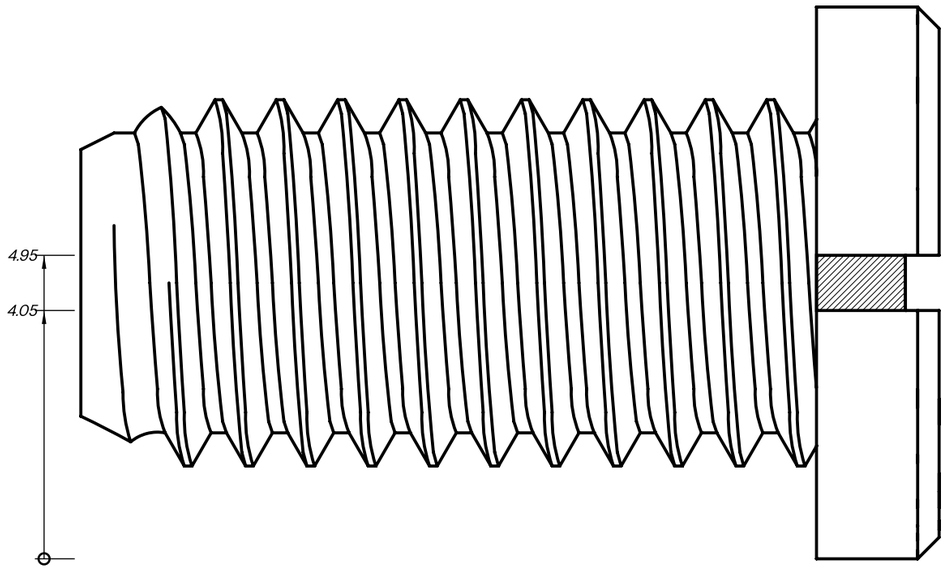
What it does (P. 51 00 02-2). Turning on its screw on the lid's front, its end under the keeper (its place estimated).

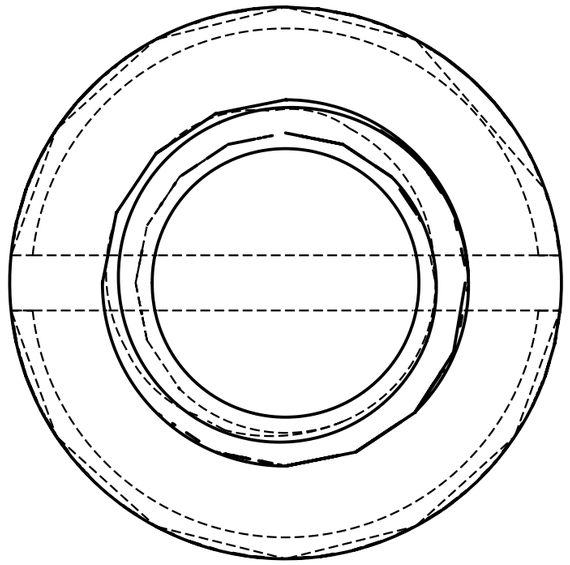
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 45.00.

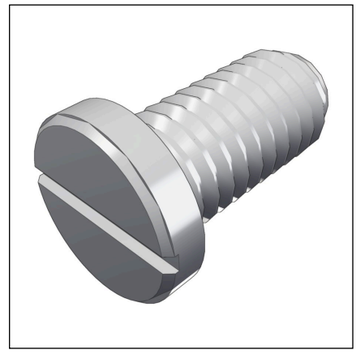
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

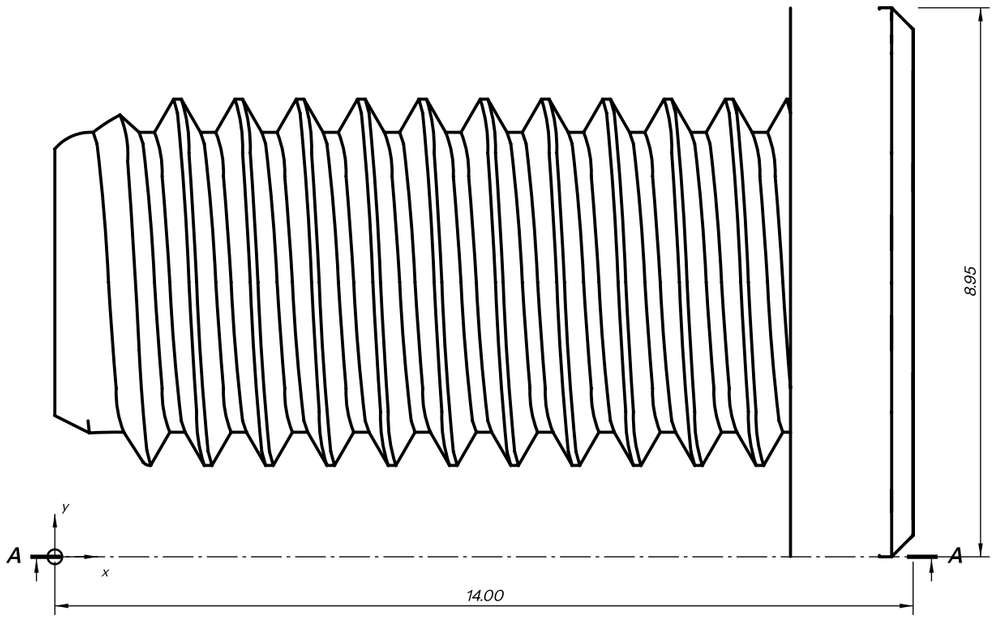
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	1.46, 4.00	10.00, 4.00		
Outline	2	line	10.00, 4.00	10.00, 0.00		
Outline	3	line	10.00, 0.00	0.00, 0.00		
Outline	4	line	0.00, 0.00	0.00, 4.00		
Outline	5	line	0.00, 4.00	1.46, 4.00		







Isometric view, not to scale



Hidden lines are left out of the plan (too many to read); the section and the hole table give the depths.

Werkstoff / Material not on the drawing	Maßstab / Scale **10:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **WN-zg** Teil / Part lid.tongueScrew
Behandlung / Treatment		Benennung / Name **Tongue screw**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Tongue screw — sheet 2

The screw the tongue turns on.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-zg	Schraube (the tongue's pivot screw, estimated)	1	not on the drawing		not measured on the model: see its function, The screw the tongue turns on.

What it does (WN-zg). The screw the tongue turns on.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 9.00.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

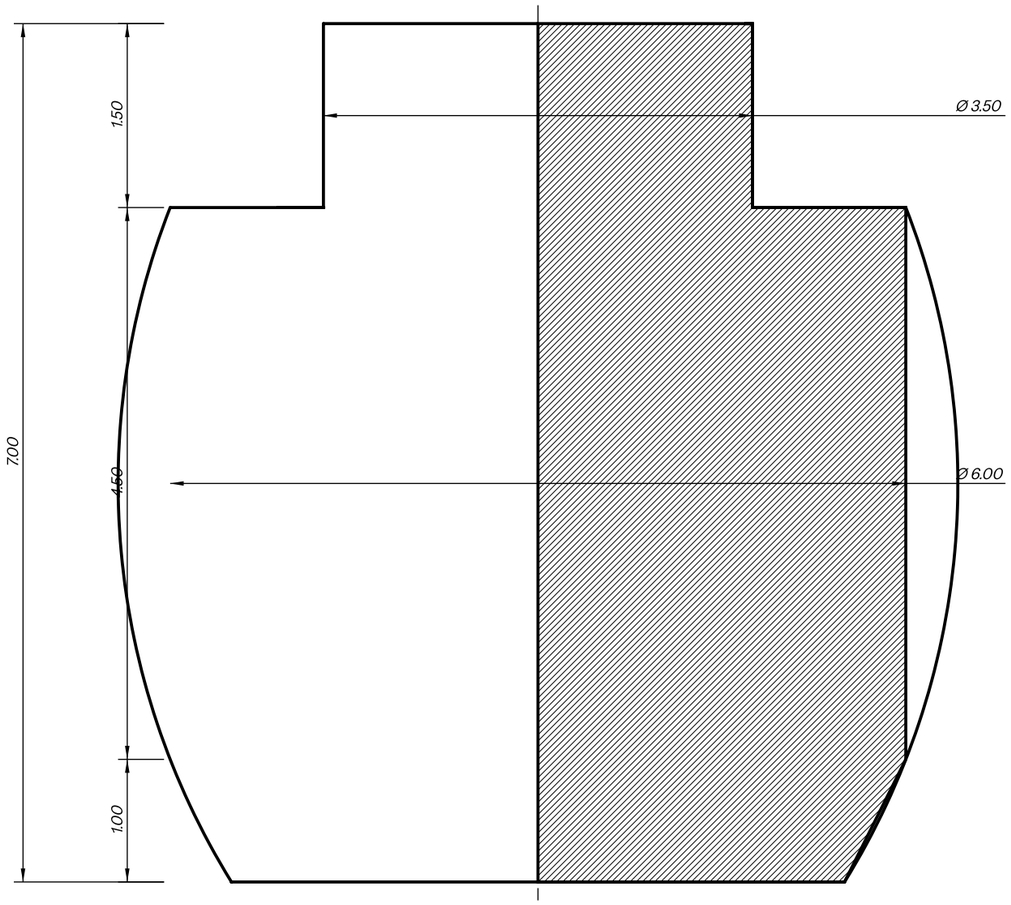
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 5.31	0.00, 6.65		
Outline	2	arc clockwise	0.00, 6.65	0.62, 6.92	0.58, 6.16	0.76
Outline	3	arc clockwise	0.62, 6.92	1.06, 7.21	2.18, 4.99	2.48
Outline	4	line	1.06, 7.21	1.38, 6.92		
Outline	5	line	1.38, 6.92	1.62, 6.92		
Outline	6	line	1.62, 6.92	1.94, 7.46		
Outline	7	line	1.94, 7.46	2.06, 7.46		
Outline	8	line	2.06, 7.46	2.38, 6.92		
Outline	9	line	2.38, 6.92	2.62, 6.92		
Outline	10	line	2.62, 6.92	2.94, 7.46		
Outline	11	line	2.94, 7.46	3.06, 7.46		
Outline	12	line	3.06, 7.46	3.38, 6.92		
Outline	13	line	3.38, 6.92	3.62, 6.92		
Outline	14	line	3.62, 6.92	3.94, 7.46		
Outline	15	line	3.94, 7.46	4.06, 7.46		
Outline	16	line	4.06, 7.46	4.38, 6.92		
Outline	17	line	4.38, 6.92	4.62, 6.92		
Outline	18	line	4.62, 6.92	4.94, 7.46		
Outline	19	line	4.94, 7.46	5.06, 7.46		
Outline	20	line	5.06, 7.46	5.38, 6.92		
Outline	21	line	5.38, 6.92	5.62, 6.92		
Outline	22	line	5.62, 6.92	5.94, 7.46		
Outline	23	line	5.94, 7.46	6.06, 7.46		
Outline	24	line	6.06, 7.46	6.38, 6.92		
Outline	25	line	6.38, 6.92	6.62, 6.92		
Outline	26	line	6.62, 6.92	6.94, 7.46		
Outline	27	line	6.94, 7.46	7.06, 7.46		
Outline	28	line	7.06, 7.46	7.38, 6.92		
Outline	29	line	7.38, 6.92	7.62, 6.92		

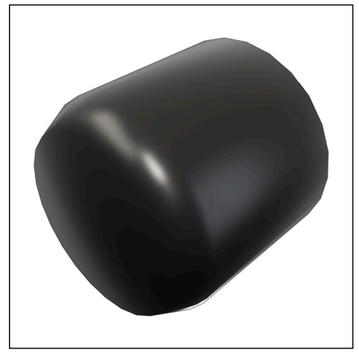
Outline	30	line	7.62, 6.92	7.94, 7.46		
Outline	31	line	7.94, 7.46	8.06, 7.46		
Outline	32	line	8.06, 7.46	8.38, 6.92		
Outline	33	line	8.38, 6.92	8.62, 6.92		
Outline	34	line	8.62, 6.92	8.94, 7.46		
Outline	35	line	8.94, 7.46	9.06, 7.46		
Outline	36	line	9.06, 7.46	9.38, 6.92		
Outline	37	line	9.38, 6.92	9.62, 6.92		
Outline	38	line	9.62, 6.92	9.94, 7.46		
Outline	39	line	9.94, 7.46	10.06, 7.46		
Outline	40	line	10.06, 7.46	10.38, 6.92		
Outline	41	line	10.38, 6.92	10.62, 6.92		
Outline	42	line	10.62, 6.92	10.94, 7.46		
Outline	43	line	10.94, 7.46	11.06, 7.46		
Outline	44	line	11.06, 7.46	11.38, 6.92		
Outline	45	line	11.38, 6.92	11.62, 6.92		
Outline	46	line	11.62, 6.92	11.94, 7.46		
Outline	47	line	11.94, 7.46	12.00, 7.46		
Outline	48	line	12.00, 7.46	12.00, 8.95		
Outline	49	line	12.00, 8.95	13.65, 8.95		
Outline	50	line	13.65, 8.95	14.00, 8.60		
Outline	51	line	14.00, 8.60	14.00, 0.35		
Outline	52	line	14.00, 0.35	13.65, 0.00		
Outline	53	line	13.65, 0.00	12.00, 0.00		
Outline	54	line	12.00, 0.00	12.00, 2.03		
Outline	55	line	12.00, 2.03	11.88, 2.03		
Outline	56	line	11.88, 2.03	11.56, 1.49		
Outline	57	line	11.56, 1.49	11.44, 1.49		
Outline	58	line	11.44, 1.49	11.12, 2.03		
Outline	59	line	11.12, 2.03	10.88, 2.03		
Outline	60	line	10.88, 2.03	10.56, 1.49		
Outline	61	line	10.56, 1.49	10.44, 1.49		
Outline	62	line	10.44, 1.49	10.12, 2.03		
Outline	63	line	10.12, 2.03	9.88, 2.03		
Outline	64	line	9.88, 2.03	9.56, 1.49		
Outline	65	line	9.56, 1.49	9.44, 1.49		
Outline	66	line	9.44, 1.49	9.12, 2.03		
Outline	67	line	9.12, 2.03	8.88, 2.03		
Outline	68	line	8.88, 2.03	8.56, 1.49		
Outline	69	line	8.56, 1.49	8.44, 1.49		
Outline	70	line	8.44, 1.49	8.12, 2.03		
Outline	71	line	8.12, 2.03	7.88, 2.03		
Outline	72	line	7.88, 2.03	7.56, 1.49		
Outline	73	line	7.56, 1.49	7.44, 1.49		
Outline	74	line	7.44, 1.49	7.12, 2.03		
Outline	75	line	7.12, 2.03	6.88, 2.03		
Outline	76	line	6.88, 2.03	6.56, 1.49		
Outline	77	line	6.56, 1.49	6.44, 1.49		

Outline	78	line	6.44, 1.49	6.12, 2.03		
Outline	79	line	6.12, 2.03	5.88, 2.03		
Outline	80	line	5.88, 2.03	5.56, 1.49		
Outline	81	line	5.56, 1.49	5.44, 1.49		
Outline	82	line	5.44, 1.49	5.12, 2.03		
Outline	83	line	5.12, 2.03	4.88, 2.03		
Outline	84	line	4.88, 2.03	4.56, 1.49		
Outline	85	line	4.56, 1.49	4.44, 1.49		
Outline	86	line	4.44, 1.49	4.12, 2.03		
Outline	87	line	4.12, 2.03	3.88, 2.03		
Outline	88	line	3.88, 2.03	3.56, 1.49		
Outline	89	line	3.56, 1.49	3.44, 1.49		
Outline	90	line	3.44, 1.49	3.12, 2.03		
Outline	91	line	3.12, 2.03	2.88, 2.03		
Outline	92	line	2.88, 2.03	2.56, 1.49		
Outline	93	line	2.56, 1.49	2.44, 1.49		
Outline	94	line	2.44, 1.49	2.12, 2.03		
Outline	95	line	2.12, 2.03	1.88, 2.03		
Outline	96	line	1.88, 2.03	1.56, 1.49		
Outline	97	arc clockwise	1.56, 1.49	1.44, 1.55	1.55, 1.63	0.14
Outline	98	line	1.44, 1.55	1.12, 2.03		
Outline	99	arc anticlockwise	1.12, 2.03	0.56, 2.02	1.00, -5.63	7.67
Outline	100	line	0.56, 2.02	0.00, 2.30		
Outline	101	line	0.00, 2.30	0.00, 5.31		
Cut-out 1	102	line	3.81, 4.48	3.69, 4.48		
Cut-out 1	103	line	3.69, 4.48	3.81, 4.48		
Cut-out 2	104	line	1.62, 4.48	1.54, 4.48		
Cut-out 2	105	line	1.54, 4.48	1.62, 4.48		

Levels

Faces in section A-A, from the bottom face: 4.05, 4.95.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **20:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **WN-zk** *Teil / Part* *lid.tongueKnob*
Behandlung / Treatment		*Benennung / Name* **Tongue knob**	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Tongue knob — sheet 2

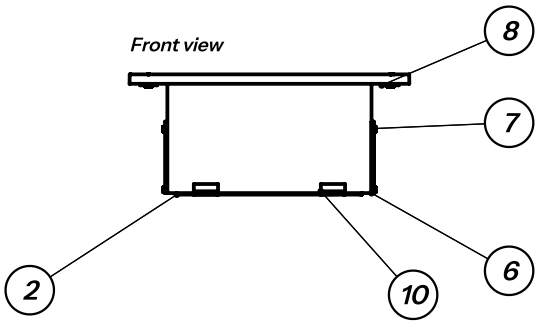
The knob in the tongue's M3.5, to lift it.

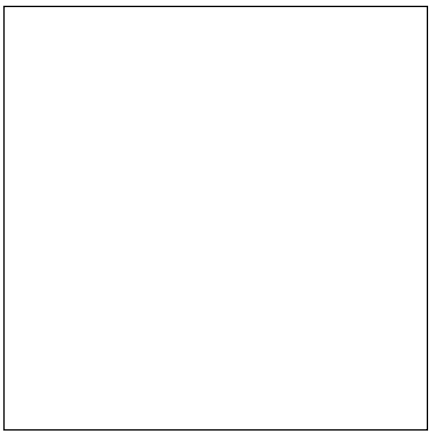
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-zk	Knopf (the tongue's knob, estimated)	1	not on the drawing		on P510002-2: 0.0181 mm

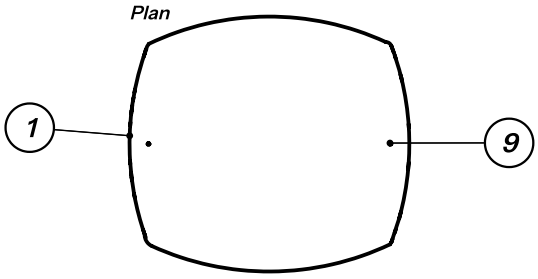
What it does (WN-zk). In the tongue's M3.5.

Turned about its axis; length 7.00, largest Ø 6.00. Diameters: Ø 3.50, Ø 6.00. Shoulders from the lower end: 0.00, 1.00, 5.50, 7.00.





Isometric view, not to scale



Not seen in either view (inside the assembly): Pos. 3, 4, 5, 11.

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
11	1	Hook eye	Hakenoese	not on the drawing
10	2	Hinges	127-556.01-05	not on the drawing
9	2	Eye screws	DIN 96	not on the drawing
8	2	Eyes	127-556.01-08	not on the drawing
7	4	Strap screws	WN-ur	not on the drawing
6	2	Strap	127-556.01-07	not on the drawing
5	1	Floor cushion	Polster.B	not on the drawing
4	2	End cushions	Polster.E	not on the drawing
3	2	Front and back cushions	Polster.S	not on the drawing
2	1	Rim strip	127-556.01-13, -14	not on the drawing
1	1	Outer box	127-556.01 (52 00 16)	„gebleichte Birke“

Werkstoff / Material: not on the drawing

Maßstab / Scale: 1:10

Behandlung / Treatment:

Blatt / Sheet: 1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

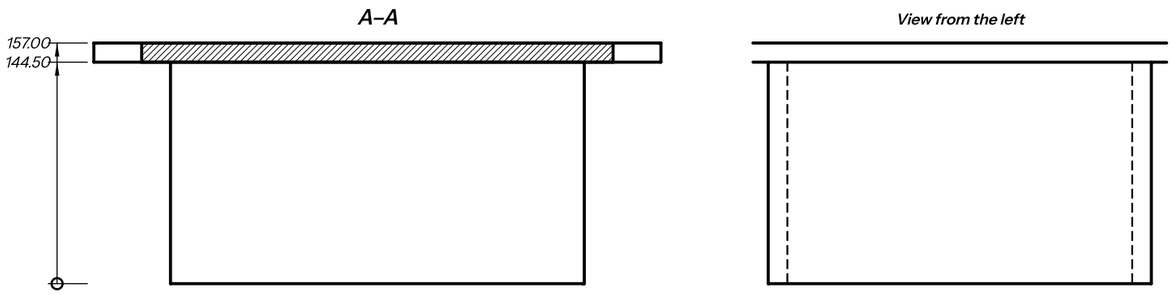
Zeichnung Nr. / Drawing no.: **Hakenoese**

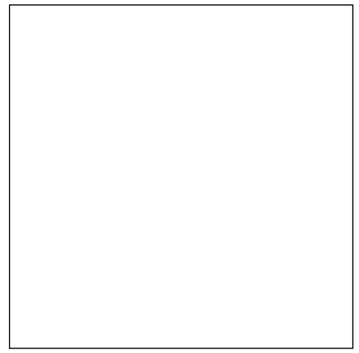
Teil / Part: overBox

Benennung / Name: **Outer box (Überkasten)**

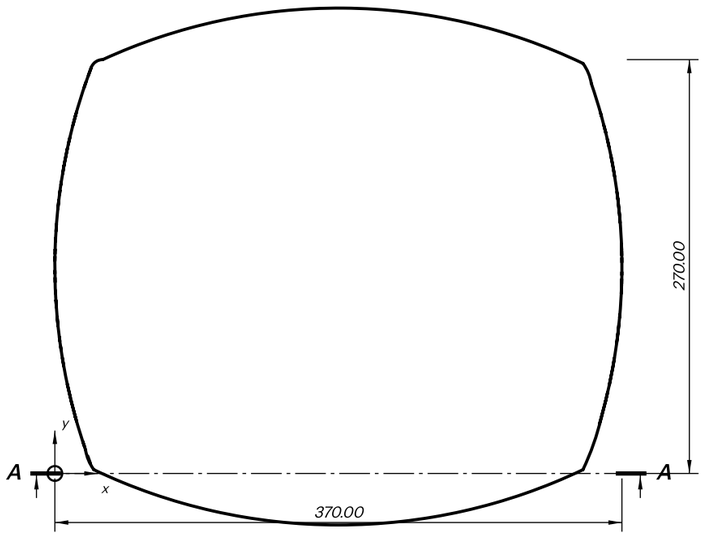
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).





Isometric view, not to scale



Hidden lines are left out of the plan (too many to read); the section and the hole table give the depths.

Werkstoff / Material „gebleichte Birke"	Maßstab / Scale **1:4**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **127-556.01 (52 00 16)**
Behandlung / Treatment none on the drawing			Teil / Part overBox.body
Blatt / Sheet **1 / 2**		Benennung / Name **Outer box**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Outer box — sheet 2

The padded box the wooden box travels in, bleached birch, dovetailed and glued. A cushion on each wall, on the floor and in the lid, each thickest in its middle, holds the wooden box with nothing to spare: 20 mm each side, its handles against the end cushions, 20 under it and 20 over it. Its base plate stands out at each end, where an eye is screwed; a strap screwed to each end wall goes over the lid to a buckle. The arrangement drawing, Boden-Chronometer (127-556. A-1, 1942), draws the two together with this lid open.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01 (52 00 16)	Überkasten (outer box): body and base plate, bleached birch	1	„gebleichte Birke“	none on the drawing	not measured on the model: see its function, Carries the wooden box for transport, cushioned all round.

What it does (127-556.01 (52 00 16)). Carries the wooden box for transport, cushioned all round.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 157.00.

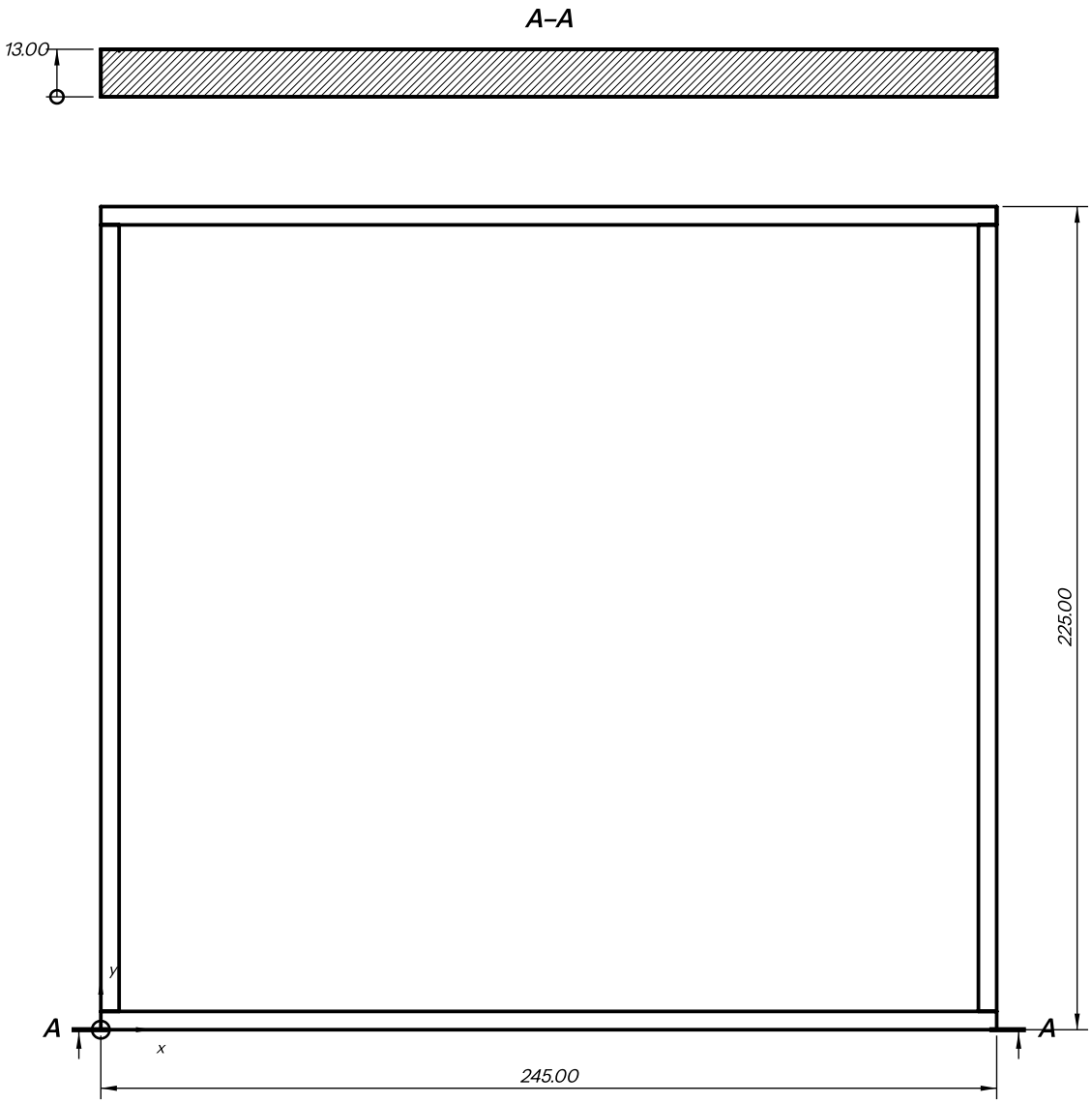
Contour in the plan

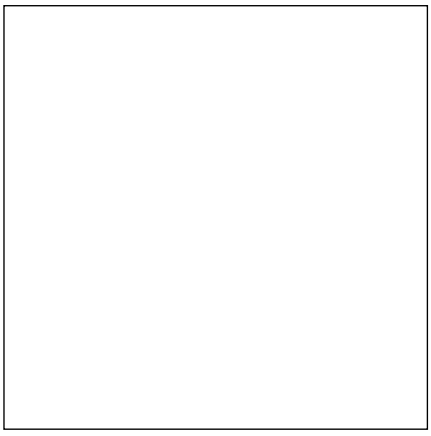
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	5.98, 69.01	0.00, 135.00	367.00, 135.00	367.00
Outline	2	arc clockwise	0.00, 135.00	23.73, 264.83	367.00, 135.00	367.00
Outline	3	arc clockwise	23.73, 264.83	31.21, 270.00	31.21, 262.00	8.00
Outline	4	arc clockwise	31.21, 270.00	340.39, 269.84	185.00, -1259.02	1536.73
Outline	5	arc clockwise	340.39, 269.84	346.27, 264.83	338.79, 262.00	8.00
Outline	6	arc clockwise	346.27, 264.83	370.00, 135.00	3.00, 135.00	367.00
Outline	7	arc clockwise	370.00, 135.00	346.27, 5.17	3.00, 135.00	367.00
Outline	8	arc clockwise	346.27, 5.17	338.79, 0.00	338.79, 8.00	8.00
Outline	9	arc clockwise	338.79, 0.00	29.61, 0.16	185.00, 1529.02	1536.73
Outline	10	arc clockwise	29.61, 0.16	23.73, 5.17	31.21, 8.00	8.00
Outline	11	arc clockwise	23.73, 5.17	5.98, 69.01	367.00, 135.00	367.00
Cut-out 1	12	line	238.11, 22.50	62.50, 57.69		
Cut-out 1	13	line	62.50, 57.69	238.11, 22.50		

Levels

Faces in section A-A, from the bottom face: 144.50, 157.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:2**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***127-556.01-13, -14*** *Teil / Part* *overBox.strip*
Behandlung / Treatment		*Benennung / Name* ***Rim strip***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Rim strip — sheet 2

A strip round the inside of the rim, standing 4 mm over the walls: the lid closes over it.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01-13, -14	Leiste (rim strip, 17, 18)	1	not on the drawing		not measured on the model: see its function, Inside the rim, 4 over the walls: the lid closes over it.

What it does (127-556.01-13, -14). Inside the rim, 4 over the walls: the lid closes over it.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 13.00.

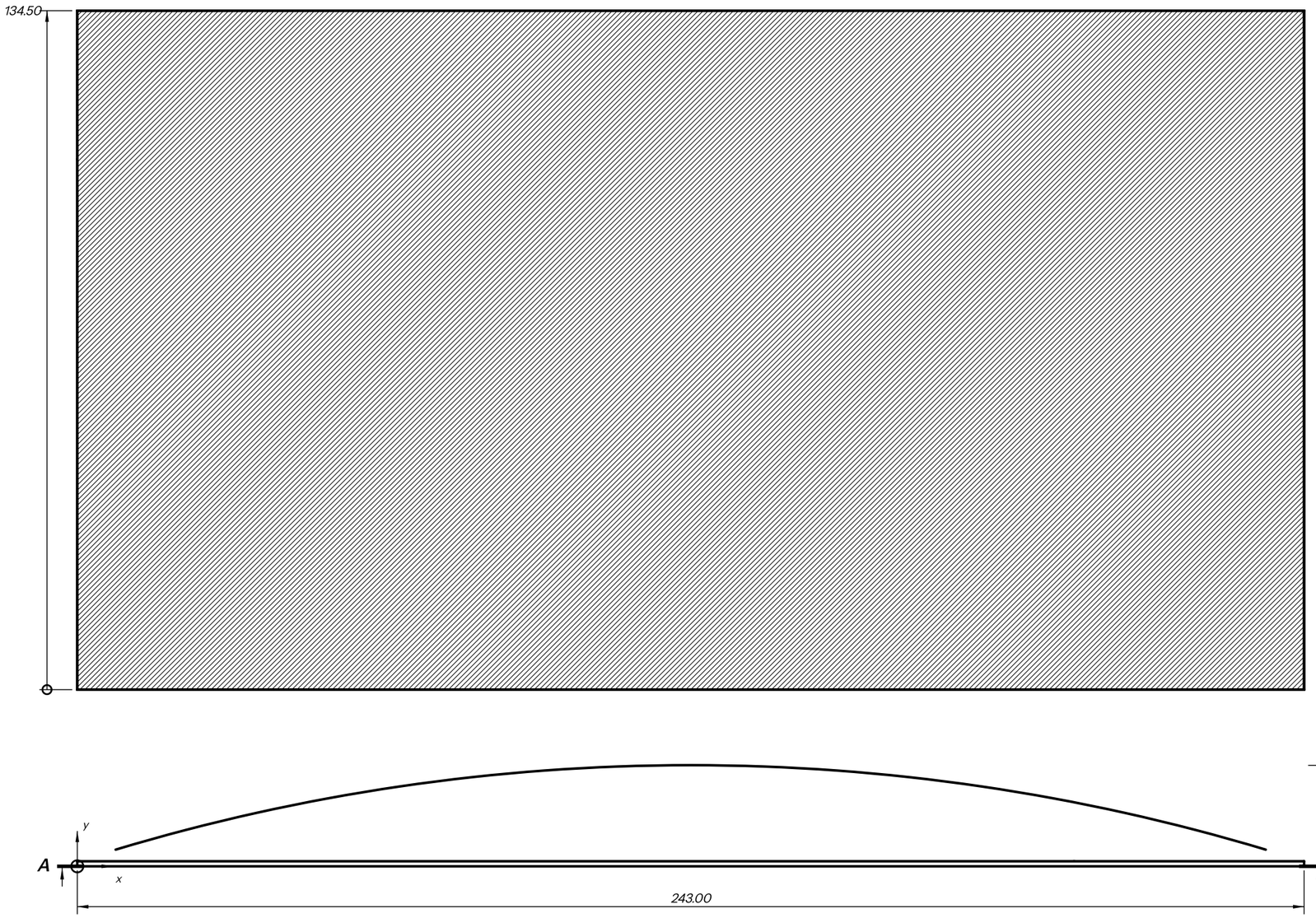
Contour in the plan

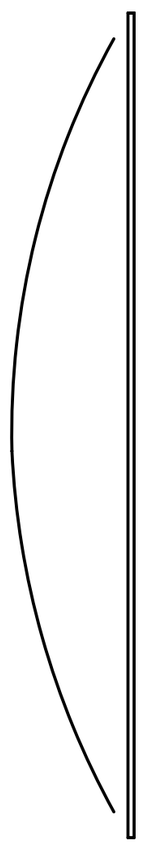
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

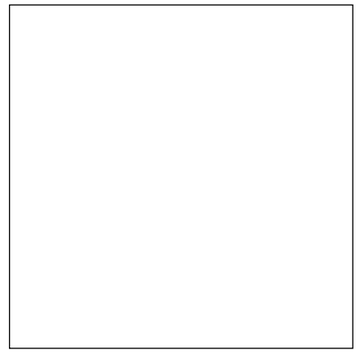
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.00	0.00, 225.00		
Outline	2	line	0.00, 225.00	245.00, 225.00		
Outline	3	line	245.00, 225.00	245.00, 0.00		
Outline	4	line	245.00, 0.00	0.00, 0.00		
Cut-out 1	5	line	5.00, 5.00	240.00, 5.00		
Cut-out 1	6	line	240.00, 5.00	240.00, 220.00		
Cut-out 1	7	line	240.00, 220.00	5.00, 220.00		
Cut-out 1	8	line	5.00, 220.00	5.00, 5.00		

Levels

Faces in section A-A, from the bottom face: 13.00.







Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***1:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Polster.S***
Behandlung / Treatment			*Teil / Part* *overBox.cushionsSide*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Front and back cushions***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Front and back cushions — sheet 2

The cushions on the long walls, against the wooden box's front and back.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Polster.S	Polster (front and back cushions)	2	not on the drawing		not measured on the model: see its function, Against the wooden box's front and back.

What it does (Polster.S). Against the wooden box's front and back.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 134.50.

Contour in the plan

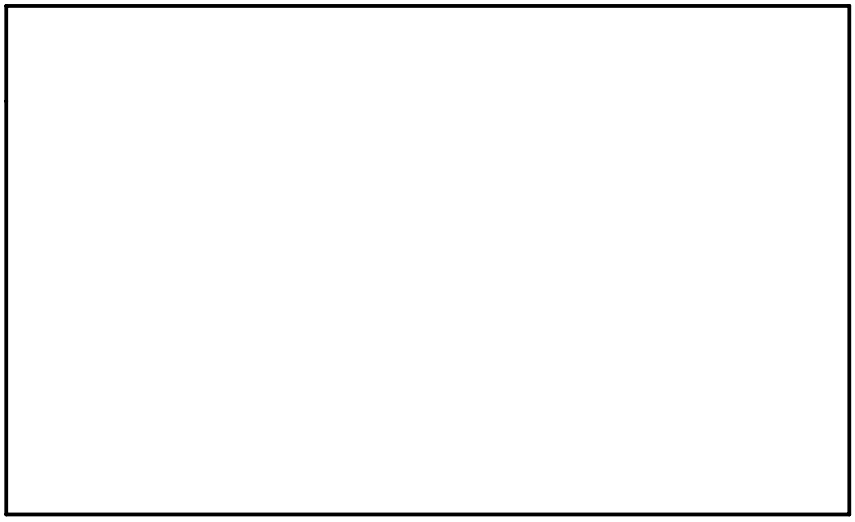
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	30.38, 9.31	113.91, 19.93	122.24, -379.00	399.02
Outline	2	arc clockwise	113.91, 19.93	205.03, 11.02	121.19, -375.36	395.36
Outline	3	arc clockwise	205.03, 11.02	243.00, 1.00	114.39, -409.31	429.99
Outline	4	line	243.00, 1.00	243.00, 0.00		
Outline	5	line	243.00, 0.00	0.00, 0.00		
Outline	6	line	0.00, 0.00	0.00, 1.00		
Outline	7	arc clockwise	0.00, 1.00	30.38, 9.31	129.44, -412.29	433.08
Cut-out 1	8	line	26.10, 7.69	26.10, 7.69		
Cut-out 1	9	arc anticlockwise	26.10, 7.69	26.10, 7.69	73.80, 8.84	47.71
Cut-out 1	10	line	26.10, 7.69	26.10, 7.69		

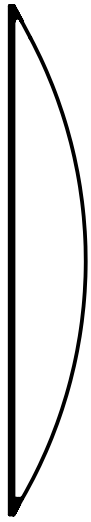
Levels

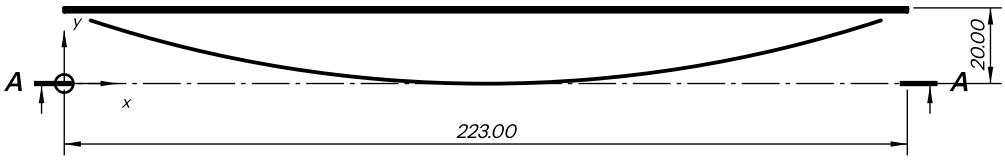
Faces in section A-A, from the bottom face: 134.50.

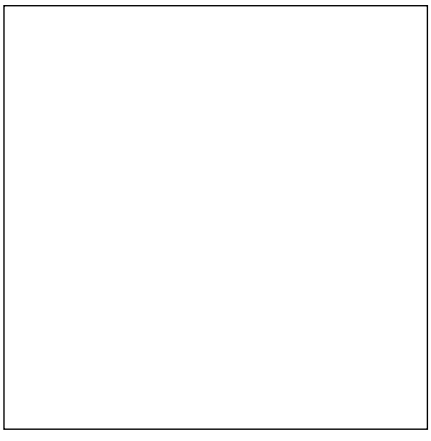
A–A



View from the left







Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **1:2**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **Polster.E** Teil / Part overBox.cushionsEnd
Behandlung / Treatment		Benennung / Name **End cushions**	
Blatt / Sheet **1 / 2**		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

End cushions — sheet 2

The cushions on the end walls, against the wooden box's handles.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Polster.E	Polster (end cushions)	2	not on the drawing		not measured on the model: see its function, Against the wooden box's handles.

What it does (Polster.E). Against the wooden box's handles.

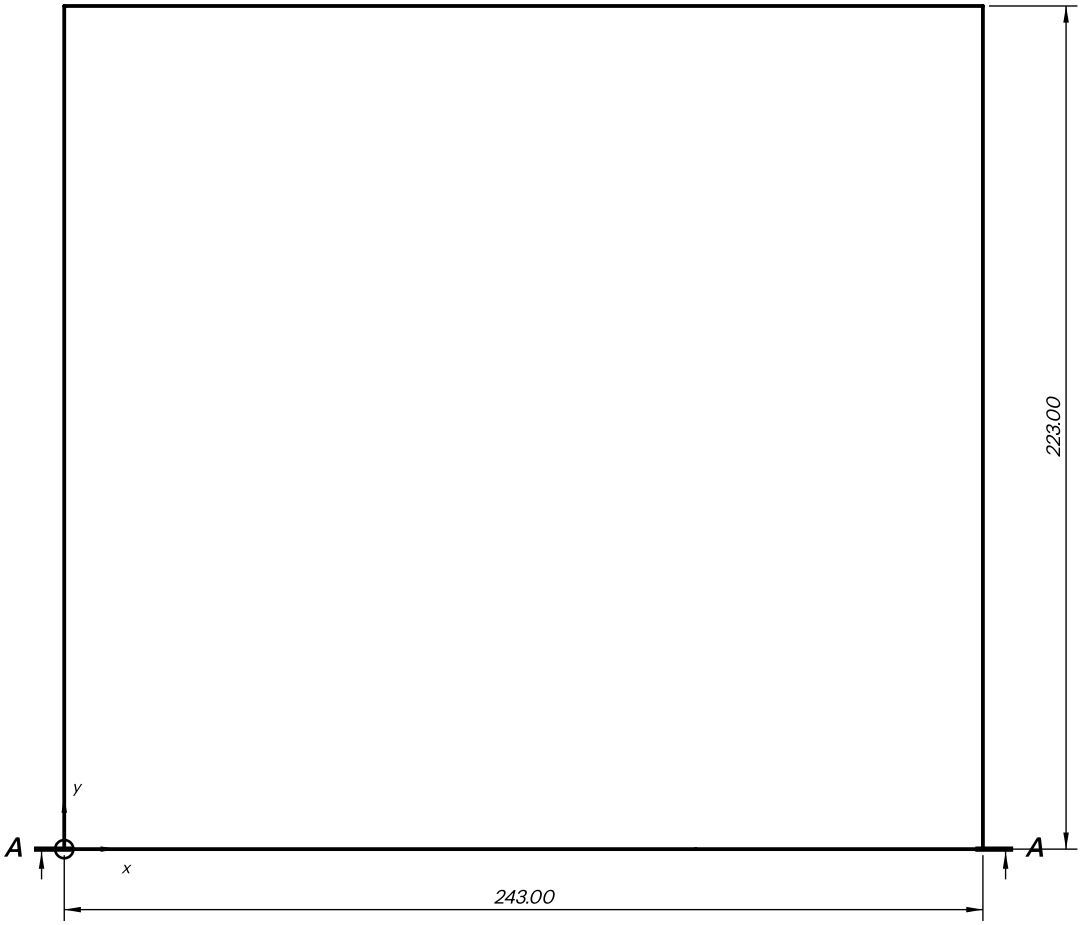
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 134.50.

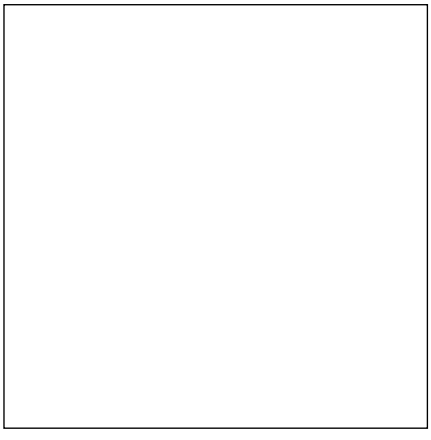
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	6.97, 20.00	223.00, 20.00		
Outline	2	line	223.00, 20.00	223.00, 19.00		
Outline	3	arc clockwise	223.00, 19.00	160.28, 3.64	106.77, 357.37	357.77
Outline	4	arc clockwise	160.28, 3.64	48.78, 6.01	111.48, 329.65	329.65
Outline	5	arc clockwise	48.78, 6.01	0.00, 19.00	117.62, 362.43	363.01
Outline	6	line	0.00, 19.00	0.00, 20.00		
Outline	7	line	0.00, 20.00	6.97, 20.00		







Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:2**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Polster.B*** *Teil / Part* *overBox.cushionFloor*
Behandlung / Treatment		*Benennung / Name* ***Floor cushion***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Floor cushion — sheet 2

The cushion on the floor the wooden box stands on.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Polster.B	Polster (floor cushion)	1	not on the drawing		not measured on the model: see its function, The wooden box stands on it.

What it does (Polster.B). The wooden box stands on it.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 20.00.

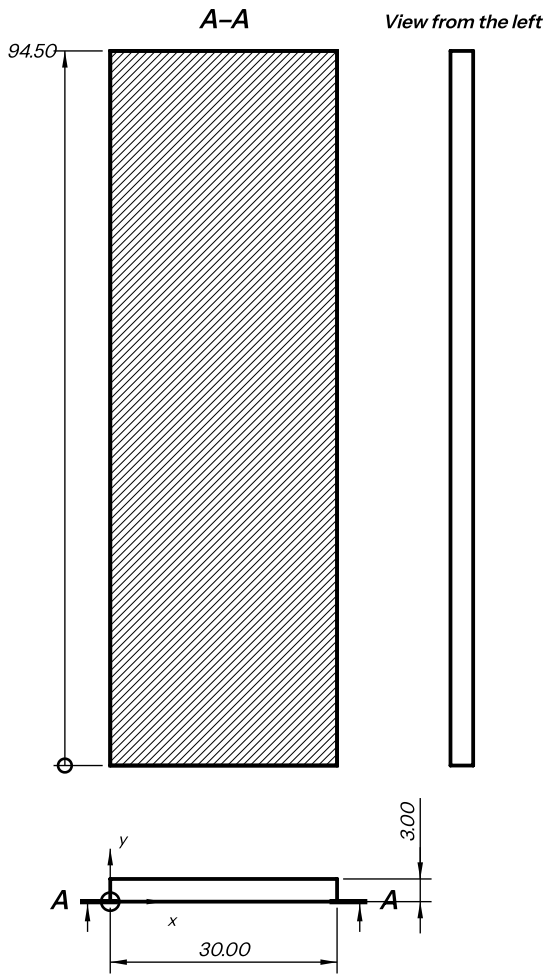
Contour in the plan

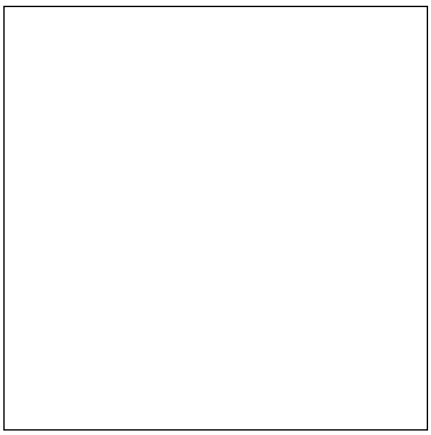
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 62.72	0.00, 223.00		
Outline	2	line	0.00, 223.00	243.00, 223.00		
Outline	3	line	243.00, 223.00	243.00, 0.00		
Outline	4	line	243.00, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 62.72		

Levels

Faces in section A-A, from the bottom face: 19.00, 20.00.





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **127-556.01-07** *Teil / Part* *overBox.strap*
Behandlung / Treatment		*Benennung / Name* ***Strap***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Strap — sheet 2

The strap's ends, screwed to the end walls: it goes over the lid to its buckle. With the lid open its loose ends fall off; they and the buckle aren't drawn.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01-07	Riemen (strap, 10): its ends on the end walls	2	not on the drawing		not measured on the model: see its function, Over the lid to its buckle (its loose ends and buckle not drawn).

What it does (127-556.01-07). Over the lid to its buckle (its loose ends and buckle not drawn).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 94.50.

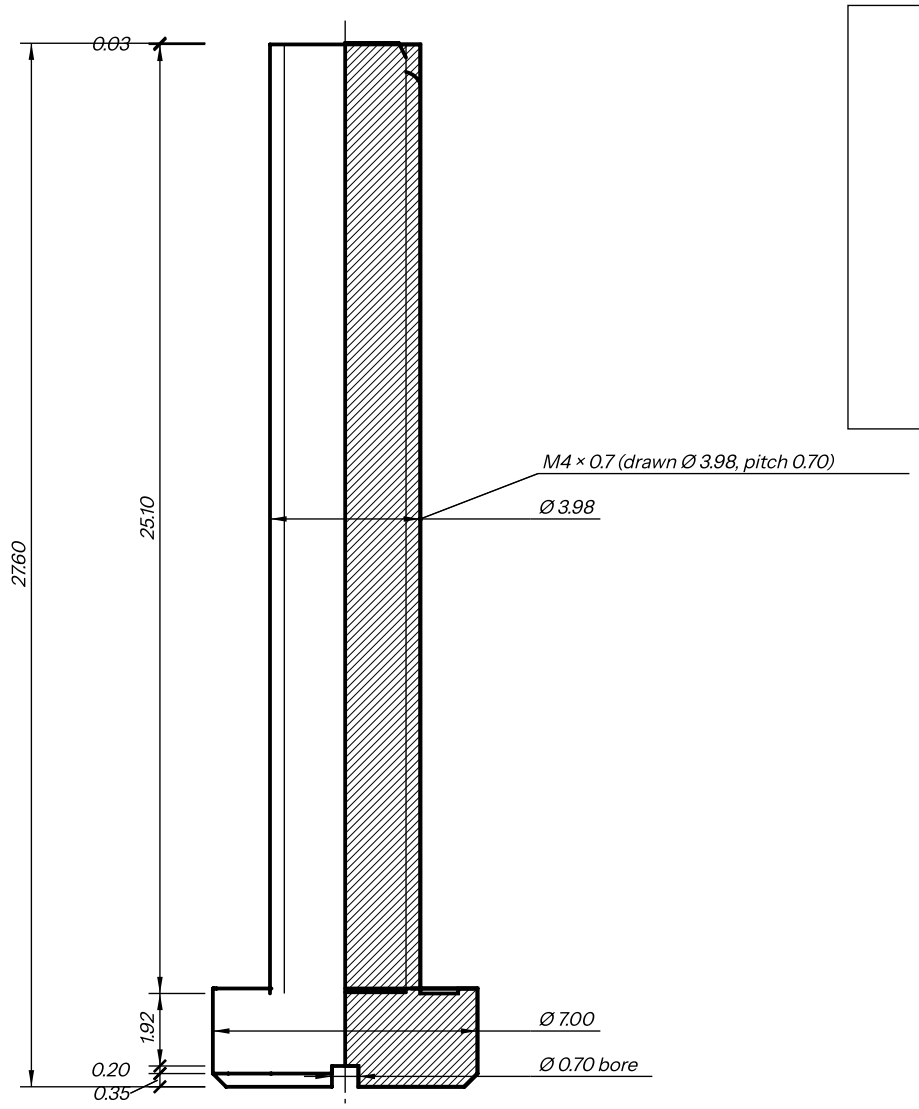
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	30.00, 0.00	0.00, 0.00		
Outline	2	line	0.00, 0.00	0.00, 3.00		
Outline	3	line	0.00, 3.00	30.00, 3.00		
Outline	4	line	30.00, 3.00	30.00, 0.00		

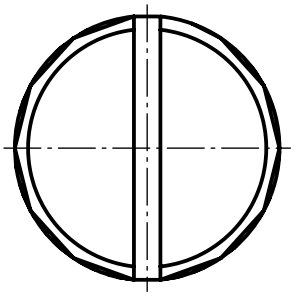
Levels

Faces in section A-A, from the bottom face: 94.50.



Isometric view, not to scale

View from below (seen from beneath, x reversed)



4 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***WN-ur*** *Teil / Part* *overBox.strapScrews*
Behandlung / Treatment		*Benennung / Name* ***Strap screws***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Strap screws — sheet 2

Two screws on each end wall through the strap (“Schraube f. Riemenbef.”).

4 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

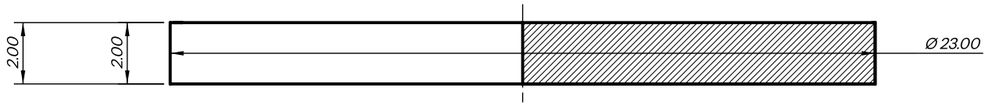
Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
WN-ur	Schraube f. Riemenbef. (the strap's screws)	4	not on the drawing		not measured on the model: see its function, Two on each end wall, 80 apart.

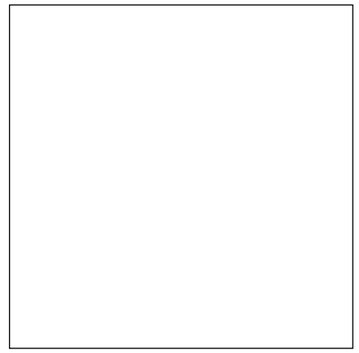
What it does (WN-ur). Two on each end wall, 80 apart.

Turned about its axis; length 27.60, largest Ø 7.00. Diameters: Ø 3.98, Ø 7.00; bores Ø 0.70. Shoulders from the lower end: 0.00, 0.35, 0.55, 2.47, 27.57, 27.60.

Thread

Outside thread 2.47–27.57 from the lower end, 25.10 long: drawn Ø 3.98 over its crests, 3.22 at its roots, pitch 0.70. Cut the ISO 261 coarse thread nearest it, M4 × 0.7, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **127-556.01-08**
Behandlung / Treatment			*Teil / Part* *overBox.eyes*
Blatt / Sheet **1 / 2**		*Benennung / Name* ***Eyes***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Eyes — sheet 2

An eye screwed on each end of the base plate.

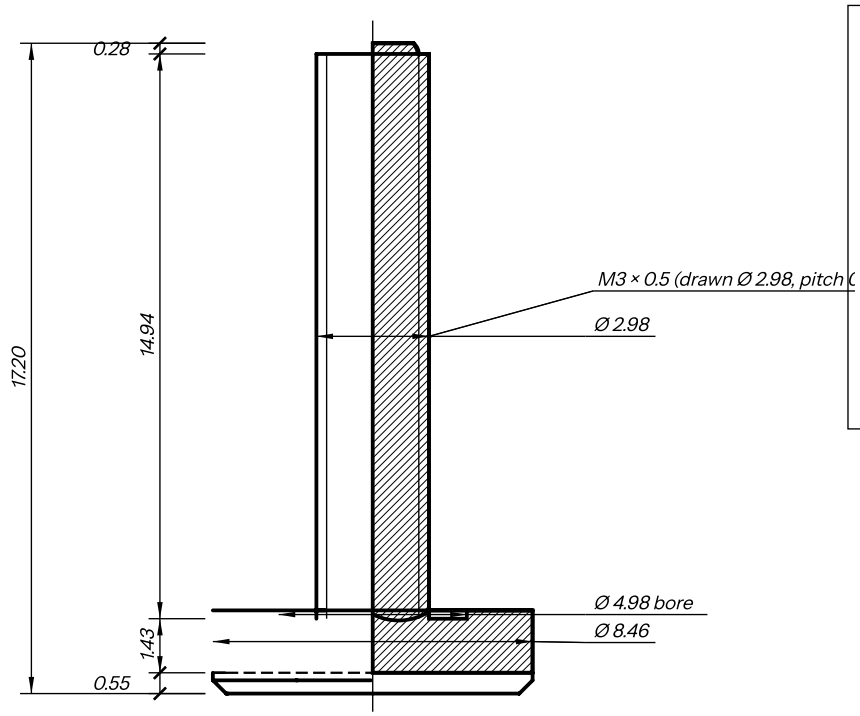
2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01-08	Öse (eye, 12)	2	not on the drawing		not measured on the model: see its function, On each end of the base plate.

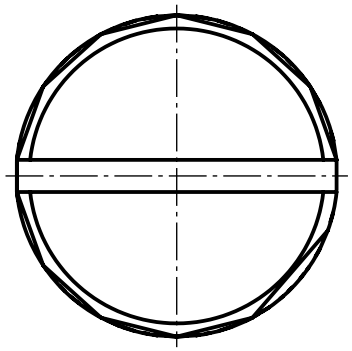
What it does (127-556.01-08). On each end of the base plate.

Turned about its axis; length 2.00, largest Ø 23.00. Diameters: Ø 23.00. Shoulders from the lower end: 0.00, 2.00.



Isometric view, not to scale

View from below (seen from beneath, x reversed)



2 made; one drawn.

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***DIN 96***
			Teil / Part *overBox.eyeScrews*
Behandlung / Treatment		*Benennung / Name* ***Eye screws***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Eye screws — sheet 2

The eyes' screws.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
DIN 96	Schraube 3 x 15 DIN 96 (11)	2	not on the drawing		not measured on the model: see its function, The eyes' screws.

What it does (DIN 96). The eyes' screws.

Turned about its axis; length 17.20, largest Ø 8.46. Diameters: Ø 2.98, Ø 8.46; bores Ø 4.98. Shoulders from the lower end: 0.00, 0.55, 1.98, 16.92, 17.20.

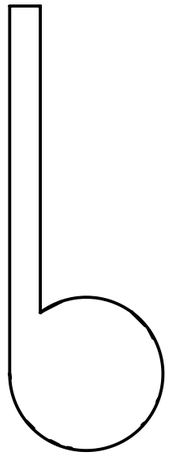
Thread

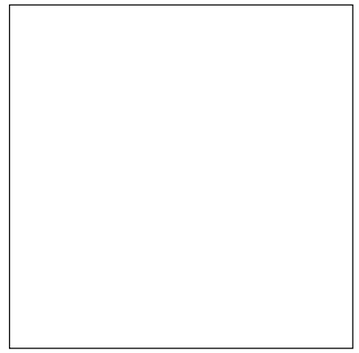
Outside thread 1.98–16.92 from the lower end, 14.94 long: drawn Ø 2.98 over its crests, 2.44 at its roots, pitch 0.50. Cut the ISO 261 coarse thread nearest it, M3 × 0.5, where the part's own sheet gives none (the Wempe-Norm screws' sheets aren't in the set).

A–A

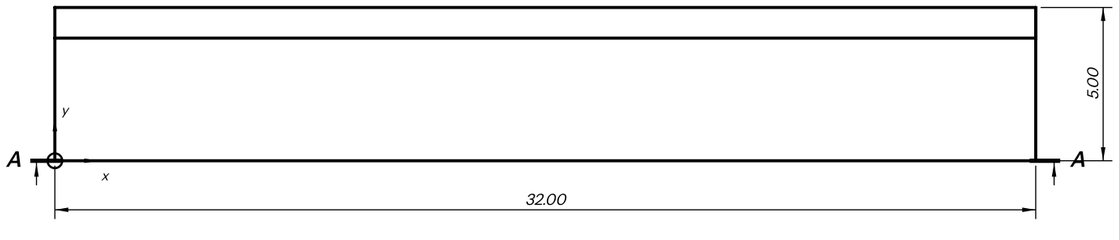


View from the left





Isometric view, not to scale



2 made; one drawn.

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **127-556.01-05**
Behandlung / Treatment			Teil / Part overBox.hinges
Blatt / Sheet 1 / 2		Benennung / Name **Hinges**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Hinges — sheet 2

Two hinges at the back joining the lid to the box.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01-05	Scharnier (outer box hinge, 7)	2	not on the drawing		not measured on the model: see its function, Join the lid to the body at the back.

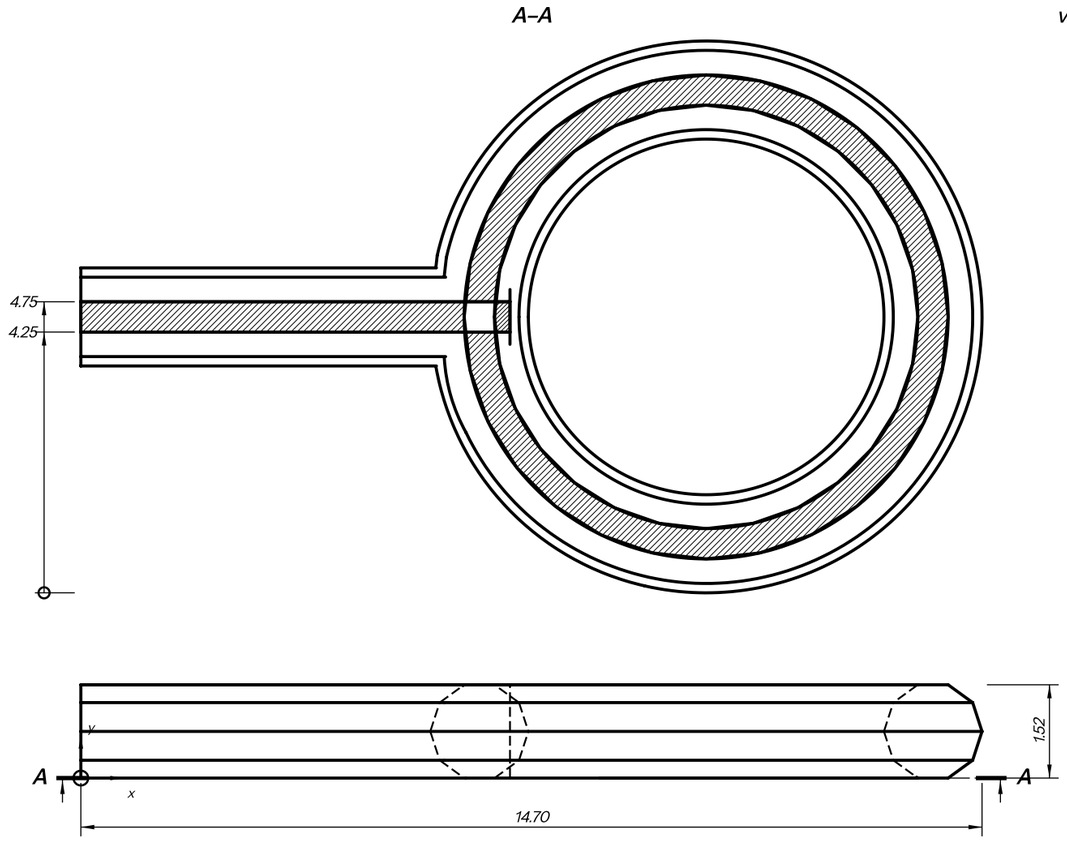
What it does (127-556.01-05). Join the lid to the body at the back.

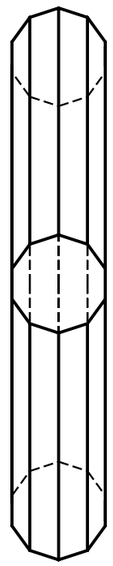
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 14.50.

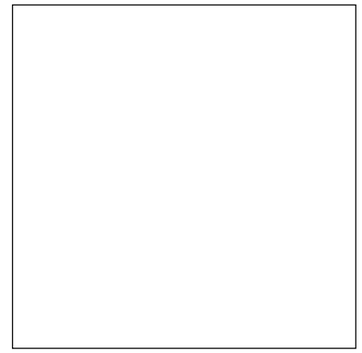
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 1.03	0.00, 5.00		
Outline	2	line	0.00, 5.00	32.00, 5.00		
Outline	3	line	32.00, 5.00	32.00, 0.00		
Outline	4	line	32.00, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 1.03		







Isometric view, not to scale

Werkstoff / Material not on the drawing	Maßstab / Scale **10:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **Hakenoese**
Behandlung / Treatment			Teil / Part overBox.eye
Blatt / Sheet **1 / 2**		Benennung / Name **Hook eye**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Hook eye — sheet 2

The eye on the box's front that the lid's hook catches.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Hakenoese	Öse (the hook's eye, 9: commercial)	1	not on the drawing		not measured on the model: see its function, On the front, under the rim.

What it does (Hakenoese). On the front, under the rim.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 9.00.

Contour in the plan

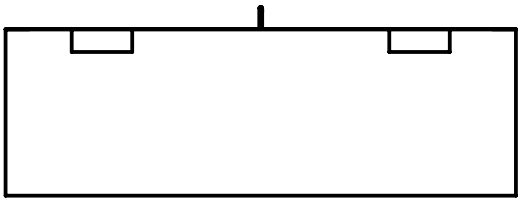
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

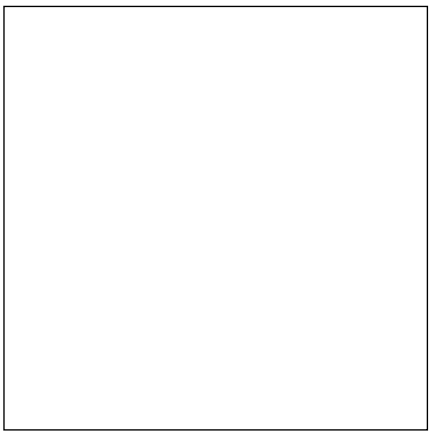
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	6.83, 0.00	0.00, 0.00		
Outline	2	line	0.00, 0.00	0.00, 1.52		
Outline	3	line	0.00, 1.52	14.15, 1.52		
Outline	4	line	14.15, 1.52	14.55, 1.23		
Outline	5	line	14.55, 1.23	14.70, 0.76		
Outline	6	line	14.70, 0.76	14.55, 0.29		
Outline	7	line	14.55, 0.29	14.15, 0.00		
Outline	8	line	14.15, 0.00	6.83, 0.00		

Levels

Faces in section A–A, from the bottom face: 4.25, 4.75.

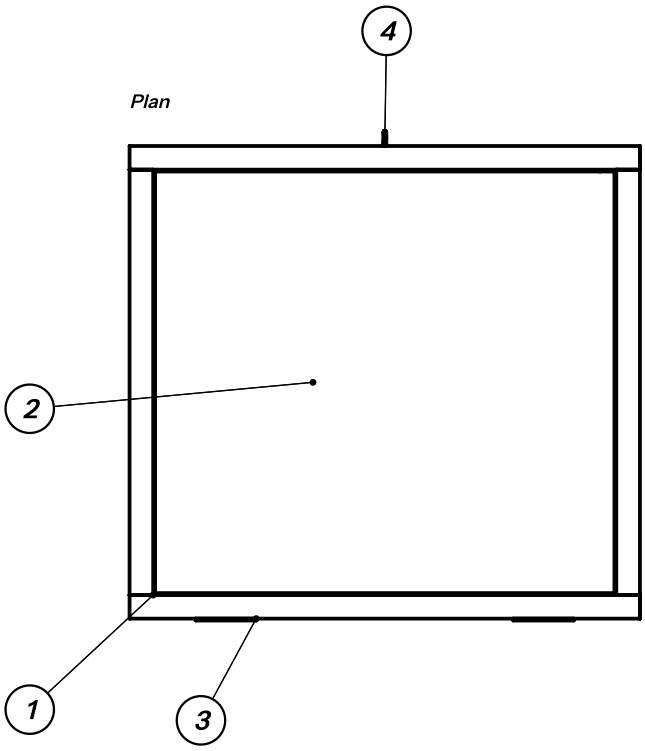
Front view





Isometric view, not to scale

Plan



Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
4	1	Hook	Haken	not on the drawing
3	2	Hinge leaves	127-556.01-05	not on the drawing
2	1	Lid cushion	127-556.01-12	not on the drawing
1	1	Outer box lid	127-556.01	not on the drawing

Werkstoff / Material: not on the drawing

Maßstab / Scale: **1:4**

Behandlung / Treatment:

Blatt / Sheet: **1 / 1**

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.: **Haken**

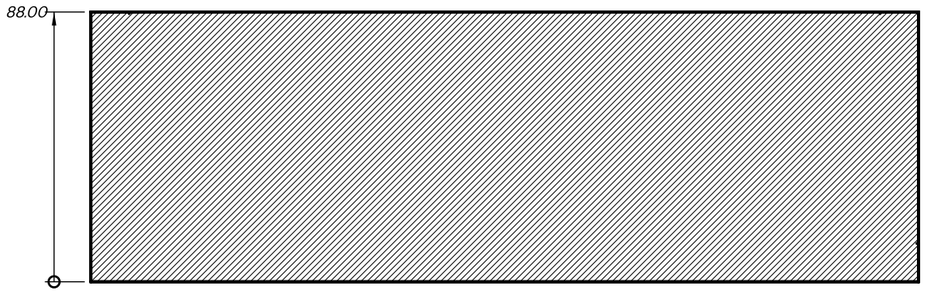
Teil / Part: overLid

Benennung / Name: **Outer box lid**

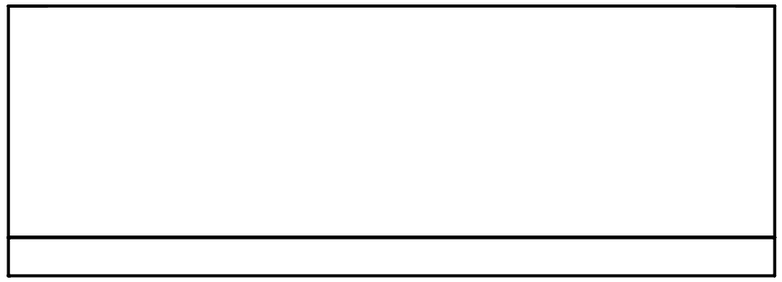
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

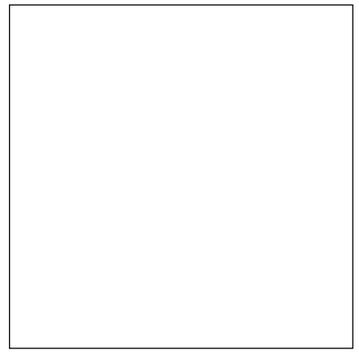
First-angle projection (DIN ISO 5456-2).

A–A

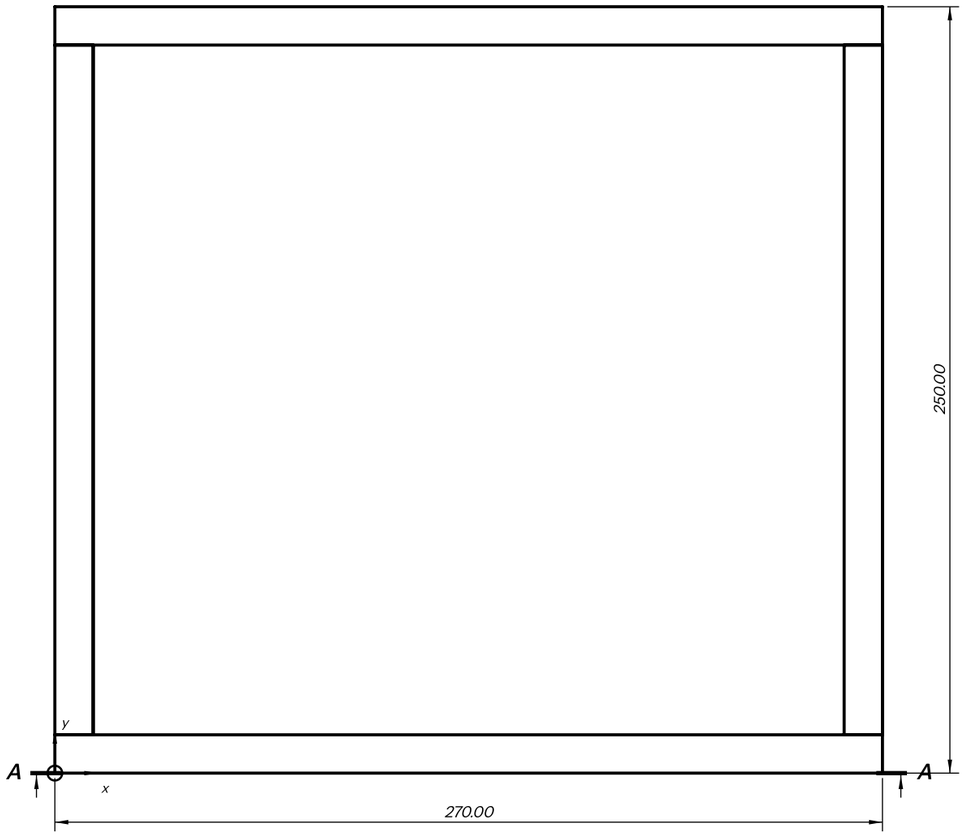


View from the left





Isometric view, not to scale



Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:2**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **127-556.01** *Teil / Part* *overLid.lid*
Behandlung / Treatment		*Benennung / Name* ***Outer box lid***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Outer box lid — sheet 2

The outer box's lid, hinged at the back on two hinges, a cushion in its top that bears on the wooden box's lid; a hook on its front catches an eye on the box.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01	Deckel (the outer box's lid)	1	not on the drawing		not measured on the model: see its function, Hinged at the back, cushioned inside.

What it does (127-556.01). Hinged at the back, cushioned inside.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 88.00.

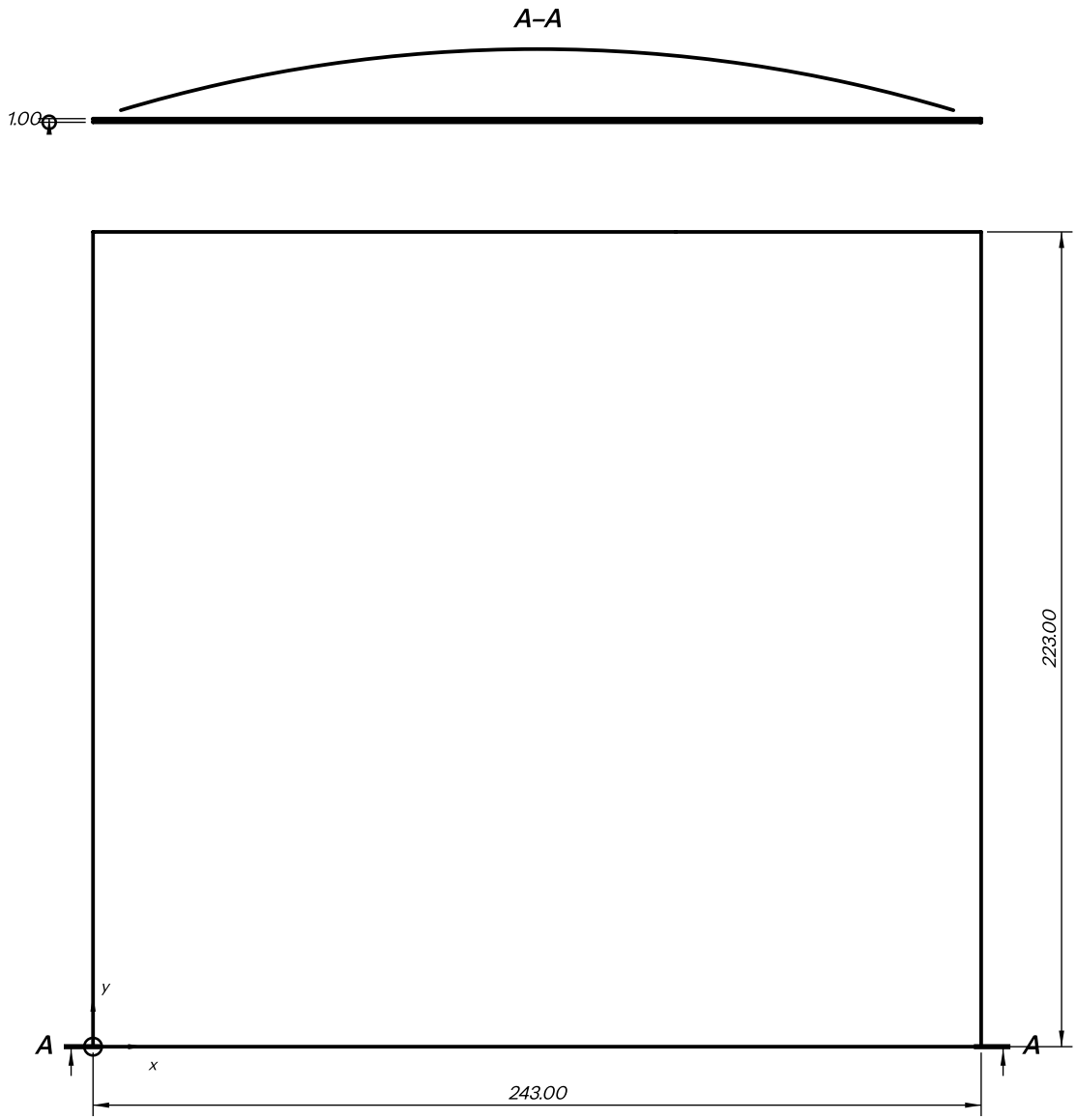
Contour in the plan

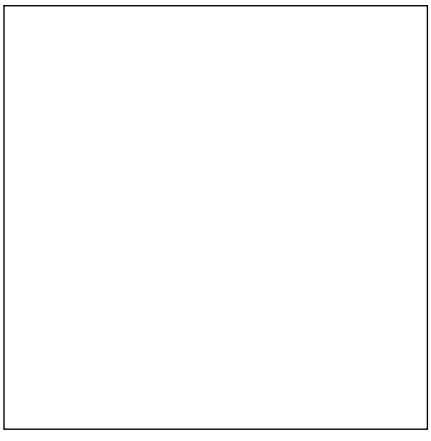
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 0.00	0.00, 250.00		
Outline	2	line	0.00, 250.00	270.00, 250.00		
Outline	3	line	270.00, 250.00	270.00, 0.00		
Outline	4	line	270.00, 0.00	0.00, 0.00		

Levels

Faces in section A-A, from the bottom face: 88.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **1:2**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* **127-556.01-12** *Teil / Part* *overLid.cushion*
Behandlung / Treatment		*Benennung / Name* ***Lid cushion***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Lid cushion — sheet 2

The cushion in the lid’s top, bearing on the wooden box’s lid.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01-12	Polster (lid cushion, 16)	1	not on the drawing		not measured on the model: see its function, Bears on the wooden box's lid.

What it does (127-556.01-12). Bears on the wooden box’s lid.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 20.00.

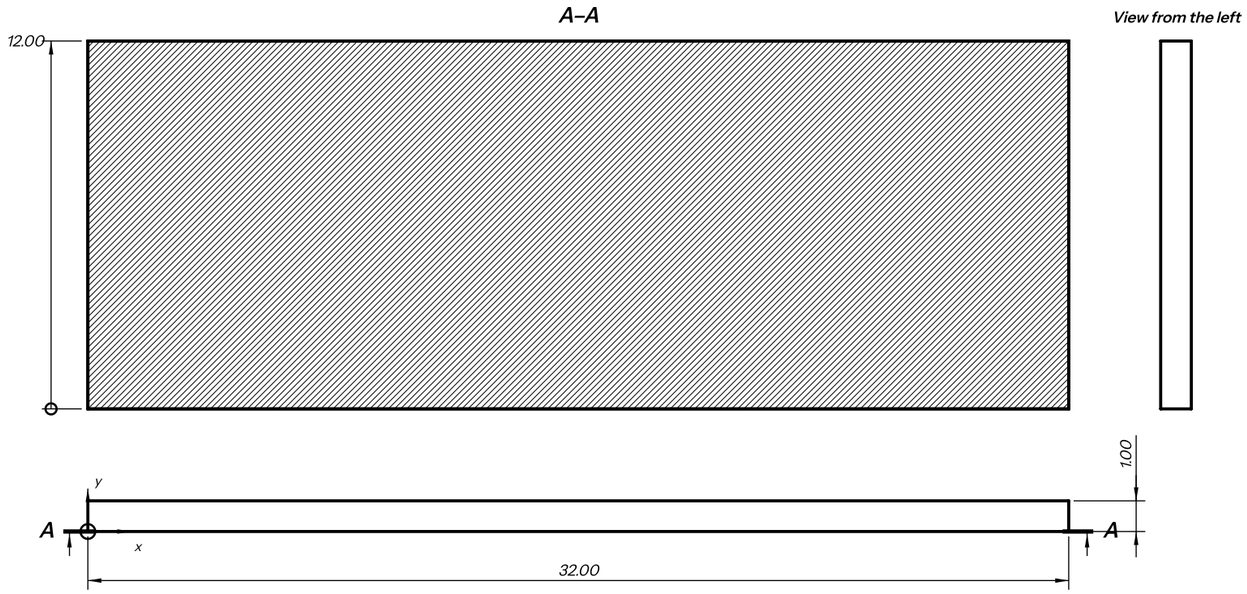
Contour in the plan

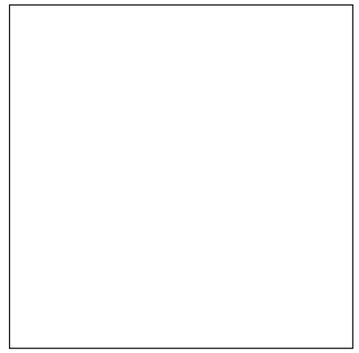
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 55.75	0.00, 223.00		
Outline	2	line	0.00, 223.00	243.00, 223.00		
Outline	3	line	243.00, 223.00	243.00, 0.00		
Outline	4	line	243.00, 0.00	0.00, 0.00		
Outline	5	line	0.00, 0.00	0.00, 55.75		

Levels

Faces in section A-A, from the bottom face: 1.00.





Isometric view, not to scale

2 made; one drawn.

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***127-556.01-05*** *Teil / Part* *overLid.hinges*
Behandlung / Treatment		*Benennung / Name* ***Hinge leaves***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Hinge leaves — sheet 2

The hinges' leaves on the lid.

2 made, all alike; the drawing shows one, in its own frame (the holes' and contour's coordinates are that one's).

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
127-556.01-05	Scharnier (outer box hinge, 7)	2	not on the drawing		not measured on the model: see its function, Join the lid to the body at the back.

What it does (127-556.01-05). Join the lid to the body at the back.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 12.00.

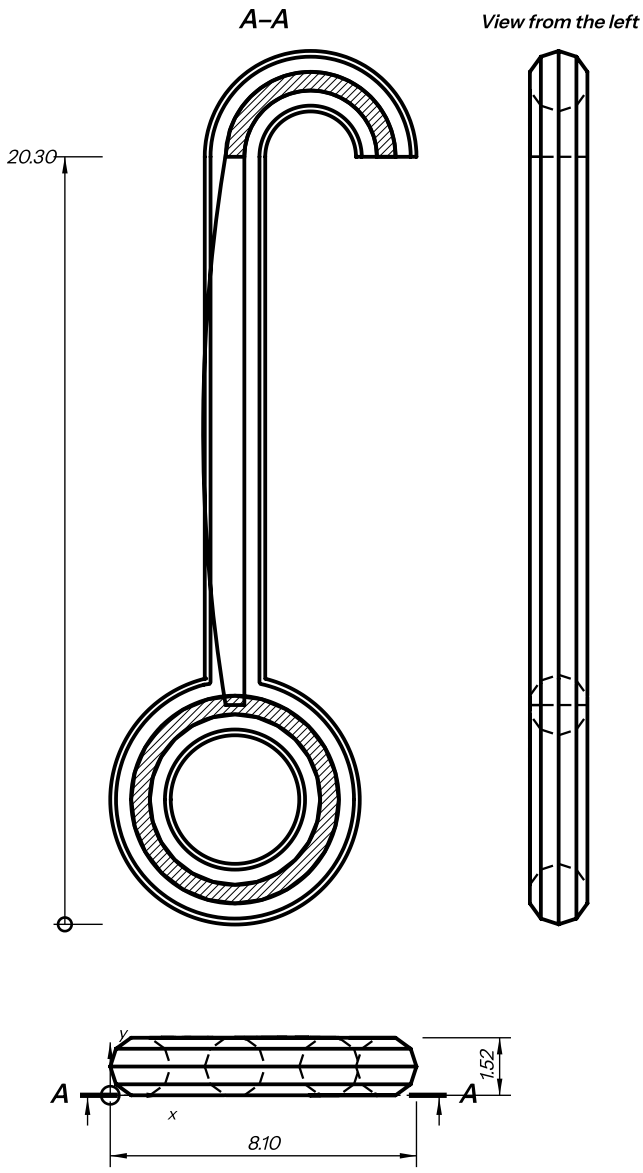
Contour in the plan

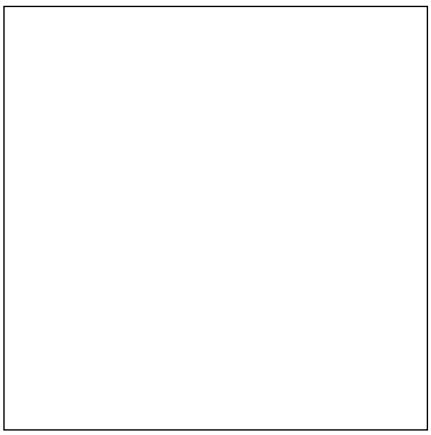
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	32.00, 1.00	32.00, 0.00		
Outline	2	line	32.00, 0.00	0.00, 0.00		
Outline	3	line	0.00, 0.00	0.00, 1.00		
Outline	4	line	0.00, 1.00	32.00, 1.00		

Levels

Faces in section A-A, from the bottom face: 12.00.





Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **5:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***Haken*** *Teil / Part* *overLid.hook*
Behandlung / Treatment		*Benennung / Name* ***Hook***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Hook — sheet 2

The hook on the lid's front, caught in the box's eye.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
Haken	Haken (hook, 9: commercial)	1	not on the drawing		not measured on the model: see its function, On the lid's front, caught in the eye.

What it does (Haken). On the lid's front, caught in the eye.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 23.10.

Contour in the plan

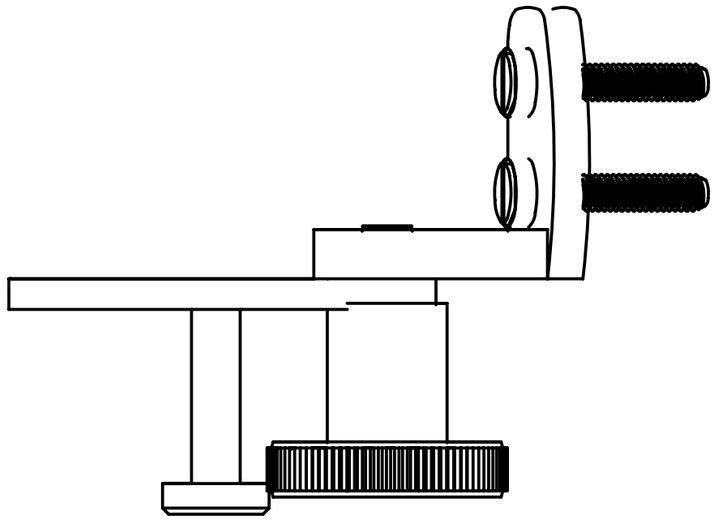
The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.92, 0.00	0.55, 0.00		
Outline	2	line	0.55, 0.00	0.15, 0.29		
Outline	3	line	0.15, 0.29	0.00, 0.76		
Outline	4	line	0.00, 0.76	0.15, 1.23		
Outline	5	line	0.15, 1.23	0.55, 1.52		
Outline	6	line	0.55, 1.52	7.55, 1.52		
Outline	7	line	7.55, 1.52	7.95, 1.23		
Outline	8	line	7.95, 1.23	8.10, 0.76		
Outline	9	line	8.10, 0.76	7.95, 0.29		
Outline	10	line	7.95, 0.29	7.55, 0.00		
Outline	11	line	7.55, 0.00	0.92, 0.00		

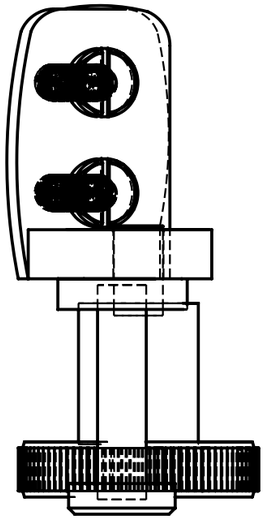
Levels

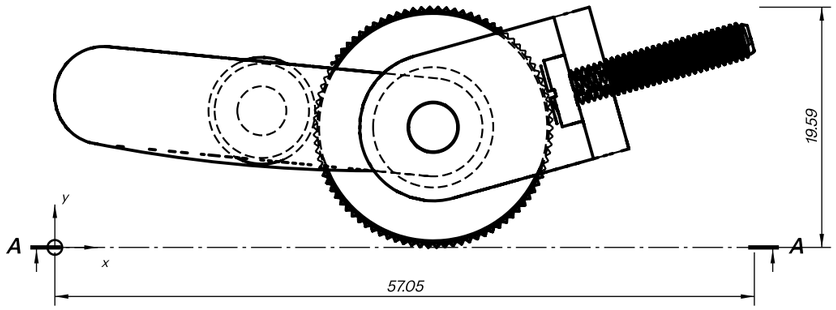
Faces in section A-A, from the bottom face: 20.30.

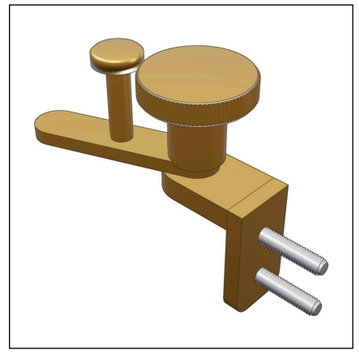
A–A



View from the left







Isometric view, not to scale

Werkstoff / Material not on the drawing	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 01 01 (05 01's list)*** *Teil / Part* *caseLock*
Behandlung / Treatment		*Benennung / Name* ***Transit lock (Feststeller)***	
Blatt / Sheet **1 / 2**		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Transit lock (Feststeller) — sheet 2

A knob on a bracket in the box's front right corner turning a lever at the pivots' height: turned in (Display's Motion, the transit lock), the lever's end passes through a slot in the gimbal ring into the slot across the locking bracket on the bowl and stops against its end, so neither gimbal can swing, for carrying the box; the knob screwed down holds it. Turned off, it lies along the front wall, clear of the ring.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 01 (05 01's list)	Feststeller (transit lock)	1	not on the drawing		not measured on the model: see its function, A knob and lever in the box's front right corner: turned in, the lever presses the ring onto the locking bracket.
DIN96	Holzschraube DIN 96, 3 × 15 (the transit lock's bracket's wood screws, estimated)	2	not on the drawing		not measured on the model: see its function, Round-headed wood screws through the bracket's wall leg into the box's right wall (the owner's case design, CAD Model Case.md §10: 2 × 3 × 15 DIN 96).

What it does (05 01 01 (05 01's list)). A knob and lever in the box's front right corner: turned in, the lever presses the ring onto the locking bracket.

What it does (DIN96). Round-headed wood screws through the bracket's wall leg into the box's right wall (the owner's case design, CAD Model Case.md §10: 2 × 3 × 15 DIN 96).

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 41.20.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	14.81, 7.32	2.83, 8.55	15.07, 68.75	61.43
Outline	2	arc clockwise	2.83, 8.55	4.37, 16.37	3.98, 12.38	4.00
Outline	3	arc anticlockwise	4.37, 16.37	15.71, 15.33	11.11, 27.66	13.16
Outline	4	arc clockwise	15.71, 15.33	18.95, 14.96	16.88, 11.18	4.32
Outline	5	line	18.95, 14.96	22.69, 14.60		
Outline	6	line	22.69, 14.60	22.58, 15.04		
Outline	7	line	22.58, 15.04	23.05, 15.13		
Outline	8	line	23.05, 15.13	22.96, 15.61		
Outline	9	line	22.96, 15.61	23.44, 15.66		
Outline	10	line	23.44, 15.66	23.39, 16.14		
Outline	11	line	23.39, 16.14	23.87, 16.16		
Outline	12	line	23.87, 16.16	23.85, 16.65		
Outline	13	line	23.85, 16.65	24.33, 16.63		
Outline	14	line	24.33, 16.63	24.34, 17.12		
Outline	15	line	24.34, 17.12	24.83, 17.07		
Outline	16	line	24.83, 17.07	24.87, 17.56		
Outline	17	line	24.87, 17.56	25.35, 17.48		
Outline	18	line	25.35, 17.48	25.43, 17.95		
Outline	19	line	25.43, 17.95	25.90, 17.84		
Outline	20	line	25.90, 17.84	26.01, 18.31		

Outline	21	line	26.01, 18.31	26.47, 18.17		
Outline	22	line	26.47, 18.17	26.61, 18.63		
Outline	23	line	26.61, 18.63	27.07, 18.45		
Outline	24	line	27.07, 18.45	27.24, 18.90		
Outline	25	line	27.24, 18.90	27.68, 18.70		
Outline	26	line	27.68, 18.70	27.88, 19.13		
Outline	27	line	27.88, 19.13	28.31, 18.90		
Outline	28	line	28.31, 18.90	28.54, 19.32		
Outline	29	line	28.54, 19.32	28.95, 19.05		
Outline	30	line	28.95, 19.05	29.21, 19.46		
Outline	31	line	29.21, 19.46	29.60, 19.16		
Outline	32	line	29.60, 19.16	29.89, 19.55		
Outline	33	line	29.89, 19.55	30.25, 19.23		
Outline	34	line	30.25, 19.23	30.57, 19.59		
Outline	35	line	30.57, 19.59	30.91, 19.25		
Outline	36	line	30.91, 19.25	31.26, 19.59		
Outline	37	line	31.26, 19.59	31.57, 19.22		
Outline	38	line	31.57, 19.22	31.94, 19.54		
Outline	39	line	31.94, 19.54	32.23, 19.15		
Outline	40	line	32.23, 19.15	32.62, 19.44		
Outline	41	line	32.62, 19.44	32.88, 19.03		
Outline	42	line	32.88, 19.03	33.28, 19.29		
Outline	43	line	33.28, 19.29	33.52, 18.86		
Outline	44	line	33.52, 18.86	33.94, 19.10		
Outline	45	line	33.94, 19.10	34.14, 18.66		
Outline	46	line	34.14, 18.66	34.58, 18.86		
Outline	47	line	34.58, 18.86	34.75, 18.41		
Outline	48	line	34.75, 18.41	35.21, 18.58		
Outline	49	line	35.21, 18.58	35.34, 18.11		
Outline	50	line	35.34, 18.11	35.81, 18.25		
Outline	51	line	35.81, 18.25	35.91, 17.78		
Outline	52	line	35.91, 17.78	36.39, 17.89		
Outline	53	line	36.39, 17.89	36.43, 17.59		
Outline	54	line	36.43, 17.59	43.59, 19.59		
Outline	55	line	43.59, 19.59	44.85, 15.09		
Outline	56	line	44.85, 15.09	44.89, 15.27		
Outline	57	line	44.89, 15.27	44.95, 15.29		
Outline	58	line	44.95, 15.29	45.18, 15.07		
Outline	59	line	45.18, 15.07	45.30, 15.11		
Outline	60	line	45.30, 15.11	45.37, 15.41		
Outline	61	line	45.37, 15.41	45.43, 15.43		
Outline	62	line	45.43, 15.43	45.66, 15.21		
Outline	63	line	45.66, 15.21	45.78, 15.24		
Outline	64	line	45.78, 15.24	45.85, 15.54		
Outline	65	line	45.85, 15.54	45.91, 15.56		
Outline	66	line	45.91, 15.56	46.14, 15.34		
Outline	67	line	46.14, 15.34	46.26, 15.38		
Outline	68	line	46.26, 15.38	46.34, 15.68		

Outline	69	line	46.34, 15.68	46.40, 15.70		
Outline	70	line	46.40, 15.70	46.62, 15.48		
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Outline	75	line	47.10, 15.61	47.22, 15.65		
Outline	76	line	47.22, 15.65	47.30, 15.95		
Outline	77	line	47.30, 15.95	47.36, 15.97		
Outline	78	line	47.36, 15.97	47.58, 15.75		
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Outline	145	line	55.48, 18.24	55.54, 18.26		
Outline	146	line	55.54, 18.26	55.77, 18.04		
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Outline	149	line	56.47, 18.09	57.05, 16.01		
Outline	150	arc clockwise	57.05, 16.01	56.79, 15.79	56.70, 16.16	0.38
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Outline	152	arc clockwise	56.67, 15.76	56.51, 15.50	56.23, 15.84	0.45
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Outline	155	line	56.18, 15.62	56.11, 15.32		
Outline	156	line	56.11, 15.32	56.05, 15.30		
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Outline	159	line	55.70, 15.49	55.63, 15.19		
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Outline	161	line	55.57, 15.17	55.34, 15.39		
Outline	162	line	55.34, 15.39	55.22, 15.35		
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Outline	164	line	55.14, 15.05	55.08, 15.03		

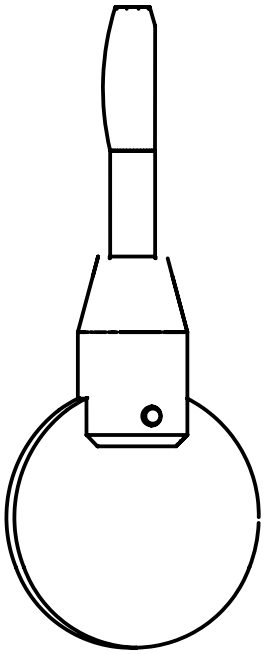
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Outline	191	line	51.85, 14.41	51.77, 14.11		
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Outline	195	line	51.37, 14.28	51.29, 13.97		
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Outline	197	line	51.23, 13.96	51.01, 14.17		
Outline	198	line	51.01, 14.17	50.89, 14.14		
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Outline	200	line	50.81, 13.84	50.75, 13.82		
Outline	201	line	50.75, 13.82	50.53, 14.04		
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Outline	208	line	49.85, 13.57	49.79, 13.55		
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Outline	219	line	48.48, 13.47	48.40, 13.16		
Outline	220	line	48.40, 13.16	48.34, 13.15		
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Outline	234	line	46.68, 12.96	46.56, 12.93		
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Outline	236	line	46.48, 12.62	46.42, 12.61		
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Outline	239	line	46.07, 12.79	46.00, 12.49		
Outline	240	line	46.00, 12.49	45.94, 12.47		
Outline	241	line	45.94, 12.47	45.71, 12.69		
Outline	242	line	45.71, 12.69	45.59, 12.66		
Outline	243	arc anticlockwise	45.59, 12.66	46.31, 9.88	48.36, 11.88	2.87
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Outline	253	line	38.75, 3.99	38.27, 3.93		
Outline	254	line	38.27, 3.93	38.32, 3.45		
Outline	255	line	38.32, 3.45	37.84, 3.43		
Outline	256	line	37.84, 3.43	37.86, 2.94		
Outline	257	line	37.86, 2.94	37.38, 2.96		
Outline	258	line	37.38, 2.96	37.37, 2.47		
Outline	259	line	37.37, 2.47	36.88, 2.52		
Outline	260	line	36.88, 2.52	36.84, 2.04		

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Outline	262	line	36.36, 2.12	36.29, 1.64		
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Outline	264	line	35.81, 1.75	35.70, 1.28		
Outline	265	line	35.70, 1.28	35.24, 1.42		
Outline	266	line	35.24, 1.42	35.10, 0.96		
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Outline	268	line	34.65, 1.14	34.47, 0.69		
Outline	269	line	34.47, 0.69	34.03, 0.90		
Outline	270	line	34.03, 0.90	33.83, 0.46		
Outline	271	line	33.83, 0.46	33.40, 0.70		
Outline	272	line	33.40, 0.70	33.17, 0.27		
Outline	273	line	33.17, 0.27	32.76, 0.54		
Outline	274	line	32.76, 0.54	32.50, 0.13		
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Outline	287	line	28.43, 0.30	28.19, 0.73		
Outline	288	line	28.19, 0.73	27.77, 0.49		
Outline	289	line	27.77, 0.49	27.57, 0.94		
Outline	290	line	27.57, 0.94	27.13, 0.73		
Outline	291	line	27.13, 0.73	26.96, 1.19		
Outline	292	line	26.96, 1.19	26.50, 1.01		
Outline	293	line	26.50, 1.01	26.37, 1.48		
Outline	294	line	26.37, 1.48	25.90, 1.34		
Outline	295	line	25.90, 1.34	25.80, 1.81		
Outline	296	line	25.80, 1.81	25.33, 1.71		
Outline	297	line	25.33, 1.71	25.25, 2.19		
Outline	298	line	25.25, 2.19	24.77, 2.11		
Outline	299	line	24.77, 2.11	24.74, 2.59		
Outline	300	line	24.74, 2.59	24.25, 2.55		
Outline	301	line	24.25, 2.55	24.25, 3.04		
Outline	302	line	24.25, 3.04	23.76, 3.03		
Outline	303	line	23.76, 3.03	23.79, 3.52		
Outline	304	line	23.79, 3.52	23.31, 3.54		
Outline	305	line	23.31, 3.54	23.37, 4.02		
Outline	306	line	23.37, 4.02	22.89, 4.08		
Outline	307	line	22.89, 4.08	22.99, 4.56		
Outline	308	line	22.99, 4.56	22.51, 4.65		

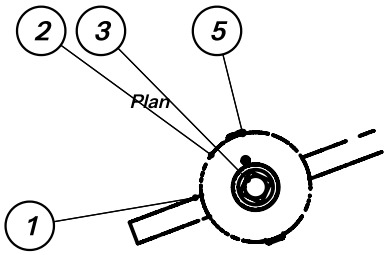
Outline	309	line	22.51, 4.65	22.64, 5.12		
Outline	310	line	22.64, 5.12	22.17, 5.25		
Outline	311	line	22.17, 5.25	22.34, 5.71		
Outline	312	line	22.34, 5.71	21.88, 5.87		
Outline	313	line	21.88, 5.87	22.07, 6.31		
Outline	314	line	22.07, 6.31	21.63, 6.50		
Outline	315	line	21.63, 6.50	21.71, 6.66		
Outline	316	arc anticlockwise	21.71, 6.66	18.05, 6.95	19.57, 3.00	4.23
Outline	317	arc clockwise	18.05, 6.95	14.81, 7.32	16.88, 11.11	4.32

Front view





Isometric view, not to scale



Not seen in either view (inside the assembly): Pos. 4.

Pos.	Units	Benennung / Name	Zeichnung / Drawing	Werkstoff / Material
5	[0, 1]	Grooved pin (Kerbstift)	51 00 11-6	not on the drawing
4	[0, 1]	Coil spring (Wendelfeder)	51 00 11-4	not on the drawing
3	[0, 1]	Key tube (Schlüsselrohr)	05 01 02-3	„Ms. 58“ (brass, CuZn40Pb, inf
2	[0, 1]	Key body (Schlüsselmittelteil)	05 01 02-2	„St. Az / Din 668“ ("St. Az" =
1	[0, 1]	Key handle (Schlüsselgriff)	51 00 11-1	not on the drawing

Werkstoff / Material

„St. Az / Din 668“ ("St. Az" = bright-drawn steel ("Automatenstahl"/"Stahl, gezogen", inferred) to DIN 668)

Maßstab / Scale

1:1

Behandlung / Treatment

„** (sure) = sand-blast, phosphate, spray black. - **Number note:** on the key assembly 51 00 11 this part is “; „sandeln, phosphatieren, schwarz spritzen“ (sand-blast, phosphate, spray black)

Blatt / Sheet

1 / 1

Wempe Einheits-Chronometer

The Marine Chronometer working model

drawn from its solids by tools/drawbook.py

Model v3.35.00 · 2026-10-26

Zeichnung Nr. / Drawing no.

05 01 02-2

Teil / Part

key

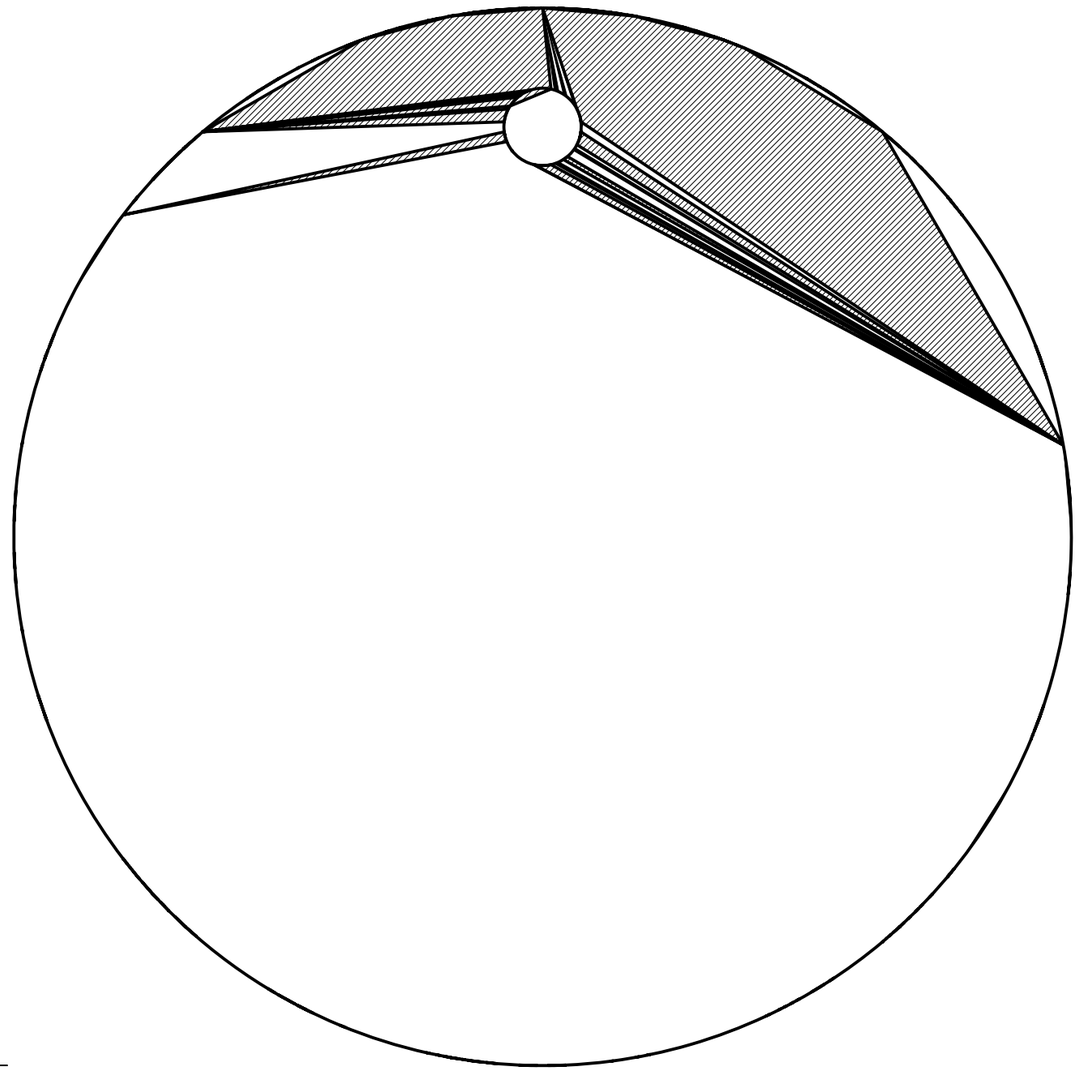
Benennung / Name

Winding key (Chronometerschlüssel)

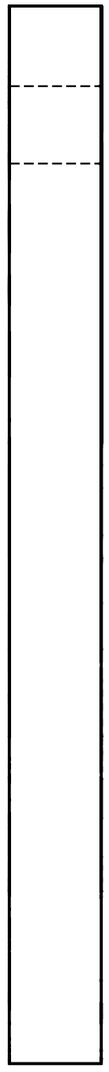
Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.

First-angle projection (DIN ISO 5456-2).

A–A



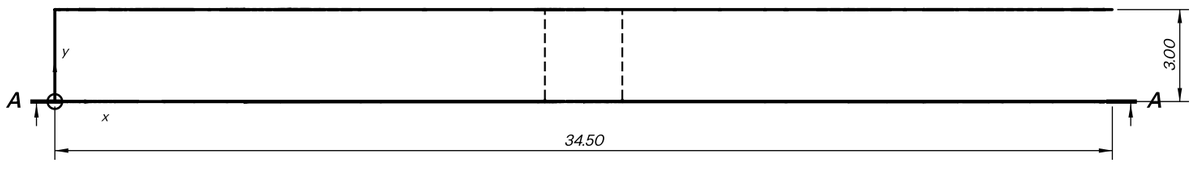
View from the left





Isometric view, not to scale

0.02



Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **51 00 11-1**
Behandlung / Treatment			Teil / Part key.paddle
Blatt / Sheet **1 / 2**		Benennung / Name **Key handle (Schlüsselgriff)**	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Key handle (Schlüsselgriff) — sheet 2

The round flat paddle, in the body's slot, held by the grooved pin.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
51 00 11-1	Schlüsselgriff (key handle)	[0, 1]	not on the drawing		not measured on the model: see its function, The key's round flat paddle.

What it does (51 00 11-1). The key's round flat paddle.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 34.50.

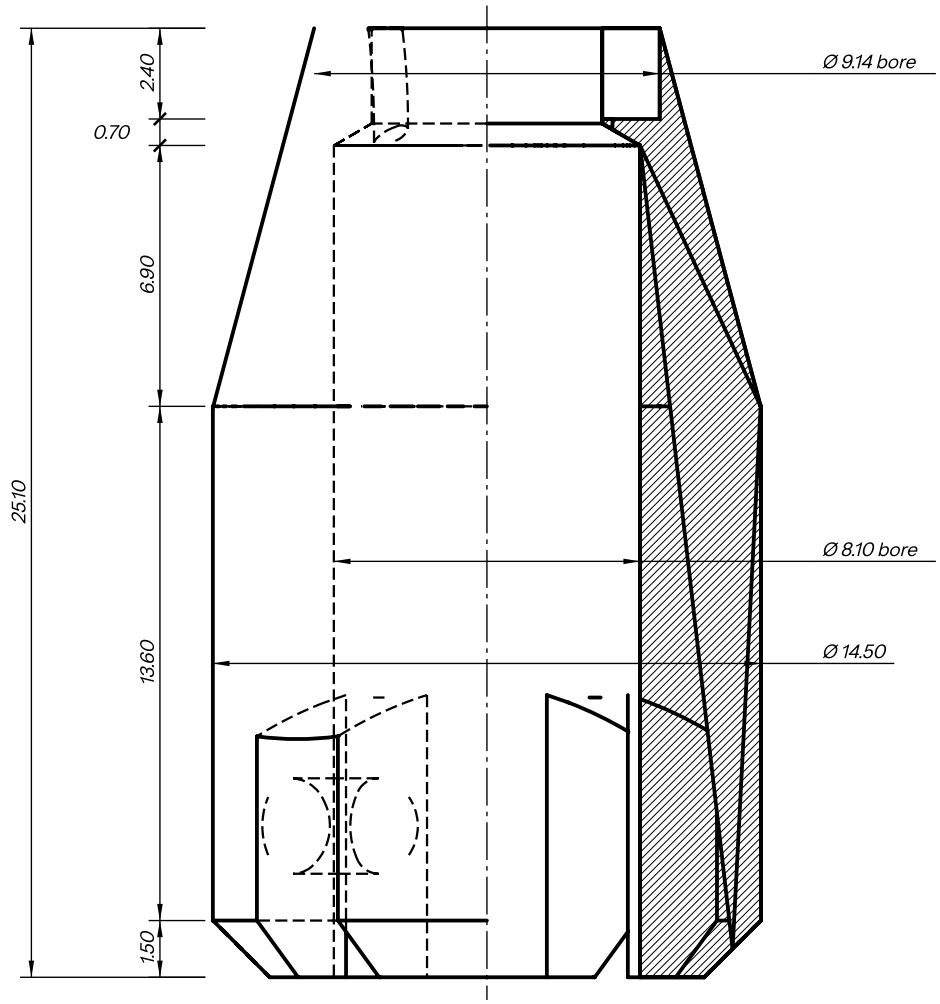
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

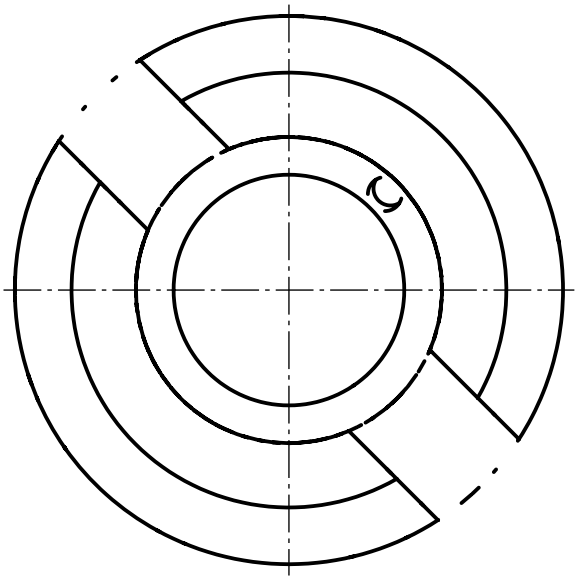
Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.07, 0.00	0.00, 0.00		
Outline	2	line	0.00, 0.00	0.00, 3.00		
Outline	3	line	0.00, 3.00	34.50, 3.00		
Outline	4	line	34.50, 3.00	34.50, 0.00		
Outline	5	line	34.50, 0.00	0.07, 0.00		
Cut-out 1	6	line	25.88, 0.00	25.88, 3.00		
Cut-out 1	7	line	25.88, 3.00	25.88, 0.00		

Levels

Faces in section A-A, from the bottom face: 0.02.



View from below (seen from beneath, x reversed)





Isometric view, not to scale

Werkstoff / Material *„St. Az / Din 668" ("St. Az" = bright-drawn steel ("Automatenstahl"/"Stahl, gezogen", inferred) to DIN 668)*	*Maßstab / Scale* ***5:1***	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 01 02-2*** *Teil / Part* *key.body*
Behandlung / Treatment *„** (sure) = sand-blast, phosphate, spray black. - **Number note:** on the key assembly 51 00 11 this part is "; „sandeln, phosphatieren, schwarz spritzen" (sand-blast, phosphate, spray black)*		*Benennung / Name* ***Key body (Schlüsselmittelteil)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Key body (Schlüsselmittelteil) — sheet 2

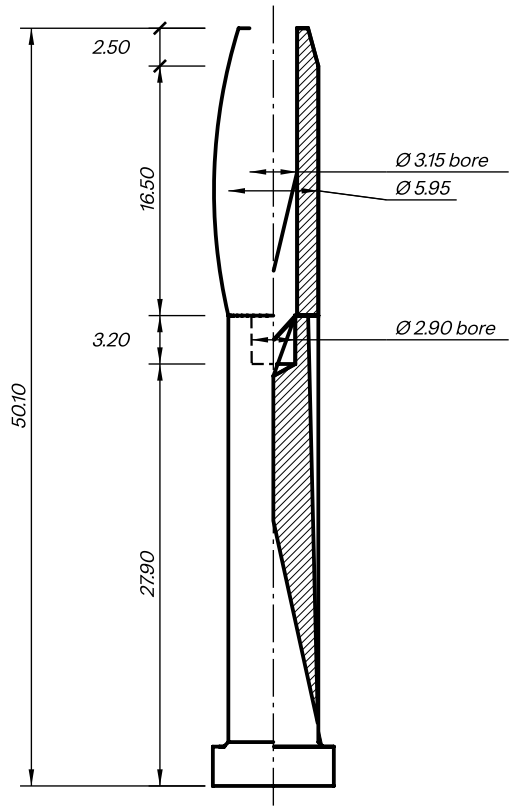
The black steel body: the tube runs in its bore, the spring round it.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 02-2	Schlüsselmittelteil (key body)	[0, 1]	„St. Az / Din 668“ ("St. Az" = bright-drawn steel ("Automatenstahl"/"Stahl, gezogen", inferred) to DIN 668)	„** (sure) = sand-blast, phosphate, spray black. - **Number note:** on the key assembly 51 00 11 this part is “; „sandeln, phosphatieren, schwarz spritzen“ (sand-blast, phosphate, spray black)	takes Schlüsselrohr (key tube): in it (free): 4.979 mm, least gap 0.0549

What it does (05 01 02-2). The key's black steel body.

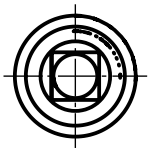
Turned about its axis; length 25.10, largest Ø 14.50. Diameters: Ø 14.50; bores Ø 8.10, Ø 9.14. Shoulders from the lower end: 0.00, 1.50, 15.10, 22.00, 22.70, 25.10.





Isometric view, not to scale

View from above



Werkstoff / Material „Ms. 58" (brass, CuZn40Pb, inferred)	*Maßstab / Scale* **2:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***05 01 02-3*** *Teil / Part* *key.tube*
Behandlung / Treatment *surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)*		*Benennung / Name* ***Key tube (Schlüsselrohr)***	
Blatt / Sheet ***1 / 2***		*Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2.* *First-angle projection (DIN ISO 5456-2).*	

Key tube (Schlüsselrohr) — sheet 2

The brass tube with the square socket for the fusee's winding square; its mouth stops on the dust plate, where the square is as wide as the socket.

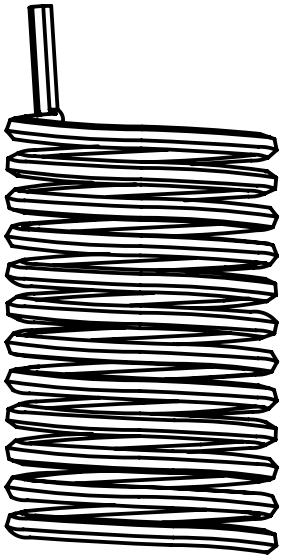
Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
05 01 02-3	Schlüsselrohr (key tube)	[0, 1]	„Ms. 58“ (brass, CuZn40Pb, inferred)	surface ▽▽ (DIN 140: ▽ roughed, ▽▽ smoothed, ▽▽▽ fine, ▽▽▽▽ polished)	in 050102-2 (free): 4.979 mm, least gap 0.0549

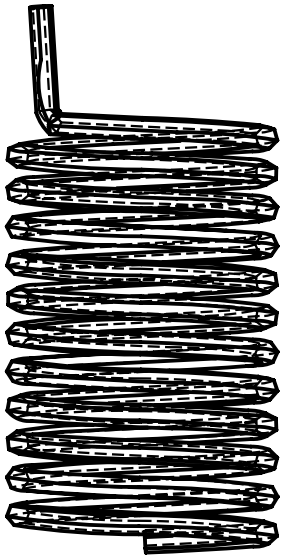
What it does (05 01 02-3). The brass tube with the square socket.

Turned about its axis; length 50.10, largest Ø 8.00. Diameters: Ø 5.95; bores Ø 2.90, Ø 3.15. Shoulders from the lower end: 0.00, 27.90, 31.10, 47.60, 50.10.

A–A

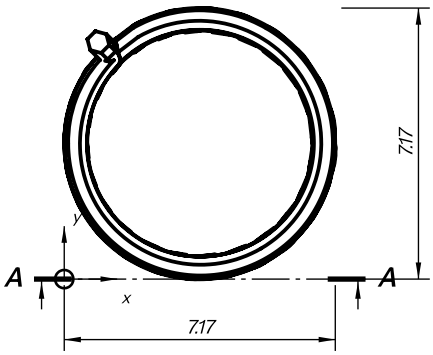


View from the left





Isometric view, not to scale



Hidden lines are left out of the plan (too many to read); the section and the hole table give the depths.

Werkstoff / Material not on the drawing	Maßstab / Scale **5:1**	**Wempe Einheits-Chronometer** The Marine Chronometer working model drawn from its solids by tools/drawbook.py Model v3.35.00 · 2026-10-26	Zeichnung Nr. / Drawing no. **51 00 11-4** Teil / Part key.spring
Behandlung / Treatment		Benennung / Name **Coil spring (Wendelfeder)**	
Blatt / Sheet 1 / 2		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. First-angle projection (DIN ISO 5456-2).	

Coil spring (Wendelfeder) — sheet 2

The coil spring wound on the tube in the body's bore, its end turned down into the body's Ø 0.9 hole: winding, the body drags its end the way that closes its coils on the tube; turned back they open and slip.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
51 00 11-4	Wendelfeder (the key's coil spring)	[0, 1]	not on the drawing		not measured on the model: see its function, Round the tube in the body's bore: it grips the tube one way and slips the other.

What it does (51 00 11-4). Round the tube in the body's bore: it grips the tube one way and slips the other.

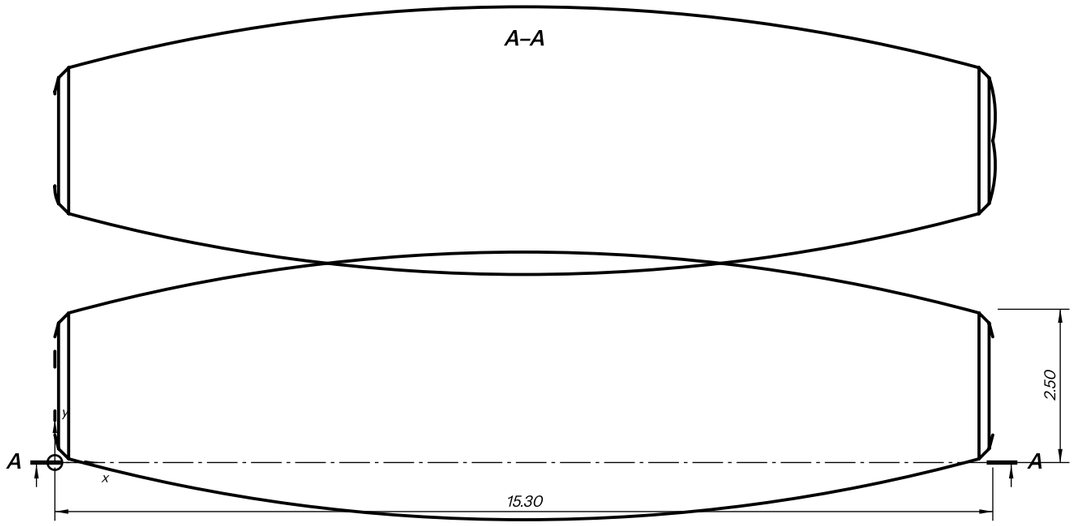
Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 14.46.

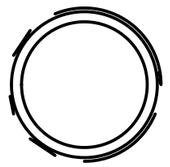
Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	arc clockwise	0.11, 2.71	0.06, 3.15	0.90, 3.03	0.85
Outline	2	arc clockwise	0.06, 3.15	0.00, 3.63	0.93, 3.49	0.94
Outline	3	line	0.00, 3.63	0.00, 3.66		
Outline	4	line	0.00, 3.66	0.13, 4.53		
Outline	5	line	0.13, 4.53	0.14, 4.59		
Outline	6	line	0.14, 4.59	0.49, 5.40		
Outline	7	line	0.49, 5.40	0.52, 5.44		
Outline	8	line	0.52, 5.44	0.87, 5.88		
Outline	9	line	0.87, 5.88	0.69, 6.06		
Outline	10	line	0.69, 6.06	0.61, 6.34		
Outline	11	line	0.61, 6.34	0.82, 6.56		
Outline	12	arc clockwise	0.82, 6.56	1.20, 6.39	0.87, 6.15	0.41
Outline	13	line	1.20, 6.39	1.29, 6.31		
Outline	14	line	1.29, 6.31	1.81, 6.70		
Outline	15	line	1.81, 6.70	1.86, 6.72		
Outline	16	line	1.86, 6.72	2.68, 7.05		
Outline	17	line	2.68, 7.05	2.73, 7.06		
Outline	18	line	2.73, 7.06	3.64, 7.17		
Outline	19	line	3.64, 7.17	4.56, 7.03		
Outline	20	line	4.56, 7.03	5.42, 6.66		
Outline	21	line	5.42, 6.66	6.16, 6.08		
Outline	22	line	6.16, 6.08	6.71, 5.33		
Outline	23	line	6.71, 5.33	7.06, 4.46		
Outline	24	line	7.06, 4.46	7.17, 3.53		
Outline	25	line	7.17, 3.53	7.03, 2.61		
Outline	26	line	7.03, 2.61	6.66, 1.75		
Outline	27	line	6.66, 1.75	6.08, 1.02		
Outline	28	line	6.08, 1.02	6.06, 0.99		

Outline	29	line	6.06, 0.99	5.31, 0.44		
Outline	30	line	5.31, 0.44	4.44, 0.10		
Outline	31	line	4.44, 0.10	3.51, 0.00		
Outline	32	line	3.51, 0.00	2.58, 0.14		
Outline	33	line	2.58, 0.14	1.77, 0.49		
Outline	34	line	1.77, 0.49	1.72, 0.52		
Outline	35	line	1.72, 0.52	1.03, 1.07		
Outline	36	line	1.03, 1.07	1.00, 1.10		
Outline	37	line	1.00, 1.10	0.47, 1.81		
Outline	38	line	0.47, 1.81	0.44, 1.86		
Outline	39	line	0.44, 1.86	0.12, 2.68		
Outline	40	line	0.12, 2.68	0.11, 2.71		
Cut-out 1	41	line	3.51, 6.56	2.85, 6.47		
Cut-out 1	42	line	2.85, 6.47	2.13, 6.19		
Cut-out 1	43	line	2.13, 6.19	1.50, 5.72		
Cut-out 1	44	line	1.50, 5.72	1.02, 5.11		
Cut-out 1	45	line	1.02, 5.11	0.71, 4.39		
Cut-out 1	46	line	0.71, 4.39	0.60, 3.62		
Cut-out 1	47	line	0.60, 3.62	0.69, 2.85		
Cut-out 1	48	line	0.69, 2.85	0.98, 2.13		
Cut-out 1	49	line	0.98, 2.13	1.45, 1.51		
Cut-out 1	50	line	1.45, 1.51	2.06, 1.02		
Cut-out 1	51	line	2.06, 1.02	2.77, 0.72		
Cut-out 1	52	line	2.77, 0.72	3.54, 0.60		
Cut-out 1	53	line	3.54, 0.60	4.31, 0.69		
Cut-out 1	54	line	4.31, 0.69	5.03, 0.98		
Cut-out 1	55	line	5.03, 0.98	5.66, 1.44		
Cut-out 1	56	line	5.66, 1.44	6.14, 2.05		
Cut-out 1	57	line	6.14, 2.05	6.45, 2.76		
Cut-out 1	58	line	6.45, 2.76	6.57, 3.54		
Cut-out 1	59	line	6.57, 3.54	6.48, 4.31		
Cut-out 1	60	line	6.48, 4.31	6.19, 5.04		
Cut-out 1	61	line	6.19, 5.04	5.72, 5.67		
Cut-out 1	62	line	5.72, 5.67	5.10, 6.16		
Cut-out 1	63	line	5.10, 6.16	4.38, 6.46		
Cut-out 1	64	line	4.38, 6.46	3.62, 6.57		
Cut-out 1	65	line	3.62, 6.57	3.51, 6.56		







Isometric view, not to scale

Werkstoff / Material *not on the drawing*	*Maßstab / Scale* **10:1**	***Wempe Einheits-Chronometer*** *The Marine Chronometer working model* *drawn from its solids by tools/drawbook.py* *Model v3.35.00 · 2026-10-26*	*Zeichnung Nr. / Drawing no.* ***51 00 11-6***
Behandlung / Treatment			*Teil / Part* *key.pin*
Blatt / Sheet ***1 / 2***		*Benennung / Name* ***Grooved pin (Kerbstift)***	
		Sizes in mm. Untoleranced sizes are the model's; fits and tolerances on sheet 2. *First-angle projection (DIN ISO 5456-2).*	

Grooved pin (Kerbstift) — sheet 2

The pin through the body and the paddle.

Parts list

Drawing	Name	Units	Material	Finish, heat treatment	Fit, as measured on the model
51 00 11-6	Kerbstift K.S. 1 (the key's grooved pin)	[0, 1]	not on the drawing		not measured on the model: see its function, Through the body and the paddle.

What it does (51 00 11-6). Through the body and the paddle.

Coordinates from the datum, the corner of its bounding box (marked on the plan), x to the right and y up the plan as drawn; depths from the face seen in the plan (the top). Thickness 2.50.

Contour in the plan

The outline and cut-outs as a CNC program runs them: each element from its start to its end, an arc with its centre and radius.

Contour	#	Element	From x, y	To x, y	Centre x, y	R
Outline	1	line	0.00, 1.25	0.00, 2.05		
Outline	2	arc clockwise	0.00, 2.05	0.45, 2.50	0.45, 2.05	0.45
Outline	3	arc clockwise	0.45, 2.50	15.08, 2.44	7.65, -24.82	28.25
Outline	4	arc clockwise	15.08, 2.44	15.30, 2.05	14.85, 2.05	0.45
Outline	5	line	15.30, 2.05	15.30, 0.45		
Outline	6	arc clockwise	15.30, 0.45	14.85, 0.00	14.85, 0.45	0.45
Outline	7	arc clockwise	14.85, 0.00	0.23, 0.06	7.65, 27.32	28.25
Outline	8	arc clockwise	0.23, 0.06	0.00, 0.51	0.46, 0.46	0.46
Outline	9	line	0.00, 0.51	0.00, 1.25		

Materials and how each part is made

Each line of the parts list by how it is made, with its material as its drawing gives it. The special ones first: what needs a hand, a special supply or both.

by hand (1)

Drawing	Name	Material	How
05 04 01-1	Gangfeder (spring detent, drawing 1436)	„Stahl“	the spring detent (1436, „Stahl“): one piece of steel, its spring (Federung) 0.07 thick (0.09 at its ends) and 5.0 long filed and stoned to its ring (the bench's test), the locking stone set in its pipe (Röhrchen); 1436 gives no hardening

by hand and special supply (2)

Drawing	Name	Material	How
05 03 01-1	Unruhreifen (balance rim with arm, nickel steel and brass)	„Nickelstahl Sonderlegierung u. Messing“ (nickel steel, special alloy, and brass)	the cut bimetallic rim: 05 03 01-1's „Nickelstahl Sonderlegierung u. Messing“, a nickel-steel special alloy with brass fused or brazed on, turned, then cut through beside the arm and its masses' holes drilled; no stock supplier makes it: a maker fuses brass onto the nickel steel or buys a balance blank, then adjusts it for temperature on the trial (wempe_thermal.js, the Wempe README's "Physics: hairspring and temperature")
05 03 02-1	Spiralfeder (helical hairspring)	„Bandstahl 0,52×0,285 poliert“ (strip steel, polished)	the helical hairspring: 05 03_6's „Bandstahl 0,52×0,285 poliert“ wound 15 turns left-hand on a form, hardened and tempered dark blue on it, its terminal curves bent to 05 03_7 by hand with curve pliers on a drawing of them; vibrated against the balance (its tolerance)

special supply (1)

Drawing	Name	Material	How
05 07 00-6	Zugfeder (mainspring)	not on the drawing	the mainspring: 05 07 00-6's strip, or the build's Nivaflex 45/5 (Straumann's, from a mainspring supplier) cut to 1240 mm, its ends formed, set

special supply and hand work (1)

Drawing	Name	Material	How
05 04 01-2	Goldfeder (gold passing spring)	„Goldblech 333 federhart“ (gold sheet, 333 fine (8 ct), spring-hard)	05 04 01-2's „Goldblech 333 federhart“: 8 ct gold sheet rolled spring-hard, from a bullion or jewellery supplier; cut, thinned at its bend (the build) and stoned

special supply or by hand (1)

Drawing	Name	Material	How
05 07 04	Kette (fusee chain)	„Bandstahl 5×0,3; 10×0,9; Präz. Rundstahl 0,6“ (strip steel 5 × 0,3 [the link plates, inferred]; 10 × 0,9 [the end hooks, inferred]; precision round steel 0,6 [the rivets, inferred])	the fusee chain (05 07 04): links blanked from thin steel strip and riveted on pins; a chain from a watch-parts supplier, or blanked with a press tool

bought in (17)

Drawing	Name	Material	How
05 05 04-2	Fassung (the seconds jewel's setting)	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 05 04-1	Lagerstein f. Sek. (the seconds arbor's lower jewel)	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 03-2	Hebestein (impulse stone, sapphire)	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 04-2	Auslösestein (discharge stone)	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 01-3	Ruhestein (locking stone)	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 08-1	Deckstein (endstone), balance, upper	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 06-2	Fassung (jewel setting), balance, upper	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made

Drawing	Name	Material	How
05 04 07-1	Lagerstein (hole jewel), balance, upper	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 06-2	Fassung (jewel setting), balance, lower	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 07-1	Lagerstein (hole jewel), balance, lower	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 08-1	Deckstein (endstone), balance, lower	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 08-1	Deckstein (endstone), escape wheel, upper	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 06-2	Fassung (jewel setting), escape wheel, upper	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 06-1	Lagerstein (hole jewel), escape wheel, upper	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 06-2	Fassung (jewel bush), escape wheel, lower	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 06-1	Lagerstein (hole jewel), escape wheel, lower	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made
05 04 08-1	Deckstein (endstone), escape wheel, lower	not on the drawing	a synthetic jewel: choose the standard hole and outside diameter nearest the drawn (the hole to the pivot plus 0.01–0.02), or have it made

bought in or made by a case maker (21)

Drawing	Name	Material	How
05 04 04-1	Auslöserolle (discharge roller)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
05 07 01-3	Federhaushaken (spring hook block)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
05 07 02-2	Federhaken (arbor hook)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
05 01 01-1	Glas (glass)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.02	Holzkasten (wooden box)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
51 00 10	Griff f. Chronometerkasten (handle)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
Scharnier	Scharnier (piano hinge, estimated)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
DIN96	Holzschraube DIN 96, 3 × 15 (the transit lock's bracket's wood screws, estimated)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.02	Deckel (the box's lid)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
Deckelglas	Deckelglas (the lid's glass, estimated)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.01-13, -14	Leiste (rim strip, 17, 18)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
Polster.S	Polster (front and back cushions)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
Polster.E	Polster (end cushions)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)

Drawing	Name	Material	How
Polster.B	Polster (floor cushion)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.01-07	Riemen (strap, 10): its ends on the end walls	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.01-08	Öse (eye, 12)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.01-05	Scharnier (outer box hinge, 7)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
Hakenoese	Öse (the hook's eye, 9: commercial)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.01	Deckel (the outer box's lid)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
127-556.01-12	Polster (lid cushion, 16)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)
Haken	Haken (hook, 9: commercial)	not on the drawing	box and case hardware, upholstery or glass: its sheet gives no material (the outer box's wood is on its sheet: „gebleichte Birke“)

bought in or turned (52)

Drawing	Name	Material	How
WN-56	Pillar screw, dial side (WN-56, by the coloured section's label)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-57	Pillar screw, train side (WN-57, by the coloured section's label)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-rs	Movement ring screw (WN number not read)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-201	Stellstift (WN-201): the plate's pin in the movement ring	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-22	Click screw, shouldered (WN-22)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-55	Barrel bridge screw (WN-55, by the coloured section's label)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-203.bb	Stellstift (WN-203), barrel bridge	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-34	Fusee bush screw (WN-34)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-55	Upper balance cock screw (hole 8; WN-55 by the coloured section's label)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-203.uc	Stellstift, upper balance cock (holes 6, 9, 50; WN not given, as WN-203)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-53	Lower balance cock screw (WN-53, by the coloured section's label)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-203.lc	Stellstift, lower balance cock (holes 22, 37; WN not given, as WN-203)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)

Drawing	Name	Material	How
WN-52	Intermediate bridge screw (WN-52)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-203.zb	Stellstift (WN-203), intermediate bridge	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-301	Messinglager WN-301 (brass bush, centre arbor's upper pivot)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-300.t38	Messinglager WN-300 (brass bush, intermediate arbor's upper pivot)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-300.t35	Messinglager WN-300 (brass bush, seconds arbor's upper pivot)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-302	Messinglager WN-302 (brass bush, centre arbor's journal)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-300.zb	Messinglager WN-300 (brass bush, intermediate arbor's lower pivot)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-hs	Bush in hole 18, the hand-setting pinion's (not named on 05 08 02)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-12	Schraube (WN-12), the mass screw	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-12a	Schraube (WN-12a), for the tapped hole in a mass's top (not fitted)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-10	Schraube (WN-10), the hub's screws	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-11	Schraube (WN-11), the stud's screw	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-14	Schraube (WN-14), the gold spring's screw	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-13	Detent foot screw, M1 (WN-13 by the coloured section's label, likely)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-205	Stellstift (WN-205 by the coloured section's label), the detent's foot pins	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-15	Detent block screw, M1.7 (WN-15 by the coloured section's label, likely)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-gkA	Screw across the block's slit upright, M1.2 (its WN number not read)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-201	Stellstift (WN-201, "Stellstift 2" by the coloured section's label), the detent block's steady pins Ø 0.9	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-55.ck	Schraube (WN-55 by the coloured section), the upper endstone's screws, M1.2	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-10	Schraube (WN-10 by the coloured section's label, its last digit 0 or 8: likely 10), the lower settings' screws, M1.2	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-es	Escape endstone screws, upper, M1.2 (their WN number not read)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)

Drawing	Name	Material	How
WN-ec	Escape endstone screws, lower, M1.2 (their WN number not read)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-19	Schraube WN-19 (click wheel screw)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-20	Schraube WN-20 (cover plate screw)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-21	Schraube WN-21 (click screw)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-10	Schraube (dial screw)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-202	Stellstift 3 (WN-202): the movement ring's radial pin to the case	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-78	Schraube WN-78 (13 on 05 01): the brackets' screws	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-71	Schraube WN-71 (7 on 05 01, the number unsure): the closing plate's screw	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-72	Schraube WN-72: the closing plate's knob	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-kr	Schraube (the ring bracket's screws, no number)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-79	Schraube WN-79 (12 on 05 01): the inner pivots	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-400	Kontermutter WN-400 (10 on 05 01), knurled brass	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-80	Schraube WN-80 (11 on 05 01): the outer pivots, M6	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-hg	Schraube (the handles' screws, estimated)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-kh	Schraube (the key holder's fitting's screws, estimated)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-zg	Schraube (the tongue's pivot screw, estimated)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-zk	Knopf (the tongue's knob, estimated)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
WN-ur	Schraube f. Riemenbef. (the strap's screws)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)
DIN 96	Schraube 3 x 15 DIN 96 (11)	not on the drawing	a standard part (Wempe-Norm, its sheet not in the set): a screw or pin of the thread and size the model draws (its card)

cast and machined (1)

Drawing	Name	Material	How
R-05 01 00-1	Gehäusetopf (bowl)	„Ms-Sandguß (Kokillenguß)“	(„Ms-Sandguß (Kokillenguß)“)

CNC milled or turned (18)

Drawing	Name	Material	How
05 08 00-1	Pfeiler (pillar)	„Rundmessing 8,5 Din 1758 Ms 58“ ("hart" struck through)	from the drawn stock („Rundmessing 8,5 Din 1758 Ms 58“ ("hart" struck through))
05 08 03-1	Unruhkloben oben (upper balance cock)	„Profilmessing 50×15×3,3×5,5 Ms58“ (the "5,5" likely)	from the drawn stock („Profilmessing 50×15×3,3×5,5 Ms58“ (the "5,5" likely))
05 08 00-2	Unruhkloben unten (lower balance cock)	„Profilmessing 20×10×8×3,3 Ms 58“	from the drawn stock („Profilmessing 20×10×8×3,3 Ms 58“)
05 04 02-2	Unruhbutzen (balance hub, brass)	„Ms.“ (brass)	from the drawn stock („Ms.“ (brass))
02 04 02	Heberolle (impulse roller, beryllium copper; 05 04 03-1 in the coloured section)	„Kupfer Beryllium, Saphir“ (beryllium copper; sapphire)	from the drawn stock („Kupfer Beryllium, Saphir“ (beryllium copper; sapphire))
05 06 04-2	Schneckenkörper (fusee body)	„Rundmessing 36 Din 1756(?) Ms 58“ ("Rundmessing 36", "Ms 58"; the DIN number reads 1756 or 1758)	from the drawn stock („Rundmessing 36 Din 1756(?) Ms 58“ ("Rundmessing 36", "Ms 58"; the DIN number reads 1756 or 1758))
05 06 00-2	Vorreiber	„Präz. Rundstahl 14 Din 175“ (precision round steel bar 14 dia, DIN 175; "Rundstahl" cut off at the box edge, likely)	from the drawn stock („Präz. Rundstahl 14 Din 175“ (precision round steel bar 14 dia, DIN 175; "Rundstahl" cut off at the box edge, likely))
05 06 00-1	Staubplatte (dust plate)	„Rundmessing 8 Din 1758 Ms58“ (round brass bar 8 dia, DIN 1758, Ms 58)	from the drawn stock („Rundmessing 8 Din 1758 Ms58“ (round brass bar 8 dia, DIN 1758, Ms 58))
05 07 01-1	Federhaustrommel (barrel drum)	„Ms. 63 3“ (brass Ms 63, 3 mm, i.e. 3 mm sheet)	from the drawn stock („Ms. 63 3“ (brass Ms 63, 3 mm, i.e. 3 mm sheet))
05 07 01-2	Federhausdeckel (barrel lid)	„Ms 58 DIN 1751“	from the drawn stock („Ms 58 DIN 1751“)
05 02 00-1	Minutenzeiger (minute hand)	„Goldin“ (a gold-coloured brass alloy)	from the drawn stock („Goldin“ (a gold-coloured brass alloy))
05 02 00-2	Stundenzeiger (hour hand)	„Goldin“	from the drawn stock („Goldin“)
05 01 00-2	Verschlußplatte (closing plate)	„Ms.“ (brass; sure)	from the drawn stock („Ms.“ (brass; sure))
R-05 01 01-2	Glasrand (bezel)	„Ms.“	from the drawn stock („Ms.“)
00 01 01-9	Firmenschild (maker’s plate), Type 3371	„Alu 10“	from the drawn stock („Alu 10“)
P. 51 00 02-2	Zunge (latch tongue)	„Eisen 10 × 4 blank gezogen“	from the drawn stock („Eisen 10 × 4 blank gezogen“)
05 01 02-2	Schlüsselmittelteil (key body)	„St. Az / Din 668“ ("St. Az" = bright-drawn steel ("Automatenstahl"/"Stahl, gezogen", inferred) to DIN 668)	from the drawn stock („St. Az / Din 668“ ("St. Az" = bright-drawn steel ("Automatenstahl"/"Stahl, gezogen", inferred) to DIN 668))
05 01 02-3	Schlüsselrohr (key tube)	„Ms. 58“ (brass, CuZn40Pb, inferred)	from the drawn stock („Ms. 58“ (brass, CuZn40Pb, inferred))

CNC milled or turned, the material the maker's choice (60)

Drawing	Name	Material	How
05 08 00-3	Werkring (movement ring)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 08 01-4	Korkring (cork ring)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 08 04-1	Zwischenbrücke (intermediate bridge; its sheet not in the set, the assembly 05 08 04)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 08 02-2	Futter (stud bush)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 00-6	Anrichtstift lg (long stud)	not on the drawing	its sheet gives no material: a maker chooses one for its function

Drawing	Name	Material	How
05 02 00-7	Anrichtstift kurz (short stud)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 03 01-4	Masse (compensation mass)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 03 01-2	Zeitschraube (timing screw)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 03 01-3	Zeitmasse (timing weight)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 02-1	Unruhwelle (balance staff)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 03 02-4	Spiralstift (hairspring pin, at the collet)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 03 02-4	Stift (hairspring pin, at the stud)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 05-3	Gangradbutzen (escape wheel collet; 05 04 10 on 05 04_7.2)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 01-4	Gangfederkloben (detent block)	not on the drawing	its sheet gives no material: a maker chooses one for its function
WN-16	Anschlagschraube 1941 (the 1941 block's banking screw, domed head and pointed tail; WN-16 by the coloured section's label)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 08-2	Fassung (endstone setting), balance, upper	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 08-2	Fassung (endstone setting), balance, lower	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 08-2	Fassung (endstone setting), escape wheel, upper	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 04 08-2	Fassung (endstone cap), escape wheel, lower	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 00-6	Kettenstift (chain pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 00-5	Vorreiberstift (Vorreiber pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 02-4	Stift (click spring pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 00-8	Stellstift f. Gegenfeder (maintaining spring's pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 05 05-2	Gegensperrwelle (its arbor)	not on the drawing	its sheet gives no material: a maker chooses one for its function
WN-27	Schraube (WN-27 by the coloured section's label, likely: the maintaining detent's wire spring's stud)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 05 06-1	Feder (the maintaining detent's wire spring)	not on the drawing	its sheet gives no material: a maker chooses one for its function

Drawing	Name	Material	How
05 06 03-1	Stellungssteg (stop bar)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 04-4	Stift (05 06 04-4: the coloured section's label and 05 06 04's parts list, likely), the stop bar's spring's pin	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 03-2	Stift (05 06 03-2, the coloured section's label), the stop bar's pin, on which its spring's end bears	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 06 03-3	Stegfeder (the stop bar's spring)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 08 01-5	Anschlagstift (stop pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 07 00-1	Steg (spring brace)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 07 03-1	Band (fusee band)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 05 03-3	Minutenrohr (cannon tube)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 02-2	Wechselbutzen (change wheel pipe)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 02-3	Wechseltrieb (change pinion)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 03-2	Stundenbutzen (hour wheel pipe)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 04-2	Auf u. Abrad-Butzen (up-and-down wheel pipe)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 00-5	Auf u. Abtrieb (up-and-down pinion)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 02 01-2	Zeigerfutter (seconds hand collet)	not on the drawing	its sheet gives no material: a maker chooses one for its function
dialPin	Stift (dial pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 01 00-3	Gehäusekloben (case bracket)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 01 00-4	Feststellkloben (locking bracket)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 01 00-7	Verschlußfeder (closing spring)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 01 00-5	Kardanring (gimbal ring)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 01 00-6	Kardanringkloben (ring bracket)	not on the drawing	its sheet gives no material: a maker chooses one for its function
Lagerscheibe	Bearing disc (no number)	not on the drawing	its sheet gives no material: a maker chooses one for its function

Drawing	Name	Material	How
Deckelstuetze	Deckelstütze (lid stay, estimated)	not on the drawing	its sheet gives no material: a maker chooses one for its function
Schluesselhalter	Schlüsselhalter (key holder, estimated)	not on the drawing	its sheet gives no material: a maker chooses one for its function
Schluesselhalter.F	Fassung (the key holder's round fitting, estimated)	not on the drawing	its sheet gives no material: a maker chooses one for its function
Schliessblech	Schließblech (keeper, estimated)	not on the drawing	its sheet gives no material: a maker chooses one for its function
Kastenstift	Pin over the outer pivot (no number; inferred a lid-locating pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
05 01 01 (05 01's list)	Feststeller (transit lock)	not on the drawing	its sheet gives no material: a maker chooses one for its function
51 00 11-1	Schlüsselgriff (key handle)	not on the drawing	its sheet gives no material: a maker chooses one for its function
51 00 11-4	Wendelfeder (the key's coil spring)	not on the drawing	its sheet gives no material: a maker chooses one for its function
51 00 11-6	Kerbstift K.S. 1 (the key's grooved pin)	not on the drawing	its sheet gives no material: a maker chooses one for its function
127-556.02-02	Körper (the 1940 key's body, with its collar)	not on the drawing	its sheet gives no material: a maker chooses one for its function
127-556.02-01	Griff (the 1940 key's flat pear-shaped handle, its tang)	not on the drawing	its sheet gives no material: a maker chooses one for its function
127-556.02-03	Schaft (the 1940 key's stem, with the socket for the fusee's square)	not on the drawing	its sheet gives no material: a maker chooses one for its function
127-556.02-05	Schraube (the 1940 key's screw beside the tang)	not on the drawing	its sheet gives no material: a maker chooses one for its function

CNC milled or wire-cut (27)

Drawing	Name	Material	How
05 08 02-1	Unterplatte (pillar plate)	„Messingblech 4 / SonderMs α-βGef.“	from the drawn sheet or strip („Messingblech 4 / SonderMs α-βGef.“)
05 09 01	Oberplatte (top plate)	„Messingblech 4 / Sonder Ms α-βGef.“ (special brass, α-β structure)	from the drawn sheet or strip („Messingblech 4 / Sonder Ms α-βGef.“ (special brass, α-β structure))
05 08 05-1	Federhausbrücke (barrel bridge)	„Messing blech 3,5 Din 1751 Ms 58“	from the drawn sheet or strip („Messing blech 3,5 Din 1751 Ms 58“)
05 08 05-2	Sperrkegel (click)	„Messingblech 3 Din 1751 Ms 58 federhart“	from the drawn sheet or strip („Messingblech 3 Din 1751 Ms 58 federhart“)
05 05 03-1	Minutenrad (centre wheel)	„Messingblech 1,5 Din 1751 Ms 58 federhart“	from the drawn sheet or strip („Messingblech 1,5 Din 1751 Ms 58 federhart“); its teeth cut to the drawn tooth form
05 05 02-1	Zwischenrad (intermediate wheel)	„Messingblech 1,2 Din 1751 Ms 58 federhart“	from the drawn sheet or strip („Messingblech 1,2 Din 1751 Ms 58 federhart“); its teeth cut to the drawn tooth form
05 05 01-1	Sekundenrad (seconds wheel)	„Messingblech 1,2 Din 1751 Ms 58 federhart“ (brass sheet 1.2 mm DIN 1751, Ms 58 spring-hard)	from the drawn sheet or strip („Messingblech 1,2 Din 1751 Ms 58 federhart“ (brass sheet 1.2 mm DIN 1751, Ms 58 spring-hard)); its teeth cut to the drawn tooth form
05 03 02-2	Spiralrolle (hairspring collet)	„Messingblech 3,5 (overwritten, likely 3,5) Din 1751 Ms 58“	from the drawn sheet or strip („Messingblech 3,5 (overwritten, likely 3,5) Din 1751 Ms 58“)
05 03 02-3	Klötzchen (hairspring stud)	„Flachmessing 15×7 Din 1759 Ms 58“	from the drawn sheet or strip („Flachmessing 15×7 Din 1759 Ms 58“)

Drawing	Name	Material	How
05 04 05-1	Gangrad (escape wheel, 15 teeth)	„Messingblech 1,6 Din 1751 Ms 63 federhart“ (brass sheet 1,6, spring-hard)	from the drawn sheet or strip („Messingblech 1,6 Din 1751 Ms 63 federhart“ (brass sheet 1,6, spring-hard)); its teeth cut to the drawn tooth form
05 06 04-3	Schneckensperrrad (click wheel)	„Ms Blech 2mm DIN 17.51 Ms 58 federhart“ (brass sheet 2 mm, DIN 1751, Ms 58 spring-hard)	from the drawn sheet or strip („Ms Blech 2mm DIN 17.51 Ms 58 federhart“ (brass sheet 2 mm, DIN 1751, Ms 58 spring-hard)); its teeth cut to the drawn tooth form
05 06 00-4	Deckplatte (cover plate)	„Bandstahl 24×1 Din 1016“ (strip steel 24 × 1, DIN 1016)	from the drawn sheet or strip („Bandstahl 24×1 Din 1016“ (strip steel 24 × 1, DIN 1016))
05 06 00-3	Schneckenrad (fusee wheel)	„Messingblech 5,5 Din 1751 Ms 58 federhart“ (brass sheet 5,5 thick, DIN 1751, Ms 58, spring-hard)	from the drawn sheet or strip („Messingblech 5,5 Din 1751 Ms 58 federhart“ (brass sheet 5,5 thick, DIN 1751, Ms 58, spring-hard)); its teeth cut to the drawn tooth form
05 06 02-1	Gegensperrrad (maintaining ratchet)	„Messingblech 3,5 Din 1751 Ms 58 federhart“ (brass sheet 3,5, DIN 1751, Ms 58 spring-hard)	from the drawn sheet or strip („Messingblech 3,5 Din 1751 Ms 58 federhart“ (brass sheet 3,5, DIN 1751, Ms 58 spring-hard)); its teeth cut to the drawn tooth form
05 06 02-3	Sperrkegel (click)	„Bandstahl 11×2 Din 1016“ (strip steel 11 × 2, DIN 1016)	from the drawn sheet or strip („Bandstahl 11×2 Din 1016“ (strip steel 11 × 2, DIN 1016))
05 06 02-2	Sperrfeder (click spring)	„Bronzeband 1,5×0,4 federhart“ (bronze strip 1,5 × 0,4, spring-hard)	from the drawn sheet or strip („Bronzeband 1,5×0,4 federhart“ (bronze strip 1,5 × 0,4, spring-hard))
05 06 01-1	Gegenfeder (maintaining spring)	„Bandstahl 35×1,75 Din 1016“ (strip steel 35 × 1,75, DIN 1016)	from the drawn sheet or strip („Bandstahl 35×1,75 Din 1016“ (strip steel 35 × 1,75, DIN 1016))
05 05 05-1	Gegensperrkegel (maintaining detent)	„Bandstahl 2,25 × 18 DIN 1016“ (steel strip 2.25 × 18)	from the drawn sheet or strip („Bandstahl 2,25 × 18 DIN 1016“ (steel strip 2.25 × 18))
05 07 00-3	Sperrad (set-up ratchet)	„Messingblech 3 Din 1751 Ms58“ (brass sheet 3, DIN 1751, Ms 58)	from the drawn sheet or strip („Messingblech 3 Din 1751 Ms58“ (brass sheet 3, DIN 1751, Ms 58)); its teeth cut to the drawn tooth form
05 07 03-2	Bandhaken rund (band's round hook)	„Bandstahl 10×0,8 Din 1016“ (strip steel 10 × 0,8, DIN 1016)	from the drawn sheet or strip („Bandstahl 10×0,8 Din 1016“ (strip steel 10 × 0,8, DIN 1016))
05 07 03-3	Bandhaken f. Trommel (band's drum hook)	„Bandstahl 10×0,8 Din 1016“ (strip steel 10 × 0,8, DIN 1016)	from the drawn sheet or strip („Bandstahl 10×0,8 Din 1016“ (strip steel 10 × 0,8, DIN 1016))
05 02 02-1	Wechselrad (change wheel)	„Messingblech 1,2 Din 1751 Ms 58 federhart“	from the drawn sheet or strip („Messingblech 1,2 Din 1751 Ms 58 federhart“); its teeth cut to the drawn tooth form
05 02 03-1	Stundenrad (hour wheel)	„Messingblech 1,2 Din 1751 Ms 58 federhart“	from the drawn sheet or strip („Messingblech 1,2 Din 1751 Ms 58 federhart“); its teeth cut to the drawn tooth form
05 02 04-1	Auf u. Abrad (up-and-down wheel)	„Messingblech 1,2 Din 1751 Ms 58 federhart“	from the drawn sheet or strip („Messingblech 1,2 Din 1751 Ms 58 federhart“); its teeth cut to the drawn tooth form
05 02 00-3	Auf u. Abzeiger (up-and-down hand)	„Bandstahl 0,6 × 15“	from the drawn sheet or strip („Bandstahl 0,6 × 15“)
05 02 01-1	Sekundenzeiger (seconds hand)	„Bandstahl 30 × 0,6 Din 1016“	from the drawn sheet or strip („Bandstahl 30 × 0,6 Din 1016“)
05 02 00-4	Zifferblatt (dial)	„Messingblech 1,3 Din 1751 Ms 58“ (1,3 struck, pencil "1,2")	from the drawn sheet or strip („Messingblech 1,3 Din 1751 Ms 58“ (1,3 struck, pencil "1,2"))

CNC turned (8)

Drawing	Name	Material	How
05 08 01-2	Schneckenfutter (fusee bush)	„Rundmessing 13,5 Din 1758 Ms 58“	from its drawn bar („Rundmessing 13,5 Din 1758 Ms 58“); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished
05 08 01-3	Staubfutter (dust bush)	„Rundm s. 15 Din 1758 Ms58“	from its drawn bar („Rundm s. 15 Din 1758 Ms58“); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished
05 05 03-2	Minutentrieb (centre pinion and arbor)	„Präz. Rundstahl 7,2 Din 175“	from its drawn bar („Präz. Rundstahl 7,2 Din 175“); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished
05 05 02-2	Zwischentrieb (intermediate pinion and arbor)	„Präz. Rundstahl 5,2 Din 175“	from its drawn bar („Präz. Rundstahl 5,2 Din 175“); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished

Drawing	Name	Material	How
05 05 01-2	Sekundentrieb (seconds pinion and arbor)	„Präz. Rundstahl 3,5 Din 175“ (precision round steel bar 3.5 mm, DIN 175)	from its drawn bar („Präz. Rundstahl 3,5 Din 175“ (precision round steel bar 3.5 mm, DIN 175)); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished
05 04 05-2	Gangtrieb (escape pinion, z 10, m 0.263, and arbor)	„Präz. Rundstahl 3,4 Din 175“ (precision round steel Ø3,4)	from its drawn bar („Präz. Rundstahl 3,4 Din 175“ (precision round steel Ø3,4)); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished
05 06 04-1	Schneckenwelle (fusee arbor)	„Präz. Rundstahl 7,5 Din 175“ (precision round steel 7,5 dia, DIN 175)	from its drawn bar („Präz. Rundstahl 7,5 Din 175“ (precision round steel 7,5 dia, DIN 175)); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished
05 07 02-1	Federkern (barrel arbor)	„Präz. Rundstahl 14 Din 175“ (precision round steel 14 dia, DIN 175)	from its drawn bar („Präz. Rundstahl 14 Din 175“ (precision round steel 14 dia, DIN 175)); pinions cut on a hobbing or wheel-cutting machine, hardened and tempered as the sheet says, pivots burnished

joinery (1)

Drawing	Name	Material	How
127-556.01 (52 00 16)	Überkasten (outer box): body and base plate, bleached birch	„gebleichte Birke“	(„gebleichte Birke“)

Tests on the bench

The analysis says the design works; only these say a chronometer made to it does. Each figure is the working model's own, for the 1950s build (tools/wempe_bench.js under MC_WREV=build; its tolerances tools/wempe_tolerances.js), or its drawings'. Where the drawings name a test (05 06_20's pull on the fusee arbor), it is theirs.

1. The mainspring's torque

When: before it goes into the barrel, and again in its barrel.

How. In its barrel, the arbor held by a lever (a 3D-printed arm 50 mm long and a hanging weight do: 1 kg at 50 mm is 0.49 N·m) or a torque gauge: wind it from slack and read the torque each half turn, then unwind and read again (the spring's friction is the difference). Leave it wound a week and read again. Once, wind it on until its coils lie tight on the arbor and count the turns from slack.

It must read. Set up ¾ turn past slack (05 06_20's "3/4 Umgang gespannt") and wound the drawn fusee's 5.08 barrel turns, the model's spring runs 546 to 1074 mN·m at the level 05 06_20 measures (the solved shape, wempe_physics.js); every reading within ×0.684–1.253 of it keeps the swing in the guard band over the wind and the strip under a mainspring's stress limit (wempe_tolerances.js). From slack to coil-bound it must have the set-up, the fusee's 5.08 turns and half a turn to spare (the model's 0.93 past full wind). Unwinding reads less than winding: the coils' friction gives 1–3 % on the solved spring, mainsprings 10–20 % (ICMFF9); over 22 %, clean and grease it. After the week, still in the band.

2. The pull on the fusee arbor, against 05 06_20

When: the chain on, the barrel set up ¾ turn, before the train goes in.

How. As Herb. Müller did in 1941 (05 06_20: "Hebellänge 262 m/m", the mean of two runs): a lever 262 mm long on the fusee arbor, and the weight (g) that holds it level read at each half turn running down from full wind (8.75 turns of the fusee, the curve's 0.05 to 8.74), twice; or a torque gauge (1 g on 262 mm is 2.57 mN·m).

It must read. Each reading within ×0.684–1.253 of the sheet's at the same turn (its readings' mean 132.7 g: 90.8–166.4 g; fully wound 138.4 g), and the run's shape the sheet's: its readings −4.7 to +4.3 % about their mean, the hump at turns 1–1.5, the dip at 4–5.75 and the rise to full wind (the solved spring through the drawn fusee fits it to 2.10 % rms). Low everywhere: the strip is thin (set up a tooth more, or a stronger strip: wempe_tolerances.js); out of shape at one end: the spring's length or set against the fusee. The sources' band, 225–270°, would want 134.3 g mean (×1.012): open (README, Unresolved).

3. The train's depths

When: the plates bored and reamed, before the bushes and jewels go in.

How. Gauge pins in each pair of holes and a micrometer, or a measuring machine, on both plates; then each mesh tried in turn with the arbors in, the escape wheel held: feel the shake at the pinion and watch the teeth through a loupe.

It must read. Centres 22.650, 18.120, 12.630, 11.850 mm (fusee–centre, centre–intermediate, intermediate–seconds, seconds–escape: the hole tables' coordinates), each within ±25 µm (gearing.js --model wempe: the seconds mesh's window less its pivots' shake; the tables give the places to 0.001 and no tolerance). A hole that errs should err wide: each mesh takes far more wide than close (−240/+436, −136/+355, −112/+281, −100/+263 µm). Closer than its window less the pivots' shake (220, 101, 62, 50 µm close): re-bush the hole, bored on the drawn centre.

4. Endshake

When: each arbor in its bearings, the plates pinned and screwed, before the escapement goes in.

How. Lift each arbor by its wheel with tweezers and watch its shoulder against a dial indicator, or feel it; the fusee arbor before its face a is re-turned (05 06 00-2: "bei Schneckenmontage … Fläche a nachdrehen").

It must read. At 20 °C, as the Soviet Navy's 1958 MX repair book sets them (Appendix 4; the Wempe's sheets give none; the TU's 4 °C takes 3 µm of it): the centre arbor 0.07–0.10 mm (drawn 0.07, the sheets' tolerances −0.02 to 0.13), the intermediate arbor 0.07–0.10 mm (drawn 0.06, the sheets' tolerances −0.09 to 0.15), the seconds arbor 0.07–0.10 mm (drawn 0.05, the sheets' tolerances −0.11 to 0.15), the escape arbor 0.05–0.07 mm (drawn 0.06, the sheets' tolerances 0.02 to 0.08), the fusee arbor 0.08–0.12 mm (drawn 0.05, the sheets' tolerances −0.09 to 0.08), the barrel arbor 0.07–0.10 mm (drawn 0.06, the sheets' tolerances −0.10 to 0.12); the balance staff 0.04–0.06 at its endstones. The drawings' stacks come out at or just under the book's floors, so expect to free most arbors a little. The pillars' seats (0.3 +0.03 in each plate) take up to 0.06 from every arbor at once: a tight one, check the seats first. Out of the band: set the bush or jewel again, re-turn face a (the fusee), stone the shoulder.

5. The balance and its hairspring vibrated

When: the spring pinned to its collet and stud, its curves bent to 05 03_7, before it goes in.

How. Vibrate the balance on its spring in a vibrating tool (or the cock on a holder) against a master or a phone's timer: count the oscillations over two minutes, the weights against their screws' heads as drawn.

It must read. 2 oscillations a second (7,200 an hour), or slow by up to 293 s a day: the timing weights stand against their screws' heads as drawn and can only be screwed in, 5.0 turns of 58.7 s a day each (wempe_tolerances.js; their size estimated). The drawn spring on the weighed balance (2,501 g·mm²) is 18 min a day fast (0.4937 s): the strip wants 0.2826 mm, not 0.285 (0.2808–0.2854 over the steel's 200–210 GPa). No section tolerance holds the period (a µm of the strip is 455 s a day, the coil's ±0.05 ±173): fast, grind the strip thinner or choose a weaker spring; slow past the weights' reach, a stronger one.

6. The balance runs down free

When: the balance, hairspring and detent in, the train held still at a lock; before the mainspring is chosen.

How. Give the balance a full swing (a twist of the movement) and let it run down, the train held. Film it from above at 30 frames a second, a card marked every 30° under it, and read the swing every few seconds (python wempe_swing.py reads a video so); time how long it takes to halve from about 207°.

It must read. The model's balance halves its swing in 60 s (its TF 111 s; a running 6MX on video, moMvyx8eXfo, has TF 102–125 s at it: wempe_startup.js, which the model's TF is set from). Halving in 46, 55, 60, 66, 79, 93 s (TF 80, 100, 111, 125, 160, 200): 05 06_20's drive swings it 202, 221, 230, 241, 265, 286° at 12 h; the guard's 180° at the wind's lowest wants ×0.860, 0.734, 0.684, 0.633, 0.544, 0.480 of it, the sources' 225° ×1.288, 1.090, 1.012, 0.932, 0.793, 0.693 (the strip as the drive's cube root). Shorter than 46 s, look for a pivot, the endstone or the detent's springs dragging.

7. The detent rings

When: the detent made with its gold spring on, before it goes in.

How. Clamp its foot as the block holds it (a printed jig with the foot's screw and two steady pins), the stone and the horn free, and flick the blade's end lightly with a hair. Record the ring with a phone a few millimetres off and read its lowest strong pitch with a spectrum app, or python benchcheck.py --ring FILE.

It must read. 93 Hz with the spring as drawn (0.070 mm, 1436's "0,07"); 80–134 Hz for a spring 0.063–0.100 mm thick, half to twice the drawn stiffness, over which the detent is home before the next tooth (dynamics.js --model wempe). The pitch goes nearly as the thickness to the 1.5 (0.050 mm 58 Hz, 0.063 mm 80 Hz, 0.060 mm 75 Hz, 0.070 mm 93 Hz, 0.080 mm 109 Hz, 0.100 mm 134 Hz, 0.090 mm 123 Hz: wempe_tripping.js's frame clamped at the foot), so it measures the spring's section, which no figure on 1436 settles (its "0,09" would ring 123 Hz). The gold spring rings near 213 Hz: the detent's own is the lower.

8. The escapement's figures and the banking

When: the detent, its gold spring and the balance in; the train's power let down.

How. Turn the balance slowly by hand with a protractor card under it and read each angle through a loupe; the depth of lock and the roller's shake with feeler gauges or under a measuring microscope. The preload: wire riders of known weight hung on the stone's pipe, the detent on its banking (as the Hamilton manual's Op. 78 hangs its 0.770 g).

It must read. 05 04_1's layout and the model: the impulse stone's face 28°10' before the line of centres at rest, the locking stone 37.0° on in the wheel's turn and 7.14 from its centre; the gold spring met at -8.8° of the balance (05 04_1's −12°), the wheel released at 4.0° (lock 12.8°), the gold spring off the stone at 6.9° (let-off 2.8°), the return swing past it at -21.6° (overall 28.5°); roller shake 0.011 mm, the teeth 0.39 mm into the roller. Set the banking (the block's cross screw, the 1954 instruction's п. 11) for a lock 0.187 mm deep at the stone, the middle of its window (the drawing's 0.235 lies 146 µm over its floor): never shallower than 0.089 (a held tooth's stone rings more than half the depth: wempe_tripping.js) nor deeper than 0.285 (the discharge stone can't lift the stone further: it doesn't unlock). A wheel at the top of its 14 +0.03 locks 0.250 deep where the drawn wheel locks 0.235: set the banking to the wheel as made. The preload at the stone: a 0.64 g rider must leave the detent on its banking and a 0.80 g one part it; aim near 0.68 g (the model's 0.8 g is an estimate: wempe_tripping.js finds a held tooth's stone ringing 44 % of the depth of lock there, under half from 0.40 g, teeth getting past at a fifth of it and under; and a 3 ms blow of 20 g across the plate lets a tooth go under 0.64 g, the floor taken here, the higher; each gram more costs about 10.4° of swing; never over 0.80 g: above it the stone, falling back after the release, lands again on the escaping tooth's back during the impulse, where the models part: dynamics.js with brass's restitution). The return pass: turn the balance back slowly by hand through the discharge with the train's power on and watch the locking stone under a loupe: the discharge stone must bend the gold spring aside without lifting the detent (wempe_tripping.js: as the model takes the stone's contact on the gold spring no tooth gets past over 333 gold's whole range of modulus, 83–110 GPa, but at 4 times that contact's stiffness 6 in 12 do, whatever the alloy: only the made spring can show it); its tip at rest 0.10–0.15 short of the discharge stone and 0.35–0.40 past the detent's end (the 1958 Navy repair book's checks, p. 60).

9. The swing

When: the movement running, fully wound, at 24 h and near run-down.

How. Read the balance's swing against marks under it, from a phone's slow-motion video (python wempe_swing.py reads the arm's sweep), or from a recording of the beat (python acoustic.py --model wempe FILE fits its clicks to the escapement in time).

It must read. In the guard band, 180–270° each way, over the whole wind (the model 223.5–237.2°, 230.3° at 12 h). The sources' band, 225–270° (the 1954 instruction, chronometers in general; the Hamilton manual 247.5–270°), is reported, not required: the model's swing is under it on 05 06_20's drive (open, README). Low at both ends with the train free: the strip doesn't suit the balance's damping (the run-down test chooses it), or the preload is high; low at one end only: the fusee's test.

10. The maintaining spring drives the train

When: the movement running, part wound.

How. First, assembling: the spring's eye pin up into the ratchet (05 06_4 A–B: "Stift nach oben"), its other pin down into the fusee wheel; the ring lies free in the wheel's recess, its eye a gap short of its square end. Running, the ratchet stands ahead of the wheel as the drive closes the ring: see its ends nearly meet, not touch. Then turn the key back a little (winding) and hold it still, the fusee off the train: time how long the balance keeps going.

It must read. Running, the ring closed 17.8–19.7° by the drive, its ends −0.74…−0.28 mm apart at the eye (they would meet after 16.7°; wempe_sustaining.js); the key held, the balance keeps going 11.4–14.6 minutes (the detent catching a whole tooth late or at once), its swing falling; the 1954 instruction gives 1–2 minutes for chronometers in general; a wind takes 10–30 s. Under 5 minutes (estimated): the ring is open (the pins the wrong way round: it bears on the recess's wall and barely gives), weak, or has taken a set; its ends touching in running: weak or set.

11. The stop work: the bar springs back, the stop holds full wind

When: the bar, its spring and pin in the fusee's small end, the cover plate off; then the fusee in its plates, the chain on and the barrel set up, the train out.

How. Cover plate off: push the bar in from its lower end with a pegwood to its full 3.6 and let go, the small end up and then down; repeat ten times. Then, assembled, wind slowly to the stop and count the fusee's turns from run-down; let the chain run back a quarter turn and wind again.

It must read. The bar springs fully out at once every time, in both positions: its spring (05 06 03-3; no source draws its section: any height 0.9–1.5, its pin's hole to the recess's floor, 0.10 thick; wempe_strength.js) pushes it 0.06–0.08 N half way at the model's 1.0 high, 41 times or more the 0.5 g bar's sliding on the cover plate (μ 0.3, estimated). Assembled: the stop pin (hole 7) stops the fusee with the chain 8.638 turns on it (8.70 from run-down), 0.112 turn short of the groove's end N, and the bar never touches the pin earlier (0.69 off a turn before: wempe_stopwork.py). Sluggish: clean the slot and the bar, and look for a burr at the slot's ends; still slow, bend the spring's free span further in, keeping it under its steel's yield (wempe_strength.js). Stopping short of 8.64 turns: the bar is caught early (its spring too weak to hold it out, or the bar's upper end too long).

12. The stop takes a strong hand, and the key is one-way

When: the stop bar and pin hardened, the movement wound to the stop; the key made.

How. File-test the stop bar and the stop pin (a file skates on hardened steel). Wound to full wind, hold the key against the stop with a torque gauge on its square (or a lever and a weight) up to 1.36 N·m, a strong hand's on the key's Ø 34.5 paddle (wempe_overwind.js). Then turn the key the wrong way on a spare square held in a vice with a gauge: it must slip.

It must read. Nothing moves or marks: hardened, the stop bar yields at 3.73 N·m on the key and the pin at 9.97; left soft the bar bends at 1.41. The key's coil spring carries the hand: wound of spring wire Ø 0.68 or more (EN 10270-1 DH; 51 00 11 scales about 0.6, which yields at 1.57), its end in the body's hole formed on a radius. The wrong way it slips at a few mN·m (the model takes 5): one that grips both ways sends the hand into the click wheel, whose teeth yield at 0.61 N·m, as the plain 1940 key would.

13. The cannon pinion's friction

When: the motion work and the hands on, the movement level.

How. Hang a weight on the minute hand 30 mm from its arbor, the hand horizontal, and see whether it turns the hand (the cannon pinion slipping on the centre arbor).

It must read. It slips between 10.3 and 247 g (wempe_motionwork.js, wempe_transmission.js's window: looser, a 45 g knock turns the hands; tighter, setting them forward drives the balance to the escapement's trip): set it near 51 g. Set the hands forward only (the 6MX's instruction: backward "can break the escape wheel's teeth"): a fit near the window's top would turn the train back against its locking stone near run-down.

14. The rate, and the 6MX TU's trial

When: cased, in its gimbals, rated.

How. Each day at the same hour compare it with a time signal to a tenth of a second and log the error (python rating.py works the daily rates). The trial: 35 days in five periods of seven at 36, 20, 4, 20, 36 °C (a wine cooler, a room, a box with a

lamp and a thermostat), the first two days of each not counted, wound daily after the comparison to "8" on the up-and-down dial, to "0" at each period's end.

It must read. The TU's figures (losing positive): E, the mean of |each counted rate − its period's mean|, within ±0.35; R = ω_V − ω_I ±2.00; D, the largest change between consecutive rates in a period, ±2.30; C = (ω36 − ω4)/32 ±0.10 s a day a °C; S = (ω36 + ω4)/2 − ω20 ±1.20; every period's mean ±3.50; N = 42E + 4.5R + 60C + 2.4S ≤ 30 (wempe_errorbudget.js passes it as adjusted). Out on C: move the four masses along the rim ("8 Körner in 60 Teilung", 6° a division) by trial: at the drawn place (30° from the cuts' line) the balance loses 2.55 s a day a °C, they null it near 19.3° from the cuts' line, a degree along the rim 0.228 s a day a °C (wempe_thermal.js). For a chronometer to navigate by (about 2 miles of longitude a month: wempe_errorbudget.js's voyage, 6 s at 30 days against 17 s with the 6MX passports' ±0.45°), go on past the TU's ±0.10: repeat the 4 °C and 36 °C periods (five counted days each, which measure C to about ±0.005) and move the masses by C / 0.228 degrees each round until |C| is within ±0.023 s a day a °C, the masses within ±0.1° of their place: 27 µm along the rim at r 15.7, both of a pair moved alike to keep the poise, against a scribed line under a 10× loupe; two or three rounds. Out on S past ±1.20: the nickel steel isn't Guillaume's integral alloy (wempe_thermal.js).